Corporate Finance for MBA Leaders

Valuation, Financing, Risk and Corporate Strategy

A Comprehensive Textbook for MBA Students

About This Book

This textbook develops the core ideas of corporate finance for MBA students and practising managers: how firms value investments, finance them, manage risk, and create or destroy value through major transactions. Each chapter contains worked examples, a case study, review questions and problems with selected answers. All companies, figures and cases are fictional and illustrative unless stated; they are for teaching only and are not investment advice. Figures were computed by the authors' working and should be verified before use in professional decisions.

Chapters are organised in six parts: Foundations; Investment Decisions; Risk, Return and the Cost of Capital; Financing and Payout Policy; Derivatives and Risk Management; and Corporate Transactions and Strategy. Appendices provide a formula sheet, a glossary and a study guide.

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Part I — Foundations

Chapter 1: The Corporation and the Finance Function

Learning Objectives

After studying this chapter you should be able to:

- Describe the three core decisions of corporate finance (investment, financing and payout) and explain how the finance function is organised to support them.
- Explain why maximising the long-run value of the firm's equity is the operating objective of corporate finance, and distinguish it from profit maximisation, earnings-per-share targets and stakeholder-neutral slogans.
- Identify the principal agency conflicts (shareholders and managers, shareholders and creditors, controlling and minority shareholders) and quantify their cost in simple settings.
- Evaluate governance mechanisms such as boards, incentive pay, creditor covenants, the market for corporate control and regulation, and judge which fit a given situation.
- Connect financial analysis to strategy using return on invested capital, economic profit and the value-driver formula.
- Describe the structure and function of financial markets and intermediaries, and interpret an announcement-day abnormal return as evidence about how markets value corporate decisions.

Opening Vignette

Illustrative scenario. Northgate Industrial Group is a fictional company.

On a Tuesday in March, the board of Northgate Industrial Group met to approve its capital budget. Northgate makes industrial pumps and valves, employs about 6,000 people and has a market capitalisation of roughly $1.2 billion and 100 million shares outstanding. The chief executive, Marguerite Ellison, presented a plan that included a $380 million acquisition of Delmar Flow Systems, a smaller rival. Delmar's standalone value, by the finance team's best estimate, was $300 million. The synergy case, built mainly on purchasing savings and cross-selling, was worth perhaps $40 million in present value. The price therefore exceeded everything the deal could plausibly deliver.

The debate that followed was revealing. The head of corporate development argued that Delmar would add 14% to revenue and make Northgate the second-largest firm in its niche. The chief operating officer pointed out that the deal would lift earnings per share in year one because Northgate could borrow cheaply. An independent director who had spent a career in private equity asked a different question: "What does a Northgate shareholder own after this transaction that she did not own before, and what did she pay for it?" The chief financial officer, Tomas Reyes, had prepared for exactly that question. He showed that Northgate would pay $80 million more than Delmar's standalone value, $40 million of it beyond any synergy, and that on a fair reading of the numbers the share price should fall by about $0.40.

Ellison's annual bonus was tied to revenue growth and to the size of the company relative to its peers. Reyes's was tied to earnings per share. Neither measure punished the board for overpaying. The independent director proposed that executive compensation and acquisition approval both be reframed around returns above the cost of the capital employed.

Nothing in this story is unusual. It contains almost every theme of this chapter: the objective of the firm, the gap between managers' interests and owners' interests, the machinery of governance that tries to close the gap, the link between finance and strategy, and the role of markets that will grade the decision within hours of its announcement. This chapter builds the framework within which the rest of the book operates.

1.1 What Corporate Finance Is, and Who Does It

1.1.1 The three decisions

Corporate finance is the study of how firms decide which real assets to acquire, how to pay for them, and how to return money to the people who supplied it. Every topic in this book, from capital budgeting to option pricing to mergers, is a refinement of one of three decisions.

1. **The investment decision (capital budgeting).** Which projects, products, acquisitions and assets should the firm commit resources to? This decision determines the left-hand side of the balance sheet: the operating assets the firm uses to generate cash flows. A manager who builds a plant, launches a brand, buys a competitor or invests in software is making an investment decision, whatever department she belongs to.
2. **The financing decision (capital structure).** How should the firm raise the money for those investments? The choices include retained earnings, new equity, bank loans, bonds, leases and hybrid securities. This decision determines the right-hand side of the balance sheet and the risk borne by each class of investor.
3. **The payout decision (dividend and repurchase policy).** When the firm generates more cash than it can profitably reinvest, how much should it return to shareholders, and in what form? Payout determines how much of the firm's cash flow stays inside the firm, under management's control, rather than going back to owners.

A fourth, operational theme underlies these: **working-capital and liquidity management**, the short-term decisions that keep the firm able to pay its bills while pursuing long-term goals. We treat it in detail later; Chapter 2 introduces the ratios that diagnose it.

The decisions interact. A firm cannot judge whether a project is worth doing without a view of the return investors require, and that return depends on how the project is financed and how risky its cash flows are. A firm that over-borrows may have to cancel good projects. A firm that hoards cash may invite bad ones. Teaching the three decisions separately is a pedagogical convenience; practising managers face them at once.

1.1.2 The organisation of the finance function

In a large corporation the finance function is typically led by a **chief financial officer (CFO)**, who reports to the chief executive and often sits on the board or attends its meetings. Beneath the CFO, responsibilities are usually divided between two broad roles.

- The **controller** (or chief accounting officer) is responsible for recording transactions, producing the financial statements, maintaining internal controls, managing tax compliance and supporting external audit. The controller looks backward and sideways: what happened, and is the record accurate?
- The **treasurer** is responsible for raising capital, managing cash and liquidity, handling banking relationships, managing foreign-exchange and interest-rate exposure and, often, investor relations. The treasurer looks forward and outward: how will we fund the plan, and what can go wrong with prices and markets?

Many firms add **financial planning and analysis (FP&A)**, a group that builds budgets, forecasts and variance analysis and supports business units; **corporate development**, which originates and evaluates mergers, acquisitions and divestitures; and **internal audit** and **risk management**, which should report with some independence from the CFO.

The modern CFO's role extends well beyond stewardship. Surveys of practice over recent decades describe CFOs as strategic partners to the chief executive: they shape portfolio decisions, communicate with investors and analysts, and challenge business-unit proposals. This is also where tension arises. A CFO who is too close to management may bless every proposal; a CFO who is too distant may be ignored. Part of the skill that this book aims to develop is the ability to ask the right question at the right moment, as the independent director did in the vignette.

1.1.3 Why non-financial leaders need finance

You do not need to become a CFO to need corporate finance. A marketing leader proposing a brand-building campaign is asking for capital that will pay back over several years; the proposal is an investment decision. An operations leader choosing between a flexible factory and a cheaper dedicated one is making a trade-off about fixed versus variable cost and operating leverage. A product leader negotiating a transfer price is influencing which division appears to create value. A start-up founder deciding how much equity to sell is making a financing decision with consequences for control. Fluency in the language of cash flows, discount rates and risk is the price of being taken seriously in resource-allocation debates.

1.2 Forms of Organisation and the Goal of the Firm

1.2.1 Why the corporation dominates large-scale business

Businesses are organised as sole proprietorships, partnerships, limited liability companies and corporations. Each trades off simplicity, taxation, liability and the ability to raise capital. The corporation's defining features explain its dominance among large firms.

- **Separate legal personality.** The corporation owns assets, signs contracts and sues or is sued in its own name. It outlives any individual owner.
- **Limited liability.** A shareholder can lose what she invested but, in general, no more. This caps downside risk, makes it rational to hold a diversified portfolio of many firms, and makes it possible for strangers to pool capital.
- **Transferable ownership.** Shares can be sold without the consent of other owners and without dissolving the firm. Liquidity in the shares lowers the return investors demand.
- **Centralised management.** Owners elect a board, which appoints managers. Specialised professionals can run the firm, and owners can hold many other things besides.

These features create the firm's strength and also its central problem. Pooling capital from thousands of dispersed owners makes large projects possible, but it separates **ownership** from **control**. The people who bear the financial consequences of decisions are not the people who make them. Adolf Berle and Gardiner Means described this separation in a classic study of the American corporation, and Michael Jensen and William Meckling formalised its consequences in 1976 in the theory of agency costs we develop in Section 1.3.

Form	Liability of owners	Ease of raising large capital	Typical tax treatment	Continuity
Sole proprietorship	Unlimited	Very low	Income taxed to the owner	Ends with the owner
General partnership	Unlimited, joint	Low	Pass-through to partners	Depends on partnership agreement
Limited liability company	Limited	Moderate	Flexible; often pass-through	Governed by operating agreement
Corporation	Limited	High	Corporate tax, then tax on distributions (jurisdiction-dependent)	Perpetual

The tax row varies across countries and over time, so treat it as a pattern rather than a rule.

1.2.2 The objective: maximise the value of the equity

If the firm has multiple stakeholders and multiple conceivable goals, how should a manager choose? Corporate finance answers with a clear objective: **take the actions that maximise the current market value of the shareholders' equity**, subject to honouring contracts and obeying the law.

The logic rests on several arguments.

1. **Shareholders are the residual claimants.** Employees are paid wages, lenders interest, suppliers invoices, governments taxes; shareholders receive whatever is left. Because the residual is uncertain, shareholders bear the risk of every decision, and they benefit most when the decision is good. Giving them the decision rule aligns the person who gains from a good decision with the person who chose it.
2. **Value maximisation is a single, measurable criterion.** Competing goals, such as growth, market share, employee satisfaction and profit, cannot be traded off without a common currency. Market value is that currency. It incorporates the size, timing and risk of all future cash flows, a combination that no accounting number captures.
3. **Value is the surplus available for everyone.** A firm that creates more total value can, in principle, pay its employees and suppliers more or invest more in the community. A firm that destroys value will eventually have fewer resources for every constituency.
4. **Markets discipline the alternatives.** A firm that pursues other goals at the expense of value is vulnerable to takeover, activist investors and higher capital costs.

This is not the same as saying that managers should ignore other stakeholders. Treating customers, employees and suppliers well is usually a route to value, because their loyalty, effort and cooperation are productive assets. Environmental and social liabilities have real cash-flow consequences. A modern view of value maximisation is therefore long-horizon and attentive to stakeholder relationships. Where stakeholder commitments cost money without producing any compensating benefit, the manager is spending the owners' money and should be transparent that she is doing so, ideally with owner consent.

1.2.3 What value maximisation is not

Several common objectives sound similar to maximising value but lead to different decisions.

Profit maximisation. Accounting profit ignores the timing of cash flows, ignores risk, ignores the capital used to earn it and can be altered by accounting choices. A firm can raise this year's profit by cutting research and maintenance, stretching suppliers, or taking on projects whose returns fall below the cost of capital but above zero.

Earnings-per-share (EPS) maximisation. EPS rises when a firm acquires a target with a lower price-to-earnings ratio using its own more highly valued stock, or when it borrows to buy back shares, even if neither action creates any real value. EPS is also silent on risk: a leveraged firm's EPS is higher, but so is its volatility.

Market-share or revenue growth. Growth is valuable only if the return on the new investment exceeds the cost of capital. Section 1.5 shows that growth at a return below the cost of capital destroys value.

Short-term share price maximisation. The goal is the current price, which reflects expectations of the entire future. Taking actions that inflate the price today at the expense of the long run is not value maximisation, and informed markets tend to see through it. The key phrase is *current* value, which already discounts long-run effects.

Example 1.1 – Value maximisation versus accounting profit. Hartwell Components, a fictional firm, has 20 million shares outstanding. It considers a project requiring a $40 million outlay today. The project will generate cash flows of $14 million at the end of each of the next five years. For simplicity assume no taxes, and depreciation of $8 million per year on a straight-line basis. The firm's cost of capital for projects of this risk is 10%.

Step 1 – Accounting view. Annual accounting profit is cash flow minus depreciation: 14 − 8 = $6 million. Average book investment is (40 + 0) / 2 = $20 million, so the accounting return on average investment is 6 / 20 = 30%. By this measure the project looks outstanding.

Step 2 – Present value of the cash flows. The five-year annuity factor at 10% is [1 − 1.10^−5] / 0.10 = 3.7908. Present value = 14 × 3.7908 = $53.07 million.

Step 3 – Net present value. NPV = 53.07 − 40 = **$13.07 million**.

Step 4 – Effect on shareholders. If the market recognises the project's value, the share price rises by 13.07 / 20 = **$0.65 per share**.

Step 5 – Internal rate of return. The discount rate that makes NPV zero is found by trial: at 22.1%, 14 × annuity factor = 40, so IRR ≈ **22.1%**.

Interpretation. The accounting return of 30% overstates the true rate of return because it divides by *average* book value and ignores the time value of money. The project is nevertheless worth doing, and the proper measure of how worth doing is the NPV of $13.07 million, the expected increase in shareholder wealth. Accounting returns can also point the wrong way: a project whose cash arrives late can show a high accounting return yet have a negative NPV, while a project with modest early profits but a large terminal value can be strongly positive. The decision rule that aligns with shareholder wealth is: accept projects with NPV greater than zero.

1.2.4 The stakeholder debate

Critics argue that maximising shareholder value is too narrow and encourages short-termism, inequality and environmental harm. Defenders reply that stakeholder maximisation without a priority gives managers no usable decision rule and no accountability: a manager who is responsible to everyone is accountable to no one. Both sides have a point, and a useful synthesis exists.

- **Enlightened value maximisation** (sometimes called enlightened stakeholder theory) holds that a firm cannot maximise long-run value if it ignores the interests of important constituencies, but also insists that managers keep value as the tiebreaker when interests conflict.
- Where the law or contracts protect stakeholders, as with workplace safety, environmental rules and creditor rights, the manager treats those protections as constraints within which she maximises value.

- Where the firm voluntarily goes beyond the law, the manager should be able to articulate how the choice affects long-run cash flows (reputation, recruitment, regulatory risk, customer loyalty) or that owners have accepted the cost for non-financial reasons.
- Some corporate forms, including public-benefit corporations, write a broader purpose into the charter explicitly. This resolves the accountability problem by changing the contract rather than ignoring it.

For the purposes of this book, the operating objective is long-run equity value, interpreted with attention to stakeholder relationships and externalities. It is the criterion against which investment, financing and payout decisions are tested in later chapters.

1.3 Agency Problems

1.3.1 The principal-agent relationship

An **agency relationship** exists whenever one party (the principal) delegates decision-making authority to another (the agent) whose interests differ. Shareholders are principals and managers are agents. The difficulty arises because (a) interests diverge, (b) the principal cannot observe everything the agent does or knows, and (c) the agent's actions affect the principal's wealth. Jensen and Meckling defined **agency costs** as the sum of three elements:

1. **Monitoring costs** borne by the principal to limit the agent's deviant behaviour (audits, boards, reporting requirements).
2. **Bonding costs** borne by the agent to assure the principal that she will behave (contractual commitments, certified accounts, restrictive covenants).
3. **Residual loss**, the decline in the principal's welfare that remains even after monitoring and bonding, because perfect alignment costs more than it is worth.

Agency theory does not require anyone to be dishonest. Rational, honest managers can still prefer larger firms, quieter lives, and safer paths than diversified owners would choose, and they can persuade themselves that these preferences are the owners' interest.

1.3.2 Shareholders and managers

The classic manager-shareholder conflicts include the following.

- **Perquisite consumption and shirking.** Executives may spend firm resources on luxuries or exert less effort than owners would wish, because they bear only a fraction of the cost.
- **Empire building.** Managers' pay, status and security often rise with firm size, which gives them an incentive to grow beyond the point where growth adds value. Acquisitions are the prime vehicle.
- **Risk aversion.** A manager's wealth is concentrated in her job and her reputation, while shareholders can diversify. She may therefore reject positive-NPV but risky projects, preferring the quiet life.
- **Horizon problems.** A manager nearing retirement has less incentive to invest in projects that pay off after she leaves, such as research and development, while one with a short-term bonus may boost near-term earnings at the expense of the long term.
- **Entrenchment.** Managers may adopt strategies that make them hard to replace, such as investing in assets that depend on their special skills, resisting takeovers, or packing the board with allies.

- **Free cash flow.** Jensen's free-cash-flow hypothesis argues that when a firm generates cash beyond its profitable investment opportunities, managers tend to retain and misspend it rather than return it. High leverage and payouts discipline this behaviour by committing cash to outsiders.

Example 1.2 – The price of empire building. Return to the Northgate acquisition from the opening vignette. Northgate has 100 million shares and a market capitalisation of $1,200 million before the deal (so $12.00 per share). Delmar's standalone value is $300 million, the present value of synergies is $40 million, and Northgate would pay $380 million in cash.

Step 1 – Value created by the combination. Standalone value of the target plus synergies: 300 + 40 = $340 million.

Step 2 – NPV to the acquirer. NPV = value acquired − price paid = 340 − 380 = **−$40 million**.

Step 3 – Premium. The premium over standalone value is 380 / 300 − 1 = 26.7%. Because synergies are only 40 / 300 = 13.3% of standalone value, the acquirer is paying away all the synergies and a further $40 million of its own shareholders' wealth.

Step 4 – Effect on the share price. Equity value after the deal, if the market is efficient and believes the estimates: 1,200 + 340 − 380 = $1,160 million. Share price = 1,160 / 100 = **$11.60**, a fall of $0.40 (3.3%).

Step 5 – Who gains? Delmar's shareholders receive $380 million for something worth $300 million standalone: a gain of $80 million. Of that, $40 million is their share of synergies they did not create alone and $40 million is a transfer from Northgate's owners.

Interpretation. The deal enlarges Northgate and may raise EPS, but it transfers $40 million of wealth from Northgate's shareholders to Delmar's. A manager rewarded on revenue and EPS has no reason to refuse; a manager rewarded on value creation would. The empirical finding that acquirers' shareholders earn around zero or slightly negative returns on average, while targets gain substantially, is consistent with this pattern, although the average hides many good deals.

1.3.3 The cost of outside equity: a Jensen-Meckling illustration

Jensen and Meckling's insight can be seen in a simple model. Suppose an owner-manager runs a firm and consumes perquisites. When she owns 100% of the firm, each dollar of perks costs her a dollar of firm value, so she consumes only until the marginal benefit of perks equals their cost. When she sells part of the firm to outsiders, she bears only her ownership fraction of each perquisite dollar, and she will consume more.

Example 1.3 – Selling equity to outsiders. An entrepreneur owns 100% of a firm worth $1,000 million if run with no excess perquisites. She sells 40% to outside investors and keeps 60%. After the sale she will consume extra perquisites costing the firm $80 million of value, but each dollar of perquisites is worth only $0.50 to her. Investors understand this and price the shares accordingly.

Step 1 – Value of the firm after perquisites. 1,000 − 80 = $920 million.

Step 2 – Price rational outsiders pay for 40%. 0.40 × 920 = **$368 million**. They do not pay 0.40 × 1,000 = $400 million because they anticipate the perquisites.

Step 3 – The entrepreneur's wealth. She holds cash of $368 million, a 60% stake worth 0.60 × 920 = $552 million, and perquisites worth to her 0.50 × 80 = $40 million. Total = 368 + 552 + 40 = **$960 million**.

Step 4 – Her wealth if she could commit to no perquisites. She would sell the 40% for $400 million and keep a 60% stake worth $600 million, total **$1,000 million**.

Step 5 – The agency cost. The difference, 1,000 − 960 = **$40 million**, is the residual loss. It is borne by the entrepreneur herself, because outsiders paid a fair price. The firm's value fell by $80 million, but she recovered $40 million through the perks she values; the net deadweight loss is $40 million.

Interpretation. The agent bears the cost of her own lack of commitment, so she has an incentive to invest in bonding and monitoring: independent audits, a board with outside members, a perquisite policy in the bylaws. Each such device is worth adopting as long as it costs less than the loss it removes. This is why firms voluntarily subject themselves to external audit and why public-company governance rules are generally acceptable to issuers, though the optimal intensity varies by firm.

1.3.4 Shareholders and creditors

Debt introduces a second conflict. Creditors have a fixed claim and receive no extra reward from success, but absorb losses if the firm fails. Shareholders, who hold the residual, benefit disproportionately when risk rises. Equity can be viewed as a call option on the firm's assets, with the face value of debt as the exercise price, and an option's value increases with the volatility of the underlying asset. This leads to three classical problems.

- **Asset substitution (risk shifting).** After raising debt, the firm switches to riskier projects, shifting value from creditors to shareholders.
- **Underinvestment (debt overhang).** A highly indebted firm may reject positive-NPV projects because the benefits would accrue largely to creditors.
- **Claim dilution and payout.** The firm may issue more debt of equal or higher priority, or pay large dividends or repurchases, leaving less for existing creditors.

Creditors anticipate these behaviours and protect themselves through higher interest rates, covenants, collateral, shorter maturities and seniority. The costs of the conflict thus return to shareholders as a higher cost of debt.

Example 1.4 – Asset substitution. A firm has debt with a face value of $100 million due in one year, and it must choose between two projects that each require the same investment. The asset values at the end of the year are:

- Project A (safe): worth $120 million with certainty.
- Project B (risky): worth $190 million with probability 50%, and $30 million with probability 50%.

Step 1 – Expected value. A: 120. B: 0.5 × 190 + 0.5 × 30 = **$110 million**. Project A is worth $10 million more to the firm as a whole.

Step 2 – Equity under A. Debt is repaid in full, so equity = 120 − 100 = **$20 million**.

Step 3 – Equity under B. In the good state, equity = 190 − 100 = 90. In the bad state, the firm is worth 30, below the debt, so equity gets 0 (limited liability). Expected equity = 0.5 × 90 + 0.5 × 0 = **$45 million**.

Step 4 – Creditors under B. Good state: 100. Bad state: 30. Expected = 0.5 × 100 + 0.5 × 30 = **$65 million**, versus $100 million under A.

Step 5 – Check. 45 + 65 = 110, the total expected value of B.

Interpretation. Shareholders prefer the inferior project B because their expected payoff rises from $20 million to $45 million, even though total value falls by $10 million. Creditors lose $35 million in expected value. Shareholders gain $25 million, creditors lose $35 million, and the net loss of $10 million is the cost of the conflict. If lenders foresee this, they will

demand covenants or a higher interest rate before lending. This is a central reason why highly leveraged firms in volatile industries face tighter contracts.

1.3.5 Controlling and minority shareholders

In many countries, including much of Europe, Asia and Latin America, large firms are not widely held. A founding family, a state, a bank or a business group holds a controlling stake. The main conflict then is not between dispersed owners and managers but between **controlling** and **minority** shareholders. Controlling holders may extract private benefits through related-party transactions, transfer pricing within a business group, dual-class share structures that separate votes from cash-flow rights, or dilutive issuance. The legal protection of minority shareholders, the quality of disclosure and the independence of directors are therefore crucial in these settings. A manager valuing a controlled company, or investing alongside a controlling shareholder, should adjust for the risk that value will not reach minority investors.

1.3.6 Information asymmetry

A related friction is **information asymmetry**: managers know more about the firm's prospects than outside investors do. Akerlof's analysis of markets with hidden quality, and later Myers and Majluf, showed that this affects financing choices. If managers know the stock is overvalued, they have an incentive to sell it, and so investors treat a new equity issue as bad news; the share price tends to fall on announcement. Firms respond by preferring internal funds, then debt, then equity (the pecking-order theory), and by using costly signals, such as dividends or insider ownership, to convey confidence. We return to these ideas in the chapters on capital structure and payout policy.

1.4 Corporate Governance

Corporate governance is the system of rules, practices and institutions by which a firm is directed and controlled, and in particular the mechanisms that reduce agency costs. No single mechanism works everywhere; they substitute for and complement one another.

1.4.1 The board of directors

The board is the owners' first line of defence. It appoints, evaluates and removes the chief executive, approves strategy, major investments and financing, oversees risk and the integrity of financial reporting, and sets executive compensation. Its effectiveness depends on several characteristics.

- **Independence.** Directors with no material ties to management are more likely to challenge proposals. Independence is not a guarantee, as friendly outsiders can be as compliant as insiders, but it is a precondition for challenge.
- **Expertise and time.** Directors need financial and industry knowledge, and enough time to read the materials. Overcommitted directors are weak monitors.
- **Information.** Management controls the information flow. Strong boards demand direct access to the CFO, internal audit and external auditors, and meet without management present.
- **Structure.** Separation of the roles of chair and chief executive is common in many jurisdictions and hotly debated in others. Audit, compensation and nomination committees staffed by independent directors are standard in listed companies.
- **Incentives.** Directors who hold meaningful equity have a stake in outcomes; those paid mainly by fees do not.

1.4.2 Executive compensation

Pay is the most direct tool for aligning interests, and also a source of problems when badly designed. Typical components are base salary, an annual bonus, long-term incentives such as restricted stock and stock options, and benefits including pensions and severance.

Principles of good design.

- *Pay for performance the executive controls and that matters to owners.* Measures should track value creation, such as economic profit or total shareholder return relative to peers, rather than size or accounting measures that can be manipulated.
- *Match the horizon.* Vesting periods and clawbacks should extend beyond the point at which the consequences of decisions become visible.
- *Avoid rewarding luck.* Raw share-price-based pay rises with the market tide. Relative performance measures and indexed options filter out market-wide movements.
- *Avoid distorting risk.* Options have convex payoffs: they reward upside and cost nothing on the downside, so an option-heavy package can encourage excess risk, much like the shareholder's position in Example 1.4. Restricted stock, in contrast, exposes the manager to losses too.
- *Manage manipulation.* Any metric that triggers a bonus creates a reason to manage the metric.

Example 1.5 – What a stock option actually pays. An executive receives 200,000 at-the-money options with a strike price of $40 per share. Her annual bonus also equals 0.5% of net income, and net income is $120 million.

Step 1 – Bonus. $0.005 \times 120{,}000{,}000 =$ **$600,000**.

Step 2 – Option payoff if the share price at exercise is $55. Intrinsic value per option = 55 − 40 = $15. Total = 200,000 × 15 = **$3,000,000**.

Step 3 – Option payoff if the share price is $35. Intrinsic value = max(35 − 40, 0) = 0. Total = **$0**.

Step 4 – Asymmetry. A share price rise of $15 earns her $3.0 million; a fall of $5 costs her nothing out of pocket. A shareholder holding the same 200,000 shares would have gained $3.0 million in the first case and lost $1.0 million (200,000 × 5) in the second.

Interpretation. The option holder faces only upside, so she may favour riskier strategies than a shareholder would. Boards often counter this by granting restricted stock, requiring share-ownership guidelines, and including downside-sensitive measures such as returns on capital in the performance mix. The bonus formula tied to net income, meanwhile, rewards profit regardless of the capital used to earn it, the same weakness that makes profit an imperfect objective.

1.4.3 Creditors, lenders and debt discipline

Creditors monitor firms through covenants, reporting requirements and the discipline of refinancing. Bank loans, in particular, involve a lender with expertise in monitoring borrowers and the power to renegotiate. Debt service also commits the firm's cash flow, which restrains free-cash-flow waste. The cost is that excessive debt creates distress risk and the conflicts in Section 1.3.4. Capital structure is therefore a governance decision as well as a financing one.

1.4.4 The market for corporate control

When internal governance fails, the firm's equity is likely to trade below its potential value, which creates an opportunity for an outsider. Hostile takeovers, proxy contests and activist campaigns are the mechanisms. The threat can discipline managers even when no bid occurs. Defences, such as poison pills, staggered boards and super-voting shares, protect incumbents. They can serve shareholders by giving the board bargaining power against a low-ball offer, or serve managers by entrenching them; the evidence is mixed, and which effect dominates depends on the quality of the board that wields the defence.

Activist investors have become a prominent form of governance. They accumulate a stake, propose changes to strategy, capital allocation or leadership, and seek support from other shareholders. Institutional investors, such as pension funds and index funds, are increasingly important voters and engage with management through voting, dialogue and stewardship codes.

1.4.5 Law, regulation and disclosure

Securities law, stock-exchange listing rules, accounting standards, external audit and enforcement shape the environment in which all other mechanisms operate. Disclosure requirements reduce information asymmetry. Fiduciary duties, which in common-law systems include duties of care and loyalty owed by directors, and anti-fraud provisions give shareholders legal recourse. Following major accounting scandals in the early 2000s and the financial crisis of 2007-2009, many jurisdictions strengthened requirements on internal control, auditor independence, audit committee composition and executive accountability, and introduced say-on-pay votes. The effectiveness of any rule depends on enforcement, and the quality of legal protection differs substantially between countries.

Mechanism	What it does	Main strength	Main weakness
Independent board	Monitors and removes management	Close, continuous oversight	Information dependence on management; social ties
Incentive pay	Aligns managers' wealth with owners'	Directly rewards value creation	Can be gamed; encourages risk; rewards luck
Debt and covenants	Commits cash flow; creditors monitor	Hard constraint on waste	Distress costs; creditor-shareholder conflicts
Takeovers and activism	External challenge to poor performance	Works when internal controls fail	Costly, uncertain; defences blunt it
Law and disclosure	Sets minimum standards; enables recourse	Universal, baseline protection	Depends on enforcement; compliance cost
Reputation and labour market	Managers' careers depend on past results	Cheap and automatic	Weak near retirement or in a boom

1.5 Finance and Strategy

1.5.1 Strategy creates value only if returns exceed the cost of capital

Strategy is about choosing where to compete and how to win. Finance supplies the test of whether the choice pays. A strategy creates value when the firm earns, on the capital it commits, a return greater than the return investors could earn elsewhere on investments of similar risk. This leads to the key measures.

Return on invested capital (ROIC) measures operating profit after tax relative to the capital invested in operations:

```
ROIC = NOPAT / Invested Capital
NOPAT = EBIT × (1 − tax rate)
Invested Capital = Operating working capital + Net fixed assets + Other operating assets
                (equivalently, Debt + Equity − Excess cash)
```

Economic profit (also known as economic value added, or EVA, a trademark of Stern Stewart, or residual income) measures the dollar surplus after charging for all capital, including equity:

```
Economic Profit = NOPAT − WACC × Invested Capital
                = (ROIC − WACC) × Invested Capital
```

where WACC is the weighted average cost of capital, the blended return required by debt and equity holders. We derive WACC in a later chapter; for now treat it as the opportunity cost of capital.

A business unit with a positive spread (ROIC − WACC) creates value; one with a negative spread destroys value even if it reports an accounting profit. Accounting net income deducts interest but not the cost of equity, which is why a firm can appear profitable while failing to earn what its owners could have earned elsewhere.

Example 1.6 – Economic profit by division. Orion Diversified, a fictional conglomerate, reports the following for three divisions. WACC differs by division because the risk of their businesses differs.

Division	NOPAT (*m*)|*Investedcapital*(m)	WACC	ROIC	Capital charge (*m*)|*Economicprofit*(m)		
Aerospace	96	800	9%	12.0%	72	+24
Marine	130	1,000	10%	13.0%	100	+30
Retail	54	600	9%	9.0%	54	0

Step 1 – ROIC. Aerospace: 96 / 800 = 12.0%. Marine: 130 / 1,000 = 13.0%. Retail: 54 / 600 = 9.0%.

Step 2 – Capital charge = WACC × invested capital. Aerospace: 0.09 × 800 = 72. Marine: 0.10 × 1,000 = 100. Retail: 0.09 × 600 = 54.

Step 3 – Economic profit = NOPAT − capital charge. Aerospace: 96 − 72 = **24**. Marine: 130 − 100 = **30**. Retail: 54 − 54 = **0**.

Step 4 – Interpretation. Marine has the highest ROIC and the highest economic profit. Aerospace has a lower ROIC than Marine, but both earn the same 3-point spread over their own costs of capital (12% − 9% and 13% − 10%); Marine's larger capital base makes its dollar economic profit larger. Retail earns exactly its cost of capital: it is profitable in accounting terms (NOPAT of $54 million) but creates no value for owners. The group's total economic profit is 24 + 30 + 0 = **$54 million**.

Management implication. Capital should flow to businesses where the spread is positive and the growth opportunity is real. The Retail division deserves a hard look: either find a way to raise its returns, or release its $600 million of capital to better uses. Note that comparing ROICs without regard to WACC would have misled: Marine's 13% looks superior, but it is the riskier business with a higher cost of capital, and its spread over that cost is the same 3 points as Aerospace's.

1.5.2 The value-driver formula: growth is not always good

A useful result links value to three drivers: the level of profit, the growth rate and the return on new investment. For a firm with next year's NOPAT, perpetual growth rate g and return on new invested capital RONIC, the value of operations is:

```
Value of operations = NOPAT₁ × (1 − g / RONIC) / (WACC − g)
```

The term (1 − g / RONIC) is the fraction of NOPAT that need not be reinvested: to grow at rate g with a return RONIC on new capital, the firm must reinvest g / RONIC of its profits, and the remainder is free cash flow. The formula is the constant-growth perpetuity applied to free cash flow, and it holds for g < WACC.

Three cases follow directly.

- If RONIC = WACC, the formula collapses to $NOPAT_1$ / WACC regardless of g. Growth at the cost of capital creates no value.
- If RONIC > WACC, higher growth increases value.
- If RONIC < WACC, higher growth reduces value.

Example 1.7 – When does growth help? A business has next-year NOPAT of $96 million and WACC of 9%. We examine its value under different growth and return assumptions.

Step 1 – Value with no growth (g = 0). FCF = NOPAT because no reinvestment is required, so value = 96 / 0.09 = **$1,066.7 million**, regardless of RONIC.

Step 2 – g = 4%, RONIC = 12%. Reinvestment rate = 0.04 / 0.12 = 33.3%. Free cash flow = 96 × (1 − 0.3333) = 64. Value = 64 / (0.09 − 0.04) = 64 / 0.05 = **$1,280.0 million**.

Step 3 – g = 6%, RONIC = 12%. Reinvestment rate = 0.06 / 0.12 = 50%. FCF = 48. Value = 48 / 0.03 = **$1,600.0 million**.

Step 4 – g = 4%, RONIC = 9% (equal to WACC). Reinvestment rate = 0.04 / 0.09 = 44.4%. FCF = 96 × 0.5556 = 53.33. Value = 53.33 / 0.05 = **$1,066.7 million**, the same as with no growth.

Step 5 – g = 4%, RONIC = 6% (below WACC). Reinvestment rate = 0.04 / 0.06 = 66.7%. FCF = 96 × 0.3333 = 32. Value = 32 / 0.05 = **$640.0 million**.

Scenario	g	RONIC	Reinvestment rate	FCF (*m*)|*Value*(m)	
No growth	0%	any	0%	96.0	1,066.7
Value-creating growth	4%	12%	33.3%	64.0	1,280.0
Faster value-creating growth	6%	12%	50.0%	48.0	1,600.0
Growth at cost of capital	4%	9%	44.4%	53.3	1,066.7
Value-destroying growth	4%	6%	66.7%	32.0	640.0

Interpretation. The same 4% growth is worth +$213 million in one case, nothing in another, and −$427 million in a third. The question is never “how fast can we grow?” but “what do we earn on the capital that growth requires?” This is the mathematical form of the warning in Section 1.2.3 against growth and market-share objectives, and it explains why strategists who care about value spend as much effort on the sources of returns as on the sources of growth.

1.5.3 Sources of returns above the cost of capital

In a competitive economy, returns above the cost of capital attract entry and erode. Persistent positive spreads require some form of **competitive advantage**: cost leadership based on scale or process, differentiation through brand or technology, switching costs, network effects, regulatory protection or privileged access to inputs. A finance leader tests a strategy by

asking three questions. What is the source of the advantage? How long will it last, given competitors' likely response? How much capital must be tied up to exploit it? The answers become the explicit forecasting assumptions about margin, growth and capital intensity in a valuation.

1.5.4 The CFO as strategic partner

Finance contributes to strategy in four main ways. First, **portfolio management**: deciding which businesses to invest in, harvest or divest, using ROIC, growth and risk. Second, **capital allocation**: setting hurdle rates, deciding the pace of investment and the balance between reinvesting, paying down debt and returning cash. Third, **financial flexibility**: maintaining the debt capacity and liquidity to seize opportunities and survive shocks. Fourth, **communication**: explaining to investors why the strategy will create value, so that the share price reflects the firm's real prospects. A strategy no investor can understand is priced at a discount.

1.6 Financial Markets and Institutions

1.6.1 What financial markets do

Financial markets channel savings from households and institutions with surplus funds to firms and governments with productive uses for them. In doing so they perform four functions.

1. **Resource allocation.** Prices and expected returns guide capital to its most productive uses.
2. **Risk sharing.** Securities with different risk and return profiles allow investors to hold the risk they prefer and diversify the rest.
3. **Liquidity.** Secondary markets let investors convert holdings to cash, which lowers the return they require from issuers.
4. **Information and price discovery.** Prices aggregate information from many participants and provide a continuous signal about the value of corporate decisions.

1.6.2 Classifying markets

Dimension	Categories	Examples and notes
Maturity	Money market (up to one year) versus capital market (longer)	Treasury bills, commercial paper, certificates of deposit versus bonds and equities
Issue stage	Primary (new securities sold by the issuer) versus secondary (trading among investors)	Initial public offerings and bond issues versus exchange trading
Claim type	Debt versus equity versus derivatives	Loans and bonds; common and preferred stock; options, futures, swaps
Trading structure	Exchange-traded (centralised, standardised) versus over-the-counter (negotiated, dealer-based)	Listed shares versus many bonds and swaps
Participation	Public (open offering, regulated disclosure) versus private (limited investors, fewer disclosures)	Listed shares versus private placements and private equity

The **primary market** is where the firm receives cash; it matters directly to corporate finance. An initial public offering (IPO) converts a private company into a public one; a seasoned equity offering issues more shares of an already listed company; a bond issue sells debt securities. Investment banks typically underwrite and distribute these issues, advising on pricing and timing, and often committing their own capital to buy any unsold portion. The **secondary market** does not provide new money to the issuer, but it is essential because it gives primary-market investors liquidity and creates the price that the board sees every day.

1.6.3 Financial intermediaries and institutions

Many savers do not lend directly to firms. **Intermediaries** such as banks, insurers, pension funds, mutual funds and exchange-traded funds stand between them, adding value through diversification, maturity transformation, screening and monitoring, and economies of scale in information and transactions. Banks take deposits and make loans, bearing credit and liquidity risk. Insurers pool risk. Asset managers aggregate savings into diversified portfolios; their voting power makes them important governance actors. **Credit-rating agencies** assess the creditworthiness of issuers, and their ratings shape borrowing costs and covenants. **Regulators and central banks** supervise the system, set the policy interest rate and act as lenders of last resort in crises. The “shadow banking” sector, comprising non-bank lenders and market-based credit, performs bank-like functions with different regulation.

The structure of financing differs across countries. Economies such as the United States rely heavily on capital markets for corporate funding; others, such as Germany and Japan, have traditionally relied more on bank lending, though markets have grown in importance. For a firm, the practical implication is that the menu of financing options and the relationship with providers vary by location and size, and a leader needs to know which markets are open to her business.

1.6.4 Market efficiency

An **efficient market** is one in which prices fully reflect available information. Eugene Fama classified efficiency into three forms according to the information set.

- **Weak form:** prices reflect all past trading information, so technical analysis of price patterns cannot generate abnormal returns.
- **Semi-strong form:** prices reflect all public information, so fundamental analysis of published data cannot generate abnormal returns, and prices adjust quickly to new announcements.
- **Strong form:** prices reflect all information, including private information, so even insiders cannot earn abnormal returns (a form widely rejected, and one that insider-trading laws presuppose to be false).

Evidence broadly supports reasonably quick price adjustment to public news in developed markets, while documenting anomalies and episodes, such as speculative bubbles, that challenge the strictest interpretations. The behavioural finance literature, associated with Kahneman, Tversky, Shiller, Thaler and others, argues that cognitive biases and limits to arbitrage can sustain mispricing. A balanced position for managers is this: the market price is the best available single estimate of value in most circumstances; it can be wrong, sometimes badly, but a manager cannot predict when, and should not base long-term strategy on the belief that she can outguess it. The implication for corporate finance is that, in the main, a decision that increases true value will, once understood, be reflected in the price.

1.6.5 Reading the market's verdict: event studies

Efficiency suggests a way to measure how the market evaluates a corporate decision: look at the share price reaction around the announcement. Because the stock also moves with the market, we measure the **abnormal return**, the actual return minus the return expected given the market's movement. The simplest tool is the market model:

```
Expected return on day t = α + β × (market return on day t)
Abnormal return (AR) = Actual return − Expected return
Cumulative abnormal return (CAR) = Σ AR over the event window
```

Example 1.8 – An event study of an acquisition announcement. Meridian Foods, a fictional firm with a market capitalisation of $2,500 million, announces the acquisition of a regional competitor. Using a market model with $\alpha = 0$ and $\beta = 1.2$ estimated from prior data, the following daily returns were observed (in percent):

Day	Meridian actual return	Market return	Expected return (1.2 × market)	Abnormal return
Announcement day (0)	4.5	1.0	1.2	3.3
Day +1	1.0	−0.5	−0.6	1.6
Day +2	−0.2	0.3	0.36	−0.56

Step 1 – Expected returns. Day 0: 1.2 × 1.0 = 1.20%. Day 1: 1.2 × (−0.5) = −0.60%. Day 2: 1.2 × 0.3 = 0.36%.

Step 2 – Abnormal returns. Day 0: 4.5 − 1.20 = **3.30%**. Day 1: 1.0 − (−0.60) = **1.60%**. Day 2: −0.2 − 0.36 = **−0.56%**.

Step 3 – Cumulative abnormal return over the three days. 3.30 + 1.60 − 0.56 = **4.34%**.

Step 4 – Dollar value. 0.0434 × 2,500 = **$108.5 million** of value attributed to the announcement.

Interpretation. The market views the acquisition as creating about $108 million for Meridian's shareholders, more than the price paid for the target would suggest as a pure transfer. Such evidence is powerful for managers because it reveals the market's assessment of both the deal and the credibility of the team announcing it. It has limits. The abnormal return includes anything else announced at the same time; the market model relies on estimated parameters; and the price reaction reflects expectations about how the deal will be executed, not realised outcomes. But the method is widely used in research and in litigation, and it illustrates why a board should ask not only "what is the NPV?" but also "how will the market read this?".

1.7 Ethics, Responsibility and the Road Ahead

1.7.1 Ethics as a financial issue

Agency problems and ethical failures are close relatives. Accounting manipulations, misleading disclosures, insider trading, mis-selling of products and bribery all use the information advantages of insiders at the expense of outsiders, and each has contributed to spectacular corporate collapses. Ethics is also a matter of self-interest. A firm's reputation for honesty is an asset that lowers the cost of capital, because investors will pay more for a firm whose numbers they trust. A single scandal can destroy years of accumulated trust; the share-price drop on revelation of fraud typically exceeds the direct legal penalty, because the market is repricing the credibility of everything the firm has ever said.

Finance leaders face particular pressures: targets that incentivise earnings management, the temptation to treat the quarter as the horizon, and requests from colleagues to adjust assumptions to justify a preferred outcome. A practical guard is to establish in advance, as the independent director in the Northgate vignette did, the criteria by which an investment will be

judged, so that proposals are tested against standards that were set before the temptation arose. Another is to separate the people who propose projects from the people who evaluate them, and to review a sample of past decisions against actual outcomes (post-audits).

1.7.2 The architecture of the book

The remaining chapters build the toolkit this chapter has motivated. We begin by learning to read the firm's financial statements and to compute the cash flows that valuation depends on (Chapter 2). We then develop the time value of money, the valuation of bonds and equities, the measurement of risk and return, the cost of capital, capital-budgeting techniques, capital-structure and payout theory, and the application of these tools in mergers, international finance and risk management. At each stage the question is the one the independent director asked at the Northgate board table: what does the shareholder own after this decision that she did not own before, and what did she pay for it?

Case Study: The Westbrook Packaging Board Room

Illustrative case. Westbrook Packaging is a fictional company.

Westbrook Packaging is a listed manufacturer of corrugated and flexible packaging, with annual revenue of about $3.1 billion, 11,000 employees and plants in nine countries. Founded by the Westbrook family, it remains 18% family-owned, with the remainder held by institutions and the public. Chief executive Dana Okafor has held the role for eleven years. Her tenure has been marked by steady expansion: revenue has doubled through eight acquisitions. Return on invested capital, however, has drifted from 11.5% to 8.1% over the same period, against an estimated weighted average cost of capital of 8.8%.

The compensation committee, which is chaired by a long-serving director who was Okafor's mentor, sets the annual bonus using three equally weighted measures: revenue growth, adjusted EBITDA growth and adjusted EPS. The long-term plan awards stock options with a three-year vesting period, struck at the grant-date price. Adjusted EBITDA excludes restructuring costs, which have been recorded in each of the last six years. The board has nine members. Five are classified as independent, but three of those have served for more than twelve years and two have business relationships with the company through consulting contracts. Okafor serves as both chair and chief executive.

Westbrook now has a proposal before it. A private flexible-packaging company, Tarrant Films, is for sale at an enterprise value of $900 million, which is 10.5 times its EBITDA. Management's analysis projects annual synergies with a present value of $70 million. Independent valuation work by the treasury team, using conservative growth and margin assumptions and Tarrant's cost of capital, places Tarrant's standalone enterprise value at $780 million. Financing would be a mix of new debt and new shares. The combined company would carry net debt of about 3.6 times EBITDA, up from 2.4 times, which would put it near the limit of its bank covenants. The treasurer has also warned that the rating agencies could place the company on negative watch.

A large institutional investor holding 6% of the shares has written to the board. It notes that ROIC has been below the cost of capital for four years, that bonus measures do not penalise capital misallocation, and that the shares trade at a discount to peers. It requests an independent chair, a bonus measure based on returns on capital, and a pause on acquisitions until a portfolio review has been completed. It has hinted at a campaign at the next annual meeting.

Okafor argues that the Tarrant deal will make Westbrook the leading flexible-packaging producer in two regions and that a pause would hand the asset to a competitor. The CFO, Aisha Rahman, privately shares the investor's concerns but also knows her own bonus is linked to EPS, which the deal would raise in year one because of the accretion from leveraged

financing. The family representative on the board supports the deal and is wary of an activist campaign. Two independent directors have asked for a second opinion from an outside adviser.

Discussion Questions

1. Using the figures in the case, compute the net present value of the Tarrant acquisition to Westbrook's shareholders, assuming the independent valuation and the management synergy estimate are correct. What would the synergies need to be worth for the deal to break even? What does your result imply about who captures the value in this transaction?
2. Identify at least four agency problems visible in Westbrook's situation, including the incentive features of the compensation plan. For each, explain how it could lead to value-destroying decisions.
3. Evaluate the institutional investor's three requests. Which would you support, which would you modify, and what risks does each carry for the company? What additional governance mechanisms would you propose?
4. Rahman faces a conflict between her private doubts and the incentive in her own pay. How should she handle the board discussion, and how should the compensation structure be redesigned so that future CFOs are not placed in the same position? Propose specific metrics and vesting terms.

Key Takeaways

- Corporate finance addresses three linked decisions, investment, financing and payout, and the quality of each is judged by its effect on the value of the firm's equity.
- The operating objective is the maximisation of long-run equity value. This differs from maximising accounting profit, EPS, revenue or market share, none of which accounts for the time value of money, risk and the capital used.
- Net present value is the measure of the change in shareholder wealth from an investment; accounting returns and payback rules can mislead.
- Separation of ownership and control creates agency problems: perquisites, empire building, excess risk aversion, short horizons and entrenchment between managers and shareholders; asset substitution and underinvestment between shareholders and creditors; and private-benefit extraction by controlling over minority holders.
- Agency costs are the sum of monitoring costs, bonding costs and residual loss. Rational parties adopt monitoring and bonding to the point where the marginal cost equals the marginal reduction in loss.
- Governance works through a combination of an independent and informed board, well-designed incentive pay, disciplined use of debt, the market for corporate control, and law and disclosure. No single device suffices, and each has typical failure modes.
- Strategy creates value only when the return on invested capital exceeds the cost of capital. Growth at a return equal to the cost of capital adds nothing, and growth at a lower return destroys value.
- Economic profit, NOPAT minus a charge for all capital, aligns divisional performance measurement and executive pay with value creation more closely than net income or EPS.
- Financial markets allocate capital, share risk, provide liquidity and generate prices that reveal how investors judge corporate decisions; event studies convert that judgement into a measurable abnormal return.
- Ethical conduct and transparent reporting reduce the cost of capital, while manipulation and fraud transfer wealth from outsiders to insiders and destroy the trust on which the cost of capital depends.

Review Questions

1. Describe the three core decisions of corporate finance and give an example of how a decision in one area constrains the others.
2. Distinguish between the roles of the treasurer and the controller. Why is it important that internal audit does not report solely to the CFO?
3. Explain why limited liability and transferable shares make large-scale enterprise possible, and how they create the separation of ownership from control.
4. Why is maximising current equity value a better corporate objective than maximising profit? Discuss at least three failures of the profit objective.
5. How would you respond to a colleague who claims that the shareholder-value objective requires managers to ignore employees, customers and the environment?
6. Define agency costs according to Jensen and Meckling and give an example of each of the three components.
7. Explain the asset-substitution problem. Why do lenders anticipate it, and how do they protect themselves?
8. What are the main weaknesses of using EPS growth as the basis for executive bonuses? Propose an alternative and discuss its own weaknesses.
9. A firm announces that its ROIC is 14% and its WACC is 10%. Under what circumstances would faster growth still fail to increase its value?
10. Distinguish weak-form, semi-strong-form and strong-form market efficiency. Why does the semi-strong form matter most for corporate decisions, and what are the limitations of using an event study to evaluate a decision?

Problems

1. **Project value.** A project requires an investment of $500 million today and returns $160 million at the end of each of the next five years. The cost of capital is 9%. The firm has 10 million shares outstanding. (a) Calculate the NPV. (b) Calculate the expected change in the share price if the market fully values the project.
2. **Acquisition gains.** A bidder offers $880 million for a target with a standalone value of $800 million. The bidder estimates the present value of synergies at $120 million. (a) Compute the NPV of the deal to the bidder. (b) How is the $120 million of synergies divided between the two sets of shareholders?
3. **Selling equity.** An entrepreneur owns 100% of a firm worth $2,000 million if run without excess perquisites. She sells 30% to outside investors. Following the sale she consumes perquisites that reduce firm value by $150 million; each dollar of perquisites is worth $0.60 to her. Outsiders anticipate this. (a) What price do they pay for their 30%? (b) What is the entrepreneur's total wealth after the sale? (c) What is the deadweight agency cost?
4. **Event study.** A firm with a market capitalisation of $3,200 million has $\beta = 0.9$ ($\alpha = 0$). Over a three-day window, its returns were 2.0%, 1.5% and −0.5%, while the market returned 1.0%, −0.4% and 0.2% respectively. Compute the abnormal return on each day, the cumulative abnormal return and its dollar value.
5. **Economic profit.** A division has NOPAT of $210 million and invested capital of $1,500 million. WACC is 11%. (a) Calculate ROIC and economic profit. (b) Management proposes to add $200 million of capital that will generate $20 million of additional NOPAT. Calculate the division's new economic profit and say whether the expansion creates value.

6. **Risk shifting.** A firm has $200 million of debt due in one year. It can choose a safe project worth $260 million for certain, or a risky project worth $400 million with probability 50% and $100 million with probability 50%. (a) Find the expected value of equity under each. (b) Find the expected value of the debt under the risky project. (c) Which project do shareholders prefer, and is it the value-maximising choice?
7. **Option pay.** An executive holds 200,000 options with a strike price of $40 and receives a bonus of 0.5% of net income, which is $120 million. Calculate (a) the bonus, (b) the option payoff if the exercise-date price is $55, and (c) the option payoff if the price is $35. How does her payoff differ from that of a shareholder who owns 200,000 shares bought at $40?
8. **Retention versus payout.** A firm has $100 million of free cash flow it can either distribute or invest in a project that earns a perpetual $6 million per year. The relevant cost of capital is 10%. (a) What is the present value of the project's cash flows? (b) What is the NPV? (c) Which decision do shareholders prefer, and what does this say about the free-cash-flow problem?

Answers to Selected Problems

1. a. The five-year annuity factor at 9% is 3.8897, so PV = 160 × 3.8897 = $622.3 million and NPV = 622.3 − 500 = **$122.3 million**. (b) 122.3 / 10 = **$12.23 per share**.
2. a. NPV = (800 + 120) − 880 = **+$40 million**. (b) Target shareholders receive 880 − 800 = **$80 million** (the premium), and the bidder keeps 120 − 80 = **$40 million**. Together the two gains sum to the $120 million of synergies, split two-thirds to the target and one-third to the bidder.
3. a. Firm value after perquisites = 2,000 − 150 = $1,850 million; outsiders pay 0.30 × 1,850 = **$555 million**. (b) Wealth = 555 + 0.70 × 1,850 + 0.60 × 150 = 555 + 1,295 + 90 = **$1,940 million**. (c) Without perquisites her wealth would be $2,000 million, so the agency cost is **$60 million**, which is equal to 150 × (1 − 0.60).
4. Expected returns: 0.9 × 1.0 = 0.90%; 0.9 × (−0.4) = −0.36%; 0.9 × 0.2 = 0.18%. Abnormal returns: 2.0 − 0.90 = **1.10%**; 1.5 + 0.36 = **1.86%**; −0.5 − 0.18 = **−0.68%**. CAR = 1.10 + 1.86 − 0.68 = **2.28%**. Dollar value = 0.0228 × 3,200 = **$72.96 million**.
5. a. ROIC = 210 / 1,500 = **14.0%**. Economic profit = 210 − 0.11 × 1,500 = 210 − 165 = **$45 million**. (b) New NOPAT = 230; new capital = 1,700; capital charge = 187; new economic profit = **$43 million**. Incremental return = 20 / 200 = 10%, below the 11% WACC, so the expansion destroys $2 million of annual economic profit (20 − 22) and should be rejected.
6. a. Safe: 260 − 200 = **$60 million**. Risky: good state equity = 400 − 200 = 200; bad state = 0 (value of 100 is less than the debt); expected equity = 0.5 × 200 = **$100 million**. (b) Debt under the risky project: 0.5 × 200 + 0.5 × 100 = **$150 million**, against $200 million under the safe project. (c) Shareholders prefer the risky project ($100 million versus $60 million), but its expected firm value is 0.5 × 400 + 0.5 × 100 = $250 million, below the safe project's $260 million, so it is not value-maximising. Creditors lose $50 million; shareholders gain $40 million; the net loss is $10 million.
7. a. 0.005 × 120,000,000 = **$600,000**. (b) 200,000 × (55 − 40) = **$3,000,000**. (c) **$0**. The shareholder gains $3,000,000 in case (b) but loses 200,000 × 5 = $1,000,000 in case (c); the option holder never loses, which can encourage risk-taking.
8. a. PV = 6 / 0.10 = **$60 million**. (b) NPV = 60 − 100 = **−$40 million**. (c) Shareholders prefer the distribution, since investing destroys $40 million of value. A manager who invests anyway because she prefers a larger firm

is demonstrating the free-cash-flow problem; commitments such as dividends, buybacks and debt service constrain this behaviour.

Chapter 2: Financial Statements, Cash Flows and Ratio Analysis

Learning Objectives

After studying this chapter you should be able to:

- Explain the purpose and structure of the balance sheet, income statement, cash flow statement and statement of changes in equity, and show how the four statements articulate with one another.
- Prepare operating cash flow from net income using the indirect method, and calculate free cash flow to the firm and free cash flow to equity from the statements.
- Compute and interpret liquidity, efficiency, leverage and profitability ratios, including the cash conversion cycle and interest coverage.
- Decompose return on equity with the three-step and five-step DuPont frameworks and use the decomposition to diagnose changes in performance.
- Apply common-size and trend analysis to identify the sources of change in a firm's financial position.
- Assess the quality of earnings using cash-based and accrual-based diagnostics, and recognise the limitations of ratio analysis.

Opening Vignette

Illustrative scenario. Helix Devices is a fictional company.

Helix Devices designs connected medical monitors and sells them to hospital systems. For three straight years it reported rising profits. Revenue grew 22% in the latest year, net income grew 31%, and the chief executive told analysts that Helix had "turned the corner." The company's lender, a regional bank, was about to renew a $60 million revolving credit facility, and the relationship manager asked a new credit analyst, Priya, to prepare a one-page recommendation.

Priya did not start with the income statement. She began with the cash flow statement and found that operating cash flow had been negative for two years even though net income was strongly positive. She then moved to the balance sheet. Accounts receivable had grown 64% while revenue grew 22%. Days sales outstanding had risen from 58 to 91. Inventory had risen faster than sales, and the company had begun to extend supplier payments from 45 to 70 days. Capital expenditure had been financed by new borrowing. The footnotes revealed that Helix had begun offering hospitals extended payment terms, and that a growing share of revenue was recognised at shipment under distributor arrangements that allowed returns.

None of this proved wrongdoing. It might simply mean a growth company stretching to win accounts. But the profit figure that had impressed investors told only part of the story. The firm was consuming cash faster than it earned it, its liquidity depended on suppliers' patience, and its reported earnings depended on judgements about collectibility and returns that were open to challenge.

Priya recommended renewing the facility with a smaller limit, tighter covenants on receivables and a requirement for quarterly cash flow reporting. The line was one she would repeat to her colleagues: "Profit is an opinion; cash is a fact."

Financial statements are the common language of management, lenders, investors and regulators. This chapter teaches you to read them as Priya did: to connect the statements, to extract cash flows, to compute ratios that summarise performance and risk, and to ask whether the numbers deserve to be trusted.

2.1 The Financial Statements and How They Fit Together

2.1.1 Purpose and framework

Financial statements summarise the economic activity of a firm in a standard format so that outsiders can compare it with other firms and with itself over time. Listed firms in most of the world prepare them under International Financial Reporting Standards (IFRS), and in the United States under US generally accepted accounting principles (US GAAP). The two frameworks differ in detail but share the same architecture: accrual accounting, going-concern assumption, a historical-cost orientation with selective use of fair value, and a set of qualitative characteristics (relevance, faithful representation, comparability, verifiability, timeliness and understandability).

Accrual accounting is the central idea. Revenue is recognised when it is earned, and expenses when the resources are consumed to earn it, irrespective of when cash changes hands. This gives a better measure of performance than cash receipts and payments over short periods, but it introduces estimates and judgement, and therefore the possibility of bias.

A complete set of annual statements has four primary components and accompanying notes:

1. The **balance sheet** (statement of financial position): a snapshot of assets, liabilities and equity at a date.
2. The **income statement** (statement of profit or loss): revenues, expenses and profit over a period.
3. The **cash flow statement**: cash inflows and outflows over a period, classified as operating, investing and financing.
4. The **statement of changes in equity**: the reconciliation of opening to closing equity through profit, dividends, share issues and other items.

The **notes** disclose accounting policies, details of major items, commitments and contingencies, segment data and related-party transactions. Many of the most useful analytical facts are found in the notes, not on the face of the statements.

2.1.2 The balance sheet

The balance sheet rests on the accounting identity:

```
Assets = Liabilities + Shareholders' Equity
```

Assets are resources controlled by the firm from which future economic benefits are expected. They are typically listed in order of liquidity or presented as current (expected to be converted to cash or used within a year or the operating cycle) and non-current. Current assets include cash, receivables, inventory and prepaid expenses. Non-current assets include property, plant and equipment (PP&E) net of accumulated depreciation, intangible assets and goodwill, long-term investments and deferred tax assets.

Liabilities are present obligations to transfer economic resources. Current liabilities include trade payables, accrued expenses, short-term debt and the current portion of long-term debt. Non-current liabilities include long-term debt, lease liabilities, pension obligations and deferred taxes.

Equity is the residual: assets minus liabilities. It comprises contributed capital (share capital and premium), retained earnings and other reserves. Book equity is a historical accounting number and can differ markedly from market capitalisation, because the balance sheet omits many valuable assets, such as brands, internally generated technology and

human capital, and reports others at depreciated historical cost.

Several measurement points matter for analysis. Inventory may be valued at cost using FIFO or weighted average (LIFO is permitted under US GAAP but not IFRS), which affects margins during inflation. PP&E is depreciated over estimated lives; longer lives defer expense. Goodwill, created in acquisitions, is tested for impairment, and impairments tend to be recognised late. Under current lease standards, most leases appear on the balance sheet as a right-of-use asset and a lease liability, which makes lease-heavy firms look more leveraged than they once did and requires care when comparing firms across periods or between standards.

2.1.3 The income statement

The income statement reports financial performance over a period. A typical layout, together with the intermediate measures analysts use, is:

```
Revenue
− Cost of goods sold (COGS)
= Gross profit
− Selling, general and administrative expenses (SG&A)
− Research and development
− Depreciation and amortisation (if shown separately)
= Operating profit (EBIT)
− Interest expense
= Profit before tax (EBT)
− Income tax expense
= Net income
```

EBITDA (earnings before interest, taxes, depreciation and amortisation) equals EBIT plus depreciation and amortisation. It is widely used as a rough proxy for operating cash generation and as the denominator in leverage ratios, but it ignores capital expenditure, working-capital investment and taxes. A business that must continually replace expensive equipment can have a high EBITDA and little free cash.

Net income is the profit attributable to the owners after all expenses including interest and tax. **Earnings per share (EPS)** is net income less preferred dividends, divided by the weighted-average shares outstanding; diluted EPS also counts options, convertibles and other potentially dilutive securities.

Some income statements classify expenses by function (as above), others by nature (materials, personnel, depreciation). Items outside normal operations, such as restructuring charges, impairments and gains on disposal, require judgement: are they truly non-recurring? A firm that reports “one-off” charges every year should be analysed as if they were recurring.

2.1.4 The cash flow statement

The cash flow statement reconciles the opening and closing cash balance by activity.

- **Operating activities:** cash generated by the firm’s principal revenue-producing activities. It starts with net income and adjusts for non-cash items and working-capital changes (the indirect method) or lists cash receipts and payments directly (the direct method).
- **Investing activities:** purchases and sales of long-term assets, acquisitions and divestitures, and investments in securities.
- **Financing activities:** issuance and repayment of debt, issuance and repurchase of shares, and dividends. (Under IFRS, interest paid and dividends paid may be classified in operating or financing, subject to consistent policy; US

GAAP places interest paid in operating activities.)

The cash flow statement is harder to manipulate than the income statement, although classification choices exist. Analysts use it to answer three questions. Does the business generate cash from its operations? Is it funding growth internally or by raising capital? Is it returning cash to owners from genuine surplus or from borrowing?

2.1.5 The statement of changes in equity and the articulation of the statements

Equity movements link the income statement to the balance sheet:

```
Closing retained earnings = Opening retained earnings + Net income − Dividends
```

(with similar lines for share issues, repurchases and other comprehensive income). The statements **articulate**: net income flows into retained earnings; the cash flow statement explains the change in the cash line of the balance sheet; the changes in other balance sheet accounts explain the adjustments in the operating, investing and financing sections. Learning to trace these connections is the best protection against error and manipulation, because an inconsistency in one statement shows up in another.

2.1.6 The running example: Calder Components

To keep the chapter concrete, we use a single fictional manufacturer, Calder Components, for most of the worked examples. All figures are in $ millions. We assume a 25% tax rate and that all depreciation relates to PP&E.

Calder Components – Income statement

Item	Year 2	Year 1
Revenue	2,400	2,200
Cost of goods sold	1,512	1,430
Gross profit	**888**	**770**
SG&A	420	396
Research and development	60	52
Depreciation and amortisation	96	88
EBIT	**312**	**234**
Interest expense	48	42
Profit before tax	**264**	**192**
Income tax (25%)	66	48
Net income	**198**	**144**
Dividends declared and paid, Year 2	48	

Calder Components – Balance sheet at year end

Item	Year 2	Year 1
Cash	180	150
Accounts receivable	360	305
Inventory	330	270
Prepaid expenses	30	25
Total current assets	**900**	**750**
Net property, plant and equipment	1,100	1,000
Goodwill and intangibles	300	300
Total assets	**2,300**	**2,050**
Accounts payable	240	210
Accrued liabilities	120	100
Short-term debt	100	90
Total current liabilities	**460**	**400**
Long-term debt	600	560
Total liabilities	**1,060**	**960**
Shareholders' equity	1,240	1,090
Total liabilities and equity	**2,300**	**2,050**

Check. Year 2: 460 + 600 + 1,240 = 2,300. Year 1: 400 + 560 + 1,090 = 2,050. Equity rose by 150, which equals net income of 198 minus dividends of 48. No shares were issued or repurchased.

2.2 Cash Flows: From Profit to Free Cash Flow

2.2.1 The indirect method

Operating cash flow (CFO) begins with net income and applies two sets of adjustments. First, add back non-cash expenses such as depreciation and amortisation (they reduced profit but not cash). Second, adjust for changes in operating working capital: an increase in an operating asset (receivables, inventory, prepaid expenses) uses cash, whereas an increase in an operating liability (payables, accruals) provides cash.

```
CFO = Net income + D&A + other non-cash charges
      − Increase in receivables − Increase in inventory − Increase in prepaids
      + Increase in payables + Increase in accrued liabilities
```

The logic is intuitive. If Calder sells goods on credit, revenue and profit rise, but the cash arrives only when the customer pays; the increase in receivables is therefore a deduction. If Calder delays paying a supplier, expense is recorded but cash is retained; the increase in payables is an addition.

Example 2.1 – Operating cash flow for Calder Components. Using the Year 2 and Year 1 balance sheets:

Step 1 – Start with net income: 198.

Step 2 – Add back depreciation and amortisation: +96. Subtotal 294.

Step 3 – Working-capital changes: - Receivables: 360 − 305 = +55 increase → −55 - Inventory: 330 − 270 = +60 increase → −60 - Prepaid expenses: 30 − 25 = +5 increase → −5 - Accounts payable: 240 − 210 = +30 increase → +30 - Accrued liabilities: 120 − 100 = +20 increase → +20

Net working-capital effect = −55 − 60 − 5 + 30 + 20 = **−70**.

Step 4 – CFO = 294 − 70 = **224**.

Step 5 – Investing. PP&E rose from 1,000 to 1,100. Net PP&E = opening + capital expenditure − depreciation, so capex = 1,100 − 1,000 + 96 = **196**. Goodwill and intangibles were unchanged. CFI = **−196**.

Step 6 – Financing. Short-term debt rose 10 and long-term debt rose 40, giving net borrowing of +50; dividends were −48. CFF = 50 − 48 = **+2**.

Step 7 – Net change in cash. 224 − 196 + 2 = **+30**, which equals the change in the balance sheet cash line (180 − 150 = 30). The statements articulate.

Calder Components – Cash flow statement, Year 2	**$m**
Net income	198
Depreciation and amortisation	96
Increase in receivables	(55)
Increase in inventory	(60)
Increase in prepaid expenses	(5)
Increase in payables	30
Increase in accrued liabilities	20
Cash flow from operations	**224**
Capital expenditure	(196)
Cash flow from investing	**(196)**
Net borrowing	50
Dividends	(48)
Cash flow from financing	**2**
Net increase in cash	**30**

Interpretation. Calder converted 224 / 198 = 113% of net income into operating cash flow, a healthy sign, but it also spent 196 on capex, nearly twice its depreciation (196 / 96 = 2.04), so internally generated cash after investment was only 28. The dividends of 48 and part of the growth were financed with debt. This is a growth firm investing ahead of sales, a pattern revisited in Section 2.7.

2.2.2 Why free cash flow matters

Net income and even CFO are not the amounts available to investors, since the firm must reinvest in fixed assets and working capital to sustain its operations. **Free cash flow (FCF)** is the cash generated after those reinvestments. It is the amount that can be paid to the providers of capital without impairing the business, and it is the quantity that discounted-cash-flow valuation (developed in later chapters) discounts.

Two versions are used.

Free cash flow to the firm (FCFF), also called unlevered free cash flow, is the cash flow available to all capital providers, debt and equity, before any financing flows:

```
FCFF = EBIT × (1 − t) + D&A − Capital expenditure − Increase in operating working capital
     = NOPAT + D&A − Capex − ΔNWC
```

Equivalent expressions use CFO: `FCFF = CFO + Interest × (1 − t) − Capex`. FCFF is independent of capital structure, because it is computed before interest and uses the tax charge on operating income (the tax shield on interest is accounted for in the discount rate, via the after-tax cost of debt).

Free cash flow to equity (FCFE) is the cash flow available to the shareholders after meeting all debt obligations:

```
FCFE = FCFF − Interest × (1 − t) + Net borrowing
     = CFO − Capex + Net borrowing
```

Operating working capital here includes only operating items (receivables, inventory, prepaids, less payables and accruals); cash and interest-bearing debt are excluded, because they are financing items.

Example 2.2 – FCFF and FCFE for Calder Components.

Step 1 – NOPAT. EBIT × (1 − t) = 312 × 0.75 = **234.0**.

Step 2 – Change in operating working capital. Increases in receivables (55), inventory (60) and prepaids (5) less increases in payables (30) and accruals (20): 55 + 60 + 5 − 30 − 20 = **70**.

Step 3 – FCFF. 234 + 96 − 196 − 70 = **64**.

Step 4 – Cross-check using CFO. Interest after tax = 48 × 0.75 = 36. FCFF = 224 + 36 − 196 = **64**. ✓

Step 5 – FCFE. FCFF − after-tax interest + net borrowing = 64 − 36 + 50 = **78**. Cross-check: CFO − capex + net borrowing = 224 − 196 + 50 = **78**. ✓

Step 6 – Use of FCFE. Dividends were 48, so 78 − 48 = 30 was added to cash, matching the change in the cash balance.

Interpretation. Despite NOPAT of 234, the firm's FCFF is only 64, because it invested 196 in fixed assets and 70 in working capital. Whether this is good or bad depends on what the investment buys. If the new capacity earns returns above the cost of capital, then low current FCFF is the price of future value. If not, it is a warning. Note also that FCFE of 78 includes 50 of borrowing; without new debt, Calder's cash after investment would have been 28, short of its dividend of 48.

A useful way of thinking about free cash flow is as the reconciliation of the *earnings* a firm reports with the *cash* it can distribute. Firms with high EBITDA and high capital intensity (telecommunications, utilities, airlines) often have low FCF; firms with low capital needs (software, consulting) often convert most of their profit to cash.

2.2.3 Common mistakes in cash-flow analysis

- **Using EBITDA as "cash flow."** EBITDA omits capex, working capital and taxes.
- **Ignoring capex versus depreciation.** Capex persistently below depreciation means that the asset base is being consumed; persistently above depreciation signals growth (or inflation of asset costs).
- **Treating borrowing as operating performance.** A firm reporting positive total cash flow may be funding operations with debt.
- **Mismatching FCFF with equity measures.** FCFF is discounted at the WACC to give enterprise value; FCFE is discounted at the cost of equity to give equity value. Mixing them leads to double counting of debt.
- **Forgetting that one year can mislead.** Working-capital swings and lumpy capex distort single-year FCF; analysts typically examine three to five years and normalise.

2.3 Common-Size and Trend Analysis

Ratios compress information; common-size and trend statements preserve the shape of the statements while making firms and years comparable.

2.3.1 Common-size statements

A **common-size income statement** expresses every line as a percentage of revenue; a **common-size balance sheet** expresses every line as a percentage of total assets. This removes the effect of scale, so that a small firm can be compared with a large one in the same industry, and it highlights structural change over time.

Example 2.3 – Common-size analysis of Calder. Dividing each line of the income statement by revenue:

Line (% of revenue)	Year 2	Year 1	Change (points)
COGS	63.0%	65.0%	−2.0
Gross profit	37.0%	35.0%	+2.0
SG&A	17.5%	18.0%	−0.5
R&D	2.5%	2.4%	+0.1
D&A	4.0%	4.0%	0.0
EBIT	13.0%	10.6%	+2.4
Interest	2.0%	1.9%	+0.1
Net income	8.25%	6.5%	+1.7

Step 1 – Calculations. Year 2: COGS 1,512 / 2,400 = 63.0%; SG&A 420 / 2,400 = 17.5%; EBIT 312 / 2,400 = 13.0%; net income 198 / 2,400 = 8.25%. Year 1: COGS 1,430 / 2,200 = 65.0%; SG&A 396 / 2,200 = 18.0%; EBIT 234 / 2,200 = 10.6%; net income 144 / 2,200 = 6.5%.

Step 2 – Growth rates. Revenue grew 2,400 / 2,200 − 1 = 9.1%. EBIT grew 312 / 234 − 1 = 33.3%, and net income grew 198 / 144 − 1 = 37.5%.

Step 3 – Where did the improvement come from? Of the 2.4-point rise in EBIT margin, 2.0 points came from gross margin (cheaper inputs, better pricing or more efficient production) and 0.5 points from SG&A leverage, partly offset by a 0.1-point increase in R&D. Profit grew about four times faster than sales, which is **operating leverage** at work: when a portion of costs is fixed, a modest increase in revenue lifts profit disproportionately. The degree of operating leverage here is 33.3% / 9.1% = **3.7**. The firm must investigate whether the gross-margin gain is sustainable (a one-time commodity price fall) or structural (a plant efficiency programme).

The common-size balance sheet reveals shifts in the structure of assets and financing:

Line (% of total assets)	Year 2	Year 1
Cash	7.8%	7.3%
Receivables	15.7%	14.9%
Inventory	14.3%	13.2%
Total current assets	39.1%	36.6%
Net PP&E	47.8%	48.8%
Goodwill and intangibles	13.0%	14.6%
Current liabilities	20.0%	19.5%
Long-term debt	26.1%	27.3%
Equity	53.9%	53.2%

Interpretation. Receivables grew 18% and inventory 22%, both faster than revenue (9.1%). The share of current assets in total assets rose by 2.5 points. This signals either a build to support growth or slower collections and slower-moving stock, a question the efficiency ratios in Section 2.4 will address. Financing is stable: equity still funds about 54% of assets, and long-term debt a little over a quarter.

2.3.2 Trend (index) analysis

Trend analysis expresses each line over several years as a percentage of a base year (base = 100). It shows which items grow faster or slower than others. If receivables index at 160 while revenue indexes at 130, receivables are rising faster than sales. A standard test of reasonableness is that receivables, inventory and payables should move broadly in line with sales unless there is a known reason (a change in credit policy, a new product with a longer cycle, a supplier negotiation). Differences between growth rates, rather than the rates themselves, are the signal.

2.3.3 Benchmarks

A ratio has meaning only relative to a benchmark. Three are customary: the firm's own history, the industry or peer group, and a theoretical or contractual standard (such as a covenant). Peer comparison is not straightforward: firms differ in accounting policies, business mix, geographic exposure and fiscal year-ends. Where possible, analysts adjust for differences (for example, converting leases consistently or reclassifying one-off items) before comparing.

2.4 Liquidity and Efficiency Ratios

2.4.1 Liquidity: can the firm pay its bills?

Liquidity ratios measure the firm's ability to meet short-term obligations from short-term resources.

```
Current ratio = Current assets / Current liabilities
Quick (acid-test) ratio = (Cash + Marketable securities + Receivables) / Current liabilities
Cash ratio = (Cash + Marketable securities) / Current liabilities
```

The current ratio counts inventory and prepaids, which may take a long time to convert to cash, or may be sold only at a discount. The quick ratio is more conservative. The cash ratio is the strictest. What counts as an adequate level depends on the industry: a supermarket that collects cash at the till and pays suppliers on 30-day terms may operate with a current ratio below 1, while a capital-goods manufacturer with long production cycles may need 2 or more. Very high liquidity is not unambiguously good; it may indicate idle cash or excessive inventory.

2.4.2 Efficiency (activity) ratios

Efficiency ratios measure how intensively the firm uses its assets, in particular the working capital that sits between buying inputs and collecting cash.

```
Receivables turnover = Revenue / Receivables            Days sales outstanding (DSO) = 365 / Receivables
turnover
Inventory turnover = COGS / Inventory                     Days inventory outstanding (DIO) = 365 /
Inventory turnover
Payables turnover = COGS / Payables                       Days payables outstanding (DPO) = 365 /
Payables turnover
Cash conversion cycle (CCC) = DSO + DIO − DPO
```

The **cash conversion cycle** is the number of days between paying suppliers and collecting from customers. A shorter cycle means less capital is tied up in working capital. A negative cycle (as in some retail and subscription businesses) means that customers and suppliers effectively finance the firm.

Where balances are seasonal or growing quickly, analysts use *average* balances rather than year-end, though year-end figures are acceptable if applied consistently. Credit sales should be used instead of total revenue where disclosed. Payables turnover is sometimes computed on purchases rather than COGS; if purchases are not disclosed, COGS is a reasonable approximation.

Broader asset-efficiency measures include **total asset turnover** (revenue / total assets) and **fixed asset turnover** (revenue / net PP&E).

Example 2.4 – Liquidity and the cash conversion cycle for Calder.

Step 1 – Liquidity, Year 2. Current ratio = 900 / 460 = **1.96**. Quick ratio = (180 + 360) / 460 = 540 / 460 = **1.17**. Cash ratio = 180 / 460 = **0.39**.

Step 2 – Liquidity, Year 1. Current ratio = 750 / 400 = **1.875**. Quick ratio = (150 + 305) / 400 = **1.14**. Cash ratio = 150 / 400 = **0.375**.

Liquidity is comfortable and slightly improved.

Step 3 – Efficiency, Year 2 (year-end balances). - DSO = 360 / 2,400 × 365 = **54.8 days** - DIO = 330 / 1,512 × 365 = **79.7 days** - DPO = 240 / 1,512 × 365 = **57.9 days** - CCC = 54.8 + 79.7 − 57.9 = **76.5 days**

Step 4 – Efficiency, Year 1. - DSO = 305 / 2,200 × 365 = **50.6 days** - DIO = 270 / 1,430 × 365 = **68.9 days** - DPO = 210 / 1,430 × 365 = **53.6 days** - CCC = 50.6 + 68.9 − 53.6 = **65.9 days**

Step 5 – Interpretation. The cycle lengthened by about 10.6 days. Customers took 4 more days to pay, inventory stayed on the shelf 11 more days, and the firm stretched suppliers by only 4 more days, which offset only a portion of the rest. The impact in cash terms: one day of sales is 2,400 / 365 = $6.58 million and one day of cost of goods is 1,512 / 365 = $4.14 million. A four-day slip in collections ties up roughly 4 × 6.58 = $26 million, and the eleven-day inventory build roughly 11 × 4.14 = $46 million.

The current ratio looks fine, yet the underlying drivers of the cycle are moving the wrong way; this is why one should examine the components of working capital, not only the aggregate liquidity ratio. Management should find out whether the longer cycle reflects deliberate strategy (higher safety stocks ahead of a product launch, more generous terms to win a big customer) or deteriorating discipline (slow-moving stock, rising disputes over invoices).

Example 2.5 – Window dressing a liquidity ratio. Calder's treasurer proposes using $100 million of cash on the last day of the year to repay payables ahead of schedule. The effect on the ratios, using the year-end figures from Example 2.4:

Step 1 – Current ratio after the payment. Current assets fall to 900 − 100 = 800; current liabilities fall to 460 − 100 = 360. New ratio = 800 / 360 = **2.22** (up from 1.96).

Step 2 – Quick ratio after the payment. Quick assets fall to 540 − 100 = 440; new ratio = 440 / 360 = **1.22** (up from 1.17).

Step 3 – Reality. No new resources have been created, and the firm has lost 100 of cash. It has less ability to meet unexpected needs, even though both ratios improved. This works because the starting ratios exceeded 1: removing equal amounts from numerator and denominator increases a ratio that is above one. The reverse is true for a ratio below one, where the same payment would worsen the ratio. Lenders know the trick, which is why covenants are often tested on average or quarterly figures, and why analysts examine cash flow, not merely ratios.

2.5 Leverage and Solvency Ratios

2.5.1 Why leverage matters

Leverage measures how much of the firm's financing is fixed-cost debt. Debt amplifies returns to shareholders when the firm earns more than its cost of debt, and magnifies losses when it does not. It also creates fixed obligations that, if unmet, trigger distress. Leverage ratios fall into two groups: **capital-structure ratios**, which compare debt with the capital base, and **coverage ratios**, which compare earnings or cash flow with the burden of servicing the debt.

2.5.2 Capital-structure ratios

```
Debt-to-equity = Total debt / Shareholders' equity
Debt-to-capital = Total debt / (Total debt + Equity)
Debt ratio = Total liabilities / Total assets
Equity multiplier = Total assets / Equity
Net debt = Total debt − Cash and marketable securities
Net debt / EBITDA
```

"Total debt" means interest-bearing obligations, including lease liabilities under IFRS 16 and similar US standards. Some analysts also add underfunded pensions and guarantees. Book values of equity may understate the market value of equity for profitable firms, so some analysts use market-value ratios as well.

2.5.3 Coverage ratios

```
Interest coverage (times interest earned) = EBIT / Interest expense
EBITDA coverage = EBITDA / Interest expense
Fixed-charge coverage = (EBIT + Lease payments) / (Interest + Lease payments)
Debt service coverage = Cash flow available for debt service / (Interest + Scheduled principal)
```

Coverage should be assessed under stress. A coverage ratio of 6 means that EBIT could fall by 83% before interest is uncovered; but for a cyclical business whose profit may fall by half in a downturn, the cushion matters more than the average. Bank covenants frequently specify a maximum Net debt / EBITDA and a minimum coverage ratio, and breaches give lenders the right to demand repayment or renegotiate terms.

Example 2.6 – Leverage analysis of Calder. Total debt in Year 2 is 100 + 600 = 700, and in Year 1 is 90 + 560 = 650. EBITDA in Year 2 is 312 + 96 = 408; in Year 1, 234 + 88 = 322.

Step 1 – Capital structure, Year 2. - Debt-to-equity = 700 / 1,240 = **0.56** - Debt-to-capital = 700 / (700 + 1,240) = 700 / 1,940 = **36.1%** - Debt ratio = 1,060 / 2,300 = **46.1%** - Equity multiplier = 2,300 / 1,240 = **1.85** - Net debt = 700 − 180 = **520**; Net debt / EBITDA = 520 / 408 = **1.27×**

Step 2 – Capital structure, Year 1. Debt-to-equity = 650 / 1,090 = **0.60**. Net debt = 650 − 150 = 500; Net debt / EBITDA = 500 / 322 = **1.55×**.

Step 3 – Coverage. Year 2: EBIT coverage = 312 / 48 = **6.5×**; EBITDA coverage = 408 / 48 = **8.5×**. Year 1: EBIT coverage = 234 / 42 = **5.6×**.

Step 4 – Average interest rate. 48 / [(700 + 650) / 2] = 48 / 675 = **7.1%**.

Step 5 – Stress test. Suppose EBIT fell 50% in a downturn, to 156. Coverage would then be 156 / 48 = 3.25×, still above 3, a typical comfort threshold, though the figure will vary with industry and rating. Debt capacity is not exhausted. If Calder had to cover the entire 100 of short-term debt from operating cash flow in addition to interest, a drop to 156 of EBIT would leave after-tax earnings (156 − 48) × 0.75 = 81 plus depreciation 96 = 177 of cash flow before capex and working capital, enough to repay the short-term debt but not to sustain 196 of capex.

Interpretation. Calder's leverage is moderate and falling in relative terms (profit grew faster than debt). It could carry more debt, but whether it should depends on the stability of its cash flows, its investment needs and its financial flexibility; those issues are developed in the capital-structure chapters.

2.6 Profitability and the DuPont Framework

2.6.1 Profitability ratios

Profitability ratios measure the return earned on sales and on capital.

```
Gross margin = Gross profit / Revenue
Operating (EBIT) margin = EBIT / Revenue
EBITDA margin = EBITDA / Revenue
Net profit margin = Net income / Revenue
Return on assets (ROA) = Net income / Total assets
Return on equity (ROE) = Net income / Shareholders' equity
Return on invested capital (ROIC) = NOPAT / Invested capital
```

ROA and ROE depend on the denominator measure: year-end, beginning or average balances. Average balances better match a flow earned over a period with a stock that changed during it; for simplicity many textbook calculations, including those in this chapter, use year-end balances unless stated, and the choice should be applied consistently.

ROIC is the preferred measure of operating performance because it is independent of capital structure. Invested capital is the capital tied up in operations: operating working capital plus fixed assets plus other operating assets (equivalently, debt plus equity less excess cash). For Calder in Year 2: operating working capital = 360 + 330 + 30 − 240 − 120 = 360; plus net PP&E 1,100 and intangibles 300 gives **1,760**. As a check: debt 700 + equity 1,240 − cash 180 = 1,760. ✓ ROIC = NOPAT / invested capital = 234 / 1,760 = **13.3%**. In Year 1, invested capital was 650 + 1,090 − 150 = 1,590 and NOPAT was 234 × 0.75 = 175.5, so ROIC = 175.5 / 1,590 = **11.0%**. Calder's ROIC improved by about 2.3 points, and, if the cost of capital is, say, 9%, the firm earns a spread of about 4.3 points in Year 2.

2.6.2 The three-step DuPont identity

The DuPont framework, developed at the DuPont corporation in the early twentieth century, breaks ROE into the product of three drivers:

```
ROE = Net income / Equity
    = (Net income / Revenue) × (Revenue / Total assets) × (Total assets / Equity)
    =     Net profit margin    ×    Asset turnover     ×    Equity multiplier
```

- **Net profit margin** captures operating efficiency and pricing power: how much profit each dollar of sales yields.
- **Asset turnover** captures asset efficiency: how many dollars of sales each dollar of assets supports.
- **Equity multiplier** captures financial leverage: how many dollars of assets each dollar of equity controls.

The decomposition shows that a high ROE can arise from different strategies. A discount retailer typically has thin margins but fast turnover; a luxury brand has high margins and slow turnover; a bank has low margins on assets and high leverage. Comparing ROE alone conceals these differences, and, importantly, it shows that ROE can be raised by leverage without any improvement in the underlying business, at the price of more risk.

2.6.3 The five-step extension

The five-step version separates the effects of taxes and interest:

```
ROE = Tax burden × Interest burden × EBIT margin × Asset turnover × Equity multiplier
    = (NI / EBT) × (EBT / EBIT) × (EBIT / Revenue) × (Revenue / Assets) × (Assets / Equity)
```

The tax burden (NI / EBT = 1 − tax rate) reflects the fraction of pre-tax profit retained after tax. The interest burden (EBT / EBIT) reflects the fraction of operating profit retained after paying lenders. The EBIT margin is a measure of operating profitability before financing and tax.

Example 2.7 – DuPont analysis of Calder.

Step 1 – Three-step, Year 2 (year-end balances). - Net profit margin = 198 / 2,400 = **8.25%** - Asset turnover = 2,400 / 2,300 = **1.043×** - Equity multiplier = 2,300 / 1,240 = **1.855×** - ROE = 0.0825 × 1.0435 × 1.8548 = **15.97%**. Check: 198 / 1,240 = 15.97%. ✓

Step 2 – Three-step, Year 1. - Net profit margin = 144 / 2,200 = **6.55%** - Asset turnover = 2,200 / 2,050 = **1.073×** - Equity multiplier = 2,050 / 1,090 = **1.881×** - ROE = 0.06545 × 1.0732 × 1.8807 = **13.21%**. Check: 144 / 1,090 = 13.21%. ✓

Step 3 – What changed? ROE rose 2.76 points. Margin rose from 6.55% to 8.25% (+1.7 points, a 26% relative improvement), asset turnover fell from 1.073 to 1.043, and the equity multiplier fell slightly from 1.881 to 1.855. All of the improvement therefore came from profitability; the asset base grew faster than sales (12.2% against 9.1%) and leverage dropped a little. The improvement is attributable to operations rather than financial engineering, which is the better kind.

Step 4 – Five-step, Year 2. - Tax burden = 198 / 264 = **0.750** - Interest burden = 264 / 312 = **0.846** - EBIT margin = 312 / 2,400 = **13.0%** - Asset turnover = **1.0435**; equity multiplier = **1.8548** - ROE = 0.750 × 0.8462 × 0.130 × 1.0435 × 1.8548 = **15.97%** ✓

Step 5 – Five-step, Year 1. - Tax burden = 144 / 192 = **0.750**; interest burden = 192 / 234 = **0.821**; EBIT margin = 234 / 2,200 = **10.64%**; turnover **1.0732**; multiplier **1.8807** - ROE = 0.75 × 0.8205 × 0.10636 × 1.0732 × 1.8807 = **13.21%** ✓

Step 6 – Interpretation. The tax burden was constant. The interest burden improved from 0.821 to 0.846, because EBIT grew faster than interest. The dominant driver is the EBIT margin, up 2.4 points. Notably, the improvement in the interest burden is a by-product of operating success rather than lower debt. A rival reporting the same ROE might get there with a 10% margin, 0.9× turnover and a 2.3× multiplier; the DuPont decomposition would reveal that its ROE depends on gearing, and would raise the question of how it would fare in a downturn.

Using average balances: Average equity is (1,240 + 1,090) / 2 = 1,165, so ROE on average equity = 198 / 1,165 = 17.0%. The level differs, but the diagnosis is the same.

2.6.4 ROE, growth and the sustainable growth rate

ROE links profitability to growth. If a firm retains a fraction **b** of earnings (the retention or plowback ratio) and earns a return ROE on its equity, then equity grows at b × ROE when ROE is calculated on *beginning* equity. This is the **sustainable growth rate**: the rate at which the firm can grow without issuing new equity or changing its leverage and profitability. When ROE is measured on year-end equity, the same result is written:

```
Sustainable growth rate g = (ROE × b) / (1 − ROE × b)      [ROE on year-end equity]
                          = ROE(beginning equity) × b
```

Example 2.8 – Sustainable growth for Calder.

Step 1 – Retention. Dividends were 48 on net income of 198, a payout ratio of 48 / 198 = 24.2%, so b = 1 − 0.2424 = **0.7576**.

Step 2 – Return on beginning equity. 198 / 1,090 = **18.17%**.

Step 3 – Growth. g = 0.1817 × 0.7576 = **13.76%**. Check: equity grew from 1,090 to 1,240, an increase of 13.76%. ✓

Step 4 – Using year-end ROE. ROE × b = 0.1597 × 0.7576 = 0.1210; g = 0.1210 / (1 − 0.1210) = **13.76%**. ✓ (The simple product 12.1% understates growth when year-end equity is used.)

Step 5 – Compare with actual sales growth. Revenue grew 9.1%, below the 13.8% sustainable rate. Calder could grow faster without raising equity, provided its margin, turnover and leverage hold. In fact, if turnover and the capital structure stay the same, assets must grow at the same rate as equity.

Interpretation. The sustainable growth rate gives a quick test of financial feasibility. A firm planning growth above its sustainable rate must raise more equity, increase leverage, improve margins or turnover, or reduce dividends. Rapidly growing businesses frequently fail not because they are unprofitable but because their growth outpaces their financing, a condition known as overtrading.

2.7 Quality of Earnings and the Limits of Ratio Analysis

2.7.1 What is meant by quality of earnings

High-quality earnings are those that are sustainable, repeatable, backed by cash and derived from conservative, transparent accounting. Low-quality earnings are inflated by one-off gains, aggressive revenue recognition, understated expenses or non-cash items that will reverse. Because management has discretion over accrual estimates and timing, no analysis of profitability is complete without an assessment of quality.

Red flags commonly include:

- **Revenue recognised early or aggressively:** receivables growing faster than sales, rising DSO, bill-and-hold arrangements, large returns or credit notes after quarter-end, channel stuffing (shipping more than distributors need).
- **Expenses deferred:** capitalisation of costs that peers expense (software development, customer-acquisition costs), lengthening depreciation lives, reduced maintenance or R&D, inadequate provisions for bad debts or warranties.
- **Reliance on non-operating items:** gains on asset sales, pension income assumptions, reversal of provisions, tax benefits.
- **Frequent "non-recurring" charges** and heavy use of adjusted or "pro forma" earnings that exclude recurring costs, such as stock-based compensation.
- **Divergence between earnings and cash flow:** profit persistently exceeding CFO.
- **Complex structures:** off-balance-sheet entities, aggressive related-party transactions, frequent changes in auditors or accounting policies, unusual fourth-quarter profits.

2.7.2 Cash-based and accrual-based diagnostics

Two simple diagnostics compare profit with cash.

```
Cash conversion (CFO / Net income)
Accruals ratio = (Net income − CFO) / Average total assets
```

A ratio of CFO to net income persistently below one, or a persistently positive accruals ratio, means that earnings rely on accruals rather than cash. Research on the accrual anomaly, notably the work of Richard Sloan, found that firms with high accruals tend to experience lower subsequent returns, consistent with investors overpricing the persistence of accrual-based earnings. A refined version scales accruals by net operating assets (NOA).

Example 2.9 – Earnings quality of Calder and of a contrast firm.

Step 1 – Calder: cash conversion. CFO / net income = 224 / 198 = **1.13**.

Step 2 – Calder: accruals ratio. (198 − 224) / [(2,300 + 2,050) / 2] = −26 / 2,175 = **−1.2%**. Negative accruals mean cash earnings exceed accounting profit, usually a good sign. Part of this is the large depreciation add-back; the working-capital investment of 70 is a drag on CFO, which is partly why the ratio is not even higher.

Step 3 – Calder: free cash flow conversion. FCFE / net income = 78 / 198 = **0.39**. Cash conversion before investment is excellent, but after heavy capex only 39% of profit is free cash.

Step 4 – Contrast firm "Vireo Systems". Vireo reports net income of 120 and CFO of 45, with average total assets of 800 and sales of 3,000. Cash conversion = 45 / 120 = **0.375**. Accruals ratio = (120 − 45) / 800 = **9.4%**, a high level that would trigger scrutiny.

Step 5 – Diagnosis. Vireo's DSO rose from 48 to 62 days. At sales of 3,000 per year (8.22 per day), receivables at 48 days were 3,000 × 48 / 365 = 394.5; at 62 days they are 3,000 × 62 / 365 = 509.6, an increase of **115.1**. That increase alone exceeds the entire gap between profit and cash (75), so the profit is, in effect, sitting in customers' accounts. Possible explanations include deliberate loosening of credit, early revenue recognition or customers in difficulty. Each implies a different risk, but none supports valuing Vireo at a multiple of reported earnings without further investigation.

2.7.3 Normalisation and adjusted earnings

Analysts often adjust reported earnings to estimate sustainable (core or recurring) earnings: remove gains and losses on disposals, restructuring and impairment charges that are genuinely unusual, and adjust for unusual tax rates. The risk lies in the adjustments themselves. A rule of thumb: if a cost recurs, it is an operating cost. Another: treat stock-based compensation as an economic expense, since the firm pays employees in claims on shareholders' wealth rather than in cash.

2.7.4 Limits of ratio analysis

Ratios are indispensable but imperfect.

- **They are backward-looking** and based on historical cost; valuation is forward-looking.
- **Accounting policies differ** across firms and countries (inventory methods, depreciation lives, capitalisation of development costs, lease treatment), reducing comparability.
- **Year-end snapshots can mislead** for seasonal businesses, or if management times transactions (see Example 2.5).
- **Aggregation hides detail.** A consolidated ratio for a diversified firm blends different businesses with different norms.
- **A ratio has no meaning in isolation.** It should be interpreted against history, peers and the firm's strategy and risk, and tied to the drivers of cash flow.
- **Inflation and growth distort** comparisons over time.
- **Non-financial drivers** (customer satisfaction, product pipeline, management quality, regulatory exposure) are invisible in the statements but often decisive for future cash flows.

The right approach is therefore to treat ratios as questions rather than answers. A deteriorating cash conversion cycle prompts a question to operations; a rising accruals ratio prompts questions to the controller and auditor; a falling interest burden prompts a question about the debt maturity profile. Used this way, ratio analysis is a disciplined method of directing management's attention.

2.7.5 Putting it together: a systematic approach

A practical sequence for analysing a firm is the following.

1. **Understand the business and the accounting.** Read the strategy, segment data and accounting policy notes; identify the key judgements.
2. **Read the cash flow statement first.** Is CFO healthy and consistent with profit? How is growth funded?
3. **Common-size and trend the statements.** Identify the lines that moved.
4. **Compute ratios by category.** Liquidity, efficiency, leverage and profitability, with DuPont to link them.
5. **Benchmark** against history and peers, adjusting for accounting differences.
6. **Test earnings quality** with the diagnostics above.
7. **Form a view**: what are the drivers of value, the key risks and the questions for management?

Category	Key ratios	Question answered	Typical warning sign
Liquidity	Current, quick, cash ratio	Can the firm pay its bills in the next year?	Falling ratios with rising short-term debt
Efficiency	DSO, DIO, DPO, CCC, asset turnover	How well are assets and working capital used?	CCC lengthening; receivables outpacing sales
Leverage	Debt/equity, net debt/EBITDA, coverage	How much fixed-claim financing, and can it be serviced?	Coverage below covenants; rising net debt/EBITDA
Profitability	Margins, ROA, ROE, ROIC	How much return is earned on sales and capital?	ROIC below WACC; margin erosion
Cash quality	CFO/NI, accruals ratio, FCF	Is profit backed by cash?	Persistent CFO below net income

Case Study: Pinecrest Outdoor Gear

Illustrative case. Pinecrest Outdoor Gear and Ridgeline Sports are fictional companies.

Pinecrest Outdoor Gear is a mid-sized designer and retailer of hiking, climbing and camping equipment. It sells through 140 company-owned stores and through wholesale agreements with independent outdoor shops. Over the past two years it has expanded rapidly, opening 35 new stores and launching a premium line of technical apparel. Pinecrest's lender, Northern Plains Bank, holds a $150 million term loan with covenants requiring net debt to EBITDA below 3.0 times and EBITDA interest coverage above 4.0 times. Its chief financial officer, Elena Marsh, has asked the finance team to prepare a review before a meeting with the bank and with the board, because she suspects the headline numbers are flattering.

The headline numbers appear in the following summary ($ millions), with a competitor, Ridgeline Sports, shown for comparison.

Item	Pinecrest Year 2	Pinecrest Year 1	Ridgeline Year 2
Revenue	1,120	900	1,050
Gross profit	448	342	399
EBITDA	140	108	126
Depreciation and amortisation	52	38	40
EBIT	88	70	86
Interest expense	30	18	14
Net income	43.5	39	54
Cash flow from operations	36	71	78
Capital expenditure	110	65	50
Receivables	118	74	62
Inventory	252	165	168
Payables	120	108	105
Total debt	380	240	150
Cash	40	55	70
Shareholders' equity	360	330	400
Total assets	880	720	770

Management's presentation to the board emphasised that revenue was up 24%, EBITDA up 30% and net income up 12%. The new apparel line had grown wholesale accounts, and the company had agreed to 75-day payment terms with several large independent retailers, compared with 45 days historically. In the final quarter, a promotional campaign had cleared some older stock, although the inventory balance at year-end was still higher than planned because of delayed deliveries of the winter range. Marsh also noted that the firm had extended the assumed useful life of its store fixtures from seven to ten years in the middle of the year, which she estimated reduced depreciation by about $6 million.

Marsh's team also found that the company had classified $9 million of store pre-opening costs as a capitalised asset, a treatment its auditors had accepted but which Ridgeline, in its own filings, expensed as incurred. The treasurer reminded her that if the covenant ratios are tested at the year-end, there is some flexibility: management could delay a shipment of capital equipment into the new year, reduce inventory purchases in the last two weeks, and ask suppliers to accept extended terms. The CEO, who is paid a bonus based on EBITDA growth and revenue growth, argued that the numbers speak for themselves and that the bank's covenants are conservative.

Marsh is not convinced. She wants to know what the cash flow statement says about the pace of growth, whether the apparent profitability is sustainable, and whether the company is approaching the limits of its debt capacity. She also wants to prepare the board for a possible conversation with the bank about covenant headroom.

Discussion Questions

1. Using the table, compute for Pinecrest and Ridgeline (Year 2): gross margin, EBIT margin, DSO, DIO, DPO, cash conversion cycle, net debt / EBITDA, EBITDA interest coverage and the accruals ratio (use total assets at year-end). Which firm is stronger and why? Is Pinecrest in compliance with its covenants, and how much headroom does it have?
2. Perform a three-step DuPont analysis of ROE for Pinecrest for both years and for Ridgeline in Year 2. What explains the difference between the firms, and what does the trend at Pinecrest tell you?
3. Estimate the impact of the change in depreciation policy and of capitalising pre-opening costs on Pinecrest's reported EBIT and EBITDA (assume a 25% tax rate for the effect on net income). Re-state the covenant ratios on the basis of Ridgeline's more conservative treatment. What does this tell you about the reliability of the headline growth numbers?
4. Evaluate the CEO's incentive and the treasurer's suggestions for managing year-end ratios. What should Marsh recommend to the board regarding the bank meeting, the bonus metrics and the pace of expansion?

Key Takeaways

- The balance sheet, income statement, cash flow statement and statement of changes in equity articulate; checking that they reconcile is the first line of defence against error and manipulation.
- Accrual accounting gives a better picture of performance than cash receipts, but it depends on estimates and judgements; profit is an opinion, while cash is a fact.
- Operating cash flow is derived from net income by adding back non-cash charges and adjusting for working-capital changes: increases in operating assets use cash and increases in operating liabilities provide it.
- Free cash flow to the firm (NOPAT + D&A − capex − ΔNWC) is the cash available to all capital providers and is the basis of enterprise valuation; free cash flow to equity subtracts after-tax interest and adds net borrowing.

- Common-size and trend analysis reveal the structure and direction of change in a firm and show whether margins, working capital and financing are moving together with growth.
- Liquidity ratios test short-term solvency, but the efficiency ratios and the cash conversion cycle explain why liquidity changes; year-end ratios can be window-dressed.
- Leverage must be judged on both capital structure (debt to equity or capital, net debt to EBITDA) and coverage (EBIT or EBITDA to interest), and stress-tested against downturns and covenants.
- The DuPont framework decomposes ROE into margin, asset turnover and leverage (and in five steps, tax burden, interest burden and operating margin), showing whether returns arise from operations or financial engineering.
- ROIC, being independent of capital structure, is the best summary of operating performance and should be compared with the cost of capital; the sustainable growth rate links ROE and retention to feasible growth.
- Quality of earnings is assessed by comparing profit with cash (CFO/NI, accruals ratio), by scrutinising receivables, capitalisation and non-recurring items, and ratios should be treated as questions that direct investigation rather than as final answers.

Review Questions

1. Why do the four primary financial statements articulate? Give an example of how an error in one statement would show up in another.
2. Explain why a growing firm can report rising profits and still run short of cash. Which line items would you examine first?
3. Using the indirect method, explain the direction of the adjustment to net income for an increase in (a) inventory, (b) accounts payable, (c) prepaid expenses and (d) depreciation.
4. What is the difference between FCFF and FCFE? Why is FCFF unaffected by the firm’s capital structure, and at what rate is each discounted?
5. Why is EBITDA a poor substitute for cash flow? In which industries is the distinction most important?
6. A company’s current ratio rises from 1.5 to 2.2, yet analysts express concern. What explanations might justify their concern?
7. Define the cash conversion cycle. How can a firm shorten it, and what risks might each approach create?
8. Explain how a firm can increase its ROE without improving operating performance. Use the DuPont identity and discuss the risks.
9. Why is ROIC preferred to ROE as a measure of operating performance, and against what benchmark should it be compared?
10. Describe five red flags of low-quality earnings and, for each, state the ratio or analysis that would help detect it.

Problems

Problems 1 to 7 use the following data for Harbor Retail, a fictional company ($ millions). Income statement: revenue 5,000; COGS 3,500; operating expenses (SG&A) 900; depreciation 150; interest 60; tax rate 30%. Balance sheet at year end: cash 200; receivables 250; inventory 600; other current assets 50; net PP&E 1,400; other non-current assets 100 (total assets 2,600); payables 400; accrued liabilities 150; short-term debt 100; long-term debt 750; equity 1,200.

1. **Liquidity and efficiency.** Calculate Harbor's current ratio, quick ratio (cash plus receivables over current liabilities), cash ratio, DSO, DIO, DPO and cash conversion cycle, using year-end balances and a 365-day year.
2. **DuPont.** Compute net income, then the net profit margin, asset turnover, equity multiplier and ROE, using year-end balances. Verify that the product of the three equals the direct ROE.
3. **Leverage.** Calculate EBIT, EBITDA, EBIT interest coverage, total debt, net debt, net debt / EBITDA, debt-to-equity and debt-to-capital.
4. **Free cash flow.** Harbor's capital expenditure was 220, the increase in operating working capital was 40 and net new borrowing was 30. Calculate NOPAT, FCFF and FCFE.
5. **Operating cash flow.** Net income, depreciation, an increase in receivables of 20, an increase in inventory of 45 and an increase in payables of 30 are the only items affecting operating cash flow. Compute CFO, free cash flow after capex of 220 (CFO less capex), and the ratio of CFO to net income.
6. **Common-size and operating leverage.** In the prior year Harbor had revenue of 4,600, COGS of 3,266, operating expenses of 874 and depreciation of 140. Compute the gross margin and EBIT margin for both years, the growth rates of revenue and EBIT, and the degree of operating leverage (percentage change in EBIT divided by percentage change in revenue).
7. **Sustainable growth.** Harbor pays out 40% of earnings as dividends. Using year-end ROE, calculate the sustainable growth rate.
8. **Earnings quality.** A firm has net income of 120, CFO of 45 and average total assets of 800. Its DSO rose from 48 to 62 days with annual sales of 3,000. Calculate the CFO / net income ratio, the accruals ratio, and the increase in receivables implied by the change in DSO. Comment on how much of the profit-cash gap it explains.

Answers to Selected Problems

1. Current assets = 200 + 250 + 600 + 50 = 1,100; current liabilities = 400 + 150 + 100 = 650. Current ratio = 1,100 / 650 = **1.69**. Quick ratio = (200 + 250) / 650 = **0.69**. Cash ratio = 200 / 650 = **0.31**. DSO = 250 / 5,000 × 365 = **18.3 days**. DIO = 600 / 3,500 × 365 = **62.6 days**. DPO = 400 / 3,500 × 365 = **41.7 days**. CCC = 18.3 + 62.6 − 41.7 = **39.1 days**.
2. EBIT = 5,000 − 3,500 − 900 − 150 = 450; EBT = 450 − 60 = 390; tax = 117; net income = **273**. Margin = 273 / 5,000 = **5.46%**; asset turnover = 5,000 / 2,600 = **1.923×**; equity multiplier = 2,600 / 1,200 = **2.167×**. ROE = 0.0546 × 1.923 × 2.167 = **22.75%**, equal to 273 / 1,200.
3. EBIT = **450**; EBITDA = 450 + 150 = **600**; coverage = 450 / 60 = **7.5×**. Total debt = 100 + 750 = **850**; net debt = 850 − 200 = **650**; net debt / EBITDA = **1.08×**; debt-to-equity = 850 / 1,200 = **0.71**; debt-to-capital = 850 / 2,050 = **41.5%**.
4. NOPAT = 450 × 0.70 = **315**. FCFF = 315 + 150 − 220 − 40 = **205**. After-tax interest = 60 × 0.70 = 42. FCFE = 205 − 42 + 30 = **193**.
5. CFO = 273 + 150 − 20 − 45 + 30 = **388**. CFO less capex = 388 − 220 = **168**. CFO / net income = 388 / 273 = **1.42**.
6. Current-year gross margin = 1,500 / 5,000 = **30.0%**; prior = (4,600 − 3,266) / 4,600 = 1,334 / 4,600 = **29.0%**. Prior EBIT = 1,334 − 874 − 140 = 320, a margin of **6.96%**; current EBIT margin = 450 / 5,000 = **9.0%**. Revenue growth = 5,000 / 4,600 − 1 = **8.70%**; EBIT growth = 450 / 320 − 1 = **40.6%**; degree of operating leverage = 40.6 / 8.70 = **4.67**.
7. Retention b = 0.60. ROE × b = 0.2275 × 0.60 = 0.1365. Sustainable growth with year-end ROE = 0.1365 / (1 − 0.1365) = **15.8%**. (The simple product, 13.65%, understates it.)

8. CFO / NI = 45 / 120 = **0.375**. Accruals ratio = (120 − 45) / 800 = **9.4%**. Receivables at 48 days = 3,000 × 48 / 365 = 394.5; at 62 days = 509.6; increase = **115.1**. The receivables build exceeds the entire profit-cash gap of 75, indicating that profit is being tied up in customers’ accounts; the other items (inventory, payables, depreciation) must offset 40 in the other direction. Such a pattern warrants investigation of credit policy and revenue recognition.

Chapter 3: Time Value of Money

Learning Objectives

After studying this chapter you should be able to:

- Explain why a dollar today is worth more than a dollar tomorrow, and translate cash flows across time using present value (PV) and future value (FV) factors.
- Value level annuities, annuities due, deferred annuities, perpetuities, growing perpetuities and growing annuities, and recognise which formula fits a given payment pattern.
- Convert between stated (nominal) rates, periodic rates and effective annual rates (EAR) for any compounding frequency, including continuous compounding.
- Build a loan amortization schedule, split each payment into interest and principal, and compute the outstanding balance at any date.
- Distinguish nominal from real cash flows and discount rates using the Fisher relationship, and keep cash flows and discount rates consistent in inflation terms.
- Apply time-value tools to managerial decisions such as lease-versus-buy, rebate-versus-cheap-financing, retirement funding and the pricing of long-lived commitments.

Opening Vignette

Illustrative scenario. Harbor Ridge Components is a fictional mid-sized manufacturer of precision valves.

On a Tuesday morning, Harbor Ridge's chief financial officer, Marisol Okafor, found three proposals on her desk, each of which looked attractive on its face.

The first came from a supplier of automated machining cells. It offered a price of $2.4 million payable in full on delivery, or "easy terms" of $600,000 per year for five years beginning at delivery. The sales representative described the second option as "only $3.0 million in total, with no interest charged." Marisol noticed that the second option was in fact more expensive in nominal terms, yet the supplier's brochure said it was "free financing." She wanted to know what interest rate was hidden inside the instalment plan.

The second proposal came from the head of human resources, who wanted to replace a defined-benefit pension promise with a package that paid each retiring employee $1,500 per month for twenty years. The question was how much money the company must set aside per employee today, given that retirement was some 25 years away, that pension assets were expected to earn 6% a year, and that the monthly benefit would eventually need to keep up with inflation.

The third came from the treasurer, who had negotiated a $5 million term loan. Two banks quoted nearly identical headline rates, but one compounded monthly and the other semi-annually, and one required level monthly payments while the other required equal principal repayments. Which was cheaper? And what would the outstanding balance be if the company chose to prepay after four years?

None of these questions can be answered by adding up cash amounts. Each requires the same underlying idea: money received at different dates is not directly comparable, and the exchange rate between dates is the interest rate. In this chapter we develop that idea carefully, because every later chapter in this book (bond valuation, capital budgeting, the cost

of capital, firm valuation) is an application of it. By the end we will return to Marisol's three problems and resolve them with the tools in hand.

3.1 Why Money Has a Time Value

3.1.1 Three reasons a dollar today beats a dollar tomorrow

A dollar received today is worth more than a dollar received a year from now for three distinct reasons, and it is useful to keep them separate because they matter differently in different applications.

1. **Opportunity cost (productive investment).** A dollar in hand can be invested at a positive return. If a safe deposit pays 4% a year, then $1.00 today becomes $1.04 in a year. Anyone offered $1.00 a year from now must therefore be compensated for what the dollar could have earned in the meantime. This is the core economic reason and it would apply even in a world with no inflation and no uncertainty.
2. **Inflation (purchasing power).** If prices rise 3% a year, a dollar next year buys less than a dollar today. Inflation erodes the real value of a fixed nominal sum, which is why lenders demand a rate that exceeds expected inflation. We treat this formally in Section 3.8.
3. **Risk and uncertainty.** A promised future payment may not be made. The more uncertain the payment, the less it is worth today. In practice the interest rate used to convert future amounts to today's amounts contains a premium for this risk, which is why a bond issued by a distressed firm is discounted at a higher rate than a government bond.

The *interest rate*, or equivalently the *discount rate*, is the single number that summarises all three effects for a particular cash flow. Time-value mathematics is simply the arithmetic of applying that number consistently.

3.1.2 The vocabulary

A handful of terms recur throughout the chapter:

- **Present value (PV):** the value today of a cash flow (or stream of cash flows) to be received in the future.
- **Future value (FV):** the value at some future date of a cash flow (or stream) that exists today.
- **Discount rate / interest rate (r):** the rate per period used to move values between dates.
- **Number of periods (n or t):** the number of compounding intervals between the two dates.
- **Compounding:** earning interest on previously earned interest. **Discounting** is the reverse operation.
- **Cash flow timeline:** a horizontal line marking dates 0, 1, 2, … with the cash flows beneath. Drawing the timeline is the single most effective defence against errors, particularly for annuities that start late or payments that come at the beginning of periods.

A convention worth fixing now: unless stated otherwise, cash flows are assumed to arrive at the *end* of each period, and date 0 is "today". We will explicitly flag annuities due, in which payments arrive at the *start* of each period.

3.1.3 Simple versus compound interest

Under **simple interest**, interest is earned only on the original principal: investing $1,000 at 8% simple interest for three years yields $1,000 × (1 + 3 × 0.08) = $1,240. Under **compound interest**, interest is earned on accumulated interest as well: $1,000 × 1.08^3 = $1,259.71. The difference is small over three years, but it grows without bound. Over 30 years the

simple-interest account would hold $3,400 while the compounded account would hold $10,062.66. Compounding is the engine of long-horizon wealth accumulation, and also the reason why debt left unpaid becomes so burdensome. Virtually every financial contract in practice uses compound interest, so that is our default.

3.2 Present Value and Future Value of a Single Cash Flow

3.2.1 The two fundamental formulas

If an amount PV is invested for n periods at a periodic rate r, its future value is

```
FV = PV × (1 + r)^n
```

The term (1 + r)^n is the **future value factor** (or compounding factor). Rearranging gives the present value of a future amount:

```
PV = FV / (1 + r)^n = FV × (1 + r)^(−n)
```

The term 1 / (1 + r)^n is the **discount factor**. It tells us how many dollars today are equivalent to one dollar at date n. At r = 7% and n = 8, the discount factor is 1 / 1.07⁸ = 1 / 1.7182 = 0.5820; one dollar eight years from now is worth about 58 cents today.

Notice that the two formulas contain four variables (PV, FV, r, n). Given any three, the fourth can be solved:

```
r = (FV / PV)^(1/n) − 1
n = ln(FV / PV) / ln(1 + r)
```

Example 3.1 – Present and future value of a lump sum. Harbor Ridge has promised to pay a supplier a $250,000 settlement in eight years. The company can earn 7% per year on a sinking fund. (a) How much must it set aside today? (b) Separately, the company invests $100,000 today at 6% for ten years; what will it be worth?

(a) Step 1: Compute the discount factor. 1.07⁸ = 1.718186, so the factor is 1 / 1.718186 = 0.58201. *Step 2:* PV = $250,000 × 0.58201 = **$145,502.28**. *Check:* $145,502.28 × 1.718186 = $250,000. ✓

(b) Step 1: 1.06¹⁰ = 1.790848. *Step 2:* FV = $100,000 × 1.790848 = **$179,084.77**.

Interpretation: the sinking fund needs to start at less than 60% of the eventual obligation because seven percent growth does the remaining work. This ratio will fall as either the rate or the horizon rises.

3.2.2 Solving for the rate and the horizon

Two of the most useful applications run the formula “backwards”. If a $100 investment becomes $150 in five years, the implied annual return is (150 / 100)^(1/5) − 1 = 1.5^0.2 − 1 = 8.447%. Notice that the answer is the *compound* annual growth rate (CAGR), which is lower than the simple average (50% / 5 = 10%) because compounding does some of the work. Managers should always compute growth over multiple years as a geometric, not arithmetic, rate when describing what happened to a single balance.

Solving for n is the basis of the well-known **rule of 72**: at a rate of r percent, money doubles in roughly 72 / r years. At 9%, the exact doubling time is ln 2 / ln 1.09 = 8.04 years, while the rule gives 72 / 9 = 8.0 years. The rule is a mental shortcut and works best for rates between about 6% and 10%.

3.2.3 Common mistakes with single cash flows

- **Mismatched periods.** Using an annual rate with a number of months, or a monthly rate with a number of years. The rate and the period count must refer to the same unit.
- **Forgetting that compounding is non-linear.** Doubling the horizon more than doubles the growth; halving the rate does not halve the final value.
- **Discounting the wrong number of periods.** A cash flow at the *end* of year 5 is discounted five periods; one at the *start* of year 5 is discounted four. Draw the timeline.
- **Comparing cash flows at different dates.** Before comparing or adding, always move everything to a common date. Any date works, because the ratio between two values moved to the same date does not depend on the date chosen. Present value (date 0) is merely the most convenient.

3.2.4 The principle of value additivity

A powerful consequence of the formulas is **value additivity**: once cash flows are expressed at the same date, they can be added. The present value of a stream is the sum of the present values of its individual payments:

$$PV = CF_1/(1+r) + CF_2/(1+r)^2 + \dots + CF_n/(1+r)^n = \Sigma\, CF_t/(1+r)^t$$

Every annuity and perpetuity formula in the rest of this chapter is just a closed-form shortcut for this sum when the cash flows follow a pattern. Whenever a formula does not fit, one can always fall back on discounting each cash flow individually, which is exactly what a spreadsheet does.

3.3 Annuities

3.3.1 Ordinary annuities

An **annuity** is a finite series of equal cash flows at equal intervals. Leases, mortgage payments, bond coupons, pension benefits and instalment contracts are all annuities. If the first payment arrives one period from now (the **ordinary annuity**), the present value of n payments of C at rate r is

```
PV = C × [1 − (1 + r)^(−n)] / r
```

The bracketed expression is the **annuity factor**, often written A(r, n). It is simply the sum of the first n discount factors. For r = 8% and n = 5 the factor is $(1 - 1.08^{-5}) / 0.08 = (1 - 0.680583) / 0.08 = 3.992710$. So five annual payments of $1 are worth $3.99 today, which is less than $5 because later payments are discounted more heavily.

The future value of the same annuity, measured at the date of the last payment, is

```
FV = C × [(1 + r)^n − 1] / r
```

and it is related to the present value by FV = PV × (1 + r)^n, as it must be.

Derivation by subtraction. The annuity formula can be understood without algebra. A perpetuity of C starting at date 1 is worth C / r. A perpetuity of C starting at date n + 1 is worth C / r at date n, or $C / [r(1 + r)^n]$ today. The difference between the two perpetuities is exactly an n-payment annuity: $C / r - C / [r(1 + r)^n] = (C / r)[1 - (1 + r)^{(-n)}]$. This “cancel the tail” idea is useful for valuing many irregular patterns.

3.3.2 Annuities due and deferred annuities

If payments are made at the *beginning* of each period (rent, insurance premiums and lease payments are typically paid in advance), the annuity is an **annuity due**. Every payment arrives one period earlier than in the ordinary case, so its value is higher by exactly one period of compounding:

```
PV(annuity due) = PV(ordinary annuity) × (1 + r)
FV(annuity due) = FV(ordinary annuity) × (1 + r)
```

A **deferred annuity** is an ordinary annuity whose first payment is more than one period away. Value it in two steps: use the annuity formula to find its value one period *before* the first payment, then discount that value back to today. For example, if the first of 20 payments occurs at date 6, the annuity formula gives the value at date 5, and this must be discounted five periods to date 0.

Example 3.2 – Annuity, annuity due and future value. A supplier offers Harbor Ridge a maintenance contract costing $40,000 per year for five years. The company's opportunity cost of funds is 8%.

(a) Ordinary annuity (paid at year-end). Annuity factor = $(1 - 1.08^{-5}) / 0.08 = (1 - 0.680583) / 0.08 = 3.992710$. PV = $40,000 × 3.992710 = **$159,708.40**.

(b) Annuity due (paid at the start of each year). PV = $159,708.40 × 1.08 = **$172,485.07**. The extra $12,776.67 is the cost of paying one year sooner on every instalment.

(c) Future value of the ordinary annuity at year 5. FV factor = $(1.08^5 - 1) / 0.08 = (1.469328 - 1) / 0.08 = 5.866601$. FV = $40,000 × 5.866601 = **$234,664.04**. *Check:* $159,708.40 × 1.469328 = $234,664.04. ✓

Managerial point: the contract that bills in advance is more expensive in present-value terms by 8%, even though the cash totals ($200,000) are identical. Negotiating the payment date is a real economic concession.

Example 3.3 – The hidden interest rate in "easy terms." Return to the machining-cell offer in the vignette: $2.4 million cash on delivery, or five annual instalments of $600,000 with the first due at delivery (an annuity due). What interest rate is implied?

Step 1: If Harbor Ridge pays cash it pays $2,400,000. If it chooses instalments, it pays $600,000 now and four more payments of $600,000 over the next four years. The financing it receives is therefore $2,400,000 − $600,000 = $1,800,000, repaid by four year-end payments of $600,000. *Step 2:* The annuity factor must satisfy $600,000 × A(r, 4) = $1,800,000, so A(r, 4) = 3.0000. *Step 3:* Try r = 12%: $A = (1 - 1.12^{-4}) / 0.12 = (1 - 0.635518) / 0.12 = 3.037349$. Try r = 13%: $A = (1 - 1.13^{-4}) / 0.13 = (1 - 0.613319) / 0.13 = 2.974471$. Interpolating: r ≈ 12% + (3.037349 − 3.000000) / (3.037349 − 2.974471) = 12% + 0.594 ≈ **12.59%**.

So the "free financing" costs about 12.6% a year. If Harbor Ridge can borrow at, say, 7%, paying cash (funded by its bank) is far cheaper. Total nominal outlay is $3.0 million versus $2.4 million, so the extra $600,000 is interest; it is not "free". Whenever a seller offers a choice between a cash price and a payment plan, compute the implied rate and compare it with your own cost of capital.

3.3.3 A reference table of annuity factors

Table 3.1 shows ordinary annuity factors A(r, n) for several rates and horizons. Such tables were once indispensable; today a calculator or spreadsheet does the same work, but the table is useful for building intuition about how factors grow with horizon and shrink with rate.

Table 3.1 – Present value annuity factors A(r, n) = [1 − (1 + r)^(−n)] / r

n (years)	r = 4%	r = 6%	r = 8%	r = 10%	r = 12%
5	4.4518	4.2124	3.9927	3.7908	3.6048
10	8.1109	7.3601	6.7101	6.1446	5.6502
20	13.5903	11.4699	9.8181	8.5136	7.4694
30	17.2920	13.7648	11.2578	9.4269	8.0552
∞ (perpetuity 1/r)	25.0000	16.6667	12.5000	10.0000	8.3333

Two lessons stand out. First, at high discount rates the distant future barely matters: at 12%, a 30-year annuity is already 96.7% of the value of a perpetuity. Second, at low rates the tail is long: at 4% the same 30-year annuity captures only 69% of the perpetuity value. This explains why long-duration assets (infrastructure, real estate, growth equities) are so sensitive to interest rates. We return to this in Chapter 4 under the heading of duration.

3.4 Perpetuities and Growing Perpetuities

3.4.1 The level perpetuity

A **perpetuity** is an annuity that never ends. Although no cash flow lasts forever, the model is a good approximation for preferred shares, endowment payouts, some government bonds historically (the British consol), and the terminal value of a business. The present value of a perpetuity paying C at the end of each period, beginning one period from now, is

```
PV = C / r
```

This follows from the annuity formula by letting n → ∞, since (1 + r)^(−n) → 0. It can also be seen intuitively: to receive C each year forever, one needs a principal P that earns exactly C at rate r, so P × r = C. The principal is never touched.

3.4.2 The growing perpetuity

If the payment grows at a constant rate g per period, with the first payment C_1 one period from now, the present value is the **Gordon growth formula**:

```
PV = C₁ / (r − g)        valid only if r > g
```

The requirement r > g is essential. If growth exceeded the discount rate, the present value would be infinite and the formula meaningless; in practice it signals that a perpetual growth assumption is unrealistic. For a firm, long-run perpetual growth above the economy's nominal growth rate is untenable, so g in terminal-value calculations should typically be bounded by long-run nominal GDP growth.

Note carefully which payment appears in the numerator: C_1, the payment *next* period, not the payment just received. A common error is to use C_0 without multiplying by (1 + g).

Example 3.4 – Perpetuity and growing perpetuity. A university endowment fund wishes to finance a chair that costs \$90,000 per year, paid at the end of each year, starting next year. The fund expects to earn 7.5%.

(a) Level cost. PV = \$90,000 / 0.075 = **\$1,200,000**.

(b) Cost that grows 3% a year (with the first payment still \$90,000). PV = \$90,000 / (0.075 − 0.03) = \$90,000 / 0.045 = **\$2,000,000**.

Interpretation: allowing the chair's cost to keep pace with 3% inflation raises the required gift by two-thirds ($800,000). In perpetuity problems the denominator is the *spread* r − g, so small changes in either input move the value dramatically. If g rose to 4.5%, the denominator would fall to 0.03 and the required fund would grow to $3,000,000. Sensitivity to the spread is why terminal values in discounted cash flow models deserve particular scrutiny.

3.5 Growing Annuities

3.5.1 Formula and logic

Many real cash flows grow for a finite period: a ten-year royalty indexed to sales, a salary profile that rises with experience, a lease with annual escalators. A **growing annuity** pays C_1 at date 1, $C_1(1 + g)$ at date 2, and so on up to $C_1(1 + g)^{(n-1)}$ at date n. Its present value is

```
PV = [C₁ / (r − g)] × [1 − ((1 + g)/(1 + r))^n]
```

The formula remains valid when r < g (the bracket and the leading term are then both negative and the product is positive), and when r = g the value is simply n × C_1 / (1 + r). The bracket can be read as "perpetuity value minus the value of the perpetuity's tail beyond n", exactly as in the level case.

The same idea gives the future value of a growing annuity at date n: PV × $(1 + r)^n$.

3.5.2 Step-by-step check

Example 3.5 – A growing annuity. A franchise agreement will pay Harbor Ridge royalties of $120,000 in year 1, growing 5% per year, for six years. The discount rate is 10%. Value the stream.

Step 1 (formula). C_1 / (r − g) = $120,000 / 0.05 = $2,400,000. *Step 2.* $((1 + g)/(1 + r))^6 = (1.05 / 1.10)^6 = 0.954545^6 = 0.756449$. *Step 3.* PV = $2,400,000 × (1 − 0.756449) = $2,400,000 × 0.243551 = **$584,522.27**.

Step 4 (cross-check by discounting each year):

Year	Cash flow	Discount factor at 10%	Present value
1	$120,000.00	0.909091	$109,090.91
2	$126,000.00	0.826446	$104,132.23
3	$132,300.00	0.751315	$99,398.95
4	$138,915.00	0.683013	$94,880.81
5	$145,860.75	0.620921	$90,568.05
6	$153,153.79	0.564474	$86,451.32
Total			**$584,522.27**

The two methods agree, which is the best advice for any new formula: verify it once by brute force. Note how the present values still decline over time even though the cash flows rise, because the discount rate (10%) exceeds the growth rate (5%).

3.6 Compounding Frequency and the Effective Annual Rate

3.6.1 Stated rates versus effective rates

Interest rates are quoted in many ways, and the quoted number is often not the rate you actually earn or pay over a year. A **stated rate** (also called the annual percentage rate, APR, or nominal annual rate) is a convention: it is the periodic rate multiplied by the number of periods in a year. A credit card that charges 1.5% per month has an APR of 18%. But it does not charge 18% per year; because interest is added every month and then itself earns interest, the true annual cost is higher.

The **effective annual rate (EAR)** is the rate that, compounded once per year, produces the same growth as the stated rate compounded m times per year:

```
EAR = (1 + APR/m)^m − 1
```

where m is the number of compounding periods per year. The periodic rate is APR / m. Always convert to an EAR before comparing any two rates with different compounding frequencies, and always use a periodic rate that matches the cash-flow frequency when valuing a stream.

3.6.2 Continuous compounding

As m grows without bound, (1 + APR/m)^m approaches e^APR. With **continuous compounding**:

```
FV = PV × e^(r × t)        EAR = e^r − 1        PV = FV × e^(−r × t)
```

Continuous compounding is mainly a mathematical convenience. It appears in option pricing (Black-Scholes), in log returns and in theoretical models where growth is treated as a flow. Because e^r exceeds any discrete compounding, it is the upper bound of the EAR for a given stated rate: the gain from moving from daily to continuous compounding is negligible in practice.

Example 3.6 – EAR under different compounding conventions. A bank advertises a stated annual rate of 12%. Compute the EAR for various compounding frequencies and the value of $1,000 after five years under annual and continuous compounding.

Step 1. Apply EAR = (1 + 0.12/m)^m − 1. - Annual (m = 1): 1.12 − 1 = 12.0000%. - Semi-annual (m = 2): $(1.06)^2 - 1$ = 12.3600%. - Quarterly (m = 4): $(1.03)^4 - 1$ = 12.5509%. - Monthly (m = 12): $(1.01)^{12} - 1$ = 12.6825%. - Daily (m = 365): (1 + 0.12/365)^365 − 1 = 12.7475%. - Continuous: e^0.12 − 1 = 12.7497%.

Step 2. Five-year values of $1,000: annual compounding gives $1{,}000 \times 1.12^5$ = $1,762.34; continuous compounding gives 1,000 × e^0.60 = $1,822.12.

Table 3.2 – Effect of compounding frequency on a 12% stated annual rate

Compounding	Periods per year (m)	Periodic rate	EAR
Annual	1	12.0000%	12.0000%
Semi-annual	2	6.0000%	12.3600%
Quarterly	4	3.0000%	12.5509%
Monthly	12	1.0000%	12.6825%
Daily	365	0.0329%	12.7475%
Continuous	∞	n/a	12.7497%

The table shows diminishing returns to more frequent compounding: the jump from annual to semi-annual adds 36 basis points, from monthly to daily only 7, and from daily to continuous barely 0.2.

3.6.3 Going backwards: from EAR to a periodic rate

Often the problem runs the other way. Suppose a financial model requires a monthly discount rate and the known annual cost of capital is an EAR of 10%. The correct monthly rate is not 10% / 12 = 0.8333%; it is the rate that compounds to 10% over twelve months:

```
r_monthly = (1 + EAR)^(1/12) − 1 = 1.10^(1/12) − 1 = 0.7974%
```

The corresponding stated annual rate, compounded monthly, is 12 × 0.7974% = 9.5690%. The difference between 9.57% and 10% is real money on a large book of loans. An analyst who divides 10% by 12 and then compounds will overstate the effective cost by about 47 basis points (10.4713% instead of 10.0000%).

3.6.4 Rates that are not annual: the APR trap

Regulators in many countries require lenders to disclose an APR, but the definition varies by jurisdiction and product, and APR typically does not capture compounding. Payday-style loans illustrate the trap: a fee of $15 per $100 borrowed for two weeks is a 15% two-week rate. Quoted as an APR it is 15% × 26 = 390%; but as an EAR it is $1.15^{26} - 1$, a number in the several-thousand-percent range. Borrowers who focus on the "15%" fee underestimate the cost by a very wide margin, and managers who run revolving trade-credit terms can make the same mistake in a milder form.

The same logic applies to trade credit terms. A supplier offering "2/10, net 30" (a 2% discount if paid within 10 days, otherwise the full amount due in 30 days) is effectively lending at a rate of 2/98 = 2.0408% for 20 days. As an EAR that is (1.020408)^(365/20) − 1 ≈ 44.6%. For any firm with access to a bank loan at 10% or less, taking the discount is a far better use of cash than paying on day 30. Treasurers should therefore treat early-payment discounts as an investment opportunity with a very high return.

3.7 Loan Amortization

3.7.1 The level-payment (annuity) loan

Most consumer and commercial term loans are **amortizing**: each payment covers the interest accrued since the last payment plus a portion of the principal, so the balance falls to zero by maturity. The standard design is a level-payment loan in which every instalment is the same. Because the payments form an annuity whose present value equals the amount borrowed, we can solve for the payment directly:

```
Payment = Loan × r / [1 − (1 + r)^(−n)]
```

where r is the periodic rate (APR divided by payments per year) and n is the total number of payments. In each period:

```
Interest_t  = Balance_(t−1) × r
Principal_t = Payment − Interest_t
Balance_t   = Balance_(t−1) − Principal_t
```

Early payments consist mostly of interest, and later payments mostly of principal, because interest is charged on the declining balance. The **outstanding balance** after k payments equals the present value of the remaining payments, which gives a closed form that avoids building the entire table:

```
Balance_k = Payment × [1 − (1 + r)^(−(n−k))] / r
```

An equivalent forward-looking form is Balance_k = Loan × (1 + r)^k − Payment × [(1 + r)^k − 1] / r.

Example 3.7 – A 30-year mortgage-style loan. Harbor Ridge finances a $300,000 building improvement with a 30-year loan at a stated rate of 6%, compounded and paid monthly. (a) Compute the payment. (b) Show the first three months of the schedule. (c) Find the balance after five years. (d) Compare with a 15-year loan at the same rate.

(a) Payment. Periodic rate r = 0.06 / 12 = 0.5% = 0.005; n = 360. $(1.005)^{-360} = 0.166042$; 1 − 0.166042 = 0.833958; annuity factor = 0.833958 / 0.005 = 166.7916. Payment = $300,000 / 166.7916 = **$1,798.65**.

(b) First three months.

Month	Beginning balance	Interest (0.5%)	Principal	Ending balance
1	$300,000.00	$1,500.00	$298.65	$299,701.35
2	$299,701.35	$1,498.51	$300.14	$299,401.21
3	$299,401.21	$1,497.01	$301.65	$299,099.56

Only about 17% of the first payment reduces the balance. Principal repayment accelerates slowly, because each month the interest portion falls by 0.5% of the previous month's principal portion, and the principal portion rises by exactly that amount.

(c) Balance after five years (k = 60). Remaining payments: 300. Using the forward form: $300{,}000 \times 1.005^{60} - 1{,}798.65 \times (1.005^{60} - 1) / 0.005$. With $1.005^{60} = 1.348850$, the first term is $404,655.0 and the second is 1,798.65 × 69.7700 = $125,491.9, so the balance = **$279,163.07** (to the cent, using unrounded payment). After 60 payments totalling $107,919 the borrower has repaid only $20,837 of principal.

(d) 15-year comparison. n = 180: payment = $300,000 × 0.005 / $(1 - 1.005^{-180})$ = **$2,531.57**. Total interest on the 30-year loan is 360 × $1,798.65 − $300,000 = **$347,514.57**; on the 15-year loan it is 180 × $2,531.57 − $300,000 = **$155,682.69**. The shorter loan raises the monthly payment by $732.92 (41%) yet saves $191,831.88 in interest.

3.7.2 Alternative repayment structures

Level payments are not the only design. A comparison of the common structures appears in Table 3.3.

Table 3.3 – Common loan repayment structures

Structure	Payment pattern	Balance behaviour	Typical use	Key feature
Level-payment amortizing	Equal instalments	Declines slowly, then faster	Mortgages, auto loans, equipment loans	Predictable cash flow; interest-heavy early payments
Equal principal	Equal principal plus falling interest	Declines linearly	Some corporate term loans	Higher initial payments; lower total interest
Interest-only (bullet)	Interest each period; principal at maturity	Constant until maturity	Bonds, bridge loans	Refinancing risk at maturity
Balloon	Small level payments; large final payment	Declines partially	Commercial real estate	Payment looks affordable; final lump sum is large
Negative amortization	Payment below interest	Balance rises	Certain flexible-payment loans	Balance can exceed the original loan

The firm's choice depends on cash-flow timing. A project with front-loaded cash needs a bullet or balloon; a project with steady cash flows suits level payments. In all cases the *present value of payments at the loan's own contract rate equals the amount borrowed*, so the structures differ in timing and risk, not in the rate charged. The comparison that matters between two lenders is their EAR (or, more precisely, the internal rate of return on the full set of cash flows including fees).

3.7.3 Why amortization matters to managers

Three practical implications deserve attention.

- **Prepayment value.** A borrower deciding whether to prepay after k payments needs only the outstanding balance, which is the present value of the remaining payments at the contract rate. If the borrower can refinance at a lower rate, the saving is the difference between the remaining payments valued at the old and new rates, less transaction costs.
- **Interest deductibility and tax timing.** Because interest declines over time, the tax shield from interest deductions is front-loaded. A schedule is therefore necessary for after-tax analysis of debt financing.
- **Covenants and balance forecasts.** Lenders' covenants often reference leverage ratios; the amortization schedule yields the debt balance on each reporting date, which should feed the firm's financial plan.

3.8 Inflation: Real versus Nominal Values

3.8.1 Nominal and real quantities

A **nominal** cash flow is measured in currency units of the date on which it occurs. A **real** cash flow is measured in constant purchasing power, that is, in currency units of some base date. If inflation is 3% a year, a nominal payment of $1,030 a year from now is equivalent to a real payment of $1,000 in today's money. Similarly, a **nominal interest rate** is the percentage growth in currency units, while the **real interest rate** is the percentage growth in purchasing power.

The link between them is the **Fisher equation** (after the economist Irving Fisher):

```
1 + nominal rate = (1 + real rate) × (1 + inflation rate)
real rate = (1 + nominal) / (1 + inflation) − 1
```

A common approximation is real ≈ nominal − inflation, which is adequate when both are small but becomes inaccurate for high inflation. With a nominal rate of 8% and inflation of 3%, the exact real rate is 1.08 / 1.03 − 1 = 4.854%, versus the approximation of 5.00%. At 80% nominal and 60% inflation the approximation would give 20%, while the exact figure is 12.5%.

Note that the Fisher equation is stated in terms of *expected* inflation when applied to forward-looking decisions. Realised real returns differ from expected real returns by the inflation surprise: if inflation turns out higher than expected, the holder of a fixed nominal claim loses purchasing power, and the borrower gains.

3.8.2 The golden rule: match cash flows and discount rates

The most frequent error in applied finance is mixing real and nominal quantities. The rule is simple:

- **Nominal cash flows must be discounted at nominal rates.**
- **Real cash flows must be discounted at real rates.**

If done consistently, both approaches give the same present value. Mixing them (for example, discounting cash flows that include inflation at a real rate) overstates value, and discounting real cash flows at a nominal rate understates it. In project analysis, the choice usually depends on which is easier to forecast. If revenues and costs are mostly contractual or fixed in nominal terms (loan payments, fixed-price leases), work in nominal terms. If they track the general price level (many utilities, indexed pensions), a real analysis is simpler. Importantly, depreciation tax shields are fixed in nominal terms, since tax depreciation is based on historical cost, so models that otherwise use real cash flows must adjust the tax shield.

Example 3.8 – Real versus nominal consistency. Harbor Ridge's engineers project that a new product line will generate after-tax cash flows of $500,000 per year in *today's* purchasing power for the next ten years (a real annuity). Expected inflation is 3% and the nominal required return is 8%.

Approach 1: real. Real rate = 1.08 / 1.03 − 1 = 4.8544%. Ten-year annuity factor = $(1 - 1.048544^{-10}) / 0.048544$. We have $1.048544^{10} = 1.606443$, so $1.048544^{-10} = 0.622491$ and the factor = 0.377509 / 0.048544 = 7.77664. PV = $500,000 × 7.77664 = **$3,888,319**.

Approach 2: nominal. Nominal cash flow in year t = $500,000 × 1.03^{t}, a growing annuity with C_1 = $515,000 and g = 3% at r = 8%: PV = [515,000 / (0.08 − 0.03)] × $[1 - (1.03/1.08)^{10}]$ = 10,300,000 × $[1 - 0.953704^{10}]$. Now $0.953704^{10} = 0.622491$, so PV = 10,300,000 × 0.377509 = **$3,888,319**.

The two approaches agree because they are the same calculation seen through different units. If the analyst had mistakenly discounted the real $500,000 stream at the nominal 8%, the result would be $500,000 × 6.7101 = $3,355,041, an understatement of about 14%; if the analyst had inflated the cash flows and then discounted at 4.85%, the value would be overstated.

3.8.3 Inflation and long-term commitments

Inflation transforms the economics of long-dated promises. A fixed-nominal pension of $1,500 per month, paid starting in 25 years, will buy far less than $1,500 does today: at 3% inflation, the purchasing power at the start of retirement is $1,500 / 1.03^{25} = $716.4 in today's money, a loss of more than half. A pension indexed to inflation, in contrast, preserves purchasing power, and costs substantially more to fund. Any firm making a long-term nominal or indexed commitment must decide whether to fund in real or nominal terms, and must match its assets accordingly; we return to this matching principle in Chapter 4 on inflation-linked bonds.

Table 3.4 summarises how a $100 balance grows in nominal and real terms under different assumptions over 20 years.

Table 3.4 – Nominal and real value of $100 invested for 20 years at 6% nominal

Annual inflation	Nominal value at year 20	Real value (today's money)	Real annual return
0%	$320.71	$320.71	6.00%
2%	$320.71	$215.83	3.92%
4%	$320.71	$146.37	1.92%
6%	$320.71	$100.00	0.00%
8%	$320.71	$68.80	−1.85%

The nominal balance is unaffected by inflation, but its purchasing power is not. With inflation equal to the nominal return, twenty years of "growth" leave the investor exactly where they started in real terms.

3.8.4 Putting it together: applications

The tools of this chapter combine naturally in practical problems. Three are worth working through.

Example 3.9 – Retirement funding with real cash flows. An employee aged 35 wants to retire at 65 and receive a real income of $80,000 per year (in today's purchasing power) for 25 years, with the first withdrawal one year after retirement. During retirement the portfolio earns a real return of 3%; during the 30 working years it earns a real return of 4%. How much must be saved each year (real, year-end) during the working years?

Step 1 – Required fund at age 65. Real annuity factor at 3% for 25 years = $(1 - 1.03^{-25}) / 0.03 = (1 - 0.477606) / 0.03 = 17.4131$. Required fund = $80,000 × 17.4131 = **$1,393,051.82**. *Step 2 – Savings.* Future value factor at 4% for 30 years = $(1.04^{30} - 1) / 0.04 = (3.243398 - 1) / 0.04 = 56.0849$. Annual saving = $1,393,051.82 / 56.0849 = **$24,838.2** in real terms.

Because the saving is real, it rises with inflation: if inflation is 3%, the first year's nominal saving is about $25,583 (that is, $24,838 × 1.03). The employee saves roughly 31% of the targeted annual retirement income each year, a sobering ratio that illustrates the cost of a 25-year payout stage. If the retirement withdrawals are made at the start of each year (annuity due), the required fund rises by 3% to $1,434,843.37.

Example 3.10 – Rebate versus cheap financing. Harbor Ridge is buying a $25,000 delivery vehicle. The dealer offers either (i) 0% financing over 48 months (monthly payments of $520.83) or (ii) a $3,000 rebate with a bank loan at 6% stated, monthly compounding, for 48 months on the remaining $22,000. Which is cheaper?

Option (ii) payment. r = 0.5% per month; annuity factor = $(1 - 1.005^{-48}) / 0.005 = 42.5803$; payment = $22,000 / 42.5803 = **$516.67**. *Comparison.* The rebate option has the lower monthly payment ($516.67 versus $520.83) and a total outlay of $24,800.19 versus $25,000. Even with a nominal borrowing rate of 6%, the discounted value of the 0% financing is only $520.83 × 42.5803 = $22,177.25, a saving of $2,822.75 on a $25,000 price, which is less than the $3,000 rebate. The rebate wins by $177.25 in present-value terms. The lower the loan rate, the less the 0% offer is worth; if the buyer's loan rate were 3%, the 0% option would save even less than at 6%. But if the buyer's rate were 9% (0.75% per month), the 0% payments would have a present value of $520.83 × 40.1848 = $20,929.57, a saving of $4,070.43, which exceeds the rebate, and the verdict would reverse. The decision therefore depends on the buyer's own borrowing rate; zero-percent financing is valuable only to the extent that the alternative rate is high.

Example 3.11 – Deferred annuity. A licence agreement entitles Harbor Ridge to 20 annual payments of $30,000, the first arriving at the end of year 6. The discount rate is 8%.

Step 1. Value at the end of year 5 (one period before the first payment): $30,000 × A(8%, 20) = $30,000 × 9.818147 = $294,544.42. *Step 2.* Discount five years: $294,544.42 / 1.469328 = **$200,461.98**.

The common mistake is to discount by six years, applying the annuity formula as though it valued the stream at date 0 and then also discounting from the date of the first payment. Remember: the annuity formula always values the stream one period before the first payment.

3.8.5 Returning to the vignette

We can now answer Marisol's three questions. For the machining cells (Example 3.3), the instalment plan's implied financing rate is 12.59%, so paying cash is preferable if the firm can borrow at a materially lower rate. For the pension, the $1,500 monthly benefit over twenty years is an annuity whose present value at retirement, at a 6% stated rate with monthly compounding (0.5% per month), is $1,500 × $(1 - 1.005^{-240}) / 0.005 = $1,500 × 139.5808 = $209,371; discounted a further 25 years at the same rate, this is a funding target of about $46,892 per employee today if the benefit is fixed in nominal

terms, and considerably more if indexed. For the loans, the treasurer should compare EARs (not stated rates) and fully account for the repayment profile: the equal-principal structure has lower total interest but higher early payments, so it is cheaper only if the firm can comfortably carry the early cash burden.

Case Study: Kestrel Freight and the Structured Settlement

Illustrative case. Kestrel Freight is a fictional regional trucking company.

Kestrel Freight operates 140 tractor units across the central part of the country. Two years ago, a paving contractor's subsidence damaged Kestrel's main depot, and after a long dispute Kestrel's insurer offered a settlement of $3.2 million. The settlement can be taken in either of two forms. Under the **lump-sum option**, Kestrel receives $3.2 million today. Under the **structured option**, the insurer pays $410,000 at the end of each of the next ten years, with the amount growing by 2% a year thereafter for an additional ten years (that is, the payment in year 11 is $410,000 × 1.02 and so on through year 20, after being level at $410,000 for years 1 to 10).

The chief financial officer, Dariusz Wen, has three concerns. First, Kestrel's board is divided. One director argues that "twenty payments totalling nearly $8.7 million is obviously better than $3.2 million." Another argues that the lump sum can be invested immediately in new tractors that earn a return on investment of 14%. A third points out that the lump sum would also let Kestrel retire a term loan carrying a stated rate of 9% compounded monthly, whose outstanding balance is $1.1 million with 60 equal monthly payments remaining.

Second, Dariusz worries about uncertainty. The insurer is rated investment grade, but it is not risk-free; the structured payments are claims against the insurer, and a failure in year 12 would leave Kestrel with an unpaid stream. Kestrel's treasury team suggests that a discount rate of 8% is appropriate for the insurer's payments, compared with 14% for the return on tractors, which are riskier. It reminds the board that the right discount rate depends on the risk of the cash flows being valued, not on the use to which the money is put.

Third, the payments are in nominal dollars. Kestrel's fuel and labour costs have been rising by about 3% annually, and the CFO wishes to know the settlement's value in purchasing-power terms. He also wants to compare the settlement's implied yield with the bank's lending rate of 6.5% (EAR) on an equipment loan, to determine whether taking the lump sum and borrowing less is better than taking the structured payments and borrowing more.

Dariusz asks the analysts to prepare a recommendation. The analysts note that the valuation is a stack of the building blocks in this chapter. Years 1 to 10 form a level ordinary annuity. Years 11 to 20 form a growing annuity that is deferred by ten years, so it must be valued as of year 10 and then discounted. The term loan is an amortizing loan for which the outstanding balance, the monthly payment, and the effective annual rate must be computed. The 3% inflation figure allows the nominal flows to be converted to real terms by using the Fisher relationship. Finally, the analysts must decide which discount rate to use and justify the choice.

The analysts also point out two further complications. The structured payments are taxed as received, whereas the lump sum may be taxed at once and then reinvested, so an after-tax comparison would require tax rates not given in the offer. And the lump sum eliminates the counterparty risk but exposes Kestrel to reinvestment risk, since the 14% return on new tractors is a forecast, not a promise. Their recommendation, they say, should therefore show how the answer changes as the discount rate moves between 6% and 12%.

Discussion questions

1. Value the structured option at discount rates of 6%, 8% and 12%, treating years 1 to 10 as a level annuity and years 11 to 20 as a deferred growing annuity. At what approximate discount rate would Kestrel be indifferent between the two options? What does the break-even rate say about the insurer's offer?
2. Evaluate the director's claim that "twenty payments totalling nearly $8.7 million is obviously better than $3.2 million." What is wrong with comparing undiscounted totals, and what does the comparison look like once the cash flows are discounted at the rate the treasury team recommends?
3. Compute the monthly payment on the term loan and its EAR. If Kestrel took the lump sum and used part of it to repay the loan in full, what rate of return would it be earning on the repayment, and how does it compare with the 8% and 14% discount rates under discussion?
4. Restate the structured option in real terms using 3% inflation. What does the real value of the year-20 payment suggest about the insurer's offer, and how should the board's view of the "growing" payments change?

Key Takeaways

- Money has a time value because of opportunity cost, inflation and risk; the discount rate summarises all three, and cash flows at different dates can be compared only after converting them to a common date.
- PV = FV / (1 + r)^n and FV = PV × (1 + r)^n are the foundations; every other formula in this chapter is a shortcut for summing discounted cash flows.
- An ordinary annuity is worth C × [1 − (1 + r)^(−n)] / r, valued one period before the first payment; an annuity due is worth (1 + r) times as much; a deferred annuity needs a second discounting step.
- A perpetuity is worth C / r, and a growing perpetuity C_1 / (r − g), requiring r > g; valuation is highly sensitive to the spread r − g, so terminal-value assumptions deserve particular scrutiny.
- A growing annuity of finite life is the perpetuity value minus the value of its tail, PV = [C_1 / (r − g)] × [1 − ((1 + g)/(1 + r))^n]; verify unfamiliar formulas by brute-force discounting once.
- Always compare rates on an effective annual basis. EAR = (1 + APR/m)^m − 1, and the periodic rate for a given EAR is (1 + EAR)^(1/m) − 1, not EAR / m.
- A level-payment loan has constant instalments but a changing mix: interest is charged on the declining balance, so early payments are mostly interest. The outstanding balance equals the present value of remaining payments.
- Seller financing "at no interest" and early-payment discounts embed implicit interest rates, often very high; compute the implied rate and compare it with the firm's own cost of funds.
- Real rate = (1 + nominal) / (1 + inflation) − 1. Discount nominal cash flows at nominal rates and real cash flows at real rates; mixing them is one of the commonest and costliest errors in valuation.
- Long-dated commitments are dominated by the discount rate and by inflation; small changes in either can change present values by tens of percent, so sensitivity analysis is part of the answer, not an add-on.

Review Questions

1. Explain the three reasons that a dollar today is worth more than a dollar in the future. Which of these would still operate in an economy with no inflation and no default risk?
2. Why can a decision-maker compare two cash flows only after moving them to the same date? Does the choice of that date affect the ranking of alternatives?

3. Describe the difference between an ordinary annuity, an annuity due and a deferred annuity. For each, state where the annuity formula places the value of the stream.
4. Why must the discount rate exceed the growth rate in the growing perpetuity formula? What does a model that violates the condition imply about its assumptions?
5. A colleague says that a 30-year annuity at 12% is “practically a perpetuity”, whereas at 4% it is not. Explain, and discuss the implication for the interest-rate sensitivity of long-duration assets.
6. Distinguish a stated annual rate, a periodic rate and an effective annual rate. Why is dividing an EAR by twelve an incorrect way to obtain a monthly rate?
7. Why does a supplier’s “2/10, net 30” credit term imply a very high annual interest rate? Under what circumstances should a firm decline the discount?
8. In a level-payment amortizing loan, why do early payments contain more interest than later ones? How would the interest and principal pattern differ under an equal-principal loan?
9. State the Fisher equation. Why is the approximation “real ≈ nominal − inflation” less reliable at high inflation, and what is the consequence of discounting real cash flows at a nominal rate?
10. A manager argues that a 0% financing offer is “free money”. Describe how to test the claim using time-value tools, and explain why the answer depends on the buyer’s own borrowing rate.

Problems

1. **Future value.** A company deposits $15,000 into an account that earns 5.5% per year, compounded annually. What will the balance be after 12 years?
2. **Annuity.** A lease requires payments of $25,000 at the end of each year for eight years. At a discount rate of 9%, compute (a) the present value of the lease, and (b) the present value if the payments are made at the start of each year instead.
3. **Perpetuities.** A preferred share pays $45,000 per year forever, starting one year from now, and the required return is 6%. (a) Compute its value. (b) If the payment instead starts at $45,000 next year and grows 2% a year forever, compute its value.
4. **Growing annuity.** A contract pays $50,000 at the end of year 1, growing at 4% per year, for ten years. Compute its present value at a discount rate of 9%.
5. **EAR.** (a) Compute the EAR of a loan with a stated rate of 9% compounded quarterly. (b) A lender wishes to offer an EAR of 8% with monthly compounding. What stated annual rate (APR) should it quote?
6. **Amortization.** A $200,000 loan is repaid with level monthly payments over 20 years at a stated rate of 5%, compounded monthly. Compute (a) the monthly payment, (b) the balance after 10 years, and (c) the total interest paid over the life of the loan.
7. **Real and nominal.** The nominal interest rate is 6.5% and expected inflation is 2.8%. (a) Compute the exact real interest rate. (b) What is $100,000 received in 10 years worth in today’s purchasing power? (c) What is the purchasing power, in today’s money, of $50,000 invested today at 6.5% for 10 years?
8. **Retirement planning.** A manager wants to receive $40,000 per year in real terms for 20 years after retirement (first withdrawal one year after retirement). The real return is 3% during retirement and 4% before retirement, with 30 years to go. Compute (a) the fund needed at retirement, (b) the real annual saving needed over 30 years, and (c) the single sum that would have to be invested today to reach the same fund.

Answers to Selected Problems

1. $15,000 × 1.055^{12} = **$28,518.11**.

2. a. $25,000 × A(9%, 8) = $25,000 × 5.534819 = **$138,370.48**. (b) Annuity due: × 1.09 = **$150,823.82**.

3. a. $45,000 / 0.06 = **$750,000**. (b) $45,000 / (0.06 − 0.02) = **$1,125,000**.

4. PV = [$50,000 / 0.05] × [1 − (1.04/1.09)^{10}] = $1,000,000 × 0.374729 = **$374,728.82**.

5. a. (1.0225)^{4} − 1 = **9.3083%**. (b) Monthly rate = 1.08^(1/12) − 1 = 0.64340%; APR = 12 × 0.64340% = **7.7208%**.

6. a. Payment = **$1,319.91**. (b) Balance after 120 payments = **$124,443.04**. (c) Total interest = 240 × $1,319.91 − $200,000 = **$116,778.75**.

7. a. 1.065 / 1.028 − 1 = **3.5992%**. (b) $100,000 / 1.028^{10} = **$75,869.79**. (c) $50,000 × (1.065 / 1.028)^{10} = **$71,209.01**.

8. a. $40,000 × A(3%, 20) = $40,000 × 14.8775 = **$595,098.99**. (b) $595,098.99 / FVA(4%, 30) = $595,098.99 / 56.0849 = **$10,610.67** per year. (c) $595,098.99 / 1.04^{30} = **$183,480.13**.

Chapter 4: Bonds and Interest Rates

Learning Objectives

After studying this chapter you should be able to:

- Price a fixed-rate bond as the present value of its coupons and principal, and explain the inverse relationship between bond prices and yields, including the pull-to-par effect and the difference between clean and dirty prices.
- Compute and interpret yield to maturity (YTM), current yield and realised return, and explain the assumptions embedded in YTM.
- Measure interest-rate risk with Macaulay duration, modified duration and convexity, and use them to estimate price changes and to build an immunized portfolio.
- Derive spot rates and forward rates from a term structure, and explain the main theories (expectations, liquidity preference, market segmentation) of why yield curves slope as they do.
- Decompose a corporate bond yield into a risk-free component and a credit spread, interpret credit ratings, and distinguish expected loss from the risk premium priced into spreads.
- Explain how inflation-linked bonds work, compute real yields and breakeven inflation, and assess when they are the right funding or investment instrument.

Opening Vignette

Illustrative scenario. Tessellate Power is a fictional regional electricity generator.

Tessellate Power’s treasurer, Anil Desmarais, faced a decision in the spring that would shape the company’s balance sheet for a decade. The company needed to raise $400 million to build a wind and storage complex. Its investment bankers had laid out three alternatives: a 5-year fixed-rate bond, a 15-year fixed-rate bond, and a 15-year bond whose principal and coupons would be indexed to consumer-price inflation, which Tessellate could service from regulated tariffs that were themselves adjusted for inflation each year.

The headline yields on the page told one story. The 5-year bond would cost 5.3%, the 15-year bond 6.4% and the indexed bond a real yield of 3.1% above inflation. But Anil knew from experience that the headline numbers were the beginning, not the end, of the analysis. Why was the 15-year yield 110 basis points higher than the 5-year yield? Partly because investors expected short rates to rise, partly because investors demand compensation for holding long bonds whose prices swing more when rates change, and partly because longer exposure to Tessellate’s credit means more time for something to go wrong. Disentangling those components would tell Anil whether the extra 110 basis points was fair compensation or an opportunity to borrow short and refinance.

The credit spread was another matter. Tessellate was rated one notch above the lowest investment-grade category. A downgrade would push the spread up sharply, since many institutional investors are barred from holding sub-investment-grade debt. Anil needed to understand how the rating agencies’ and bond markets’ views would be affected by the choice of structure. A larger share of long-dated debt would reduce refinancing risk but raise the sensitivity of the debt’s market value to interest rates.

Lastly, the inflation-indexed alternative looked cheap on a real basis, but its nominal cost would depend on future inflation. If inflation averaged 2.5%, the all-in nominal cost would be about 5.7%; at 4% it would be above 7%. Anil's tariffs rose with inflation, so the real exposure matched his revenues; but what would the board think of a liability that could balloon with an inflation shock?

This chapter provides the tools for each of Anil's questions: how to price a bond, how to read a yield, how to measure interest-rate risk, how to interpret the term structure, what a credit spread contains, and how inflation-linked instruments work. We return to Tessellate in the case study.

4.1 Bond Basics and Pricing

4.1.1 What a bond is

A **bond** is a tradable debt contract in which the issuer promises to pay the holder a stream of fixed cash flows. The main features are:

- **Face value (par value, principal):** the amount repaid at maturity, usually $1,000 for corporate bonds in the United States.
- **Coupon rate:** the annual interest paid as a percentage of face value. A 5% coupon on a $1,000 bond pays $50 a year, often in two semi-annual instalments of $25.
- **Maturity:** the date on which the principal is repaid.
- **Issuer and seniority:** governments, agencies, municipalities and corporations all issue bonds. Senior secured debt ranks ahead of senior unsecured, which ranks ahead of subordinated debt, in the event of default.
- **Embedded options:** callable bonds let the issuer redeem early; putable bonds let the holder require early repayment; convertible bonds can be exchanged for shares. In this chapter we treat bonds as option-free, because the options matter chiefly for pricing models treated in a later chapter on derivatives and hybrid securities.

A **zero-coupon bond** pays no coupons and sells at a discount to face value; its entire return is the difference between price and par. Treasury bills and strips are zero-coupon bonds. Because a zero-coupon bond has a single cash flow, it is the building block of all other bonds: any coupon bond can be viewed as a portfolio of zero-coupon bonds, one for each payment date. This observation underpins the term-structure analysis in Section 4.5.

4.1.2 The pricing formula

A bond is worth the present value of its promised cash flows, discounted at the rate investors demand for bonds of similar risk and maturity. For a bond with face value F, periodic coupon C, n remaining periods and a periodic yield y:

```
P = C/(1+y) + C/(1+y)² + … + (C+F)/(1+y)ⁿ
  = C × [1 − (1+y)^(−n)] / y  +  F / (1+y)ⁿ
```

This is exactly the annuity-plus-lump-sum structure of Chapter 3: the coupons form an ordinary annuity and the principal is a single lump sum. The yield y is the market's required return, which is a property of the market, not of the bond's coupon. The coupon is set at issuance and is fixed, whereas the yield changes every day.

Three relationships follow directly from the formula and should be internalised.

1. **Price and yield move in opposite directions.** Higher required returns reduce the present value of fixed cash flows. This is the single most important fact about fixed-income markets.

2. **A bond sells at par when coupon rate equals yield, at a discount when coupon rate is below yield, and at a premium when coupon rate exceeds yield.** The coupon rate is only a label; its relation to the yield decides the price.
3. **The relationship is convex, not linear.** A fall in yield raises the price by more than an equal rise in yield lowers it. We quantify this in Section 4.4.

Example 4.1 – Pricing an annual-coupon bond at three yields. Tessellate has outstanding $1,000 face value bonds with a 5% annual coupon and ten years to maturity. Compute the price at yields of 6%, 5% and 4%.

Step 1 – at y = 6%. Annuity factor = $(1 - 1.06^{-10}) / 0.06 = (1 - 0.558395) / 0.06 = 7.360087$. PV of coupons = $50 × 7.360087 = $368.00. PV of principal = $1,000 × 0.558395 = $558.39. Price = **$926.40** (a discount).

Step 2 – at y = 5%. Because the coupon equals the yield, price = **$1,000.00** exactly (par).

Step 3 – at y = 4%. Annuity factor = $(1 - 1.04^{-10}) / 0.04 = (1 - 0.675564) / 0.04 = 8.110896$. PV of coupons = $50 × 8.110896 = $405.54. PV of principal = $1,000 × 0.675564 = $675.56. Price = **$1,081.11** (a premium). (The rounded components sum to a cent less; the prices quoted are computed from unrounded factors.)

A 1-percentage-point rise in the yield (5% → 6%) cut the price by $73.60 (7.36%), while a 1-point fall (5% → 4%) raised it by $81.11 (8.11%). The asymmetry is convexity.

Table 4.1 – Price of a 10-year, 5% annual-coupon bond (face value $1,000)

Yield	Price	Change from par	Status
3%	$1,170.60	+17.06%	Premium
4%	$1,081.11	+8.11%	Premium
5%	$1,000.00	0.00%	Par
6%	$926.40	−7.36%	Discount
7%	$859.53	−14.05%	Discount
8%	$798.70	−20.13%	Discount

4.1.3 Semi-annual coupons and bond-market conventions

Most U.S. corporate and Treasury bonds pay coupons semi-annually, so a 6% coupon bond pays $30 every six months. The yield is quoted as a **bond-equivalent yield**: the periodic six-month yield multiplied by two, rather than the compounded effective annual rate. A quoted yield of 7% therefore means a periodic rate of 3.5% and an effective annual rate of $1.035^2 - 1 = 7.1225\%$. The number of periods doubles and the rate halves.

Example 4.2 – Semi-annual pricing. A five-year bond has a 6% coupon, paid semi-annually, and a quoted yield of 7%. Price it.

Step 1. Periodic coupon C = $1,000 × 6% / 2 = $30; periodic yield = 3.5%; n = 10 periods. *Step 2.* Annuity factor = $(1 - 1.035^{-10}) / 0.035 = (1 - 0.708919) / 0.035 = 8.316605$. *Step 3.* PV of coupons = $30 × 8.316605 = $249.50. PV of principal = $1,000 × 0.708919 = $708.92. *Step 4.* Price = **$958.42**.

If the same bond were valued by mistakenly treating the coupon as annual ($60 a year for 5 years at 7%), the price would be $959.00; the error is small here but grows with maturity and with the level of rates.

4.1.4 Clean price, dirty price and accrued interest

Between coupon dates, a buyer who purchases the bond will receive the *full* next coupon, even though the seller held the bond for part of the period. By convention the buyer compensates the seller with **accrued interest**, the pro-rata share of the coming coupon earned by the seller:

```
Accrued interest = Coupon × (days since last coupon / days in coupon period)
Dirty (full) price = Clean (quoted) price + Accrued interest
```

Bond prices are *quoted* clean, so that quoted prices do not jump by the amount of the coupon every six months. The buyer pays the dirty price. If 60 days have passed in a 180-day period on the bond in Example 4.2, the accrued interest is \$30 × 60 / 180 = \$10.00, so a buyer whose quoted price is \$958.42 would pay \$968.42.

4.1.5 Pull to par

If a bond's yield stays constant, its price drifts toward par as maturity approaches, because the discount (or premium) is amortized away. A 10-year 5% bond priced at \$926.40 at a 6% yield will be priced at \$931.98 after one year (nine years remaining), \$944.18 after three years, \$957.88 after five years and \$981.67 after eight years, reaching \$1,000 at maturity. The increase is the **accretion** that partly compensates the investor for the below-market coupon. This steady drift is the reason why a bond's total return over a holding period is not simply the coupon: it is coupon plus price change, even if market yields do not move.

4.1.6 Common mistakes in bond pricing

- Using the coupon rate as the discount rate.
- Forgetting to halve the rate and double the periods for semi-annual bonds.
- Confusing clean and dirty prices, particularly when settling trades or recording a bond in the accounts.
- Treating the maturity yield as a guaranteed return, which we examine next.

4.2 Yield to Maturity and Measures of Return

4.2.1 Definition of YTM

The **yield to maturity** is the single discount rate that equates the present value of a bond's promised cash flows to its current market price. It is the bond's internal rate of return if held to maturity. Unlike the pricing problem, there is no closed-form solution for y; it must be found by trial and error, interpolation, or a calculator or spreadsheet (the IRR or RATE functions).

```
Price = Σ C_t/(1+YTM)^t + F/(1+YTM)^n      (solve for YTM)
```

A useful approximation for a quick check is

```
YTM ≈ [C + (F − P)/n] / [(F + P)/2]
```

which averages the annual coupon income and the annual amortization of the discount, divided by the average investment.

Example 4.3 – Solving for YTM. A bond with a 4% annual coupon, eight years to maturity and a face value of \$1,000 trades at \$950. Find its YTM.

Step 1 – approximation. [40 + (1,000 − 950)/8] / [(1,000 + 950)/2] = [40 + 6.25] / 975 = 46.25 / 975 = **4.744%**. *Step 2 – refine by pricing at trial yields.* At y = 4.75%: price = $951.03. At y = 4.77%: price = $949.77. *Step 3 – interpolate.* The target price of $950.00 lies 1.03/1.26 = 0.82 of the way from 4.75% to 4.77%, so YTM = 4.75% + 0.82 × 0.02% = **4.766%**. *Step 4 – check.* At y = 4.766%, price = $950.02. ✓

The bond trades at a discount because its 4% coupon is below the 4.77% the market demands. The investor earns 4% in coupons plus a capital gain of $50 over eight years.

4.2.2 Hidden assumptions of YTM

YTM is a convenient single number, but it relies on three assumptions that managers should be able to name.

1. **The bond is held to maturity.** If sold earlier, the return depends on the price at the sale date, which depends on yields then.
2. **The issuer pays all promised cash flows in full and on time.** For risky bonds, the promised yield overstates the expected yield. The gap is discussed in Section 4.6.
3. **Coupons are reinvested at the YTM itself.** If market rates fall, the coupons are reinvested at lower rates and the realised return falls short of the YTM; if rates rise, reinvestment is more rewarding but the bond's price has fallen. This is the **reinvestment risk** that offsets **price risk**; we exploit the offset when we immunize a portfolio.

Example 4.4 – Realised compound yield. An investor buys a five-year, 5% annual-coupon bond at a 6% yield and holds it to maturity, reinvesting every coupon at 4%.

Step 1 – purchase price. Price at 6% = $957.88. *Step 2 – terminal wealth.* Coupons reinvested at 4%: 50 × (1.04⁴ + 1.04³ + 1.04² + 1.04 + 1) = 50 × 5.416323 = $270.82. Add the principal: $1,000 + $270.82 = $1,270.82. *Step 3 – realised yield.* (1,270.82 / 957.88)^(1/5) − 1 = 1.326706^0.2 − 1 = **5.817%**.

The realised return of 5.82% falls short of the 6% YTM because coupons were reinvested at 4% rather than at 6%. The shortfall of 18 basis points is small for a five-year bond with a modest coupon, but it widens with maturity and coupon size; for a 30-year high-coupon bond, reinvestment income can be most of the total return.

4.2.3 Other yield measures

- **Current yield** = annual coupon / current price. For the bond in Example 4.1 at a 6% yield, 50 / 926.40 = 5.397%. It ignores capital gain or loss, so it is useful for income comparisons only.
- **Yield to call (YTC)** is calculated like YTM but to the first call date at the call price. For a callable bond selling at a premium, the lower of YTM and YTC (the **yield to worst**) is the more realistic return measure.
- **Holding-period return** over a specific horizon = (coupons + ending price − beginning price) / beginning price. It depends on yields at the end of the horizon.

A manager issuing debt should note an asymmetry: the YTM that investors demand is the issuer's pre-tax cost of borrowing (before fees and ignoring default risk). This is the correct input for the cost of debt in the weighted-average cost of capital (discussed in a later chapter), not the coupon on existing debt.

4.3 Interest-Rate Risk and Duration

4.3.1 Why maturity alone is not enough

All fixed-rate bonds fall in price when yields rise, but not by the same amount. A 30-year zero-coupon bond falls much more than a 2-year bond for the same change in yield. Maturity is a rough guide, but it ignores the timing of coupons. A 10-year bond paying a 12% coupon returns much of its value early, and is less sensitive than a 10-year zero-coupon bond. We need a measure that takes the whole cash-flow profile into account. This is **duration**.

4.3.2 Macaulay duration

Macaulay duration (named for Frederick Macaulay) is the present-value-weighted average time to receipt of a bond's cash flows:

```
D = Σ [ t × PV(CF_t) ] / P
```

where $PV(CF_t) = CF_t / (1 + y)^t$ and P is the bond's price. It is measured in years. Properties:

- A zero-coupon bond's duration equals its maturity.
- A coupon bond's duration is less than its maturity, because early coupons pull the average forward.
- Duration falls as the coupon rate rises, and falls as the yield rises (higher yield means the distant cash flows are discounted more heavily).
- Duration generally rises with maturity, though at a decreasing rate; for a perpetuity it is (1 + y)/y, a finite number (17.67 years at 6%).

4.3.3 Modified duration and the price approximation

Differentiating the price formula with respect to the yield gives the key practical result. **Modified duration** is

```
D_mod = D / (1 + y/m)          (m = coupons per year; m = 1 for annual)
```

and for a small change in yield Δy:

```
ΔP / P ≈ − D_mod × Δy
```

Modified duration is therefore the percentage price change for a one-unit (100%) change in yield, which we normally read as the percentage change in price for a one-percentage-point change in yield. A bond with a modified duration of 7.57 loses about 7.57% of its price if yields rise by 1 percentage point. A related measure is **dollar duration** (or the DV01, "dollar value of an 01"), the dollar change in price for a one-basis-point change in yield: DV01 = D_mod × P × 0.0001.

Example 4.5 – Computing duration. Compute the Macaulay duration, modified duration and approximate price change for a three-year bond with a 6% annual coupon, face value $1,000, priced at a 5% yield.

Step 1 – cash flows and present values.

Year t	Cash flow	PV at 5%	t × PV
1	$60	$57.14	$57.14
2	$60	$54.42	$108.84
3	$1,060	$915.67	$2,747.01
Sum		**$1,027.23**	**$2,913.00**

Step 2 – Macaulay duration. D = 2,913.00 / 1,027.23 = **2.836 years**. *Step 3 – modified duration.* D_mod = 2.836 / 1.05 = **2.701**. *Step 4 – price change for a +1 percentage point move.* ΔP ≈ −2.701 × 0.01 × $1,027.23 = **−$27.74**. For a −1 point move: +$27.74. *Step 5 – exact.* At 6%, the price is $1,000.00 (a change of −$27.23); at 4% the price is $1,055.50 (a change of +$28.27).

The approximation is close but overstates the decline and understates the gain; the missing second-order effect is convexity.

Table 4.2 – Duration depends on maturity and coupon (5% yield, annual coupons)

Maturity	Macaulay duration, 5% coupon bond	Macaulay duration, zero-coupon bond	Modified duration, 5% coupon	Modified duration, zero
2 years	1.95	2.00	1.86	1.90
5 years	4.55	5.00	4.33	4.76
10 years	8.11	10.00	7.72	9.52
20 years	13.09	20.00	12.46	19.05
30 years	16.14	30.00	15.37	28.57

The table shows two effects. The coupon bond's duration grows far more slowly than its maturity, and the gap between a 5% coupon bond and a zero-coupon bond widens with maturity. A 30-year zero-coupon bond has a modified duration nearly double that of a 30-year coupon bond, and consequently is much more volatile.

4.3.4 Portfolio duration and managerial use

The duration of a portfolio is the market-value-weighted average of the durations of its holdings. Likewise, a firm can compute the duration of its assets and of its liabilities. A firm with a long-lived asset base funded by short-term debt has a **duration gap**: a rise in rates reduces the value of assets by more than liabilities, or raises the cost of rolling over debt. Banks and insurers manage duration gaps formally; non-financial firms with large pension liabilities or long-dated receivables should do the same.

Duration is also a risk-budgeting tool. A treasury policy might limit the portfolio's DV01 to a given dollar amount, or require that the weighted-average duration of debt remain within a range.

4.4 Convexity and Immunisation

4.4.1 The curvature of the price–yield relationship

Modified duration is the slope of the price–yield curve at the current yield; it is a linear approximation to a curved relationship. The price–yield relationship is convex: the curve lies above its tangent line. **Convexity** measures this curvature. For annual cash flows:

```
Convexity = [ Σ t(t+1) × CF_t/(1+y)^(t+2) ] / P
```

and the price change including the second-order term is

```
ΔP / P ≈ − D_mod × Δy + ½ × Convexity × (Δy)²
```

Because the second term is positive whatever the sign of Δy, convexity helps the holder: a bond gains more when yields fall than it loses when yields rise. All else equal, investors prefer higher convexity and will pay for it; this is why two bonds with identical durations can have different yields. Zero-coupon bonds have the highest convexity for a given maturity;

callable bonds can have *negative* convexity in some yield ranges, because the issuer's right to call caps the price gain when rates fall.

Example 4.6 – Duration and convexity together. Consider the 10-year, 5% annual-coupon bond of Example 4.1, priced at a 6% yield: P = \$926.40, Macaulay duration = 8.0225 years, modified duration = 7.5684, convexity = 72.57. Estimate and compare the price change for yield shifts of ±1 and ±2 percentage points.

Step 1 – duration-only estimate. For Δy = +1%: −7.5684 × 0.01 × 926.40 = −\$70.11. For Δy = +2%: −\$140.23. For Δy = −1%: +\$70.11; for −2%: +\$140.23. *Step 2 – convexity term.* ½ × 72.57 × $(0.01)^2$ × 926.40 = \$3.36 for ±1%; ½ × 72.57 × $(0.02)^2$ × 926.40 = \$13.45 for ±2%. *Step 3 – combine.* For +1%: −70.11 + 3.36 = **−\$66.75**; for −1%: 70.11 + 3.36 = **+\$73.48**; for +2%: −140.23 + 13.45 = **−\$126.78**; for −2%: 140.23 + 13.45 = **+\$153.67**. *Step 4 – compare with exact repricing.* The actual changes are −\$66.87, +\$73.60, −\$127.70 and +\$154.71.

Table 4.3 – Duration and convexity approximations versus exact price changes (10-year, 5% bond at a 6% yield; P = \$926.40)

Yield change	Exact ΔP	Duration only	Duration + convexity	Error, duration only (approx. − exact)	Error, duration + convexity (approx. − exact)
+2%	−\$127.70	−\$140.23	−\$126.78	−\$12.53	+\$0.92
+1%	−\$66.87	−\$70.11	−\$66.75	−\$3.24	+\$0.12
−1%	+\$73.60	+\$70.11	+\$73.48	−\$3.49	−\$0.12
−2%	+\$154.71	+\$140.23	+\$153.67	−\$14.48	−\$1.04

Adding the convexity term cuts the error by an order of magnitude. For large moves, such as the 200-basis-point shifts seen during rate cycles, ignoring convexity is a material mistake.

4.4.2 Immunisation

An investor or firm with a known future liability can use duration to lock in the ability to meet it, a technique called **immunization**. The idea is to offset price risk and reinvestment risk. If yields rise, the bonds lose market value but reinvested coupons earn more; if yields fall, the opposite occurs. Choosing a portfolio whose duration equals the liability's horizon makes the two effects cancel for small, parallel shifts in the yield curve.

Two conditions are required:

1. The present value of assets equals the present value of liabilities.
2. The duration of assets equals the duration of liabilities.

A third condition, that asset convexity exceed liability convexity, guarantees that the portfolio remains ahead for large shifts in either direction.

Example 4.7 – Immunising a single liability. A company must pay \$1,000,000 in six years (for example, a legal settlement). The market yield is 5% for all maturities. The treasurer will invest in two zero-coupon bonds, maturing in 3 and 10 years. How should she split the funds?

Step 1 – present value of the liability. \$1,000,000 / 1.05^6 = **\$746,215.40**. The liability has a duration of 6 years because it is a single payment. *Step 2 – solve for weights.* Let w be the weight in the 3-year bond: 3w + 10(1 − w) = 6, so 10 − 7w = 6, giving w = 4/7 = 0.5714. *Step 3 – dollar amounts.* 3-year zero: 0.5714 × \$746,215.40 = **\$426,408.80**. 10-year zero: 0.4286

× \$746,215.40 = **\$319,806.60**. *Step 4 – face values purchased.* 3-year: \$426,408.80 × 1.05^3 = \$493,621.48. 10-year: \$319,806.60 × 1.05^{10} = \$520,931.25. *Step 5 – test with immediate parallel shifts.*

Yield	Asset value	Liability value	Surplus
3%	\$839,355	\$837,484	+\$1,871
4%	\$790,750	\$790,315	+\$436
5%	\$746,215	\$746,215	\$0
6%	\$705,339	\$704,961	+\$379
7%	\$667,757	\$666,342	+\$1,415

In every scenario the surplus is non-negative: the barbell portfolio has higher convexity than the single-maturity liability, so it is ahead after any parallel shift. The portfolio must be rebalanced as time passes and yields change, since durations drift. Immunisation also fails when the yield curve changes shape rather than moving in parallel, which is why practitioners use key-rate durations or cash-flow matching (a **dedicated portfolio** whose coupons and maturities match the liability schedule) when exactness is needed.

4.5 The Term Structure of Interest Rates

4.5.1 Spot rates and the yield curve

So far we have used a single yield to discount all of a bond's cash flows. But the market does not charge the same rate for money lent for one year as for ten. The **term structure of interest rates** describes how yields vary with maturity. The **yield curve** plots the YTM of otherwise comparable bonds (typically government securities) against their maturities.

The cleanest building block is the **spot rate** s_t, the yield on a zero-coupon bond maturing at date t. A cash flow at date t should be discounted at s_t, not at some blended YTM:

```
P = Σ CFₜ / (1 + sₜ)ᵗ
```

The YTM of a coupon bond is a complicated average of the spot rates applicable to its individual payments, weighted by their present values. Two bonds with identical maturities and different coupons will therefore have different YTMs when the curve is not flat (the "coupon effect"), even though they are priced from the same spot curve. Spot rates are not directly observable beyond the bills market, but they can be extracted ("bootstrapped") from the prices of coupon bonds and strips.

4.5.2 Forward rates

A **forward rate** is the interest rate, implied by today's spot curve, for a loan that starts at a future date. Consider two ways to invest \$1 for two years: buy a two-year zero at s_2, or buy a one-year zero at s_1 and reinvest the proceeds for a second year at the rate $f_{1,2}$ agreed today. To rule out riskless arbitrage the two must produce the same result:

```
(1 + s₂)² = (1 + s₁)(1 + f₁,₂)      →      f₁,₂ = (1 + s₂)² / (1 + s₁) − 1
```

In general, the forward rate between dates t − 1 and t is

```
f(t−1, t) = (1 + sₜ)ᵗ / (1 + s(t−1))^(t−1) − 1
```

and the forward rate for a multi-period loan starting at date a and ending at date b is [(1 + s_b)^b / (1 + s_a)^a]^(1/(b−a)) − 1. Forward rates are the marginal cost of extending maturity: they show what the market charges for the *additional* year of borrowing. When the spot curve rises, forward rates lie above spot rates; when it falls, forwards lie below.

Example 4.8 – From spot rates to forwards and bond prices. The government zero-coupon curve is: 1-year 3.0%, 2-year 3.5%, 3-year 4.0%, 4-year 4.4%.

Step 1 – discount factors. $1/1.03 = 0.970874$; $1/1.035^2 = 0.933511$; $1/1.04^3 = 0.888996$; $1/1.044^4 = 0.841779$.

Step 2 – one-year forward rates. - $f_{1,2} = 1.035^2 / 1.03 - 1 = 1.071225 / 1.03 - 1 =$ **4.002%**. - $f_{2,3} = 1.04^3 / 1.035^2 - 1 = 1.124864 / 1.071225 - 1 =$ **5.007%**. - $f_{3,4} = 1.044^4 / 1.04^3 - 1 = 1.187960 / 1.124864 - 1 =$ **5.609%**. - Three-year forward rate starting in one year: $[(1.044^4) / 1.03]^{(1/3)} - 1 = (1.153360)^{(1/3)} - 1 =$ **4.871%**.

Step 3 – price a 4-year 5% annual-coupon bond off the curve. $P = 50 \times 0.970874 + 50 \times 0.933511 + 50 \times 0.888996 + 1{,}050 \times 0.841779 = 48.54 + 46.68 + 44.45 + 883.87 =$ **$1,023.54**.

Step 4 – its YTM. The single rate that discounts the bond's flows to $1,023.54 is **4.346%**. This is lower than the 4-year spot rate (4.4%) because the YTM averages in the lower short-term spot rates through the coupon payments.

A manager who wants to lock in a rate today for borrowing that will start in one year can ask a bank for a forward-rate agreement, or construct the position from zero-coupon bonds; the cost is the forward rate, here 4.871% for three years.

4.5.3 Why the curve slopes: three theories

The pure expectations theory. Forward rates are unbiased predictions of future spot rates. An upward-sloping curve means the market expects short rates to rise; a downward-sloping one means it expects them to fall. If this were the whole story, investors would be indifferent between a long bond and a rolled-over series of short ones. The theory is elegant, but empirically forward rates tend to exceed subsequently realised spot rates, on average, which suggests that something else is also priced.

Liquidity preference (term premium). Investors dislike price risk, and long bonds are more volatile than short ones (higher duration). Investors therefore require a premium, the *term premium*, to hold long bonds, while borrowers are willing to pay it to lock in long-term funds. Forward rates equal expected future spot rates *plus* a term premium that rises with maturity. This explains why yield curves are upward-sloping more often than not.

Market segmentation and preferred habitat. Different investors have natural maturities: pension funds and life insurers want long assets to match long liabilities; money-market funds want short ones. In the pure segmentation view, each maturity has its own supply and demand. The preferred-habitat variant, developed by Modigliani and Sutch, allows investors to move out of their habitat if the extra return is sufficient, so arbitrage links the segments but imperfectly. Central bank asset purchases, which remove long bonds from the market, are a modern example of a segmentation effect on the curve.

In practice the curve embodies all three: expectations of the central bank's policy path, term premia for duration risk, and supply-and-demand pressures in particular maturities.

Table 4.4 – Common yield-curve shapes and their usual interpretation

Shape	Description	Usual interpretation	Managerial implication
Normal (upward-sloping)	Long yields above short yields	Expected rate rises and/or positive term premium	Short-term funding is cheaper now, but refinancing risk is higher
Flat	Similar yields across maturities	Transition; mixed expectations	Little yield sacrifice for extending maturity
Inverted	Short yields above long yields	Expected rate cuts, often tight policy; historically associated with slowdowns	Long-term funding may be attractive; short-term investments earn more
Humped	Medium maturities highest	Expected rate rise followed by fall	Medium-term borrowing is the most expensive point

An **inverted** curve gets attention because it has often preceded economic slowdowns, but it should be treated as one input into a forecast, not as a mechanical prediction.

4.5.4 Using the curve in corporate decisions

The term structure bears on managers in several concrete ways. First, project cash flows at different horizons should in principle be discounted using rates for matching maturities, though in practice a single risk-adjusted rate is usually applied. Second, the shape of the curve guides the maturity decision for borrowing: the extra yield for extending from 5 to 15 years is the term premium plus expected rate rises, and the forward rate is the benchmark against which any view of future rates should be tested. Anil's 110-basis-point premium is worth paying for a 15-year bond only if the firm values the removal of refinancing risk more than the expected saving from borrowing short and rolling over. Third, for any floating-rate liability, the forward curve gives the expected path of interest cost, and the swap market can fix it.

4.6 Credit Risk, Spreads and Ratings

4.6.1 Yield as risk-free rate plus credit spread

A government bond in its own currency is treated as free of default risk. A corporate bond is not. The extra yield that investors demand over a comparable government bond is the **credit spread**:

```
Corporate yield = Risk-free yield (same maturity) + Credit spread
```

The spread compensates for (i) **expected loss** from default, equal to the probability of default (PD) times the loss given default (LGD = 1 − recovery rate), (ii) a **risk premium** for bearing uncertainty about default and spread changes, which investors cannot fully diversify, particularly in recessions when defaults cluster, and (iii) in some markets a **liquidity premium**, since corporate bonds trade less than government bonds, and sometimes a tax effect. A rough rule is

```
Spread ≈ PD × LGD + risk premium + liquidity premium
```

Therefore the promised yield on a risky bond overstates its expected return. An investor who buys a bond with a 250-basis-point spread should not expect to earn 250 bp more than the government, since part of the spread is the expected cost of default.

Example 4.9 – Extracting default probability from a spread. A one-year zero-coupon government bond yields 4.0%. A one-year zero-coupon bond from Tessellate Power yields 6.5% and recovers 40% of face value in default. What default probability is implied (the risk-neutral probability)? How much of the 250 bp spread is expected loss if the analyst's own

estimate of the true default probability is 2.0%?

Step 1 – no-arbitrage condition. In a risk-neutral setting the expected payoff of the corporate bond must be worth the same as the government bond. Payoff per \$1 of face value: \$1 with probability 1 − q, \$0.40 with probability q. Setting the expected payoff discounted at the risk-free rate equal to the corporate price (1/1.065):

$$(1 - q) + 0.40\,q = 1.04 / 1.065 = 0.976526$$

Step 2 – solve. $1 - 0.60\,q = 0.976526$, so $q = 0.023474 / 0.60 =$ **3.912%**. *Step 3 – compare with the simple rule.* PD × LGD = 3.912% × 0.60 = 2.347%, which is close to the actual 2.5% spread (the difference arises from discounting). *Step 4 – decompose with the analyst's estimate.* If the true default probability is 2.0%, expected loss is 2.0% × 0.60 = 1.20% (120 bp). The remaining spread of about 130 bp is a premium for risk and illiquidity. The risk-neutral probability exceeds the real-world probability because investors are risk-averse and want compensation for default risk that is correlated with bad economic states.

The distinction matters in practice: to value a bond, use the market's risk-neutral probability, which is embedded in the price; to estimate the expected return on a bond portfolio, use the real-world probability.

4.6.2 Credit ratings

Credit-rating agencies (Standard & Poor's, Moody's and Fitch are the best known) publish opinions on the creditworthiness of issuers and individual bonds. Ratings are ordinal categories, not precise probabilities, and they reflect the agency's judgement about the issuer's business risk, financial leverage, cash-flow coverage, liquidity, covenants and, for bonds, seniority. The scale runs from the highest quality to default.

Table 4.5 – Structure of long-term credit rating scales

Category	S&P / Fitch	Moody's	General meaning	Market segment
Highest quality	AAA	Aaa	Extremely strong capacity to meet obligations	Investment grade
High quality	AA	Aa	Very strong capacity	Investment grade
Upper medium	A	A	Strong, but more exposed to adverse conditions	Investment grade
Lower medium	BBB	Baa	Adequate capacity; adverse conditions may weaken it	Investment grade (lowest tier)
Speculative	BB	Ba	Faces major uncertainties	High yield ("junk")
Highly speculative	B	B	Vulnerable; currently able to pay	High yield
Substantial risk	CCC and below	Caa and below	Currently vulnerable; reliant on favourable conditions	High yield / distressed
Default	D (S&P)	C (Moody's, in the lowest category)	Payment default or bankruptcy filing	Defaulted

Each category from AA to CCC is subdivided with modifiers (+/− at S&P and Fitch; 1, 2, 3 at Moody's). The boundary between BBB−/Baa3 and BB+/Ba1 is the key line between **investment grade** and **speculative grade**. It is more than a label: many pension funds, insurers and index funds have mandates that restrict them to investment-grade debt, so a downgrade across that line (a "fallen angel") can force selling, widen spreads sharply and raise the issuer's cost of capital across all its borrowing. Managers therefore treat the investment-grade boundary as a constraint on capital structure, and often target a rating that leaves some headroom.

Several cautions apply. Ratings lag market information, so bond spreads often move before rating changes. Ratings are through-the-cycle opinions and are therefore smoother than market-implied credit risk. Rating agencies were widely criticised for the high ratings assigned to certain structured securities before the financial crisis of 2007–2009, a reminder that ratings are fallible opinions and not guarantees. Investors should use them alongside market spreads and their own analysis.

4.6.3 Spread risk and spread duration

Spreads change over time, typically widening in recessions and narrowing in expansions. For a holder of corporate bonds, a spread move is another source of price risk, added to the movement in government yields. The sensitivity of a bond's price to its spread is approximated by its **spread duration**, which for a fixed-rate bond is close to its modified duration.

Example 4.10 – Spread widening. Tessellate has a 7-year bond with a 5% annual coupon. The government 7-year yield is 4.0% and the bond's spread is 250 bp, giving a yield of 6.5%. The market becomes nervous about the sector, and the spread widens by 100 bp to 350 bp while government yields are unchanged.

Step 1 – price at 6.5%. Annuity factor = $(1 - 1.065^{-7}) / 0.065 = (1 - 0.643506) / 0.065 = 5.4845$; PV of coupons = \$50 × 5.4845 = \$274.23; PV of principal = \$1,000 × 0.643506 = \$643.51. Price = **\$917.73** (computed from unrounded factors). *Step 2 – price at 7.5%.* Price = **\$867.58**. *Step 3 – change.* −\$50.15, or −5.46%. *Step 4 – check against duration.* Modified duration at 6.5% = 5.66, so the approximate change is −5.66% × 1 = −5.66%, or −\$51.9 (convexity makes the actual loss slightly smaller).

The company's own borrowing cost is unaffected by this change in the market value of the bond if it holds to maturity, but a widening spread signals a higher cost on any new issue, and affects the fair value of any liabilities the firm marks to market.

4.6.4 Managing credit risk as an issuer

For the issuer the spread is a function of leverage, coverage, asset quality and market conditions. Actions that reduce spreads include lowering debt, extending maturities to remove clustering of refinancing, offering collateral or seniority, accepting covenants, and building a reputation for conservative financial policy. Each has costs: lower leverage forgoes tax shields; covenants limit flexibility; and collateral drains the pool of unencumbered assets. The efficient choice depends on the trade-offs developed in the chapters on capital structure.

4.7 Inflation-Linked Bonds

4.7.1 How they work

A conventional bond promises fixed nominal payments, so its real value is exposed to inflation. An **inflation-linked bond** adjusts its cash flows for realised inflation so that the investor's purchasing power is preserved. The best-known examples are U.S. Treasury Inflation-Protected Securities (TIPS) and the index-linked gilts of the United Kingdom, but many

governments and some corporations and utilities issue them. Design details vary, but the typical TIPS structure is:

- The **principal is multiplied by an index ratio**, the current consumer-price index divided by the index at issuance.
- The **coupon rate is fixed and real**; each coupon equals that rate times the *inflation-adjusted* principal at the coupon date.
- At maturity the holder receives the adjusted principal, and in the U.S. design not less than the original principal (a deflation floor on redemption).

The quoted yield on such a bond is a **real yield**: the yield to maturity of the real cash flows, measured in purchasing-power units. The nominal return depends on realised inflation:

```
(1 + nominal return) = (1 + real yield) × (1 + realised inflation)
```

Example 4.11 – Cash flows of an inflation-linked bond. An investor buys a five-year inflation-linked bond with $1,000 original principal and a 1.5% real coupon, paid annually. Inflation turns out to be a constant 3.0% a year.

Year	Index ratio	Adjusted principal	Coupon (1.5% × adjusted principal)
1	1.0300	$1,030.00	$15.45
2	1.0609	$1,060.90	$15.91
3	1.0927	$1,092.73	$16.39
4	1.1255	$1,125.51	$16.88
5	1.1593	$1,159.27	$17.39

At maturity the holder receives $1,159.27 of principal plus the final coupon of $17.39. The nominal coupons rise steadily, and the redemption amount compensates for the cumulative inflation of 15.93%. The holder's nominal return is approximately (1.015)(1.03) − 1 = 4.545% per year, which is the real coupon rate compounded with inflation. A conventional bond with a fixed 4.5% coupon would have paid $45 every year and returned $1,000, and would have lost to the indexed bond if inflation had averaged more than about 3%.

(Check: coupon in year 4 is 1.5% × $1,125.51 = $16.88; year 5, 1.5% × $1,159.27 = $17.39.)

4.7.2 Real yields and breakeven inflation

Comparing a conventional government bond with an indexed bond of the same maturity yields the market's implied expectation of inflation, the **breakeven inflation rate**:

```
Breakeven inflation = (1 + nominal yield) / (1 + real yield) − 1  ≈  nominal yield − real yield
```

This is the Fisher equation (Chapter 3) applied to market yields. If the 10-year nominal yield is 4.5% and the 10-year real yield is 1.8%, breakeven inflation is 1.045 / 1.018 − 1 = **2.652%** (the simple difference gives 2.7%). An investor who believes inflation will average more than 2.65% should prefer the indexed bond; one who thinks it will be lower should hold the nominal bond. The breakeven rate is not a pure inflation forecast. It also includes an *inflation risk premium* (compensation to nominal bondholders for inflation uncertainty), which raises it, and the lower liquidity of indexed bonds, which raises their real yield and thereby pushes the breakeven down. Interpreting breakeven rates as forecasts therefore requires caution.

The price of an indexed bond is the present value of its real cash flows at the real yield, multiplied by the index ratio. For a five-year bond with a 1.5% real coupon and a real yield of 1.2%, the real price per $1,000 of original principal is $1,014.47 per unit of index ratio. The bond trades above par because its coupon exceeds the real yield.

4.7.3 Who benefits, and the risks

For investors, indexed bonds hedge inflation risk: pension funds with inflation-linked liabilities, retirees wanting a steady real income, and diversified portfolios all benefit. Indexed bonds are not riskless, though. Their real yields move, so a rise in real yields lowers their prices, and with long maturities the real-rate duration is substantial. Realised inflation affects cash flows with a lag, because of the index publication lag. In some markets, taxation of the inflation accrual can create "phantom income": taxable income that is not received in cash until maturity.

For issuers, the trade-off is the reverse. An issuer whose revenues track inflation, such as a regulated utility whose tariffs are adjusted annually or a toll-road operator whose tolls are indexed, can match its liabilities to its revenues and avoid a mismatch. It may also save the inflation risk premium that investors build into nominal yields. An issuer whose revenues are fixed in nominal terms faces the danger of an inflation surprise, as Tessellate's board worried: servicing costs rise but revenues do not.

Table 4.6 – Nominal versus inflation-linked funding for an issuer (3.1% real yield; illustrative inflation outcomes)

Average inflation	All-in nominal cost of indexed bond	Cost of 6.4% fixed bond	Indexed cheaper?
1.0%	4.13%	6.40%	Yes
2.0%	5.16%	6.40%	Yes
2.5%	5.68%	6.40%	Yes
3.2%	6.40%	6.40%	Break-even
4.0%	7.22%	6.40%	No
5.0%	8.26%	6.40%	No

(Cost = 1.031 × (1 + inflation) − 1.) The breakeven inflation for Tessellate is 1.064 / 1.031 − 1 = 3.20%. For a revenue stream that is itself indexed, the comparison is misleading, because higher inflation raises revenue as well as cost; what matters is the *real* margin, which is the same under all inflation outcomes if the structure is matched.

4.8 Managerial Implications

Bond markets are not just a source of funds. They are a source of information and a place where several of the firm's risks are priced. The main managerial takeaways of the chapter are as follows.

Cost of debt. The correct cost of debt for valuation is the current market yield on the firm's debt (the YTM on comparable-risk, comparable-maturity bonds), not the coupon on the debt it issued years ago. In the weighted-average cost of capital the cost of debt is then adjusted for the tax deductibility of interest.

Maturity and refinancing. A firm with long-lived assets that funds with short-term debt bears refinancing risk: when a large block of debt matures, the firm must refinance at whatever rate and spread prevail then. Spreading maturities ("laddering") reduces this exposure. Extending maturity costs the term premium, but buys certainty.

Interest-rate exposure. Duration provides the language for managing the sensitivity of debt and pension liabilities to rate changes. Firms can alter exposure by choosing fixed versus floating-rate debt, by using swaps to convert one to the other, or by matching the duration of pension assets to that of the liabilities.

Credit rating as a strategic variable. Rating thresholds affect the investor base, the cost of debt and access to commercial paper markets. Managers may deliberately hold lower leverage than the tax-optimal level to preserve a rating, a decision that should be explicit and justified.

Reading the market. The yield curve and breakeven inflation provide market-based evidence on rate and inflation expectations for use in planning. Credit spreads on the firm's own bonds and on its peers' provide an early warning of changing investor perceptions.

Matching principle. A recurring theme is to match the characteristics of liabilities with those of assets and revenues: maturity, rate type (fixed or floating), currency and inflation linkage. Mismatches are not always wrong, but they should be taken deliberately and measured.

Case Study: Tessellate Power's Financing Choice

Illustrative case. Tessellate Power is a fictional company.

Tessellate Power generates and sells electricity to about 900,000 customers under a regulatory framework in which tariffs are reset annually to reflect a measure of consumer-price inflation, with a lag of one year. The company has $2.1 billion of debt outstanding, with a weighted average maturity of 6.5 years and a weighted average coupon of 4.8%. It is rated BBB by one agency and Baa2 by another. The company's principal covenant is a limit on net debt to earnings before interest, tax, depreciation and amortization (EBITDA).

The board has approved a $400 million wind and storage project. The treasurer, Anil Desmarais, must choose between three structures that investment bankers have priced on the same day.

- **Option A**: 5-year fixed-rate bond, 5.3% annual coupon, issued at par.
- **Option B**: 15-year fixed-rate bond, 6.4% annual coupon, issued at par.
- **Option C**: 15-year inflation-linked bond, real coupon 3.1%, principal indexed to consumer prices.

The government yield curve that day shows a 5-year yield of 4.0%, a 15-year yield of 4.6% and a 15-year real (inflation-linked government) yield of 1.9%. Thus the credit spreads implied are 130 bp for Option A and 180 bp for Option B, and 120 bp for Option C measured on a real basis. The bankers note that the spread for Option B is wider partly because of the longer exposure to Tessellate's credit and because the BBB rating is only one notch above the investment-grade boundary.

Anil gathers more information. Tessellate's cash flows are expected to be flat in real terms for the first six years, since the project ramps up slowly, and then to rise as the project reaches full capacity. The company's EBITDA is about 70% regulated and indexed, and 30% from merchant sales whose prices are volatile and have no inflation link. Existing debt is mostly fixed-rate and nominal, with a modest maturity wall of $600 million due in year 5. If the company were downgraded to BB+, Anil estimates that spreads on all new issuance would widen by at least 150 bp, and several large investors would sell.

The bankers' view is that the 5-year bond is cheapest now, but they warn that refinancing the whole $400 million in five years would coincide with the $600 million maturity wall, so that $1 billion would need to be refinanced in one year. The chief executive prefers Option B because "we can fix the cost for 15 years and forget about it." The head of risk argues for

Option C, noting that regulated revenues are indexed, and that Option C reduces the interest-rate mismatch. But the chief financial officer fears that inflation, which averaged 2.5% in the recent past, could be volatile and that a serious spike would raise the nominal burden on the 30% of cash flow that is not indexed.

Anil's analysts also compute that, at a 6.4% yield, a 15-year 6.4% bond would have a modified duration of about 9.5, whereas the 5-year bond at 5.3% would have a duration of about 4.3. A 100 bp rise in market yields would lower the market value of the 15-year bond by roughly 9.5%, but, because Tessellate intends to hold the debt to maturity and not mark it to market, the effect would show up only in disclosures and in the opportunity cost of having locked in the rate. The analysts also point out that the all-in cost of Option C is 5.7% in nominal terms if inflation averages 2.5%, and exceeds 6.4% if inflation averages more than 3.2%.

Discussion questions

1. Compute the breakeven inflation rate on the government curve (15-year nominal 4.6%, 15-year real 1.9%) and on Tessellate's own bonds (Options B and C). What does the comparison imply about whether Tessellate is buying inflation protection cheaply or dearly, and which components of breakeven (expected inflation, inflation risk premium, liquidity) might explain the difference?
2. Assess the refinancing risk of Option A given the maturity wall in year 5. Propose a debt-maturity profile (a mix of the three options or an alternative) that spreads refinancing risk, and discuss the cost of that choice using the shape of the yield curve and the term premium.
3. Using the information on revenue (70% indexed, 30% merchant), discuss how well each option matches Tessellate's cash flows. How might the firm hedge the remaining mismatch, and how would you decide between the hedge and a simpler fixed-rate structure?
4. The downgrade threshold is a close constraint. How should the potential for a downgrade influence the choice, and what disclosures or financial policies (such as leverage targets) might Tessellate adopt to protect its investment-grade rating? Estimate the added annual interest cost on $400 million of a 150 bp widening.

Key Takeaways

- A bond is the present value of its coupons (an annuity) and principal (a lump sum), discounted at the market yield; the coupon rate is a label fixed at issue while the yield moves with the market.
- Bond prices and yields move in opposite directions; a bond sells at a premium, par or discount according to whether its coupon is above, equal to or below its yield, and its price is pulled toward par as maturity approaches.
- YTM is the bond's internal rate of return if held to maturity, with full payment and reinvestment of coupons at the YTM itself. Realised returns differ when rates change, and for risky bonds promised yields exceed expected yields.
- Quoted bond prices are clean prices; the buyer pays the dirty price, equal to the clean price plus accrued interest.
- Macaulay duration is the present-value-weighted average time to cash flows; modified duration ($D / (1 + y)$) gives the approximate percentage price change per unit change in yield. Duration rises with maturity and falls with coupon and yield.
- Convexity captures the curvature of the price–yield relationship and improves the duration estimate for large rate moves; investors value positive convexity and the options embedded in callable bonds can make it negative.
- Immunisation sets asset duration equal to the liability horizon (and present values equal) so that price and reinvestment effects offset; it requires rebalancing and protects only against parallel shifts.

- The term structure is described by spot rates, and forward rates implied by the spot curve are the market-implied marginal cost of extending maturity; slopes reflect rate expectations, term premia and segmentation.
- A credit spread compensates for expected loss, a risk premium and illiquidity; the rating scale's investment-grade boundary is a practical constraint on issuers because many institutional investors cannot hold speculative-grade debt.
- Inflation-linked bonds pay a real yield on an inflation-adjusted principal; the difference between nominal and real yields (the breakeven rate) reflects expected inflation plus risk and liquidity premia, and indexed funding is attractive mainly when the issuer's revenues are themselves indexed.

Review Questions

1. Why must the price of a fixed-rate bond fall when market yields rise? Explain in terms of present values.
2. A bond has a coupon of 7% and a yield of 5%. Will it trade at par, at a premium or at a discount? What will happen to its price if its yield remains at 5% until maturity?
3. Explain the difference between clean and dirty prices. Why do bond markets quote clean prices?
4. List the three assumptions underlying the YTM calculation. Which is most likely to be violated, and what is the consequence for a long-dated, high-coupon bond?
5. Why is duration a better measure of interest-rate risk than maturity? Under what conditions does duration equal maturity?
6. Explain why convexity is valuable to a bondholder. Why can a callable bond exhibit negative convexity?
7. Describe how a company with a known future liability could immunize it. What are the limitations of the technique?
8. Define a forward rate and explain how it is derived from spot rates. What does it mean if the forward rate is above the expected future spot rate?
9. Distinguish expected loss from the credit spread. Why is the risk-neutral default probability higher than the real-world default probability?
10. Explain how breakeven inflation is calculated from nominal and real yields, and why it should not be read as a pure forecast of inflation.

Problems

1. **Bond pricing.** A $1,000 face value bond has a 7% annual coupon and six years to maturity. Compute its price at yields of (a) 8% annual, (b) 8% bond-equivalent with semi-annual coupons, and (c) 6% annual.
2. **YTM.** A five-year bond with a 6% annual coupon sells for $1,080. Compute its yield to maturity.
3. **Duration and convexity.** A four-year bond with an 8% annual coupon is priced at a 7% yield. Compute (a) its price, (b) its Macaulay duration, (c) its modified duration and convexity, and (d) the estimated price change for a yield increase of 50 basis points, with and without the convexity adjustment, together with the exact change.
4. **Spot and forward rates.** The zero-coupon spot rates are 2.0% (1 year), 2.8% (2 years) and 3.4% (3 years). Compute (a) the price of a three-year zero-coupon bond with face value $1,000, (b) the forward rates $f_{1,2}$ and $f_{2,3}$, and (c) the price of a three-year 4% annual-coupon bond.
5. **Credit spread.** A one-year government zero yields 3.5%, and a one-year corporate zero yields 5.9%. If recovery in default is 50%, compute the risk-neutral default probability. What is the spread, and how does it compare with PD × LGD?

6. **Inflation-linked bond.** The 10-year nominal yield is 5.2% and the 10-year real yield is 2.1%. (a) Compute breakeven inflation. (b) An inflation-linked bond has an original principal of $1,000 and a 2.0% real annual coupon; if inflation is 2.5% each year, what is the principal and coupon in year 2?
7. **Immunisation.** A firm owes $2,000,000 in five years. The yield is 6% for all maturities. Design a portfolio of 2-year and 8-year zero-coupon bonds that is immunized: compute the present value of the liability and the amount to invest in each bond.
8. **Portfolio and zero-coupon duration.** A portfolio holds 40% of its value in a bond with a modified duration of 3.2 and 60% in a bond with a modified duration of 7.5. (a) Compute the portfolio modified duration. (b) Estimate the percentage change in portfolio value if yields rise by 75 basis points. (c) Compute the price of a 10-year zero-coupon bond at a 5% yield, and the percentage change if the yield rises to 5.5%.

Answers to Selected Problems

1. a. **$953.77**. (b) Semi-annual coupons of $35 for 12 periods at 4%: **$953.07**. (c) **$1,049.17**.
2. YTM = **4.193%**.
3. a. **$1,033.87**. (b) Macaulay duration **3.585 years**. (c) Modified duration **3.350**; convexity **15.06**. (d) Duration only: $-3.350 \times 0.005 \times 1{,}033.87 =$ **−$17.32**; with convexity: **−$17.12**; exact: price at 7.5% = $1,016.75, a change of **−$17.13**.
4. a. $1{,}000 / 1.034^3 =$ **$904.56**. (b) $f_{1,2} = 1.028^2 / 1.02 - 1 =$ **3.606%**; $f_{2,3} = 1.034^3 / 1.028^2 - 1 =$ **4.611%**. (c) $40/1.02 + 40/1.028^2 + 1{,}040/1.034^3 =$ **$1,017.81**.
5. $q = (1 - 1.035/1.059) / 0.5 =$ **4.533%**. Spread = 5.9% − 3.5% = 2.4% (240 bp); PD × LGD = 4.533% × 0.5 = 2.266%, close to the spread.
6. a. $1.052 / 1.021 - 1 =$ **3.036%**. (b) Year 2 principal = $1{,}000 \times 1.025^2 =$ **$1,050.63**; coupon = 2% × $1,050.63 = **$21.01**.
7. PV of liability = $2{,}000{,}000 / 1.06^5 =$ **$1,494,516.35**. Weight in the 2-year zero: $(8 - 5)/(8 - 2) = 0.5$, so **$747,258.17** in each.
8. a. $0.4 \times 3.2 + 0.6 \times 7.5 =$ **5.78**. (b) $\approx -5.78 \times 0.0075 =$ **−4.34%**. (c) Price = $1{,}000 / 1.05^{10} =$ **$613.91**; at 5.5% the price falls by about **4.64%** (duration approximation: −4.76%).

Part II — Investment Decisions

Chapter 5: Equity Valuation

Learning Objectives

After studying this chapter you should be able to:

- Derive the dividend discount model (DDM) from the principle that a share is worth the present value of the cash it will deliver to its owner, and apply it over finite and infinite horizons.
- Use the constant-growth (Gordon) model to value a share, to back out the market's implied required return or implied growth rate, and to explain why the model breaks down when growth approaches the discount rate.
- Link dividend growth to corporate fundamentals through the retention ratio and return on equity, and judge whether growth creates or destroys value.
- Build and interpret a multi-stage valuation, and explain why the terminal value usually dominates the answer.
- Decompose a share price into the value of assets in place and the present value of growth opportunities (PVGO), and use this to interpret price-earnings (P/E) ratios.
- Value preferred stock as a perpetuity, and describe how the mechanics of stock markets (order types, spreads, margin, indices) shape the prices and returns investors actually experience.

Opening Vignette

Illustrative scenario. Verdant Grid Systems is a fictional company.

Verdant Grid Systems designs control software for regional electricity distributors. After a decade of rapid expansion it has become profitable, and its board is debating two questions that look unrelated but are really the same question. The first comes from the chief financial officer, Marisol Quade: "Our shares trade at $62. Are we cheap, fair, or expensive, and how would we know?" The second comes from the head of strategy, Dev Anand: "We can reinvest 70 percent of earnings in new grid-analytics products, or we can raise the payout and buy back stock. Which one makes the shares worth more?"

Marisol begins with what she can observe. Next year's earnings are expected to be $4.00 per share, and the company currently pays out 40 percent of them. Analysts expect growth of roughly 6 percent for the foreseeable future. Investors in comparable software-and-infrastructure companies appear to demand about 10 percent. She calculates in a few lines that a $1.60 dividend growing at 6 percent, discounted at 10 percent, is worth $40 per share. The market is paying $62. Either investors are more optimistic than the analysts, or they see growth opportunities that the simple formula has not captured, or the required return is lower than 10 percent.

Dev's question turns out to be sharper. If new products earn 14 percent on the money invested, retention creates value. If they earn 8 percent, then every dollar retained is worth less than a dollar to shareholders, and a buyback would be better. Neither executive can answer without a framework that connects dividends, growth, risk and the return on reinvested capital.

This chapter provides that framework. We begin with the logic that makes a share worth the present value of its payouts, move to the models that practitioners use, and examine what price-earnings ratios say about growth. We then look at preferred stock and the market machinery through which shares trade. By the end, Marisol's puzzle ($62 versus $40) will be something you can decompose into specific, testable assumptions.

5.1 From Cash Flows to Share Prices: The Dividend Discount Model

5.1.1 The valuation principle

Chapter 4 valued bonds by discounting contractual cash flows at a rate that reflects risk. Stocks follow the same principle with two complications: the cash flows are not promised, and they have no maturity date. A share of common stock is a residual claim. Shareholders receive whatever is left after creditors, taxes and reinvestment needs, and they hold that claim for as long as the firm survives. Despite these complications, the economic logic is unchanged: the value of any asset is the present value of the cash its owner expects to receive, discounted at the return available on alternative investments of equivalent risk.

The expected cash flows to a shareholder are dividends while the share is held and a sale price at the end. Because the next buyer will also value the share as the present value of her own expected dividends and resale price, the sale price is itself a present value of later dividends. Substituting repeatedly, the resale prices disappear and the price today depends only on dividends.

5.1.2 One-period model

Suppose an investor plans to hold a share for one year. Let D_1 be the dividend expected at the end of the year, P_1 the expected price then, and r the required return (the cost of equity). The price today is

```
P₀ = (D₁ + P₁) / (1 + r)
```

Equivalently, the required return is the sum of the expected dividend yield and the expected capital gain:

```
r = D₁ / P₀ + (P₁ − P₀) / P₀
```

This second form is useful because it shows what the market must offer to compensate investors: income, appreciation, or some mix. A utility paying out most of its earnings offers mostly yield; a young technology company paying nothing must offer all of its return as expected appreciation.

Example 5.1 – One-year holding period. An investor expects a share of Verdant Grid to pay a dividend of $2.10 in one year and to trade at $45.00 at that time. The investor requires a return of 9 percent.

Step 1: Add the expected cash received at the end of the year: 2.10 + 45.00 = $47.10.

Step 2: Discount one year at 9 percent: P_0 = 47.10 / 1.09 = $43.21.

Step 3: Check by computing the implied return if the share is bought at $43.21: dividend yield = 2.10 / 43.21 = 4.86 percent; capital gain = (45.00 − 43.21) / 43.21 = 4.14 percent; total = 9.00 percent. If the market price were $46, the expected return would fall to (47.10 / 46) − 1 = 2.39 percent, and the investor would pass.

5.1.3 Extending the horizon

If the investor holds for two years, P_1 is replaced by the present value at date 1 of D_2 and P_2, and so on. After N years:

```
P₀ = D₁/(1+r) + D₂/(1+r)² + … + D_N/(1+r)^N + P_N/(1+r)^N
```

Letting N grow without bound, the last term shrinks toward zero provided the price grows more slowly than the discount rate (an assumption that excludes explosive bubbles), and the price becomes the present value of an infinite stream:

```
P₀ = Σ (t = 1 to ∞) D_t / (1 + r)^t
```

This is the **dividend discount model (DDM)**. Its message is deceptively simple and often misunderstood. It does not say that investors care only about dividends in the short run, nor that a company that pays no dividend today is worthless. It says that, ultimately, the only cash a shareholder can extract from a firm is distributions: dividends, repurchases, or a final liquidating payment. A firm that never distributed anything to anyone would be worth nothing. Companies that pay no dividends now are, in effect, promising larger distributions later.

5.1.4 Estimating the required return

The discount rate r is the cost of equity capital. Chapter 3 developed the capital asset pricing model, in which

```
r = r_f + β × (market risk premium)
```

Suppose the risk-free rate is 4.0 percent, the market risk premium is 5.5 percent and a company's equity beta is 1.2. Then r = 4.0 + 1.2 × 5.5 = 10.6 percent. The estimate is only as good as its inputs. Beta estimated from five years of monthly returns carries a wide standard error, and the market risk premium is a judgment rather than a measurement. Managers should treat the discount rate as a range, a point we revisit repeatedly because valuation is highly sensitive to it.

5.1.5 Why dividends, and not earnings?

A common objection is that earnings, not dividends, measure what the firm produces. But earnings that are retained are reinvested, and investors benefit only if the reinvestment earns at least the required return. Valuing earnings directly would double count: it would credit shareholders with earnings that the firm spent on new assets, and then credit them again with the later earnings those assets generate. The DDM avoids double counting by valuing only what leaves the firm. We will see in Section 5.5 how the earnings-based and dividend-based views reconcile exactly, through the concept of growth opportunities.

Two practical remarks complete the foundation. First, when firms return cash through repurchases rather than dividends, the DDM can be applied to total payout (dividends plus repurchases) divided by the current number of shares, or equivalently to free cash flow to equity. Second, the DDM values minority stakes in publicly traded firms where the investor cannot control the payout. An acquirer who controls the firm values the underlying cash flows rather than the dividends the current board chooses to pay; that approach is developed in later chapters on enterprise valuation.

5.2 The Constant-Growth (Gordon) Model

5.2.1 Derivation and formula

Forecasting an infinite series of dividends is impossible, so we need simplifying assumptions. The simplest is that dividends grow at a constant rate g forever, so that D_t = D_0(1 + g)^t. The infinite sum is then a growing perpetuity, which has a closed form when r > g:

```
P₀ = D₁ / (r − g)
```

where $D_1 = D_0 \times (1 + g)$. This is the **Gordon growth model**, named after Myron Gordon, who popularized it (it is also called the Gordon-Shapiro model). It is the most heavily used equation in equity valuation and also among the most abused.

Three features deserve emphasis. First, the formula uses next year's dividend, D_1, not the one just paid. Mixing these up is the single most common error in student work. Second, the formula gives the price at the date one period before the first dividend; it values a stream that starts one year from now. Third, the denominator r − g is a small number obtained by subtracting two uncertain ones, so tiny changes in either input cause large changes in value.

Example 5.2 – Gordon growth valuation and implied return. Verdant Grid's peer, Meridian Instruments (fictional), just paid an annual dividend of $2.00 per share. Dividends are expected to grow at 4 percent per year indefinitely. The required return is 9 percent.

Step 1: Next year's dividend: $D_1 = 2.00 \times 1.04 = \$2.08$.

Step 2: Apply the model: $P_0 = 2.08 / (0.09 - 0.04) = 2.08 / 0.05 = \$41.60$.

Step 3: Suppose the stock actually trades at $50. Solve for the return the market is implicitly pricing: $r = D_1/P_0 + g = 2.08 / 50 + 0.04 = 4.16\% + 4.00\% = 8.16$ percent. A buyer at $50 earns an expected 8.16 percent if growth is really 4 percent. Alternatively, holding r at 9 percent, the market price implies growth of $g = r - D_1/P_0$ approximately; solving exactly, $50 = 2.00(1+g)/(0.09 - g)$ gives $g = (50 \times 0.09 - 2.00) / (50 + 2.00) = 2.50 / 52 = 4.81$ percent.

Step 4: Interpretation. At $50 the market is either demanding about 0.8 percentage points less than 9 percent or expecting almost 0.8 points more growth. Both views are defensible, and the analyst's job is to decide which is more plausible.

5.2.2 Sensitivity to r and g

The table below shows the value of a share with $D_1 = \$2.00$ for different combinations of required return and growth. Notice how the price doubles from $50 to $100 when g moves from 4 percent to 6 percent at an 8 percent discount rate.

g \ r	8%	9%	10%	11%	12%
2%	33.33	28.57	25.00	22.22	20.00
3%	40.00	33.33	28.57	25.00	22.22
4%	50.00	40.00	33.33	28.57	25.00
5%	66.67	50.00	40.00	33.33	28.57
6%	100.00	66.67	50.00	40.00	33.33

Table 5.1 – Gordon model price with $D_1 = \$2.00$ (illustrative).

The lesson is humility. A one-point error in the spread (r − g) can change the answer by 25 to 50 percent. Responsible practitioners therefore present valuation as a grid or a range, never a single number, and they ask which assumptions the answer depends on most.

5.2.3 Limits of the model

The model requires $r > g$. If $g \geq r$ the formula gives a negative or infinite price, which is a signal that the assumption of perpetual growth at that rate is impossible. No firm can grow faster than the economy forever, because it would eventually become larger than the economy. In practice, the long-run g should not exceed the long-run growth of nominal GDP, and in many analyses it is set near expected inflation plus a small real component.

The model also fits only firms whose dividends are stable, positive and growing smoothly: regulated utilities, consumer staples, mature banks. It does poorly for young firms that pay nothing, cyclical firms with erratic payouts, and firms in transition. For these, the multi-stage models of Section 5.4 or the earnings-based approach of Section 5.5 are more appropriate.

Finally, the Gordon model is a **price model that holds at every date**, not only today. If dividends grow at g forever, then the price also grows at g: $P_1 = P_0(1 + g)$. The total return is therefore the dividend yield plus the growth rate, $r = D_1/P_0 + g$. The model should be read as an equilibrium relationship between yield, growth and required return. Any one of the three can be inferred from the other two.

5.2.4 Common mistakes

- Using D_0 in the numerator instead of D_1, which understates the value by a factor of (1 + g).
- Choosing a growth rate above the long-run growth of the economy, then concluding that the stock is hugely undervalued.
- Using a recent burst of growth (say, 18 percent over three years) as the perpetual g.
- Using a discount rate that does not match the risk of the dividends, such as a bond yield for a volatile equity stream.
- Forgetting that the model delivers price at one period before the first dividend. If the first dividend is received two years from now, the formula gives the price one year from now, and it must be discounted once more.

5.3 Growth from Fundamentals: Retention, ROE and Value Creation

5.3.1 Where does growth come from?

Growth is not a free parameter; it must be financed. A firm that retains a fraction b of its earnings (the **retention ratio**, or plowback ratio) and reinvests at a return on equity of ROE can grow earnings and dividends at

```
g = b × ROE
```

provided the firm’s capital structure is stable and the new investments earn the same ROE as the old. The payout ratio is 1 − b. Dividends are $D_1 = (1 - b) \times EPS_1$. Substituting into the Gordon model:

```
P₀ = (1 − b) × EPS₁ / (r − b × ROE)
```

This is a powerful formula because it ties the stock price to three operating decisions: how much to reinvest, how profitably to reinvest it, and how risky the business is.

5.3.2 Does retention create value?

The derivative of price with respect to b depends on the comparison of ROE with r. Increasing retention lowers current dividends but raises growth. If ROE > r, the extra growth more than compensates for the lost dividends and value rises. If ROE = r, retention is a wash: shareholders could have earned the same return themselves, so price does not change. If ROE < r, retention destroys value. A dollar reinvested at 8 percent when shareholders require 10 percent is worth less than a dollar.

Example 5.3 – The effect of retention depends on ROE. Verdant Grid expects EPS_1 = \$5.00 and has a required return of 10 percent. We compare three scenarios in which the company retains 60 percent of earnings (b = 0.6) but earns different returns on reinvested equity.

Step 1: Dividend = (1 − 0.6) × 5.00 = \$2.00 in each case.

Step 2: Growth rates: ROE 8 percent gives g = 0.6 × 0.08 = 4.8 percent; ROE 10 percent gives 6.0 percent; ROE 15 percent gives 9.0 percent.

Step 3: Prices: with ROE 8 percent, P_0 = 2.00 / (0.10 − 0.048) = 2.00 / 0.052 = \$38.46; with ROE 10 percent, P_0 = 2.00 / 0.04 = \$50.00; with ROE 15 percent, P_0 = 2.00 / (0.10 − 0.09) = 2.00 / 0.01 = \$200.00.

Step 4: Benchmark. If the firm paid out all earnings and did not grow, price would be EPS_1 / r = 5.00 / 0.10 = \$50.00. Retention at ROE = r leaves the price at \$50; retention at ROE 8 percent cuts it to \$38.46; retention at 15 percent quadruples it.

The extreme sensitivity of the last case (a price of $200 from a spread of one point) is itself a warning: a 15 percent return on 60 percent reinvestment is hard to sustain, because attractive opportunities are scarce and competition erodes returns. Managers presenting growth plans should be asked how long the high ROE can be maintained, which leads naturally to the multi-stage model.

5.3.3 Practical considerations

Real firms do not hold a constant ROE. Accounting ROE is an average over old and new assets and is affected by depreciation methods, goodwill and buybacks. For valuation, the relevant figure is the **return on new investment** (marginal ROE). Firms with persistent competitive advantages (strong brands, network effects, regulatory franchises) can sustain high marginal returns; firms in commodity industries typically cannot. The distinction between growth and *profitable* growth is central to managerial decisions on capital allocation and payout policy. A manager who follows the rule "grow at all costs" without comparing ROE with r is destroying shareholder value while reporting rising earnings.

A related point concerns leverage. ROE can be raised by borrowing, but so does the required return r, since equity becomes riskier. A valid comparison of ROE with r must hold the capital structure fixed, otherwise the apparent value creation is just transferred risk.

5.4 Multi-Stage Growth Models

5.4.1 The logic of stages

Few firms grow at one rate forever. A typical life cycle features a high-growth phase in which opportunities are abundant, a transition in which competition compresses growth, and a mature phase in which growth settles near the economy's. Multi-stage models capture this by assigning different growth rates to different periods. In the simplest, the **two-stage model**, dividends grow at g_1 for N years and at a sustainable g_2 thereafter:

```
P₀ = Σ (t = 1 to N) D_t / (1+r)^t  +  [D_{N+1} / (r − g₂)] / (1+r)^N
```

The second term is the present value of the **terminal value** (or continuing value): the Gordon price at date N of all the dividends from year N + 1 onward. Many analysts also use a three-stage variant in which the growth rate declines linearly from g_1 to g_2 over a transition period, which avoids the unrealistic cliff at the end of the high-growth phase.

5.4.2 A worked two-stage valuation

Example 5.4 – Two-stage dividend valuation. Helios Storage Systems (fictional) just paid a dividend of $1.50 per share ($D_0$). Dividends are expected to grow at 15 percent per year for the next four years, and at 5 percent per year thereafter. The required return is 10 percent.

Step 1: Dividends during the high-growth phase.

Year	Calculation	Dividend ()|*Discountfactor*1/1.10^t|*Presentvalue*()		
1	1.5000 × 1.15	1.7250	0.9091	1.5682
2	1.7250 × 1.15	1.9838	0.8264	1.6395
3	1.9838 × 1.15	2.2813	0.7513	1.7140
4	2.2813 × 1.15	2.6235	0.6830	1.7919
Total				6.7135

Table 5.2 – Present value of high-growth dividends (rounded).

Step 2: First dividend of the mature phase: $D_5 = 2.6235 \times 1.05 = \$2.7547$.

Step 3: Terminal value at the end of year 4: $TV_4 = D_5 / (r - g_2) = 2.7547 / (0.10 - 0.05) = 2.7547 / 0.05 = \$55.09$.

Step 4: Discount the terminal value to today: $PV(TV) = 55.09 / 1.10^4 = 55.09 / 1.4641 = \$37.63$.

Step 5: Share price: $P_0 = 6.71 + 37.63 = \$44.34$.

Step 6: Interpretation. The terminal value accounts for 37.63 / 44.34 = 84.9 percent of the total. Even though the company enjoys four years of 15 percent growth, most of the value comes from what happens after year four. This is typical of growth stock valuations. If the analyst had used the Gordon model with the initial 15 percent growth, the model would have been unusable (g > r); had she used 5 percent from the start, she would have obtained 1.5 × 1.05 / 0.05 = \$31.50, understating by \$12.84 the value of the temporary growth.

5.4.3 Why the terminal value dominates

The terminal value dominates because it is a perpetuity discounted over a finite horizon. The shorter the high-growth phase, the larger the share of value in the terminal term. Three practical consequences follow.

First, the terminal growth rate and the discount rate must be chosen with great care, since they drive most of the value. Second, the terminal growth rate should be internally consistent with the terminal payout and ROE. If a firm is assumed to grow at 5 percent forever with an ROE of 10 percent, then b = g / ROE = 50 percent must be retained, and the terminal dividend must reflect that. Models that assume high growth with low retention quietly assume that growth is free. Third, terminal-value cross-checks are worthwhile. The implied terminal P/E in Example 5.4 can be computed from the terminal price and expected year-five earnings; if it far exceeds the multiples of mature companies, the assumptions should be reconsidered.

5.4.4 When dividends are not the right cash flow

Many growth companies pay no dividends for years. The two-stage model can still be applied by treating the early dividends as zero and valuing a terminal price when the firm is expected to begin paying. A cleaner approach is to value **free cash flow to equity**: cash generated for shareholders after investment and after net borrowing, whether or not it is distributed. Under the assumption that the firm eventually distributes all free cash flow to equity, the two approaches coincide. For firms that hoard cash (retaining without a plan to invest), the dividend approach gives a lower value than the FCFE approach, and the difference measures the agency cost of idle cash. We return to this theme in the chapters on payout policy and corporate governance.

5.5 Price-Earnings Ratios and Growth Opportunities

5.5.1 The P/E ratio

The P/E ratio is the share price divided by earnings per share. Analysts use trailing P/E (price over last year’s EPS) and forward P/E (price over expected EPS_1). From the constant-growth model, the forward P/E is

```
P₀ / EPS₁ = (1 − b) / (r − g)
```

so the ratio depends positively on the payout ratio and on growth, and negatively on the required return. A high P/E can mean high expected growth, low perceived risk, or both. The reciprocal, EPS_1/P_0, is the **earnings yield**. It equals r only for a no-growth firm that pays out all earnings.

5.5.2 Decomposing price: assets in place plus PVGO

A firm that never invested in anything new and paid out all earnings would be worth

```
P₀ (no growth) = EPS₁ / r
```

This is the value of **assets in place**: the existing business perpetually earning what it earns now. If the firm has opportunities to invest in positive-NPV projects, the price will exceed this benchmark. The difference is the **present value of growth opportunities**:

```
P₀ = EPS₁ / r + PVGO
```

PVGO is the net present value, as of today, of all the future investments the firm will make. Its sign and size are governed by the same logic as in Section 5.3: growth projects that earn more than r contribute positively; those that earn less contribute negatively. A firm may well have a negative PVGO if management has a record of undertaking poor projects, in which case the market price is below the value of simply liquidating the earnings stream, and investors may press for higher payouts.

Example 5.5 – Decomposing the price and interpreting the P/E. Return to the 15 percent ROE case of Example 5.3. EPS_1 = \$5.00, r = 10 percent, b = 0.6, ROE = 15 percent, g = 9 percent, price \$200.

Step 1: Value of assets in place: 5.00 / 0.10 = \$50.00.

Step 2: PVGO = 200.00 − 50.00 = \$150.00.

Step 3: Share of price from growth: 150 / 200 = 75 percent.

Step 4: Forward P/E = 200 / 5 = 40. A no-growth firm with the same required return would have P/E = 1/r = 10. We can therefore write P/E = (1/r) × (1 + PVGO / (EPS_1/r)) = 10 × (1 + 150/50) = 10 × 4 = 40.

Step 5: Contrast with the ROE = 8 percent scenario: price 38.46, assets in place 50.00, PVGO = −11.54. The company's growth plan subtracts \$11.54 per share. Shareholders would be better off if management stopped investing and paid out earnings.

The following table summarizes the four scenarios (EPS_1 = 5.00, r = 10%, b = 0.6).

ROE	g = b × ROE	Price ()|*Assetsinplace*()	PVGO (\$)	Forward P/E	
8%	4.8%	38.46	50.00	−11.54	7.7
10%	6.0%	50.00	50.00	0.00	10.0
12%	7.2%	71.43	50.00	21.43	14.3
15%	9.0%	200.00	50.00	150.00	40.0

Table 5.3 – Growth opportunities and P/E (illustrative).

5.5.3 Reading P/E ratios sensibly

P/E ratios are useful as shorthand but dangerous as standalone evidence. Several cautions apply.

Earnings quality. Accounting earnings are affected by depreciation methods, one-off charges, stock-based compensation and discretionary accruals. Two companies with identical economics may report different EPS and hence different P/Es. Analysts normalize earnings for non-recurring items before comparing.

Cyclicality. At the trough of a cycle earnings are depressed and the P/E looks very high; at the peak, earnings are inflated and the P/E looks low. Cyclical companies are better assessed on normalized or average earnings.

Capital structure. P/E ratios are affected by leverage because earnings are after interest. Enterprise multiples such as EV/EBITDA are more comparable across firms with different debt levels.

Comparable-company analysis. In practice, analysts value a company by applying the multiples of similar firms to its own earnings. This presupposes that the comparables share similar growth, risk and payout. The constant-growth formula is a reminder of what must be comparable: the spread r − g and the payout ratio. Applying a peer's P/E of 20 to a company whose growth is half the peer's would badly overvalue it.

Interpretation in Marisol's puzzle. Verdant Grid's CFO obtained $40 from a $1.60 dividend growing at 6 percent with a 10 percent discount rate. The market price of $62 implies $22 per share for something the simple model does not capture. If earnings of $4.00 with no growth would be worth 4.00 / 0.10 = $40.00, then the Gordon price of $40 corresponds to PVGO of zero: the 6 percent growth at a 40 percent payout implies ROE = g / b = 0.06 / 0.60 = 10 percent, exactly equal to r. The market is therefore pricing in PVGO of $22, which requires either a higher return on reinvestment than 10 percent or a longer period of above-average growth. That is a concrete, debatable claim about Verdant's future, which is exactly what valuation should deliver.

5.5.4 PVGO and corporate strategy

The PVGO view clarifies strategic questions. A mature firm whose value is mostly assets in place has stable cash flows and should return excess cash to shareholders, since little growth is available at attractive returns. A young firm whose value is mostly PVGO is vulnerable to changes in the discount rate and in competitive position, because its value rests on future decisions and distant cash flows. When interest rates rise, high-PVGO stocks tend to fall more than low-PVGO stocks, because a larger share of their value lies in the far future. Managers should understand which type of firm they run, because the investor base, financing strategy and payout policy appropriate to each differ.

5.6 Preferred Stock and Other Equity Claims

5.6.1 Characteristics of preferred stock

Preferred stock is a hybrid security that stands between debt and common equity. It pays a stated dividend, usually fixed as a percentage of par (face) value or as a dollar amount per share, and it ranks ahead of common stock but behind all debt in claims on income and on assets in liquidation. Unlike interest on debt, a preferred dividend is not a legal obligation: the board may suspend it without triggering bankruptcy. Common features include:

- **Cumulative dividends.** Unpaid preferred dividends accumulate and must be paid before any dividend on common stock. Non-cumulative preferred lacks this protection and is therefore riskier.
- **Priority in liquidation.** Preferred holders are paid before common holders, up to par or a stated liquidation preference.
- **Limited or no voting rights**, unless dividends are in arrears for a specified period.
- **Optional features.** Some issues are callable by the company, convertible into common shares, or carry a sinking fund. Others are participating, sharing in extra profits.
- **Tax treatment.** In many jurisdictions the dividends are not deductible to the issuer, which makes preferred stock more costly than debt after tax. In some jurisdictions, corporate holders receive a partial exclusion on dividend

income, which makes preferred stock attractive to corporate investors and explains why they are the main buyers. Treatment varies by country and changes over time; check current rules before relying on any generalization.

Because the dividend is fixed and has no maturity, non-convertible, non-callable preferred stock is a perpetuity, and valuation is simple.

5.6.2 Valuation

```
P₀ = D_p / r_p
```

where D_p is the annual preferred dividend and r_p is the required return on preferred stock of comparable risk. The rate r_p lies above the yield on the firm's senior debt, because preferred is junior, and below the cost of common equity, because it has a fixed payout and priority over the common. The implied yield of an existing issue is D_p / P_0. If the preferred stock is cumulative and dividends are in arrears, the present value of the arrears must be added to the claim, and the market price will reflect the probability that arrears are paid.

Example 5.6 – Valuing preferred stock. Meridian Instruments has outstanding $100 par preferred stock with a 5 percent dividend, so D_p = $5.00 per year. Comparable preferred issues yield 6.25 percent.

Step 1: Value = 5.00 / 0.0625 = $80.00.

Step 2: If the market yield on comparable issues rises to 8 percent, the value falls to 5.00 / 0.08 = $62.50, a drop of 21.9 percent. If it falls to 5 percent, the value rises to $100.00, which is par.

Step 3: Compare with common stock. A common stock paying a dividend that grows at 3 percent with the same $5.00 first-year dividend at r = 9 percent would be worth 5.00 / (0.09 − 0.03) = $83.33. The preferred share trades below that because it forgoes any growth, but its price is far less volatile if the company's fortunes change.

Step 4: A call feature. If the issuer can call the preferred at $100 and the yield falls to 5 percent, investors cannot expect the price to rise much above the call price; callable preferred has capped upside, as callable bonds do.

5.6.3 Why firms issue preferred

For issuers, preferred stock provides capital that does not dilute control (non-voting), does not create default risk (dividends can be omitted), and may be treated partially as equity by rating agencies. It is used by banks and utilities to meet capital requirements and by growing companies seeking a form of financing between debt and equity. Venture capital investors typically receive convertible preferred shares with special rights, which gives them downside protection in a liquidation and the upside of conversion into common if the venture succeeds. The cost to the issuer is the higher required return and the non-deductibility of dividends in many tax systems.

5.6.4 Warrants, convertibles and the broader equity picture

Common stock can also be affected by other equity-linked claims: options granted to employees, warrants, and convertible securities. When these are exercised they dilute existing holders. A value per share computed from total equity value must be divided by the fully diluted share count, which includes in-the-money options and convertibles on an as-converted basis. Ignoring dilution is a frequent source of overvaluation in technology and start-up companies where stock compensation is heavy. The valuation of options themselves is the subject of a later chapter.

5.7 Stock Market Mechanics

Valuation models give intrinsic value; the market sets the price at which you can trade. Understanding how trading works matters to managers because it determines the cost of raising equity, the information that prices convey, and the way investors earn returns.

5.7.1 Primary and secondary markets

In the **primary market** firms sell new shares to raise capital, through an initial public offering (IPO) or a seasoned equity offering. Investment banks underwrite the issue, advise on pricing, and distribute the shares. Underwriting costs and the common phenomenon of "underpricing" (shares rising on the first day of trading) are real costs to the issuing firm, although the size of the effect varies across time and markets. In the **secondary market** investors trade existing shares among themselves. The company receives no cash from these trades, but the secondary market is crucial: it gives shareholders liquidity, which lowers the return investors demand when buying new shares, and it produces a continuous price that serves as a performance signal and as currency for acquisitions and employee compensation.

5.7.2 Exchanges and trading venues

Shares are traded on organized exchanges, in electronic trading networks, and in dealer markets. In an **order-driven** (auction) market, buy and sell orders are matched by price and time priority through a central order book. In a **quote-driven** (dealer) market, market makers post prices at which they stand ready to buy and sell. Most modern markets combine both. Trading increasingly occurs on multiple venues linked by electronic routing, and a significant volume is executed by algorithms that split large orders into small pieces. Regulators in each jurisdiction require trade reporting and rules on fairness, and exchanges set listing standards for disclosure and governance. The details differ by country and change often, so a manager should consult current local rules.

5.7.3 Quotes, spreads and order types

Dealers and the order book express two prices: the **bid**, at which one can sell immediately, and the **ask** (or offer), at which one can buy immediately. The difference is the **bid-ask spread**, which compensates market makers for risk and the cost of providing immediacy. For liquid large-company stocks the spread may be a fraction of a percent; for thinly traded small companies it can be several percent, which is a direct cost of owning illiquid shares and part of why investors demand a higher return from them.

The main order types are:

- **Market order**: buy or sell immediately at the best available price. Execution is nearly certain but the price is not.
- **Limit order**: buy or sell only at a specified price or better. The price is controlled but execution is not guaranteed.
- **Stop (stop-loss) order**: becomes a market order when the stop price is reached. It is used to limit losses but can execute at prices well beyond the stop in fast markets.
- **Day, good-till-cancelled and fill-or-kill instructions** specify how long an order stays active and whether it must be filled completely.

For a stock quoted at bid $49.90 and ask $50.10, a market buy order executes near $50.10 and a market sell near $49.90. An investor who buys and immediately sells loses the $0.20 spread, or about 0.4 percent of the price, before any commissions.

5.7.4 Buying on margin

Investors can borrow part of the purchase price from a broker. The **initial margin** is the proportion of the purchase that must be funded with the investor's own money, and the **maintenance margin** is the minimum equity percentage that must be maintained afterwards. If the equity falls below maintenance, the broker issues a **margin call** requiring additional funds or the sale of securities. Margin magnifies returns in both directions.

Example 5.7 – Margin purchase. An investor buys 200 shares at $50 on 50 percent initial margin. Interest and commissions are ignored for simplicity.

Step 1: Purchase value = 200 × 50 = $10,000. Investor's equity = 50% × 10,000 = $5,000; loan = $5,000.

Step 2: Price rises to $60. Position value = 200 × 60 = $12,000. Equity = 12,000 − 5,000 = $7,000. Return on equity = (7,000 − 5,000) / 5,000 = 40 percent, compared with a 20 percent return on the stock.

Step 3: Price falls to $40. Position value = $8,000; equity = 8,000 − 5,000 = $3,000; equity ratio = 3,000 / 8,000 = 37.5 percent; return on equity = (3,000 − 5,000) / 5,000 = −40 percent, double the 20 percent fall in the stock.

Step 4: Margin call price. With maintenance margin of 30 percent, the equity ratio is (200P − 5,000) / (200P). Setting this equal to 0.30: 200P − 5,000 = 0.30 × 200P, so 140P = 5,000 and P = $35.71. If the price falls below $35.71, the investor must add funds or sell. A 28.6 percent fall in the share price (from 50 to 35.71) triggers the call.

Margin trading amplifies not only losses but also market volatility, because forced selling during downturns pushes prices lower and triggers further calls. Regulators therefore set limits on margin lending in many jurisdictions.

5.7.5 Short selling

A short seller borrows shares, sells them, and later buys them back to return to the lender, hoping to profit if the price falls. The potential loss is unlimited, because the price can rise without bound, and the seller must pay any dividends to the lender. Short sellers add liquidity and help bring prices toward fundamental values by expressing negative information, though they are periodically criticized, and some jurisdictions restrict shorting in market stress.

5.7.6 Stock market indices

An index summarizes the performance of a group of stocks. The two main constructions are price-weighted and value-weighted.

A **price-weighted index** (such as the Dow Jones Industrial Average, which is a well-known example) is the sum of the component prices divided by a divisor. High-priced stocks have greater influence regardless of company size. When a component splits, the divisor is adjusted so the index does not change. A **value-weighted index** (such as the S&P 500, whose weights are further adjusted for the number of freely traded shares) weights each stock by its market capitalization, so large companies dominate, and splits require no adjustment.

Example 5.8 – Price-weighted and value-weighted indices. Three fictional stocks: A at $90, B at $50 and C at $25, with 10 million, 40 million and 100 million shares outstanding.

Step 1: Price-weighted index at the start: (90 + 50 + 25) / 3 = 165 / 3 = 55.00.

Step 2: Stock B splits 2-for-1: its price becomes $25, with 80 million shares. The sum of prices becomes 90 + 25 + 25 = 140. To keep the index at 55.00 the divisor must change: new divisor = 140 / 55 = 2.5455.

Step 3: Next day, the prices are A $93, B $26 (equivalent to $52 before the split) and C $24. Sum of prices = 93 + 26 + 24 = 143. Index = 143 / 2.5455 = 56.18, an increase of 2.15 percent, driven mostly by A, the highest-priced stock.

Step 4: Value-weighted index. Day 0 market capitalizations (in millions): A 10 × 90 = 900; B 40 × 50 = 2,000; C 100 × 25 = 2,500; total 5,400. Day 1: A 10 × 93 = 930; B 80 × 26 = 2,080 (the split doubled the shares and halved the price, so capitalization is unaffected); C 100 × 24 = 2,400; total 5,410. Change = 5,410 / 5,400 − 1 = 0.19 percent, so a base of 100 becomes 100.19.

Step 5: Interpretation. The same price moves give +2.15 percent in the price-weighted index but only +0.19 percent in the value-weighted index. The value-weighted index is dominated by C, a $2.5 billion company whose price fell, whereas the price-weighted index is dominated by the $90 stock A, which rose. The choice of index construction therefore affects how "the market" is described.

5.7.7 Market efficiency and what prices mean

Eugene Fama's efficient markets hypothesis states that prices reflect available information. In its weak form, past prices cannot be used to earn abnormal profits; in the semi-strong form, all public information is already in prices; in the strong form, even private information is reflected. The evidence indicates that large, liquid markets are reasonably efficient in the semi-strong sense, so that it is hard for investors to consistently beat the market after costs, though there is ongoing debate about anomalies and the role of behavioral biases. For managers, the main implications are practical: the market price is generally a better estimate of value than an executive's private view, so managers should be skeptical of claims that the stock is "obviously undervalued" and should be cautious about timing equity issuance. At the same time, market prices can deviate from fundamentals for extended periods, especially for stocks with high PVGO, hard-to-value intangible assets, or limited analyst coverage.

5.8 Managerial Implications

Cost of equity is a hurdle rate. The required return that discounts dividends in this chapter is the same cost of equity that enters the weighted average cost of capital and project evaluation. A manager who understands how the market derives value from dividends and growth can see how project decisions feed back into the share price.

Growth must clear the hurdle. Section 5.3 showed that retention is valuable only when the return on new investment exceeds the cost of equity. Policies that reward managers on earnings growth or on size encourage over-investment. Metrics that compare returns with the cost of capital (such as economic profit) align incentives better.

Payout policy and valuation. In the Modigliani-Miller framework with perfect markets, the choice between dividends and repurchases does not by itself change value, because what matters is the stream of free cash flow; but in practice payouts convey information and discipline management. Cutting the dividend often triggers a price fall because investors read it as a signal of lower future cash flows.

Communication. The market price depends on expectations about growth and risk. Clear communication about strategy, capital allocation and return targets reduces uncertainty and, other things equal, the required return.

Valuation as a range. Because small changes in r and g produce large changes in value (Table 5.1), a sound valuation reports a range, tests multiple methods (DDM, multiples, discounted free cash flow) and explains the gap between the model value and the market price in terms of specific assumptions.

Case Study

Harbor Foods and the Dividend Debate. *Illustrative, fictional case.*

Harbor Foods is a regional packaged-foods company that has traded on a national exchange for twenty years. Its shares trade at $36.00. It recently paid an annual dividend of $1.20 per share. After a successful launch of a premium frozen-meals line, management expects earnings and dividends to grow at 12 percent a year for the next five years. Thereafter, as the line matures and competitors enter, growth is expected to settle at 4 percent, in line with long-term nominal growth of the packaged foods industry. Harbor’s equity beta is 0.8; the risk-free rate is 3.5 percent and the market risk premium is 5.0 percent.

Harbor’s chief executive, Ingrid Solheim, faces pressure from two directions. An activist shareholder, owning 4 percent of the stock, argues that the company is hoarding cash. Harbor retains about half its earnings and invests mostly in plant upgrades whose returns, according to the activist, average only 7 percent. The activist demands that the company double its payout ratio and use the freed cash for a repurchase. Meanwhile, the head of corporate development wants to retain even more earnings to fund the acquisition of a regional competitor, arguing that “growth is what the market rewards.”

The analysts’ consensus is mixed. One brokerage sets a target of $44 using a peer P/E multiple of 22 applied to expected EPS of $2.00. Another values the stock at around $30 using a conservative dividend model. Ingrid asks the finance team to prepare a valuation and to examine the activist’s claim.

The finance team estimates the cost of equity with the CAPM: r = 3.5% + 0.8 × 5.0% = 7.5 percent. However, the team also notes that Harbor’s earnings are more cyclical than its beta suggests, because of commodity input costs, and that a risk adjustment of two percentage points is appropriate, giving r = 9.5 percent for valuation purposes. They then run a two-stage dividend model. The dividends are $1.344, $1.505, $1.686, $1.888 and $2.115 in years 1 to 5. The present value of these at 9.5 percent is $6.42. The terminal value at year 5 is 2.115 × 1.04 / (0.095 − 0.04) = $39.99, and its present value is $25.40. The estimated intrinsic value is therefore about $31.83.

A sensitivity analysis shows the fragility of the result. With a cost of equity of 10.5 percent the value is $26.79; with terminal growth of 3.5 percent it is $29.60; with terminal growth of 3.0 percent it is $27.71. The market price of $36.00 is therefore above every one of these cases, though not far above the base case if the cost of equity were lower than 9.5 percent.

On the activist’s claim, the team compares returns. If the plant upgrades truly earn 7 percent while the cost of equity is 9.5 percent, each dollar retained and reinvested is worth less than a dollar to shareholders, and the company’s 12 percent growth forecast cannot be financed by the existing investment pipeline: with 50 percent retention and a 7 percent return, sustainable growth would be only 3.5 percent. The team notes that the 12 percent near-term growth rests on the frozen-meals line, which presumably earns a much higher return, and that the rest of the capital budget may be earning poor returns.

Ingrid must decide whether to accept the activist’s proposal, to proceed with the acquisition, or to present a third path to the board.

Discussion Questions

1. Verify the sensitivity of the intrinsic value to the cost of equity: why does the choice between 7.5 percent (pure CAPM) and 9.5 percent (adjusted) matter so much, and which would you defend to the board? What happens to the Gordon-type terminal value at 7.5 percent?
2. Reconcile the brokerage target of $44 (P/E of 22 on EPS of $2.00) and the $30 figure with the intrinsic estimate of $31.83. What would each analyst have to believe about growth and the required return?

3. Evaluate the activist's argument using the relationship g = b × ROE and the PVGO concept. Under what conditions would doubling the payout raise the share price, and under what conditions would it lower it?
4. Should Harbor pursue the acquisition? Describe the information you would need to determine whether it adds to PVGO, and explain how financing the acquisition by retaining earnings rather than paying dividends interacts with your answer.

Key Takeaways

- The value of a share is the present value of the cash distributions expected by its owner, discounted at the opportunity cost of equity; resale prices are themselves present values of later dividends.
- The Gordon model $P_0 = D_1 / (r - g)$ values a perpetually growing dividend stream, requires $r > g$, uses next year's dividend, and is extremely sensitive to the spread $r - g$.
- Market price can be read in reverse: $r = D_1/P_0 + g$ gives the return implied by the price, and $g = r - D_1/P_0$ the growth implied by it.
- Sustainable growth equals retention times return on equity ($g = b \times ROE$); retention creates value only when the return on new investment exceeds the cost of equity.
- Multi-stage models allow growth to fade from high levels to a sustainable long-run rate; the terminal value commonly represents most of the total, so terminal assumptions deserve the greatest scrutiny.
- Price equals the value of assets in place (EPS_1/r) plus the present value of growth opportunities; a high P/E signals large PVGO, low risk, or both.
- Growth opportunities can be negative: investing at a return below the cost of capital destroys value even while earnings and dividends grow.
- Non-convertible preferred stock is a perpetuity valued as D_p / r_p; its price moves inversely with market yields and is capped when the issue is callable.
- Trading mechanics (spreads, order types, margin and index construction) determine the costs and risks that investors actually bear, and margin amplifies both gains and losses.
- Market prices are generally informative estimates of value, so managers should treat private convictions about undervaluation with caution and focus on investments that earn more than the cost of capital.

Review Questions

1. Explain why the dividend discount model is consistent with valuing a firm that pays no dividend today. What assumptions are needed?
2. Why must the growth rate in the Gordon model be less than the required return? What does it mean in economic terms if an analyst's inputs violate this condition?
3. A student values a stock using D_0 in the numerator of the Gordon formula. State the error, its direction, and its approximate size.
4. Describe two ways in which the market price of a stock can be used to infer something about investors' expectations, using the constant-growth model.
5. Show how the sustainable growth rate depends on retention and ROE, and explain why a growth strategy can lower value even though earnings rise.

6. Why does the terminal value dominate in a two-stage model? What practical steps can an analyst take to avoid an unreasonable terminal value?
7. Define PVGO and explain how it relates to the P/E ratio. Why might a firm have a negative PVGO?
8. List four reasons why comparing P/E ratios across companies can be misleading.
9. Compare preferred stock with debt and common equity in terms of claim priority, cash-flow certainty, tax treatment and valuation.
10. A trader buys stock on margin and the price falls sharply. Explain how a margin call arises and how forced selling can affect market prices.

Problems

1. **Gordon model.** A company is expected to pay a dividend of $1.80 next year. Dividends are expected to grow at 3 percent indefinitely. (a) What is the price if the required return is 8 percent? (b) What is the price if the required return is 9 percent? (c) By what percent does the price change?
2. **Implied return.** A stock just paid a dividend of $3.00, dividends are expected to grow at 5 percent forever, and the stock trades at $60. What return does the market expect?
3. **Two-stage model.** A company just paid $2.00. Dividends will grow at 20 percent for three years, then at 4 percent forever. The required return is 11 percent. Find the value per share and the proportion of that value due to the terminal value.
4. **PVGO.** A company expects EPS of $8.00 next year, retains 50 percent, earns 14 percent on reinvested equity, and has a required return of 10 percent. Compute the growth rate, the price, the value of assets in place, PVGO and the forward P/E.
5. **Preferred stock.** A preferred share pays a dividend of $4.50 per year and trades at $60. (a) What is its yield? (b) A $100 par 6 percent preferred is priced to yield 8 percent. What is the price?
6. **Margin.** An investor buys 300 shares at $40 with 60 percent initial margin. (a) How much is borrowed? (b) If the price rises to $48, what is the percentage return on the investor’s equity (ignore interest)? (c) At what price does a margin call occur if the maintenance margin is 30 percent?
7. **Index.** A price-weighted index has three stocks priced at $80, $40 and $60 with an initial divisor of 3. The third stock undergoes a 3-for-1 split. (a) What is the new divisor? (b) If the next-day prices are $82, $41 and $21, what is the index?
8. **CAPM and DDM.** A stock has a beta of 0.9, the risk-free rate is 3.5 percent and the market risk premium is 6 percent. The next dividend is $1.60 and is expected to grow at 4.5 percent. Compute the cost of equity and the price.

Answers to Selected Problems

1. a. 1.80 / 0.05 = $36.00. (b) 1.80 / 0.06 = $30.00. (c) The price falls by 16.7 percent.
2. $D_1 = 3.00 \times 1.05 = 3.15$; r = 3.15 / 60 + 0.05 = 5.25% + 5.00% = 10.25 percent.
3. Dividends: 2.40, 2.88, 3.456; present values at 11 percent: 2.162, 2.337, 2.527, total 7.03. $D_4 = 3.456 \times 1.04 = 3.594$; $TV_3 = 3.594 / 0.07 = 51.35$; $PV = 51.35 / 1.11^3 = 37.54$. Value = $44.57; terminal share = 37.54 / 44.57 = 84.2 percent.
4. g = 0.5 × 0.14 = 7 percent; $D_1 = 4.00$; $P_0 = 4.00 / (0.10 - 0.07) = \$133.33$; assets in place 8.00 / 0.10 = $80.00; PVGO = $53.33; forward P/E = 16.7.

5. a. 4.50 / 60 = 7.5 percent. (b) Dividend = $6.00; price = 6.00 / 0.08 = $75.00.

6. a. Cost = 12,000; equity 7,200; loan $4,800. (b) Value at $48 = 14,400; equity = 9,600; return = (9,600 − 7,200) / 7,200 = 33.3 percent. (c) (300P − 4,800) / 300P = 0.30 gives 210P = 4,800, so P = $22.86.

7. a. Initial sum 180 / 3 = 60. After the split the third price is $20, so the sum is 140 and the new divisor = 140 / 60 = 2.3333. (b) Sum = 82 + 41 + 21 = 144; index = 144 / 2.3333 = 61.71.

8. r = 3.5 + 0.9 × 6.0 = 8.9 percent; P_0 = 1.60 / (0.089 − 0.045) = 1.60 / 0.044 = $36.36.

Chapter 6: Net Present Value and Investment Criteria

Learning Objectives

After studying this chapter you should be able to:

- Apply the net present value (NPV) rule, explain why it maximizes shareholder wealth, and construct and read an NPV profile.
- Calculate the internal rate of return (IRR) and identify the situations (nonconventional cash flows, mutually exclusive projects, differences in scale and timing) in which the IRR rule misleads.
- Compute the modified internal rate of return (MIRR) and explain how it addresses the reinvestment-rate and multiple-IRR problems.
- Calculate payback and discounted payback periods, and assess when these rules are useful and when they are dangerous.
- Use the profitability index (PI) to rank projects, and explain why it can fail when projects are indivisible.
- Choose among mutually exclusive projects of different scale, timing and life using incremental analysis and equivalent annual annuities, and allocate a limited budget under capital rationing.

Opening Vignette

Illustrative scenario. Cascade Rail Components is a fictional company.

Cascade Rail Components makes brake assemblies and couplers for freight railcars. At its annual capital-planning meeting the finance team presents the controller, Priya Raman, with a table of eight proposals competing for a limited budget. She notices that the project ranking changes depending on the column she reads. The automation line for the coupler plant has the highest internal rate of return, over 30 percent, and pays back in just over two years. A larger expansion of the brake-assembly plant has a lower IRR and a longer payback, but the highest net present value. A third proposal, a retrofit of a leased facility, has two IRRs and a negative cash flow in its final year when the site must be restored.

The chief operating officer, Tomas Weller, favors the quick-payback projects: “Our customers’ orders are volatile and I want my capital back fast.” The treasurer favors IRR because it “can be compared with our borrowing costs.” The chief executive asks Priya a deceptively simple question: “Which projects, in total, make our shareholders richest?” Priya realizes that the question has a precise answer only if the firm uses a single, consistent measure of value, and that each criterion on the table is a different approximation of it.

This chapter builds that answer. We begin with the NPV rule, the benchmark against which every other criterion is judged. We then examine the IRR, MIRR, payback and profitability index, showing precisely where each agrees with NPV and where it departs. Finally we turn to the two hardest practical problems: choosing among mutually exclusive projects, and allocating a budget when the firm cannot fund every positive-NPV opportunity. By the end you should be able to take Priya’s table and produce a defensible capital plan.

6.1 The Net Present Value Rule

6.1.1 Definition and logic

The **net present value** of a project is the present value of its expected future cash flows, discounted at the project's opportunity cost of capital, minus the initial investment:

```
NPV = CF₀ + CF₁/(1+r) + CF₂/(1+r)² + … + CF_T/(1+r)^T = Σ (t = 0 to T) CF_t / (1+r)^t
```

where CF_0 is typically negative (the outlay), CF_t are the net cash flows in year t, and r is the discount rate. The **NPV rule** states: accept every project with NPV > 0, reject projects with NPV < 0, and, when projects are mutually exclusive, choose the one with the highest NPV.

The rule rests on three observations. First, a dollar today is worth more than a dollar tomorrow, because it can be invested at the return available in capital markets. The discount rate reflects that opportunity cost and the project's risk. Second, NPV measures the increase in wealth, in today's dollars, that the project creates for the firm's owners. A project with NPV of $1 million is worth $1 million more than it costs; if the firm takes it, the value of the firm's equity should rise by that amount (assuming the market understands the project). Third, NPVs are **additive**: the NPV of a package of projects is the sum of the individual NPVs. This property lets managers evaluate projects one at a time and sum the results, and it is not shared by IRR or other ratio measures.

The NPV rule is consistent with the separation principle of corporate finance. Whatever the owners' preferences for consumption now versus later, they will all agree that the firm should accept positive-NPV projects, because the surplus can be converted into consumption at any date through borrowing and lending in capital markets. NPV is thus the criterion that all shareholders can agree on.

6.1.2 Choosing the discount rate

The correct discount rate is the expected return on alternative investments with the same risk as the project, not the rate at which the firm happens to borrow, and not the firm's historical return. For a project similar in risk to the firm's existing business, the weighted average cost of capital (WACC) is appropriate; for projects with different risk (a new market, a different technology), a rate that reflects the project's own risk is needed. If two divisions have very different risk, using one company-wide rate will lead the firm to accept too many risky projects and reject safe ones. The determination of discount rates was addressed in earlier chapters on risk and return and is developed in the chapter on the cost of capital.

6.1.3 Cash flows, not accounting profits

The inputs to NPV are **incremental after-tax cash flows**: the cash flows the firm will have if it takes the project minus the cash flows it would have without it. Several principles govern their estimation.

- **Include opportunity costs.** If the project uses a building the firm already owns, the foregone rental or sale value is a cost.
- **Ignore sunk costs.** Money already spent and not recoverable (past R&D, a feasibility study already paid for) is irrelevant to the go/no-go decision.
- **Include changes in working capital.** Growth in receivables and inventory ties up cash; it is an investment recovered at the end of the project.
- **Include side effects.** If a new product cannibalizes sales of an existing one, the lost margin is a cost; if it generates sales of complementary products, the gain is a benefit.

- **Treat financing separately.** Interest payments and debt principal are not included in project cash flows, because their cost is in the discount rate. Counting both would double count.
- **Count taxes properly.** Depreciation is not a cash flow, but it reduces taxes, and the tax shield (depreciation × tax rate) is a cash inflow.
- **Be consistent about inflation.** Nominal cash flows are discounted at nominal rates; real flows at real rates.

6.1.4 A worked NPV calculation

Example 6.1 – Machine replacement with NPV. Cascade Rail is considering a new forming machine that costs $250,000 and is expected to generate incremental after-tax cash flows of $70,000, $80,000, $90,000, $85,000 and $75,000 in years 1 through 5. The cost of capital is 12 percent.

Year	Cash flow ()|*Discountfactorat*12)		
0	−250,000	1.0000	−250,000
1	70,000	0.8929	62,500
2	80,000	0.7972	63,776
3	90,000	0.7118	64,060
4	85,000	0.6355	54,019
5	75,000	0.5674	42,557
Sum of PV of inflows			286,912
NPV			36,912

Table 6.1 – NPV of the forming machine (discount factors rounded to four decimals; present values computed without rounding).

Step 1: Discount each inflow at 12 percent: 70,000 / 1.12 = 62,500; 80,000 / 1.12^2 = 63,776; 90,000 / 1.12^3 = 64,060; 85,000 / 1.12^4 = 54,019; 75,000 / 1.12^5 = 42,557.

Step 2: Sum the present values: 62,500 + 63,776 + 64,060 + 54,019 + 42,557 = $286,912 (to the nearest dollar).

Step 3: Subtract the investment: NPV = 286,912 − 250,000 = **$36,912**.

Step 4: Decision. NPV is positive, so the machine should be accepted. Interpretation: after covering the 12 percent return that investors could earn elsewhere, the machine adds about $37,000 of value.

6.1.5 The NPV profile

The **NPV profile** plots NPV against the discount rate. At a rate of zero, NPV is simply the sum of the undiscounted cash flows; as r rises, NPV falls (for conventional projects, with an outlay followed by inflows). The point at which the profile crosses the horizontal axis is the IRR. Profiles are valuable for three reasons: they show how sensitive the decision is to the discount rate, they locate the IRR, and they reveal conflicts between projects.

For the machine of Example 6.1 the NPV is 36,912 at 12 percent, 51,996 at 10 percent, 16,425 at 15 percent, −1,821 at 18 percent and −12,895 at 20 percent. The profile crosses zero near 17.7 percent. The decision is therefore robust to errors of several points in the cost of capital, which is reassuring; a project whose IRR is barely above the hurdle rate deserves more caution.

NPV also has a shortcoming: it is expressed in dollars and so does not directly show the efficiency of capital use or the margin of safety. This is the legitimate attraction of rate-based measures such as IRR, to which we now turn.

6.2 The Internal Rate of Return

6.2.1 Definition

The **internal rate of return** is the discount rate at which NPV equals zero:

```
0 = CF₀ + CF₁/(1+IRR) + CF₂/(1+IRR)² + … + CF_T/(1+IRR)^T
```

There is generally no closed-form solution; the IRR is found by trial and error, by a financial calculator or by a spreadsheet function. The **IRR rule** says to accept a project if IRR exceeds the opportunity cost of capital and to reject it otherwise. For a conventional project with a single outlay followed by inflows, the IRR rule gives the same accept/reject decision as the NPV rule, because the NPV profile slopes down and crosses zero once: the NPV is positive exactly when r is below the IRR.

IRR is intuitive: it is the project's rate of return on invested capital, and it can be compared directly with the cost of capital, with the cost of debt, or with the returns on other opportunities. This explains its popularity in practice. Many managers find "a 20 percent return" easier to grasp than "an NPV of $37,000."

Example 6.2 – NPV profile and IRR. A project requires an investment of $100,000 and returns $30,000, $35,000, $40,000 and $45,000 in years 1 through 4. The cost of capital is 10 percent.

Step 1: Present values at 10 percent: 30,000 / 1.10 = 27,273; 35,000 / 1.21 = 28,926; 40,000 / 1.331 = 30,053; 45,000 / 1.4641 = 30,736. Total = 116,987.

Step 2: NPV = 116,987 − 100,000 = $16,987.

Step 3: NPV at other rates.

Discount rate	NPV ($)
0%	50,000
5%	31,893
10%	16,987
15%	4,582
20%	−5,845
25%	−14,688

Table 6.2 – NPV profile of Example 6.2.

Step 4: The NPV changes sign between 15 and 20 percent. By interpolation, IRR ≈ 15 + 5 × 4,582 / (4,582 + 5,845) = 15 + 2.2 = 17.2 percent; a more exact solution gives 17.09 percent. (Linear interpolation slightly overstates because the profile is convex.)

Step 5: The IRR of 17.09 percent exceeds the 10 percent hurdle, so the project is accepted, in agreement with the NPV.

6.2.2 Pitfall 1: Nonconventional cash flows and multiple IRRs

A polynomial of degree T can have as many as T roots. By Descartes' rule of signs the number of positive IRRs cannot exceed the number of changes in the sign of the cash flows. A conventional project (one sign change) has at most one IRR. A project with negative flows later in life (restoration costs for a mine, decommissioning of a plant, a major refit in mid-life, or a deferral of tax payments) can have several.

Example 6.3 – Two IRRs. A project has cash flows of −$100 million today, +$230 million in year 1 and −$132 million in year 2. The second outflow represents site restoration.

Step 1: NPV at several rates (in $ million): at 0 percent, −100 + 230 − 132 = −2.00. At 5 percent: −100 + 230/1.05 − 132/1.1025 = −100 + 219.05 − 119.73 = −0.68. At 10 percent: −100 + 209.09 − 109.09 = 0.00. At 15 percent: −100 + 200.00 − 99.81 = +0.19. At 20 percent: −100 + 191.67 − 91.67 = 0.00. At 25 percent: −100 + 184.00 − 84.48 = −0.48.

Step 2: The NPV is zero at both 10 percent and 20 percent. The project has two IRRs. Between 10 and 20 percent the NPV is positive; outside that range it is negative.

Step 3: Decision. At a cost of capital of 8 percent, NPV is negative (about −0.21), so reject; at 15 percent, NPV is +0.19 million and the project is marginally acceptable. The IRR rule cannot deliver this verdict: with IRRs of 10 and 20 percent, "accept if IRR > r" gives contradictory answers for any cost of capital between them (a cost of 15 percent is above one IRR and below the other). Only NPV is reliable. At 25 percent the NPV is again negative, because the large final outflow is discounted less heavily than the earlier inflow that funds it.

Besides multiple IRRs, some nonconventional projects have no IRR at all: for example, cash flows of −4, +9, −6 have no discount rate that sets NPV to zero, so the IRR rule offers no guidance, yet NPV is well defined (and negative at every rate).

6.2.3 Pitfall 2: Lending versus borrowing

For a project in which the firm receives cash first and pays later (a customer deposit, a lease financing, a sale of future production), the sign pattern is reversed: positive then negative. The NPV profile slopes upward, and a project is attractive only when its IRR is *below* the cost of capital because the firm is effectively borrowing. A project with flows of +100 now and −120 in a year is a loan at 20 percent. If the firm's opportunity cost of capital is 10 percent, the mechanical rule (IRR 20 percent > 10 percent, accept) says yes, but NPV = 100 − 120 / 1.10 = −9.09, so the firm should refuse: it would be paying 20 percent for money it values at 10 percent. A manager must look at the sign of the first cash flow.

6.2.4 Pitfall 3: The reinvestment-rate assumption

Mathematically, NPV discounts intermediate cash flows at the cost of capital, which is equivalent to assuming they are reinvested at that rate. IRR implicitly assumes intermediate cash flows are reinvested at the IRR itself. A project with a 40 percent IRR that returns cash early is assumed to find fresh opportunities earning 40 percent for that cash, which is rarely realistic in a competitive economy. The reinvestment assumption inflates IRR for high-return projects and is the reason the modified IRR in Section 6.3 was developed. The NPV assumption (reinvestment at the cost of capital, the return available in the market) is the more defensible.

6.2.5 Pitfall 4: Mutually exclusive projects

When only one of several projects can be chosen, the IRR rule can rank them differently from NPV, because the IRR ignores scale (how much is invested) and timing (when the cash arrives). These issues are treated in Section 6.6. The key lesson is that, among mutually exclusive projects, one should not choose the highest IRR; one should choose the highest NPV, or equivalently examine the IRR of the incremental cash flows.

6.2.6 Pitfall 5: A single discount rate for all periods

The IRR rule compares one number with the cost of capital, which presumes the cost of capital is the same in every year. If short-term and long-term rates differ, or if the project's risk changes over time, the comparison becomes ambiguous. NPV handles term structure and changing risk simply, by using a different discount rate for each year's cash flow.

6.3 The Modified Internal Rate of Return

6.3.1 Construction

The **modified IRR (MIRR)** corrects the reinvestment assumption and the multiple-roots problem by an explicit three-step procedure:

1. Compound all positive cash flows (inflows) forward to the end of the project's life at a stated reinvestment rate (usually the cost of capital). The result is the *terminal value* TV.
2. Discount all negative cash flows (outflows) back to time 0 at a stated financing rate (also usually the cost of capital). The result is the *present value of costs* PVC.
3. The MIRR is the single discount rate that equates them:

```
MIRR = (TV / PVC)^(1/T) − 1
```

Because the procedure gives exactly one answer for any cash flow pattern with at least one positive and one negative flow, the multiple-IRR problem disappears. The MIRR also reflects a realistic reinvestment assumption: cash does not earn the project's own IRR, but the rate available in the market.

Example 6.4 – MIRR. Use the project of Example 6.2: −$100,000, then +$30,000, +$35,000, +$40,000 and +$45,000. The reinvestment and financing rate is 10 percent.

Step 1: Compound each inflow to year 4: 30,000 × 1.10^3 = 30,000 × 1.331 = 39,930; 35,000 × 1.10^2 = 42,350; 40,000 × 1.10 = 44,000; 45,000 × 1 = 45,000. TV = 39,930 + 42,350 + 44,000 + 45,000 = $171,280.

Step 2: PVC = $100,000 (the only outflow, already at time 0).

Step 3: MIRR = (171,280 / 100,000)^(1/4) − 1 = (1.7128)^0.25 − 1 = 1.1440 − 1 = **14.40 percent**.

Step 4: Compare with the 17.09 percent IRR. The MIRR is lower because it assumes reinvestment at 10 percent rather than 17.09 percent. Because MIRR exceeds the 10 percent hurdle, the project is still acceptable. Note that MIRR and NPV are directly linked: MIRR exceeds r exactly when NPV is positive (using the same r for both reinvestment and financing).

6.3.2 Limitations

MIRR is not a cure-all. It depends on an assumed reinvestment rate, which means different analysts can obtain different MIRRs from the same project. It can still rank mutually exclusive projects incorrectly where scale differs, because it is a rate and not a dollar amount. And it is less familiar than IRR. Its best use is as a supplement to NPV when managers want a rate-based figure that is free of the multiple-root and overstatement problems. Some finance scholars argue that it is not really a distinct criterion at all, since when reinvestment and financing rates equal the cost of capital it is just a transformation of NPV.

6.4 Payback and Discounted Payback

6.4.1 The payback period

The **payback period** is the time required for the cumulative undiscounted cash inflows to recover the initial investment. The rule: accept if the payback period is less than a cutoff chosen by management. Payback is simple, requires no discount rate, and emphasizes liquidity and speed of recovery, which matter to firms with limited access to capital or in rapidly

changing industries where long-lived forecasts are unreliable. Surveys of corporate practice regularly find that firms use payback, often alongside NPV and IRR.

Its flaws are fundamental:

- It **ignores the time value of money** within the payback period (a dollar in year 1 counts the same as a dollar in year 3).
- It **ignores all cash flows after the cutoff**, so it penalizes long-lived projects with large late payoffs and favors short-lived ones.
- The cutoff is **arbitrary**, with no connection to value.
- It is **biased against** long-term investments such as R&D and infrastructure.
- It does not add: the payback of a package is not the sum of the paybacks.

6.4.2 The discounted payback period

The **discounted payback** applies the cost of capital to each cash flow before cumulating. It corrects the first flaw: a project that pays back on a discounted basis within its life has a positive NPV by the end of life. However, it still ignores cash flows after the cutoff and is arbitrary, so it can reject positive-NPV projects with long lives. It is best seen as a measure of risk (how long capital is exposed) and not as a measure of value.

Example 6.5 – Payback and discounted payback. Using the project of Example 6.2 with a 10 percent discount rate:

Year	Cash flow ()|*Cumulativecashflow*()	PV at 10% ()|*CumulativePV*()		
0	−100,000	−100,000	−100,000	−100,000
1	30,000	−70,000	27,273	−72,727
2	35,000	−35,000	28,926	−43,802
3	40,000	5,000	30,053	−13,749
4	45,000	50,000	30,736	16,987

Table 6.3 – Payback and discounted payback.

Step 1: Payback. Cumulative cash flow turns positive during year 3. At the start of year 3 there remains 35,000 to recover, and year 3 provides 40,000. Payback = 2 + 35,000 / 40,000 = **2.875 years**.

Step 2: Discounted payback. Cumulative PV turns positive during year 4. At the start of year 4 the unrecovered amount is 13,749; year 4's PV is 30,736. Discounted payback = 3 + 13,749 / 30,736 = **3.45 years**.

Step 3: Decision with a three-year cutoff. The ordinary payback (2.875 years) passes; the discounted payback (3.45 years) fails, although the NPV is a healthy +$16,987. The discounted payback rule would reject a profitable project solely because of the arbitrary cutoff.

Step 4: Compare the machine of Example 6.1: cumulative cash flows are −180,000, −100,000, −10,000, +75,000, +150,000, so payback = 3 + 10,000 / 85,000 = 3.12 years; discounted payback = 4 + 5,645 / 42,557 = 4.13 years.

6.4.3 When payback is informative

Payback is a legitimate supplementary indicator when the future is hard to forecast beyond a few years (technology products with short life cycles), when capital is scarce and liquidity matters, or when a company has weak access to financing. It is also a rough screening device for small projects where the cost of full analysis exceeds the stakes. The danger is when it replaces NPV in decisions with long-lived cash flows. A sound practice is to calculate payback as a measure of risk exposure and to use NPV as the decision criterion.

6.5 The Profitability Index

6.5.1 Definition

The **profitability index (PI)**, also called the benefit-cost ratio, is the present value of future cash flows divided by the initial investment:

```
PI = PV of future cash flows / Initial investment = 1 + NPV / Initial investment
```

A project has NPV > 0 exactly when PI > 1, so the accept/reject decision agrees with NPV for a standalone project. PI measures value created per dollar invested, which is useful when capital is limited, as discussed in Section 6.7. A related measure, the *NPV index*, is NPV divided by the investment (PI − 1).

Example 6.6 – Computing PI for four projects. A firm with a 10 percent cost of capital evaluates four independent projects (in $ million). Each has an initial outlay at time 0 followed by the cash flows shown.

Project	Outlay	Inflows (years 1–n)	PV of inflows	NPV	PI	IRR
P1	50	30, 30, 30	74.61	24.61	1.492	36.3%
P2	100	50, 50, 50, 50	158.49	58.49	1.585	34.9%
P3	40	25, 25, 25	62.17	22.17	1.554	39.5%
P4	60	30, 35, 20	71.22	11.22	1.187	21.0%

Table 6.4 – Profitability index of four independent projects.

Step 1: For P1, the three-year annuity factor at 10 percent is $(1 - 1.10^{-3}) / 0.10 = 2.4869$. PV of inflows = 30 × 2.4869 = 74.61. NPV = 74.61 − 50 = 24.61. PI = 74.61 / 50 = 1.492.

Step 2: For P2, the four-year annuity factor is 3.1699. PV = 50 × 3.1699 = 158.49. NPV = 58.49; PI = 1.585.

Step 3: For P3, PV = 25 × 2.4869 = 62.17; NPV = 22.17; PI = 62.17 / 40 = 1.554.

Step 4: For P4, PV = 30/1.10 + 35/1.21 + 20/1.331 = 27.27 + 28.93 + 15.03 = 71.22 (rounded); NPV = 11.22; PI = 1.187.

Step 5: Observations. All four have positive NPV and PI above 1, so all would be accepted without a budget limit. The PI ranking (P2, P3, P1, P4) differs from the NPV ranking (P2, P1, P3, P4) and from the IRR ranking (P3, P1, P2, P4). The three measures answer different questions: NPV is the total value created, PI the value per dollar of capital, and IRR the rate of return. Which one matters depends on the constraint the firm faces.

6.5.2 Limitations of PI

Like IRR, the PI is a ratio, so it ignores scale and can mislead when ranking mutually exclusive projects: a small project with a high PI can have a lower NPV than a large project with a modest PI. It also depends on how the outlays are defined. If some cash outflows occur after time 0, an analyst can either net them against inflows or add them to the initial investment, giving different PIs for the same project. The PI is most useful as a tool for rationing a single constrained resource (Section 6.7) and not as a general-purpose selection criterion.

6.6 Mutually Exclusive Projects

Projects are **independent** if the acceptance of one does not affect the others, and **mutually exclusive** if accepting one rules out the others (two designs for the same plant, two sites for a warehouse, buy-versus-lease). With independent projects and no budget limit, NPV, IRR and PI all give the same accept/reject decisions for conventional cash flows. With mutually exclusive projects the criteria can conflict, and the NPV rule must prevail. Conflicts arise from three sources: differences in scale, differences in timing, and differences in project life.

6.6.1 Differences in scale

A small project can have a high rate of return on a small base, and a large project a modest return on a large base. The firm cares about the dollar amount of value created, not the percentage.

Example 6.7 – Scale conflict. Cascade Rail can build a small packaging line (X) or a large automated line (Y) in the same space. The cost of capital is 10 percent.

- X: invest $1,000 (thousands), receive $600 per year for 3 years.
- Y: invest $5,000, receive $2,200 per year for 4 years.

Step 1: NPV of X = 600 × 2.4869 − 1,000 = 1,492.11 − 1,000 = $492.11. IRR of X solves 600 × annuity factor(3 years) = 1,000, annuity factor = 1.6667, giving IRR = 36.31 percent.

Step 2: NPV of Y = 2,200 × 3.1699 − 5,000 = 6,973.70 − 5,000 = $1,973.70. IRR of Y solves 2,200 × annuity factor(4 years) = 5,000, giving 27.18 percent.

Step 3: Conflict. IRR prefers X (36.3 percent versus 27.2 percent); NPV prefers Y ($1,973.70 versus $492.11).

Step 4: Resolve with incremental analysis. The extra cash flows from choosing Y rather than X are: year 0, −5,000 − (−1,000) = −4,000; years 1 to 3, 2,200 − 600 = +1,600; year 4, 2,200 − 0 = +2,200. The NPV of the increment at 10 percent is 1,600 × 2.4869 + 2,200 / 1.4641 − 4,000 = 3,978.96 + 1,502.63 − 4,000 = $1,481.59, which equals 1,973.70 − 492.11 as it must. The IRR of the increment is 25.33 percent, which exceeds 10 percent.

Step 5: Interpretation. The incremental investment of $4,000 earns 25.3 percent, well above the cost of capital, so the additional spending on Y is worthwhile. The rule is: **take the larger project if the incremental IRR exceeds the cost of capital**, which is equivalent to taking the one with higher NPV. The key question, unavoidably, is whether the firm has something better to do with the extra $4,000 than earn 25 percent on it; if it did, that opportunity should enter the comparison. If capital is rationed, the scale question changes, and we address it in Section 6.7.

6.6.2 Differences in timing

Two projects of the same size can differ in when the cash arrives. One returns cash early; the other later, but in greater total amount. At a high discount rate, the early cash is worth relatively more; at a low discount rate, the later, larger cash flows dominate. The NPV profiles therefore cross at a **crossover rate** (Fisher's rate of intersection). For discount rates below the crossover, the "later" project has higher NPV; above it, the "earlier" project has higher NPV. IRR, which ignores the actual cost of capital in the ranking, always prefers the project with the quicker payoff.

Example 6.8 – Timing conflict and crossover rate. Projects A and B each cost $1,000 (time 0) and have the following cash flows.

Year	A ()|*B*()	A − B ($)	
0	−1,000	−1,000	0
1	1,000	100	900
2	300	300	0
3	200	500	−300
4	100	900	−800

Table 6.5 – Cash flows of mutually exclusive projects A and B.

Step 1: IRRs. A: 36.60 percent. B: 20.59 percent. IRR ranks A first.

Step 2: NPVs at different rates:

Discount rate	NPV of A ()|*NPVofB*()	Better by NPV	
0%	600.00	800.00	B
5%	479.53	539.70	B
10%	375.59	329.21	A
15%	285.09	157.14	A
20%	205.63	15.05	A

Table 6.6 – NPV of A and B at various discount rates.

Step 3: Crossover. The IRR of the difference (A − B) is the rate at which the two NPVs are equal: solving $900/(1+r) - 300/(1+r)^3 - 800/(1+r)^4 = 0$ gives r = 7.66 percent. Check at 7.66 percent: the discount factors for years 1, 3 and 4 are 0.9289, 0.8014 and 0.7444, so the present values are 900 × 0.9289 = 836.0, −300 × 0.8014 = −240.4 and −800 × 0.7444 = −595.5, which sum to zero within rounding.

Step 4: Decision. If the cost of capital is 10 percent, which is above the crossover rate of 7.66 percent, NPV and IRR agree: choose A. If the cost of capital were 5 percent, NPV would choose B ($539.70 versus $479.53) while IRR would still choose A. Managers who rely on IRR will therefore choose the wrong project whenever the cost of capital is below the crossover rate.

Step 5: The conflict also tells us something about sensitivity. B is a longer-duration project and is more sensitive to the discount rate: its NPV falls from 800 to 15 as the rate rises from 0 to 20 percent, whereas A's falls from 600 to 206. The economic reasoning is the one that underlies the interest-rate sensitivity of long-term bonds.

6.6.3 Differences in project life

When mutually exclusive projects have different lives, comparing NPVs directly favors the longer project simply because it generates cash for more years, even if the projects will be repeated. If the machines will be replaced at the end of their lives, the fair comparison is between *chains* of replacements, or between the *equivalent annual annuities* (EAA) of the two options. The EAA is the constant annual amount that has the same present value as the project's NPV:

```
EAA = NPV / A(r, T)   where A(r, T) = [1 − (1+r)^(−T)] / r
```

The EAA expresses the NPV as an annual flow, so projects with different lives become comparable. It is valid when the projects can be replicated indefinitely at the same cash flows, which is a restrictive assumption but reasonable for equipment replacement. If technology is changing or demand is uncertain, a replicated chain is not realistic and one should instead model the specific decision at the end of each project's life.

Example 6.9 – Unequal lives. Cascade Rail can buy Machine Q (3-year life) or Machine R (5-year life). Each costs $10,000. Q generates $4,500 per year; R generates $3,000 per year. The cost of capital is 10 percent. Assume each machine can be replaced at the end of its life by an identical one.

Step 1: NPV of Q = 4,500 × 2.4869 − 10,000 = 11,190.83 − 10,000 = $1,190.83. IRR = 16.65 percent.

Step 2: NPV of R = 3,000 × 3.7908 − 10,000 = 11,372.36 − 10,000 = $1,372.36. IRR = 15.24 percent.

Step 3: Naively, R has higher NPV ($1,372 versus $1,191) and Q has higher IRR. They conflict.

Step 4: EAA. For Q: 1,190.83 / 2.4869 = $478.85 per year. For R: 1,372.36 / 3.7908 = $362.03 per year.

Step 5: Verify with a replacement chain over 15 years (the lowest common multiple of 3 and 5). Q is bought five times, with NPV 1,190.83 recurring at times 0, 3, 6, 9 and 12: total NPV = 1,190.83 × (1 + 1.10^{-3} + 1.10^{-6} + 1.10^{-9} + 1.10^{-12}) = 1,190.83 × 3.0586 = $3,642. R is bought three times (times 0, 5, 10): total NPV = 1,372.36 × (1 + 1.10^{-5} + 1.10^{-10}) = 1,372.36 × 2.0064 = $2,754.

Step 6: Decision. Machine Q is better: its EAA ($478.85) beats R's ($362.03), and over the common 15-year horizon its chain NPV ($3,642) exceeds R's ($2,754). The original NPV comparison was misleading because R's higher NPV came from using capital for two extra years, not from higher productivity.

An alternative viewpoint is the cost-minimization problem. When machines produce identical output (so inflows are the same and only costs differ), the choice is the one with the lowest *equivalent annual cost*. The same calculation applies to the PV of costs divided by the annuity factor.

6.6.4 A decision procedure

For mutually exclusive projects:

1. Check that the cash flows and discount rates are correct, and that the projects are really comparable (same time horizon or replication assumption).
2. Compute NPV for each at the appropriate discount rate; rank by NPV (or EAA when lives differ).
3. If management prefers a rate-of-return view, compute the incremental IRR of the larger or later project over the smaller or earlier one, and compare it with the cost of capital.
4. Never rank mutually exclusive projects by IRR, payback or PI alone.

6.7 Capital Rationing

6.7.1 Soft and hard rationing

The NPV rule assumes the firm can fund every positive-NPV project at the cost of capital. In reality, a firm may face a budget limit. **Soft rationing** is a limit imposed internally, for example by headquarters allocating a fixed budget to divisions, in order to control expansion, conserve management attention, or discipline divisional managers who might overstate project benefits. **Hard rationing** arises from external constraints, such as when a firm cannot raise additional capital at a reasonable cost because of poor credit, covenant restrictions or a depressed market. Corporate finance theory suggests that hard rationing should be rare for sound firms in well-functioning markets; if a firm cannot fund a positive-NPV project, something is wrong with its disclosure, or the project's NPV is overstated. But soft rationing is common in practice, and managers must decide how to allocate within it.

6.7.2 The profitability-index approach

With a single-period budget constraint, the goal is to select the set of projects that maximizes total NPV subject to the budget. If projects are *divisible* (each can be undertaken in part), the optimal strategy is to rank by PI and fund in order until the budget is exhausted, since PI measures NPV per dollar of the scarce resource. If projects are *indivisible*, as is usual, the PI ranking can leave part of the budget unused and not produce the best combination. In that case the problem is an integer program, which can be solved by enumeration for a small set of projects, or with linear/integer programming for a large set.

Example 6.10 – Rationing with indivisible projects. Using the projects of Example 6.6 (in $ million) with a total budget of $150 million.

Project	Outlay	NPV	PI
P1	50	24.61	1.492
P2	100	58.49	1.585
P3	40	22.17	1.554
P4	60	11.22	1.187

Table 6.7 – Projects competing for a $150 million budget.

Step 1: Total outlay of all four is 250, so rationing is binding.

Step 2: PI ranking: P2 (1.585), P3 (1.554), P1 (1.492), P4 (1.187). Fund P2 (100) and P3 (40): total 140, leaving 10 unspent. P1 (50) and P4 (60) do not fit. Total NPV = 58.49 + 22.17 = $80.66 million.

Step 3: Enumerate feasible combinations within 150: P1 + P2 = 150 with NPV 24.61 + 58.49 = $83.10 million; P2 + P3 = 140 with $80.66 million; P1 + P3 + P4 = 150 with 24.61 + 22.17 + 11.22 = $58.00 million; P2 + P4 = 160, infeasible; P1 + P3 = 90 with $46.78 million.

Step 4: The best feasible set is P1 + P2, with NPV $83.10 million, which is higher than the PI-ranking selection ($80.66 million) by $2.44 million. The PI rule failed because P3, which ranked second by PI, was indivisible and left an unusable $10 million that P1's size could have absorbed.

Step 5: Opportunity cost of the budget. Suppose the budget were raised from 150 to 160. The combination P2 + P4 (160) has NPV $69.71 million, still below P1 + P2 at $83.10 million, and no feasible combination at 160 beats P1 + P2. The extra $10 million is therefore worth nothing in this small example, whereas an extra $40 million (budget 190) would allow P1 + P2 + P3 for $105.27 million. Shadow prices in integer problems are lumpy, which is why the answer is found by examining combinations rather than by applying a single rule of thumb.

6.7.3 Multi-period rationing and interactions

If budgets are limited in several periods, or if projects have interdependencies (one project requires another, or two are substitutes), the problem requires linear programming with constraints for each period and for each interdependency. The results include **shadow prices** on each budget constraint, which tell managers what an additional dollar of capital is worth and thus whether it would be worthwhile to seek additional funding. Large organizations use such models to allocate capital among divisions.

6.7.4 Managerial perspective on rationing

Three practical points deserve emphasis. First, rationing is an *internal* decision and should not change the valuation of projects; it only affects which of the positive-NPV projects are selected. Second, if the budget is persistently binding and the excluded projects have high NPVs, the firm should ask whether it should raise more capital: the cost of leaving value on the table can exceed the cost of financing. Third, internal budgets can lead managers to inflate forecasts to win funding. A disciplined process includes post-audits comparing realized with forecast cash flows, and holds sponsors accountable for their projections.

6.8 Using the Criteria in Practice

6.8.1 A comparison of the criteria

Criterion	Measures	Uses time value	Uses all cash flows	Handles scale	Main weakness
NPV	Dollar value created	Yes	Yes	Yes	Requires an estimate of the discount rate; not a ratio
IRR	Rate of return	Yes	Yes	No	Multiple roots, reinvestment assumption, conflicts for exclusive projects
MIRR	Rate of return with stated reinvestment	Yes	Yes	No	Depends on the assumed reinvestment rate
Payback	Speed of recovery	No	No	No	Ignores value after cutoff and time value
Discounted payback	Speed of recovery in PV terms	Yes	No	No	Ignores value after cutoff
PI	Value per dollar invested	Yes	Yes	No (ratio)	Misleads for exclusive or indivisible projects

Table 6.8 – Comparison of investment criteria.

6.8.2 What do firms actually do?

Surveys of corporate practice, taken in many countries over several decades, have found that most large firms use NPV and IRR as primary tools, with payback still widely used as a secondary measure, particularly by smaller firms and in jurisdictions with less developed capital markets. The popularity of IRR despite its flaws is usually attributed to its intuitive form. The persistence of payback is attributed to its simplicity, to concerns about liquidity and risk, and to managers' skepticism about distant forecasts. A sensible approach combines the criteria: use NPV as the decision rule, use IRR and PI to communicate return and capital efficiency, and use payback as an indicator of risk and liquidity exposure.

6.8.3 Sensitivity, scenarios and real options

The NPV is only as good as the cash-flow forecast. Before accepting a project, managers should run **sensitivity analysis** (varying one input at a time, such as volume, price or cost), **scenario analysis** (varying a coherent set of inputs, such as a boom, base and recession case), and **break-even analysis** (the level of an input at which NPV equals zero). For the machine of Example 6.1, which has NPV of $36,912 on an investment of $250,000, a 10 percent shortfall in every cash flow lowers PV of inflows by 28,691 to 258,221 and still leaves NPV positive at $8,221; a 13 percent shortfall would eliminate the NPV entirely (since 36,912 / 286,912 = 12.9 percent). The project therefore has a fairly thin margin of safety, a fact that the NPV figure alone conceals.

The NPV rule also ignores the managerial flexibility to expand, delay or abandon a project in response to events. Where flexibility is valuable, the value of these **real options** should be added to the static NPV; a project with a slightly negative NPV may be worth undertaking because it creates an option on future growth, while a positive-NPV project may be worth delaying if waiting resolves uncertainty. The valuation of options is treated in a later chapter.

6.8.4 Behavioral and organizational issues

Several biases degrade capital budgeting in practice: optimism in forecasts (sponsors overstate benefits and understate costs), escalation of commitment (continuing to fund failing projects because of sunk costs), and the "empire building" tendency of managers who prefer large projects. Controls include independent review of forecasts, stage-gated funding (releasing capital in tranches tied to milestones), reference-class forecasting (comparing a project with the outcomes of similar past projects), and compensation tied to value creation rather than to size or reported earnings. The criteria in this chapter are tools; the quality of the decision depends on the quality of the assumptions and the integrity of the process that produces them.

Case Study

Cascade Rail Components: The Capital Plan. *Illustrative, fictional case.*

Cascade Rail Components, whose controller Priya Raman opened this chapter, must finalize next year's capital plan. The company's weighted average cost of capital, for projects similar in risk to the core business, is 11 percent. The board has set a hard internal limit of $10 million for new projects in the coming year, partly because of a covenant on the company's term loan and partly because the board wants to see operating results before committing more.

Five proposals remain after screening (all figures in $ million; outlay at time 0, followed by annual after-tax cash flows).

- **Alpha**, a major expansion of the brake-assembly plant: outlay 8.0; inflows 2.6, 2.9, 3.1, 3.1 and 2.8 in years 1 to 5.
- **Beta**, a smaller upgrade of the same plant: outlay 3.0; inflows of 1.4 in each of years 1 to 4. Alpha and Beta are mutually exclusive because they use the same site.

- **Gamma**, a new coupler-testing line: outlay 4.0; inflows of 1.6 in each of years 1 to 4.
- **Delta**, an energy-efficiency retrofit: outlay 2.0; inflows of 1.0 in each of years 1 to 3.
- **Epsilon**, a warehouse-automation project: outlay 5.0; inflows of 2.0 in each of years 1 to 4.

Priya’s team produces the following summary at 11 percent.

Project	Outlay	NPV	IRR	PI	Payback (years)
Alpha	8.0	2.67	23.3%	1.333	2.81
Beta	3.0	1.34	30.7%	1.448	2.14
Gamma	4.0	0.96	21.9%	1.241	2.50
Delta	2.0	0.44	23.4%	1.222	2.00
Epsilon	5.0	1.20	21.9%	1.241	2.50

Tomas Weller, the chief operating officer, wants to apply a strict two-and-a-half-year payback cutoff and fund Beta first, since it has the fastest payback, the highest IRR and the highest PI among the mutually exclusive pair. The treasurer observes that the incremental investment in Alpha over Beta is $5.0 million, with incremental cash flows of 1.2, 1.5, 1.7, 1.7 and 2.8 in years 1 to 5, an incremental IRR of 19.8 percent and an incremental NPV of $1.32 million. The chief executive wants to know whether the entire budget should go into Alpha plus Delta, which together use exactly $10 million, or whether the company is better off spreading the money across smaller projects.

Priya enumerates the feasible portfolios. Alpha and Delta together have NPV of $3.11 million. Beta, Delta and Epsilon (outlay $10.0 million) have NPV of $2.99 million. Beta, Gamma and Delta (outlay $9.0 million) have NPV of $2.75 million. Alpha alone has $2.67 million. A ranking by profitability index would put Beta first (1.448); with Alpha excluded, Gamma and Epsilon tie (1.241) and Delta follows (1.222). Funding in that order gives Beta, Gamma and Delta, because Epsilon no longer fits, and that portfolio is worse than the NPV-maximizing one.

She also notes two cautions. The cash-flow forecasts of Alpha rely on a continuing high volume of freight-car orders, while the other projects are cost-saving and less dependent on volume. A 15 percent shortfall in Alpha’s inflows would reduce its PV of inflows from 10.67 to 9.07, making its NPV about +1.07 million. Second, Beta has a four-year life while Alpha has five, so the comparison should consider what happens in year 5 if Beta is chosen.

Discussion Questions

1. Explain why the rankings by NPV, IRR, PI and payback differ for Alpha and Beta. Which criterion should decide between them, and what role does the incremental IRR of 19.8 percent play?
2. Evaluate Tomas Weller’s proposed two-and-a-half-year payback cutoff. What does it accomplish, what does it cost, and what better way is there to address his concern about volatile orders?
3. Under the $10 million limit, which portfolio would you recommend? Show why the PI ranking does not give the best answer here, and explain what the budget limit implies about the opportunity cost of capital.
4. Discuss how you would test the robustness of the recommendation, including the risk that Alpha’s volume assumptions are optimistic, and whether the firm should seek additional funding for the projects left out.

Key Takeaways

- NPV measures the wealth a project creates in today's dollars; it uses all cash flows, respects the time value of money, is additive across projects, and is the benchmark against which other criteria are judged.
- Project cash flows must be incremental, after-tax and free of financing charges; sunk costs are ignored and opportunity costs, working capital and side effects are included.
- The IRR is the discount rate that sets NPV to zero; for conventional, independent projects it gives the same accept/reject decision as NPV.
- IRR fails in several situations: multiple or nonexistent IRRs with sign changes in the cash flows, projects that resemble borrowing, an unrealistic reinvestment assumption, and conflicts among mutually exclusive projects.
- MIRR restores a single answer by compounding inflows at a stated reinvestment rate and discounting outflows at a financing rate, but it remains a rate and so is not a cure for scale conflicts.
- Payback and discounted payback measure speed of recovery and exposure; they ignore cash flows after the cutoff and can reject valuable long-lived projects, so they are better used as risk indicators than as decision rules.
- The profitability index measures value per dollar invested and is useful for rationing capital, but it can mislead for mutually exclusive projects and for indivisible projects.
- Among mutually exclusive projects, choose the highest NPV (or the highest equivalent annual annuity when lives differ), or take the larger or later project if its incremental IRR exceeds the cost of capital.
- The crossover rate is the IRR of the difference between two projects' cash flows; below it, the longer-duration project has the higher NPV.
- Under capital rationing with indivisible projects, enumerate or optimize over feasible portfolios to maximize total NPV; a PI ranking may leave budget unused and miss the best combination.

Review Questions

1. Explain why the NPV rule maximizes shareholder wealth, and why shareholders with different consumption preferences would still agree on it.
2. What is meant by "incremental" cash flows? Give three examples of items that should be included and three that should be excluded when estimating them.
3. Under what conditions do the NPV and IRR rules give the same accept/reject decision for an independent project? Illustrate with an NPV profile.
4. Why can a project have more than one IRR? What determines the maximum number of IRRs, and how should a manager decide whether to accept such a project?
5. Explain the reinvestment-rate assumption implicit in NPV and in IRR. Which is more defensible, and why?
6. Describe how the MIRR is calculated. What problems does it solve and what problems does it leave unsolved?
7. List three weaknesses of the payback rule and describe a situation in which a payback cutoff might nevertheless be reasonable.
8. Why does the profitability index sometimes give the wrong ranking for mutually exclusive projects? When is it a useful tool?
9. What is the crossover rate? Explain why IRR and NPV rank two projects differently when the cost of capital is below it.

10. Distinguish soft from hard capital rationing. Why might a PI ranking fail to identify the optimal set of projects when projects are indivisible?

Problems

1. **NPV, IRR and payback.** A project costs $80,000 and returns $25,000, $30,000, $35,000 and $28,000 in years 1 to 4. The cost of capital is 9 percent. Compute NPV, IRR and payback period. Should the project be accepted?
2. **NPV and IRR.** A project requires $500 now and returns $200, $250 and $300 in years 1, 2 and 3. Compute the NPV at 12 percent and the IRR.
3. **MIRR.** A project has cash flows −$1,000, +$400, +$450 and +$500 in years 0 to 3. The IRR is 15.98 percent. Compute the MIRR using a reinvestment rate of 8 percent, and the NPV at 8 percent. Why do the MIRR and IRR differ?
4. **Capital rationing.** At a 10 percent cost of capital, the NPVs and outlays of five independent projects are (outlay in $ million, NPV in $ million): A (120, 29.21), B (100, 42.64), C (80, 44.34), D (150, 71.89), E (60, 5.51). The budget is $300 million. Compute the PI of each project and find the portfolio that maximizes NPV.
5. **Scale.** Two mutually exclusive projects at a 10 percent cost of capital: Small costs $200 and returns $90 per year for 4 years; Large costs $600 and returns $230 per year for 4 years. Compute NPV and IRR of each, the incremental IRR, and state which to choose.
6. **Unequal lives.** Machine M1 costs $30,000 and returns $12,000 per year for 3 years; Machine M2 costs $45,000 and returns $11,000 per year for 6 years. The cost of capital is 9 percent and each can be replaced indefinitely. Compute NPV and EAA of each, and choose.
7. **Nonconventional cash flows.** A project has cash flows −$200, +$500 and −$310 in years 0, 1 and 2. Find the IRRs, compute the NPV at 10 percent, 12 percent, 20 percent and 25 percent, and state the range of discount rates for which the project is acceptable.
8. **Payback measures.** A project costs $60,000 and returns $20,000, $25,000, $30,000, $10,000 and $10,000 in years 1 to 5. The cost of capital is 10 percent. Compute payback, discounted payback and NPV. If the firm’s cutoff is 2.5 years, what do the two payback rules say?

Answers to Selected Problems

1. PV of inflows = 25,000/1.09 + 30,000/1.09² + 35,000/1.09³ + 28,000/1.09⁴ = 22,936 + 25,250 + 27,026 + 19,836 = 95,049 (from unrounded values), so NPV = $15,049. IRR = 17.05 percent. Cumulative cash flows are −55,000, −25,000, +10,000, so payback = 2 + 25,000/35,000 = 2.71 years. Accept (NPV > 0, IRR > 9 percent).
2. PV = 200/1.12 + 250/1.2544 + 300/1.404928 = 178.57 + 199.30 + 213.53 = 591.40; NPV = $91.40. IRR = 21.65 percent.
3. Terminal value at year 3 = 400 × 1.08² + 450 × 1.08 + 500 = 466.56 + 486.00 + 500 = 1,452.56. MIRR = (1,452.56/1,000)^(1/3) − 1 = 13.25 percent. NPV at 8 percent = $153.09. The MIRR is lower than the IRR (15.98 percent) because it assumes reinvestment at 8 percent instead of at the IRR; it still exceeds 8 percent, consistent with a positive NPV.
4. PIs: A 1.243, B 1.426, C 1.554, D 1.479, E 1.092. Ranking by PI: C, D, B, A, E; C (80) + D (150) = 230, B (100) does not fit, A (120) does not fit, E (60) fits, giving C + D + E at 290 with NPV 44.34 + 71.89 + 5.51 = $121.74

million (121.75 with unrounded NPVs). Checking alternatives (C + D = 116.23; A + B + C = 116.20; B + D = 114.53) confirms that C + D + E is best.

5. Small: NPV = 90 × 3.1699 − 200 = $85.29; IRR = 28.49 percent. Large: NPV = 230 × 3.1699 − 600 = $129.07; IRR = 19.60 percent. Incremental flows: −400, then +140 per year for 4 years; incremental NPV = $43.78; incremental IRR = 14.96 percent, above 10 percent. Choose Large (higher NPV), despite its lower IRR.
6. M1: NPV = 12,000 × 2.5313 − 30,000 = $375.54; EAA = 375.54/2.5313 = $148.36. M2: NPV = 11,000 × 4.4859 − 45,000 = $4,345.10; EAA = 4,345.10/4.4859 = $968.61. Choose M2 (higher EAA and higher NPV).
7. Solving $-200 + 500x - 310x^2 = 0$ with $x = 1/(1+r)$ gives x = 0.8786 and 0.7343, so IRRs are 13.82 percent and 36.19 percent. NPV at 10 percent = −1.65; at 12 percent = −0.70; at 20 percent = +1.39; at 25 percent = +1.60. The project is acceptable only for discount rates between about 13.8 and 36.2 percent; the IRR rule alone is ambiguous.
8. Cumulative cash flows: −40,000, −15,000, +15,000, so payback = 2 + 15,000/30,000 = 2.5 years. PVs: 18,182; 20,661; 22,539; 6,830; 6,209. Cumulative PV after year 2 = −21,157, so discounted payback = 2 + 21,157/22,539 = 2.94 years. NPV = 18,182 + 20,661 + 22,539 + 6,830 + 6,209 − 60,000 = $14,422. With a 2.5-year cutoff, ordinary payback is exactly at the cutoff (accept or marginal), while discounted payback (2.94 years) fails; yet the NPV is clearly positive.

Chapter 7: Capital Budgeting and Project Cash Flows

Learning Objectives

By the end of this chapter you should be able to:

- Identify the incremental, after-tax cash flows of a project, and distinguish them from accounting profits, allocated costs and sunk costs.
- Build a project cash flow forecast that correctly handles depreciation tax shields, salvage value, taxes on asset sales and investment in net working capital.
- Apply the opportunity cost and cannibalization principles to decide which existing resources and sales belong in a project's analysis.
- Keep inflation consistent by matching real cash flows with real discount rates and nominal cash flows with nominal discount rates.
- Use the equivalent annual annuity (EAA) method and the replacement chain method to compare mutually exclusive projects with unequal lives.
- Evaluate an equipment replacement decision, separating the genuinely relevant cash flows from book values and historical costs.

Opening Vignette

(Illustrative scenario; the company is fictional.)

Larchmont Beverages is a regional bottler with a strong portfolio of sparkling waters. In January its product development team presents a proposal to the investment committee: a new line of cold-brew coffee in recyclable cans. The proposal is enthusiastic. The team has spent $150,000 on market research, a pilot recipe and a consumer taste panel. It forecasts that the new line will earn "$780,000 of operating profit a year before depreciation" and points out that the company already owns an unused warehouse on the edge of town, which it describes as "free space, since we have already paid for it." Finance has also allocated $120,000 of corporate overhead to the project because "every product carries its share."

The chief financial officer, Marguerite Oyelaran, reads the memo twice and begins marking it up. The $150,000 research cost has already been spent whether or not the line goes ahead, so it should not weigh on the decision. The warehouse is not free: a logistics firm has offered to rent it for $90,000 a year, so using it for coffee means giving up that rent. The overhead allocation will be paid regardless of the coffee line, so it is not an incremental cost. Meanwhile, nobody has asked what the coffee line will do to sales of Larchmont's existing iced-tea range, which sits on the same shelf and appeals to the same shoppers. Nor has anyone counted the inventory and receivables the new business will tie up, or the fact that the sales forecast is in today's money while the equipment will be depreciated over five years at historical cost.

By the time Marguerite finishes, the project looks quite different. This chapter teaches the discipline she applied: forecasting only incremental after-tax cash flows, treating every resource at its true opportunity cost, accounting for working capital and depreciation tax shields, keeping inflation consistent, and comparing projects with unequal lives on a common footing. The objective is to turn an enthusiastic narrative into a defensible number.

7.1 The Principle of Incremental Cash Flows

7.1.1 Cash flows, not accounting profit

Chapters on valuation established that the value of any asset is the present value of the cash flows it generates. A capital budgeting decision is simply a valuation of a proposed project. The analyst's central task is therefore to forecast the cash flows the project causes and to discount them at a rate that reflects their risk. The net present value rule says to accept the project if NPV > 0.

Accounting profit is a poor guide to these cash flows for three reasons. First, accounting income deducts depreciation, a non-cash charge, and so understates cash generation. Second, accounting income ignores the cash invested in inventory and receivables and the cash released when they unwind. Third, accounting income records revenue and expense when earned or incurred rather than when cash moves. A project can show a healthy accounting profit while consuming cash, and the reverse can occur.

7.1.2 The "with versus without" test

The governing rule for capital budgeting is that **only incremental cash flows matter**. An incremental cash flow is a change in the firm's total cash flow that occurs as a direct consequence of accepting the project. The test is a comparison of two worlds:

```
Incremental CF(t) = Firm's total CF(t) with the project − Firm's total CF(t) without the project
```

Note the phrasing "without," not "before." The relevant baseline is what would happen if the project were rejected, which may not equal today's status quo. If a competitor will launch a similar product regardless, then sales that the firm would lose by not entering are a benefit of the project, even though they are not an increase over the present.

7.1.3 The standard layout

For a typical project the cash flow in year t is built from four components:

```
Project CF(t) = Operating CF(t) − Capital expenditure(t) − ΔNWC(t) + After-tax salvage(t)

Operating CF = (Revenue − Cash operating costs − Depreciation) × (1 − T) + Depreciation
             = (Revenue − Cash operating costs) × (1 − T) + Depreciation × T
```

where T is the marginal corporate tax rate and ΔNWC is the change in net working capital. The two equivalent expressions for operating cash flow are used throughout this chapter. The first starts from net operating profit after tax and adds back the non-cash depreciation charge. The second, sometimes called the tax shield approach, takes after-tax cash revenue less cost and adds the tax saved by depreciation. They always agree, and you should verify one against the other as a check.

Financing costs (interest, dividends, principal repayment) are left out of the project cash flows. The standard approach, developed in the chapters on cost of capital, captures financing through the discount rate, the weighted average cost of capital. Subtracting interest from the cash flows and also discounting at a rate that reflects the cost of debt would count the cost of debt twice.

7.1.4 Timing conventions

Most textbook analyses place the initial outlay at time 0 and treat each year's operating cash flow as arriving at the end of the year. This is a simplification, since cash actually arrives throughout the year, but it is conventional and conservative when flows are positive. The sub-annual refinements matter for fast-moving projects and can be handled by moving to

quarterly or monthly periods, but the logic does not change.

Example 7.1 – A full project cash flow and NPV.

Larchmont evaluates a simple project, stripped of the complications in the vignette, to illustrate the mechanics. A new canning line costs $2,000,000 and will be depreciated straight-line to zero over five years. It is expected to generate revenue of $3,000,000 per year and cash operating costs of $2,100,000 per year for five years. At the end of year 5 the line will be sold for $200,000. The project requires an up-front investment of $300,000 in net working capital, which will be fully recovered at the end of year 5. The tax rate is 25% and the required return is 10%.

Step 1 – Depreciation. $2,000,000 / 5 = $400,000 per year.

Step 2 – Operating cash flow.

```
EBIT                  = 3,000,000 − 2,100,000 − 400,000 = 500,000
Tax at 25%            = 125,000
Net operating profit  = 375,000
Operating CF          = 375,000 + 400,000 = 775,000
```

Check by the tax shield approach: (3,000,000 − 2,100,000) × 0.75 + 400,000 × 0.25 = 675,000 + 100,000 = 775,000. ✓

Step 3 – Terminal cash flows. The line is fully depreciated, so book value is zero. Selling for $200,000 produces a taxable gain of $200,000, and tax is 0.25 × 200,000 = $50,000. After-tax salvage is $150,000. The $300,000 of working capital is released.

Step 4 – Cash flow timeline.

Year	Capex	ΔNWC	Operating CF	Salvage (after-tax)	Net CF
0	−2,000,000	−300,000	0	0	−2,300,000
1	0	0	775,000	0	775,000
2	0	0	775,000	0	775,000
3	0	0	775,000	0	775,000
4	0	0	775,000	0	775,000
5	0	+300,000	775,000	150,000	1,225,000

Step 5 – Discount at 10%. The present value of the four-year level annuity of $775,000 is 775,000 × (1/1.10 + 1/1.10² + 1/1.10³ + 1/1.10⁴) = 775,000 × 3.16987 = $2,456,650. The year-5 flow is 1,225,000 × 0.620921 = $760,628. Total PV of inflows is $3,217,278 against an outlay of $2,300,000.

```
NPV = −2,300,000 + 2,456,650 + 760,628 ≈ $917,274
```

(The unrounded figure is $917,274.) The project creates about $917,000 of value and should be accepted. Notice that the working capital outlay and its recovery are included: ignoring them would have overstated the NPV by $300,000 × (1 − 0.6209) = about $113,700, the cost of tying up $300,000 for five years at 10%.

7.1.5 What counts as incremental

A useful checklist for the “with versus without” test follows.

Item	Treatment	Reason
Capital outlay for new assets	Include	Cash leaves the firm because of the project
Cash revenue and cash operating cost changes	Include	Direct effect of the project
Depreciation	Include only through its tax shield	Non-cash; matters because it reduces taxes
Change in net working capital	Include	Cash tied up or released
After-tax salvage value	Include	Cash received when the project ends
Interest and financing flows	Exclude	Captured in the discount rate
Sunk costs	Exclude	Already spent; unaffected by the decision
Allocated overhead that does not change	Exclude	Not caused by the project
Opportunity costs of existing assets	Include	Value of the best alternative use is forgone
Cannibalization and synergies	Include	Effects on other products' cash flows

The remainder of this chapter takes each item in turn.

7.2 Depreciation, Taxes and the Depreciation Tax Shield

7.2.1 Why depreciation matters for cash

Depreciation is not a cash flow. When a firm buys a machine for $2,000,000, the cash leaves on the purchase date, not over the machine's accounting life. Yet tax law allows the firm to deduct a portion of the cost each year, and deductions reduce taxable income and hence tax paid. The cash value of depreciation is therefore the tax it saves:

```
Depreciation tax shield (year t) = Depreciation(t) × T
```

A dollar of depreciation is worth T dollars of cash. The firm cannot raise the total amount it deducts, which is limited to the asset's cost less salvage, but it can influence *when* the deductions arrive, and because of the time value of money earlier deductions are worth more.

7.2.2 Straight-line versus accelerated depreciation

Straight-line depreciation spreads the depreciable base evenly over the asset's life. Many tax systems instead prescribe accelerated schedules that front-load the deductions. The best known is the United States Modified Accelerated Cost Recovery System (MACRS), which assigns an asset to a recovery class and gives a table of annual percentages. The principle is the same in every jurisdiction that allows accelerated allowances, though the specific rates and classes differ and should be taken from the current tax code when used in practice. For illustration this chapter uses the widely published five-year class percentages under the half-year convention: 20.00%, 32.00%, 19.20%, 11.52%, 11.52% and 5.76% across six tax years.

Accelerated depreciation does not change total deductions, but it raises the present value of the tax shield.

Example 7.2 – PV of depreciation tax shields: straight-line versus accelerated.

A firm buys equipment costing $1,000,000 with a five-year recovery class. The tax rate is 25% and the discount rate is 10%. Compare the present value of the tax shield under (a) straight-line over five years and (b) the six-year MACRS five-class percentages given above.

(a) Straight-line. Annual depreciation is $200,000, annual tax shield $50,000 for five years. PV = 50,000 × 3.790787 = **$189,539**.

(b) Accelerated.

Year	MACRS %	Depreciation	Tax shield (25%)	Discount factor	PV
1	20.00%	200,000	50,000	0.909091	45,455
2	32.00%	320,000	80,000	0.826446	66,116
3	19.20%	192,000	48,000	0.751315	36,063
4	11.52%	115,200	28,800	0.683013	19,671
5	11.52%	115,200	28,800	0.620921	17,883
6	5.76%	57,600	14,400	0.564474	8,128
Total	100.00%	1,000,000	250,000		**193,315**

Both schedules deduct $1,000,000 in total and generate undiscounted tax savings of $250,000. But the accelerated schedule is worth $193,315 against $189,539, a gain of about **$3,776** per million dollars of equipment. The advantage grows with the discount rate, the tax rate and the length of the asset's life, and it is why governments use accelerated allowances as an investment incentive: the interest-free loan from the tax authority is worth more the earlier it is granted.

7.2.3 Tax on the sale of assets

When an asset is sold, the tax authority compares the sale price with the asset's **tax book value** (original cost less accumulated tax depreciation).

```
Taxable gain (or loss) = Sale price − Tax book value
After-tax proceeds     = Sale price − T × (Sale price − Tax book value)
```

If the sale price exceeds book value, the excess is taxed (so after-tax proceeds are lower than the price). If the sale price is below book value, the loss generates a tax saving, and after-tax proceeds are higher than the price. In Example 7.1, book value was zero and the whole $200,000 was taxed. A machine with a book value of $300,000 sold for $120,000 realises a $180,000 loss and, at a 25% tax rate, a tax saving of $45,000, for after-tax proceeds of $165,000. Real tax codes can treat gains and losses differently (for example, capital versus ordinary treatment, or recapture rules), so the analyst must follow the applicable rules; the simple symmetric treatment used here is the standard teaching convention.

7.2.4 Depreciation choices and financial reporting

A common confusion arises from the difference between *book* (financial-reporting) depreciation and *tax* depreciation. Companies often use straight-line depreciation in their published accounts and an accelerated schedule for tax returns. For capital budgeting, only the tax schedule matters, because only tax depreciation affects cash taxes. The book schedule affects reported earnings and can influence managerial behaviour, but it does not alter value. A manager who rejects a positive-NPV project because the depreciation charge would depress reported earnings is trading shareholder value for a cosmetic accounting number.

7.2.5 Common mistakes

- **Subtracting depreciation as if it were cash.** Depreciation reduces taxes, but the firm does not pay it out. The cash cost of the asset is the capital expenditure at the time of purchase.

- **Forgetting the terminal tax effect.** Salvage value must be taxed (or the loss credited) relative to tax book value, not relative to the purchase price.
- **Using average rather than marginal tax rates.** The relevant rate is the rate on the incremental income or deduction. For a profitable firm this is the statutory marginal rate; for a loss-making firm, tax shields may be deferred and worth less.
- **Mixing book and tax schedules.** Use the tax schedule for cash flows.

7.3 Net Working Capital

7.3.1 Why working capital belongs in the analysis

Few projects can grow without financing the day-to-day operating cycle. A new product line needs raw material and finished goods in stock, extends credit to customers, and receives credit from suppliers. Net working capital (NWC) is the operating current assets minus the operating current liabilities, excluding cash held as a financial buffer and interest-bearing debt:

```
NWC = Accounts receivable + Inventory − Accounts payable (and other operating accruals)
```

When sales rise, NWC usually rises, and the increase is a cash *outflow*, because cash is tied up in stock and receivables. When the project winds down, NWC falls and the cash is released. These flows are real, though no income statement shows them, and ignoring them overstates value. Because the investment is made early and recovered late, the cost to the project is the time value of money on the balance.

7.3.2 Forecasting NWC

Analysts usually model NWC as a percentage of sales, drawing on the firm's experience with comparable products:

```
NWC(t) = w × Sales(t)              where w is the NWC-to-sales ratio
ΔNWC(t) = NWC(t) − NWC(t−1)        (positive ΔNWC = cash outflow)
Project cash flow(t) includes −ΔNWC(t)
```

Three points are worth emphasising. First, the investment typically precedes the sales that justify it: inventory must be built before it is sold, so NWC at the end of year 0 may already be positive. Second, because NWC is driven by the *level* of sales while the cash flow depends on the *change*, a fast-growing project absorbs the most cash in its early years even when its profitability is highest. Third, the working-capital release at the end of the project is not a certain return of all that was invested: obsolete stock may be written down and some receivables may prove uncollectable. Conservative analysts assume less than 100% recovery.

7.3.3 Payables as a source of financing

Trade credit from suppliers reduces NWC because accounts payable is a liability. A firm that negotiates 60-day payment terms rather than 30-day terms lowers its working capital requirement for the same sales, thereby raising project value. For businesses with long production cycles and thin margins, working capital management can be worth more than a modest improvement in the price of the product.

Example 7.3 – A project with a working-capital build-up (amounts in $000).

A new product requires equipment of $1,200 (straight-line to zero over five years, no salvage). Sales are forecast at 1,000, 1,500, 2,000, 2,000 and 1,200 in years 1 to 5. The cash margin before fixed costs is 40% of sales, and fixed cash costs are 100 per year. NWC is 15% of the year's sales. NWC is recovered in full at the end of year 5. Tax rate is 25%; required return is 10%.

Step 1 – Depreciation. 1,200 / 5 = 240 per year.

Step 2 – Operating cash flow. EBITDA = 0.40 × Sales − 100.

Year	Sales	EBITDA	EBIT (− 240)	Tax (25%)	Operating CF = EBITDA − tax
1	1,000	300	60	15	285
2	1,500	500	260	65	435
3	2,000	700	460	115	585
4	2,000	700	460	115	585
5	1,200	380	140	35	345

Step 3 – NWC schedule.

Year	NWC (15% of sales)	ΔNWC	Cash effect
0	0	0	0
1	150	+150	−150
2	225	+75	−75
3	300	+75	−75
4	300	0	0
5	180 → 0 (final recovery)	−120, then −180	+120 +180 = +300

In year 5, sales fall to 1,200, so NWC drops from 300 to 180, releasing 120. At the end of the project the remaining 180 is also released, so the total year-5 working capital inflow is 300.

Step 4 – Net cash flows.

Year	Capex	Operating CF	NWC effect	Net CF
0	−1,200	0	0	−1,200
1	0	285	−150	135
2	0	435	−75	360
3	0	585	−75	510
4	0	585	0	585
5	0	345	+300	645

Step 5 – NPV at 10%.

$$\begin{aligned}\text{NPV} &= -1{,}200 + 135/1.10 + 360/1.10^2 + 510/1.10^3 + 585/1.10^4 + 645/1.10^5 \\ &= -1{,}200 + 122.73 + 297.52 + 383.17 + 399.55 + 400.49 \\ &\approx 403.48 \quad (\$403{,}476)\end{aligned}$$

For comparison, ignoring NWC entirely gives net cash flows of −1,200, 285, 435, 585, 585, 345 and an NPV of **471.89**. Working capital therefore costs the project about 68.4 of value (14.5% of the NPV omitted-NWC figure). The project survives, but a more aggressive working-capital assumption (say 25% of sales) would erode the margin of safety, a point we revisit through sensitivity analysis in the next chapter.

7.4 Sunk Costs, Opportunity Costs and Side Effects

7.4.1 Sunk costs: ignore them

A **sunk cost** is a cost that has already been incurred and cannot be recovered whatever the firm decides next. The $150,000 Larchmont spent on market research is sunk. The decision it faces is whether the *future* cash flows from the line justify the *future* cash outlays. If the research cost is included as a negative cash flow, the project is penalised for a decision that was already taken and cannot be reversed. If the project would fail only because of the sunk cost, rejecting it destroys value.

Sunk costs appear in subtler forms. Consider a half-finished factory on which $40 million has been spent and which needs $25 million more to complete. The $40 million is irrelevant; the only question is whether the completed factory is worth more than $25 million (in present value terms) to the firm. Managers find this hard to accept, which is why the *sunk cost fallacy* is among the best documented errors in decision-making: the instinct to "throw good money after bad" in order to justify earlier spending. A disciplined capital budgeting process makes a rule of treating each stage-gate as a fresh decision.

A related question is whether research costs that precede the project can be recovered if the project proceeds. They cannot; so the proper way to evaluate *the research programme itself* is ex ante, when the spending decision is still open. This links directly to the real-option perspective in Chapter 8.

7.4.2 Opportunity costs: include them

An **opportunity cost** is the value of the best alternative use of a resource that the firm already owns. If the project uses an asset the firm could otherwise sell, rent or use elsewhere, the project must be charged for it, though no cash changes hands, because the project causes the firm to give up that value. The idle warehouse in the vignette is a classic case: its purchase price is sunk, but its rental value is a real forgone cash flow. If the next best use is a $90,000 pre-tax annual rent, the after-tax opportunity cost is $90{,}000 \times (1 - 0.25) = \$67{,}500$ per year. If the next best use is to sell the asset now, the opportunity cost is the after-tax sale proceeds in year 0.

The same logic applies to people and information: a manager transferred from another profitable activity, or a proprietary technology that could be licensed to a third party, carries an opportunity cost equal to the value forgone elsewhere.

7.4.3 Allocated costs: include only what changes

Corporate overhead is typically allocated to divisions by formula, as a share of sales or headcount. For capital budgeting what matters is whether the *total* overhead of the firm rises because of the project. If the head-office costs are the same with or without the project, the allocation is not incremental and must be excluded. If the project requires, say, an additional accountant, her salary is incremental, even though it is not labelled "overhead allocation." The test is always the with-versus-without comparison, never the accounting label.

7.4.4 Side effects: cannibalization and synergy

New products can reduce the sales of existing ones (**cannibalization** or erosion) or increase them (**synergy** or complementarity). Both are incremental cash flows and must be reflected. If the cold-brew coffee line takes sales from Larchmont's iced tea, the lost contribution margin from iced tea is charged to the coffee project. If a competitor would have

launched a rival coffee product anyway and taken those iced-tea sales regardless, only the *additional* cannibalization caused by Larchmont's own entry counts, because the baseline is the "without" world and not the status quo. Strategic arguments of this kind need explicit evidence, since it is tempting to claim that "we would be cannibalized anyway" for any product.

Positive side effects deserve the same scrutiny: a new gaming console that boosts game sales, or a loss-leading product that increases customer retention. They must be supported by an estimate and included in the cash flow forecast; they should not be left as a vague qualitative justification.

Example 7.4 – Applying the incremental cash flow filter.

Larchmont's coffee line, cleaned up, has these inputs. Equipment costs $1,500,000 depreciated straight-line to zero over five years ($300,000 per year). Revenue is $2,400,000 per year and cash operating costs are $1,500,000 per year. Launch will cannibalize iced-tea sales, costing the company $200,000 per year of pre-tax contribution. The warehouse could be rented out for $90,000 per year pre-tax. The market research ($150,000) is sunk. A corporate overhead charge of $120,000 per year is allocated to the project but total head-office costs do not change. The tax rate is 25% and the discount rate is 10%. The project lasts five years with no salvage and no NWC (to keep the focus on the filter).

Step 1 – Identify the relevant pre-tax cash items.

Item	Amount	Include?
Revenue	+2,400,000	Yes
Cash operating costs	−1,500,000	Yes
Cannibalized iced-tea contribution	−200,000	Yes (side effect)
Forgone warehouse rent	−90,000	Yes (opportunity cost)
Corporate overhead allocation	−120,000	No (not incremental)
Market research	−150,000	No (sunk)

Step 2 – Pre-tax incremental operating cash flow. 2,400,000 − 1,500,000 − 200,000 − 90,000 = **610,000**.

Step 3 – After-tax operating cash flow.

```
OCF = 610,000 × (1 − 0.25) + 300,000 × 0.25 = 457,500 + 75,000 = 532,500
```

Step 4 – NPV. PV of a five-year annuity at 10% (factor 3.790787) is 532,500 × 3.790787 = $2,018,594. Subtract the $1,500,000 outlay:

```
NPV = 2,018,594 − 1,500,000 = $518,594
```

Step 5 – What the naive analysis would say. The original memo's approach (charging overhead, ignoring cannibalization and rent, and either leaving out or including the sunk research) gives pre-tax flow 2,400,000 − 1,500,000 − 120,000 = 780,000 and OCF = 780,000 × 0.75 + 75,000 = 660,000. NPV excluding research = 660,000 × 3.790787 − 1,500,000 = $1,001,919; including research as a year-0 cost = $851,919. Neither is right. The naive figures are hugely different from the true NPV of $518,594, because the errors do not cancel: the omitted cannibalization and rent (together $290,000 pre-tax) outweigh the spurious overhead charge ($120,000). Even so, the project remains attractive, a reminder that cleaning up the analysis does not necessarily kill a project, though it can easily change the ranking among projects.

7.4.5 Summary of relevance

Concept	Rule	Typical error
Sunk cost	Exclude	Including past R&D or feasibility study costs
Opportunity cost	Include at best alternative use	Treating owned assets as free
Allocated overhead	Include only if total cost changes	Charging fixed allocations
Cannibalization	Include lost contribution of existing products	Ignoring erosion of the core business
Synergy	Include with evidence	Using unquantified strategic benefits to justify weak projects
Financing costs	Exclude from cash flows	Double-counting with the discount rate
Working capital	Include build-up and release	Treating inventory as if it were free

7.5 Inflation in Capital Budgeting

7.5.1 Nominal versus real

Inflation raises the money price of goods over time. Cash flows measured in the money of the day are **nominal**; cash flows adjusted for the general price level, in the money of a base date, are **real**. Discount rates come in the same two forms. The nominal rate includes compensation for expected inflation; the real rate does not. The two are linked by the Fisher relationship:

```
(1 + nominal rate) = (1 + real rate) × (1 + inflation rate)
```

For example, with a real rate of 5% and expected inflation of 3%, the nominal rate is 1.05 × 1.03 − 1 = 8.15%. (The approximation “nominal ≈ real + inflation” gives 8%, close but not exact.)

7.5.2 The consistency rule

The cardinal rule is simple: **discount nominal cash flows at nominal rates and real cash flows at real rates.** If both are done correctly, the NPV is identical. The most common error is a mismatch: forecasting cash flows in constant (today’s) prices and then discounting at a market-derived nominal rate, which builds inflation into the denominator but not the numerator and understates value.

Example 7.5a – Consistency between real and nominal.

A project yields a real cash flow of $100 per year for four years (in today’s prices) for an investment whose value we study in isolation. The real discount rate is 5% and expected inflation is 3%.

Real approach. PV = 100 × annuity factor(5%, 4) = 100 × 3.545951 = **$354.60**.

Nominal approach. The nominal rate is 8.15%. Nominal cash flows: 100 × 1.03 = 103.00, 100 × 1.03^2 = 106.09, 100 × 1.03^3 = 109.27, 100 × 1.03^4 = 112.55. Discount at 8.15%:

```
PV = 103.00/1.0815 + 106.09/1.0815² + 109.27/1.0815³ + 112.55/1.0815⁴ ≈ $354.60
```

The two methods agree. *Mixing* real flows with the nominal rate gives 100 × annuity factor(8.15%, 4) ≈ $330.11, understating value by about 6.9%.

7.5.3 Where inflation truly changes the answer

If everything inflated at the same rate, the choice between real and nominal analysis would be a matter of taste. But depreciation tax shields are fixed in nominal terms: they are based on historical cost and do not rise with prices. As inflation increases, the real value of the depreciation shield falls, which makes projects with large depreciation shields less attractive in high-inflation environments. Similarly, working capital rises with inflation, because the same physical inventory costs more, so inflation increases the cash investment in NWC. Some costs (wages, say) may adjust at a different rate from selling prices, and a good forecast models these separately rather than applying one inflation rate to everything. Finally, if the project's revenues are contractually fixed in nominal terms, inflation erodes their real value.

Example 7.5b – The inflation penalty on depreciation shields.

A firm invests $400 in equipment, straight-line over four years ($100 per year). Real pre-tax operating cash flow is $220 per year in today's prices, rising with 3% inflation. The tax rate is 30%, the real discount rate is 5%, and the nominal rate is 8.15%.

Nominal approach.

Year	Pre-tax CF (nominal)	After-tax CF = pre-tax × 0.70	Dep. shield (100 × 0.30)	Net nominal CF
1	226.60	158.62	30.00	188.62
2	233.40	163.38	30.00	193.38
3	240.40	168.28	30.00	198.28
4	247.61	173.33	30.00	203.33

NPV at 8.15% = −400 + 188.62/1.0815 + 193.38/1.0815² + 198.28/1.0815³ + 203.33/1.0815⁴ = −400 + 174.41 + 165.33 + 156.76 + 148.63 ≈ **−400 + 645.11 = 245.11**.

Naive "real" approach. If the analyst uses the constant real cash flow of 220 × 0.70 + 30 = 184 each year and discounts at the real 5%, NPV = −400 + 184 × 3.545951 = **252.45**.

The difference of about 7.3 arises solely because the depreciation shield of $30 per year was treated as inflating when in fact it is fixed in nominal terms. The *correct* real analysis would deflate the nominal shield; doing so reproduces the 245.11. The lesson: wherever a cash flow is fixed in nominal terms (depreciation, fixed-rate contracts, nominal debt service), convert it carefully rather than assuming it moves with prices.

7.5.4 Practical guidance

Most practitioners prefer to forecast in nominal terms because taxes, depreciation, debt and market quotations for discount rates are all nominal. If long-horizon projects in high-inflation economies are forecast in real terms for convenience, the analyst should still check the tax and working-capital effects explicitly. For multinational projects in different currencies, the same logic applies with exchange rates linked through purchasing power parity, as discussed in the chapters on international finance.

7.6 Projects with Unequal Lives: The Equivalent Annual Annuity

7.6.1 The problem

NPV is the right criterion for independent projects. For mutually exclusive projects with equal lives, the project with the higher NPV wins. But when projects have different lives, comparing NPVs directly can mislead. A project that lasts ten years may have a larger NPV than one that lasts three simply because it runs longer, even if the short project is more profitable per year and could be repeated.

This problem is most acute for *cost-minimisation* decisions, in which the machines or vehicles under comparison deliver the same service but differ in purchase price, running costs and durability. A cheap machine that wears out in three years must be compared with an expensive one that lasts five.

7.6.2 Two equivalent methods

Replacement chain (common life) method. Assume each project is repeated until both reach a common horizon, usually the least common multiple of their lives. Compare the NPVs over that horizon. A three-year and a five-year machine would be compared over fifteen years: five cycles of the first against three cycles of the second.

Equivalent annual annuity (EAA) method. Convert each project's NPV to the level annual payment, over the project's own life, that has the same present value:

```
EAA = NPV / Annuity factor(r, n)          where Annuity factor = [1 − (1 + r)^(−n)] / r
```

For cost-only projects the equivalent figure is called the **equivalent annual cost (EAC)**, and the lower the EAC the better. The EAA method gives the same ranking as the replacement chain whenever the projects can be repeated indefinitely with the same cash flows, and it is far less laborious. Indeed, repeating a project forever creates a perpetuity of EAA, whose value is EAA / r, so ranking by EAA is ranking by the value of the infinite chain.

7.6.3 Assumptions and limits

EAA assumes that (i) the service is needed indefinitely (or at least for the common horizon), (ii) replacement is possible on identical terms, and (iii) costs do not change over time. Where technology is improving, replacement will probably be cheaper and better, and the long-lived project should be penalised relative to the EAA ranking. Where prices are inflating, the replacement cost will be higher in nominal terms, which may favour the longer-lived machine. If the horizon is truly finite (a three-year contract, say), the common-life assumption is not valid and the analysis should use the actual horizon, perhaps with residual values for the leftover machine life.

Example 7.6 – Choosing between two machines.

A packaging firm needs a machine to perform a task. Machine A costs \$100,000, lasts 3 years, and costs \$30,000 per year to operate. Machine B costs \$150,000, lasts 5 years, and costs \$20,000 per year to operate. Ignore taxes and salvage for clarity; the required return is 10%. Revenue is the same with either machine.

Step 1 – PV of costs over each machine's own life.

```
PV(A) = 100,000 + 30,000 × annuity factor(10%, 3) = 100,000 + 30,000 × 2.486852 = 174,606
PV(B) = 150,000 + 20,000 × annuity factor(10%, 5) = 150,000 + 20,000 × 3.790787 = 225,816
```

A is cheaper in total PV, but B lasts two years longer, so the raw comparison is misleading.

Step 2 – Equivalent annual cost.

```
EAC(A) = 174,606 / 2.486852 = 70,211
EAC(B) = 225,816 / 3.790787 = 59,570
```

Machine B has the lower EAC, saving about $10,641 per year.

Step 3 – Verify by the replacement chain over 15 years. A is replaced at the start of years 4, 7, 10 and 13 (five purchases in all). The PV of the chain is 174,606 × [1 + 1.1^{-3} + 1.1^{-6} + 1.1^{-9} + 1.1^{-12}] = **$534,034**. B is repurchased at the start of years 6 and 11 (three purchases in all): 225,816 × [1 + 1.1^{-5} + 1.1^{-10}] = **$453,091**. B is cheaper by $80,943, and the ranking agrees with the EAC method. (Cross-check: 70,211 × annuity factor(10%, 15) = 70,211 × 7.606080 = 534,034. ✓)

Measure	Machine A	Machine B
Initial cost	100,000	150,000
Annual operating cost	30,000	20,000
Life (years)	3	5
PV of costs, own life	174,606	225,816
Equivalent annual cost	70,211	59,570
PV of costs over 15-year chain	534,034	453,091

The decision rule for cost comparison with different lives is to choose the lowest EAC. The naive comparison favoured the machine that was worse.

7.7 Replacement Decisions

7.7.1 The structure of the decision

A replacement decision asks whether to retire an existing asset and buy a new one. It is easily confused with expansion decisions, but it has a particular structure because the cash flows from the old asset are *given up* when it is replaced. The framework is the same incremental one:

```
Incremental CF = CF(with replacement) − CF(keep old asset)
```

The key elements are as follows.

1. **Initial outlay** = cost of the new asset − after-tax sale proceeds of the old asset (and any NWC change).
2. **Operating savings** = after-tax reduction in operating costs (or increase in revenue) from the new asset.
3. **Change in depreciation tax shield** = T × (new depreciation − old depreciation). The old asset's remaining depreciation is lost if it is sold, but this is captured by comparing the two schedules.
4. **Terminal values** = after-tax salvage value of the new asset minus the after-tax salvage value of the old asset at the same horizon.

7.7.2 The book value trap

The old machine's **book value** is a sunk, historical number. It is irrelevant as a measure of what the machine is worth, and a manager who says "we can't scrap it yet because it still has $300,000 of book value" is making a decision on accounting, not economics. Book value matters only for the *tax effect* of a sale: selling below book value creates a loss that shields other income from tax, selling above book value creates a taxable gain. The relevant price of the old machine is its market value (after tax), the opportunity cost of continuing to use it.

7.7.3 Matching horizons

The new and old assets must be compared over the same horizon. If the old machine has three years of remaining life and the new one has five, either compare over the old machine's remaining life with the new asset's residual value at that date, or use EAC to compare each alternative on a per-year basis. The simplest case is to choose a horizon equal to the remaining life of the old machine (or a common horizon) and include an appropriate terminal value.

Example 7.7 – Should Larchmont replace its old filler?

Larchmont's old bottle filler has a tax book value of $300,000 with three years of straight-line depreciation remaining ($100,000 per year). It could be sold today for $120,000, has three years of life left, and costs $140,000 a year to operate; its salvage value in three years is zero. A new filler costs $360,000, would be depreciated straight-line to zero over three years ($120,000 per year), costs $40,000 per year to run and can be sold in three years for $30,000. The tax rate is 25% and the required return is 10%. Output is identical with either machine.

Step 1 – Initial outlay.

```
Purchase of new machine              −360,000
Sale of old machine (market value)   +120,000
Tax saving on loss: (300,000 − 120,000) × 25% = +45,000
Net initial incremental CF           −195,000
```

The $300,000 book value never enters directly: only its tax consequence does.

Step 2 – Annual incremental cash flow (years 1 to 3).

```
After-tax operating savings = (140,000 − 40,000) × (1 − 0.25) = 75,000
Change in depreciation shield = (120,000 − 100,000) × 0.25   =  5,000
Incremental annual CF                                          = 80,000
```

Step 3 – Terminal flow at year 3. The new machine's tax book value is zero, so the $30,000 sale is fully taxed: after-tax salvage = 30,000 × 0.75 = $22,500. The old machine would have had zero salvage, so the incremental terminal flow is $22,500.

Step 4 – NPV.

```
NPV = −195,000 + 80,000 × 2.486852 + 22,500 / 1.10³
    = −195,000 + 198,948 + 16,905
    = +20,853
```

Replacement adds about $20,900 of value, a modest positive NPV that depends heavily on the old machine's running cost.

Step 5 – Break-even analysis. What old-machine operating cost makes the firm indifferent? Set NPV = 0: the required annual incremental CF is (195,000 − 16,905) / 2.486852 = 71,617. Subtracting the $5,000 depreciation effect leaves 66,617 of after-tax savings, i.e. 66,617 / 0.75 = $88,823 of pre-tax savings, so the old machine's operating cost would have to fall below about $128,823 before replacement stops paying. The decision is therefore sensitive to the cost forecast: an 8% overestimate of old-machine costs flips the answer. This type of break-even reasoning, developed formally in Chapter 8, is often more valuable to a manager than the point estimate itself.

7.7.4 Timing and “when to replace”

The replacement question also arises in timing: given that a machine will eventually be replaced, should it be replaced this year or next? The right test compares the *cost of keeping the old machine for one more year* (forgone sale proceeds, higher maintenance, lost efficiency, and the delayed tax effects) with the *equivalent annual cost of the new machine*. If the incremental cost of keeping the old asset one more year is less than the EAC of the new one, keep the old; otherwise replace it now. Because the old machine’s costs typically rise as it ages while the new machine’s EAC is stable, a replacement date eventually arrives.

7.8 Managerial Implications and Good Practice

7.8.1 Building a trustworthy forecast

A project analysis is only as good as its cash flow forecast, and forecasts are subject to systematic biases. Project sponsors are enthusiastic by nature, and capital rationing creates an incentive to inflate benefits and understate costs. Several practices improve discipline:

- **Separate the forecaster from the sponsor** where possible, or have finance review the key assumptions independently.
- **Document every assumption**, particularly the sales volume, price, margin and working capital ratios, along with the evidence for them.
- **Use post-audits** to compare actual results with forecasts and learn from systematic errors.
- **Check the cash flows against the strategic story.** If the product is a market leader because of a unique capability, the forecast margins should be consistent with that advantage and with the competitive response.
- **Stress the terminal value.** Many projects derive most of their NPV from the final year or a continuing value; assume reasonably conservative resale or continuing values.

7.8.2 A checklist before the committee meeting

1. Are all cash flows incremental, assessed with-versus-without the project?
2. Are sunk costs excluded and opportunity costs included?
3. Are taxes computed on the right base using tax depreciation, with a correct treatment of asset sales?
4. Is working capital built up and released over time?
5. Are cannibalization and synergies counted?
6. Is the inflation treatment consistent between flows and rates?
7. Are mutually exclusive alternatives compared over comparable lives?
8. Is the discount rate appropriate for the project’s *risk* rather than for the firm’s average risk?
9. Have the key assumptions been stress-tested?

The last item is the subject of the next chapter. Point estimates of cash flows are only central tendencies; managers must understand how wrong the forecast can be, and what they can do about it.

7.8.3 Beyond NPV mechanics

The mechanics in this chapter are the grammar of capital budgeting. Good judgement requires more: strategic fit, organizational capacity, competitive response and the flexibility to adapt. The next chapter shows how to quantify uncertainty and the value of flexibility. For now, remember that a rigorous cash flow forecast accomplishes two things: it prevents obvious analytic errors (sunk costs, double counting, inconsistent inflation) and it forces the project team to articulate exactly what must be true for the project to create value.

Case Study

Harborline Marine Components: The Gelcoat Line Decision

(Fictional company; all figures illustrative.)

Harborline Marine Components manufactures fibreglass hulls and deck parts for boat builders. Its plant in a coastal town runs two moulding lines. The newer line, bought eight years ago, is in good order. The older line, Line 1, was bought twelve years ago and has a tax book value of $240,000, with two years of straight-line depreciation remaining at $120,000 per year. It could be sold today for $150,000. Its running costs have crept up: maintenance, scrap and energy now total $410,000 per year, and the plant manager expects them to rise by roughly $30,000 per year as components age. Line 1 would have no salvage value in three years.

Last spring, Harborline paid a consultancy $85,000 to assess automated gelcoat spraying equipment. The recommendation is to buy a robotic line, Line 3, for $900,000. It would be depreciated straight-line to zero over six years for tax and would be sold at the end of year 6 for an estimated $90,000. Line 3's running costs would be $230,000 per year, and it would reduce scrap enough to save materials worth $60,000 per year relative to Line 1. The robotic line would also lift capacity: the sales director believes that Harborline could win a long-term contract with a builder of fishing boats worth $700,000 per year in additional revenue, with a contribution margin of 35% before any fixed cost, but admits that the contract might be awarded to a competitor if Harborline delays. Winning the contract would raise inventories and receivables by an amount equal to 12% of the extra revenue, to be recovered at the end of the horizon.

The finance director, Tomás Adeyemi, has found three items in the project file that trouble him. The consultancy fee has been included as a year-0 cost by the project team. The plant's general overhead of $90,000 per year, allocated by floor area, has been charged against Line 3 at a rate of $15,000 per year because the new line occupies the floor space of the old one. And the sales director has included the full $700,000 of contract revenue in the base case as certain, whereas Tomás believes the probability of winning it is closer to 60% whether or not Harborline buys the robot, because the builder has also invited a bid for a smaller contract with Line 2 only.

There is a further complication. If Harborline sells Line 1, its employees must be retrained at a cost of $40,000 after tax, spent in year 1, but the retained staff would also let the plant avoid an expensive temporary-labour contract worth $25,000 per year before tax. The firm faces a 28% tax rate and uses 11% for projects of this risk.

Tomás wants an analysis that his board can rely on. He also wonders whether to wait a year, since a rival supplier might release a cheaper robotic line, but that rival's product is unproven, and the contract is time-sensitive. A colleague asks whether the analysis should simply use the six-year life of Line 3 against the three remaining years of Line 1, but Tomás points out that Line 1 will have to be replaced by something at the end of its life, and it is not obvious that the comparison should favour either.

Discussion Questions

1. Which of the items in the project file (consultancy fee, overhead allocation, contract revenue, retraining, temporary-labour savings, sale of Line 1) are incremental cash flows, and how should each be treated? Explain the role of the old line's book value.
2. How should Tomás handle the question of different lives (six years for Line 3 against three remaining years for Line 1)? Describe two methods, and explain what assumptions each method makes about replacement of Line 1.
3. The contract is assumed to have a 60% probability of award regardless of the robot. What does this imply about whether the contract revenue is an incremental benefit of Line 3? Under what circumstances would it be incremental?
4. Discuss the option to wait a year. What would you need to know to value it, and what are the costs of waiting? (Chapter 8 develops the real-option framework for this decision.)

Key Takeaways

- Capital budgeting values projects by their incremental after-tax cash flows: the difference between the firm's cash flows with and without the project.
- Depreciation is a non-cash expense whose only cash effect is the tax it saves; accelerated depreciation raises the present value of those savings without changing total deductions.
- After-tax salvage must be computed relative to tax book value; a sale above book value is taxed and a sale below it generates a tax saving.
- Net working capital is a real investment: the build-up is a cash outflow, the release at the end is a cash inflow, and ignoring it overstates NPV.
- Sunk costs are irrelevant to forward-looking decisions; the sunk cost fallacy leads firms to throw good money after bad.
- Opportunity costs, such as forgone rent on an owned asset, must be charged to a project even when no cash changes hands; allocated overhead is relevant only if total costs change.
- Cannibalization and synergies are incremental effects on other products' cash flows and belong in the analysis, measured against the "without" baseline.
- Discount nominal cash flows at nominal rates and real cash flows at real rates; mismatches understate value, and fixed-nominal items such as depreciation shields lose real value as inflation rises.
- For mutually exclusive projects with unequal lives, compare equivalent annual annuities (or equivalent annual costs) or use a common-life replacement chain.
- In a replacement decision the old asset's book value matters only through taxes; its market value is the opportunity cost of keeping it.

Review Questions

1. Why is accounting profit an unreliable proxy for project cash flow? Give three distinct reasons.
2. Explain the "with versus without" test and why the baseline is not necessarily the status quo.
3. Why are interest payments excluded from project cash flows when calculating NPV with the weighted average cost of capital?
4. Describe the depreciation tax shield. Why does accelerated depreciation increase project NPV even though total depreciation is unchanged?

5. A machine is sold for less than its tax book value. Explain the tax consequence and its effect on after-tax proceeds.
6. How does net working capital enter a project cash flow? Why can a fast-growing project be cash-negative in its early years despite being profitable?
7. A manager argues that a feasibility study costing $200,000 must be recovered by the project. Evaluate the argument.
8. Distinguish an opportunity cost from a sunk cost with an example of each involving an owned building.
9. What is the Fisher relationship, and what is the most common error in applying it to project analysis?
10. Why can NPV mislead when comparing mutually exclusive projects with unequal lives, and how does the equivalent annual annuity method fix this? State two limitations of the method.

Problems

Problem 7.1. A firm buys equipment for $800,000, depreciated straight-line to zero over four years. The project adds $1,500,000 revenue and $1,100,000 cash costs per year for four years. NWC of $80,000 is required at the start and recovered at the end of year 4. The equipment is sold at the end of year 4 for $100,000. Tax is 30% and the required return is 12%. Compute the annual operating cash flow and the NPV.

Problem 7.2. Equipment costs $500,000. Compare the PV of depreciation tax shields at a 30% tax rate and 9% discount rate under (a) straight-line over five years and (b) an accelerated schedule of 33.33%, 44.45%, 14.81% and 7.41% over four years.

Problem 7.3. A project's real cash flows are $50 per year for three years, with an initial outlay of $120 (today's prices). The real discount rate is 6% and inflation is 4%. (a) Find the nominal discount rate. (b) Compute NPV with nominal cash flows at the nominal rate. (c) Compute NPV with real flows at the real rate. (d) Compute the NPV that results from the mismatch of discounting real flows at the nominal rate.

Problem 7.4. Machine X costs $80,000, lasts four years and has annual operating costs of $12,000. Machine Y costs $120,000, lasts six years and has annual operating costs of $8,000. Ignoring taxes, compute the equivalent annual cost of each at 8% and recommend one.

Problem 7.5. An old machine has a tax book value of $200,000 with four years of straight-line depreciation remaining ($50,000 per year) and a market value of $90,000; it costs $100,000 per year to run and has no salvage. A new machine costs $300,000, is depreciated straight-line to zero over four years, costs $45,000 per year to run and sells for $40,000 at the end of year 4. Tax is 30%, the required return is 10%. Compute the NPV of replacing.

Problem 7.6. A firm owns land bought long ago. A new plant would cost $2,000,000 (straight-line to zero over six years) and use the land, which could otherwise be leased for $100,000 per year pre-tax. The plant produces pre-tax operating cash flow of $900,000 per year before the lease consideration. Tax is 30% and the discount rate 11%. Compute the NPV (a) ignoring the land's opportunity cost and (b) charging it. Does the decision change?

Problem 7.7. A project's sales are 800, 1,200, 1,000 and 600 in years 1 to 4 ($000). NWC is 18% of the year's sales. NWC is fully recovered at the end of year 4. Compute the annual ΔNWC cash flows and the present value of the working-capital cash flows at 10%.

Problem 7.8. A machine with a tax book value of $100,000 is sold at a 30% tax rate. Compute the after-tax proceeds if the sale price is (a) $60,000 and (b) $140,000. Explain the difference in tax treatment.

Answers to Selected Problems

- **7.1.** Depreciation = 200,000; operating CF = (1,500,000 − 1,100,000 − 200,000) × 0.70 + 200,000 = $340,000 per year. Year 4 adds NWC recovery 80,000 and after-tax salvage 100,000 × 0.70 = 70,000, giving 490,000. NPV at 12% ≈ **$248,000** (248,026 with unrounded factors; in $ thousands, −880 + 340 × 3.037349 + 150 × 0.635518 ≈ 248).
- **7.2.** (a) 30% × 100,000 = 30,000 per year; PV ≈ **$116,690**. (b) Shields 49,995; 66,675; 22,215; 11,115 → PV ≈ **$127,014**. The accelerated schedule is worth about $10,325 more.
- **7.3.** (a) 1.06 × 1.04 − 1 = **10.24%**. (b) Nominal flows 52.00, 54.08, 56.24; NPV ≈ **13.65**. (c) 50 × 2.673012 − 120 = **13.65**. (d) 50 × annuity(10.24%, 3) − 120 ≈ **3.82**; the mismatch understates NPV by about 9.8.
- **7.4.** PV(X) = 80,000 + 12,000 × 3.312127 = 119,746 → EAC ≈ **$36,154**; PV(Y) = 120,000 + 8,000 × 4.622880 = 156,983 → EAC ≈ **$33,958**. Choose Y.
- **7.5.** Initial: −300,000 + 90,000 + 0.30 × (200,000 − 90,000) = −177,000. Annual: (100,000 − 45,000) × 0.70 + (75,000 − 50,000) × 0.30 = 46,000. Terminal: 40,000 × 0.70 = 28,000. NPV = −177,000 + 46,000 × 3.169865 + 28,000 × 0.683013 ≈ **−$12,062**. Do not replace.
- **7.6.** Operating CF without land cost = (900,000 − 333,333) × 0.70 + 333,333 = 730,000; NPV ≈ 730,000 × 4.230538 − 2,000,000 ≈ **$1,088,293**. With the lease charge: pre-tax 800,000 → CF = 660,000, NPV ≈ **$792,155**. The project is accepted either way, but the true value is about $296,000 lower.
- **7.7.** NWC levels: 144, 216, 180, 108 → flows −144, −72, +36, +180 (including final recovery). PV at 10% ≈ **−$40.4** (i.e. working capital costs about $40,400 of present value).
- **7.8.** (a) Loss of 40,000 gives a tax saving of 12,000; after-tax proceeds **$72,000**. (b) Gain of 40,000 is taxed 12,000; after-tax proceeds **$128,000**.

Chapter 8: Project Analysis under Uncertainty and Real Options

Learning Objectives

By the end of this chapter you should be able to:

- Explain why a single-point NPV is an incomplete summary of a project, and distinguish forecasting risk, market risk and project-specific risk.
- Perform and interpret one-variable sensitivity analysis, tornado-style rankings and scenario analysis, and recognise their limitations.
- Compute accounting, cash and financial break-even points and relate them to operating leverage.
- Describe how Monte Carlo simulation works, specify input distributions and correlations, and read its output responsibly.
- Build and roll back a decision tree for a sequential investment decision, and compute the value of information.
- Identify and value the real options embedded in projects (to expand, abandon and delay) and explain why flexibility adds value that static NPV misses.

Opening Vignette

(Illustrative scenario; the company is fictional.)

Kestrel Energy Storage is evaluating a $40 million plant to manufacture grid-scale battery cabinets. The finance team's model shows a base-case NPV of $6.2 million at the company's 11% cost of capital, and the chief executive, Priya Ramanathan, is inclined to approve. But the board's risk committee asks a different question: "How wrong can this be, and what can we do about it if we are wrong?"

Kestrel's analysts go back to the model. The base case assumes that the plant sells 12,000 cabinets a year at $4,100 each. A one-line sensitivity run reveals that a 10% fall in the selling price turns the NPV deeply negative, while a 10% fall in volume merely halves it. Price, not volume, is the variable that matters, and price depends on a competitor's capacity decisions that Kestrel cannot control. A three-scenario analysis (a glut of imported cabinets, the base case, and a subsidy-driven boom) produces NPVs ranging from deeply negative to far above the base case. A simulation that lets price, volume, input cost and the cost of capital vary together suggests roughly a one-in-three chance of a negative NPV.

The mood in the boardroom shifts from "should we do it?" to "how should we do it?" The operations director suggests building half the capacity now and reserving the site and permits for a second phase that would be triggered only if prices hold above a threshold. The chief financial officer points out that, in the language of this chapter, the project is not a single bet but a package: a small plant plus an *option to expand*. If the market disappoints, Kestrel can stop at phase one, perhaps even sell the equipment second-hand (an *option to abandon*). If it is unsure about the regulatory environment, it could wait a year for the subsidy rules to be settled (an *option to delay*), at the cost of ceding first-mover advantage.

None of these possibilities appears in the static discounted cash flow model, because the model assumes a fixed plan from which management never deviates. This chapter shows how to measure uncertainty (sensitivity, scenarios, break-even, simulation), how to analyse sequential decisions (decision trees), and how to put a value on managerial flexibility (real

options).

8.1 From Point Estimates to Understanding Risk

8.1.1 The problem with a single number

Chapter 7 showed how to turn forecasts into a net present value. But each input to that calculation (unit sales, price, costs, working capital, the life of the project, the discount rate) is an estimate. An NPV of $1.2 million computed from uncertain inputs is not a fact; it is a point on a distribution. Two projects with the same expected NPV can be very different propositions: one may be nearly certain, the other may have a one-in-three chance of destroying value. Capital budgeting under uncertainty has two purposes: to learn *which assumptions matter most*, so that management can focus its effort on them, and to learn *how much could go wrong* so that the firm can decide whether it can tolerate that outcome and whether it can reshape the project to reduce it.

8.1.2 Three kinds of uncertainty

It is useful to distinguish the sources of risk because they call for different responses.

- **Forecasting risk** (estimation risk) is the possibility that the forecasts are wrong because the analyst lacks information, such as a new product with no sales history. It can be reduced by research, pilots and market testing, and its resolution has value, as the decision tree section will show.
- **Market (systematic) risk** is the exposure of project cash flows to the economy as a whole. This is the risk that investors price and that the discount rate should reflect, through the capital asset pricing model or similar. It cannot be diversified away by shareholders.
- **Project-specific (idiosyncratic) risk** is risk that is unique to this project. In a well-diversified portfolio it does not affect the required return, but it matters to managers because it affects the likelihood of failure, the cost of financial distress and the value of managerial flexibility.

The tools in this chapter mostly address the first and third kinds. They do not replace the discount rate; they complement it. A common error is to *raise the discount rate* to handle every kind of doubt. That penalises distant cash flows more heavily than near ones regardless of whether the risk is actually concentrated in the far future, and it fails to recognise that management can respond to bad news. A better practice is to set the discount rate for systematic risk and use scenario, simulation and option analysis to explore the rest.

8.1.3 A running example

Several sections of this chapter use the same base-case project so that the techniques can be compared. Cirrus Components, a fictional manufacturer, is evaluating a new product line.

Input	Base-case value
Investment (equipment)	$4,000,000, straight-line over 5 years, no salvage
Annual unit sales	100,000
Price per unit	$60
Variable cost per unit	$35
Fixed cash costs	$800,000 per year
Project life	5 years
Tax rate	30%
Required return	12%
Working capital	Ignored (to keep the calculations focused)

The base-case calculation follows the method of Chapter 7.

```
Contribution       = 100,000 × (60 − 35)              = 2,500,000
Less fixed cash costs                                  = 800,000
EBITDA                                                 = 1,700,000
Less depreciation (4,000,000 / 5)                      = 800,000
EBIT                                                   = 900,000
Tax at 30%                                             = 270,000
Net operating profit after tax                         = 630,000
Operating cash flow = 630,000 + 800,000                = 1,430,000

Annuity factor (12%, 5 years) = [1 − 1.12^(−5)] / 0.12 = 3.604776
NPV = −4,000,000 + 1,430,000 × 3.604776                = $1,154,830
```

The base NPV is a positive $1.15 million. The remaining sections ask how robust it is.

8.2 Sensitivity Analysis

8.2.1 The method

Sensitivity analysis changes one input at a time, holding all others at their base-case values, and records the effect on NPV. Its steps are:

1. Build the base-case model with every input in a separate, labelled cell.
2. Choose a plausible range for each input (such as ±10% and ±20% of its base value, or a range based on the analyst's judgement of the best and worst cases).
3. Recompute NPV at each end of each range.
4. Rank the inputs by the size of the swing in NPV they produce.

A chart of the ranked swings is known as a **tornado diagram**, because the widest bar sits at the top and the shape narrows downward. Tabulated results serve the same purpose, and Example 8.1 does so. A related summary is the **elasticity** or **sensitivity coefficient**, the percentage change in NPV for a one-percent change in an input.

8.2.2 What sensitivity analysis tells us

Sensitivity analysis answers the question “which assumptions matter most?” It identifies where research should be concentrated: if a 10% change in selling price swings NPV by three million dollars but the same change in fixed costs by only 400,000, then precise knowledge of price is worth far more than a precise estimate of overhead. It also reveals hidden fragility: a project that looks comfortable in the base case may have an NPV that is wiped out by a small adverse change in one variable.

8.2.3 Limitations

The method has real weaknesses, and a good analyst keeps them in mind.

- **Variables are not independent.** A price decline typically comes with a volume change (if demand is elastic), and an economic slowdown moves price, volume and costs together. One-at-a-time changes ignore correlations and can paint an unrealistic picture.
- **The ranges are subjective.** A ±10% swing in each input treats them as equally uncertain. In practice, the price of a commodity may have a range of ±40% and fixed costs ±5%. Sensible analysis uses ranges that reflect real uncertainty, which means the ranking depends heavily on how those ranges are chosen.
- **No probabilities.** The analysis does not say how likely each outcome is.
- **Management response is ignored.** It assumes the firm continues unchanged, even when conditions would prompt it to cut costs or close the project.

Example 8.1 – Sensitivity analysis of the Cirrus project.

Starting from the base case (NPV = $1,154,830), each input is varied by ±10% in turn. For each case the NPV is recalculated from the full model.

Price +10% (to $66). Contribution = 100,000 × (66 − 35) = 3,100,000. EBIT = 3,100,000 − 800,000 − 800,000 = 1,500,000. After-tax = 1,050,000. OCF = 1,050,000 + 800,000 = 1,850,000. NPV = −4,000,000 + 1,850,000 × 3.604776 = **$2,668,836**.

Price −10% (to $54). Contribution = 100,000 × 19 = 1,900,000. EBIT = 300,000; after-tax 210,000; OCF = 1,010,000. NPV = −4,000,000 + 1,010,000 × 3.604776 = **−$359,176**.

Volume ±10%. At 110,000 units contribution is 2,750,000; EBIT = 1,150,000; after-tax 805,000; OCF = 1,605,000; NPV = **$1,785,666**. At 90,000 units contribution is 2,250,000; EBIT = 650,000; after-tax 455,000; OCF = 1,255,000; NPV = **$523,994**.

Variable cost ±10%. At $38.50 the unit contribution is $21.50 and total contribution 2,150,000; EBIT = 550,000; after-tax 385,000; OCF = 1,185,000; NPV = **$271,660**. At $31.50 the unit contribution is $28.50, so total contribution is 2,850,000; EBIT = 1,250,000; after-tax 875,000; OCF = 1,675,000; NPV = **$2,038,000**.

Fixed costs ±10%. Each $80,000 change in fixed cost changes after-tax cash flow by 80,000 × 0.70 = 56,000; the NPV changes by 56,000 × 3.604776 = $201,867 in the opposite direction. NPV at +10% = **$952,963**; at −10% = **$1,356,697**.

Investment ±10%. At $4,400,000 the depreciation is $880,000; EBIT = 820,000; after-tax 574,000; OCF = 1,454,000; NPV = −4,400,000 + 1,454,000 × 3.604776 = **$841,345**. At $3,600,000, depreciation is $720,000; EBIT = 980,000; after-tax 686,000; OCF = 1,406,000; NPV = **$1,468,315**.

The results are summarised in the following table, ordered by the size of the swing.

Input (±10%)	NPV at unfavourable	NPV at favourable	Swing (range)	% change in NPV for 10% adverse move
Price	−359,176	2,668,836	3,028,012	−131.1%
Variable cost	271,660	2,038,000	1,766,340	−76.5%
Volume	523,994	1,785,666	1,261,672	−54.6%
Investment	841,345	1,468,315	626,970	−27.1%
Fixed costs	952,963	1,356,697	403,734	−17.5%

Interpretation. Price is by far the most important variable: a 10% fall in price takes the project from +$1.15 million to −$0.36 million. This is the effect of *operating leverage*: because costs are largely independent of price, every dollar of price decline falls straight through to profit. Unit variable cost is second, and volume third; fixed costs and even the size of the investment matter much less. The managerial implication is clear. Spend the research budget on pricing power and competitive response, and negotiate hard on variable costs, rather than shaving the capital budget. Note that the ranking would change if the plausible range for volume were ±40% while price were known within ±3%; the table depends on the ranges chosen, not only on the model's structure.

8.3 Scenario Analysis

8.3.1 Moving variables together

Scenario analysis addresses the main weakness of sensitivity analysis by changing several inputs *simultaneously* in a internally consistent story. A typical set has three scenarios: a pessimistic (downside), a base and an optimistic (upside) case. Each scenario is a coherent narrative about the future, not just a set of numbers: "a recession cuts demand, the competitor undercuts our price, and input costs rise." The consistency of the story matters, because it is the correlation among inputs that drives the extremes.

Scenarios may be assigned subjective probabilities, which allow a probability-weighted expected NPV and a measure of dispersion:

$$\text{Expected NPV} = \sum p_i \times \text{NPV}_i$$
$$\sigma(\text{NPV}) = \sqrt{\left[\sum p_i \times (\text{NPV}_i - \text{Expected NPV})^2\right]}$$

8.3.2 Building good scenarios

Several practical principles apply. First, tie scenarios to *drivers* of the business, such as regulation, technology adoption or competitor behaviour, instead of arbitrary percentage shocks. Second, avoid making the worst case an implausible catastrophe or the best case a fantasy; the point is to cover the plausible range. Third, include management responses where plausible, as the firm will not mechanically continue a failing project. Fourth, resist giving excessive precision to the probabilities: they are judgemental, and the exercise is more valuable for what it reveals about the structure of risk than for the arithmetic.

The weakness of scenario analysis is that it considers only a handful of discrete outcomes, whereas reality offers a continuum. It also depends on the analyst's ability to imagine the relevant scenarios. Scenario planning exercises in practice often miss the event that actually occurs, which is why they complement but do not replace simulation.

Example 8.2 – Three scenarios for the Cirrus project.

The product team defines three scenarios.

Input	Pessimistic	Base	Optimistic
Units per year	70,000	100,000	130,000
Price ()|54|60|64||*Variablecost*()	38	35	33
Fixed cash cost ($)	900,000	800,000	750,000
Probability	25%	50%	25%

Pessimistic. Contribution = 70,000 × (54 − 38) = 1,120,000. EBIT = 1,120,000 − 900,000 − 800,000 = −580,000. Assuming the firm can use the loss against profits elsewhere, tax is −174,000, so net operating profit = −406,000 and OCF = −406,000 + 800,000 = 394,000. NPV = −4,000,000 + 394,000 × 3.604776 = **−$2,579,718**.

Base. NPV = **$1,154,830** (from Section 8.1).

Optimistic. Contribution = 130,000 × (64 − 33) = 4,030,000. EBIT = 4,030,000 − 750,000 − 800,000 = 2,480,000. After-tax = 1,736,000. OCF = 2,536,000. NPV = −4,000,000 + 2,536,000 × 3.604776 = **$5,141,712**.

Expected value and dispersion.

```
Expected NPV = 0.25 × (−2,579,718) + 0.50 × 1,154,830 + 0.25 × 5,141,712
             = −644,930 + 577,415 + 1,285,428 = $1,217,913

Variance = 0.25 × (−3,797,631)² + 0.50 × (−63,083)² + 0.25 × (3,923,799)²
         ≈ 3.6055×10¹² + 0.0020×10¹² + 3.8492×10¹² = 7.4567×10¹²
σ(NPV)   ≈ $2,730,667
```

The project has a positive expected NPV of about $1.22 million but a standard deviation of about $2.73 million, more than twice the expected value. There is a 25% chance (the probability assigned to the pessimistic story) of losing $2.58 million. Contrast this with the sensitivity analysis: the scenario spread is far wider than the ±10% swings, because the scenarios combine adverse moves in price, volume and costs. A board that considered only the base case would see a comfortable project; a board that sees the scenarios must ask whether the company can absorb a $2.6 million loss and whether anything can be done to reduce its probability.

8.4 Break-Even Analysis

8.4.1 Three break-even concepts

Break-even analysis asks: how bad can things get before the project fails? It reports the value of one input at which a performance measure just reaches its threshold. Three kinds of break-even are used in capital budgeting, and distinguishing them avoids a classic confusion.

- **Accounting break-even** is the sales level at which accounting profit (EBIT or net income) is zero. At that level the project covers all operating costs and depreciation, but earns nothing for its capital.
- **Cash break-even** is the sales level at which operating cash flow is zero. Because depreciation is non-cash, this level is lower than the accounting break-even. A project at cash break-even covers its cash costs but never recovers its investment. (The standard textbook convention assumes no tax credit on losses here, so that OCF equals EBITDA.)
- **Financial break-even** is the sales level at which NPV is zero, so the project earns exactly the required return. This is the economically meaningful one and it is always higher than the accounting break-even for a project with positive investment, because it also requires a return on the capital tied up.

The formulas, with Q the volume, P price, v unit variable cost, FC fixed cash costs, D depreciation and T the tax rate, are:

```
Accounting break-even:  Q = (FC + D) / (P − v)
Cash break-even:        Q = FC / (P − v)                          (OCF = EBITDA = 0)
Financial break-even:   Q = [ FC + D + (EOCF − D) / (1 − T) ] / (P − v)
   where EOCF = I / Annuity factor(r, n) is the equivalent annual operating cash flow needed for NPV = 0
```

Break-even points can also be computed for price, variable cost or any other input, holding the rest constant. The distance between the base-case value and the break-even point is a **margin of safety**, often expressed as a percentage.

8.4.2 Operating leverage

Fixed costs magnify the effect of sales changes on profit. The **degree of operating leverage (DOL)** measures this:

```
DOL = Contribution / EBIT = % change in EBIT / % change in sales
```

For Cirrus, DOL = 2,500,000 / 900,000 = 2.78: a 1% change in volume changes EBIT by 2.78%. High operating leverage means a large profit when volumes are high and a rapid collapse when they fall. It also means that break-even is closer to the base case. Capital-intensive and technology-heavy projects, such as chemical plants, airlines or software with large development costs, generally have high operating leverage, and their managers should look at the margin of safety closely.

Example 8.3 – Break-even points for the Cirrus project.

Fixed cash costs are $800,000, depreciation is $800,000, unit contribution is P − v = $25 and the required annual operating cash flow for NPV = 0 is found as follows.

Accounting break-even. Q = (800,000 + 800,000) / 25 = **64,000 units**. At this volume EBIT is zero, OCF = depreciation = $800,000 and the NPV is −4,000,000 + 800,000 × 3.604776 = **−$1,116,179**. The project is accounting-profitable and still destroys over a million dollars of value, a vivid illustration that accounting break-even is not a safe criterion.

Cash break-even. Q = 800,000 / 25 = **32,000 units**. At this volume OCF is zero and NPV = **−$4,000,000**: the firm loses the entire investment.

Financial break-even. The annual OCF required for NPV = 0 is 4,000,000 / 3.604776 = $1,109,639. Since OCF = EBIT × (1 − T) + D, the required EBIT is (1,109,639 − 800,000) / 0.70 = $442,341. Then contribution required is EBIT + FC + D = 442,341 + 800,000 + 800,000 = $2,042,341, so

```
Q = 2,042,341 / 25 = 81,694 units
```

The margin of safety on volume is (100,000 − 81,694) / 100,000 = **18.3%**.

Other break-evens (holding other inputs at base).

Variable	Base value	Break-even (NPV = 0)	Margin of safety
Units per year	100,000	81,694	18.3%
Price per unit	$60.00	$55.42	7.6%
Variable cost per unit	$35.00	$39.58	13.1%
Fixed cash costs	$800,000	$1,257,659	57.2%

The table reinforces the sensitivity analysis. The project can lose almost 18% of volume or almost 13% of cost-side slack but only 7.6% of price before value is destroyed, while the fixed cost has a very thick cushion. The three break-even definitions for volume (32,000, 64,000 and 81,694 units) are ordered cash < accounting < financial, and only the last

answers the question the board is really asking.

8.5 Monte Carlo Simulation

8.5.1 The concept

Sensitivity analysis varies one input; scenario analysis a handful of combinations. **Monte Carlo simulation** takes the idea to its natural conclusion: it specifies a probability distribution for each uncertain input, draws a random value from each, computes the project's NPV for that draw, and repeats this thousands of times. The collection of NPVs is an empirical probability distribution of project value. The method is named after the casino and is attributed to work on numerical problems in the 1940s, but its logic is just repeated sampling.

The steps are:

1. **Build the deterministic model** with all inputs on separate lines (as in Chapter 7).
2. **Choose the uncertain inputs** (the ones the sensitivity analysis shows matter) and assign each a distribution: normal for quantities with symmetrical errors, triangular for variables where the analyst can state a minimum, most likely and maximum, lognormal for strictly positive variables with a right skew, and so on.
3. **Specify correlations** among the inputs, because variables that move together make extreme outcomes more likely than independent draws suggest.
4. **Run many trials** (typically thousands). In each, draw every uncertain input, compute the cash flows and the NPV.
5. **Analyse the output:** the mean and standard deviation of NPV, the probability that NPV is negative, percentiles, and a histogram. Rank the inputs by their correlation with NPV to see which uncertainty drives the result.

In pseudo-code:

```
for trial = 1 to N:
    units = draw Normal(100,000; 20,000), floored at 0
    price = draw Triangular(min 52, mode 60, max 66)
    vcost = draw Triangular(min 32, mode 35, max 41)
    fixed = draw Normal(800,000; 60,000)
    OCF   = ((units × (price − vcost) − fixed − 800,000) × 0.70) + 800,000
    NPV[trial] = −4,000,000 + OCF × 3.604776
report mean(NPV), stdev(NPV), share of trials with NPV < 0, percentiles
```

8.5.2 Strengths and cautions

Simulation can handle many inputs, skewed distributions, non-linear relationships and path-dependent rules (for example, "if cumulative sales fall below a threshold, cut capacity"). Its output speaks in the language of risk: "there is a one-in-three chance of a loss." But it is no better than its inputs. The main cautions are these.

- **Garbage in, garbage out.** The distributions are subjective, and an elaborate simulation can lend a false sense of precision to guesses. Document why each distribution was chosen.
- **Correlation is hard to specify and easy to omit.** Assuming independence generally understates the spread of outcomes, because in practice good and bad years cluster.
- **Fat tails.** Normal distributions understate the probability of extreme events. If the real risk includes rare disasters, choose distributions that allow them.

- **Misleading discounting.** The simulation's mean NPV should be computed with a discount rate that reflects only market risk; using a high "risk-adjusted" rate on top of a simulation that already includes the project's variability double-counts risk.
- **It cannot choose for you.** The result is a distribution, not a decision. The firm must still decide how much risk it can bear.

Example 8.4 – A simulation of the Cirrus project.

Cirrus's analysts specify these distributions for the uncertain inputs, drawn independently: units per year Normal(mean 100,000; σ 20,000) floored at zero; price Triangular(52, 60, 66); variable cost Triangular(32, 35, 41); fixed cash costs Normal(800,000; 60,000). Depreciation, the tax rate and the discount rate are fixed. A 10,000-trial simulation (random seed 42 in a standard numerical library) produced the results below. Because this is a random sample, a rerun with a different seed would produce slightly different figures; the pattern, not the third digit, is what matters.

Statistic	Result
Mean NPV	$727,072
Standard deviation of NPV	$1,481,661
5th percentile	−$1,560,379
25th percentile	−$300,039
Median	$641,708
75th percentile	$1,672,323
95th percentile	$3,307,583
Probability NPV < 0	32.4%
Probability NPV < −$1,000,000	11.4%
Probability NPV > $2,000,000	18.9%

Why is the mean below the deterministic NPV of $1,154,830? The triangular distributions are skewed against the firm: the average price is (52 + 60 + 66)/3 = $59.33 and the average variable cost is (32 + 35 + 41)/3 = $36.00, so the expected unit margin is $23.33 rather than $25. With 100,000 expected units, the expected contribution is about $2.33 million, not $2.5 million, which alone lowers NPV by about $0.43 million. This is a general point: NPV evaluated at the *expected* inputs is not always the *expected* NPV when distributions are asymmetric, and the simulation's mean is the better estimate of expected value if the distributions are sound.

What drives the risk? The correlation of each input draw with the resulting NPV across the trials is approximately 0.80 for units, 0.49 for price, −0.32 for variable cost and −0.09 for fixed costs. In the sensitivity analysis price had the largest effect per 10% move, but under realistic uncertainty ranges, volume (with a 20% standard deviation) dominates, while price (a narrow triangular range) comes second. The two tools answer different questions: sensitivity asks what the model is most sensitive to, simulation asks what matters given how uncertain we actually are. A manager should keep both in mind. It is also noteworthy that a project with a deterministic NPV of +$1.15 million has a probability of about 32% of destroying value once the uncertainty is acknowledged.

8.6 Decision Trees and Sequential Decisions

8.6.1 Why a tree?

The tools so far treat the investment as a one-time, all-or-nothing choice. Real projects unfold in stages: a pilot, a regulatory ruling, a market test, a first-phase build. At each stage management learns something and can change course. A **decision tree** models this sequence explicitly. It has two kinds of nodes.

- **Decision nodes** (conventionally drawn as squares) are points where management chooses among alternatives, such as invest, wait, expand or abandon.
- **Chance nodes** (circles) are points where uncertainty resolves, with a probability attached to each branch.

The tree is solved by **backward induction** (rollback): start at the end of the tree, compute the value of each terminal payoff, take the probability-weighted expected value at each chance node, choose the best alternative at each decision node, and work back to the present, discounting along the way. Because the decision made at a later node depends on information revealed earlier, the tree values the *policy*, not just a single path.

8.6.2 Value of information

A pilot, a test market or a feasibility study buys information. The tree shows what that information is worth: the extra expected value from being able to choose the best action in each state of the world, instead of committing in advance. If the information is worth more than it costs, the study is justified. Here the contrast with Chapter 7's treatment of sunk costs is instructive: *after* the money is spent it is irrelevant; *before* it is spent, the study is an investment decision that deserves its own NPV.

8.6.3 Cautions

Decision trees are easy to draw and difficult to parameterise. Three issues recur. First, the probabilities are usually subjective. Second, the discount rate should reflect the risk of the cash flows at each branch, and a single rate applied throughout is an approximation; the more rigorous approach, used for options below, adjusts probabilities rather than rates. Third, trees grow quickly: realistic problems with many stages and outcomes become unmanageable, which is one reason analysts prefer simulation or option pricing for continuous uncertainty.

Example 8.5 – Pilot plant or immediate launch? (amounts in $ millions)

Orchard Biomaterials has a new process for compostable packaging film. Technical feasibility is uncertain. Management can (a) build a pilot plant now at a cost of $1.0 million, learning after one year whether the process works commercially, or (b) commit to the full-scale plant ($8.0 million) immediately. The cost of capital is 10%.

The facts are as follows. The process works with probability 0.40. If it works, the present value (at year 1) of the full-scale plant's future cash flows is $14.0 million. If it fails, the plant would be worth only $3.0 million. The full-scale investment of $8.0 million is made at year 1 if the pilot is run (after the result is known), or is made at once (with the outcome revealed afterwards) if the firm skips the pilot. For comparability, value the immediate launch with the same year-1 payoffs.

Strategy A – Immediate full launch. The expected year-1 value of the plant is $0.40 \times 14.0 + 0.60 \times 3.0 = 5.6 + 1.8 = 7.4$. Present value = 7.4 / 1.10 = 6.727. NPV = −8.0 + 6.727 = **−$1.273 million**. The firm would reject it.

Strategy B – Pilot first. Roll back the tree.

Node	Branch	Probability	Payoff at year 1	Decision
Pilot result (chance)	Works	0.40	Invest 8.0; value 14.0; net 6.0	Invest
Pilot result (chance)	Fails	0.60	Invest 8.0; value 3.0; net −5.0	Do not invest; payoff 0

Expected payoff at year 1 = 0.40 × 6.0 + 0.60 × 0 = 2.40. Present value = 2.40 / 1.10 = 2.182. Subtract the pilot cost: NPV = 2.182 − 1.0 = **+$1.182 million**.

Comparison. The pilot strategy has NPV +1.18 million; immediate launch has −1.27 million; doing nothing has 0. The pilot turns a project that fails the NPV test into one that creates $1.18 million, because it lets management avoid the $5.0 million loss that the failure branch would otherwise bring. The pilot's gross value (its worth before cost) is $2.18 million, so a pilot costing up to $2.18 million would still be justified.

Break-even probability. The pilot pays if 6.0p / 1.10 ≥ 1.0, i.e. p ≥ 0.183. If the chance of technical success were below 18.3%, even the pilot would not be worth running. The following table shows the sensitivity.

Probability of success	Pilot strategy NPV (m)|$ImmediatelaunchNPV$(m)	
0.20	0.09	−3.27
0.30	0.64	−2.27
0.40	1.18	−1.27
0.50	1.73	−0.27

(Immediate-launch NPV = −8.0 + [p × 14.0 + (1 − p) × 3.0] / 1.10. For p = 0.50 this is −8.0 + 8.5 / 1.10 = −8.0 + 7.727 = −0.273.) Even at a 50% chance of success, immediate launch is still slightly negative, and the pilot strategy continues to dominate. In every row the staged approach is better than going all in, an illustration of a general principle: when a large irreversible commitment can be preceded by a cheap test, the test is valuable in proportion to the uncertainty it resolves and the loss it avoids.

8.7 Real Options

8.7.1 Flexibility has value

A **real option** is the right, but not the obligation, to take an action on a real asset (to expand, contract, delay, switch inputs or abandon) at a predetermined cost, within a specified time. The term was developed by analogy with financial options, whose valuation is covered in Chapter 18, and the central insight is the same: *the right to choose is valuable, and that value rises with uncertainty.* A static NPV analysis assumes that management sets the plan at the outset and never revises it. A project with embedded flexibility is worth more than that analysis shows:

```
Project value = Static (passive) NPV + Value of embedded real options
```

Different kinds of flexibility give different options.

Option	What management can do	Financial analogue	Typical example
Delay (timing)	Wait to invest until uncertainty resolves	Call option	Waiting for a licence, a price recovery or a regulatory decision
Expand (growth)	Invest more if conditions prove favourable	Call option	Pilot plant with permitted capacity for a second phase
Abandon	Sell the assets or stop if conditions are poor	Put option	Resale value of aircraft or equipment
Contract/scale down	Reduce scale to save costs	Put option	Mothballing a mine in a price slump
Switch	Change inputs or outputs	Option to exchange	Dual-fuel power plant; flexible production line
Compound (staged)	Each stage creates the option to start the next	Option on an option	R&D programmes, drug trials

8.7.2 Why options matter more under uncertainty

Option value comes from asymmetry: management captures the upside of favourable developments and limits the downside of unfavourable ones. The more volatile the underlying value, the more valuable the right to decide after seeing the outcome. This contradicts the instinct that uncertainty is always bad. For an inflexible commitment, higher volatility increases the risk of loss without improving the average outcome, but for a flexible project, higher volatility increases value. It also explains why firms in uncertain industries often favour modular, staged or reversible investments, accept a lower static NPV for a design that keeps options open, and why a standard NPV rule ("invest if NPV > 0") can mislead when waiting is possible.

8.7.3 Valuation approaches

Three approaches are common, in increasing order of sophistication.

1. **Decision-tree analysis** with probabilities and a risk-appropriate discount rate, as in Example 8.5. It is transparent but the choice of discount rate is difficult: the risk of a project with an option differs from that of the passive project.
2. **Binomial (risk-neutral) valuation.** Model the underlying value of the project as moving up or down over each period. Find the risk-neutral probability q that makes the expected return equal to the risk-free rate, compute the option's payoffs, and discount at the risk-free rate:

```
u = up factor, d = down factor, rf = risk-free rate per period
q = (1 + rf − d) / (u − d)          (risk-neutral probability of the up move)
Option value = [ q × Payoff(up) + (1 − q) × Payoff(down) ] / (1 + rf)
```

If the underlying pays out cash (so that holding the option means giving up cash flows), the expected growth in the underlying value must be reduced by the payout yield δ, and q = (1 + rf − δ − d) / (u − d). This method avoids the problem of the risky discount rate by moving the risk adjustment into the probabilities. 3. **Black-Scholes and its extensions** apply when the option is European-style, with a known expiry and continuous movement in value. The mapping is: the underlying asset price S is the present value of the cash flows from the project, the exercise price K is the investment cost, T is the time to the decision deadline, σ is the volatility of the project's value, r is the risk-free rate, and the dividend yield is the fraction of project value lost per period by waiting (the cash flows forgone or competitive erosion).

The practical limits are real. The inputs (especially σ and the "price" of the underlying, which is not traded) are hard to observe, many options are shared with competitors or interact with each other (the value of two options together is less than the sum if they are substitutes), and option thinking can become an excuse for projects that cannot pass the NPV test on merit. A fair test is whether the claimed option is real: can management actually exercise it, at the stated cost, at the stated time, and will it be rational to do so?

8.7.4 The option to expand

An option to expand arises when an initial investment creates the possibility of a larger follow-on investment. A first-phase plant, a market entry in one country, a licence for a new technology, or a research project all have this feature. The option has the payoff of a call: expand only if the value created exceeds the cost.

Example 8.6 – A first-phase plant with an expansion option (amounts in $ millions).

Kestrel's phase-one plant costs $10.0 million. The present value of its cash flows today, discounted at the appropriate rate, is $9.4 million, so the static NPV is 9.4 − 10.0 = **−$0.6 million**: on its own, the project fails. However, one year from now Kestrel has the right to spend $4.0 million to add capacity equal to 50% of the original plant's value. Over the next year the value of the original project will either rise by a factor of u = 1.40 or fall by a factor of d = 0.80. The risk-free rate is 5%.

Step 1 – Risk-neutral probability. q = (1 + 0.05 − 0.80) / (1.40 − 0.80) = 0.25 / 0.60 = **0.4167**.

Step 2 – Values of the project after one year.

```
Up state:   V = 9.4 × 1.40 = 13.16 ; expansion adds 0.50 × 13.16 = 6.58 ; cost 4.0 ; payoff = 2.58
Down state: V = 9.4 × 0.80 =  7.52 ; expansion adds 0.50 ×  7.52 = 3.76 ; cost 4.0 ; payoff = 0 (do not
expand)
```

Step 3 – Option value.

```
Expansion option = [0.4167 × 2.58 + 0.5833 × 0] / 1.05 = 1.075 / 1.05 = $1.024 million
```

Step 4 – Total. The project's value with the option is the static NPV plus the option: −0.6 + 1.024 = **+$0.424 million**. The plant should be built even though its stand-alone NPV is negative: the firm is in effect paying $0.6 million for an option worth $1.024 million. The logic transfers directly to the vignette: building in phases with a reserved site can turn an unattractive all-at-once plan into an attractive one. The decision rule is not "NPV greater than zero" but "static NPV plus option value greater than zero." Note also what the example does not say: the expansion will be exercised in only one of the two states, so it will not be exercised about 58% of the time on a risk-neutral basis. The option has value precisely because the firm is not forced to expand when conditions are poor.

8.7.5 The option to abandon

If assets can be sold or redeployed, the firm has a floor under the project's value: it can walk away and receive the salvage value if the future looks bleak. This is a put option on the project. The option is more valuable when salvage values are high and secure, when the outcomes are uncertain, and when information arrives early enough to act on it. It is most valuable for assets with active secondary markets (trucks, aircraft, standard machinery) and least valuable for specialised assets such as a custom-built plant that has no other buyer. Note that firms often cannot costlessly abandon in practice: there are contractual obligations, severance costs and reputational damage, which should be deducted from the salvage value.

Example 8.7 – Valuing the option to abandon with a decision tree (amounts in $ millions).

A project requires an investment of $12.0 million today. Cash flows are uncertain: in year 1 the market reveals itself to be strong (probability 0.60) or weak (probability 0.40). If strong, the project generates $4.0 million in each of years 1 to 5. If weak, it generates $1.0 million in each of years 1 to 5. At the end of year 1, after receiving the year-1 cash flow, management has the right to abandon and sell the assets for a net $8.0 million. The cost of capital is 10%. The annuity factor for five years at 10% is 3.790787, and for four years is 3.169865.

Without the option (commit to five years).

```
Strong: PV = 4.0 × 3.790787 = 15.163
Weak:   PV = 1.0 × 3.790787 =  3.791
Expected PV = 0.60 × 15.163 + 0.40 × 3.791 = 9.098 + 1.516 = 10.614
NPV = 10.614 − 12.0 = −1.386
```

The project fails the NPV test.

With the option. In the strong state, continue. In the weak state, compare continuing with abandoning at year 1. Continuing yields years 2-5 cash flows with a value at year 1 of 1.0 × 3.169865 = 3.170, whereas abandoning yields 8.0. Abandon. The weak-state value at time 0 is therefore the year-1 cash flow plus the sale proceeds, discounted one year: (1.0 + 8.0) / 1.10 = 8.182.

```
Expected PV with option = 0.60 × 15.163 + 0.40 × 8.182 = 9.098 + 3.273 = 12.371
NPV with option = 12.371 − 12.0 = +0.371
```

Interpretation. The abandonment option is worth 12.371 − 10.614 = **$1.756 million**, and turns a project with NPV −$1.386 million into one with NPV +$0.371 million. The decision depends on the salvage value: the option is worth exercising in the weak state only if the salvage exceeds 3.170, and the project as a whole breaks even if the salvage value is about $6.98 million (solve 0.60 × 15.163 + 0.40 × (1.0 + S)/1.10 = 12.0 to get S ≈ 6.98). A manager considering this project should therefore ask how reliable the $8.0 million salvage estimate is, whether a buyer would exist in the weak market state (when other firms are also selling), and whether the firm can make the signal of weakness arrive early. This analysis uses subjective probabilities and a single discount rate and is therefore only an approximation; it is intended to show the logic, which is the same under a more rigorous risk-neutral approach.

8.7.6 The option to delay

Many investment opportunities do not expire immediately. A firm that owns a licence, a mineral claim, a patent or a site may invest now or wait. If waiting resolves uncertainty (a regulatory outcome, a commodity price, a technology standard), the firm can avoid investing in the bad states. The cost of waiting is the cash flows forgone, plus the risk that competitors will pre-empt the opportunity. The option to delay is a call option on the project, with exercise price equal to the investment cost.

The key insight is that a project with positive NPV should not necessarily be started at once. The correct rule is to invest now only if the NPV is large enough to exceed the value of waiting; otherwise the firm should wait. Said differently, the hurdle for an irreversible investment in an uncertain environment is higher than “NPV > 0”: the project value must exceed the investment cost by a *premium* that compensates for the lost option. This helps explain why firms sometimes appear to sit on seemingly profitable projects in volatile markets, and why investment can fall sharply when uncertainty rises even though expected profitability does not change.

Example 8.8 – Invest now or wait a year? (amounts in $ millions)

A firm can build a facility costing $50.0 million. The present value of its operating cash flows is $52.0 million, so the static NPV is +$2.0 million. Over the next year, the value of the facility will either rise to 1.30 times its present level or fall to 0.80 times. If the firm waits, it forgoes a payout (the first-year cash flow) equal to 4% of project value, and the risk-free rate is 5%. Either way, the firm may invest at year 1 for the same $50.0 million.

Step 1 – Risk-neutral probability, adjusted for the forgone payout.

```
q = (1 + 0.05 − 0.04 − 0.80) / (1.30 − 0.80) = 0.21 / 0.50 = 0.42
```

Step 2 – Values at year 1.

```
Up:   V = 52.0 × 1.30 = 67.6 ; invest : payoff = 67.6 − 50.0 = 17.6
Down: V = 52.0 × 0.80 = 41.6 ; do not invest (41.6 < 50.0) : payoff = 0
```

Step 3 – Value of waiting. [0.42 × 17.6 + 0.58 × 0] / 1.05 = 7.392 / 1.05 = **$7.04 million**.

Step 4 – Compare. Investing today is worth $2.0 million; waiting is worth $7.04 million. The firm should wait, even though NPV is positive today. Investing now "kills" the option, so the opportunity cost of acting early is the $7.04 million of option value, which must be deducted from the $2.0 million of NPV, giving a net value of acting now of −$5.04 million relative to the best alternative.

Step 5 – Where is the trigger? Repeating the calculation for different current values of the project gives the following.

Project value V today	NPV today (V − 50)	Value of waiting	Best action
52.0	2.0	7.04	Wait
55.0	5.0	8.60	Wait
58.0	8.0	10.16	Wait
60.0	10.0	11.20	Wait
62.0	12.0	12.24	Wait
65.0	15.0	14.90	Invest now
70.0	20.0	19.71	Invest now

The two lines cross at V = 62.5, where setting V − 50 = (1.01V − 50) / 1.05 gives V = 62.5, a project value 25% above the investment cost. Below that level the firm should wait; above it, the value of the option no longer exceeds the cost of delay. The trigger of 62.5 versus the naive threshold of 50 is the "option premium" that this stylised case implies. Its size depends on volatility (a wider range of outcomes raises the premium) and on the cost of waiting (a higher forgone payout, or a faster competitor, lowers it).

8.7.7 Interactions and strategy

Real options are not only a valuation device but also a way of thinking about strategy. Several principles follow.

- **Design for flexibility.** Choose modular technology, reserve land or permits, and negotiate contracts with exit and extension clauses. Flexibility has a price, so the question is whether the option is worth more than its premium.
- **Resolve uncertainty cheaply.** Pilot projects, trial launches and staged capital commitments are expensive in time but inexpensive in capital, and they let the firm learn before it commits.
- **Beware of competition.** Options to wait are valuable only if they are proprietary. If a competitor can preempt, the option to delay is worth much less and the case for investing early is stronger (a game-theoretic perspective beyond

this chapter).

- **Do not double count.** If the discount rate or the cash-flow forecast already includes a "flexibility premium," do not add the option value again.
- **Check exercisability.** Many "options" cited in strategy documents (e.g. "this gives us a platform for the future") are not options at all because there is no defined right, cost or time. Insist on specifying the exercise conditions.

8.8 Managerial Implications

8.8.1 A toolkit, not a single answer

Each technique in this chapter answers a different question, and a thorough capital budgeting review uses several together.

Technique	Question answered	Main strength	Main weakness
Sensitivity analysis	Which input matters most?	Simple, directs attention and research	One variable at a time; ranges arbitrary
Scenario analysis	What if a coherent set of things goes wrong or right?	Captures correlated moves; tells a story	Few outcomes; subjective probabilities
Break-even analysis	How far can we fall before failing?	Concrete margins of safety	Single variable; threshold only
Monte Carlo simulation	What is the distribution of NPV?	Handles many inputs, correlation, skewness	Needs defensible distributions
Decision tree	What should we do at each stage?	Values information and sequential choices	Subjective probabilities; discount rate issues
Real options	What is flexibility worth?	Values asymmetry, staging and timing	Hard inputs; risk of over-claiming

8.8.2 From analysis to action

The purpose of risk analysis is not to produce numbers for their own sake; it is to change what the firm does. Four kinds of action follow.

1. **Focus the research.** Spend the forecasting budget on the variables with the greatest effect, as sensitivity and simulation reveal. Test the price assumption, not the cost of office furniture.
2. **Reshape the project.** If a project's risk is concentrated in a variable the firm can influence, reduce it: by contracts (long-term supply and offtake agreements), by lowering fixed costs, by staging capital commitments or by pursuing partnerships.
3. **Build in options.** Prefer designs that allow expansion, abandonment and switching, and quantify what they are worth.
4. **Decide how much risk to take.** The distribution of outcomes should be compared with the firm's capacity to bear loss. A positive expected NPV with a 25% chance of a loss that would breach a debt covenant may be unacceptable, and the same project with a smaller loss exposure would be fine.

8.8.3 Organizational pitfalls

- **False precision.** Reporting a simulated probability of loss to one decimal place conveys more certainty than the inputs support.

- **Optimism bias and strategic misrepresentation.** Project sponsors may choose scenarios and distributions that favour approval. A review function that challenges input ranges is essential.
- **Anchoring on the base case.** Decision-makers tend to treat the base case as the expected outcome. Show the whole distribution.
- **Option inflation.** Claims of "strategic option value" can justify any project. Require an explicit exercise condition and a numerical estimate.
- **Neglect of correlation with the firm's other businesses.** A project that is risky stand-alone may reduce the firm's overall risk if its returns offset those of existing units; the reverse also holds. Portfolio effects are treated in the chapters on risk and return.

Case Study

Tidewater Water Technologies: The Desalination Module Decision

(Fictional company; all figures illustrative.)

Tidewater Water Technologies designs and operates small desalination plants for island resorts and coastal municipalities. Its board is considering whether to build a manufacturing facility for a standardised, containerised reverse-osmosis module that could be sold into several regional markets. The facility would cost $60 million, depreciated straight-line over ten years, and the company's cost of capital for this kind of risk is 10%.

The planning team's base case assumes that Tidewater sells 90 modules a year at an average price of $1.2 million, with variable cost of $0.75 million per module and fixed cash operating costs of $14 million per year. At a 25% tax rate, the model yields an NPV of roughly $11 million over a ten-year life. The chief financial officer, Imani Okafor, is not satisfied with a single number. She asks the team to explore the risks and to say what the company could do about them.

The team's first finding is that the forecast is far more sensitive to price than to anything else. Module prices are set in competitive tenders, and two Asian manufacturers have announced capacity expansions. A one-at-a-time analysis shows that a 5% reduction in price eliminates more than half the NPV, while a 5% reduction in volume reduces it by about a fifth. The team then builds three scenarios. In the "price war" scenario, new capacity floods the market: volume stays at 90 but price falls to $1.0 million and the plant's costs do not change; in this case the NPV is substantially negative. In the "steady growth" scenario, the base case holds. In the "water stress" scenario, droughts and rising tourist demand lift demand to 130 modules at $1.3 million. The team assigns probabilities of 30%, 45% and 25%. The probability-weighted NPV is positive but the standard deviation is about twice the expected value.

A Monte Carlo simulation, in which price, volume and unit cost are drawn from skewed distributions and price and volume are positively correlated (both fall in downturns), puts the probability of a negative NPV at roughly 35%. The board's risk committee notes that Tidewater's loan covenants limit the company's ability to absorb a loss of more than about $30 million.

Imani then raises several design alternatives. The first is to build the facility in two phases: a $35 million first phase with capacity for 60 modules a year, and a second phase costing $30 million, to be decided two years later, which would raise capacity to 140. The site's permits and utility connections would be sized for the larger plant at a premium of $2 million up front. The static NPV of phase one alone is only marginally positive, about $3 million, and the extra $2 million of permits

does not appear in the base case. The operations director argues that the phased design is “obviously better,” but the finance team points out that the second phase would be built only if prices hold above $1.1 million, and they want to know what the right to make that decision is worth.

A second alternative is to lease, rather than own, the production equipment, with a termination clause that allows Tidewater to return the equipment after year four on payment of a fee. This would raise annual costs but cap the downside. A third is to delay the whole project for 18 months until the outcome of a regulatory review of desalination subsidies in two major markets is known. The delay would cost Tidewater the first-mover relationship with two buyers that have signalled willingness to sign framework agreements, but it would resolve the largest uncertainty about price support.

Finally, the head of strategy points out that the facility’s modules are standardised and could be sold into the used-equipment market at perhaps 40% of cost, which provides a floor. The operations director disagrees: used modules already circulate at discounts of more than 50%, and sales volume would be thin in a downturn, precisely when Tidewater would need to sell.

Discussion Questions

1. Using the evidence in the case, rank the sources of uncertainty. What would you want the sensitivity analysis to show, and what additional scenarios or simulation features (such as correlation) matter, and why?
2. Describe how you would value the two-phase design. Which real option does it contain, what are its inputs, and how would you decide whether the $2 million permit premium is justified?
3. Evaluate the lease-with-termination alternative and the claim that used-equipment values provide an abandonment floor. What are the conditions under which an abandonment option is genuinely valuable?
4. How would you advise the board on the 18-month delay? State the trade-off, the information you would want, and the threshold at which you would recommend investing now.

Key Takeaways

- A single NPV is a point estimate; managers need to know how it could be wrong, which assumptions drive it, and what can be done about the risk.
- Use the discount rate for systematic risk and use sensitivity, scenario, simulation and option analysis for project-specific uncertainty; do not inflate the discount rate to cover every doubt.
- Sensitivity analysis ranks inputs by their effect on NPV, but ignores correlation and depends on the ranges chosen; it directs research effort rather than measuring risk.
- Scenario analysis varies several inputs together in coherent stories and can attach probabilities to give an expected NPV and dispersion, at the cost of covering only a few outcomes.
- Accounting break-even (EBIT = 0) is below financial break-even (NPV = 0); a project can be accounting-profitable and still destroy value, so the financial break-even and its margin of safety are what matter.
- Operating leverage amplifies the effect of sales changes on profit; high fixed costs push break-even closer to the base case.
- Monte Carlo simulation turns input distributions into a distribution of NPV and probabilities of loss, but depends critically on the distributions and the correlations assumed.
- Decision trees value sequential decisions by backward induction and show the value of information; a cheap test before a large commitment can turn a rejected project into an accepted one.

- Real options (to expand, abandon, delay) are embedded rights whose value rises with uncertainty; project value is static NPV plus option value.
- The hurdle for an irreversible investment that can be postponed exceeds zero NPV, and option claims must be tested for a defined right, cost and time of exercise.

Review Questions

1. Why can two projects with the same expected NPV be very different investments? Illustrate with the measures introduced in this chapter.
2. Distinguish forecasting risk, market risk and project-specific risk. Which should be reflected in the discount rate?
3. Explain how a tornado diagram is built and why its ranking depends on the ranges assumed for each input.
4. What are the limitations of one-at-a-time sensitivity analysis? How does scenario analysis address one of them, and what new weakness does it introduce?
5. Define accounting, cash and financial break-even. Why is the financial break-even the relevant one for an investment decision?
6. What does the degree of operating leverage measure, and how does it affect the margin of safety?
7. Outline the steps in a Monte Carlo simulation. Why does omitting correlation among inputs usually understate risk?
8. Why can the mean of a simulated NPV distribution differ from the NPV computed at the expected values of the inputs?
9. Explain backward induction in a decision tree and how a tree can be used to value information such as a pilot study.
10. Why does greater uncertainty increase the value of a real option? Identify one condition under which an option to wait is worth little.

Problems

Unless stated otherwise, use the Cirrus-style model of this chapter: operating cash flow = (Units × (Price − Variable cost) − Fixed cash costs − Depreciation) × (1 − T) + Depreciation, with no working capital or salvage.

Problem 8.1 (Sensitivity). A project costs $2,000,000 (straight-line over 5 years, no salvage) and sells 50,000 units a year at $80, with variable cost $50 per unit and fixed cash costs $600,000 per year. The tax rate is 30% and the required return is 10% (annuity factor for 5 years = 3.790787). Compute the base-case NPV and the NPV when (a) price is 10% lower and 10% higher, and (b) variable cost is 10% higher and 10% lower. Which variable is the NPV more sensitive to?

Problem 8.2 (Scenarios). For the project in Problem 8.1, define three scenarios. Pessimistic (probability 30%): 35,000 units, price $74, variable cost $54, fixed costs $650,000. Base (50%): as above. Optimistic (20%): 60,000 units, price $84, variable cost $48, fixed costs $580,000. Compute the NPV in each scenario, the expected NPV, the standard deviation and the probability of a negative NPV (assume losses generate tax credits).

Problem 8.3 (Break-even). For the project in Problem 8.1, compute (a) the accounting break-even volume, (b) the cash break-even volume (assuming no tax credit on losses, so OCF = EBITDA), (c) the financial break-even volume and the margin of safety relative to 50,000 units, and (d) the NPV at the accounting break-even volume.

Problem 8.4 (Decision tree, $ millions). A firm can spend $0.6 million on a test market now. With probability 0.50 the test succeeds; then at year 1 the firm invests $5.0 million in a launch whose year-1 value is $9.0 million. If the test fails, the launch would be worth $2.0 million, so the firm would not invest. The alternative is to launch immediately (investing $5.0 million now) with the same year-1 values and probabilities. The discount rate is 8%. Find the NPV of each strategy and the value of the test relative to immediate launch.

Problem 8.5 (Expansion option, $ millions). A project costs $20.0 million and has a present value of cash flows of $18.0 million. After one year it will have gone up by a factor of 1.5 or down by 0.7. At that time the firm can pay $8.0 million to add capacity equal to 50% of the project's then-current value. The risk-free rate is 4%. Compute the risk-neutral probability, the value of the expansion option and the total value of the project including the option.

Problem 8.6 (Abandonment option, $ millions). A project costs $14.0 million and has a present value of $15.0 million. After one year its value will have risen by a factor of 1.4 or fallen to 0.6. At that time the firm can sell the project for $11.0 million. The risk-free rate is 5%. Compute the value of the abandonment option and the total project value.

Problem 8.7 (Delay option, $ millions). A firm may build a plant costing $100.0 million. Its present value of cash flows is $105.0 million. In a year the value will be 1.25 or 0.80 times its present level. Waiting forgoes a payout equal to 3% of value; the risk-free rate is 5%. Compute the value of waiting, compare it with investing now, and state your recommendation.

Problem 8.8 (Simulation output). A simulation shows that a project's NPV is approximately normally distributed with mean $800,000 and standard deviation $500,000. Estimate (a) the probability that NPV is negative, (b) the probability that NPV exceeds $1,000,000, and (c) the 5th percentile of NPV. (Use the standard normal cumulative probabilities: $\Phi(-1.60) = 0.0548$; $\Phi(0.40) = 0.6554$; z at 5% = −1.645.)

Answers to Selected Problems

- **8.1.** Base: contribution 50,000 × 30 = 1,500,000; EBIT = 1,500,000 − 600,000 − 400,000 = 500,000; after-tax 350,000; OCF = 750,000; NPV = −2,000,000 + 750,000 × 3.790787 = **$843,090**. (a) Price $72: contribution 1,100,000, EBIT 100,000, OCF 470,000, NPV = **−$218,330**; price $88: NPV = **$1,904,510**. (b) Variable cost $55: contribution 1,250,000, EBIT 250,000, OCF 575,000, NPV = **$179,702**; $45: NPV = **$1,506,478**. The swing from price (±10%) is about $2.12 million versus about $1.33 million from variable cost, so NPV is more sensitive to price.
- **8.2.** Pessimistic: contribution 35,000 × 20 = 700,000; EBIT = 700,000 − 650,000 − 400,000 = −350,000; after-tax −245,000; OCF = 155,000; NPV = **−$1,412,428**. Base: **$843,090**. Optimistic: contribution 60,000 × 36 = 2,160,000; EBIT = 1,180,000; after-tax 826,000; OCF 1,226,000; NPV = **$2,647,505**. Expected NPV = 0.3 × (−1,412,428) + 0.5 × 843,090 + 0.2 × 2,647,505 = **$527,318**. $\sigma \approx$ **$1,441,416**. Probability of negative NPV = **30%** (the pessimistic scenario only).
- **8.3.** (a) (600,000 + 400,000) / 30 = **33,333 units**. (b) 600,000 / 30 = **20,000 units**. (c) Required OCF = 2,000,000 / 3.790787 = 527,595; EBIT = (527,595 − 400,000) / 0.70 = 182,279; contribution = 182,279 + 1,000,000 = 1,182,279; Q = **39,409 units**; margin of safety = (50,000 − 39,409) / 50,000 = **21.2%**. (d) At 33,333 units OCF = 400,000, NPV = −2,000,000 + 400,000 × 3.790787 = **−$483,685**.
- **8.4.** Test strategy: −0.6 + 0.5 × (9.0 − 5.0) / 1.08 = −0.6 + 1.852 = **+$1.252 million**. Immediate launch: −5.0 + (0.5 × 9.0 + 0.5 × 2.0) / 1.08 = −5.0 + 5.093 = **+$0.093 million**. The test adds about **$1.159 million** relative to immediate launch.

- **8.5.** q = (1.04 − 0.70) / (1.50 − 0.70) = **0.425**. Up: V = 27.0, expansion payoff = 0.5 × 27.0 − 8.0 = 5.5. Down: V = 12.6, expansion payoff = 6.3 − 8.0 < 0, so 0. Option = 0.425 × 5.5 / 1.04 = **$2.248 million**. Static NPV = −2.0; total = **+$0.248 million**.
- **8.6.** q = (1.05 − 0.60) / (1.40 − 0.60) = 0.5625. Down value = 9.0, payoff = 11.0 − 9.0 = 2.0; up value 21.0, payoff 0. Option = (1 − 0.5625) × 2.0 / 1.05 = **$0.833 million**. Static NPV = +1.0; total = **+$1.833 million**.
- **8.7.** q = (1.05 − 0.03 − 0.80) / (1.25 − 0.80) = 0.4889. Up: 131.25, payoff 31.25; down: 84.0, no investment. Waiting = 0.4889 × 31.25 / 1.05 = **$14.55 million** versus an NPV of **$5.0 million** now. Wait: investing now sacrifices about $9.55 million of option value.
- **8.8.** (a) z = (0 − 800,000) / 500,000 = −1.60, probability ≈ **5.5%**. (b) z = 0.40, so P(NPV > 1,000,000) = 1 − 0.6554 = **34.5%**. (c) 800,000 − 1.645 × 500,000 = **−$22,500**.

Part III — Risk, Return and the Cost of Capital

Chapter 9: Risk and Return

Learning Objectives

After studying this chapter, you should be able to:

- Compute and interpret holding-period, arithmetic, geometric, real, and log returns, and explain when each average is the appropriate one to use.
- Measure the risk of an asset using variance, standard deviation, and scenario analysis, and recognize the limits of these measures when returns are not symmetric.
- Calculate covariance and correlation, and explain why they, rather than the stand-alone risk of an asset, drive the risk of a portfolio.
- Derive and apply the two-asset and n-asset portfolio variance formulas, and quantify how much risk diversification can and cannot remove.
- Separate the total risk of a security into systematic (market) and idiosyncratic (firm-specific) components, and explain why only the former earns a risk premium.
- Describe how historical risk premia are estimated, why they are measured imprecisely, and how managers should use them in setting required returns.

Opening Vignette

The following scenario is illustrative; Alder Ridge Holdings is a fictional company.

Alder Ridge Holdings is a family-controlled maker of agricultural equipment with annual revenue of about $1.8 billion. For three generations the family has reinvested nearly every dollar of profit in the business, and the family office now holds roughly 85 percent of its net worth in Alder Ridge shares. In a recent board meeting, the new chief financial officer, Priya Raman, presented two proposals that appeared unrelated but turned out to rest on the same idea.

The first proposal was a $120 million expansion into precision-farming software. The expansion's projected cash flows were attractive, but their volatility was extreme: in the downside scenario the unit lost money for three years. A director objected, "We cannot afford a project that swings that much. Our own earnings are volatile enough." Priya replied that the relevant question was not how much the software unit's profits swung in isolation but how they moved relative to everything else the company's owners held. Software demand, she argued, was driven by data and subscription economics, not by commodity prices and weather, which dominate the equipment business.

The second proposal came from the family office: sell 30 percent of the family's Alder Ridge shares and invest the proceeds in a broad index of public equities. Some family members were offended. "Why would we sell what we know best?" Priya's answer was blunt. The family earns no extra compensation for bearing risks that a diversified investor would not bear. Any risk that is specific to one company, no matter how well understood, can be reduced at almost no cost by holding many securities.

Both proposals hinge on a distinction at the heart of modern finance: the difference between the risk of an asset on its own and the risk it adds to a portfolio, and the related distinction between risk that diversification can remove and risk that it cannot. This chapter builds the tools to make those distinctions precise. We begin with the measurement of return, move to

the measurement of risk, develop covariance and correlation, show mathematically how diversification works, split risk into systematic and idiosyncratic parts, and finish with what history can and cannot tell us about the reward investors demand for bearing risk. Chapter 10 then builds on these results to develop portfolio theory and the Capital Asset Pricing Model.

9.1 Measuring Returns

Before we can talk about risk, we must agree on what a return is and how to summarize a history of returns in a single number. These sound like trivial questions. In practice, mistakes in averaging returns are among the most common and expensive errors in financial analysis, because the wrong average can overstate wealth accumulation by a large margin.

9.1.1 The holding-period return

The **holding-period return** (HPR) is the total percentage gain on an investment over the period it is held, including both price change and any cash distributions:

```
HPR = (P₁ − P₀ + D) / P₀
```

where P_0 is the beginning price, P_1 is the ending price, and D is the cash received (dividends or interest) during the period. The ratio $(P_1 - P_0)/P_0$ is the *capital gain* component, and D/P_0 is the *income* (or yield) component. An investor who buys a share at $40.00, receives $1.20 in dividends, and sells at $43.60 earns an HPR of (43.60 − 40.00 + 1.20) / 40.00 = 4.80 / 40.00 = 12.0 percent, of which 9.0 percent is capital gain and 3.0 percent is dividend yield.

It is often convenient to work with the **gross return** or wealth relative, 1 + R, which is the factor by which wealth grows over the period. A 12 percent return is a gross return of 1.12. Multi-period returns compound by multiplying gross returns:

```
1 + R(0→T) = (1 + R₁)(1 + R₂)...(1 + R_T)
```

Returns must be measured on a consistent basis. Managers comparing a project’s internal rate of return to a security’s return should be careful that both are expressed per year and with the same compounding convention, a theme from the time-value-of-money chapters.

9.1.2 Nominal, real, and excess returns

Returns quoted in the market are **nominal**: they measure growth in currency units, not in purchasing power. The **real return** removes inflation, using the Fisher relationship:

```
1 + r_real = (1 + r_nominal) / (1 + inflation)
```

If a bond returns 9.5 percent in a year when inflation is 4.0 percent, the real return is 1.095 / 1.04 − 1 = 5.29 percent. The common shortcut, nominal minus inflation, gives 5.50 percent. The approximation is adequate when rates are low and becomes poor when inflation is high.

The **excess return** of an asset is its return minus the risk-free rate for the same period. Excess returns are the natural currency of risk analysis because they measure the payment received specifically for bearing risk. The expected excess return on the market portfolio is the **market risk premium**, which will appear throughout this chapter and the next.

9.1.3 Arithmetic versus geometric averages

Suppose we observe a sequence of annual returns and want one number that characterizes the history. There are two natural choices.

The **arithmetic average** is the simple mean:

```
R̄_A = (1/T) Σ R_t
```

The **geometric average** (also called the compound annual growth rate) is the constant annual return that would have produced the same terminal wealth:

```
R̄_G = [ Π (1 + R_t) ]^(1/T) − 1
```

The two answer different questions. The arithmetic average is the best estimate of the *expected return in a single future period*, provided that returns are independent draws from the same distribution. The geometric average describes *what actually happened to wealth* over the sample. The geometric average is always less than or equal to the arithmetic average, and the gap widens with volatility. A useful approximation is:

```
R̄_G ≈ R̄_A − σ² / 2
```

The intuition is the arithmetic of losses: a 50 percent loss requires a 100 percent gain to recover. A volatile sequence therefore compounds more slowly than its simple average suggests. We will use the volatility drag again in Section 9.6 when we discuss which average to use for a historical risk premium.

Example 9.1 – Arithmetic and geometric average returns. A diversified fund reports the following annual returns over six years: +12%, −8%, +20%, +5%, +15%, −15%.

Step 1 – Arithmetic average. The sum is 12 − 8 + 20 + 5 + 15 − 15 = 29 percentage points. Dividing by 6 gives 29 / 6 = **4.83%**.

Step 2 – Terminal wealth per $1 invested. Multiply the gross returns:

```
1.12 × 0.92 = 1.030400
1.030400 × 1.20 = 1.236480
1.236480 × 1.05 = 1.298304
1.298304 × 1.15 = 1.493050
1.493050 × 0.85 = 1.269092
```

Cumulative return over six years = 26.91 percent.

Step 3 – Geometric average. (1.269092)^(1/6) − 1. Since ln(1.269092) = 0.2383, dividing by 6 gives 0.03972, and exp(0.03972) = 1.0405, so the geometric average is **4.05%**.

Check. $1 × (1.0405)⁶ = $1.2691, which matches the terminal wealth. An investor who believed the fund had earned “4.83 percent per year” would expect $1.00 × 1.0483⁶ = $1.327 after six years, overstating the actual outcome by about 5.8 cents per dollar invested. The difference of 0.78 percentage points per year is the volatility drag. The approximation gives σ²/2 = 0.01886 / 2 = 0.94 percent (using the sample variance computed in Example 9.2), reasonably close.

Interpretation. Use 4.83 percent as the best single-period expectation if you believe the sample is representative of the future; use 4.05 percent to describe the compound wealth growth actually delivered.

9.1.4 Log returns

Analysts often work with **continuously compounded (log) returns**, defined as $r = \ln(1 + R) = \ln(P_1/P_0)$ for returns without distributions. Log returns have a convenient property: multi-period log returns are simply the *sum* of single-period log returns, so averaging and scaling by time are simple. If annual log returns are independent with mean μ and standard deviation σ, the T-year log return has mean μT and standard deviation $\sigma\sqrt{T}$. Log returns cannot fall below −100 percent, which matches the limited liability of equity, and they are more nearly symmetric than simple returns. The price paid for these conveniences is that the log return of a *portfolio* is not the weighted average of the log returns of its components, so portfolio formulas in this chapter are stated in terms of simple returns.

9.1.5 Common mistakes in measuring returns

Several errors recur in practice:

- **Averaging the wrong way.** Reporting the arithmetic mean as the “annual growth rate” overstates compounding. Reporting the geometric mean as the “expected return for next year” understates it.
- **Ignoring distributions.** Comparing price returns across stocks with different dividend policies produces misleading rankings. Always use total return.
- **Mixing nominal and real quantities.** Discounting real cash flows at a nominal rate (or vice versa) is a classic capital-budgeting error that can swing a project from acceptable to unacceptable.
- **Survivor bias.** Samples that include only firms or funds that exist today systematically overstate historical returns, a point we return to in Section 9.6.

9.2 Measuring Risk

Return is what we hope to earn; risk is the possibility that we earn something different. To manage and price risk, we need a measure of the dispersion of possible outcomes.

9.2.1 Variance and standard deviation

For a set of T historical returns, the **sample variance** is

```
s² = [1 / (T − 1)] Σ (R_t − R̄)²
```

and the **standard deviation** is its square root, s. Dividing by T − 1 rather than T corrects for the fact that deviations are measured from the sample mean rather than the true mean, which makes the sample mean “too close” to the data. Standard deviation is expressed in the same units as returns, which makes it easier to interpret than variance. Variance, however, is the quantity that combines additively in portfolio calculations, so we will move between the two constantly.

Example 9.2 – Sample variance and standard deviation. Using the six fund returns from Example 9.1 (arithmetic mean 4.8333%):

Year	Return R	Deviation (R − 4.8333%)	Squared deviation
1	+12%	+7.1667%	0.005136
2	−8%	−12.8333%	0.016469
3	+20%	+15.1667%	0.023003
4	+5%	+0.1667%	0.000003
5	+15%	+10.1667%	0.010336
6	−15%	−19.8333%	0.039336
Sum			**0.094283**

Sample variance = 0.094283 / (6 − 1) = **0.018857**. Sample standard deviation = √0.018857 = **13.73%**.

If we had instead treated the six observations as the entire population, the variance would be 0.094283 / 6 = 0.015714 and the standard deviation 12.54 percent. With only six observations the difference is substantial (about 1.2 percentage points); with sixty it would be negligible. The fund's return of 4.83 percent per year with 13.73 percent annual volatility illustrates the typical equity pattern: the standard deviation is nearly three times the mean, so a single year's outcome is dominated by noise.

9.2.2 Expected returns and risk from scenarios

Managers evaluating a project or a security rarely have a long history. Instead they build scenarios, assign probabilities, and compute the **expected return** and variance of the probability distribution:

```
E(R) = Σ p_s R_s
σ² = Σ p_s [R_s − E(R)]²
```

Scenario analysis is only as good as the scenarios and probabilities. Its key advantage is that it forces explicit thought about what could happen, including states that history has not yet produced.

Example 9.3 – Expected return and risk from scenarios. Two securities, X and Y, have the following return distribution over the next year.

State of the economy	Probability	Return on X	Return on Y
Recession	0.20	−10%	+12%
Normal	0.50	+8%	+6%
Boom	0.30	+25%	−2%

Expected returns.

```
E(R_X) = 0.20(−10%) + 0.50(8%) + 0.30(25%) = −2.0% + 4.0% + 7.5% = 9.5%
E(R_Y) = 0.20(12%) + 0.50(6%) + 0.30(−2%) = 2.4% + 3.0% − 0.6% = 4.8%
```

Variance and standard deviation of X. Deviations from 9.5%: −19.5%, −1.5%, +15.5%. Squares: 0.038025, 0.000225, 0.024025.

```
σ²_X = 0.20(0.038025) + 0.50(0.000225) + 0.30(0.024025)
     = 0.007605 + 0.0001125 + 0.0072075 = 0.014925
σ_X = √0.014925 = 12.22%
```

Variance and standard deviation of Y. Deviations from 4.8%: +7.2%, +1.2%, −6.8%. Squares: 0.005184, 0.000144, 0.004624.

```
σ²_Y = 0.20(0.005184) + 0.50(0.000144) + 0.30(0.004624)
     = 0.0010368 + 0.000072 + 0.0013872 = 0.002496
σ_Y = √0.002496 = 5.00%
```

Interpretation. X has the higher expected return and the higher risk. Notice also that X does well when Y does badly: in the recession state, X loses 10 percent while Y gains 12 percent. This opposite pattern is the raw material of diversification and is the subject of Section 9.3.

9.2.3 Reading standard deviation: the normal approximation

If returns are roughly normally distributed, about 68 percent of outcomes fall within one standard deviation of the mean, about 95 percent within two, and about 99.7 percent within three. For an equity fund with an expected return of 8 percent and a standard deviation of 18 percent, a rough 95 percent range for next year's return is 8% ± 2 × 18% = −28% to +44%. The width of that range is the main message: even a well-diversified equity portfolio has an enormous range of plausible one-year outcomes. Risk shrinks as a fraction of the mean only slowly as the horizon extends, because the mean grows with T while the standard deviation grows with √T, so the ratio improves by only √T. This is why long horizons reduce the probability of a loss but do not eliminate the possibility of large dollar losses.

9.2.4 Beyond standard deviation

Standard deviation treats upside and downside deviations alike and is a complete description of risk only if returns are symmetric, as in the normal distribution. Real return distributions often depart from this in two ways.

- **Skewness** measures asymmetry. Negative skewness means a longer left tail: the occasional very large loss. Strategies that sell insurance, write options, or lend to risky borrowers often collect small steady gains and suffer rare large losses, which makes their standard deviations look deceptively low.
- **Kurtosis** measures the heaviness of the tails relative to a normal distribution. Excess kurtosis means extreme outcomes occur more often than the normal curve predicts. Financial crises are a reminder that "three-sigma" events happen more frequently than a bell curve suggests.

Practitioners complement standard deviation with several downside measures. **Value at Risk (VaR)** states the loss that will not be exceeded with a given probability over a given horizon, for example "a 1-day 99 percent VaR of $4 million," which means that on 99 days out of 100 the loss should be smaller than $4 million. VaR says nothing about how bad the loss is on the remaining days. **Expected shortfall** (or conditional VaR) fills the gap by reporting the average loss on those worst days. **Maximum drawdown** measures the largest peak-to-trough decline over a period, which matters to investors who may be forced to liquidate after a loss.

For the purposes of corporate finance, standard deviation and its portfolio relatives remain the workhorse, partly because they aggregate neatly and partly because the theory of the next chapter is built on them. But a manager should always ask whether a project or an investment strategy has a payoff pattern that makes variance a poor summary, such as a drug-development program with a small probability of a very large payoff, or a project with exposure to a catastrophic downside.

9.3 Covariance and Correlation

An asset's standard deviation tells us how much it moves. It does not tell us how it moves *with other assets*. For portfolio risk, the second question is the one that matters.

9.3.1 Covariance

The **covariance** between the returns of assets i and j measures the tendency of their returns to move together:

```
Cov(R_i, R_j) = σ_ij = E[(R_i − E(R_i))(R_j − E(R_j))]
```

For historical data the sample covariance is

```
s_ij = [1 / (T − 1)] Σ (R_i,t − R̄_i)(R_j,t − R̄_j)
```

A positive covariance means that when asset i is above its average, asset j tends to be above its average as well. A negative covariance means they tend to move in opposite directions. A covariance near zero means there is no linear tendency either way. Note that the covariance of an asset with itself is its variance: Cov(R_i, R_i) = σ_i².

Covariance has a drawback: its magnitude depends on the units and on the volatility of the two assets, so a number like 0.0032 is hard to interpret on its own.

9.3.2 Correlation

The **correlation coefficient** standardizes covariance by the two standard deviations:

```
ρ_ij = σ_ij / (σ_i σ_j)
```

Correlation always lies between −1 and +1. A value of +1 means returns are perfectly positively linearly related; −1 means perfectly negatively related; 0 means linearly unrelated. Because it is dimensionless, correlation allows comparison across pairs of assets. Rearranging gives a formula we will use constantly: **σ_ij = ρ_ij σ_i σ_j**.

Three cautions about correlation are essential for practitioners.

1. **Correlation measures linear association only.** Two variables can be perfectly dependent in a nonlinear way (for example, a call option and its underlying stock over large moves) and still show modest correlation.
2. **Correlations are unstable.** Estimated correlations change over time and, notoriously, rise during market crises when investors sell many different assets at once. A diversification benefit estimated in calm markets may shrink just when it is needed.
3. **Estimates are noisy.** With a short sample, a correlation estimate can be far from the true value. When many assets are involved the number of correlations grows quickly (n(n − 1)/2), and estimation error accumulates.

Example 9.4 – Computing covariance and correlation from data. The annual returns of two companies, A (a retailer) and B (a consumer staples producer), over five years are:

Year	R_A	R_B
1	10%	6%
2	4%	3%
3	−6%	−2%
4	14%	9%
5	8%	7%

Step 1 – Means. $\bar{R}_A = (10 + 4 - 6 + 14 + 8)/5 = 30/5 = 6.0\%$. $\bar{R}_B = (6 + 3 - 2 + 9 + 7)/5 = 23/5 = 4.6\%$.

Step 2 – Deviations and products.

Year	R_A − 6.0%	R_B − 4.6%	Product	(Dev A)²	(Dev B)²
1	+0.04	+0.014	0.00056	0.0016	0.000196
2	−0.02	−0.016	0.00032	0.0004	0.000256
3	−0.12	−0.066	0.00792	0.0144	0.004356
4	+0.08	+0.044	0.00352	0.0064	0.001936
5	+0.02	+0.024	0.00048	0.0004	0.000576
Sum			**0.01280**	**0.0232**	**0.00732**

Step 3 – Sample covariance and standard deviations.

```
s_AB = 0.01280 / 4 = 0.0032
s_A  = √(0.0232 / 4) = √0.0058 = 7.62%
s_B  = √(0.00732 / 4) = √0.00183 = 4.28%
```

Step 4 – Correlation.

```
ρ_AB = 0.0032 / (0.0762 × 0.0428) = 0.0032 / 0.003258 = 0.982
```

Interpretation. The two stocks moved almost in lockstep over this short sample. If the relationship were true over the long run, combining them would offer almost no diversification. A manager should be wary of reading too much into five data points, but the exercise shows how the machinery works. In a real application, one would use dozens of observations at least, and one would also ask *why* the two firms should be related; here, both are consumer businesses exposed to household income.

9.3.3 Sources of correlation

Economic reasoning should accompany statistics. Firms in the same industry share demand shocks, input costs, and regulation, so their returns are positively correlated. Firms in different industries still share exposure to interest rates, inflation, and the business cycle. Bonds and equities may move together when discount rates drive both, and in opposite directions during flights to safety. The most useful way to think about correlation is as the extent to which two assets are exposed to *common sources of risk*. This idea is formalized in the single-index and multifactor models of Chapter 10.

9.4 Portfolio Risk and Diversification

9.4.1 Portfolio return and risk of two assets

A portfolio's return is the weighted average of the returns of its components, with weights equal to the proportion of wealth invested in each:

```
R_p = w₁R₁ + w₂R₂,  with w₁ + w₂ = 1
E(R_p) = w₁E(R₁) + w₂E(R₂)
```

The expected return is linear in the weights. The variance is not:

```
σ_p² = w₁²σ₁² + w₂²σ₂² + 2 w₁ w₂ σ₁₂
     = w₁²σ₁² + w₂²σ₂² + 2 w₁ w₂ ρ₁₂ σ₁ σ₂
```

The last term is the crucial one. If $\rho_{12} = +1$, then $\sigma_p = w_1\sigma_1 + w_2\sigma_2$ and the portfolio's standard deviation is just the weighted average of the components' standard deviations: no diversification. For any $\rho_{12} < 1$, σ_p is strictly less than the weighted average. If $\rho_{12} = -1$ the risk can be eliminated entirely with the right weights.

The following example uses a pair of assets and shows how the benefit varies with correlation.

Example 9.5 – Two-asset portfolio risk and the role of correlation. Asset 1 has $\sigma_1 = 30\%$ and asset 2 has $\sigma_2 = 20\%$. A portfolio is 60% in asset 1 and 40% in asset 2.

Stand-alone terms.

```
w₁²σ₁² = (0.6)²(0.30)² = 0.36 × 0.09 = 0.0324
w₂²σ₂² = (0.4)²(0.20)² = 0.16 × 0.04 = 0.0064
2w₁w₂σ₁σ₂ = 2(0.6)(0.4)(0.30)(0.20) = 0.48 × 0.06 = 0.0288
```

The last term is multiplied by ρ. The weighted-average standard deviation is 0.6(30%) + 0.4(20%) = 26%.

Portfolio variance as a function of ρ.

ρ_{12}	Portfolio variance	Portfolio σ	Reduction vs. 26%
+1.0	0.0324 + 0.0064 + 0.0288 = 0.0676	26.00%	0.00 pts
+0.5	0.0324 + 0.0064 + 0.0144 = 0.0532	23.07%	2.93 pts
0.0	0.0324 + 0.0064 + 0 = 0.0388	19.70%	6.30 pts
−0.5	0.0324 + 0.0064 − 0.0144 = 0.0244	15.62%	10.38 pts
−1.0	0.0324 + 0.0064 − 0.0288 = 0.0100	10.00%	16.00 pts

Interpretation. Even with a positive correlation of 0.5, the portfolio's volatility is about three percentage points lower than the average of its components. At zero correlation, the portfolio is less risky than the safer of the two assets alone (19.70 percent versus 20 percent), even though 60 percent of it is in the riskier asset. This is the often-surprising feature of diversification: adding a *risky* asset to a portfolio can reduce its risk, provided the new asset is not too highly correlated with what is already held.

9.4.2 The n-asset portfolio

For a portfolio of n assets with weights w_i, the variance is the sum over *all* pairs, including each asset paired with itself:

$$\sigma_p^2 = \sum_i \sum_j w_i w_j \sigma_{ij}$$
$$= \sum_i w_i^2 \sigma_i^2 + \sum_i \sum_{j \neq i} w_i w_j \sigma_{ij}$$

There are n variance terms and n(n − 1) covariance terms. As n grows, the covariance terms dominate: with 30 assets there are 30 variances but 870 covariances. This arithmetic is the whole story of diversification. The variance terms get multiplied by w_i^2, which shrinks quickly as weights shrink, while the covariance terms retain their influence in aggregate.

To see this explicitly, take the simple case of **equal weights** $w_i = 1/n$. Let $\bar{\sigma}^2$ denote the average variance and $\bar{\sigma}_{ij}$ the average covariance. Then

$$\sigma_p^2 = (1/n)\,\bar{\sigma}^2 + (1 - 1/n)\,\bar{\sigma}_{ij}$$

As $n \to \infty$, the first term vanishes and the portfolio variance converges to the average covariance. Put differently, **the risk of a well-diversified portfolio is determined by how its assets covary, not by how volatile each is individually**.

Example 9.6 – Diversification with n identical assets. Each of n stocks has a standard deviation of 30 percent (variance 0.09) and every pair has a correlation of 0.25 (covariance 0.25 × 0.09 = 0.0225). A portfolio holds all n stocks in equal proportions.

Formula: $\sigma_p^2 = 0.09/n + (1 - 1/n)(0.0225)$.

n	Variance of portfolio	Standard deviation	Share of single-stock variance remaining
1	0.0900	30.00%	100.0%
2	0.0563	23.72%	62.5%
5	0.0360	18.97%	40.0%
10	0.0293	17.10%	32.5%
30	0.0248	15.73%	27.5%
100	0.0232	15.22%	25.8%
1,000	0.0226	15.02%	25.1%
∞	0.0225	15.00%	25.0%

Worked checks: for n = 5, $\sigma_p^2 = 0.09/5 + (0.8)(0.0225) = 0.018 + 0.018 = 0.036$ and $\sigma_p = 18.97\%$. For n = 30, $\sigma_p^2 = 0.003 + (0.96667)(0.0225) = 0.003 + 0.021750 = 0.02475$, so $\sigma_p = 15.73\%$.

Interpretation. Half of the diversifiable risk disappears with just two stocks and about 80 percent of it by the time five are held; by n = 30 the portfolio standard deviation is within about 0.7 percentage points of its floor of 15 percent. Beyond about thirty stocks, the benefit of adding names is real but small. The floor (15 percent, or half the single-stock risk) exists because the stocks share a common component: 25 percent of each stock's variance is "shared", reflected in the correlation of 0.25. No amount of additional diversification within this market can remove it.

This stylized result is consistent with many empirical studies, which find that a few dozen randomly selected stocks capture most of the risk reduction available, although more are needed when idiosyncratic volatility is high, as it has been in recent decades, or when investors want to guard against the wide dispersion of outcomes of any single portfolio. The exact number depends on the market, the period, and the tolerance for tracking error.

9.4.3 Diversification across asset classes and borders

The same mathematics applies when the "assets" are asset classes. Equities, government bonds, real estate, commodities, and foreign assets have different exposures to growth, inflation, and interest rates, so their correlations are well below one. International diversification offered large benefits when markets were segmented; as economies and capital markets have become more integrated, correlations have risen, particularly in crises, and the incremental gains have fallen though not disappeared. Diversification across currencies, regulatory regimes, and sovereign risks still reduces some exposures that cannot be removed within a single country.

9.4.4 What diversification does not do

It is worth being precise about what diversification accomplishes.

- It does not change expected return in the simple two-asset case: the portfolio's expected return is the weighted average. The benefit is the reduction in risk for a given return, or, equivalently, a higher return for a given risk. (Chapter 10 develops this idea into the efficient frontier.)
- It cannot remove risk that is common to all assets.
- It does not protect against estimation errors: if the correlations used to construct a portfolio are wrong, the portfolio can be much riskier than planned.
- It is not free in practice: transaction costs, taxes, and monitoring costs rise with the number of holdings, and in the corporate context diversification across unrelated businesses has costs of its own (Section 9.7).

9.5 Systematic and Idiosyncratic Risk

9.5.1 Two kinds of risk

The preceding section suggests dividing the total variance of any security into two parts.

Systematic risk (also called market risk or non-diversifiable risk) is the part that stems from economy-wide influences: interest rates, inflation, economic growth, exchange rates, energy prices, and the other forces that affect virtually all firms. It cannot be eliminated by holding more securities, because all securities respond to it to some degree.

Idiosyncratic risk (also called unsystematic, firm-specific, or diversifiable risk) stems from events particular to a company or small group of companies: a product recall, a successful drug trial, a lost contract, a management change, a lawsuit, a fire at a single plant. In a large portfolio these events are roughly independent and offset one another.

The formal decomposition comes from a **single-index model**, in which the return on security i is

```
R_i = α_i + β_i R_m + ε_i
```

where R_m is the return on a broad market index, β_i measures the security's sensitivity to the market, α_i is a constant, and ε_i is a firm-specific disturbance that is uncorrelated with R_m and has zero mean. Taking variances,

```
σ_i² = β_i² σ_m²  +  σ²(ε_i)
       systematic     idiosyncratic
```

The fraction of a stock's variance explained by the market is the R^2 of a regression of R_i on R_m, and equals $\beta_i^2 \sigma_m^2 / \sigma_i^2$. Equivalently $R^2 = \rho^2_{i,m}$, the squared correlation of the stock with the market.

For a diversified portfolio p of many stocks, the idiosyncratic terms average out, because they are uncorrelated across stocks, and $\sigma_p^2 \to \beta_p^2 \sigma_m^2$ where $\beta_p = \sum w_i \beta_i$. The beta of a portfolio is the weighted average of the betas of its components, which makes beta an additive measure of risk contribution, unlike standard deviation.

Example 9.7 – Decomposing the risk of a single stock. A stock has a beta of 1.2. The market's annual standard deviation is 18 percent, and the stock's total annual standard deviation is 30 percent.

Systematic variance: $\beta^2\sigma_m^2 = (1.2)^2(0.18)^2 = 1.44 \times 0.0324 = 0.046656$, which is a standard deviation of $\sqrt{0.046656} = 21.6\%$.

Total variance: $0.30^2 = 0.0900$.

Idiosyncratic variance: $0.0900 - 0.046656 = 0.043344$, which is a standard deviation of $\sqrt{0.043344} = 20.82\%$.

R^2*:* $0.046656 / 0.0900 = 0.5184$, or **51.8%** of the stock's variance is systematic. The implied correlation with the market is $\sqrt{0.5184} = 0.72$.

Interpretation. Roughly half of this stock's variance can be diversified away. An undiversified investor holding only this stock bears 30 percent volatility; a diversified investor, who removes the idiosyncratic portion, bears only the 21.6 percent associated with market exposure. Competition among diversified investors pushes the stock's price to the level at which it offers just the return that a diversified investor requires for its 21.6 percent of market-driven risk. The extra 20.82 percent of idiosyncratic volatility, which the undiversified investor must bear, is therefore not priced.

9.5.2 Why only systematic risk is rewarded

The argument for this central result can be stated in three steps.

1. Investors can eliminate idiosyncratic risk at very low cost by holding diversified portfolios.
2. Because it is costless to avoid, no investor needs to be paid for bearing it. If a security offered a premium for its idiosyncratic risk, diversified investors would buy it, driving its price up and its expected return down, until the premium disappeared.
3. Systematic risk cannot be avoided by diversification, so investors must be compensated for bearing it, and the compensation is proportional to the amount of systematic risk the security carries.

The implication is that **the required return on a security depends on its beta, not its total volatility**. A start-up biotechnology stock may be extraordinarily volatile, but if its returns are mostly driven by trial outcomes that are unrelated to the market, its required return can be low. Conversely, a stable utility can have a modest standard deviation but a substantial beta if its earnings are sensitive to interest rates. Chapter 10 turns this logic into the Capital Asset Pricing Model.

9.5.3 Sources of systematic and idiosyncratic risk in a firm

For a corporate manager, it is useful to relate this distinction to business risk. Sensitivity to the economic cycle is a major driver of beta: firms selling discretionary goods or capital equipment have high operating leverage to GDP, while food retailers and utilities tend to have lower betas. **Operating leverage** (fixed costs relative to variable costs) magnifies the effect of sales fluctuations on profits and thus raises beta. **Financial leverage** raises the beta of equity further because debt holders have a fixed claim and equity absorbs all the variation. These links are developed in the capital-structure chapters, and they show that a firm's beta is a managerial outcome, not a fixed trait.

Idiosyncratic risks, by contrast, include execution risks, legal events, and competitive moves. They matter enormously for the firm's survival, for its employees and managers, and for the family that owns most of it. But for a well-diversified shareholder they are noise, and they should not enter the discount rate used to value the firm's cash flows. They do belong in the *expected cash flows*, where a probability-weighted view of failure belongs.

9.6 Historical Risk and Risk Premia

9.6.1 What history shows, qualitatively

Researchers have assembled long records of returns on broad classes of securities: short-term government bills, long-term government bonds, corporate bonds, and equities. The qualitative pattern is robust across countries and periods and does not depend on any particular data set:

- Short-term government bills have had the lowest average returns and the lowest volatility, and are the usual empirical proxy for the risk-free rate.
- Long-term government bonds have earned higher average returns than bills, along with higher volatility, because their prices respond to changes in interest rates.
- Equities have earned the highest average returns and exhibited the highest volatility, with occasional deep drawdowns.
- Small-company stocks have historically been more volatile than large-company stocks and have, in many samples, earned higher average returns.

Table 9.1 sketches this pattern with stylized, entirely illustrative figures. They are not statistics from any actual data set; their purpose is to show the shape of the risk-return trade-off.

Table 9.1 – Stylized risk-return pattern across asset classes (illustrative only).

Asset class	Illustrative average annual return	Illustrative standard deviation	Illustrative premium over bills
Short-term government bills	3%	2%	0%
Long-term government bonds	5%	9%	2%
Corporate bonds	6%	8%	3%
Large-company equities	10%	18%	7%
Small-company equities	12%	28%	9%

The central lesson is that average returns rise with risk, but not proportionally. The ratio of premium to standard deviation (a Sharpe ratio, defined in Chapter 10) is far from constant across classes, and the strength of the relation varies by period. In some multi-decade stretches, riskless bonds have outperformed equities; the premium is an expectation, not a promise.

9.6.2 The equity risk premium

The **equity risk premium** (ERP) is the expected excess return of the market portfolio over the risk-free rate. It is the single most important input to estimating the cost of equity and hence to valuation. It is unobservable. The three families of approaches are:

1. **Historical (ex post) averages.** Average the realized market excess return over a long sample. This is the most common approach and the subject of the rest of this section.

2. **Implied (ex ante) estimates.** Take the current market index level, forecasts of dividends or earnings, and solve for the discount rate that equates the present value of expected payouts with the index. Subtract the risk-free rate. This is forward-looking and responds to market conditions, but depends on growth forecasts.
3. **Surveys and fundamentals.** Ask investors, CFOs, or professors, or derive the premium from models linking it to consumption, risk aversion, and growth.

Each has weaknesses, and sensible practitioners triangulate. Estimates in practice often range across several percentage points depending on the method and horizon.

9.6.3 Estimating the premium: how precise can it be?

Suppose annual market excess returns are independent with standard deviation σ. The sample mean of N annual observations has a **standard error**

```
SE(R̄) = σ / √N
```

For equity markets, σ is large relative to the plausible premium, so the standard error is large unless N is very large. The result is sobering.

Example 9.8 – Precision of a historical risk premium. Using a long sample, an analyst estimates an average annual market excess return of 7.0 percent with an annual standard deviation of 20 percent.

100 years of data (N = 100). SE = 20% / √100 = 20% / 10 = **2.0%**. A 95 percent confidence interval is 7.0% ± 1.96 × 2.0% = 7.0% ± 3.92%, or roughly **3.1% to 10.9%**.

50 years of data (N = 50). SE = 20% / √50 = 20% / 7.071 = **2.83%**, giving 7.0% ± 5.54%, or 1.5% to 12.5%.

How much data to reach a standard error of 1 percent? Solve σ / √N = 1%: N = (20 / 1)² = **400 years**.

Effect on valuation. Consider a stock with a beta of 1.0 and a risk-free rate of 3 percent. Using the low end of the interval, the cost of equity is 6.1 percent; using the high end, 13.9 percent. For a growing perpetuity with a 2 percent growth rate, the value-to-cash-flow multiple is 1 / (0.061 − 0.02) = 24.4 at the low end versus 1 / (0.139 − 0.02) = 8.4 at the high end, a factor of nearly three. The uncertainty about the premium is a first-order problem for any valuation, not a detail.

Interpretation. Statistical power is the issue. Extending the sample backward helps, but older data come from different economies, and the risk premium itself probably changes over time. Shortening the window to focus on recent experience makes the estimate depend on the luck of the period.

9.6.4 Arithmetic versus geometric premium

Which average should be used to estimate the premium for a *one-year* discount rate? In theory, the arithmetic mean, because it is an unbiased estimate of the expected return in a single year. For discounting cash flows that arrive many years in the future, there is an argument that the correct compound rate lies between the two averages, because estimation error in the arithmetic mean compounds. Practitioners largely use the arithmetic average for the premium used in single-period CAPM calculations, and are aware that using geometric averages understates the required return by roughly σ²/2. For a market with a standard deviation of 20 percent that gap is 0.20² / 2 = 2.0 percentage points, as we saw in Section 9.1.3: a geometric premium of 5 percent corresponds to an arithmetic premium of about 7 percent. Whichever is used, the analyst should be consistent with how the premium is used downstream.

9.6.5 Biases in historical estimates

- **Survivorship bias.** Markets that did well continue to be studied and those that suffered catastrophe or were closed (for example, through nationalization or hyperinflation) vanish from data sets. A premium measured on the survivors overstates what an investor in a randomly chosen market could have expected.
- **Look-back and period-selection bias.** A premium estimated over a period that began at a market trough, or ended at a peak, will be unusually high.
- **Changing risk and discount rates.** If investors' required return falls over time, realized returns during the decline are higher than expected returns, so history overstates what the future will provide. A secular fall in discount rates produces a capital gain that will not repeat.
- **Measurement of the risk-free rate.** The choice of the bill rate versus a long-term bond yield as the risk-free benchmark changes the measured premium by the term premium.

9.6.6 Using the premium in practice

For a corporate manager the practical advice is to use a defensible, stable premium rather than to re-estimate with every market swing; to document the source and the approach; and to test valuation conclusions against a range of premia, as Example 9.8 suggests. It is the *relative* ranking of projects by systematic risk that matters most for capital allocation, and this ranking is less sensitive to the premium than the absolute value.

9.7 Managerial Implications

9.7.1 Which risk should the firm price?

If shareholders hold diversified portfolios, they care about the contribution of a project to their portfolio's risk, which is its systematic risk. The firm should therefore discount a project's expected cash flows at a rate that reflects the project's beta, and ignore its idiosyncratic volatility when setting the discount rate. This is why Priya Raman, in the vignette, could argue for the software unit even though it was individually volatile. If its cash flows are weakly correlated with the market and with the company's core business, its contribution to systematic risk may be small. This principle underlies the use of divisional or project-specific hurdle rates, developed in the cost-of-capital chapters, and requires the manager to identify "pure-play" comparables whose betas reflect the project's business risk.

9.7.2 Caveats: when idiosyncratic risk does matter to the firm

The diversified-shareholder logic has limits, and a thoughtful manager should know them.

- **Financial distress costs.** Idiosyncratic volatility can drive a firm into bankruptcy, and bankruptcy imposes real costs (legal fees, lost customers, forced asset sales) on all claimants. These costs reduce firm value even if shareholders are diversified, which is a reason to manage volatility.
- **Undiversified stakeholders.** Owner-managers, employees with firm-specific human capital, suppliers, and the family in our vignette cannot diversify. Their preferences may legitimately affect decisions, but the owners should recognize that such risk-aversion has a cost: the family office in the vignette pays for concentration with a lower risk-adjusted return, which is not compensated by the market.
- **Convex taxes and capital-market frictions.** If taxes are progressive or if external financing is costly, smoothing cash flows can add value. Hedging can ensure that investment funds are available when good projects arise.

- **Managerial incentives.** Managers whose pay depends on stock price or on survival may take less (or more) risk than shareholders would like. Compensation design must account for this.

9.7.3 Corporate diversification versus investor diversification

Investors can diversify cheaply on their own. A conglomerate that diversifies across unrelated businesses delivers to its shareholders little that they could not obtain by buying shares of focused firms, and it can destroy value through cross-subsidization of weak divisions and weakened managerial accountability. The "conglomerate discount" observed in many markets reflects this concern. Diversification by the firm can nevertheless be justified where it creates real operational synergies, reduces expected distress costs, or provides access to capital in markets where individual investors cannot diversify freely.

9.7.4 Hedging and risk management

If the firm's exposure to a risk (an exchange rate, a commodity price) is purely a market risk that investors can diversify, hedging does not by itself create value, since investors can manage the exposure on their own. Hedging does create value when it reduces expected distress costs, protects the firm's ability to fund valuable investments, or lowers expected taxes. A coherent risk-management policy distinguishes between risks the firm has a comparative advantage in bearing (its core business risks), and risks that should be transferred or hedged.

9.7.5 The role of the Sharpe-style trade-off

Managers who evaluate business units or investments on return alone, ignoring risk, will over-allocate capital to the riskiest activities. A simple and powerful discipline is to compare the reward per unit of the relevant risk across alternatives. Chapter 10 formalizes this with the Sharpe ratio, the Treynor ratio, and the Security Market Line.

Case Study: Northgate Instruments Considers Diversifying

Northgate Instruments is a fictional company created for this case.

Northgate Instruments is a family-controlled manufacturer of industrial measurement equipment, with annual revenue of roughly $900 million and a stock market capitalization of $1.4 billion. About 60 percent of the shares are held by the founding family, whose members have most of their wealth tied to the company. Northgate's profits follow the capital-spending cycle of its customers in oil and gas, mining, and heavy manufacturing. Over the past decade the stock has had an annual standard deviation of about 25 percent and an estimated beta of 1.3.

The board is evaluating a proposal from the CEO, Dana Whitfield, to acquire Meridian Analytics, a software-as-a-service firm that provides predictive maintenance subscriptions, for $280 million. Meridian is growing rapidly but is unprofitable, and its stock-like returns would be very volatile. Using Meridian's recent history and data from comparable listed software companies, the finance team has produced the following estimates for the return on the *new line of business* if it were operated as a stand-alone unit:

Quantity	Core business	Meridian unit
Expected return (annual)	11%	14%
Standard deviation	25%	40%
Beta (relative to a broad equity index)	1.3	0.9
Correlation between the two units	0.10	
Share of combined firm value	80%	20%

The risk-free rate is 3 percent and the market risk premium used by the firm is 6 percent. The index has an annual standard deviation of 18 percent.

Whitfield argues that the acquisition will "smooth" Northgate's earnings: when capital spending drops, subscription revenue continues. The CFO, Samuel Okafor, reports that the combined unit's standard deviation would be sqrt(0.64 × 0.0625 + 0.04 × 0.16 + 2 × 0.8 × 0.2 × 0.10 × 0.25 × 0.40), which equals 22.3 percent, below the 25 percent of the core business alone. The CFO also notes that the combined beta would be 0.8 × 1.3 + 0.2 × 0.9 = 1.22.

Opposition emerges from two directions. A family director, Helen Northgate, objects that the family "knows measurement hardware, not software" and fears that integration will distract management from the core. She also argues that a 40 percent standard deviation is "unacceptable" for a company whose lenders have covenants on earnings volatility. A second director, an institutional investor named Raj Patel, takes the opposite view: the family and the company's institutional shareholders can diversify on their own, so Northgate should not pay a control premium for a business only to reduce the volatility that its owners could reduce by buying software stocks. He proposes instead that Northgate return $280 million to shareholders via a share repurchase.

Okafor points out that the acquisition price embeds an expected return of 14 percent against a required return given Meridian's beta of 3% + 0.9 × 6% = 8.4%, suggesting a positive spread if the projections are reliable. But he cautions that Meridian's forecasts are optimistic, that the 14 percent figure is an expectation resting on a short history, and that integration risks are idiosyncratic to Northgate and not captured in a beta estimated from comparables.

The board must decide whether the acquisition creates value, how to think about the company's high exposure to the capital-spending cycle, and what role the family's concentrated wealth should play in the decision.

Discussion questions

1. Verify the CFO's calculation of the combined standard deviation and explain, in terms of the portfolio variance formula, why the combined risk is lower than the core business's risk even though 20 percent of the firm is placed in a much riskier unit. How does the answer change if the correlation is 0.60?
2. Is the board correct to use standard deviation or beta to judge the acquisition? Compute the required return for Meridian and for the core business using the information given, and discuss whether the 14 percent expected return justifies the acquisition.
3. Evaluate Patel's argument that diversification by the firm creates no value for shareholders. Under what conditions would the diversification benefit be real? Consider the family's position separately from that of institutional holders.
4. Identify the idiosyncratic risks in the proposal that do not appear in beta, and describe how they should be handled in the valuation. How reliable are the estimates in the table, and how would you stress-test them?

Key Takeaways

- The holding-period return combines capital gain and income; always use total returns, and match nominal returns with nominal discount rates and real with real.
- The arithmetic average estimates a single-period expected return, while the geometric average measures realized compound growth. The gap between them is approximately $\sigma^2/2$.
- Standard deviation measures total risk, but it is an incomplete description when return distributions are skewed or have fat tails; supplement it with downside measures such as VaR, expected shortfall, and drawdown.

- Covariance and correlation determine how assets combine. The portfolio variance is a double sum over all covariances, and with many assets the covariances dominate.
- Combining assets with correlation below one reduces portfolio risk below the weighted average of component risks; adding a risky asset can even reduce portfolio risk if it is sufficiently uncorrelated with existing holdings.
- Diversification removes idiosyncratic risk but not systematic risk. The variance of a large equal-weighted portfolio converges to the average covariance among its assets.
- A security's total variance equals $\beta^2\sigma_m^2$ plus idiosyncratic variance; the R^2 of the market regression gives the share that is systematic.
- Because diversified investors can eliminate idiosyncratic risk at low cost, only systematic risk earns a premium. Required returns should reflect beta rather than total volatility.
- Historical risk premia are noisy: with a 20 percent standard deviation, even a century of data yields a standard error of about 2 percentage points, and surveys, implied estimates, and judgment are all needed.
- Idiosyncratic risk still matters to firms through financial distress costs, undiversified stakeholders, and incentives, but it should be handled in cash flows and risk-management policy, not by inflating the discount rate.

Review Questions

1. Why does the geometric average return never exceed the arithmetic average, and what determines the size of the gap?
2. Explain why variance, rather than standard deviation, is the quantity that adds in portfolio calculations. What does the covariance term represent?
3. What is the difference between covariance and correlation? Why do analysts prefer to report correlation, and what are its limitations?
4. A portfolio manager says, "I never add a risky stock to a portfolio of safer holdings." Use the two-asset variance formula to explain when this policy is mistaken.
5. Describe why the variance of an equal-weighted portfolio approaches the average covariance as the number of assets grows. What does this imply about the benefits of the fifty-first stock compared with the second?
6. Distinguish systematic from idiosyncratic risk and give two examples of each for a retail company.
7. Why does the market not pay a premium for idiosyncratic risk? What would happen to prices if it tried?
8. Why do correlations tend to rise in market crises, and what does this imply for how diversification benefits should be estimated and stress-tested?
9. Explain why the equity risk premium is hard to estimate precisely and list three biases that can affect historical estimates.
10. A conglomerate argues that combining unrelated businesses reduces its earnings volatility and so should lower its cost of capital. Evaluate this argument from the viewpoint of a diversified shareholder, and note one circumstance where the argument has merit.

Problems

Problem 9.1. A stock has annual returns of +10%, −5%, +18%, and +7% over four years. Compute (a) the arithmetic average return and (b) the geometric average return.

Problem 9.2. A stock has the following return scenarios: −12% with probability 0.25, +6% with probability 0.50, and +20% with probability 0.25. Compute the expected return, variance, and standard deviation.

Problem 9.3. A portfolio is invested 50% in stock A (σ = 25%) and 50% in stock B (σ = 15%), with a correlation of 0.30 between them. Compute the portfolio variance and standard deviation.

Problem 9.4. Two assets have standard deviations of 12% and 20% and a correlation of −0.40. Compute their covariance.

Problem 9.5. An equal-weighted portfolio holds 20 stocks, each with a standard deviation of 40% and a pairwise correlation of 0.30. Compute (a) the portfolio standard deviation and (b) the limiting standard deviation as the number of stocks goes to infinity.

Problem 9.6. A stock has a beta of 0.9, the market's standard deviation is 16%, and the stock's total standard deviation is 24%. Compute (a) the systematic variance, (b) the proportion of variance that is systematic, and (c) the idiosyncratic standard deviation.

Problem 9.7. An analyst estimates the market risk premium from N = 81 years of data with an annual standard deviation of 18 percent. Compute the standard error of the sample mean and the half-width of a 95 percent confidence interval.

Problem 9.8. (a) A bond returns 9.5% nominal in a year with 4.0% inflation. Compute the real return. (b) An equity portfolio has an arithmetic average return of 8% and a standard deviation of 20%. Estimate the geometric average using the $\sigma^2/2$ approximation, and compare the value of \$10,000 compounded for 20 years at the geometric rate with that at the arithmetic rate.

Answers to Selected Problems

- **9.1.** (a) Arithmetic average = (10 − 5 + 18 + 7) / 4 = 7.50%. (b) Product of gross returns: 1.10 × 0.95 × 1.18 × 1.07 = 1.31942; fourth root = 1.0718, so the geometric average is approximately **7.18%**.
- **9.2.** E(R) = 0.25(−12) + 0.50(6) + 0.25(20) = −3 + 3 + 5 = **5.0%**. Deviations: −17, +1, +15; squares 289, 1, 225 (in %²). Variance = 0.25(289) + 0.50(1) + 0.25(225) = 72.25 + 0.5 + 56.25 = 129 %², i.e. **0.0129**; σ = **11.36%**.
- **9.3.** σ_p^2 = 0.25(0.0625) + 0.25(0.0225) + 2(0.5)(0.5)(0.30)(0.25)(0.15) = 0.015625 + 0.005625 + 0.005625 = **0.026875**; σ_p = **16.39%**.
- **9.4.** Cov = $\rho\sigma_1\sigma_2$ = −0.40 × 0.12 × 0.20 = **−0.0096**.
- **9.5.** (a) σ_p^2 = 0.16/20 + (19/20)(0.30)(0.16) = 0.008 + 0.0456 = 0.0536; σ_p = **23.15%**. (b) Limiting variance = 0.30 × 0.16 = 0.048; σ = √0.048 = **21.91%**.
- **9.6.** (a) $\beta^2\sigma_m^2$ = 0.81 × 0.0256 = **0.020736**. (b) Total variance 0.0576; share = 0.020736 / 0.0576 = **36.0%**. (c) Idiosyncratic variance = 0.0576 − 0.020736 = 0.036864; σ = **19.2%**.
- **9.7.** SE = 18% / √81 = 18% / 9 = **2.0%**; half-width = 1.96 × 2.0% = **3.92%**.
- **9.8.** (a) Real return = 1.095 / 1.04 − 1 = **5.29%**. (b) Geometric ≈ 8% − (0.20)²/2 = 8% − 2% = **6%**. \$10,000 × 1.06^{20} = **\$32,071** versus \$10,000 × 1.08^{20} = **\$46,610**; using the arithmetic rate overstates 20-year wealth by about \$14,540, or 45 percent.

Chapter 10: Portfolio Theory and the CAPM

Learning Objectives

After studying this chapter, you should be able to:

- Construct the opportunity set of a two-asset portfolio, find the minimum-variance portfolio, and describe the efficient frontier of risky assets.
- Explain how introducing a risk-free asset yields the tangency portfolio and the Capital Market Line, and apply the separation theorem to a client's choice of risk level.
- State the assumptions of the Capital Asset Pricing Model, derive the Security Market Line, and use it to compute required returns and detect over- and under-valued securities.
- Estimate beta by regression and by the covariance formula, interpret its statistical precision, and adjust and re-lever betas from comparable firms for use in project valuation.
- Evaluate portfolio performance with the Sharpe ratio, Treynor ratio, and Jensen's alpha, and explain why they can rank managers differently.
- Describe the Arbitrage Pricing Theory, multifactor models, and the Fama-French factors, and discuss the empirical and practical limits of the CAPM and its alternatives.

Opening Vignette

The following scenario is illustrative; Brightline Logistics is a fictional company.

Brightline Logistics operates a regional trucking and warehousing network with annual revenue of about $2.4 billion. Its CFO, Marcus Oyelaran, has a problem that sounds technical but is worth tens of millions of dollars: what discount rate should Brightline use for three very different investments on the capital plan?

The first is a $90 million expansion of its core trucking fleet. The second is a $60 million automated cold-storage warehouse serving pharmaceutical distributors. The third is a $40 million stake in a freight-matching technology start-up. The treasury team's initial suggestion was to discount all three at Brightline's company-wide cost of capital of 9.5 percent. Marcus objected: "That makes the start-up look cheap and the warehouse look expensive, or the other way around, depending on which cash flows you believe. If I discount different risks at the same rate, I am giving away value in one direction or the other."

At the same time, Brightline's pension fund committee was meeting in the next room. Its members were debating whether the fund's equity manager, who had beaten the index by 2 percentage points last year, was skilled or simply running a riskier portfolio. A committee member proposed comparing the manager's return to that of a benchmark with the same sensitivity to the market. Another asked how the fund's equity allocation should be set in the first place, and whether the fund should hold a single optimal portfolio of risky assets and merely dial total risk up or down with cash and borrowing.

These questions share a framework. Harry Markowitz showed how to combine assets to obtain the best trade-off between expected return and risk. William Sharpe, John Lintner, and Jan Mossin took this portfolio theory and asked what prices would result if all investors followed it, producing the Capital Asset Pricing Model (CAPM), a simple statement that required return rises linearly with an asset's exposure to market risk. Later work, from Stephen Ross's Arbitrage Pricing

Theory to the factor models of Eugene Fama and Kenneth French, loosened the CAPM's single-factor structure. This chapter develops the mathematics of portfolio choice, derives and applies the CAPM, shows how to estimate and adjust beta, evaluates performance, and then surveys multifactor models and their implications for the managers' hurdle rates.

10.1 Portfolio Choice with Two Risky Assets

10.1.1 The investment opportunity set

Chapter 9 showed that the expected return of a portfolio is a weighted average of the expected returns of its components, whereas its variance depends on the covariances among them. Combining those two results gives the **investment opportunity set**: the collection of all (risk, return) pairs attainable by varying the weights.

For two risky assets, 1 and 2, with weights w and 1 − w:

$$E(R_p) = w\,E(R_1) + (1-w)\,E(R_2)$$
$$\sigma_p^2 = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho_{12}\sigma_1\sigma_2$$

Plotting σ_p on the horizontal axis and E(R_p) on the vertical axis for all w produces a curve. When ρ = +1 the curve is a straight line between the two assets. When ρ = −1 it consists of two straight lines that touch the vertical axis at the zero-risk portfolio. For intermediate correlations it bows to the left: the lower the correlation, the greater the bow and the larger the diversification benefit. Allowing negative weights (short sales) extends the curve beyond the two assets in both directions.

10.1.2 The minimum-variance portfolio

The portfolio with the lowest possible variance is found by setting the derivative of σ_p^2 with respect to w equal to zero. For two assets the weight on asset 1 in the **minimum-variance portfolio** (MVP) is

$$w_1^* = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$

Notice that this weight depends only on risks and covariance, not on expected returns. Notice also that it can exceed 1 or fall below 0 for extreme parameter values, signifying that the minimum-variance portfolio requires a short position.

Example 10.1 – The two-asset opportunity set and the minimum-variance portfolio. Consider an equity fund E and a bond fund D with the following parameters:

	Expected return	Standard deviation
Equity fund E	12%	20%
Bond fund D	6%	10%
Correlation ρ(E, D)	0.20	

Covariance. $\sigma_{ED} = 0.20 \times 0.20 \times 0.10 = 0.0040$.

Minimum-variance weight on E.

$$w_E^* = \frac{\sigma_D^2 - \sigma_{ED}}{\sigma_E^2 + \sigma_D^2 - 2\sigma_{ED}} = \frac{0.0100 - 0.0040}{0.0400 + 0.0100 - 0.0080} = \frac{0.0060}{0.0420} = 0.1429$$

So the MVP is 14.29% in E and 85.71% in D.

MVP return and risk.

```
E(R) = 0.1429(12%) + 0.8571(6%) = 1.714% + 5.143% = 6.857%
σ²   = (0.1429)²(0.04) + (0.8571)²(0.01) + 2(0.1429)(0.8571)(0.004)
     = 0.000816 + 0.007347 + 0.000980 = 0.009143
σ    = √0.009143 = 9.56%
```

The MVP has a lower standard deviation (9.56 percent) than even the safer asset, D, held alone (10 percent). Here is the opportunity set for a range of weights:

Weight in E	Expected return	Variance	Standard deviation
0%	6.00%	0.01000	10.00%
10%	6.60%	0.00922	9.60%
14.29% (MVP)	6.86%	0.00914	9.56%
20%	7.20%	0.00928	9.63%
30%	7.80%	0.01018	10.09%
40%	8.40%	0.01192	10.92%
50%	9.00%	0.01450	12.04%
60%	9.60%	0.01792	13.39%
80%	10.80%	0.02728	16.52%
100%	12.00%	0.04000	20.00%

Interpretation. Portfolios with less than 14.29 percent in E (such as 100 percent in D) lie on the *inefficient* lower branch of the opportunity set: there is a portfolio with a higher expected return and lower risk (e.g. the 10 percent E portfolio improves on D on both counts: return 6.60 versus 6.00 percent, risk 9.60 versus 10.00 percent). Only the portion of the curve from the MVP upward, where more risk buys more expected return, is efficient. An investor holding all bond fund would be wise to move at least to the MVP.

10.1.3 Dominance and efficiency

Portfolio A **dominates** portfolio B if A has an expected return at least as high and a standard deviation no higher, with at least one strict inequality. The **efficient set** consists of portfolios that no other attainable portfolio dominates. Mean-variance investors choose only among efficient portfolios, picking a point on the efficient set according to their risk tolerance. This is the key insight of Markowitz: the *portfolio*, not the individual security, is the unit of analysis, and an asset should be judged by what it adds to the portfolio's risk and return.

10.2 The Efficient Frontier, the Risk-Free Asset, and the Capital Market Line

10.2.1 The efficient frontier of many risky assets

With n risky assets, the set of attainable portfolios is a region (a bullet-shaped area) in risk-return space, bounded on the left by a curve called the **minimum-variance frontier**. The upper half of that boundary, running from the global minimum-variance portfolio upward, is the **efficient frontier**. Every portfolio on it offers the highest expected return available for its level of risk.

Finding the frontier requires solving a quadratic optimization problem, minimizing σ_p^2 subject to a target expected return and weights summing to one, using inputs of n expected returns, n variances, and $n(n-1)/2$ covariances. For 100 assets that is 100 + 100 + 4,950 = 5,150 inputs, each estimated with error. This leads to a well-known practical problem: optimizers magnify estimation errors. A small error in an expected return can produce large, unstable weights and extreme

positions, a phenomenon sometimes described as “error maximization”. Practical remedies include imposing weight limits, shrinking estimates toward a prior, using factor models to reduce the number of parameters, and combining the optimizer with market-implied returns, as in the Black-Litterman approach. The main text of this chapter focuses on the theory, but the practitioner should be modest about its numerical precision.

10.2.2 Adding a risk-free asset

Now allow investors to lend and borrow at a risk-free rate r_f. A risk-free asset has zero variance and zero covariance with every risky asset, so a portfolio that places a fraction x of wealth in a risky portfolio P and 1 − x in the risk-free asset has

```
E(R_c) = r_f + x [E(R_P) − r_f]
σ_c    = x σ_P
```

Eliminating x gives a straight line in risk-return space:

```
E(R_c) = r_f + [(E(R_P) − r_f) / σ_P] · σ_c
```

The slope of this line is the risky portfolio’s **Sharpe ratio**, or reward-to-variability ratio: the excess return per unit of standard deviation. Values of x between 0 and 1 represent lending; x > 1 means borrowing at the risk-free rate to invest more than one’s wealth in P.

Among all lines from the risk-free rate through points in the risky opportunity set, the steepest is the one tangent to the efficient frontier. The point of tangency is the **tangency portfolio** T, the risky portfolio with the highest possible Sharpe ratio. Every investor, whatever their risk tolerance, who has the same beliefs about returns and risks would hold some combination of the risk-free asset and T. This is the **separation theorem** (James Tobin): the investment decision splits into two independent steps. First, identify the optimal risky portfolio T, which depends only on the market’s risks and returns. Second, decide how much to allocate to T versus the risk-free asset, which depends on the investor’s risk aversion.

For two risky assets the tangency weights have a closed form. Define excess returns $e_1 = E(R_1) - r_f$ and $e_2 = E(R_2) - r_f$:

```
w₁ = [e₁σ₂² − e₂σ₁₂] / [e₁σ₂² + e₂σ₁² − (e₁ + e₂)σ₁₂]
```

with $w_2 = 1 - w_1$.

Example 10.2 – The tangency portfolio and the Capital Market Line. Continue with E and D from Example 10.1 and introduce a risk-free rate of 3 percent.

Excess returns. e_E = 12% − 3% = 9%; e_D = 6% − 3% = 3%.

Tangency weight on E.

```
Numerator   = e_E σ_D² − e_D σ_ED = 0.09(0.0100) − 0.03(0.0040)
            = 0.00090 − 0.00012 = 0.00078
Denominator = e_E σ_D² + e_D σ_E² − (e_E + e_D) σ_ED
            = 0.00090 + 0.03(0.0400) − 0.12(0.0040)
            = 0.00090 + 0.00120 − 0.00048 = 0.00162
w_E = 0.00078 / 0.00162 = 0.4815,   w_D = 0.5185
```

Tangency portfolio return and risk.

```
E(R_T) = 0.4815(12%) + 0.5185(6%) = 5.778% + 3.111% = 8.889%
σ_T²   = (0.4815)²(0.04) + (0.5185)²(0.01) + 2(0.4815)(0.5185)(0.004)
       = 0.009273 + 0.002689 + 0.001997 = 0.013959
σ_T    = 11.81%
```

Sharpe ratio of T. (8.889% − 3%) / 11.81% = **0.498**. For comparison, the Sharpe ratio of E alone is 9% / 20% = 0.45 and that of D alone is 3% / 10% = 0.30. The tangency portfolio is better than either.

The Capital Market Line for this economy:

```
E(R_c) = 3% + 0.498 × σ_c
```

Application 1 – A conservative client who wants a risk of 6%. x = 6% / 11.81% = 0.508. Invest 50.8% in T and 49.2% in the risk-free asset. Expected return = 3% + 0.508 × 5.889% = **5.99%**. Dollar allocation to the funds: E gets 0.508 × 0.4815 = 24.5%, D gets 0.508 × 0.5185 = 26.3%, risk-free 49.2%.

Application 2 – An aggressive client who wants an expected return of 10%. x = (10% − 3%) / 5.889% = 1.1887. Borrow 18.87% of wealth at 3% and invest 118.87% in T. Risk = 1.1887 × 11.81% = **14.04%**. Net positions: E = 1.1887 × 0.4815 = 57.2%, D = 1.1887 × 0.5185 = 61.6%, risk-free = −18.9%.

Comparison with no risk-free asset. On the risky-only opportunity set, a 10 percent expected return required a mix of about 66.7 percent E and 33.3 percent D (0.667 × 12 + 0.333 × 6 = 10%), with a risk of 14.38 percent. Allowing the client to borrow at the risk-free rate reduces the standard deviation to 14.04 percent for the same return.

Interpretation. Both clients hold the same mix of risky funds, about 48/52 E to D; they differ only in the amount of cash or leverage. The investor's risk tolerance affects *how much* risk, not *which* risky assets.

10.2.3 The Capital Market Line

If all investors share the same expectations and the same risk-free rate, they all identify the same tangency portfolio. For markets to clear, the sum of all investors' holdings of risky assets must equal the supply of risky assets, so T must be the **market portfolio** M: the value-weighted portfolio of all risky assets. The line from the risk-free rate through M is the **Capital Market Line** (CML):

```
E(R_c) = r_f + [(E(R_M) − r_f) / σ_M] · σ_c
```

The CML describes the equilibrium trade-off between risk (standard deviation) and expected return for *efficient portfolios*. Its slope, (E(R_M) − r_f)/σ_M, is the market price of risk: the extra expected return investors demand for each additional unit of total risk on an efficient portfolio. The CML applies to efficient portfolios only. Individual securities and inefficient portfolios lie below it, because they carry diversifiable risk that earns no reward.

10.2.4 Practical consequences for managers and investors

Although the idea of a single optimal risky portfolio is a theoretical simplification, it has practical force.

- **Passive indexing.** If the market portfolio is the tangency portfolio, then a low-cost index fund combined with cash or leverage is an efficient strategy. Active managers must earn enough to cover their fees.
- **Asset-allocation policy.** A pension fund's primary decision is the risk-level mix between a diversified risky portfolio and low-risk assets, not the selection of individual securities. Differences in strategic allocation typically explain most of the variation in fund returns.

- **Constraints.** Many investors cannot borrow at the risk-free rate, which flattens the upper part of the line and gives high-beta assets an extra attraction; this is one explanation (discussed in Section 10.8) for empirical deviations from the CAPM.

10.3 The Capital Asset Pricing Model

10.3.1 Assumptions

The CAPM, developed independently by Sharpe, Lintner, and Mossin in the 1960s, derives equilibrium prices from the portfolio theory above. It rests on a set of idealized assumptions:

1. Investors are risk-averse, care only about the mean and variance of portfolio returns over a single period, and maximize expected utility.
2. All investors have identical (homogeneous) expectations about returns, variances, and covariances.
3. There is a risk-free asset at which all investors can lend and borrow without limit.
4. Markets are frictionless: no taxes, no transaction costs, and all assets are infinitely divisible and tradable.
5. Investors are price takers, and the information is costlessly available to all.

These assumptions are obviously unrealistic. The model is valuable not because they are true but because the structure that results is simple, testable, and a useful first approximation. The relaxation of each assumption has been studied, and generally the central message (priced risk is market-related risk) survives in modified form.

10.3.2 Deriving the Security Market Line

Under the assumptions, every investor holds the market portfolio M, plus or minus cash. In equilibrium, M is the tangency portfolio and the expected return of every asset must be consistent with its contribution to the risk of M. The marginal contribution of asset i to the variance of the market portfolio is proportional to its covariance with the market, σ_iM. In equilibrium the ratio of an asset's risk premium to its marginal risk contribution must be the same for all assets, including M itself:

```
[E(R_i) − r_f] / σ_iM  =  [E(R_M) − r_f] / σ_M²
```

Rearranging gives the **Security Market Line** (SML):

```
E(R_i) = r_f + β_i [E(R_M) − r_f]
β_i = σ_iM / σ_M² = Cov(R_i, R_M) / Var(R_M)
```

The **beta** of a security is the ratio of its covariance with the market to the variance of the market. It measures how many percentage points the security's return tends to move per one-point move in the market. The market portfolio has beta 1 by construction; the risk-free asset has beta 0. A portfolio's beta is the weighted average of the betas of its constituents.

The SML is the central empirical and practical implication of the CAPM:

> The expected return on any asset equals the risk-free rate plus beta times the market risk premium.

Two points deserve emphasis. First, **only beta matters**: an asset's own standard deviation and its idiosyncratic risk do not enter the required return. Second, the **SML is not the CML**. The CML plots *efficient portfolios* with standard deviation on the horizontal axis; the SML plots *any asset or portfolio* with beta on the horizontal axis. For a fully diversified portfolio,

the two coincide in the sense that its standard deviation equals $\beta_p \sigma_M$.

10.3.3 The SML as a hurdle rate and a pricing benchmark

An asset's SML return is the return required by diversified investors for its systematic risk. It plays two roles.

1. **Cost of equity.** For a firm, the required return on its equity is $r_E = r_f + \beta_E \times MRP$. This is the discount rate for equity cash flows and a component of the weighted average cost of capital (WACC).
2. **A benchmark for pricing.** An asset whose expected return (given by the analyst's forecasts) lies above the SML is underpriced: it offers a positive **alpha**, $\alpha = E(R_i) - [r_f + \beta_i(E(R_M) - r_f)]$. An asset below the line is overpriced and has negative alpha. If prices are in equilibrium, all alphas are zero.

Example 10.3 – Required returns on the SML and alpha. The risk-free rate is 3 percent and the market risk premium is 6 percent.

Required returns:

Stock	Beta	Required return = 3% + β × 6%
Utility U	0.60	3% + 3.6% = 6.6%
Market index	1.00	3% + 6.0% = 9.0%
Cyclical manufacturer C	1.40	3% + 8.4% = 11.4%

Alpha screening. An analyst forecasts the following expected returns for two other stocks:

Stock	Beta	Analyst's expected return	SML required return	Alpha
Q	1.30	12.0%	3% + 1.3 × 6% = 10.8%	+1.2%
R	0.80	7.0%	3% + 0.8 × 6% = 7.8%	−0.8%

Interpretation. Q plots above the SML: it offers 1.2 percentage points more than its beta justifies, so it appears underpriced and attractive. R plots below the line and appears overpriced. The analyst should ask what could be wrong with the forecasts before acting, since the typical difference between forecast and equilibrium return is dominated by forecasting error, and a modest alpha can vanish with a small change in beta or in the market risk premium. If Q's beta were 1.5 instead of 1.3, its required return would be 12.0% and its alpha would be zero.

10.3.4 Using the CAPM in corporate finance

For a firm with a well-estimated equity beta of 1.15, a risk-free rate of 4 percent, and a market risk premium of 5.5 percent, the cost of equity is 4% + 1.15 × 5.5% = 10.325%. Inputs deserve careful choice.

- **Risk-free rate.** Use a government yield whose maturity matches the horizon of the cash flows being valued, typically a long-term bond yield for long-lived projects. The currency must match the currency of the cash flows.
- **Market risk premium.** Use a forward-looking, defensible estimate (Chapter 9), consistently applied across the firm's valuations.
- **Beta.** Estimated from historical regression or from comparable firms, with the adjustments in Section 10.4.

The CAPM's resulting discount rate applies to the *systematic* risk of cash flows. Idiosyncratic risk should be incorporated in the expected cash flows and scenario analysis, not by raising the discount rate. A common error is to add a "company-specific premium" to compensate for an uncertain forecast: this double-counts risk if the cash flows are already probability-weighted and penalizes smaller or less-diversified firms arbitrarily. (There are legitimate reasons for adjustments, such as illiquidity in a private-company valuation, but they should be explicit and justified.)

10.4 Estimating Beta

10.4.1 The characteristic line

In practice, beta is estimated by regressing the excess returns of a security on the excess returns of a market index:

$$R_{i,t} - r_{f,t} = \alpha_i + \beta_i (R_{M,t} - r_{f,t}) + \varepsilon_{i,t}$$

This regression line is called the **characteristic line**. The slope coefficient is the estimate of beta; it equals the sample covariance of the security's and the market's returns divided by the sample variance of the market's returns:

$$\hat{\beta} = \frac{\sum (R_{M,t} - \bar{R}_M)(R_{i,t} - \bar{R}_i)}{\sum (R_{M,t} - \bar{R}_M)^2}$$

The intercept is the estimate of Jensen's alpha, and the R^2 of the regression is the proportion of variance that is systematic (Chapter 9). The standard error of the slope tells us how precisely beta is measured.

Example 10.4 – Estimating beta by regression. To keep the arithmetic visible we use eight quarterly observations (real applications use at least 60 monthly or 100 weekly observations). Returns are in excess of the risk-free rate, so the market return below is already a market excess return.

Quarter	Market excess return R_M	Stock excess return R_i
1	4%	8%
2	−2%	−5%
3	6%	5%
4	1%	4%
5	−5%	−7%
6	8%	13%
7	3%	1%
8	−1%	2%

Step 1 – Means. $\bar{R}_M = 14 / 8 = 1.75\%$. $\bar{R}_i = 21 / 8 = 2.625\%$.

Step 2 – Deviations and sums. Market deviations: 2.25, −3.75, 4.25, −0.75, −6.75, 6.25, 1.25, −2.75 (in %). Stock deviations: 5.375, −7.625, 2.375, 1.375, −9.625, 10.375, −1.625, −0.625 (in %).

$$\sum (\text{dev}_M)^2 = 0.013150 \quad \text{(in decimal units)}$$
$$\sum (\text{dev}_M)(\text{dev}_i) = 0.017925$$
$$\sum (\text{dev}_i)^2 = 0.029788$$

Step 3 – Beta and alpha.

$$\hat{\beta} = 0.017925 / 0.013150 = 1.363$$
$$\hat{\alpha} = \bar{R}_i - \hat{\beta}\bar{R}_M = 2.625\% - 1.363 \times 1.75\% = 2.625\% - 2.386\% = 0.24\% \text{ per quarter}$$

Step 4 – Fit and precision. Computing each residual as $R_i - \hat{\alpha} - \hat{\beta} R_M$ gives +2.31%, −2.51%, −3.42%, +2.40%, −0.42%, +1.86%, −3.33%, +3.12%. The residual sum of squares (RSS) is 0.005354 in decimal units.

```
R² = 1 − RSS/TSS = 1 − 0.005354 / 0.029788 = 0.820
Correlation = 0.906
Residual standard deviation = √(0.005354 / 6) = 2.99% per quarter
SE(β̂) = √[ (RSS/(n−2)) / Σ(dev_M)² ] = √[ (0.005354/6) / 0.013150 ] = 0.260
t-statistic = 1.363 / 0.260 = 5.23
95% confidence interval (t₆ = 2.447): 1.363 ± 2.447 × 0.260 = 0.73 to 2.00
```

Interpretation. The point estimate suggests a stock that is 36 percent more sensitive than the market, and 82 percent of its variance is systematic. But the confidence interval is wide: the true beta could plausibly lie anywhere from 0.73 to 2.00. Even with 60 observations and a better fit, beta intervals of ±0.3 are typical for individual stocks. A manager who plans to use this beta to price a project should treat it as an estimate, not a fact.

10.4.2 Statistical and practical issues

Estimating beta involves many choices, and they matter.

- **Return interval and window.** Monthly returns over five years are the most common convention; weekly returns over two years are another. Shorter intervals produce more observations but increase bias for thinly traded stocks, whose prices adjust to market moves with a lag. The choice can move beta by 0.1 to 0.3.
- **Market index.** The CAPM market portfolio includes all risky assets (equities, bonds, real estate, human capital), but the available proxy is an equity index. This is the foundation of Roll's critique (Section 10.8).
- **Thin trading.** Stocks that trade infrequently have betas biased downward because their recorded prices do not reflect market movements on the same day.
- **Structural change.** A firm's beta changes when it changes its business mix or leverage, and a historical regression reflects the *past* firm.
- **Estimation error and mean reversion.** Betas estimated from a historical regression regress toward 1 over time: extreme high betas tend to be overestimates, and extreme low betas underestimates. Practitioners therefore commonly use an **adjusted beta**, often the Blume adjustment:

```
Adjusted β = (2/3) × raw β + (1/3) × 1.0
```

For the raw beta of 1.363 in Example 10.4, the adjusted beta is 0.667 × 1.363 + 0.333 = 1.243. Vendors differ in their weights and methods, and the analyst should know which version is being used.

10.4.3 Betas from comparable firms: unlevering and relevering

A single firm's regression beta is often too noisy to use. A better estimate of a business's risk is the average beta across a group of **pure-play** comparable firms, adjusted for their capital structures. Equity beta reflects two kinds of risk: the risk of the business assets, and the extra risk of financial leverage. The **asset (unlevered) beta**, β_U, isolates business risk. Under the common assumptions that debt is risk-free and the debt-to-equity ratio is held constant (the Hamada relation):

```
β_E = β_U × [1 + (1 − T)(D/E)]        β_U = β_E / [1 + (1 − T)(D/E)]
```

where T is the corporate tax rate. The procedure is to (1) unlever each comparable's equity beta using its own D/E and tax rate, (2) average the asset betas, and (3) relever at the target firm's (or project's) intended capital structure.

Example 10.5 – Unlevering and relevering a comparable's beta. A listed pure-play comparable to Brightline's cold-storage project has an equity beta of 1.40, a debt-to-equity ratio of 0.50, and faces a 25 percent tax rate. Brightline plans to finance the project with a debt-to-equity ratio of 0.25.

Unlever:

```
β_U = 1.40 / [1 + (1 − 0.25)(0.50)] = 1.40 / 1.375 = 1.018
```

Relever at D/E = 0.25:

```
β_E = 1.018 × [1 + 0.75 × 0.25] = 1.018 × 1.1875 = 1.209
```

Cost of equity (r_f = 3%, MRP = 6%): 3% + 1.209 × 6% = **10.25%**. Had the manager used the comparable's raw equity beta of 1.40, the cost of equity would have been 11.4%, a difference of 1.15 percentage points arising purely from the capital-structure mismatch. Using Brightline's company-wide beta rather than the project's business beta would introduce a different error, as the next section discusses.

Interpretation. Asset betas strip out financing choices so that business risk can be transferred across firms. A real application should also consider that debt has some beta, so the simple formula tends to overstate the leverage effect when debt is risky; practitioners with highly leveraged comparables use extended versions of the formula that allow a positive debt beta.

10.4.4 Project and divisional betas

The discount rate should reflect the risk of the *project*, not the firm that undertakes it. A company whose average beta is 1.1 may have a trucking division with a beta of 1.3, a cold-storage division with 0.85, and a technology investment with 1.8. Discounting all three at the same rate causes the firm to accept too many high-risk projects, which have a positive NPV only because their risk is understated, and to reject too many safe ones. Over time the firm drifts toward higher risk without earning compensating returns. The solution is to estimate the beta of each business line from pure-play comparables, as in Example 10.5, or to use a risk-adjusted hurdle rate scheme with several risk categories.

10.5 Performance Measurement

The CAPM supplies benchmarks for evaluating portfolios, since raw returns cannot be compared without adjusting for risk. Three standard measures differ in the risk concept they use.

Sharpe ratio (reward-to-variability): total excess return per unit of total risk.

```
Sharpe = (R̄_p − r_f) / σ_p
```

It is appropriate for evaluating a portfolio that represents an investor's entire risky holdings, because total risk is what that investor bears.

Treynor ratio (reward-to-volatility): excess return per unit of systematic risk.

```
Treynor = (R̄_p − r_f) / β_p
```

It is appropriate for evaluating a portfolio as one component of a larger diversified portfolio, where idiosyncratic risk is diversified away by other holdings.

Jensen's alpha: the difference between the portfolio's average return and what the SML predicts for its beta.

```
α_p = R̄_p − [r_f + β_p (R̄_M − r_f)]
```

Alpha is a measure in return units, directly interpretable as “excess return after adjusting for beta.”

Example 10.6 – Three performance measures can disagree. The risk-free rate is 3 percent. The market return is 9 percent with a standard deviation of 15 percent (so the market’s Sharpe ratio is 0.40 and its Treynor ratio is 6 percent). Over the same period, three funds report:

Fund	Average return	Standard deviation	Beta
A	14%	22%	1.30
B	11%	12%	0.80
C	9%	9%	0.50

Sharpe ratios.

```
A: (14 − 3) / 22 = 0.500
B: (11 − 3) / 12 = 0.667
C: (9 − 3) / 9   = 0.667
```

Treynor ratios.

```
A: (14 − 3) / 1.30 = 8.46%
B: (11 − 3) / 0.80 = 10.00%
C: (9 − 3) / 0.50  = 12.00%
```

Jensen’s alphas.

```
A: 14 − [3 + 1.30 × 6] = 14 − 10.8 = +3.2%
B: 11 − [3 + 0.80 × 6] = 11 − 7.8  = +3.2%
C: 9  − [3 + 0.50 × 6] = 9  − 6.0  = +3.0%
```

Rankings.

Measure	Ranking (best to worst)
Sharpe	B = C, then A
Treynor	C, then B, then A
Jensen’s alpha	A = B, then C

Interpretation. All three funds beat the market on every measure (Sharpe above 0.40, Treynor above 6 percent, positive alpha), but they rank differently. Fund A has the highest absolute return and the highest alpha, but it earns the least per unit of total risk, indicating that it bears considerable idiosyncratic risk. If A were the investor’s only holding, the Sharpe ratio would be the relevant statistic and A would rank last. If A were one of many funds in a large diversified portfolio, the idiosyncratic risk would be diversified and alpha would be the relevant statistic, and A would rank at the top. The right measure depends on the role the fund plays in the investor’s portfolio. Note, finally, that a few years of results rarely distinguish skill from luck: for a fund with 5 percent tracking error, an annual alpha of 3 percent measured over five years has a t-statistic of only $3 / (5 / \sqrt{5}) = 1.34$.

10.6 Arbitrage Pricing Theory and Multifactor Models

10.6.1 The idea of factor models

The CAPM says there is one source of priced risk, the return on the market portfolio. The real world appears richer. Stocks respond differently to inflation shocks, interest-rate changes, industrial production, and commodity prices, even after controlling for their sensitivity to the equity index. A **factor model** posits that returns are generated by exposure to a small number of common factors plus an idiosyncratic term:

$$R_i = E(R_i) + b_{i1} F_1 + b_{i2} F_2 + \ldots + b_{iK} F_K + \varepsilon_i$$

Here each F_k is a *surprise* (unexpected change) in a systematic factor with mean zero, b_{ik} is the sensitivity (factor loading) of asset i to factor k, and ε_i is idiosyncratic noise with mean zero that is uncorrelated with the factors and across assets. The single-index model of Chapter 9 is the special case with one factor, the market return.

10.6.2 Arbitrage Pricing Theory

Stephen Ross's **Arbitrage Pricing Theory (APT)** derives a pricing relation from a weaker set of assumptions than the CAPM. It assumes only that returns follow a factor structure, that there are enough securities to diversify away idiosyncratic risk, and that markets do not allow arbitrage (riskless profit with no net investment). The intuition is that if two portfolios have the same exposure to every factor, they have the same risk once idiosyncratic risk is diversified, and thus must have the same expected return; otherwise an investor could buy the cheap one, sell the expensive one, and earn a riskless profit. The result is a linear relation between expected return and factor loadings:

$$E(R_i) = r_f + b_{i1} \lambda_1 + b_{i2} \lambda_2 + \ldots + b_{iK} \lambda_K$$

where λ_k is the **factor risk premium**: the expected excess return on a portfolio with unit exposure to factor k and zero exposure to all other factors. The CAPM emerges as a special case when the only priced factor is the market portfolio.

APT differs from the CAPM in important ways.

Feature	CAPM	APT
Source of risk	One factor (market portfolio)	K factors (unspecified by the theory)
Key assumptions	Mean-variance investors, homogeneous expectations, market portfolio observable	No arbitrage, factor structure, many assets
Role of market portfolio	Central	Not required
Pricing statement	Holds for every asset in equilibrium	Holds for almost all assets in a large economy
Practical guidance	Specifies the factor and its premium	Does not say which factors or what the premia are

The last row is the main weakness. APT does not tell us what the factors are. The analyst must choose them, and different choices lead to different estimates. Three approaches exist: **macroeconomic factor models** (using variables such as GDP growth, inflation, and term spreads), **statistical models** (extracting factors from the covariance matrix of returns by factor analysis or principal components), and **fundamental or characteristic-based models** (using firm attributes such as size, value, and momentum).

10.6.3 Using a macroeconomic multifactor model

Example 10.7 – Expected and realized return in a three-factor APT model. A stock has the following sensitivities and the economy has these factor risk premia, with a risk-free rate of 3%:

Factor	Loading b	Factor risk premium λ
Unexpected GDP growth	1.20	4.0%
Unexpected inflation	0.50	2.0%
Unexpected change in long-term interest rates	−0.30	3.0%

Expected (required) return.

$$E(R) = 3\% + 1.20(4\%) + 0.50(2\%) + (-0.30)(3\%)$$
$$= 3\% + 4.8\% + 1.0\% - 0.9\% = 7.9\%$$

Realized return in a year of surprises. Suppose that unexpected GDP growth was −1.5 percentage points, unexpected inflation was +0.5 points, and the unexpected change in interest rates was +1.0 point, and the firm had no idiosyncratic shock (ε = 0). Each surprise is measured in the same units as the loading:

$$R = E(R) + b_1F_1 + b_2F_2 + b_3F_3 + \varepsilon$$
$$= 7.9\% + 1.20(-1.5\%) + 0.50(0.5\%) + (-0.30)(1.0\%) + 0$$
$$= 7.9\% - 1.8\% + 0.25\% - 0.30\% = 6.05\%$$

Interpretation. The stock's required return is 7.9 percent. The negative loading on interest rates means the stock does well when rates rise unexpectedly, so it acts as a partial hedge and lowers the required return. The realized return of 6.05 percent is below expectation, due to the GDP shortfall, to which the stock is highly exposed. A CAPM analysis with a single beta would summarize the same risks in one number and cannot distinguish a stock that is cyclical from one that is rate-sensitive, even if the two have the same market beta. For a manager, a multifactor model identifies *which* macroeconomic risks the project carries and helps in designing hedges.

10.6.4 Evidence and use of multifactor models

Multifactor models fit the cross-section of returns better than a single-factor CAPM, but their added complexity comes with costs: more parameters to estimate, instability of factor premia over time, and the risk of data mining (testing many candidate factors until one appears to work). Practitioners use multifactor models mainly in portfolio construction and risk management (to decompose a portfolio's exposures to sources of risk, to control tracking error, and to hedge unintended bets) and, less often, in setting corporate discount rates.

10.7 The Fama-French Factors

10.7.1 Anomalies that challenged the CAPM

By the late 1970s and 1980s, researchers had documented patterns in average returns that beta alone could not explain. Small-company stocks earned higher average returns than their betas predicted (the size effect), and stocks with high book-to-market ratios ("value" stocks) earned more than stocks with low ratios ("growth" stocks), again beyond what beta could account for. Eugene Fama and Kenneth French's influential work in the early 1990s showed that, in U.S. data, firm size and book-to-market ratio explained cross-sectional return differences better than beta did.

10.7.2 The three-factor model

Fama and French responded with a three-factor model built on tradable factor portfolios:

```
R_i − r_f = α_i + β_i (R_M − r_f) + s_i SMB + h_i HML + ε_i
```

- **MKT (R_M − r_f)**: the market excess return, as in the CAPM.
- **SMB (small minus big)**: the return on a portfolio that is long small-capitalization stocks and short large-capitalization stocks. A positive loading s_i means the stock behaves like a small company.
- **HML (high minus low)**: the return on a portfolio long high book-to-market (value) stocks and short low book-to-market (growth) stocks. A positive loading h_i means the stock behaves like a value stock.

The expected return implied by the model is

```
E(R_i) = r_f + β_i E(MKT) + s_i E(SMB) + h_i E(HML)
```

where E(SMB) and E(HML) are the average factor premia, typically estimated from long historical samples and with the same imprecision (Chapter 9, Section 9.6) as the market premium.

Subsequent research added further factors. Mark Carhart added **momentum** (UMD or WML: winners minus losers over the past year), reflecting the tendency of recent winners to continue to outperform in the short run. Fama and French later proposed a five-factor model adding **profitability** (RMW: robust minus weak operating profitability) and **investment** (CMA: conservative minus aggressive asset growth). Many other "factors" have been proposed, including low volatility, quality, and liquidity.

10.7.3 Interpreting the factors: risk or mispricing?

Why do small and value stocks earn higher average returns? Two broad explanations exist, and the debate remains unsettled.

- **Rational (risk-based) explanations.** The factors may proxy for risks that investors dislike but that the market beta does not capture: for example, value firms may be financially distressed or inflexible in downturns, and small firms may be more vulnerable to credit conditions. Investors demand a premium for bearing these risks, particularly if they bear on states of the world in which investors' incomes fall.
- **Behavioral and market-friction explanations.** Investors may extrapolate recent success, overpaying for glamorous growth stocks and underpaying for unloved value stocks, with limits to arbitrage preventing correction. Alternatively, the premia may reflect compensation for illiquidity, institutional constraints on leverage, or simply data mining.

This matters for corporate managers: if the factor premia are compensation for risk, then a firm with high loadings should use a higher discount rate. If they are mispricing, the discount rate for valuation purposes should arguably not rise, although the firm's actual cost of raising capital may.

Example 10.8 – Expected return and alpha under the Fama-French model. The risk-free rate is 3%; the assumed factor premia are MKT = 6%, SMB = 2%, and HML = 4%. Two stocks have the following estimated loadings:

Stock	β (MKT)	s (SMB)	h (HML)
S: a growth-oriented larger firm	1.10	0.50	−0.30
V: a small value firm	0.90	−0.20	0.60

Recall the sign conventions: a positive s means small-firm behavior and a negative s large-firm behavior, while a negative h means growth-stock behavior.

Stock S.

```
E(R_S) = 3% + 1.10(6%) + 0.50(2%) + (−0.30)(4%)
       = 3% + 6.6% + 1.0% − 1.2% = 9.4%
```

Stock V.

```
E(R_V) = 3% + 0.90(6%) + (−0.20)(2%) + 0.60(4%)
       = 3% + 5.4% − 0.4% + 2.4% = 10.4%
```

Comparison with the CAPM, which uses beta only:

```
CAPM for S: 3% + 1.10(6%) = 9.6%
CAPM for V: 3% + 0.90(6%) = 8.4%
```

Interpretation. The CAPM says S requires more than V because of its higher beta (9.6% versus 8.4%). The three-factor model reverses the ranking (S 9.4% versus V 10.4%), because V's value tilt is heavily rewarded (+2.4 points) and S's growth tilt penalizes its required return (−1.2 points). Using CAPM alone would understate V's cost of capital by 2.0 points and overstate S's by 0.2 points. Because the SMB and HML premia are themselves measured with large standard errors, the three-factor numbers should not be treated as more precise merely because they come from a richer model.

Alpha evaluation. If a fund with the same loadings as S earned an average of 13% over the period, then its alpha against the three-factor benchmark is 13% − 9.4% = **+3.6%**; against the CAPM benchmark, it would be 13% − 9.6% = +3.4%. A fund that earns its returns by tilting toward small and value stocks would show a positive CAPM alpha and a near-zero three-factor alpha, which is why factor models have become the standard for separating "skill" from a systematic style exposure that investors could replicate cheaply.

10.8 Critiques, Evidence, and Practical Application

10.8.1 Empirical tests of the CAPM

Early tests (Black, Jensen, and Scholes; Fama and MacBeth) found that average returns are positively related to beta, as the CAPM predicts, but that the relation is *flatter* than the theory implies: low-beta stocks earned more than predicted and high-beta stocks earned less. Later studies found that beta has weak explanatory power once other characteristics (size, value, momentum, profitability) are controlled. The **low-volatility anomaly**, the finding that lower-risk stocks have earned comparable or higher returns than higher-risk stocks, is a further challenge. A common explanation is that leverage constraints lead investors to seek high-beta stocks as a substitute for borrowing, bidding up their prices and lowering their subsequent returns, consistent with a version of the CAPM in which borrowing is restricted (the Black zero-beta model).

10.8.2 Roll's critique

Richard Roll pointed out that the CAPM's central prediction, that the true market portfolio is mean-variance efficient, is in principle untestable, because the true market portfolio includes all assets (real estate, human capital, private businesses, foreign and non-traded assets), and any index used as a proxy may itself be inefficient or efficient, leading to rejections or

acceptances that say nothing about the theory. The critique does not mean the CAPM is wrong; it means that tests of the model are joint tests of the model and the choice of market proxy. It also helps explain why the estimated betas are sensitive to the index chosen.

10.8.3 Other extensions

Theoretical extensions relax the CAPM's assumptions while keeping the spirit of the model.

- **Zero-beta CAPM (Black).** If riskless borrowing is unavailable, the SML is flatter, with a zero-beta portfolio replacing the risk-free asset.
- **Intertemporal CAPM (Merton).** When investment opportunities change over time, investors hedge against shifts, and additional state-variable risks are priced.
- **Consumption CAPM.** Risk is measured by covariance with aggregate consumption growth rather than the market return, providing a link between asset prices and the macroeconomy.
- **Liquidity-adjusted CAPM.** Assets that are illiquid, or whose liquidity declines when the market falls, require higher returns.

10.8.4 What should a manager use?

Despite its empirical shortcomings, the CAPM remains the most widely used model for the cost of equity in corporate practice, for a combination of reasons. It is simple and transparent; its inputs are observable or estimable; its central lesson (diversified investors price only systematic risk) is robust; and the alternatives require more inputs that are also estimated with error. A reasonable practitioner approach is:

1. Use the CAPM as the base case, with carefully estimated and adjusted betas, a defensible risk-free rate and market premium, and business-specific betas for projects.
2. Cross-check using a multifactor model, particularly for firms with extreme size or value characteristics, or for high-stakes valuations.
3. Examine the range of outcomes, not just a point estimate, using sensitivity analysis on beta and the market premium.
4. Compare the implied cost of capital to other evidence: yields on the firm's debt plus an equity risk spread, survey-based estimates, or market-implied costs of capital.
5. Never let the sophistication of the model substitute for judgment about the cash flows. A 0.5 percentage point error in the discount rate is typically much less costly than an unrealistic cash-flow forecast.

10.8.5 Summary of the three pricing frameworks

Framework	Risk measure	Number of priced factors	Strength	Limitation
CAPM	Beta relative to market	1	Simple, intuitive, widely used	Weak empirical fit; market proxy problem
APT / macro multifactor	Loadings on macro factors	K (chosen by analyst)	Flexible, identifies risk sources	Factors and premia not specified by theory
Fama-French and extensions	Loadings on market, size, value (and others)	3 to 6	Better empirical fit	Risk-versus-mispricing debate; unstable premia

Case Study: The Hurdle Rate Debate at Brightline Logistics

Brightline Logistics is a fictional company created for this case.

Brightline Logistics is a regional trucking and warehousing company whose shares trade on a national exchange. Its CFO, Marcus Oyelaran, has to submit a capital plan to the board, and the plan includes three major projects. The investment committee has so far used a single company-wide hurdle rate of 9.5 percent for all projects, and Marcus believes this practice is distorting capital allocation. He asks his analyst, Yuki Tanaka, to prepare a case for project-specific discount rates.

Yuki collects the following information. The risk-free rate (the yield on a long-term government bond) is 3.0 percent, and the finance department uses a market risk premium of 6.0 percent. Brightline's own historical equity beta, from a regression of five years of monthly returns on a broad index, is 1.20 with a standard error of 0.22. Its debt-to-equity ratio is 0.40, its tax rate is 25 percent, and its pre-tax cost of debt is 5.0 percent. Using the company-wide beta, its cost of equity is 3.0% + 1.20 × 6.0% = 10.2%, and its after-tax WACC computed from these inputs is about 8.4 percent (0.714 × 10.2% + 0.286 × 5.0% × 0.75). The investment committee has nonetheless applied a flat 9.5 percent hurdle to all projects for many years, as a conservative policy rule.

The three projects are as follows.

1. *Fleet expansion* ($90 million). This is a core activity, and Yuki believes its risk is similar to the firm's existing business.
2. *Automated cold-storage warehouse* ($60 million). Yuki identifies four listed pure-play cold-chain operators with an average equity beta of 0.95, an average D/E of 0.60, and a 25 percent tax rate. The project's financing would be 30 percent debt, 70 percent equity (D/E = 0.4286).
3. *Freight-matching technology stake* ($40 million). Comparable listed logistics-technology companies have an average equity beta of 1.70, are almost entirely equity financed, and show a wide dispersion of individual betas (from 1.1 to 2.4).

Projected returns on investment (the internal rates of return on forecast cash flows) are 10.0 percent for the fleet, 9.0 percent for the warehouse, and 15.0 percent for the technology stake. Under the flat 9.5 percent hurdle, the fleet (10.0 percent) and the technology stake (15.0 percent) would be accepted and the warehouse (9.0 percent) rejected.

Marcus raises two other issues. First, the pension fund's equity manager has delivered an average return of 11.0 percent with a beta of 1.15 and a standard deviation of 17 percent; the market's average return was 9.0 percent, which makes his market premium 6.0 percent, and the market's standard deviation was 15 percent. A committee member wants to renew the manager's mandate because the manager "beat the market by 2 points." Second, a board member who reads the financial press asks why the company uses the CAPM at all, since "academics have shown that beta doesn't work and that size and value factors explain returns." Brightline's market capitalization places it among mid-sized firms, and its stock has a mild value tilt, with an estimated HML loading of 0.35 and an SMB loading of 0.20.

Marcus wants a recommendation that is technically defensible and can be explained to the board in plain language. He is aware that the analysis involves statistical uncertainty in the betas, a subjective market risk premium, and genuine disagreement among academics about the right model. The board will meet in two weeks.

Discussion questions

1. Unlever the cold-storage comparables' average beta and relever it at the project's financing mix. Compute the project's cost of equity using the CAPM, and discuss whether the warehouse project, with a projected 9.0 percent return, should be accepted or rejected. (You may approximate the project discount rate by weighting the cost of equity with the after-tax cost of debt at the project's 30/70 financing mix.)
2. Compute the cost of equity for the technology stake and discuss how the dispersion in comparable betas should affect the decision. Is the technology stake's 15.0 percent projected return sufficient? What additional analysis would you require?
3. Evaluate the pension manager using Jensen's alpha, the Treynor ratio, and the Sharpe ratio (use a risk-free rate of 3 percent). Does "beat the market by 2 points" demonstrate skill?
4. Respond to the board member's critique. How would the three-factor model change the cost of equity for Brightline, using factor premia of SMB = 2 percent and HML = 4 percent? How should the CFO decide whether to rely on the CAPM or the multifactor model?

Key Takeaways

- Portfolios, not individual assets, are the unit of analysis. Combining imperfectly correlated assets yields a better risk-return trade-off, and the minimum-variance portfolio can have lower risk than any single asset.
- The efficient frontier contains the portfolios with the highest expected return for each level of risk; inputs are estimated with error, and optimizers can amplify those errors.
- With a risk-free asset, the efficient set becomes a straight line from the risk-free rate through the tangency portfolio. The separation theorem says that all investors hold the same risky portfolio and differ only in how much they lend or borrow.
- In equilibrium, the tangency portfolio is the market portfolio, and the line is the Capital Market Line. The CML prices the total risk of efficient portfolios; it is not the same thing as the SML.
- The CAPM implies that expected return equals the risk-free rate plus beta times the market risk premium. Only systematic risk is priced, and an asset plotting above the SML has a positive alpha.
- Beta is estimated by regression, and the estimate is imprecise: a confidence interval of plus or minus 0.3 or more is typical. Adjust betas for mean reversion and be explicit about the return interval, window, and index.
- Business risk should be priced by comparable pure-play asset betas, unlevered and relevered to the project's capital structure. A single company-wide hurdle rate leads to systematic misallocation toward risky projects.
- The Sharpe, Treynor, and Jensen measures answer different questions and can rank managers differently; the relevant measure depends on whether the portfolio is held in isolation or as part of a diversified whole.
- The APT replaces the market portfolio with multiple factors and the no-arbitrage condition, but it does not tell the analyst which factors to use or what their premia are.
- The Fama-French factors (size, value, and later profitability, investment, and momentum) explain cross-sectional returns better than beta alone. Whether they represent risk or mispricing is unresolved, and good practice uses the CAPM as a base case, cross-checked against a multifactor model and sensitivity analysis.

Review Questions

1. Why can the minimum-variance portfolio of two risky assets have lower risk than either asset? Under what correlation does this fail?
2. Distinguish between the opportunity set, the minimum-variance frontier, and the efficient frontier.
3. What does the separation theorem say, and how does it simplify the advice an investment adviser gives to clients with different risk tolerances?
4. What are the slope and intercept of the Capital Market Line, and why does an individual stock not lie on it?
5. Explain the difference between the CML and the SML. On which axes is each plotted, and to which assets does each apply?
6. State three assumptions of the CAPM and discuss the consequences if each is violated.
7. Why does the CAPM imply that a firm's total (stand-alone) risk is irrelevant to its cost of equity? Give two reasons why managers nonetheless care about total risk.
8. Why is a beta estimate subject to considerable error? Describe three practical decisions in the regression that affect the result.
9. Explain how to obtain a project's discount rate from pure-play comparables. Why must one unlever and relever betas?
10. Compare the CAPM, APT, and the Fama-French three-factor model. For what purposes would a CFO prefer each?

Problems

Problem 10.1. Asset X has $\sigma = 25\%$ and asset Y has $\sigma = 15\%$, with a correlation of 0.10. Compute the weight in X, the variance, and the standard deviation of the minimum-variance portfolio.

Problem 10.2. The risk-free rate is 4% and the market risk premium is 5.5%. (a) Compute the required return on a stock with a beta of 1.15. (b) If an analyst forecasts a return of 12% for that stock, compute its alpha and say whether the stock appears under- or overpriced.

Problem 10.3. (a) The covariance of a stock's returns with the market is 0.0405 and the market variance is 0.0324 ($\sigma_M = 18\%$). Compute the beta. (b) A comparable firm has an equity beta of 1.35, a debt-to-equity ratio of 0.60, and a tax rate of 30%. Compute its unlevered beta, relever it at $D/E = 0.20$, and compute the resulting cost of equity with $r_f = 4\%$ and MRP = 5.5%.

Problem 10.4. A portfolio is invested 30% in a stock with a beta of 1.4, 50% in a stock with a beta of 0.9, and 20% in Treasury bills. Compute the portfolio beta and the required return ($r_f = 4\%$, MRP = 5.5%).

Problem 10.5. The market portfolio has an expected return of 9% and a standard deviation of 15%; the risk-free rate is 3%. (a) Write the Capital Market Line. (b) An investor wants a portfolio with a standard deviation of 21%. What weights in the market and the risk-free asset achieve this, and what is the expected return?

Problem 10.6. A portfolio has an average return of 12%, a standard deviation of 20%, and a beta of 1.1. The risk-free rate is 3% and the market risk premium is 6%. Compute the Sharpe ratio, the Treynor ratio, and Jensen's alpha.

Problem 10.7. A two-factor APT model has $r_f = 4\%$, a factor premium of 3% on factor 1, and 2% on factor 2. A stock has loadings of 1.5 on factor 1 and 0.8 on factor 2 and -0.4 on a third factor with premium 1%. Compute the expected return.

Problem 10.8. In a Fama-French three-factor model, r_f = 4%, MKT premium = 5.5%, SMB = 2%, HML = 4%. A fund has loadings of 1.00 (MKT), 0.40 (SMB), and 0.30 (HML) and earned 16%. Compute the expected return and the fund's alpha.

Answers to Selected Problems

- **10.1.** Cov = 0.10 × 0.25 × 0.15 = 0.00375. w_X = (0.0225 − 0.00375) / (0.0625 + 0.0225 − 0.0075) = 0.01875 / 0.0775 = **0.2419** (24.19% in X, 75.81% in Y). Variance = $0.2419^2(0.0625) + 0.7581^2(0.0225) + 2(0.2419)(0.7581)(0.00375)$ = 0.003658 + 0.012931 + 0.001375 = **0.01796**; σ = **13.40%**.
- **10.2.** (a) 4% + 1.15 × 5.5% = 4% + 6.325% = **10.325%**. (b) Alpha = 12% − 10.325% = **+1.675%**; the stock appears underpriced if the forecast is reliable.
- **10.3.** (a) β = 0.0405 / 0.0324 = **1.25**. (b) β_U = 1.35 / [1 + 0.70 × 0.60] = 1.35 / 1.42 = **0.9507**. Relevered: 0.9507 × [1 + 0.70 × 0.20] = 0.9507 × 1.14 = **1.0838**. Cost of equity = 4% + 1.0838 × 5.5% = **9.96%**.
- **10.4.** β_p = 0.30(1.4) + 0.50(0.9) + 0.20(0) = 0.42 + 0.45 = **0.87**. Required return = 4% + 0.87 × 5.5% = **8.785%**.
- **10.5.** (a) E(R_c) = 3% + 0.40 σ_c (slope = 6% / 15% = 0.40). (b) x = 21 / 15 = 1.40, so **140% in the market and −40% in the risk-free asset** (borrowing). Expected return = 3% + 0.40 × 21% = **11.4%**; check: 1.4(9%) − 0.4(3%) = 12.6% − 1.2% = 11.4%.
- **10.6.** Sharpe = (12 − 3) / 20 = **0.45**. Treynor = 9 / 1.1 = **8.18%**. Alpha = 12 − [3 + 1.1 × 6] = 12 − 9.6 = **+2.4%**.
- **10.7.** E(R) = 4% + 1.5(3%) + 0.8(2%) − 0.4(1%) = 4 + 4.5 + 1.6 − 0.4 = **9.7%**.
- **10.8.** E(R) = 4% + 1.00(5.5%) + 0.40(2%) + 0.30(4%) = 4 + 5.5 + 0.8 + 1.2 = **11.5%**. Alpha = 16% − 11.5% = **+4.5%**.

Chapter 11: The Cost of Capital

Learning Objectives

After studying this chapter, you should be able to:

- Explain why the cost of capital is an opportunity cost determined by the risk of the investment rather than by the identity or financing history of the firm that undertakes it.
- Estimate the cost of equity using the Capital Asset Pricing Model, the dividend growth model, and the bond-yield-plus-premium approach, and reconcile the resulting estimates.
- Estimate the cost of debt from market yields, adjust it for default risk and taxes, and explain why the coupon rate on existing debt is the wrong number.
- Compute the weighted average cost of capital (WACC) using market-value weights and target capital structure, and state the conditions under which WACC is a valid discount rate.
- Unlever and relever equity betas, build pure-play estimates for divisions and projects, and set divisional and project-specific hurdle rates that differ from the corporate WACC.
- Incorporate flotation costs correctly, and recognize the most common pitfalls in cost-of-capital practice, including book-value weights, a single company-wide hurdle rate, and mismatches between cash flows and discount rates.

Opening Vignette

The following scenario is illustrative; the company and its figures are fictional.

Harlow Ridge Industries is a mid-sized manufacturer of commercial refrigeration equipment with a stable, mature business that earns steady margins. Three years ago, the board approved the creation of a cloud-connected monitoring division, “Harlow Cloud,” which sells subscription software that tracks temperature data from refrigeration units in supermarkets and pharmacies. Harlow Cloud is small today but is growing at a pace the legacy business cannot match.

At the annual capital budgeting meeting, the chief financial officer, Imani Okafor, announces that the corporate hurdle rate remains 8.9 percent, the weighted average cost of capital she calculated from the firm’s stock beta, its bond yields, and its blended capital structure. Every project proposal will be evaluated against that single number.

The head of the legacy business presents an equipment-replacement project with an internal rate of return of 8.5 percent. It is rejected. Minutes later, the head of Harlow Cloud presents a new analytics platform with an internal rate of return of 10.0 percent. It is approved. Both executives leave the room satisfied that the process was objective.

Imani is uneasy. She has watched the pattern repeat for three years: the safe division keeps losing capital allocation fights, and the risky division keeps winning them. When she runs a quick comparison, she notes that listed software-monitoring competitors have equity betas well above those of listed equipment makers, and that even after adjusting for their different debt levels, the underlying business risk differs substantially. A flat 8.9 percent hurdle is too high for refrigeration replacement and too low for cloud analytics. Over time, the firm is migrating capital toward the riskiest uses without being paid for the risk.

This chapter gives Imani the tools to fix the problem. It develops the cost of equity and cost of debt, assembles them into WACC, shows how to unlever and relever betas so that comparable-company evidence can be transferred to divisions and projects, and catalogues the errors that make a clean-looking discount rate quietly destructive of value.

11.1 What the Cost of Capital Is, and What It Is Not

11.1.1 An opportunity cost, set by risk

Every capital budgeting decision in Chapter 10 required a discount rate. The *cost of capital* is that rate: the expected return that investors could earn on other investments of equivalent risk. It is an opportunity cost. When Harlow Ridge accepts a project, it commits funds that investors could otherwise have deployed in the capital market at the market's price for that risk. A project that returns less than the opportunity cost reduces the wealth of the firm's owners, even if its accounting profit is positive.

Three implications follow immediately, and they organize the entire chapter.

First, the cost of capital depends on the *risk of the cash flows being discounted*, not on the risk of the company's other businesses. A firm's overall cost of capital is an average of the costs of its parts; it is not the right rate for any specific part unless that part happens to be average.

Second, the cost of capital is *forward looking and market based*. It is what investors require today, observable (imperfectly) in current prices. It is not the interest rate printed on the bonds the firm issued ten years ago, nor the accounting return on equity last year.

Third, the cost of capital is *not the cost of the particular funds raised for the project*. If Harlow Ridge finances a software project by drawing on a 5 percent bank line, the project's discount rate is still determined by the risk of software cash flows, not by 5 percent. Financing choices matter, but they matter through tax shields, issuance costs, and the capital structure targets discussed later, not by letting the cheapest funding source set the hurdle.

11.1.2 Required return versus expected return versus cost

Three related terms are often confused. The *expected return* on a security is the probability-weighted average of its possible returns. The *required return* is what investors demand for bearing its risk; in equilibrium, expected and required returns coincide. The *cost of capital* is the required return viewed from the firm's side: what the firm must earn on its investments in order to leave investors no worse off. In practice, all of these must be estimated, because none is directly observed. The analyst's task is to build estimates from market data in a way that is internally consistent.

11.1.3 The building blocks

The firm raises capital from debt holders, preferred shareholders, and common shareholders. Each supplier of capital has a required return. The *weighted average cost of capital* combines those required returns using the proportions in which each source finances the firm's assets:

```
WACC = (E/V) × rE + (P/V) × rP + (D/V) × rD × (1 − T)

where
  E, P, D = market values of equity, preferred, and debt
  V = E + P + D
  rE, rP, rD = required returns on equity, preferred, debt
  T = marginal corporate tax rate
```

The after-tax adjustment on debt appears because interest is deductible in most tax systems, so each dollar of interest costs the firm only (1 − T) dollars after tax. Everything in this chapter is either an input to this formula or an argument about when this formula should or should not be used.

11.1.4 When WACC works as a discount rate

WACC is the correct discount rate for a project's free cash flows when three conditions hold approximately:

1. The project has the same business risk as the firm's existing operations (or the WACC has been adjusted to reflect the project's risk).
2. The project supports the same financial leverage, in market-value terms, as the firm's target capital structure.
3. The cash flows are unlevered free cash flows, calculated before interest payments and after taxes computed as if the firm had no debt, with the tax benefit of debt captured in the discount rate.

When these conditions fail, as they do for diversified conglomerates, special financing arrangements, or projects that change the firm's debt capacity, WACC must be modified, or an alternative method such as adjusted present value (APV) must be used. Sections 11.5 and 11.6 take this up.

11.2 The Cost of Equity

Equity is the hardest component to estimate because shareholders receive a residual, not a contractual payment, so their required return is never stated. Three families of methods are used in practice. A competent analyst applies at least two and reconciles them.

11.2.1 The CAPM approach

The Capital Asset Pricing Model, developed by Sharpe, Lintner, and Mossin building on Markowitz's portfolio theory, states that the expected return on a stock equals the risk-free rate plus a premium proportional to the stock's systematic risk:

```
rE = rf + β × (E[rM] − rf)
```

Here β measures the sensitivity of the stock's returns to market returns, and (E[rM] − rf) is the market risk premium (MRP). Each input requires a judgment.

The risk-free rate. Use a government yield whose maturity matches the horizon of the cash flows being discounted. For long-lived corporate investments, a ten-year or longer government bond yield is customary. The currency must match the cash flows: Euro cash flows are discounted at a rate built on a Euro risk-free rate. Using a short-term bill rate for a twenty-year project mixes a short-horizon yield with long-horizon cash flows and is inconsistent unless the term structure is flat.

The market risk premium. This is the most contested input in corporate finance, because the expected premium is not observable. Analysts draw on three sources: historical averages of realized stock returns over bond returns, which are noisy and sensitive to the sample period; surveys of academics, practitioners, and corporate managers; and implied estimates that back out the premium from current index prices and forecast cash flows. Reasonable premiums for developed markets are frequently placed in a range of roughly 4.5 to 6.5 percent, but the right choice is the one you can defend and apply consistently. The premium is a *market-wide* number: all projects in a given currency should use the same MRP, so that the variation in hurdle rates across projects comes from beta and capital structure, not from analyst mood.

Beta. Beta is the slope coefficient from regressing the stock's returns on market returns. Three practical points matter. Regression betas have large standard errors, often giving a 95 percent confidence interval of plus or minus 0.3 or more for an individual firm. Betas drift toward 1.0 over time, so many practitioners use an adjusted beta (a common convention is two-thirds of the raw beta plus one-third of 1.0). And the beta of a firm reflects its *current* mix of businesses and its *current* leverage, which may not be what the analysis requires. For that reason, industry-average betas, unlevered and relevered, are often more reliable than the firm's own regression beta. Section 11.4 develops this approach.

Example 11.1 – Cost of equity from CAPM, dividend growth, and bond-yield-plus-premium.

Harlow Ridge's equity beta is 1.20. The ten-year government yield is 4.0 percent and the market risk premium is 5.5 percent. The stock trades at $40.00, the next annual dividend is expected to be $2.40, and dividends are expected to grow at 5 percent indefinitely. The firm's long-term debt yields 7.0 percent, and the analyst's judgmental equity premium over the firm's own debt is 4.0 percent.

CAPM:

```
rE = 4.0% + 1.20 × 5.5%
   = 4.0% + 6.6%
   = 10.6%
```

Dividend growth model:

```
rE = D1 / P0 + g
   = 2.40 / 40.00 + 0.05
   = 0.06 + 0.05
   = 11.0%
```

Bond-yield-plus-premium:

```
rE = 7.0% + 4.0% = 11.0%
```

The three estimates are 10.6 percent, 11.0 percent, and 11.0 percent. The range is narrow, and Imani would reasonably adopt 10.6 percent from CAPM as her primary figure because it is the most explicit about risk, and cite the other two as corroboration. If the DDM had produced 14 percent, she would investigate: perhaps the assumed 5 percent growth is too high, or the market is pricing a lower risk than the beta implies. Disagreement among methods is information, not a nuisance.

11.2.2 The dividend growth approach

If dividends are expected to grow at a constant rate g forever, the Gordon growth formula P0 = D1 / (rE − g) can be inverted to obtain the cost of equity, as in Example 11.1. The approach is intuitive and requires no beta, but it has well-known weaknesses. It assumes stable growth, which rules out young firms and those with erratic payout policy. It is extremely sensitive to g: a one-point error in g is a one-point error in rE. And it is unusable for firms that pay no dividends, though a version using free cash flow to equity can sometimes substitute. The approach works best for regulated utilities and other mature dividend payers, which is precisely where practitioners use it.

11.2.3 The bond-yield-plus-premium approach

This approach adds a judgmental risk premium (typically 3 to 5 percent) to the yield on the firm's own long-term debt. The logic is that equity holders are junior to debt holders and therefore demand more. It is crude because the premium is not tied to any theory, and it fails to respond properly when the firm's leverage changes. Its main use is as a sanity check, as in

Example 11.1, or for firms where beta is unobservable, such as private companies.

11.2.4 Equity cost depends on leverage

An essential point, developed fully in Section 11.4, is that the cost of equity is not a fixed characteristic of a firm. It rises with financial leverage, because debt holders take a fixed claim first, making the residual claim of shareholders riskier. Two firms in the same industry with the same operating risk will have different equity betas and costs of equity if their debt-to-equity ratios differ. Any procedure that treats the cost of equity as given, while changing the capital structure, is wrong.

11.3 The Cost of Debt

11.3.1 Use the current yield, not the coupon

The relevant cost of debt is the rate at which the firm could borrow *today* on long-term debt of similar seniority and maturity. The coupon on bonds issued years ago reflects past market conditions and the firm's past credit standing; it is a sunk fact, not a forward-looking cost. For publicly traded bonds, the yield to maturity (YTM) is the market's current required return, computed as the discount rate that equates the bond's price to the present value of its promised payments. For firms without traded debt, the analyst can use the yield on bonds of the same credit rating and maturity, or the rate on a recent bank term loan.

Example 11.2 – Cost of debt from a bond price, and the default adjustment.

Harlow Ridge has a ten-year bond outstanding with a 6.0 percent annual coupon and face value of $100. It currently trades at $94.00. We solve for the YTM that satisfies:

```
94.00 = Σ (t = 1 to 10) 6 / (1 + y)^t  +  100 / (1 + y)^10
```

Trying y = 6.85 percent: the discount factor is 1.0685^(−10) = 0.5155, so the annuity factor is (1 − 0.5155) / 0.0685 = 7.0725, giving a coupon value of 6 × 7.0725 = 42.44; and the principal is worth 100 × 0.5155 = 51.55. Total = 93.99, which rounds to the $94.00 price. So y ≈ 6.85 percent. (A numerical solver gives 6.848 percent.)

The *pre-tax cost of debt* is therefore about 6.85 percent, not the 6.0 percent coupon. With a 25 percent marginal tax rate, the after-tax cost is:

```
rD × (1 − T) = 6.85% × 0.75 = 5.14%
```

Now suppose we recognize that the promised yield exceeds the expected return, because bonds can default. Take a bond with a promised yield of 7.2 percent, an annual default probability of 2 percent, and a recovery rate of 40 percent of the claim. The expected loss rate is 2% × (1 − 0.40) = 1.2 percent per year, so the *expected* return on the bond is approximately:

```
7.2% − 1.2% = 6.0%
```

For investment-grade borrowers the gap between promised and expected yield is small, and the YTM is an acceptable proxy for the cost of debt. For speculative-grade borrowers the gap can be large, and using the promised yield as the cost of debt overstates the true cost to the firm and would, if the debt were part of the WACC for a project, make the debt look more expensive than its true expected return to bondholders. In a rigorous valuation of a distressed firm, the expected return on debt (and the debt beta) should be used.

11.3.2 Tax shield considerations

The (1 − T) factor presumes that the firm earns enough taxable income to use the deduction. For a firm with persistent losses, the effective marginal tax rate may be near zero, and the after-tax cost of debt is the pre-tax cost. Interest-deductibility limits (for example, caps tied to a percentage of earnings) also reduce the benefit at high leverage. The marginal rate on the next dollar of interest, not the average effective rate on past income, is the right T.

11.3.3 Other forms of financing

Preferred stock has a fixed dividend, no maturity, and no tax deductibility in most systems, so its cost is dividend divided by price: rP = DP / PP. *Convertible bonds* and *warrants* embed options, and the clean approach is to separate them into straight-debt and option components rather than treating the low coupon as a low cost. *Leases and other debt-like obligations* belong in the debt component at the rates implied by their terms. Treating any of these off-balance-sheet obligations as free capital understates WACC.

11.3.4 Short-term debt

Accounts payable and accrued expenses are generally treated as part of operating working capital, with their cost implicitly in the cash flows, not as interest-bearing capital. Short-term bank borrowing is included in debt if it is a permanent part of the funding structure; seasonal borrowing that is fully repaid each year is generally excluded from the long-run weights.

The three equity-cost approaches differ in what they assume and where they fail, as Table 11.1 summarizes.

Table 11.1 – Methods for estimating the cost of equity

Method	Core input(s)	Strength	Main weakness	Best used for
CAPM	rf, β, market risk premium	Explicit link to systematic risk; adjusts for leverage via β	β and MRP are estimated with error	Most listed firms; divisions via pure-play betas
Dividend growth	D1, P0, g	Simple; needs no beta	Very sensitive to g; fails for non-payers	Mature dividend payers, utilities
Bond yield plus premium	Own debt yield, judgmental premium	Easy; works without a beta	Premium is arbitrary	Sanity check; private firms
Multi-factor models	Factor loadings and factor premiums	Capture size, value, momentum effects	More parameters, more estimation error	Academic work; some cross-checks

11.4 Betas, Leverage, and the Unlever–Relever Procedure

11.4.1 Two sources of risk in an equity beta

A firm’s equity beta reflects two things: the systematic risk of its operating assets (*business risk*) and the extra risk that fixed debt payments impose on shareholders (*financial risk*). The beta of the firm’s assets, which would be observed if the firm had no debt, is the *asset beta* or *unlevered beta*, βU. The beta of equity in a levered firm, βL, is higher.

To compare firms with different leverage, or to apply evidence from one firm to another, we must strip out the financial risk to isolate business risk, then add back the financial risk appropriate to the target.

11.4.2 Modigliani–Miller with taxes, and the Hamada relationship

Modigliani and Miller showed that, without taxes, the value of a firm is independent of its capital structure, and the cost of equity rises linearly with the debt-to-equity ratio:

```
rE = rU + (rU − rD) × (D/E)
```

With corporate taxes and a constant, permanent debt level, the debt tax shield creates value, and the relationship becomes:

```
rE = rU + (rU − rD) × (1 − T) × (D/E)
```

If debt is riskless (βD = 0), the equivalent statement in betas, associated with Hamada's application of CAPM to this framework, is:

```
βL = βU × [1 + (1 − T) × D/E]
βU = βL / [1 + (1 − T) × D/E]
```

These are the working formulas of the unlever–relever procedure. They rest on assumptions worth knowing: debt is held at a constant ratio or amount in perpetuity, the tax shield is as risky as the debt, and debt beta is zero. When debt carries default risk (βD > 0), a fuller formula applies:

```
βU = [E/(E + D(1−T))] × βL  +  [D(1−T)/(E + D(1−T))] × βD
```

For investment-grade borrowers, βD is typically small (0.0 to 0.2), and the simple Hamada form is adequate. For highly levered or distressed firms, ignoring βD overstates the unlevered beta of the business, because some of what looks like equity risk is actually credit risk borne by lenders. Likewise, if the firm's leverage is expected to change over time, the relationship must be applied period by period rather than with a single D/E.

Some textbooks and practitioners drop the tax term, using βL = βU × (1 + D/E), reasoning that debt tax shields are riskier than debt itself or that the tax benefit is partially offset by personal taxes. The with-tax version is more common in corporate practice, and consistency is more important than the choice: unlever and relever with the same formula.

11.4.3 The procedure

To estimate the equity beta for a target business with a target capital structure:

1. Identify *comparable (pure-play)* firms: listed companies whose operating risk resembles the target's.
2. For each comparable, obtain its equity beta and its market-value D/E ratio and tax rate.
3. Unlever each comparable's beta to obtain its βU.
4. Average (or take the median of) the βU values. Averaging across firms reduces the estimation error that plagues individual betas.
5. Relever the average βU at the target's D/E (using the target's tax rate).
6. Substitute the relevered beta into CAPM.

Example 11.3 – Unlevering comparable betas and relevering for a target.

Harlow Ridge's analyst identifies three listed equipment makers as comparables for the legacy refrigeration division. The tax rate is 25 percent for all.

Table 11.2 – Comparable-firm data and unlevered betas (T = 25%)

Comparable	Equity beta (βL)	D/E (market)	1 + (1−T)(D/E)	Unlevered beta (βU)
Comp A	1.30	0.50	1 + 0.75 × 0.50 = 1.3750	1.30 / 1.3750 = 0.9455
Comp B	1.10	0.25	1 + 0.75 × 0.25 = 1.1875	1.10 / 1.1875 = 0.9263
Comp C	1.50	0.80	1 + 0.75 × 0.80 = 1.6000	1.50 / 1.6000 = 0.9375
Average				**(0.9455 + 0.9263 + 0.9375) / 3 = 0.9364**

Notice how the raw equity betas range from 1.10 to 1.50, a spread of 0.40, but after unlevering, they cluster tightly between 0.93 and 0.95. The dispersion in raw betas was mostly financial risk, not business risk. The unlevered values tell us that the business risk of equipment making is a beta of about 0.94.

Suppose the legacy division would be financed at a target D/E of 0.40. Relever:

```
βL = 0.9364 × [1 + 0.75 × 0.40]
   = 0.9364 × 1.30
   = 1.2174
```

The cost of equity at rf = 4.0 percent and MRP = 5.5 percent:

```
rE = 4.0% + 1.2174 × 5.5% = 4.0% + 6.696% = 10.70%
```

If the same business were all-equity financed, the cost of equity would equal the unlevered cost: 4.0% + 0.9364 × 5.5% = 9.15 percent. The additional 1.55 percentage points reflect the financial risk of a 0.40 debt-to-equity ratio.

11.4.4 Leverage, the cost of equity, and WACC

Table 11.3 illustrates how, under the Hamada framework, the cost of equity climbs as leverage rises and how WACC responds. We use βU = 0.9364, rf = 4.0%, MRP = 5.5%, T = 25%, and allow the pre-tax cost of debt to increase with leverage: 5.0 percent at D/E = 0.40 and 6.5 percent at D/E = 1.0.

Table 11.3 – Leverage, cost of equity, and WACC

D/E	D/V	Equity beta	Cost of equity	Pre-tax cost of debt	WACC
0.00	0.0%	0.936	9.15%	n/a	9.15%
0.40	28.6%	1.217	10.70%	5.0%	8.71%
1.00	50.0%	1.639	13.01%	6.5%	8.94%

Three lessons emerge. First, the cost of equity increases with leverage in every case; there is no “cheap” way to substitute debt for equity, because the equity that remains is riskier. Second, WACC falls initially because of the tax shield on debt, then rises as the rising cost of debt overwhelms the shield. This is the trade-off theory of capital structure in numerical form: there is a leverage region where WACC is minimized and firm value maximized. Third, the apparent attractiveness of debt (5.0 percent versus 10.7 percent) is an illusion if one forgets to charge equity for the risk that the debt creates. Analysts who compute WACC by weighting a fixed 10.6 percent equity cost with an increasing share of cheap debt will conclude that more leverage is always better, which is wrong.

(A technical caveat: because the cost of debt in Table 11.3 rises with leverage, the debt is not riskless, so the pure Hamada formula slightly overstates the equity beta at higher leverage. The pattern is nonetheless the right one for illustration.)

11.4.5 Practical issues in beta estimation

Estimation window and frequency. Common conventions use five years of monthly returns or two years of weekly returns. Shorter windows have more noise; longer windows may include a business that no longer exists.

Choice of market index. The index should reflect the investor base. Using a domestic index for a firm whose investors are global yields a beta that does not represent the risk those investors bear.

Thin trading. Infrequently traded stocks have betas biased downward, because stale prices lag market movements. Adjustments (such as including lagged market returns) are available.

Operating leverage. Two firms in the same industry can have different business risk because of different fixed-cost structures. High operating leverage (a large share of fixed costs) raises βU. When comparables differ greatly from the target in cost structure, adjust the unlevered beta judgmentally, or restrict the comparable set.

Cash. Large cash balances lower the equity beta, because cash has a beta near zero. For firms with very large cash holdings, some analysts unlever for net debt (debt minus cash) rather than gross debt, or compute the beta of the operating assets separately.

11.5 The Weighted Average Cost of Capital in Practice

11.5.1 Market-value weights and target structure

The weights in WACC are *market-value* proportions, because WACC is the opportunity cost of the capital currently committed, and the market values are what that capital is worth today. Book values record the historical cost of past financing. A firm whose stock trades at three times book value has a far larger equity proportion in market terms than its balance sheet suggests, and a WACC built from book weights will understate the share of the (more expensive) equity.

The weights should also be *target* weights, reflecting the capital structure the firm expects to maintain over the life of the investments, not today's snapshot if that is temporarily distorted. If a firm has just made a large acquisition funded by debt that it intends to repay, the right weights are those for its long-run policy.

Debt weights use the market value of debt. For most firms, the book value of debt is a reasonable approximation of market value if interest rates have not moved much; if the firm has substantial long-term fixed-rate debt and rates have changed, the difference can be significant. Debt includes interest-bearing borrowings, capitalized leases, and often unfunded pension deficits and other debt-like claims; excluding them overstates equity's share. Some analysts use *net debt* (debt less excess cash) for weights, in which case the equity beta must be consistent with that treatment.

11.5.2 The calculation

Example 11.4 – Computing WACC.

Brightwater Foods (a fictional firm) has the following data. Market value of equity is $600 million; market value of debt is $400 million. Equity beta is 1.20, risk-free rate is 4.0 percent, MRP is 5.5 percent. Its pre-tax cost of debt (current yield on its bonds) is 6.0 percent and the marginal tax rate is 25 percent.

Step 1: Cost of equity. $r_E = 4.0\% + 1.20 \times 5.5\% = 10.6\%$.

Step 2: After-tax cost of debt. $r_D(1 - T) = 6.0\% \times 0.75 = 4.5\%$.

Step 3: Weights. $V = 600 + 400 = 1{,}000$. $E/V = 60\%$; $D/V = 40\%$.

Step 4: WACC.

```
WACC = 0.60 × 10.6% + 0.40 × 4.5%
     = 6.36% + 1.80%
     = 8.16%
```

Now suppose the board's long-run policy is to hold debt at 30 percent of capital. Using the industry unlevered beta of 0.9364 from Example 11.3, the target D/E is 0.30/0.70 = 0.4286. Relevering:

```
βL = 0.9364 × [1 + 0.75 × 0.4286] = 0.9364 × 1.3214 = 1.2374
rE = 4.0% + 1.2374 × 5.5% = 10.81%
```

With debt cost assumed to fall slightly to 5.5 percent at the lower leverage:

```
WACC = 0.70 × 10.81% + 0.30 × 5.5% × 0.75
     = 7.567% + 1.2375%
     = 8.80%
```

The two answers (8.16 percent and 8.80 percent) differ for a reason that managers should understand. The first uses Brightwater's own observed beta; its implied unlevered beta is 1.20 / (1 + 0.75 × 0.667) = 0.80, lower than the industry's 0.94. The second uses an industry-based unlevered beta. If Brightwater's business is genuinely safer than the average comparable, 8.16 percent is better; if its own regression beta is merely a noisy estimate, 8.80 percent is more reliable. The analyst should look at the underlying reason before choosing, and the choice should be documented.

11.5.3 Nominal versus real, and currency consistency

Discount rates and cash flows must be in the same terms. Nominal cash flows (including expected inflation) are discounted at nominal rates; real cash flows at real rates. For multinational projects, cash flows in a foreign currency are discounted at a rate for that currency, or converted at forecast exchange rates and discounted at the home-currency rate. The expected inflation difference between two currencies is reflected in the difference between their risk-free rates, an interest-rate parity principle; ignoring it, for example discounting high-inflation-currency cash flows at a low-inflation-currency rate, overvalues the project.

11.5.4 Is the WACC stable?

The WACC changes when interest rates, market risk premiums, tax rates, or capital structure change. Large firms usually reassess at least annually and when conditions shift. But hurdle rates that bounce each quarter with the market create planning instability, so many firms set a hurdle rate for the year, adjusting only when the underlying drivers have moved materially. Some also apply a "spread over WACC" policy for different project categories, which is the topic of the next section.

11.6 Divisional and Project-Specific Hurdle Rates

11.6.1 The problem with one rate for everything

Imani's dilemma in the opening vignette is the classic one. A firm-wide WACC is an average of the required returns on its various businesses. Applied uniformly, it is too low for projects riskier than average and too high for those safer than average. Because managers learn to propose projects that clear the hurdle, the effect over time is *adverse selection within the firm*: risky projects are accepted that destroy value, and safe, value-creating projects are rejected. The firm drifts toward

higher risk without a corresponding increase in expected return. Division managers also learn to game: they overstate cash flows in the risky division and abandon investment in the safe division, and the capital-allocation system rewards the former.

11.6.2 Building divisional hurdle rates

The solution is to assign each division (or each category of project) its own discount rate, derived from the risk of that business. The method is the pure-play approach of Section 11.4:

1. Choose comparable firms for each division.
2. Unlever their betas and average them to get the division's βU.
3. Relever at the division's target leverage (which may differ from the corporate average, since debt capacity depends on asset tangibility and cash flow stability).
4. Compute rE by CAPM, rD from the division's debt capacity and credit characteristics, and WACC using the division's target weights.

Example 11.5 – Divisional hurdle rates versus a single corporate rate.

Harlow Ridge has two divisions. The Equipment division (70 percent of firm value) has an unlevered beta of 0.80 and supports 40 percent debt (D/V = 0.40, so D/E = 0.6667); its debt costs 5.5 percent. Harlow Cloud (30 percent of firm value) has an unlevered beta of 1.30 and supports only 20 percent debt (D/E = 0.25); its debt costs 6.5 percent. Use rf = 4.0%, MRP = 5.5%, T = 25%.

Equipment division:

```
βL  = 0.80 × [1 + 0.75 × 0.6667] = 0.80 × 1.50 = 1.20
rE  = 4.0% + 1.20 × 5.5% = 10.6%
WACC = 0.60 × 10.6% + 0.40 × 5.5% × 0.75
     = 6.36% + 1.65% = 8.01%
```

Cloud division:

```
βL  = 1.30 × [1 + 0.75 × 0.25] = 1.30 × 1.1875 = 1.54375
rE  = 4.0% + 1.54375 × 5.5% = 12.49%
WACC = 0.80 × 12.49% + 0.20 × 6.5% × 0.75
     = 9.99% + 0.975% = 10.97%
```

Firm-wide blended rate (value weights 70/30):

```
WACC = 0.70 × 8.01% + 0.30 × 10.97% = 5.607% + 3.290% = 8.90%
```

This is the 8.9 percent hurdle that Imani's CFO office used. Now the two projects. Each costs $100 million and produces level annual after-tax cash flows for ten years. The Equipment replacement has an IRR of 8.5 percent, which requires annual cash flows of 100 / 6.5613 = $15.24 million (the 10-year annuity factor at 8.5% is 6.5613). The Cloud analytics platform has an IRR of 10.0 percent, which requires $16.27 million per year (annuity factor at 10% is 6.1446).

Table 11.4 – Decisions under a single rate versus divisional rates ($ millions)

Project	IRR	Single hurdle 8.90%: NPV	Decision	Divisional hurdle: NPV	Correct decision
Equipment replacement	8.5%	−1.76	Reject	+2.22 (at 8.01%)	Accept
Cloud analytics	10.0%	+4.91	Accept	−4.02 (at 10.97%)	Reject

The single-rate system gets both decisions wrong. It rejects a project worth +$2.2 million and accepts one that destroys $4.0 million of value. Notably, neither error is subtle; the margins are 0.4 and 1.0 percentage points, well within the noise of ordinary forecasting, yet cumulatively over many projects such errors reshape the firm.

11.6.3 Beyond divisions: project-specific rates

Divisions are themselves aggregates. A project's discount rate should reflect its own risk, and when a project's risk differs materially from the division's (a new product in a new market, an expansion into a regulated sector, a cost-saving investment with near-certain cash flows), a project-specific rate is warranted. Practical approaches include:

- **Pure-play comparables for the project.** If the project is the entry into a new line of business, find firms in that line and use their unlevered betas.
- **Risk categories (risk buckets).** Many firms define three to five categories, for example cost-savings and replacement (WACC minus 1 percentage point), expansion of existing lines (WACC), new products (WACC plus 2), and new-market or strategic ventures (WACC plus 4). The spreads should be calibrated to betas for the categories rather than chosen arbitrarily; otherwise, they become negotiated numbers.
- **Cash flow adjustment instead of rate adjustment.** For specific risks that are diversifiable (a plant fire, a unique regulatory event), the correct treatment is to adjust expected cash flows, not to inflate the discount rate. Adding a premium to the discount rate for unsystematic risk double-counts and penalizes long-lived projects more than short-lived ones, because the premium compounds.
- **Operating leverage and cyclicality.** A project with high fixed costs has higher beta than one with the same revenues and variable costs. Higher cyclicality of revenues raises beta.

A frequent mistake is to add arbitrary premiums for "uncertainty" or "strategic importance." Uncertainty that is diversifiable by shareholders does not raise the required return. Risk that moves with the market does. The test for any adjustment is: *Does this feature make the project's cash flows covary more with the overall economy?* If not, it belongs in the forecast cash flows (as scenarios or probabilities), not in the discount rate.

11.6.4 Adjusted present value as an alternative

When a project's financing is special, WACC is awkward because the capital structure is project-specific and changes over time. The *adjusted present value* method values the project in two steps: first, at the unlevered cost of equity as if all equity financed, then add the present value of financing side effects such as interest tax shields and issuance costs.

Example 11.6 – APV for a project with its own debt.

A project costs $1,000 and produces after-tax unlevered free cash flows of $95 per year in perpetuity. The unlevered cost of capital for this business is 9.15 percent ($\beta U = 0.9364$). The firm will finance it with $300 of permanent debt at 5.0 percent, the tax rate is 25 percent, and issuing the debt costs 2 percent of face value.

```
Base-case NPV = 95 / 0.0915 − 1,000 = 1,038.25 − 1,000 = +38.25
PV of tax shield (permanent debt) = T × D = 0.25 × 300 = +75.00
Debt issuance cost = 0.02 × 300 = −6.00
APV = 38.25 + 75.00 − 6.00 = +107.25
```

The project is worth undertaking; its value comes principally from the tax shield, with operations adding only 38.25. A manager reading this should note two things. First, the operating NPV is modest, so a small error in the 9.15 percent rate, for example 10.0 percent, would reduce 95/0.10 − 1,000 to −50, and the project would depend entirely on the tax shield. Second, the tax shield here relies on permanent debt of $300 and the firm's ability to use the deductions in perpetuity. If the debt will be repaid in five years, the tax shield is worth far less. APV makes these assumptions visible, which is its main advantage over embedding them in a single WACC percentage.

11.7 Flotation Costs and the Marginal Cost of Capital

11.7.1 What flotation costs are

Raising external capital is not free. Underwriting fees, legal and registration costs, and, for seasoned equity offerings, price discounts to attract buyers consume part of the proceeds. These *flotation costs* vary with the type of security and the size of the issue: debt issues are cheapest, in the order of a percent or two of proceeds for large issuers; seasoned equity offerings are more expensive, often several percentage points; and small issues, initial public offerings, and issuers in less developed markets pay more. (The percentages in this chapter are illustrative.)

Internal funds, retained earnings, carry no flotation cost, but they are not free either: retained earnings belong to shareholders, who expect the same return on them as on new equity.

11.7.2 Two ways to treat them, and one wrong way

Method 1 – Adjust the cash flows (preferred). Treat flotation costs as an additional initial outlay: if a project costs $I and will be financed with a mix of securities with weighted-average flotation cost f, then the amount to be raised is I / (1 − f) and the flotation cost is I / (1 − f) − I. Subtract that from project NPV. This is theoretically cleaner because the cost is a one-time cash outflow at the start, not a permanent premium on the discount rate, and it only applies to the funds actually raised externally.

Method 2 – Adjust the discount rate. Inflate the cost of each source: rE* = D1 / (P0 × (1 − fE)) + g, for example. This spreads the cost across the project life as a rate premium and is usually applied incorrectly; it overcharges long-lived projects and undercharges short ones, and it applies the charge to the entire financing when only part may be external.

Wrong way – Ignoring flotation costs on the grounds that the firm uses retained earnings. The cash flows of the project are valuable to shareholders only after the cost of capital is covered, and if the firm would otherwise have paid retained earnings out as dividends, or if marginal projects require outside funds, the flotation cost is real for the marginal project.

Example 11.7 – Adjusting project NPV for flotation costs.

Harlow Ridge's capital structure target is 70 percent equity and 30 percent debt. New external equity costs 5.0 percent of the gross proceeds, and new debt costs 1.5 percent. A project requires an outlay of $100 million, and the present value of its cash flows (discounted at the appropriate hurdle rate) is $103.5 million, so the unadjusted NPV is +$3.5 million.

Weighted-average flotation cost:

```
f = 0.70 × 5.0% + 0.30 × 1.5% = 3.50% + 0.45% = 3.95%
```

Gross amount that must be raised so that net proceeds equal $100 million:

```
Gross = 100 / (1 − 0.0395) = 100 / 0.9605 = $104.11 million
Flotation cost = 104.11 − 100 = $4.11 million
```

Adjusted NPV:

```
Adjusted NPV = 103.5 − 104.11 = −$0.61 million
```

A project that looked attractive before flotation costs is slightly value-destroying after them. The result is sensitive: with f = 2.0 percent, the gross amount is $102.04 million and adjusted NPV is +$1.46 million. Managers should recognize that flotation costs raise the effective hurdle for externally funded projects, and that this is one reason capital structure and financing flexibility matter for investment. Firms with internal cash generation exceeding their investment needs avoid these costs, which partly explains pecking-order behavior.

If the project is financed internally, the flotation cost is not incurred, but the opportunity cost of the retained funds (the shareholders' required return) still applies, and the discount rate remains the one reflecting project risk.

11.7.3 The marginal cost of capital schedule

The cost of capital is not constant as the firm raises more funds. When retained earnings are exhausted, the firm must issue new equity, which carries flotation costs and may be priced at a discount. Debt capacity is finite, and beyond some level, lenders demand higher spreads. The *marginal cost of capital* (MCC) is the cost of the last dollar raised. Plotted against the total amount of capital raised, it is stepwise upward-sloping, with breaks where the funding source changes (the "break point" at which retained earnings run out equals retained earnings available divided by the equity weight). Combined with the firm's investment opportunity schedule (projects ranked by IRR), the MCC indicates the optimal capital budget: accept projects until the IRR of the marginal project equals the MCC. In practice, a firm that has sufficient internal funds and moderate growth works with a single WACC, and the MCC schedule matters mostly for rapidly growing firms and for firms near financing constraints.

11.8 Pitfalls, Judgment, and Governance of the Hurdle Rate

11.8.1 Common errors

Most damage in practice comes not from sophisticated modeling choices but from basic errors repeated for years. Table 11.5 collects the most common, with the consequence of each.

Table 11.5 – Common cost-of-capital errors

Error	Why it is wrong	Typical consequence
Using one company-wide rate for all projects	Ignores differences in business risk	Over-investment in risky units, starvation of safe ones
Book-value weights	Weights should reflect the market value of capital supplied	Understates equity share and WACC for firms with market value above book
Coupon rate as cost of debt	Past conditions; not the current required return	Understates (or overstates) cost of debt
Using the cheapest source as the project's cost	Project risk, not funding source, determines the rate	Hurdle too low; value destruction hidden
Not adjusting equity cost when leverage changes	rE rises with D/E	False conclusion that debt is "cheap"
Mixing nominal cash flows with real rates (or vice versa)	Inflation inconsistently treated	Systematic over- or under-valuation
Pre-tax cash flows discounted at after-tax rate (or the reverse)	Tax treatment must match	Wrong NPV sign in marginal projects
Adding premiums for diversifiable risk	Only systematic risk is priced	Rejects good long-term projects
Mismatched horizon: short-term rates for long projects	Term structure matters	Understates the hurdle when yield curve slopes up
Treating WACC as constant after major strategy change	Business mix and leverage shift	Stale hurdle rates

11.8.2 A worked illustration of weight error

Example 11.8 – Book versus market weights.

A firm has book equity of $300 million and debt of $500 million (book and market value the same). The market value of its equity is $900 million. Its cost of equity is 10.6 percent, pre-tax cost of debt is 6.0 percent, and the tax rate is 25 percent.

Using book weights: E/V = 300 / 800 = 37.5%; D/V = 62.5%.

```
WACC(book) = 0.375 × 10.6% + 0.625 × 6.0% × 0.75
           = 3.975% + 2.8125%
           = 6.79%
```

Using market weights: E/V = 900 / 1,400 = 64.29%; D/V = 35.71%.

```
WACC(market) = 0.6429 × 10.6% + 0.3571 × 6.0% × 0.75
             = 6.814% + 1.607%
             = 8.42%
```

The book-weight WACC is 1.63 percentage points too low. For a level perpetuity, the difference between a 6.79 percent and an 8.42 percent discount rate is a factor of about 1.24 in capitalized value (1/0.0679 = 14.7 versus 1/0.0842 = 11.9), enough to make clearly marginal projects look attractive. In addition, a firm whose book debt ratio is high only because of accumulated losses or buybacks would be wrongly told that it has a low-cost capital structure.

11.8.3 Judgment, precision, and sensitivity

The cost of capital is estimated with error. Betas have standard errors of a few tenths, the market risk premium is uncertain by at least a point, and the target capital structure is a policy choice. A reported WACC of "8.16 percent" is a false precision; the honest statement is "roughly 8 to 9 percent." Good practice therefore includes sensitivity analysis: show the

project NPV at the base rate and at rates one or two percentage points above and below, and flag decisions that flip within that band. Projects that are acceptable across the plausible range are robust; projects that depend on the third decimal place of the discount rate are marginal and should be decided on strategic and qualitative grounds, with eyes open.

Terminal values in long-lived projects deserve particular scrutiny. In a perpetuity-growth terminal value, a 1 percentage-point error in the discount rate can change terminal value by 15 to 25 percent when r − g is small. The discount rate and the terminal growth assumption must be consistent with one another and with the competitive structure of the industry: a growth rate above the economy's long-run growth rate is a claim that the firm will eventually become larger than the economy.

11.8.4 Governance: who owns the hurdle rate?

Because hurdle rates determine capital allocation, they create incentives for gaming. Division managers lobby for low rates for their units, and corporate planners may be tempted to set conservative (high) rates to reject marginal proposals without explicit reasoning. Good governance separates responsibility: a central treasury or corporate finance group estimates the market-based components (risk-free rate, MRP, comparables' betas, cost of debt) using a documented methodology, while divisions are accountable for cash-flow forecasts. Hurdle-rate methodology should be reviewed periodically, assumptions should be recorded, and post-investment audits should compare realized returns to projections so that systematic optimism is detected. A hurdle rate that no one can reproduce from public data is a negotiated number, not a cost of capital.

11.8.5 Beyond WACC

For some situations, WACC is the wrong tool: leveraged buyouts where capital structure changes rapidly, project finance with non-recourse debt, financial institutions where debt is an operating input rather than a financing choice, and firms with significant real options (patents, expansion rights, abandonment options) whose value is captured poorly by a single discount rate. In these cases, APV, equity-cash-flow methods (discount free cash flow to equity at the cost of equity), or option-pricing methods are more appropriate. WACC remains the workhorse for ordinary corporate investment, but a manager who understands its assumptions can tell when to put it aside.

Case Study: Calder Maritime and the Port Terminal Decision

This case is fictional and illustrative; all figures are invented for teaching purposes.

Calder Maritime Group is a publicly traded company with two businesses. The first, Calder Shipping, operates a fleet of bulk carriers under medium-term charters and spot contracts. Its revenues are strongly tied to global trade volumes and freight rates, and its earnings swing widely from year to year. The second, Calder Terminals, operates three container-handling terminals under long-term concession contracts with minimum-volume guarantees from anchor customers. Terminal cash flows are stable, inflation-linked, and resemble those of infrastructure assets. Shipping accounts for about 55 percent of Calder's enterprise value and Terminals for 45 percent.

Calder's chief financial officer, Tobias Verhagen, has used a single company WACC of 9.6 percent for all investment decisions since the firm's listing. He built it from Calder's regression beta of 1.15, a risk-free rate of 4.0 percent, a market risk premium of 5.5 percent, a pre-tax cost of debt of 6.5 percent, a 25 percent tax rate, and market value weights of 62 percent equity and 38 percent debt. His computation:

```
rE   = 4.0% + 1.15 × 5.5% = 10.325%
WACC = 0.62 × 10.325% + 0.38 × 6.5% × 0.75 = 6.40% + 1.85% = 8.25%
```

When the finance team re-examined the file, they discovered that 9.6 percent was not the product of this arithmetic. It had been inherited from an earlier consultant’s report, and no one had documented how it was derived. The recomputed 8.25 percent, using the company’s own inputs, is more than a full point lower.

Calder now faces a large decision. A port authority in a neighboring country has offered a 25-year concession to build and operate a new container terminal. The initial outlay is $600 million, and the project’s expected after-tax unlevered free cash flows, in nominal terms, are projected to be $62 million in the first year, growing at 4 percent per year for the 25-year concession, with no residual value. Independent forecasters regard the volume guarantees from anchor customers as unusually strong.

The team has gathered comparable-company data. Listed pure-play terminal operators have an average equity beta of 0.85 and an average market-value debt-to-equity ratio of 0.90. Listed bulk-shipping companies have an average equity beta of 1.45 and an average D/E of 0.60. Both groups face roughly a 25 percent tax rate. Terminal operators can sustain debt at 45 percent of capital at a cost of 5.5 percent because of contracted revenues; shipping companies can sustain 30 percent at 7.0 percent.

A junior analyst, Priya Nambiar, argues for using the pure-play terminal beta, unlevered and relevered at a 45 percent debt target, for the new terminal. A senior treasury manager, Joaquin Estrada, objects: the new terminal is in a developing market with political and currency risk that the established terminals do not face, and Calder’s board will expect a premium. He proposes adding 3 percentage points to the discount rate “to be safe.” Priya answers that the currency risk is largely diversifiable and that the cash flows are in a currency different from Calder’s reporting currency, so the real issue is making the cash flows and discount rates consistent. The concession payments are indexed to hard-currency tariffs for most of the revenue, but about a quarter of operating costs are local-currency.

The financing team adds a further consideration. The project requires $270 million of new equity, and a seasoned offering of that size would cost Calder an estimated 4.5 percent in underwriting and related fees. Calder expects to borrow $330 million for the project in the bond market at an issuance cost of 1.2 percent. Management also notes that the new terminal, which would double Terminals’ EBITDA over five years, will change the firm’s mix: Terminals would rise to roughly 60 percent of enterprise value.

Tobias must present a recommended discount rate and a go/no-go decision to the board next month. He is aware that using 9.6 percent, as always, might give a different answer than a careful analysis, and that board members trust the number because they have always seen it.

Discussion Questions

1. Using the comparable data, compute unlevered betas for the terminal and shipping groups, then derive divisional WACCs for Terminals (45 percent debt target) and Shipping (30 percent debt target). How do these compare with the 8.25 percent firm-wide figure and the undocumented 9.6 percent? What does this imply for how Calder has been evaluating projects in each division?
2. Evaluate the dispute between Priya and Joaquin. Which components of the political and currency risk should be treated in the cash flows, and which, if any, in the discount rate? What tests would you apply to decide whether a premium is justified?
3. Compute the weighted flotation cost on the project’s financing. By how much does it raise the required initial outlay, and how would you incorporate it in NPV? Does the answer change if Calder could fund the equity from retained cash?

4. Discuss the governance failure revealed by the undocumented 9.6 percent. Propose a process and a set of controls so that Calder's board can trust the hurdle rates it uses, and discuss how the shift to 60 percent Terminals by value should affect the firm-wide cost of capital over time.

Key Takeaways

- The cost of capital is an opportunity cost: the return investors could earn elsewhere on investments of the same risk. It is set by the risk of the project's cash flows, not by the identity of the firm or the source of the funds.
- Estimate the cost of equity with at least two methods, normally CAPM plus a cross-check such as the dividend growth model, and investigate any large disagreements rather than averaging them away.
- The cost of debt is today's market yield on comparable debt, not the historical coupon, adjusted for default risk if the firm is speculative grade and for the marginal tax rate on the deduction.
- Equity beta embeds financial risk. Unlever comparable betas to isolate business risk, average them to reduce noise, and relever at the target capital structure; the cost of equity rises with leverage and debt is never "cheap" in isolation.
- WACC uses market-value weights and a target capital structure, with after-tax cost of debt. It is valid only when the project's risk and financing resemble those embedded in the rate.
- A single company-wide hurdle rate systematically favors high-risk divisions and starves low-risk ones; assign divisional and, where warranted, project-specific rates built from pure-play evidence.
- Add to the discount rate only systematic risk. Diversifiable and project-specific uncertainties belong in the cash-flow forecasts through scenarios and probabilities.
- Treat flotation costs as an addition to the initial outlay, scaled to the funds actually raised externally, and not as a permanent increment to the discount rate.
- When financing is project-specific or changes over time, adjusted present value makes the value of tax shields and issuance costs explicit and is often more reliable than a constant WACC.
- Most hurdle-rate errors are elementary: book weights, coupon rates, inconsistent inflation or currency treatment, and undocumented numbers. Document the method, test sensitivity, and make estimation independent of the people whose projects are being judged.

Review Questions

1. Explain why the cost of capital depends on the risk of the project rather than the source of financing. What would go wrong if a firm used the interest rate on a cheap bank loan as the discount rate for a risky venture?
2. Distinguish between expected return, required return, and cost of capital. In what sense are they equal in equilibrium?
3. Describe the main weaknesses of the dividend growth model as a method for estimating the cost of equity. For what kinds of firms is it most defensible?
4. Why is the yield to maturity on a firm's outstanding bonds a better measure of the cost of debt than the coupon rate? When does even the yield to maturity overstate the true expected cost?
5. A firm replaces some equity with debt and observes that its WACC, computed using the old cost of equity, falls sharply. What error has it made, and what does the corrected analysis typically show?
6. Describe the steps of the unlever–relever procedure. Why is it usually preferable to use the average of several comparables instead of the firm's own regression beta?[216]

7. What is the effect of operating leverage on a project's beta? Give an example of two projects with identical revenues but different betas.
8. Explain, with an example, why applying a single corporate WACC to all projects results in adverse selection within the firm.
9. When would you prefer the APV method to the WACC method? What additional information does APV require?
10. Why are flotation costs better treated as part of the project's initial cash outflow than as an increment to the cost of equity? Under what circumstances would a firm incur no flotation costs?

Problems

Problem 11.1. A firm's equity has a beta of 0.90. The risk-free rate is 3.5 percent and the market risk premium is 6.0 percent. Compute the cost of equity using CAPM.

Problem 11.2. A mature utility just paid a dividend of $1.80 per share, which is expected to grow at 4 percent per year indefinitely. The stock price is $30. Compute the cost of equity using the dividend growth model. (Remember that D1 = D0 × (1 + g).)

Problem 11.3. A comparable firm has an equity beta of 1.60, a market debt-to-equity ratio of 0.80, and a tax rate of 30 percent. (a) Compute its unlevered beta. (b) Compute the equity beta for a firm with the same business risk, a D/E ratio of 0.25, and a 30 percent tax rate.

Problem 11.4. A firm has market equity of $450 million and debt of $150 million. The cost of equity is 12 percent, the pre-tax cost of debt is 7 percent, and the tax rate is 28 percent. Compute WACC.

Problem 11.5. A firm plans to fund a $50 million project (net of flotation) with 60 percent new equity and 40 percent new debt. Flotation costs are 6 percent on equity and 2 percent on debt. Compute (a) the weighted-average flotation cost, (b) the gross amount that must be raised, and (c) the dollar flotation cost.

Problem 11.6. A division has an unlevered beta of 0.70 and a target debt-to-capital ratio of 25 percent. Its pre-tax cost of debt is 5.0 percent, the risk-free rate is 3.5 percent, the market risk premium is 5.5 percent, and the tax rate is 25 percent. Compute the division's beta, cost of equity, and WACC.

Problem 11.7. A firm's capital consists of common equity with market value $500 million and required return 11 percent, preferred stock with market value $100 million and a 7.5 percent yield, and debt of $400 million with a pre-tax cost of 6 percent. The tax rate is 25 percent. Compute WACC.

Problem 11.8. A project costs $200 million and produces after-tax cash flows of $58 million per year for five years. The corporate WACC is 9 percent, but the project is in a risk category warranting a 15 percent discount rate. Compute the NPV at each rate and state the correct decision. (Annuity factors: 3.8897 at 9 percent, 3.3522 at 15 percent.)

Answers to Selected Problems

- **11.1.** rE = 3.5% + 0.90 × 6.0% = 3.5% + 5.4% = 8.9%.
- **11.2.** D1 = 1.80 × 1.04 = 1.872; rE = 1.872 / 30 + 0.04 = 6.24% + 4.0% = 10.24%.
- **11.3.** (a) βU = 1.60 / (1 + 0.70 × 0.80) = 1.60 / 1.56 = 1.0256. (b) βL = 1.0256 × (1 + 0.70 × 0.25) = 1.0256 × 1.175 = 1.2051.
- **11.4.** E/V = 75%, D/V = 25%. WACC = 0.75 × 12% + 0.25 × 7% × 0.72 = 9.00% + 1.26% = 10.26%.

- **11.5.** (a) f = 0.6 × 6% + 0.4 × 2% = 3.6% + 0.8% = 4.4%. (b) Gross = 50 / 0.956 = $52.30 million. (c) Flotation cost = $2.30 million.
- **11.6.** D/E = 0.25/0.75 = 0.3333; β = 0.70 × (1 + 0.75 × 0.3333) = 0.875; rE = 3.5% + 0.875 × 5.5% = 8.3125%; WACC = 0.75 × 8.3125% + 0.25 × 5.0% × 0.75 = 6.234% + 0.9375% = 7.17%.
- **11.7.** V = 1,000. WACC = (500 × 11% + 100 × 7.5% + 400 × 6% × 0.75) / 1,000 = (55 + 7.5 + 18) / 1,000 = 8.05%.
- **11.8.** NPV at 9% = 58 × 3.8897 − 200 = 225.60 − 200 = +$25.60 million. NPV at 15% = 58 × 3.3522 − 200 = 194.43 − 200 = −$5.57 million. At the correct 15 percent hurdle the project should be rejected; the corporate WACC would have wrongly accepted it.

Chapter 12: Market Efficiency and Behavioral Finance

Learning Objectives

After studying this chapter, you should be able to:

- State the three forms of the efficient market hypothesis, explain the economic mechanism that is supposed to produce efficiency, and explain why testing it always involves a joint hypothesis about a model of expected returns.
- Design and interpret an event study, including the market model, abnormal returns, cumulative abnormal returns, and test statistics for single events and for samples of events.
- Describe the main empirical anomalies (momentum, value, size, post-earnings-announcement drift, and others), and evaluate whether each reflects mispricing, compensation for risk, or the product of data mining.
- Explain the limits to arbitrage, including noise-trader risk, short-selling constraints, and transaction costs, and why they allow mispricing to persist without creating easy profits.
- Identify the principal behavioral biases (overconfidence, anchoring, loss aversion, mental accounting, herding, representativeness) and trace how they affect investors and corporate managers.
- Draw practical implications for financial managers: when to trust market prices, how to time financing and repurchases honestly, how to communicate, and how to guard capital budgeting against its own biases.

Opening Vignette

The following scenario is illustrative; the company and its figures are fictional.

Marlowe Instruments is a maker of laboratory measurement devices. After a strong run, its share price has doubled in eighteen months to $64, even though its earnings have grown by only a third. The chief executive, Dana Whitfield, is having a difficult week. The board has two proposals before it.

The first, from the head of corporate development, is to use the high share price as currency to buy a rival, Stonebridge Optics, in an all-stock deal. “Our stock is richly valued,” he argues. “We are paying with expensive paper for real assets. The market is giving us a gift.” The second, from the treasurer, is to launch a $300 million buyback, because analysts have recently downgraded the sector over concerns about a cyclical downturn, and he believes Marlowe’s shares are, if anything, undervalued relative to the firm’s contracted backlog.

The two proposals pull in opposite directions: one assumes Marlowe is overvalued, the other undervalued. Dana wants to know which view is right. She asks her chief financial officer, Rafael Quintero, a series of unsettling questions. Is the market price an unbiased estimate of Marlowe’s value, in which case neither a stock-for-stock deal nor a buyback creates value for existing shareholders? Or is the price systematically distorted by investor optimism or pessimism, in which case management’s choices about when and how to raise or return capital can transfer wealth between shareholders who stay and those who leave? And if the market can be wrong, how would management ever know it, given that it has the same fallible judgment as everyone else and a strong personal interest in believing that the stock is worth more than it trades for?

Rafael recalls two competing traditions from his MBA program. One, the efficient markets view, holds that prices incorporate available information so well that deviations are small and fleeting. The other, behavioral finance, holds that psychological biases and limits on arbitrage can cause prices to drift from fundamentals for long periods. This chapter lays out both, the evidence on each side, and what the evidence implies for those who manage firms rather than trade their shares.

12.1 The Efficient Market Hypothesis

12.1.1 The idea and the mechanism

In a competitive market, many analysts and traders compete to profit from information. If a stock is underpriced relative to what available information implies, the first investors to notice buy it, pushing the price up until the opportunity disappears. If it is overpriced, they sell or short it, pushing the price down. The result, as articulated by Eugene Fama and others in the 1960s, is that prices adjust quickly to new information and, at any moment, reflect what is known. The *efficient market hypothesis* (EMH) says that the price of a security is an unbiased estimate of its intrinsic value given the information available, so that no strategy based on that information earns returns in excess of those justified by risk, after costs.

The mechanism matters. Efficiency does not require that all investors are rational, or that all know the right answer. It requires that enough capital is controlled by informed, profit-seeking traders who can act on mispricing, that their actions move prices, and that the irrational trades of others are random and cancel out or are absorbed. Market efficiency is therefore a statement about the *outcome* of competition, and its plausibility depends on how effective arbitrage is, a point to which Section 12.5 returns.

Two implications follow. Prices follow something like a *random walk*: if prices already reflect all current information, then price changes reflect only new information, which by definition is unpredictable. And, because investors cannot predictably beat the market after costs, a rational investor with no special information should hold a diversified portfolio at low cost, the logic behind index funds.

Notice what efficiency does *not* say. It does not say that prices equal true value with certainty, only that errors are unbiased and not predictable from available information. It does not say that every investor earns the market return; skilled or lucky investors may do better. And it does not say that prices are stable: prices should move substantially as news arrives and as the discount rates that investors apply to risky cash flows change.

12.1.2 The three forms

Fama classified efficiency by the information set that prices are assumed to reflect. Table 12.1 summarizes the forms, the strategies each rules out, and the kinds of evidence used to test them.

Table 12.1 – Forms of market efficiency

Form	Information reflected in prices	Strategies it rules out	Typical tests
Weak	Past prices and trading volume	Technical analysis, chart patterns, simple trading rules based on past returns	Serial correlation of returns; runs tests; trading-rule simulations
Semi-strong	All public information (financial statements, announcements, news, macro data)	Fundamental analysis based on public data; trading on announced earnings, splits, or dividends	Event studies; return predictability using public variables; performance of mutual funds
Strong	All information, public and private (including insider information)	Any profit from private information	Studies of insider trading, specialist and analyst profits

The forms are nested: if the strong form held, the semi-strong and weak forms would hold also. Most economists believe the strong form fails in practice, because corporate insiders trading on private knowledge of earnings or takeovers can earn abnormal profits (which is why insider trading is generally illegal). The weak form is the most widely supported, at least for large, liquid markets. The central debate concerns the semi-strong form.

12.1.3 The joint hypothesis problem

Every test of market efficiency is a test of two hypotheses at once: that prices reflect information, and that the model used to define "normal" or required returns is right. If a strategy appears to earn abnormal profit, it might be that the market is inefficient, or that the model of expected returns (for example, CAPM) omits a priced risk factor. This *joint hypothesis problem*, emphasized by Fama, means that efficiency can never be proven or refuted by a single test. A reported anomaly is simply a pattern of returns not explained by the chosen benchmark. Whether the pattern is mispricing or compensation for a risk the benchmark ignores is a matter of interpretation, and it is the dividing line between the efficient-markets and behavioral camps.

12.1.4 What efficiency means for corporate finance

If markets are semi-strong efficient, several principles of corporate finance simplify:

- The market price is the best available estimate of value. A manager need not guess whether shares are over- or undervalued when deciding whether to issue or repurchase them, because on average the transaction occurs at fair value.
- Financing decisions are not sources of value in themselves. Issuing equity at a fair price transfers no wealth between existing and new shareholders; the NPV of the financing transaction is zero apart from fees and taxes.
- Accounting choices that do not affect cash flows (for example, the choice between inventory methods when taxes are unaffected) should not move prices, because investors see through them.
- Investors cannot be fooled by cosmetic changes such as stock splits per se.

If markets are not efficient, each of these has qualifications, and managers face the more difficult task of taking a view on mispricing while knowing that their own biases may distort it. The remainder of the chapter explores that tension.

12.2 Testing Efficiency: Weak Form, Event Studies, and Strong Form

12.2.1 Weak-form tests

If past returns contain exploitable information, returns should be correlated across time. The first-order autocorrelation coefficient measures the relationship between today's return and tomorrow's:

```
ρ1 = Σ (rt − r̄)(rt+1 − r̄) / Σ (rt − r̄)²
```

Under the random walk hypothesis, ρ1 should be approximately zero. With n observations, the standard error of an estimated autocorrelation is about 1/√n, so values beyond ±2/√n are statistically significant at roughly the 5 percent level.

Example 12.1 – Testing for serial correlation in returns.

An analyst collects twelve daily returns (in percent) for a small-cap stock: 1.2, −0.4, 0.9, −1.1, 0.6, 0.3, −0.8, 1.0, −0.2, 0.5, −0.7, 0.4.

Step 1: Mean. The sum is 1.7, so r̄ = 1.7 / 12 = 0.1417.

Step 2: Deviations from the mean. 1.0583, −0.5417, 0.7583, −1.2417, 0.4583, 0.1583, −0.9417, 0.8583, −0.3417, 0.3583, −0.8417, 0.2583.

Step 3: Denominator, the sum of squared deviations. 6.4092.

Step 4: Numerator, the sum of products of consecutive deviations. (1.0583)(−0.5417) + (−0.5417)(0.7583) + (0.7583)(−1.2417) + … + (−0.8417)(0.2583) = −4.3142.

Step 5: Autocorrelation.

```
ρ1 = −4.3142 / 6.4092 = −0.673
```

Step 6: Significance. The 5 percent band is ±2/√12 = ±0.577. The estimate of −0.673 lies outside it, so the null hypothesis of no autocorrelation is rejected for this sample: a positive return tends to be followed by a negative one.

A careful analyst would not conclude that the market is weak-form inefficient. Twelve observations are far too few; one sample of one stock proves little; and strong negative first-order autocorrelation in small, thinly traded stocks is typically caused by *bid-ask bounce*, in which recorded prices alternate between bid and ask quotes without any change in value. A trader who tried to exploit it would pay the spread on each trade and lose the apparent gain. This illustrates a recurring theme: statistical predictability is not the same as economically exploitable profit. The EMH is a claim about profits after costs and risk.

For large stocks and for indexes, the long-standing finding is that daily and weekly return autocorrelations are small. There is evidence of short-horizon reversal in some segments, of intermediate-horizon continuation (momentum, Section 12.3), and of long-horizon reversal, which are the focus of the anomalies literature.

12.2.2 Event studies

The workhorse method for testing semi-strong efficiency, and an important practical tool for managers evaluating the value consequences of corporate actions, is the *event study*. It measures how the price of a security responds to an event (an earnings announcement, an acquisition, a dividend change, a regulatory decision) by comparing actual returns with the returns expected in the absence of the event.

The procedure has six steps:

1. **Define the event and the event date** (the day the information became public; day 0). Where announcements occur after the market closes, day 0 is the next trading day.
2. **Select the event window**, typically a few days around day 0 (for example, from day −1 to day +1 or −2 to +2) to capture the reaction and any leakage or delayed adjustment.
3. **Estimate normal returns** using data from an *estimation window* well before the event (for example, 120 days ending 20 days before), usually via the *market model*: $R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}$.
4. **Compute abnormal returns** in the event window: $AR_{i,t} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{m,t})$.
5. **Cumulate** abnormal returns over the window: $CAR_i = \Sigma AR_{i,t}$.
6. **Test significance**, using the standard deviation of the residuals from the estimation period (for a single event) or the cross-sectional dispersion of CARs (for a sample of events).

Example 12.2 – An event study for a single firm.

Marlowe Instruments announces a major product recall on day 0. From the estimation window, the market model has $\hat{\alpha} = 0.02$ percent per day (0.0002), $\hat{\beta} = 1.1$, and residual standard deviation $\sigma = 1.2$ percent per day. Returns around the event are:

Table 12.2 – Abnormal returns around the announcement

Day	Market return Rm	Stock return Ri	Expected return 0.0002 + 1.1 × Rm	Abnormal return AR
−2	0.40%	0.60%	0.0002 + 0.0044 = 0.46%	+0.14%
−1	−0.20%	0.20%	0.0002 − 0.0022 = −0.20%	+0.40%
0	1.00%	4.50%	0.0002 + 0.0110 = 1.12%	+3.38%
+1	0.30%	1.20%	0.0002 + 0.0033 = 0.35%	+0.85%
+2	−0.50%	0.10%	0.0002 − 0.0055 = −0.53%	+0.63%

(The dates and the narrative are illustrative; the positive reaction, for the sake of the example, reflects a market that had feared something worse.)

Day-0 abnormal return: 4.50% − 1.12% = 3.38%. Its t-statistic is 3.38 / 1.2 = 2.82, significant at the 1 percent level.

Cumulative abnormal return over the window (−2, +2):

```
CAR = 0.14 + 0.40 + 3.38 + 0.85 + 0.63 = 5.40%
```

Under independence across days, the standard deviation of a 5-day CAR is $\sigma\sqrt{5} = 1.2\% \times 2.236 = 2.683\%$, so:

```
t(CAR) = 5.40 / 2.683 = 2.01
```

This is just significant at the 5 percent level. Three observations arise. The largest response occurs on day 0, consistent with fast price adjustment. But the abnormal return before day 0 (+0.54 percent on days −2 and −1) is positive, which could indicate leakage or simply noise (neither pre-event abnormal return is individually significant, with t-statistics of 0.12 and

0.33). And the abnormal return after day 0 (+1.48 percent on days +1 and +2) is positive, which, if systematic, would suggest slow incorporation of news; but with a single event, it cannot be separated from noise.

12.2.3 Samples of events

A single event is noisy. Studies of corporate actions use many events and average the CARs. The average CAR across N events, and its t-statistic based on the cross-sectional standard deviation s of the CARs, is:

```
CAR-bar = (1/N) Σ CARi
t = CAR-bar / (s / √N)
```

Example 12.3 – A cross-sectional event study.

A study of 25 announcements of share repurchase programs finds a mean 3-day CAR of 1.8 percent with a cross-sectional standard deviation of 4.0 percent.

```
Standard error = 4.0% / √25 = 4.0% / 5 = 0.80%
t = 1.8% / 0.80% = 2.25
```

The t-statistic of 2.25 exceeds the 5 percent critical value of about 2.06 for 24 degrees of freedom, so the average announcement effect is statistically different from zero. The interpretation depends on the market's view: it is consistent with efficiency if the market quickly processes the signal that managers consider shares undervalued, or that the firm has surplus cash; the abnormal return is the *price of the news*, not an exploitable profit, since a trader cannot know the announcement in advance. Event studies thereby provide evidence that prices react quickly, and an inefficiency would show instead as drift in CARs after the event, which can be tested over long windows.

Long-horizon event studies (one to five years) are far less reliable. Because the benchmark model matters more as the window lengthens, the joint hypothesis problem bites hard; the choice between buy-and-hold returns and cumulative returns, and the way cross-correlated events cluster in time, can overturn conclusions. Short-window evidence supports efficiency in processing news; long-window evidence on anomalies is contested.

For managers, event studies have a practical use: to learn how the market values their actions. Studies of announcements, such as mergers, show that target shareholders typically gain substantially, while acquirers on average earn returns close to zero or slightly negative, a pattern suggestive of competitive bidding and of overpayment driven by managerial overconfidence, as discussed in Section 12.6.

12.2.4 Strong-form evidence and the performance of professionals

The strong form is the easiest to reject, since people with genuine private information, such as corporate insiders, can earn abnormal returns. More relevant for most investors is whether *professional* managers who rely on public information outperform. Studies of actively managed mutual funds tend to find that, after fees, the average fund underperforms its benchmark, and that past outperformance is only weakly persistent. This is consistent with a market in which competition among skilled professionals eliminates most exploitable profits, leaving rewards for skill roughly equal to the costs of obtaining and trading on it. Grossman and Stiglitz made the logical point: if markets were perfectly efficient, no one would have an incentive to gather information, so prices could not reflect it. Markets must therefore be *efficient enough* that the marginal gains from information roughly equal the marginal costs of acquiring it: efficiency is a matter of degree.

12.3 Anomalies and Their Interpretation

An *anomaly* is a pattern in returns that cannot be explained by the standard model of risk and return, or that appears exploitable. Researchers have catalogued dozens. Table 12.3 lists a selection, described only qualitatively; magnitudes differ across periods, markets, and studies, and the most important point about many of them is that they have weakened after publication.

Table 12.3 – Selected anomalies and competing explanations

Anomaly	Pattern	Risk-based explanation	Behavioral / frictions explanation
Momentum (Jegadeesh and Titman)	Stocks with high returns over the past 3 to 12 months tend to keep outperforming over the next several months	Time-varying risk; exposure to crash risk	Underreaction to news; herding; slow diffusion of information
Value (Fama and French)	Stocks with high book-to-market ratios have higher average returns than low book-to-market stocks	Distress risk; compensation for a risk factor	Extrapolation of past growth; overreaction to good and bad news
Size	Small firms have earned higher average returns than large firms, after CAPM adjustment	Liquidity and information risk	Neglect; limits to arbitrage in small stocks
Post-earnings-announcement drift	Returns continue in the direction of an earnings surprise for weeks or months	Few risk explanations	Investor underreaction; anchoring on prior expectations
Accruals	Firms with high accruals relative to cash flow tend to underperform	Weak	Investors fixate on earnings and underweight cash flow information
IPO underperformance	Recent IPOs tend to underperform comparable firms over multiple years	Benchmark and measurement problems	Investor over-optimism; issuers time the market
Low-risk (beta) anomaly	Low-beta stocks have earned returns close to or higher than high-beta stocks	Leverage constraints distort the security market line	Preference for lottery-like payoffs; benchmarking by managers
Calendar effects	Seasonal patterns, such as turn-of-the-year strength in small stocks	Tax-loss selling; liquidity	Window dressing; investor sentiment

12.3.1 Three ways to read an anomaly

When confronted with a documented pattern, the analyst can adopt one of three readings, each with different consequences.

It is compensation for risk. The pattern is real but not an arbitrage: stocks with high book-to-market ratios might be more exposed to financial distress, a risk that investors demand compensation to bear and which the CAPM omits. The efficient markets view survives; the model of expected returns needs more factors. This is the stance of the Fama–French multi-factor models.

It is mispricing. The pattern reflects systematic errors by investors, such as underreaction to earnings news or overreaction to long strings of good news. Expected returns on such stocks differ from the cost of capital that the CAPM implies, and the market is not efficient in the semi-strong sense.

It is an artifact. The pattern arose from searching over many possible strategies and reporting the one that worked in the sample (data snooping), from survivorship bias, from ignoring trading costs, or from the idiosyncrasies of a particular period. It will disappear out of sample.

Distinguishing among these requires out-of-sample evidence, tests across countries and periods, an understanding of the economic story, and attention to whether profits survive costs. Anomalies that are large, persistent across markets, concentrated where arbitrage is hard, and associated with an economic mechanism are the most credible.

12.3.2 Data snooping: the multiple-testing problem

Suppose thousands of researchers test hundreds of trading signals on the same historical data set. Even if no signal had any predictive power, some would appear significant by chance. A significance level of 5 percent means that 1 in 20 useless signals will pass.

Example 12.4 – The probability of a false discovery.

An analyst tests 100 independent trading strategies, none of which has any true predictive power, each at the 5 percent significance level.

Expected number of false positives: 100 × 0.05 = 5 strategies will appear to "work."

Probability of at least one false positive:

```
1 − (1 − 0.05)^100 = 1 − 0.95^100 = 1 − 0.0059 = 0.994
```

There is a 99.4 percent chance that at least one spurious strategy is found. The simple Bonferroni correction demands a stricter threshold for each test: 0.05 / 100 = 0.0005. At that level, the probability of at least one false positive across the 100 tests is:

```
1 − (1 − 0.0005)^100 = 1 − 0.9512 = 0.0488 ≈ 4.9%
```

which restores the intended overall error rate of about 5 percent. A two-sided p-value of 0.0005 corresponds to a t-statistic of about 3.5 for large samples, which is why modern researchers argue that a t-statistic of 2 is too lenient for newly discovered patterns. For the manager reading a vendor's back-tested "alpha," the lesson is to ask how many strategies were tried before this one was presented.

12.3.3 Do anomalies survive?

Evidence suggests that many anomalies weaken or vanish after they are published and exploited, which is consistent with the idea that market participants learn and arbitrage capital moves in, but also with the idea that some of the in-sample returns reflected data mining. Momentum, value, and post-earnings-announcement drift are among the more durable patterns, though with episodes of severe losses. Calendar effects are generally weaker than originally reported. A fair reading is that markets are *mostly* efficient in the sense that simple, widely known strategies do not offer large, reliable, riskless profits after costs, while sizeable pockets of apparent mispricing exist where arbitrage is costly or risky.

12.4 Limits to Arbitrage

12.4.1 Why mispricing can persist

The efficient markets argument relies on arbitrageurs to eliminate mispricing. Shleifer and Vishny and others have shown that in practice arbitrage is limited, so a mispricing can last long enough that betting against it is dangerous. The limits include:

Fundamental risk. Few arbitrages are riskless. Buying an undervalued stock exposes the investor to the risk that the stock falls because of news about its fundamentals, and the hedge (shorting a similar stock) is imperfect.

Noise-trader risk. Even when an investor is right about fundamentals, irrational traders (noise traders) can push prices further from value before they return. De Long, Shleifer, Summers, and Waldmann showed that this risk, the risk that sentiment worsens before it recovers, deters arbitrageurs, particularly those with short horizons or leverage.

Short-selling constraints. Overpricing is harder to correct than underpricing, because shorting requires borrowing shares, which may be expensive, unavailable, or subject to recall, and sometimes banned by regulators. Pessimistic investors can sell what they own but cannot easily express larger bearish views.

Implementation costs. Bid-ask spreads, commissions, price impact for large trades, and taxes consume profits.

Agency problems. Arbitrageurs often manage other people's money. If a mispricing widens, investors may withdraw capital at precisely the worst time, forcing liquidation: "the market can stay irrational longer than you can stay solvent."

These limits imply that the market can be both *almost efficient* in the sense that it is hard to make money from it and *not fully efficient* in the sense that prices may deviate persistently from value. The absence of free lunches does not prove that prices are right.

Example 12.5 – Gross versus net returns on a momentum strategy.

A fund finds that a long-short momentum strategy (buying past winners and shorting past losers) earned a gross return of 9.0 percent per year in a backtest. Implementation costs are: trading costs from portfolio turnover of 2.0 percent per year; short borrowing fees on loser stocks, which are often expensive to borrow, of 3.0 percent per year on the portfolio; and price impact at the size the fund wishes to trade of 1.5 percent per year.

```
Net return = 9.0% − 2.0% − 3.0% − 1.5% = 2.5% per year
```

Nearly three-quarters of the gross return disappears into implementation costs (6.5 / 9.0 = 72%). If the strategy also has periodic crashes, with large losses in rebounds after market declines, the risk-adjusted net return could be negative. This is how a pattern can appear in the data while the market still passes the test of "no profit after costs and risk." Arbitrage is limited because the visible profit is a gross figure, while the costs and risks that determine whether anyone will bother to trade on it appear only net.

Example 12.6 – Noise-trader risk and forced liquidation.

An arbitrage fund identifies two securities with identical cash flows trading at a persistent spread: the cheap one (A) trades 4 percent below the expensive one (B). It buys $100 of A and shorts $100 of B, expecting convergence within a year, which would yield a profit of 4 percent of $100 = $4. The fund posts $20 of its own capital and borrows the rest (five times leverage), with a covenant that requires liquidation if capital falls to $14 (a 30 percent loss).

Case 1: convergence without interruption. Profit is $4 on $20 of capital, a 20 percent return (before financing costs).

Case 2: sentiment worsens first. Noise traders push the spread to 10 percent before it narrows. The mark-to-market loss is (10% − 4%) × \$100 = \$6, which is 30 percent of the fund's \$20. Its covenant is triggered, forcing liquidation at the widest spread. The final result is a loss of \$6, a −30 percent return, even though the original thesis (spreads converge) turned out right afterwards.

The arbitrageur was correct about value but wrong about timing, and leverage converted a temporary price move into a permanent loss. This is why historical episodes in which closely related securities traded at persistent discounts, such as the use of dual-listed company structures or the case of a parent and its partly-spun-off subsidiary valued oddly relative to each other, could last for long periods without being fully exploited.

12.4.2 Evidence of apparent mispricing under frictions

Several well-known episodes are presented as evidence of limits to arbitrage. Closed-end funds frequently trade at discounts or premiums to the net asset value of their holdings, even though the holdings are observable. Dual-listed companies that share cash flows under a fixed formula have at times traded at prices that deviated from the formula, and bound together only slowly. During the late-1990s technology boom, certain subsidiaries that had been partially listed traded for more than the parent that held a controlling share, implying that the rest of the parent's operations had negative value, an apparent violation of limited liability that was hard to arbitrage because the subsidiary shares were difficult to borrow. These cases are instructive because they show mispricing in settings where fundamental value is transparent, and they trace failures of arbitrage to concrete frictions.

12.5 Behavioral Finance: How Real Decision-Makers Depart from the Textbook

Behavioral finance combines the limits to arbitrage with a second pillar: evidence from psychology that investors, and managers, systematically depart from the fully rational, utility-maximizing agent of classical theory. If those departures are correlated across many investors, and arbitrage is limited, prices can be pushed away from fundamentals.

12.5.1 Prospect theory

Kahneman and Tversky's prospect theory, developed from experiments, describes how people evaluate risky choices. Three features matter for finance. People evaluate outcomes relative to a *reference point* (often the purchase price or current wealth), not in terms of final wealth. They are *loss averse*: a loss hurts more than an equivalent gain pleases, commonly by a factor of about two in experiments. And they are risk-averse over gains but risk-seeking over losses, so they may gamble to avoid realizing a loss. Applied to investing, loss aversion helps explain the *disposition effect*, the tendency to sell winning stocks too early and hold losers too long, which is contrary to the tax advantage of realizing losses.

12.5.2 Principal biases

Table 12.4 summarizes the best-known biases, their effects on investors, and their parallel effects on managers.

Table 12.4 – Common biases and their managerial analogues

Bias	Description	Investor consequence	Managerial consequence	Possible safeguard
Overconfidence	Overestimating the precision of one's knowledge and ability	Excessive trading; underdiversification	Overinvestment; overpaying for acquisitions; narrow forecast ranges	Reference-class forecasting; post-audits
Anchoring	Over-reliance on an initial number	Slow updating of price targets	Budgets and forecasts stick to last year's figures	Independent estimates before sharing anchors
Confirmation bias	Seeking evidence that supports prior views	Ignoring bad news	Selective use of data to support a favored project	Devil's-advocate review
Representativeness	Judging by resemblance to a stereotype; ignoring base rates	Extrapolating recent growth; chasing winners	Assuming a new venture will resemble the best success stories	Base-rate data on similar projects
Loss aversion / sunk cost	Pain of loss outweighs equal gain; reluctance to realize losses	Disposition effect	Escalation of commitment to failing projects	Pre-set stop criteria; rule of ignoring sunk costs
Mental accounting	Treating money differently by label	Separate "safe" and "speculative" pots	Treating cash windfalls or "free" retained earnings as costless	Single-pool cost of capital
Herding	Following the crowd	Momentum, bubbles	Imitating competitors' investments and acquisitions	Independent valuation
Framing	Decisions change with presentation	Different choices on equivalent gambles	Project approval depends on how costs and benefits are packaged	Standardized decision templates

12.5.3 Sentiment and its effect on prices

Investor sentiment is a broad term for the optimism or pessimism of the marginal investor not justified by fundamentals. Baker and Wurgler, among others, documented that returns on speculative, hard-to-value stocks (young, unprofitable, high-volatility firms) are predictably lower following high-sentiment periods and higher following low-sentiment periods, while stable, easy-to-value stocks are less affected. The pattern fits the logic of limits to arbitrage: sentiment matters most where valuation is subjective and arbitrage costly.

Heuristics and extrapolation. Many investors project recent earnings growth into the future (representativeness) while the empirical record shows that extreme growth tends to mean-revert. A stock with several consecutive years of rapid growth is priced as though the growth will continue, and it disappoints on average. Models by Barberis, Shleifer, and Vishny, and by Daniel, Hirshleifer, and Subrahmanyam, formalize how underreaction to some news (conservatism) can coexist with overreaction to patterns of news (extrapolation), generating both momentum and long-horizon reversal.

12.5.4 Rational responses and the debate

Defenders of efficiency argue that the biases are well documented in laboratory settings but may not survive in markets, where competition, learning, and incentives weed out costly errors, and that behavioral explanations are too flexible: with enough biases, any pattern can be rationalized after the fact. Behavioralists reply that the same is true of rational models

with enough risk factors, and that behavioral theories make specific, falsifiable predictions, such as the pattern of sentiment effects across stocks. Most scholars now treat the two approaches as complementary: markets are largely efficient in the sense that prices are hard to beat, while psychology and frictions explain some departures and episodes.

12.6 Behavioral Biases in Corporate Decisions

Managers are as subject to psychological forces as investors, but with larger stakes and weaker feedback. Two aspects matter most: biased forecasts in capital budgeting, and biased judgments in mergers and financing.

12.6.1 The winner's curse in project selection

When a firm chooses the best-looking projects from a large pool of uncertain estimates, the selected projects are disproportionately those whose estimates are too optimistic. This *selection bias*, or winner's curse, makes the average realized NPV of selected projects fall short of their estimated NPV, even if forecasts are unbiased on average.

Example 12.7 – Shrinking an estimated NPV.

A firm evaluates many projects. Suppose true NPVs across projects are distributed normally with mean 0 and standard deviation \$10 million, and each project's estimated NPV equals true NPV plus an independent forecasting error with mean 0 and standard deviation \$10 million. The best Bayesian estimate of true NPV given an estimate of E is:

```
E[true NPV | estimate] = estimate × σ²true / (σ²true + σ²error)
                       = estimate × 100 / (100 + 100) = 0.5 × estimate
```

A project with an estimated NPV of \$12 million is expected to deliver only 0.5 × 12 = \$6 million. A project that is just barely positive on paper, such as one with an estimate of +\$2 million, has an expected true NPV of \$1 million, and given that the standard deviation of the posterior is √(100 × 100 / 200) = \$7.07 million, there is roughly a 44 percent chance that it actually destroys value. Projects chosen from the top of a ranking are less valuable than they appear. The remedy is to shrink forecasts toward base rates of similar past projects (reference-class forecasting), to demand a margin of safety in the hurdle, and to audit actual outcomes against forecasts so that the error variance can be estimated.

If forecast errors are larger relative to true dispersion, the shrinkage is more severe. With error standard deviation of \$20 million (variance 400), the shrinkage factor becomes 100 / (100 + 400) = 0.20, and the \$12 million estimate is worth only \$2.4 million in expectation.

12.6.2 Overconfidence, optimism, and mergers

Overconfident managers overestimate their ability to create value from acquisitions, underestimate integration risk, and believe their firm's stock is undervalued, which makes them reluctant to issue equity and more likely to finance acquisitions with internal cash or debt. Roll's *hubris hypothesis* proposes that acquirers systematically overpay because the managers' valuation includes errors, and the winner of a competitive bid is the one with the most optimistic error. The finding that acquirers on average earn roughly zero abnormal returns around announcements while targets earn large gains is consistent with this account, though also with competitive markets for corporate control. Managers of overconfident firms also tend to invest more, and their investment is more sensitive to internal cash flow.

12.6.3 Escalation of commitment

Sunk costs are irrelevant to decisions, yet managers frequently continue to fund failing projects because stopping would require admitting a mistake. The cure is institutional: reviews by people who did not sponsor the original project, explicit abandonment criteria set at approval, and incentive systems that do not penalize reporting bad news early.

12.7 Bubbles, Crashes, and Valuation Discipline

12.7.1 What is a bubble?

A *bubble* is a sustained deviation of price from fundamental value, typically accompanied by rapid price appreciation, widespread optimism, heavy trading, and an extrapolative narrative ("this time is different"). Because fundamental value is not observed, bubbles can be identified with confidence only afterward. History offers numerous episodes in which prices of an asset class rose dramatically and then fell sharply: the Dutch tulip mania of the 1630s, the South Sea Company in 1720, the technology boom and bust around 2000, and the housing and credit boom ending in 2007 to 2008. Scholars disagree on how many of these were pure irrationality; some argue the rise was explicable by rational expectations of high growth that did not materialize.

Several mechanisms are cited for how bubbles survive. Limits to arbitrage allow rational traders to wait rather than bet against the trend, because shorting is risky and costly (Section 12.4). Rational traders may even "ride the bubble," anticipating that others will keep buying. Positive feedback trading, where rising prices attract more buyers, extrapolating past returns, amplifies the swing. Leverage and credit expansion supply the funds. Financial innovation and new-era narratives supply the justification. And the incentives of professionals, who are judged against peers and who may lose clients if they stand aside while competitors profit, reward herding.

12.7.2 Using valuation arithmetic to test a price

Even without knowing the "true" value, a manager can ask what a price implies. The Gordon growth model, P = D1 / (r − g), can be solved for the growth rate that the price requires.

Example 12.8 – Implied growth and a two-stage valuation.

A stock trades at $120. Next year's dividend is expected to be $2.40, and the required return is 9 percent.

Implied perpetual growth:

```
g = r − D1 / P = 9.0% − 2.40 / 120 = 9.0% − 2.0% = 7.0%
```

A perpetual dividend growth rate of 7 percent is well above the long-run growth rate of most economies, so the price demands that the firm outgrow the economy forever. If a more plausible long-run growth rate of 4 percent applies, the value is:

```
P = 2.40 / (0.09 − 0.04) = 2.40 / 0.05 = $48.00
```

The price is 2.5 times the value implied by plausible growth. Now allow for an extended high-growth period: dividends grow 12 percent per year for years 2 to 5 (D1 = 2.40), then 4 percent forever.

Dividends: D1 = 2.400; D2 = 2.688; D3 = 3.011; D4 = 3.372; D5 = 3.776.

Present value of years 1 to 5 at 9%: $2.400/1.09 + 2.688/1.09^2 + 3.011/1.09^3 + 3.372/1.09^4 + 3.776/1.09^5 = 2.202 + 2.263 + 2.325 + 2.389 + 2.454 = \$11.63$.

Terminal value at year 5: D6 = 3.776 × 1.04 = 3.928; P5 = 3.928 / 0.05 = $78.55. Present value: $78.55 / 1.09^5 = 78.55 / 1.5386 = \$51.05$.

```
P0 = 11.63 + 51.05 = $62.68
```

Even with five years of 12 percent growth, the stock is worth about $63, roughly half the quoted price. A price of $120 requires either much longer or faster growth, or a much lower required return, than ordinary assumptions allow. This does not prove a bubble, because the inputs are uncertain, but it quantifies how much optimism the market price embeds and invites the question whether management shares it.

12.7.3 Lessons for firms that find themselves inside a bubble

A firm whose shares are inflated faces temptation: issue equity at attractive prices, make acquisitions with overvalued stock, expand capacity to justify the valuation, and reward executives in stock options that pay off from the rise. All of these can be harmful if the bubble bursts. The disciplined response is to separate value from price: estimate intrinsic value with explicit cash-flow forecasts, maintain the cost of capital and hurdle rates that would apply without the bubble (do not lower hurdle rates because the stock is high), and limit leverage so the firm survives the bust.

12.8 Implications for Financial Managers

12.8.1 A practical stance: trust prices, verify claims

The evidence supports a pragmatic position. For large, liquid, widely followed firms, market prices are an unbiased-enough estimate of value that managers should be skeptical of their own ability to out-guess them, and should regard a persistent belief that “the market is wrong” as a hypothesis requiring evidence. Yet markets are not infallible, and for hard-to-value firms, in times of high sentiment, or when arbitrage is costly, prices may deviate meaningfully from value. The right stance is therefore calibrated: assume the market price is informative, treat deviations as possible, and design decisions that do not depend on guessing the direction of mispricing.

12.8.2 Financing and the market-timing temptation

If prices can deviate from value, then financing choices are no longer neutral. When a firm issues shares that are overvalued, existing shareholders gain at the expense of new ones; when it repurchases undervalued shares, continuing shareholders gain at the expense of those who sell. Survey and empirical evidence suggest firms issue equity after periods of high returns and repurchase after low ones, consistent with attempts to time the market; Baker and Wurgler argue that cumulative timing has lasting effects on capital structure.

Example 12.9 – Wealth transfers from repurchases and issuances when price differs from value.

Alder Instruments (a fictional firm) has 100 million shares outstanding. Management’s best estimate of intrinsic equity value is $5,000 million ($50 per share), while the market price is $40.

(a) Repurchase of undervalued shares. The firm spends $400 million to buy back shares at $40:

```
Shares repurchased = 400 / 40 = 10 million
Remaining shares   = 90 million
Intrinsic value of equity after buyback = 5,000 − 400 = 4,600
Intrinsic value per remaining share     = 4,600 / 90 = $51.11
```

Continuing shareholders gain $1.11 per share, or 90 × 1.11 = $100 million, and the sellers lose exactly $100 million (10 million shares × the $10 gap between value and price). The buyback is a wealth transfer from sellers to stayers equal to the mispricing times the shares retired. If the shares had been fairly priced at $50, repurchasing 8 million shares for $400 would leave 4,600 / 92 = $50.00 per share: no gain.

(b) Issuance of overvalued shares. Suppose instead the intrinsic value is $30 per share ($3,000 million) while the price is $40, and the firm issues $400 million of equity:

```
New shares issued = 400 / 40 = 10 million; total shares = 110 million
Intrinsic value after issue = 3,000 + 400 = 3,400
Intrinsic value per share   = 3,400 / 110 = $30.91
```

Existing holders' per-share value rises from $30.00 to $30.91, a gain of $0.91 × 100 = $91 million, borne by the new investors who paid $40 for $30.91 of value. The lessons are symmetric. But note the practical constraints: the intrinsic value is the manager's estimate, not a fact; investors know managers have information and may infer overvaluation from an equity issue (which is why share prices on average fall on announcement of seasoned equity offerings); and managers with a stake in the firm, short horizons, or biased judgment may misjudge the gap. A firm that repeatedly "times" the market on the basis of unreliable views may destroy value instead.

12.8.3 Capital budgeting under potential mispricing

A key practical principle is that firms should evaluate projects on fundamentals and use a discount rate based on risk rather than adjusting it to current market conditions. Three consequences follow.

Do not lower the hurdle rate because the stock is rising. A high share price does not make the capital cheaper in an opportunity cost sense; the cost of capital as defined in Chapter 11 reflects the long-run expected return required for risk. Managers who think of equity as cheap when the price is high tend to over-invest at the top.

Do not invest merely to please the market. Catering theories suggest managers may adopt policies, such as paying dividends or adopting fashionable strategies, because investors reward them temporarily. Where this reflects genuine mispricing, a long-term shareholder is better served by decisions based on NPV; where it signals information, the effect is legitimate. Managers need to know which.

Beware of the tail of the distribution. If mispricing is in the direction of overvaluation, an NPV-driven firm will still sometimes find that the cheapest source of capital is new equity; issuing then and holding the proceeds in safe assets, if the investment pipeline is empty, is a defensible response and preserves flexibility.

12.8.4 Communication, disclosure, and investor relations

If prices respond to information quickly but investors have limited attention, managers have an interest in clear, timely, and credible disclosure that reduces information asymmetry and the discount for uncertainty. Event-study evidence on announcements shows that the market responds to genuinely new information. Evidence on accounting manipulation shows that markets are partially but not completely fooled by earnings management; reputational damage when manipulation is discovered is large. Managers should aim to provide information that is hard for competitors to replicate but that helps investors forecast, avoiding earnings guidance that creates short-term pressure, and ensure that incentive pay is tied to long-term performance rather than the short-run share price.

12.8.5 Governance against managerial biases

Because the managers' own biases are a larger practical problem than market inefficiency for most firms, governance mechanisms matter. These include forecasting from outside views and base rates, mandatory post-audits of large investments, separation of project sponsorship from approval, red-team reviews of acquisitions, scenario analysis with

explicit downside planning, and compensation that rewards long-run value creation. Each is a response to the bias of a particular kind identified in Table 12.4. Managers can learn from behavioral finance by treating it first as a diagnosis of their own decision processes, and only second as a guide to exploiting others.

Case Study: Greenfield Components Considers a Share Buyback

This case is fictional and illustrative; all figures are invented for teaching purposes.

Greenfield Components is a supplier of precision parts to the automotive industry, listed on a national exchange. It has 80 million shares outstanding, trading at $25, giving a market capitalization of $2.0 billion. The firm has $350 million of cash beyond operating needs and $600 million of debt, which is at a comfortable leverage level.

Eight months ago, the share price was $38. Over that period, the auto sector sold off sharply on concerns about electric-vehicle disruption. Analysts' consensus target price dropped from $40 to $27, with a few downgrades to "sell." Greenfield's own order backlog, however, has risen 6 percent over the period, and the firm has begun to win contracts for components used in electric vehicles that now account for 12 percent of backlog, up from 4 percent.

Greenfield's chief executive, Mirela Costa, has been a vocal bull about the firm's strategy. Last month she purchased $2 million of stock personally, and she is pushing the board to authorize a $300 million buyback. The chief financial officer, Elias Brandt, supports returning some cash but cautions about the logic: "We have been wrong about timing before. Two years ago we issued $200 million of equity at $36 to fund a plant that we now think was oversized, and we have not fully used the capacity. Today, we believe the stock is undervalued by 30 percent. I am not certain our confidence is justified; I know it is hard to distinguish conviction from wishful thinking."

An internal team prepared a discounted cash flow valuation, with a base-case equity value of $3.0 billion ($37.50 per share). They acknowledge that the forecast depends heavily on the pace of conversion to electric-vehicle components. In a downside scenario, where the conversion is slower and margins compress, equity value is $1.7 billion ($21.25 per share); in an upside scenario it is $4.0 billion ($50 per share). The team assigns probabilities of 30 percent to the downside, 50 percent to the base case, and 20 percent to the upside, giving an expected equity value of $2.81 billion ($35.13 per share) (0.30 × 1.7 + 0.50 × 3.0 + 0.20 × 4.0). The team also notes that its previous forecasts of the three prior years exceeded actual results by an average of about 15 percent.

A board member, a retired investment manager, raises questions. "The market has priced in the headwinds. What do we know that thousands of investors with access to the same filings do not? And if the stock is cheap, are we better off spending the money on a bolt-on acquisition, or on the capacity that EV customers want?" Another director notes that executive bonuses are tied in part to earnings per share, which a buyback would raise mechanically, and asks whether that creates a conflict.

The treasurer adds a practical complication. Greenfield's shares are moderately liquid. A $300 million repurchase would be about 15 percent of market capitalization and, executed over six months, would probably move the price upward, implying an average price above the current $25. And if the downside scenario materialized after the buyback, the firm would have less cash to absorb it.

Discussion Questions

1. Compute the expected value per share under the team's scenario weights, then adjust it for the team's historical forecast optimism (15 percent). Using the adjusted value, estimate the gain or loss per continuing share from a $300 million buyback at an average price of $27. How sensitive is your answer to the assumptions?

2. Which behavioral biases might be affecting Mirela, Elias, and the internal team? Classify each using Table 12.4 and suggest governance practices to counter them.
3. Interpret the CEO's personal share purchase and the broader market's reaction to the buyback announcement using the semi-strong-form framework. What would an event study of the announcement test, and how would you separate "information" from "mispricing correction"?
4. Evaluate alternatives to the buyback (acquisition, capacity expansion, special dividend, debt reduction) from the standpoint of value, risk, and signaling. Does the EPS-linked bonus plan create a conflict, and what change would you recommend?

Key Takeaways

- The efficient market hypothesis claims that prices reflect available information so that no strategy earns abnormal returns after costs and risk. It comes in weak, semi-strong, and strong forms, defined by the information set.
- All tests of efficiency are joint tests of the information hypothesis and an asset-pricing model; a pattern in returns cannot by itself distinguish mispricing from unmodeled risk.
- Event studies measure the price reaction to news using abnormal returns relative to a model such as the market model; they are the standard tool for judging the value impact of corporate actions and for testing the speed of price adjustment.
- Anomalies such as momentum, value, and post-earnings drift are well documented, but many shrink after publication, and data snooping makes new discoveries suspect; demand stricter statistical thresholds for newly found patterns.
- Arbitrage is limited by fundamental risk, noise-trader risk, short-selling constraints, transaction costs, and the incentives of leveraged intermediaries, so mispricing can persist without implying easy profits.
- Investors and managers exhibit systematic biases, including overconfidence, anchoring, loss aversion, herding, and representativeness, that affect trading, forecasting, and capital allocation.
- The winner's curse means that projects chosen for their high estimated NPVs are on average worth less than estimated; shrink estimates toward base rates and audit outcomes.
- When price and intrinsic value diverge, repurchases and issuances transfer wealth between continuing and exiting or entering shareholders, but managers' estimates of intrinsic value are fallible and the market reads the action as a signal.
- Test any extreme price by asking what growth or discount rate it implies; do not lower the hurdle rate because the share price is high.
- The practical stance is to trust market prices as informative, evaluate projects on fundamentals with risk-based discount rates, guard against one's own biases through governance, and avoid decisions that depend on being smarter than the market.

Review Questions

1. State the three forms of the efficient market hypothesis and give an example of a strategy ruled out by each.
2. What is the joint hypothesis problem, and why does it prevent a decisive test of market efficiency?

3. Explain why a statistically significant autocorrelation in daily returns does not necessarily imply a profitable trading strategy.
4. Describe the steps in an event study. Why is a market model used rather than raw returns?
5. Why are long-horizon event studies less reliable than short-horizon ones?
6. Distinguish three interpretations of an apparent anomaly. What evidence would help choose among them?
7. Explain the multiple-testing problem and how a Bonferroni correction addresses it.
8. List the main limits to arbitrage and explain how each can allow mispricing to persist.
9. How does loss aversion contribute to the disposition effect and to escalation of commitment in firms?
10. Under what conditions would a share repurchase create value for continuing shareholders, and what risks does a manager face in judging that condition is met?

Problems

Problem 12.1. A stock's market model has $\alpha = 0.03$ percent per day and $\beta = 1.25$. On an announcement day, the market return is 1.2 percent and the stock return is 3.2 percent. Compute the expected return and the abnormal return.

Problem 12.2. The abnormal returns on three consecutive days around an event are +0.8%, +1.9%, and −0.5%. The residual standard deviation of the market model is 1.1 percent per day. Compute the CAR and its t-statistic (assume independence across days).

Problem 12.3. An event study of 36 firms finds a mean CAR of 2.4 percent with cross-sectional standard deviation of 6.0 percent. Compute the t-statistic. Is the mean CAR significant at the 5 percent level (critical value about 2.03)?

Problem 12.4. A researcher examines 20 independent signals at the 5 percent level, none of which has real predictive power. (a) What is the probability that at least one appears significant? (b) What per-test significance level restores a 5 percent overall error rate under the Bonferroni correction?

Problem 12.5. A firm has 60 million shares and an intrinsic equity value of \$3,000 million (\$50 per share). The market price is \$40. It repurchases \$200 million of stock at \$40. Compute the number of shares retired, the intrinsic value per remaining share, and the total gain to continuing shareholders.

Problem 12.6. A stock trades at \$80, with an expected next-year dividend of \$2.00 and a required return of 8 percent. (a) Compute the perpetual growth rate implied by the price. (b) Compute the stock's value if long-run growth is 3.5 percent.

Problem 12.7. A long-short strategy has a gross expected return of 7.0 percent per year. Trading costs are 1.2 percent, short-borrowing fees are 2.5 percent, and price impact is 1.0 percent. Compute the net return and the share of the gross return consumed by costs.

Problem 12.8. True project NPVs have a standard deviation of \$10 million (mean zero), and the estimation error has a standard deviation of \$8 million. A project is estimated to have an NPV of \$15 million. Compute the shrinkage factor and the expected true NPV.

Answers to Selected Problems

- **12.1.** Expected return = 0.03% + 1.25 × 1.2% = 0.03% + 1.50% = 1.53%. Abnormal return = 3.2% − 1.53% = +1.67%.

- **12.2.** CAR = 0.8 + 1.9 − 0.5 = 2.2%. Standard deviation of the CAR = 1.1 × √3 = 1.905%. t = 2.2 / 1.905 = 1.155, not significant at 5 percent.
- **12.3.** Standard error = 6.0 / √36 = 1.0%. t = 2.4 / 1.0 = 2.40, which exceeds 2.03, so the mean CAR is significant.
- **12.4.** (a) 1 − 0.95^20 = 1 − 0.3585 = 64.2%. (b) 0.05 / 20 = 0.25% (0.0025) per test.
- **12.5.** Shares retired = 200 / 40 = 5 million; remaining = 55 million. Intrinsic value per remaining share = (3,000 − 200) / 55 = $50.91. Gain = $0.91 per share, or about 55 × 0.909 = $50 million in total.
- **12.6.** (a) g = 8.0% − 2.00 / 80 = 8.0% − 2.5% = 5.5%. (b) P = 2.00 / (0.08 − 0.035) = 2.00 / 0.045 = $44.44.
- **12.7.** Net return = 7.0 − 1.2 − 2.5 − 1.0 = 2.3% per year. Costs consume 4.7 / 7.0 = 67% of the gross return.
- **12.8.** Shrinkage factor = 100 / (100 + 64) = 0.610. Expected true NPV = 0.610 × 15 = $9.15 million.

Part IV — Financing and Payout Policy

Chapter 13: Raising Equity: Venture Capital, IPOs and Seasoned Offerings

Learning Objectives

After studying this chapter, you should be able to:

- Describe the stages of equity financing from founder capital through venture rounds, the initial public offering (IPO) and later seasoned offerings, and match each stage to the firm's risk profile and information environment.
- Value a start-up with the venture capital method, and translate a target return into a required ownership stake and a pre-money valuation.
- Read a term sheet critically: pre- versus post-money valuation, option-pool mechanics, liquidation preferences, anti-dilution protection, board control and other economic and governance terms.
- Calculate dilution across multiple financing rounds and the effect of down rounds under weighted-average and full-ratchet protection.
- Explain the IPO process (underwriting, book-building, pricing, allocation, stabilization) and quantify the total cost of going public, including underpricing.
- Evaluate seasoned equity offerings and rights issues, compute the theoretical ex-rights price and the value of a right, and explain why announcements of equity issuance tend to depress share prices.

Opening Vignette

Illustrative scenario; the company and all figures are fictional.

Marisol Vega is the co-founder and chief executive of Tidewell Systems, a six-year-old software firm that sells predictive maintenance tools to regional water utilities. Tidewell has grown revenue to $42 million, is modestly profitable on an operating basis, and has a backlog of utility contracts that it cannot staff fast enough to serve. Marisol needs roughly $60 million to hire engineers, fund a sales expansion into two new regions and acquire a small analytics competitor.

Her board presents three roads. The first is a late-stage private round: a growth equity fund has offered $60 million at a $240 million pre-money valuation, with a 1x liquidation preference and a board seat. The second is an IPO, which bankers say could raise $150 million at a valuation near $500 million, but which would require two more quarters of audited growth, a permanent investor-relations function and quarterly scrutiny. The third is a bank loan, which Tidewell can afford only up to about $15 million before its covenants start to bind.

Marisol's problem is not simply "which has the lowest cost." The private round is quick but carries preference terms she must model carefully. The IPO offers permanent capital and currency for acquisitions, yet bankers warn that the shares will probably be priced at a discount to where they trade on day one, and the founders' stakes will be diluted and locked up for months. Existing venture investors, who own 38% of the company and have held their shares for seven years, are impatient for liquidity. Each path changes who controls the firm, who bears risk and what the founders eventually take home.

This chapter gives you the tools to answer her questions. We start with the life cycle of equity financing, work through venture valuation and term-sheet mechanics, then follow a company through its IPO and its later equity issues, quantifying the economics at each step.

13.1 The Life Cycle of Equity Financing

Equity is the residual claim on a firm's assets and cash flows. Unlike debt, it has no maturity, no promised payment and no default trigger, which makes it the natural source of capital for firms whose cash flows are uncertain, distant or intangible. But equity is also the most expensive source of capital because investors bear the most risk, and it dilutes the ownership and control of those who already hold it. A financing strategy is therefore a sequence of choices about how much equity to raise, from whom, at what price, and with what rights attached.

13.1.1 Stages of private financing

Young companies move through recognizable stages. The labels vary by market, but the economics are consistent.

Founder capital and "friends and family." The earliest money is typically the founders' own savings, together with small amounts from relatives and friends. There is rarely a formal valuation, and the main risk is whether the idea can be built at all.

Angel investors. Wealthy individuals, often former entrepreneurs, invest small amounts (commonly tens of thousands up to a few hundred thousand dollars) in exchange for equity or convertible instruments. Angels bring advice and networks, and often invest earlier than institutions will.

Seed financing. A seed round funds product development and the first evidence of market demand. Instruments at this stage are frequently convertible notes or simple agreements for future equity, which postpone the difficult valuation question until a priced round.

Venture capital rounds (Series A, B, C, and beyond). Institutional venture capital funds invest in successive priced rounds. A Series A typically funds the move from prototype to repeatable sales; Series B and C fund scaling; later rounds are sometimes called growth or crossover rounds. Each round is priced on the milestones achieved since the last.

Private equity and growth equity. Later-stage funds invest in more mature private firms, sometimes buying existing shares (secondary transactions) as well as supplying new capital. Buyouts, discussed in other chapters, use substantial debt; growth equity typically does not.

Public markets. An IPO sells shares to the public for the first time. After listing, the firm may raise more equity through seasoned offerings or rights issues.

Table 13.1 summarizes how the character of the financing changes along this path.

Table 13.1 – Typical characteristics of financing stages (illustrative ranges)

Stage	Typical investors	Main risk	Valuation basis	Instruments
Founder / friends and family	Founders, relatives	Can it be built?	None or informal	Common shares, loans
Angel / seed	Angels, seed funds	Does anyone want it?	Judgment, comparables	Convertible notes, SAFEs, common
Series A–B	Venture funds	Can sales be repeated and scaled?	Milestones, VC method	Convertible preferred
Series C and later	Growth, crossover funds	Can it reach scale and profit?	Revenue multiples, early earnings	Convertible preferred
IPO	Public institutions, retail	Will public markets pay for the growth story?	Comparable multiples, DCF	Common shares
Seasoned offering	Public institutions	Is the firm using proceeds well?	Market price less a discount	Common shares, rights

13.1.2 Why the sequence is staged

Staged financing is not an accident of fashion; it is a response to two problems familiar from the theory of the firm. First, **information asymmetry**: the founders know much more about the venture's prospects than outside investors do. Second, **moral hazard**: once money is in the firm, entrepreneurs may take actions that favor themselves. By releasing capital in tranches tied to milestones, a venture investor retains the option to stop funding a failing venture and keeps founders focused on measurable progress. Each round also reveals information: the price at which a new investor is willing to invest is a signal of the venture's progress that existing investors and employees can read.

Staging also explains why valuations step up or, painfully, down. A company that has doubled its recurring revenue since the last round has de-risked the business and can sell new shares at a higher price. A company that has missed its plan may have to raise a **down round**, with consequences for founders and earlier investors that we analyze in Section 13.4.

13.1.3 The venture capital fund as an intermediary

Most institutional venture capital is raised by **general partners (GPs)** who form a fund with capital from **limited partners (LPs)** such as pension funds, endowments and insurers. The fund has a finite life, commonly around ten years, with an investment period of the first several years. GPs typically earn an annual management fee on committed capital, plus **carried interest**, a share of profits (conventionally 20%) after LPs have received their capital back and, in many funds, a preferred return. This structure matters for entrepreneurs because it shapes investor behavior. A fund needs a few outsized winners to return the whole fund; therefore VCs prefer high-variance bets, will tolerate many failures, and push companies toward growth and a liquidity event (a trade sale or IPO) within the fund's life. Founders whose preferences differ, for example one who wants a stable, profitable, independent company, should understand this before accepting venture money.

Managerial implication. Choosing an equity partner is choosing a set of incentives. A venture fund in year eight of a ten-year fund will care more about near-term exit than one in year two. Ask about the fund's vintage, reserves for follow-on investment and track record in down markets, not only about the headline valuation.

13.2 Valuing a Venture: The Venture Capital Method

Standard discounted cash flow valuation struggles with start-ups: cash flows are negative for years, forecasts are speculative, and the discount rate that reflects the true chance of failure is hard to estimate. Venture investors therefore use a shortcut that works backward from an expected exit, the **venture capital method**.

13.2.1 Mechanics

The method has four steps.

1. Forecast the company's value at exit (year *T*), typically by applying a multiple to projected revenue or earnings, using public or transaction comparables.
2. Choose a **target rate of return** (or target multiple of money) that reflects the investment's risk. Venture targets are high, often 30% to 60% or more per year for early stages, because they must compensate for the many ventures that fail, for illiquidity, and for the optimism embedded in projections.
3. Compute the required ownership at exit, adjusting for expected future dilution.
4. Translate the required stake into a post-money and pre-money valuation today.

In formulas:

```
Required value at exit of the investment = Investment × (1 + target return)^T
Required ownership at exit = Required value at exit / Projected exit value
Required ownership today = Required ownership at exit / Retention ratio
Retention ratio = Π (1 − dilution in each future round)
Post-money valuation = Investment / Required ownership today
Pre-money valuation = Post-money valuation − Investment
```

Example 13.1 – The venture capital method. A venture fund is considering a $5 million Series A investment in a medical-device start-up. The fund expects the company to be worth $200 million in five years (an acquisition by a larger device firm). The fund's target return is 50% per year. The founders and the fund expect that later rounds and option grants will dilute today's investors by a total of 20% before exit (retention ratio 0.80).

Step 1: Required value of the $5 million at exit. Multiple of money = (1.50)^5 = 7.59375 Required value = $5,000,000 × 7.59375 = $37,968,750

Step 2: Required ownership at exit. $37,968,750 / $200,000,000 = 18.98%

Step 3: Adjust for dilution. The fund must own more today so that after 20% dilution it still holds 18.98%. Required ownership today = 18.984% / 0.80 = 23.73%

Step 4: Valuation. Post-money = $5,000,000 / 0.2373 = $21.07 million Pre-money = $21.07 million − $5 million = $16.07 million

The fund should therefore be willing to invest $5 million for a 23.7% stake only if the pre-money valuation is no higher than about $16.1 million. If the founders insist on a $25 million pre-money, the fund's expected return falls below 50%; the fund can accept a lower return, restructure terms (for example, a liquidation preference) or pass.

13.2.2 Interpreting the target return

A 50% hurdle can look absurd next to a public-market cost of equity of 10% or so. The explanation is not that VCs are greedy. The projection of a $200 million exit is a *success-case* forecast, not an expected value. Suppose that of ten comparable ventures, six fail and return nothing, two return the invested money, and two reach the success case. The expected value of the portfolio is far below the success-case figure, and the required return on the success case must be high enough to compensate for this. A rough link is:

```
Target return on success case ≈ (1 + required expected return) / probability of success  − 1  (in
multiple-of-money terms)
```

For instance, with a 15% required expected annual return over five years (a multiple of 2.01) and a 25% probability of reaching the success case, the success-case multiple must be about 2.01 / 0.25 = 8.05 times the investment, or an annual return of roughly 51.7%. This explains why target returns of 40% to 60% are common for early-stage deals. Later-stage companies have a higher probability of reaching their projections, so required returns fall toward 20% to 30%.

13.2.3 Limitations and common mistakes

- **Double-counting risk.** If you use a high target return to reflect failure risk and also haircut the projections, you penalize the venture twice.
- **Ignoring dilution.** Omitting the retention ratio understates the stake an investor must own today to earn the target return after later rounds, and so overstates the valuation the investor can afford to accept. Always model the option pool and later rounds.
- **Exit-multiple optimism.** Applying a hot-sector multiple at exit assumes the sector will still be hot in five years. Test the valuation with a lower exit multiple.
- **Precision illusion.** A model that produces a pre-money valuation of $16.07 million gives a false sense of accuracy. In practice the output is a negotiating range.
- **Treating all equity as common.** Venture investors hold preferred shares with senior rights. The effective value of a preferred share exceeds that of a common share, which is why headline valuations in venture deals should be read with care.

The last point leads directly to the term sheet.

13.3 Reading a Term Sheet

A **term sheet** is a short, mostly non-binding document that summarizes the principal terms on which an investor proposes to invest. The binding agreements (share purchase agreement, investors' rights agreement, shareholders' agreement and amended charter) follow and track it closely, so the term sheet is where the real negotiation takes place. Headline valuation attracts attention, but a founder who focuses only on it can lose more value through the other terms than she gains from a higher price.

Terms fall into two families: **economic terms**, which determine who receives how much cash at each outcome, and **control terms**, which determine who decides.

13.3.1 Pre-money, post-money and price per share

```
Post-money valuation = Pre-money valuation + New investment
Investor ownership (immediately after) = New investment / Post-money valuation
Price per share = Pre-money valuation / Pre-money fully diluted shares
```

The phrase **fully diluted** matters. It includes not only issued shares but also outstanding options and warrants, converting securities and, critically, any unallocated option pool the investor requires to be in place. The way the option pool is treated is one of the most common sources of disagreement between what a headline valuation appears to be and what founders actually receive.

13.3.2 The option pool shuffle

Investors want a pool of options reserved for future hires so that the new money does not have to be diluted again when the company recruits. They commonly require the pool to be created or enlarged **before** the investment and counted in the pre-money. Then the dilution from the pool falls entirely on existing shareholders, not on the new investor, and the **effective pre-money valuation** (what the existing holders' shares are worth) is lower than the headline.

Example 13.2 – The option pool shuffle. Atlas Robotics has 6,000,000 founder shares and no options. A venture fund offers $10 million at a $30 million pre-money valuation, with the condition that after closing, an unallocated option pool equal to 10% of the fully diluted post-money capitalization exists.

Step 1: Post-money valuation = $30 million + $10 million = $40 million. The investor owns $10M / $40M = 25% after closing.

Step 2: Founders own the remainder after the investor's 25% and the pool's 10%: Founders' share = 100% − 25% − 10% = 65%.

Step 3: Total fully diluted shares after closing: 6,000,000 / 0.65 = 9,230,769 shares.

Step 4: Investor shares = 25% × 9,230,769 = 2,307,692. Pool = 10% × 9,230,769 = 923,077.

Step 5: Price per share = $10,000,000 / 2,307,692 = $4.333.

Step 6: Compare with the naive price if there were no pool: $30,000,000 / 6,000,000 = $5.00. With the pool, the price per share is lower by 13.3%.

Step 7: Effective pre-money for the founders = price × founder shares = $4.333 × 6,000,000 = $26.0 million. The "$30 million" pre-money is in practice $26 million for the founders, because $4 million of the pre-money value is attributed to the new pool (923,077 × $4.333 = $4.0 million).

The pool is not "free" to the founders. Of course, a well-sized pool does have value for them, because options motivate and retain employees. The lesson is to bargain over the pool's size using a hiring plan (how many senior hires in the next 18 months, at what grants), not to accept a standard percentage, and to compare offers on **effective pre-money**, not on headline pre-money.

13.3.3 Preferred stock and liquidation preference

Venture investors almost always buy **convertible preferred stock**, not common. Preferred shares carry a **liquidation preference**: when the company is sold or liquidated, preferred holders are paid first, before common holders. Three variants matter.

- **1x non-participating.** The holder receives the *greater* of (a) the original investment (the 1x preference) or (b) the amount it would receive if it converted into common and shared pro rata. It is a downside-protection feature that costs the holder nothing in high-value exits because the holder simply converts.
- **Participating (full).** The holder receives its preference *and then also* shares pro rata with common holders in the remainder, on an as-converted basis. This is a "double dip."
- **Participating with a cap.** The holder participates until total proceeds reach a stated multiple (for example 3x of the investment); beyond that, it converts if conversion yields more.

Multiples above 1x (such as 2x) and seniority rules (the latest round paid first, or all preferred paid *pari passu*) change the allocation in downside outcomes.

Example 13.3 – Liquidation waterfall. Quantum Grid raised $10 million from a Series A investor in exchange for 25% of the fully diluted equity. All other holders (founders and employees) hold the remaining 75% as common stock. Compare three preference structures at various sale prices (in $ millions).

Non-participating 1x. The investor receives max(preference, 25% × exit), but never more than the exit price. It converts when 25% × exit exceeds $10M, i.e., when exit > $40M.

Participating, uncapped. Investor receives $10M + 25% × (exit − $10M).

Participating, capped at 3x ($30M total). Investor receives the lesser of the participating amount and $30M, or 25% × exit if conversion yields more. Conversion beats the cap when 25% × exit > $30M, i.e., exit > $120M.

Calculations:

- Exit $20M. Non-participating: max($10M, $5M) = $10M. Participating: $10M + 25% × $10M = $12.5M. Capped: same as participating, $12.5M.
- Exit $60M. Non-participating: max($10M, $15M) = $15M (convert). Participating: $10M + 25% × $50M = $22.5M. Capped: $22.5M (below $30M cap).
- Exit $120M. Non-participating: $30M (convert). Participating: $10M + 25% × $110M = $37.5M. Capped: min($37.5M, $30M) = $30M, equal to conversion value.
- Exit $200M. Non-participating: $50M. Participating: $10M + 25% × $190M = $57.5M. Capped: converts, $50M.

Table 13.2 – Investor and common proceeds under three preference structures ($ millions)

Exit value	Non-part.: investor	Non-part.: common	Participating: investor	Participating: common	Capped 3x: investor	Capped 3x: common
20	10.0	10.0	12.5	7.5	12.5	7.5
40	10.0	30.0	17.5	22.5	17.5	22.5
60	15.0	45.0	22.5	37.5	22.5	37.5
120	30.0	90.0	37.5	82.5	30.0	90.0
200	50.0	150.0	57.5	142.5	50.0	150.0

At a $60 million sale, a participating preference moves $7.5 million from common holders to the investor relative to a non-participating one, which is 16.7% of the $45 million the common holders would otherwise receive. In a $200 million exit the difference is a smaller percentage of the total but still $7.5 million. Participation is therefore expensive for the common holders at every exit value, and it is most punishing at moderate outcomes, which are the most likely outcomes for many ventures.

13.3.4 Anti-dilution protection

If the company later sells shares at a lower price than a prior preferred round, the earlier investors' stake loses value. **Anti-dilution provisions** adjust the conversion price of the earlier preferred downward, so that each preferred share converts into more common.

- **Full ratchet.** The conversion price drops to the new, lower price, whatever the size of the new round. It is very investor-friendly and rare in negotiated deals except in distressed financings.
- **Broad-based weighted average.** The new conversion price (NCP) is a weighted average of the old price and the new price:

```
NCP = OCP × (A + B) / (A + C)
```

where OCP is the old conversion price; A is the fully diluted shares outstanding before the new issue; B is the number of shares that the new money would have purchased at OCP (new money / OCP); and C is the number of new shares actually issued. The smaller the new round relative to the company, the smaller the adjustment.

We apply these mechanisms in Example 13.5 below, after we master the plain dilution arithmetic in Section 13.4.

13.3.5 Control and governance terms

- **Board composition.** Typically a mix of founder directors, investor directors and independents. Who holds the swing vote at the board level determines who can replace the CEO.
- **Protective provisions.** Investor approval is required for specified actions: selling the company, issuing senior securities, changing the charter, taking on significant debt, paying dividends, or changing the board size.
- **Drag-along and tag-along rights.** A drag-along lets a majority force minority holders to join a sale; a tag-along lets minority holders join a sale made by a majority holder on the same terms.
- **Pre-emptive rights (pro rata rights).** Existing investors can buy their share of any new issue to avoid being diluted.
- **Vesting and good/bad leaver terms.** Founders' shares often vest over four years with a one-year cliff, so investors are not left funding a team that departs. Leaver provisions set the price at which the company can buy back unvested or vested shares from departing founders.
- **Redemption, dividends and information rights.** Preferred shares may carry accruing dividends (payable at exit or conversion), redemption rights after a number of years, and rights to financial information.
- **Exclusivity ("no-shop") and costs.** The binding parts of most term sheets: the company agrees not to solicit competing offers for a period, and often to bear some deal costs.

Common mistake. Treating the term sheet as a price negotiation. A $60 million pre-money with a 2x participating preference and a full ratchet may be worth less to the founders than a $45 million pre-money with a plain 1x non-participating preference. Model the founders' proceeds across a range of exit values (Table 13.2 is a template) before choosing.

13.4 Dilution Mathematics

Every new issue of shares reduces the percentage ownership of existing holders. Dilution itself is not harmful if the money raised creates more value than the ownership given up; it is "a smaller slice of a larger pie." The relevant questions are how fast ownership is eroding and whether value per share is growing faster than the ownership is shrinking.

13.4.1 The basic relationship

```
New investors' ownership = New money / Post-money valuation
Existing holders' retention = Pre-money valuation / Post-money valuation = 1 − New investors' ownership
Price per share = Pre-money valuation / Pre-money fully diluted shares
New shares issued = New money / Price per share
```

For a sequence of rounds, retention ratios multiply. If round k sells fraction d_k of the post-money company, a holder owning s before the first round owns $s \times \Pi\,(1 - d_k)$ after all rounds, unless it buys more.

Example 13.4 – Dilution across three rounds. Three founders hold 6,000,000 shares in total. The company raises: a seed round of $1.5 million at a $4.5 million pre-money; a Series A of $8 million at a $24 million pre-money; and a Series B of $20 million at a $60 million pre-money. (For clarity we ignore option pools; adding them would increase founder dilution.)

Seed: Price = $4,500,000 / 6,000,000 = $0.75 per share. New shares = $1,500,000 / $0.75 = 2,000,000. Total = 8,000,000. Post-money = $6.0 million; seed investors own 2,000,000 / 8,000,000 = 25%.

Series A: Price = $24,000,000 / 8,000,000 = $3.00. New shares = $8,000,000 / $3.00 = 2,666,667. Total = 10,666,667. Post-money = $32 million; Series A investors own 25%.

Series B: Price = $60,000,000 / 10,666,667 = $5.625. New shares = $20,000,000 / $5.625 = 3,555,556. Total = 14,222,222. Post-money = $80 million; Series B investors own 25%.

Table 13.3 – Capitalization table after each round

Holder	After seed (shares / %)	After Series A (shares / %)	After Series B (shares / %)
Founders	6,000,000 / 75.0%	6,000,000 / 56.25%	6,000,000 / 42.19%
Seed investors	2,000,000 / 25.0%	2,000,000 / 18.75%	2,000,000 / 14.06%
Series A investors	–	2,666,667 / 25.0%	2,666,667 / 18.75%
Series B investors	–	–	3,555,556 / 25.0%
Total	8,000,000	10,666,667	14,222,222

Retention check: founders keep 0.75 × 0.75 × 0.75 = 42.19%. Seed investors keep 25% × 0.75 × 0.75 = 14.06%. Value check at the Series B price of $5.625: founders' 6,000,000 shares are worth $33.75 million, versus $4.5 million at the seed price ($0.75 per share). Although the founders' percentage fell from 100% to 42.19%, the value of their stake grew 7.5-fold from the seed price. This is the essential logic of dilution: what matters is price per share relative to the original, not ownership percentage.

13.4.2 Down rounds and anti-dilution

A down round is a financing at a lower price per share than the previous round. It is damaging for founders and employees (their ownership falls *and* the value per share falls), and it triggers anti-dilution rights for earlier preferred holders, shifting more dilution onto common holders.

Example 13.5 – Anti-dilution in a down round. Pinewood Labs has 20,000,000 fully diluted shares outstanding, including 5,000,000 shares from the Series A investors' preferred stock, which was bought for $10 million, i.e., at $2.00 per share (the OCP). The company then raises $4 million at $1.00 per share, issuing 4,000,000 new shares. Compare three cases.

No protection. Series A converts 1:1 into 5,000,000 shares. Total fully diluted after the round = 20,000,000 + 4,000,000 = 24,000,000. Series A owns 5/24 = 20.83%; new investors own 4/24 = 16.67%; everyone else (15,000,000 shares) owns 62.5%.

Broad-based weighted average. A = 20,000,000. B = $4,000,000 / $2.00 = 2,000,000. C = 4,000,000. NCP = $2.00 × (20,000,000 + 2,000,000) / (20,000,000 + 4,000,000) = $2.00 × 22/24 = $1.8333. Series A now converts into $10,000,000 / $1.8333 = 5,454,545 shares (an extra 454,545). Total fully diluted = 15,000,000 + 5,454,545 + 4,000,000 = 24,454,545. Series A owns 22.30%; new investors 16.36%; the others 61.34%.

Full ratchet. NCP = $1.00. Series A converts into $10,000,000 / $1.00 = 10,000,000 shares. Total = 15,000,000 + 10,000,000 + 4,000,000 = 29,000,000. Series A owns 34.48%; new investors 13.79%; the others 51.72%.

The new investors, who set the $1.00 price, own less under the full ratchet (13.79% against 16.67% without protection), which is why they typically insist on negotiating the anti-dilution mechanics as part of the round. The full ratchet cuts the non-Series-A holders from 62.5% to 51.7%, while the weighted average costs them only about 1.2 percentage points. This difference explains why broad-based weighted average has become the market norm in most healthy financings.

13.4.3 Convertible notes and SAFEs

Before a priced round, many start-ups raise money with **convertible notes** (debt that converts to equity at the next priced round) or **simple agreements for future equity (SAFEs)**, which are not debt and carry no interest or maturity. Both usually convert at a discount to the next round price (e.g., 20%) or at a **valuation cap**, whichever gives the holder more shares. They defer the valuation negotiation, but the deferral is not free: if founders raise several uncapped or capped instruments before a priced round, dilution accumulates invisibly until conversion. Model the post-conversion cap table before signing a second or third instrument.

13.4.4 Managerial rules of thumb

- Model dilution on a fully diluted basis, including the pool, convertible instruments and warrants.
- Plan the *sequence*: a round priced too aggressively today can become a down round tomorrow if growth disappoints, with cascading anti-dilution effects.
- Keep enough ownership among founders and key employees to sustain incentives through the exit; if the cap table becomes top-heavy with preferred shares carrying large preferences, common stock may be worth little in moderate outcomes and incentives collapse (see Table 13.2).

13.5 The Initial Public Offering

An IPO is the first sale of a company's shares to the public, accompanied by the listing of those shares on an exchange. It is a financing event, a liquidity event for early investors and employees, and a governance event that subjects the firm to continuous disclosure.

13.5.1 Why go public, and why not

Reasons to go public.

- *Access to capital.* Public markets offer a deep pool of investors for large equity raises, now and later through seasoned offerings.
- *Liquidity and diversification for existing owners.* Venture funds must return capital to LPs; founders and employees can sell some shares over time.
- *Currency for acquisitions and employee compensation.* Listed shares have an observable price and are accepted in deals.
- *Visibility and credibility.* Public listing can help with customers, suppliers and lenders.
- *Valuation.* In some periods, public markets pay higher multiples than private buyers do.

Reasons to stay private.

- *Disclosure costs and competitive information.* Public firms must publish financial results and material risks, including to competitors.

- *Short-term pressure.* Quarterly earnings expectations may push managers away from long-term investment.
- *Direct and ongoing costs.* Legal, audit, listing, investor-relations and compliance costs are significant; directors' and officers' liability insurance is higher.
- *Loss of control.* Founders may be diluted and may face activist shareholders or takeover threats, although dual-class structures are used in some markets to preserve founder control.
- *Alternatives.* Large private equity and growth funds now supply late-stage capital at scale, so the capital-access argument for listing is weaker than it used to be.

13.5.2 The IPO process

An IPO typically takes six to twelve months of preparation. The main steps are:

1. **Readiness.** Audited financial statements for the required number of years (prepared under the accounting standards of the listing market), a public-company-grade finance function and controls, a board with independent directors and audit committee, and clean legal and capitalization records.
2. **Selecting underwriters.** The firm chooses one or more lead banks (bookrunners) after a "bake-off" in which banks pitch valuation, distribution strength and research coverage. A syndicate of other banks joins to widen distribution.
3. **Due diligence and the prospectus.** Underwriters and lawyers investigate the company and prepare the registration statement and prospectus, which the securities regulator reviews. The document must disclose material risks; misstatements create liability for the issuer, directors and underwriters.
4. **Marketing: the roadshow.** Management presents to institutional investors, often over one to two weeks. A preliminary prospectus may include an indicative price range.
5. **Book-building.** Investors submit non-binding indications of interest, with quantities and price limits. The bookrunners assemble a demand curve. If the book is heavily oversubscribed, the range may be raised; if weak, lowered or the deal postponed.
6. **Pricing and allocation.** On the evening before trading begins, the issuer and bookrunners set the final offer price and allocate shares. Allocation is discretionary: the banks favor long-term, price-sensitive institutions that supply information during book-building, which is a way of compensating them for revealing demand.
7. **Stabilization and the over-allotment option.** After listing, the lead underwriter supports the price if it falls (see below).
8. **Quiet period, analyst coverage and lock-up expiry.** Research coverage begins after a quiet period. Pre-IPO holders are generally barred from selling (a **lock-up**, typically around 90 to 180 days) so that a flood of insider sales does not hit the market immediately.

13.5.3 Underwriting arrangements

In a **firm commitment** underwriting, the banks buy the shares from the issuer at the offer price less the **gross spread** and resell to investors; they bear the risk of unsold shares. In a **best efforts** offering, the bank agrees only to try to sell, and the issuer bears the risk; this arrangement is more common for small, risky issues. The gross spread is the underwriters' compensation, often in the range of 5% to 7% for mid-sized U.S. IPOs and lower for very large offerings, and is shared among the management, underwriting and selling roles in the syndicate.

The **over-allotment option** (the "greenshoe", named after the company that first used it) lets underwriters sell up to about 15% more shares than the base deal and, if demand is strong, buy those shares from the issuer at the offer price to cover the short position. If the price falls below the offer, underwriters instead cover the short by buying in the open market, which supports the price. The mechanism smooths early trading.

13.5.4 Valuing the issuer and setting the price

Bookrunners mainly use **comparable-company multiples** (enterprise value to EBITDA or revenue, price to earnings), supplemented by discounted cash flow and, for recent issues, an examination of how comparable IPOs have traded after listing. They then recommend an **IPO discount** (often 10% to 20%) to the "fair" value, to attract investors, to compensate them for risk and for information acquired, and to leave momentum in the stock.

Example 13.6 – Valuing and pricing an IPO. Corvane Logistics has projected next-year EBITDA of $120 million. Listed comparables trade at 12.0 times EBITDA. Corvane has net debt of $150 million and 90 million shares outstanding before the IPO. It wants to raise $300 million of primary capital (new shares), and bankers recommend a 15% IPO discount.

Step 1: Enterprise value = 12.0 × $120M = $1,440M. Step 2: Pre-IPO equity value at "fair" market value = $1,440M − $150M = $1,290M. Step 3: Apply the 15% discount: $1,290M × 0.85 = $1,096.5M. Step 4: Offer price = $1,096.5M / 90M shares = $12.18 per share. Step 5: New shares to raise $300M gross: $300M / $12.18 = 24.62 million shares. Step 6: Post-IPO shares = 90M + 24.62M = 114.62M; new investors own 24.62 / 114.62 = 21.48%.

Existing shareholders have effectively paid for the discount: at the fair value of $1,290M the shares would be worth $14.33 each ($1,290M / 90M), and the discount has set the price $2.15 lower. After the IPO, if the market in fact values the company at the comparable multiple, the post-IPO equity value will be $1,290M + $300M (cash received, ignoring fees) = $1,590M, or $13.87 per share on 114.62M shares. The stock would rise 13.9% above the offer price, a gain for the new investors that is funded by the existing owners. Whether such a transfer is necessary is the subject of the next section.

Note the use of *pre-money* and *post-money* in the IPO context: the offer price times pre-IPO shares is the pre-money equity value; adding the primary proceeds gives the post-money value.

13.6 IPO Underpricing and the Total Cost of Going Public

13.6.1 The empirical regularity

Across markets and eras, IPO shares have, on average, closed their first day of trading above the offer price. This **underpricing**, measured as `(first-day close − offer price) / offer price`, can be large: averages in the range of 10% to 20% are common in developed markets, with much larger numbers in some periods and in many emerging markets, and substantial variation across deals. Underpricing is a real cost to the issuer, because the shares were sold for less than the market was willing to pay immediately afterward. The money "left on the table" equals `(first-day close − offer price) × shares sold`.

13.6.2 Theories of underpricing

No single theory explains the phenomenon, and the following are complementary.

- **Winner's curse (Rock).** Informed investors bid only for good issues, while uninformed investors receive a larger share of the bad ones (and a share of the good ones that is rationed down). To keep uninformed investors in the market, issuers must underprice on average so that those investors earn a normal return despite the adverse selection.

- **Information revelation or book-building (Benveniste and Spindt).** Underwriters want institutions to reveal their true demand. Deliberate underpricing and preferential allocation compensate those who reveal positive information, so that the issue price reflects it.
- **Signaling.** High-quality firms may underprice deliberately to signal quality and recoup the loss in later seasoned offerings at better prices. Evidence for this is mixed.
- **Insurance against legal liability.** A lower price reduces the likelihood of a post-listing price fall that would trigger lawsuits.
- **Agency and bargaining.** Underwriters may prefer underpricing to ease selling and to reward favored clients; issuers whose managers hold few shares at stake may not bargain hard. Evidence suggests that issuers who realize the extent of underpricing may be more willing to negotiate prices in later deals.
- **Behavioral explanations.** Retail enthusiasm and sentiment raise first-day prices above fundamental value for hot issues.

13.6.3 Beyond underpricing: the other costs

The full cost of an IPO has several components: the underwriters' gross spread; direct costs such as legal, accounting, printing, listing and roadshow expenses; underpricing; and the indirect costs of ongoing compliance and disclosure. In addition, there is **long-run underperformance**: studies of IPOs have often found that, on average, IPO firms have earned below-benchmark returns over three to five years, although the magnitude depends on methodology.

Example 13.7 – The true cost of an IPO. Hartwick Biologics sells 10 million new shares at an offer price of $20. The gross spread is 7%. Other direct costs (legal, audit, listing, printing) are $3.0 million. The shares close the first day at $26.

Step 1: Gross proceeds = 10,000,000 × $20 = $200 million. Step 2: Underwriting spread = 7% × $200M = $14.0 million. Net proceeds before other costs = $186.0 million. Step 3: Net proceeds after other direct costs = $186.0M − $3.0M = $183.0 million. Step 4: Underpricing = ($26 − $20) / $20 = 30%. Step 5: Money left on the table = ($26 − $20) × 10,000,000 = $60.0 million. Step 6: Total cost = $14.0M + $3.0M + $60.0M = $77.0 million, or 38.5% of gross proceeds.

Of this cost, only 8.5% of gross proceeds (the $14.0M spread plus $3.0M expenses = $17.0M) appears on any invoice. The economic cost to existing shareholders is much larger and mostly invisible in the accounts. This also holds for the greenshoe: with strong demand, underwriters exercise the option for 1.5 million additional shares at $20 less spread, giving the company further proceeds of 1,500,000 × $20 × 0.93 = $27.9 million, but those shares were also sold at a price below their day-one value (a further $9.0 million of "underpricing" for the shares issued under the option).

Caveat. The figure above treats the full difference between the first-day close and the offer price as a cost. If part of that jump reflects a true post-listing improvement in value, perhaps because the market resolves uncertainty, the economic cost is smaller than the headline. A common-sense benchmark is whether the issuer could have obtained the close price for those shares at all. Because a higher offer price would have reduced demand, the "left on the table" calculation overstates the true loss, but it remains the standard measure.

13.6.4 Alternatives to the traditional IPO

- **Direct listing.** The company lists existing shares without issuing new ones; it raises no primary capital, and an opening auction sets the reference price.
- **Auction IPOs.** Investors bid; the clearing price sets the offer price. This reduces discretion in allocation and often underpricing, but is rarely used.

- **Special purpose acquisition companies (SPACs).** A shell with cash raised in its own IPO merges with a private firm, giving it a listing. Costs include sponsor promote and warrants that dilute the target's holders; results for investors have been mixed.
- **Dual-track processes.** The company prepares simultaneously for an IPO and a sale, and chooses the better outcome.

13.6.5 IPO timing and market cycles

IPO volume moves in waves. Firms rush to list when valuations are high ("hot issue" markets) and are absent when markets are weak. Issuers may be rationally exploiting windows of opportunity, and managers also may be timing the market. The managerial implication is to keep the company IPO-ready, with audited statements, controls and governance, so that the board can decide in a favorable window, but never to base the business plan on the assumption that a window will open.

13.7 Seasoned Equity Offerings

A **seasoned equity offering (SEO)**, or follow-on, is an issue of additional shares by a firm that is already public. It can be a **primary** offering (the company sells new shares and receives the proceeds), a **secondary** offering (existing holders sell and the company receives nothing), or a combination. SEOs may be marketed in a full roadshow or done quickly as an overnight **bought deal** or **accelerated bookbuild**, in which a bank buys the whole block at a fixed price and then resells it.

13.7.1 Costs and price effects

Typical direct costs are lower than for IPOs, as the company's information is already in the public domain: spreads of perhaps 3% to 5%. More significant is the **offer discount**: SEO shares are usually sold at a discount to the last close, often in the range of 2% to 5%. And the **announcement effect**: on average, share prices fall when a firm announces a seasoned equity offering, commonly in the range of 2% to 3% in studies of U.S. industrial firms, with smaller effects for utilities and larger for firms with greater information asymmetry.

The standard explanation comes from Myers and Majluf. Managers know more than outside investors about the firm's true value. If managers act in the interest of existing shareholders, they issue equity when they believe it is overvalued and avoid issuing it when it is undervalued. Rational investors infer from the announcement that the shares may be overvalued, and the price falls. This reasoning also produces the **pecking order** of financing: internal funds first, then debt, and equity only as a last resort. Other reasons for negative announcement effects include the reduction in leverage and the loss of control, and the revelation that the firm's investment opportunities are less valuable than thought.

Example 13.8 – Economics of a seasoned offering. Brightwater Energy has 80 million shares at $50, a market capitalization of $4.0 billion. It issues 8 million new shares (10% of the existing count) at an offer price of $48.50 (a 3% discount to the last close). Underwriting and other costs are 4% of gross proceeds. Announcement of the offer causes the market price to fall 2% before pricing, to $49.00.

Step 1: Gross proceeds = 8,000,000 × $48.50 = $388.0 million. Step 2: Costs = 4% × $388.0M = $15.52 million. Net proceeds = $372.48 million. Step 3: Market value of existing equity after announcement = 80,000,000 × $49.00 = $3,920 million. (Loss for existing holders from the announcement effect = 80M × $1.00 = $80 million.) Step 4: Post-issue market value = $3,920M + $372.48M = $4,292.48 million, assuming the proceeds are worth their face value (that is, no extra value from the project). Step 5: Post-issue shares = 88,000,000. Value per share = $4,292.48M / 88M = $48.78.

Existing shareholders therefore own shares worth $48.78 versus $50.00 before the announcement: a loss of $1.22 per share (2.4%), of which $1.00 is the announcement effect and $0.22 results from selling shares at a discount and paying fees. Now consider the new project: if the $372.48 million is invested at an NPV of, say, $60 million, post-issue value rises to $4,352.48M and value per share to $49.46, which shows why a negative announcement effect can be offset by a clearly communicated, credible value-adding use of proceeds.

13.7.2 Managerial guidance for SEOs

- Issue after good news, not to fix a problem: the market discounts issues by firms that need cash urgently.
- Communicate the use of proceeds with specifics.
- Consider alternatives: debt, internal cash, asset sales, or a convertible bond, which may carry a smaller announcement effect.
- For a large issue relative to existing market capitalization, consider a rights issue (below) or a pre-marketed structure to limit the discount.
- Time the issue relative to scheduled earnings and the information the market is waiting for; issuing before an announcement can be exploited by arbitrageurs and can raise legal risk.

13.8 Rights Issues

A **rights issue** offers new shares first to existing shareholders in proportion to their holdings, usually at a discount to the market price. It is the dominant form of seasoned equity issuance in the United Kingdom, much of continental Europe and many Asian markets, and relatively rare in the United States. Because existing shareholders are given the first right to buy, they can protect their percentage ownership: those who do not wish to invest can sell their rights.

13.8.1 Mechanics

Each existing share receives one right. A ratio such as “1 for 5” means that five rights entitle the holder to buy one new share at the **subscription price**. Rights are typically tradable during the offer period. Because the subscription price is set below the market price, the rights have positive value, and after the ex-rights date, the share price drops to reflect the dilution.

```
Theoretical ex-rights price (TERP) = (N × P_cum + K) / (N + 1)
Value of one right = P_cum − TERP
```

where N is the number of existing shares needed per new share (5 in a 1-for-5 issue), P_cum is the cum-rights share price (before the ex-date) and K is the subscription price. Equivalently, the value of a right is `(TERP − K) / N`.

Example 13.9 – A rights issue. Solana Retail has 50 million shares priced at $40 (market capitalization $2,000 million). It announces a 1-for-5 rights issue at a subscription price of $32.

Step 1: New shares = 50,000,000 / 5 = 10,000,000. Proceeds = 10,000,000 × $32 = $320 million. Step 2: TERP = (5 × $40 + $32) / 6 = $232 / 6 = $38.667. Step 3: Value of a right = $40 − $38.667 = $1.333. Check: ($38.667 − $32) / 5 = $1.333. Step 4: Check market capitalization after the issue: 60,000,000 × $38.667 = $2,320 million = $2,000M + $320M.

Shareholder A takes up the rights. She holds 1,000 shares ($40,000). She receives 1,000 rights and needs 5 per new share, so she can subscribe for 200 new shares by paying 200 × $32 = $6,400. She then holds 1,200 shares worth 1,200 × $38.667 = $46,400, which equals her original $40,000 plus the $6,400 invested. She is exactly as well off as before.

Shareholder B sells the rights. He keeps 1,000 shares worth $38,667 and sells 1,000 rights at $1.333 for $1,333. Total = $40,000. He is also unharmed, but his ownership percentage has fallen from 1,000/50M to 1,000/60M.

Shareholder C ignores the rights. She holds 1,000 shares at $38,667 and lets the rights lapse, losing $1,333 (3.3%). This is why issuers usually arrange for unexercised rights to be sold on behalf of the holders (and the proceeds paid out), and why the process is explicit about deadlines.

13.8.2 The subscription price and the logic of a deep discount

A common puzzle is why issuers use deep discounts (10% to 40% below the market price) in rights issues, when the discount in a fully subscribed issue is not a cost: the transfer goes from shareholders to themselves. The answer is that the discount makes full take-up likely: it reduces the chance that the market price falls below the subscription price during the offer period, in which case no one would subscribe and underwriters would have to buy the shares. The real cost to a holder arises only when she cannot or will not subscribe or sell her rights. With a deeper discount, the issuer raises the same amount by issuing more shares, but TERP is lower and the value of each right is larger, leaving existing holders' wealth unchanged. In other words, the **subscription price does not change shareholder wealth** in a fully subscribed issue; it only changes the number of shares. Underwriting costs, however, are higher for narrower discounts because the underwriter bears more risk of an unsold issue.

13.8.3 Rights issues versus underwritten public offers

Rights issues protect existing shareholders against ownership dilution and wealth transfer, and can reduce the cost of selling very large amounts of equity relative to market capitalization. On the other hand, they take longer to execute (several weeks), expose the issuer to market movements during the offer period, and demand administrative effort. In markets where pre-emption rights are statutory or customary, issuing shares for cash to outsiders without first offering them to existing holders requires shareholder approval, and the rights issue is the default. In markets lacking such customs, firms prefer the speed of the bought deal.

Table 13.4 – Comparison of equity issuance methods (illustrative ranges; actual values vary by market and period)

Feature	IPO	SEO (underwritten public offer)	Rights issue
Information asymmetry	High	Moderate	Moderate
Typical discount	10%–20% to fair value	2%–5% to last close	10%–40% to market (no wealth effect if fully taken up)
Direct fees (typical)	5%–7% spread plus expenses	3%–5%	Lower, mainly for standby underwriting
Speed	Months	Days to weeks	Weeks
Ownership dilution	New investor base	Existing holders diluted unless they buy	Holders can maintain share

Case Study: Northgate Biosystems Chooses Its Next Round

Illustrative case; Northgate Biosystems and all figures are fictional.

Northgate Biosystems develops diagnostic instruments for hospital laboratories. Founded eight years ago by Dr. Anjali Rao (CEO) and Tomas Lindqvist (CTO), it has revenue of $55 million, growing 35% a year, and an operating loss of $6 million as it invests in a second product line. It holds $9 million in cash and burns about $1 million a month, so it has roughly nine

months of runway.

The cap table before any new financing shows 5.0 million founder shares (50%), 2.5 million shares held by early venture investors from a Series A and B (25%, total invested $18 million, 1x non-participating preference), 1.5 million shares held by employees and an existing option pool (15%) and 1.0 million shares of unallocated pool (10%). Total fully diluted: 10.0 million shares.

Two offers are on the table.

Offer 1: Series C from Meridian Growth Partners. $40 million at a $120 million pre-money valuation, with these terms: a 1x participating preferred with a 2.5x cap, senior to earlier preferred; a refreshed unallocated option pool equal to 12% of post-money, counted in the pre-money; broad-based weighted-average anti-dilution; one board seat with a veto over any sale below $300 million; and exclusivity for 60 days. Meridian says the company can raise a further round within 24 months if it meets revenue milestones.

Offer 2: IPO preparation. Bankers say the company could list in about nine months if it audits two more years under public-company standards and shows a path to profit. They indicate a value range of $200 million to $260 million pre-money, a base deal of $70 million primary and $15 million secondary, a gross spread of 6.5%, other costs of about $4 million, and an expected IPO discount of 15%. The 15-month timeline exceeds Northgate's cash runway, so it would need a $12 million bridge, which is available from existing investors as a convertible note at a 20% discount to the IPO price.

Dr. Rao favors the IPO because the offer would give her currency for acquisitions and public visibility. Lindqvist worries about quarterly scrutiny and fears that the second product line, which will be loss-making for two more years, will be punished by public investors. The Series A and B funds, nine and seven years into their holdings, have strong preferences for an exit within 24 months, and one has hinted it may not offer new money in a private round.

The board must decide within three weeks. Meridian's lawyers note that the term sheet's valuation is "headline" and invite Northgate to compare its effective pre-money after the pool. A banker adds that if the IPO market weakens, Northgate could find itself forced into a down round or an unfavorable sale. A third alternative, raising a smaller $20 million round from existing investors while cutting the burn, has been raised but not priced.

Discussion questions

1. Calculate Northgate's effective pre-money valuation to existing holders under Offer 1, taking into account the option pool required by Meridian. What price per share does Meridian pay, and how many shares does it receive? (Hint: consider the 12% post-money pool as a share of the total, and the unallocated pool that already exists.)
2. Compare the common stockholders' proceeds under Offer 1 at exit values of $150 million, $300 million and $500 million, using Table 13.2 as a template. At which exits does the participation feature matter most, and what does this imply about the founders' incentives?
3. Estimate the all-in cost of the IPO route at the midpoint of the range, including the spread, other costs and IPO discount. How much dilution would the bridge note create, and why might the existing investors prefer the IPO even if the cost is higher?
4. What does each path imply for control, information disclosure, strategic flexibility and risk if markets weaken? Which would you recommend, and what negotiating asks would you make on Offer 1?

Key Takeaways

- Equity financing proceeds in stages, from founders and angels through venture rounds to an IPO and seasoned offerings; each stage reflects a different mix of risk, information and investor expectations.
- Staged venture funding is a response to information asymmetry and moral hazard: capital is released as milestones are met, and each round's price carries information.
- The venture capital method works backward from a projected exit value, a target return and expected future dilution to a required ownership stake and valuation; high target returns largely compensate for the probability of failure.
- Headline valuations mislead. Compare offers by effective pre-money after the option pool, and model founders' proceeds across exit values given the liquidation preference, participation and cap.
- Dilution is not the enemy; low price per share relative to value is. Ownership percentages multiply across rounds, and value per share is what matters to existing holders.
- Down rounds trigger anti-dilution adjustments: full ratchet moves much more value to earlier investors than broad-based weighted average, at the expense of common holders and the new investors' stake.
- An IPO is priced by banks using comparables and book-building, with an intended discount; the full cost includes the spread, other expenses and underpricing, and typically far exceeds the visible fees.
- Theories of underpricing (winner's curse, information revelation, signaling, liability insurance, behavioral demand) are complementary rather than competing.
- Seasoned offerings typically depress share prices on announcement because of adverse selection (Myers and Majluf); the response is consistent with the pecking order.
- In a fully subscribed rights issue, the subscription price does not change shareholder wealth; the theoretical ex-rights price and the value of a right follow from simple averaging, and holders who neither subscribe nor sell their rights lose value.

Review Questions

1. Explain why venture capital is provided in stages rather than in a single large investment. Which problems does staging mitigate?
2. Why are venture capital target returns of 40% to 60% consistent with a much lower expected return on a diversified portfolio of ventures?
3. Distinguish between pre-money and post-money valuation. Why does an option pool included in the pre-money reduce the effective valuation received by existing holders?
4. Compare non-participating, participating and capped participating liquidation preferences. In which circumstances does each most favor the investor over common holders?
5. What is a down round and what does anti-dilution protection do? Why is the full ratchet considered more aggressive than broad-based weighted average?
6. Why might a company that could remain private choose to go public? Why might it choose not to?
7. Describe the book-building process. How does it help set the offer price, and how are allocation decisions linked to information production?
8. List three theories of IPO underpricing and describe the evidence one would look for in each.

9. Why do equity issues by listed firms tend to be accompanied by a fall in the share price? What does this imply for the timing and sizing of issues?
10. In a rights issue, why is the subscription price largely irrelevant to existing shareholders' wealth if all rights are exercised or sold? Who loses if rights lapse?

Problems

1. **Venture capital method.** A fund invests $3 million and expects the company to be worth $90 million in four years. Its target return is 40% per year, and it expects 15% dilution from later rounds (retention 0.85). Compute the required ownership today, post-money valuation and pre-money valuation.
2. **Option pool.** Cobalt Labs has 4,000,000 founder shares. An investor offers $5 million at a $20 million pre-money valuation and requires an unallocated pool of 12% of the post-money fully diluted shares. Compute the price per share, the number of new investor shares and the effective pre-money valuation to the founders.
3. **Liquidation preference.** An investor puts in $15 million for 30% of the equity with a 1x preference. For an exit of $40 million, compute the proceeds to the investor and to common holders if the preference is (a) non-participating and (b) fully participating. At what exit value would the non-participating investor convert to common?
4. **Dilution.** A company's founders own 100% initially. A seed round sells 20% of the company, a Series A sells 20% and a Series B sells 15% (each percentage is of the post-money company). What percentage do the founders own after all three rounds? What is the percentage held by seed investors after the Series B?
5. **Weighted-average anti-dilution.** Series A preferred was bought for $4 million at a conversion price of $5.00 (800,000 shares). The company has 10,000,000 fully diluted shares. It then issues new shares at $2.50 to raise $3 million. Compute the new conversion price under broad-based weighted average and the number of common shares into which the Series A now converts.
6. **IPO cost.** A firm sells 5 million shares at $16, with a gross spread of 6% and other direct costs of $1.8 million. The first-day close is $21. Calculate net proceeds, underpricing, money left on the table and the total cost as a percentage of gross proceeds.
7. **Rights issue.** A firm with 40 million shares priced at $25 announces a 1-for-4 rights issue at $20. Compute proceeds, the theoretical ex-rights price and the value of a right. A shareholder with 2,000 shares takes up her rights in full: show that her wealth is unchanged.
8. **SEO.** A firm has 30 million shares at $60. It sells 3 million new shares at $57, with costs equal to 3.5% of gross proceeds. The announcement causes the share price to fall by 2.5% before the offer is priced (to $58.50). Compute the net proceeds and the post-issue value per share, assuming the proceeds are worth their face value.

Answers to Selected Problems

1. Required multiple = 1.4^4 = 3.8416; required exit value of the investment = $11.52 million; ownership at exit = 11.52 / 90 = 12.81%; ownership today = 12.81% / 0.85 = 15.07%. Post-money = 3 / 0.1507 = $19.91 million; pre-money = $16.91 million.
2. Post-money = $25 million; investor 20%, pool 12%, founders 68%. Total shares = 4,000,000 / 0.68 = 5,882,353; investor shares = 1,176,471; price = $5,000,000 / 1,176,471 = $4.25. Effective pre-money = $4.25 × 4,000,000 = $17.0 million (versus $20 million headline).

3. a. Non-participating: max($15M, 30% × $40M = $12M) = $15M to the investor; $25M to common. (b) Participating: $15M + 30% × $25M = $22.5M; common receives $17.5M. Conversion for non-participating holder occurs when 30% × exit > $15M, i.e., exit > $50M.
4. Founders: 0.80 × 0.80 × 0.85 = 54.4%. Seed investors: 20% × 0.80 × 0.85 = 13.6%.
5. A = 10,000,000; B = $3,000,000 / $5.00 = 600,000; C = $3,000,000 / $2.50 = 1,200,000. NCP = $5.00 × (10,000,000 + 600,000) / (10,000,000 + 1,200,000) = $4.732. Series A converts into $4,000,000 / $4.732 = 845,283 shares (an increase of 45,283).
6. Gross = $80 million; spread = $4.8 million; net proceeds = $80M − $4.8M − $1.8M = $73.4 million. Underpricing = (21 − 16)/16 = 31.25%. Money left on the table = $5 × 5,000,000 = $25.0 million. Total cost = $4.8M + $1.8M + $25.0M = $31.6 million = 39.5% of gross proceeds.
7. New shares = 10 million; proceeds = $200 million. TERP = (4 × 25 + 20) / 5 = $24.00. Value of a right = $25 − $24 = $1.00. The shareholder holds 2,000 rights, subscribes for 500 shares at $20 = $10,000; she then holds 2,500 shares worth $60,000 = original $50,000 + $10,000 invested; wealth unchanged.
8. Gross = $171.0 million; net = $171.0M × 0.965 = $165.015 million. Post-announcement market value of existing equity = 30M × $58.50 = $1,755M. Post-issue value = $1,755M + $165.015M = $1,920.015M over 33 million shares = $58.18 per share (a fall of $1.82, or 3.0%, from $60).

Chapter 14: Debt Financing

Learning Objectives

After studying this chapter, you should be able to:

- Describe the main features of debt contracts (seniority, security, maturity, rate basis, amortization) and choose among bank loans, bonds, leases, convertibles and securitized funding for a given financing need.
- Compute loan payment schedules and bond prices, yields and interest-rate sensitivity (duration), and calculate the after-tax cost of debt.
- Explain the purpose of financial covenants, calculate covenant headroom and assess how covenants shift control and risk between borrower and lender.
- Evaluate a lease-versus-buy decision using the net advantage to leasing, and explain the effect of IFRS 16 on the lessee’s balance sheet.
- Value a convertible bond as straight debt plus an equity option, and explain why firms issue convertibles.
- Perform a basic credit analysis (ratios, expected loss, a Z-score) and estimate debt capacity from leverage, coverage and cash-flow constraints.
- Describe how securitization transforms a pool of receivables into tranched securities, and how credit enhancement allocates losses.

Opening Vignette

Illustrative scenario; the company and all figures are fictional.

Kestrel Aggregates is a regional producer of crushed stone and ready-mix concrete. It has EBITDA of $120 million, a fleet of trucks and plants worth more than $600 million at cost and a reputation for steady if unspectacular performance. Its chief financial officer, Imani Okafor, must fund a $180 million expansion: a new quarry, a rail spur and a fleet of delivery trucks.

Her treasury team lists the options. The firm’s relationship banks will provide a five-year term loan at a floating rate over the interbank benchmark, with a leverage covenant. The bond market would accept a ten-year senior note at a fixed coupon, but the issue must be large enough to be tradable, and a rating agency has hinted that the firm’s current leverage places it at the borderline of investment grade. Equipment dealers offer lease financing for the trucks on terms that appear cheap but would add to reported liabilities. A convertible note could reduce the coupon but would add potential dilution. A receivables securitization might release working capital but depends on the quality of customer invoices.

Okafor also knows that more debt is not always better. A downturn in construction would cut volumes quickly because the firm has high fixed costs. If the debt service were too heavy, a covenant breach could hand control of decisions to lenders at the worst moment.

Her challenge is to select the mix of instruments, maturities and covenants that gives the company enough capital at a reasonable cost, preserves flexibility, and matches obligations to the cash flows of the assets they finance. In this chapter we build the toolkit: how each instrument works and is priced, how lenders think about credit, and how a firm estimates how much debt it can responsibly carry.

14.1 The Landscape of Debt Financing

Debt is a contractual claim: the borrower promises to pay interest and repay principal on specified dates, and lenders have legal remedies if it does not. Unlike equity, debt does not share in upside, interest is generally tax deductible, and default can transfer control to creditors. Chapter-level questions about the optimal mix of debt and equity are treated elsewhere in the capital-structure material; here we concentrate on how debt is raised, priced and managed.

14.1.1 Dimensions of a debt contract

Every debt instrument is a bundle of choices on several dimensions.

- **Maturity.** Short-term debt (under a year) funds working capital; medium-term (three to seven years) funds equipment and projects; long-term (ten years or more) funds infrastructure. A general principle is **matching**: finance an asset with liabilities whose repayment profile matches the asset's cash flows, so the firm is not forced to refinance in a bad market.
- **Rate basis.** *Fixed-rate* debt provides certainty of interest cost but exposes the borrower to opportunity cost if rates fall. *Floating-rate* debt pays a benchmark rate plus a margin, so interest cost follows the market; it suits borrowers whose revenues move with rates, or who hedge with swaps.
- **Currency.** Borrowing in the currency of the firm's cash flows avoids exchange risk. Borrowing in a cheaper foreign currency is risky unless hedged.
- **Seniority and security.** *Senior* debt is paid before *subordinated* (junior) debt in bankruptcy. *Secured* debt has a claim on specific collateral. *Unsecured* debt ranks equally with other general creditors. More senior or better collateralized debt has lower expected loss and so a lower required yield.
- **Amortization.** *Bullet* debt repays all principal at maturity; *amortizing* debt repays principal gradually; a *sinking fund* provision requires scheduled retirements.
- **Optionality.** Call provisions let the issuer repay early; put provisions let the lender demand repayment; conversion rights allow exchange into equity. These features are options and are priced into the coupon.
- **Covenants.** Promises restricting the borrower's actions or requiring maintenance of financial ratios (Section 14.2).

14.1.2 Sources of debt compared

Table 14.1 – Principal debt instruments

Instrument	Typical borrower	Typical maturity	Rate	Main strengths	Main drawbacks
Bank term loan	Small to large firms	3–7 years	Floating (often)	Flexible, negotiable, fast	Covenants, refinancing risk, bank capacity
Revolving credit facility	Most corporates	3–5 years	Floating	Liquidity on demand; pay only on use	Commitment fees; covenants
Syndicated loan	Large firms	3–7 years	Floating	Large amounts, relationship banks	Syndication complexity
Public bond	Large, rated issuers	5–30 years	Fixed (often)	Long tenor, larger sums, looser covenants	Fixed costs, disclosure, rating dependence
Private placement	Mid-sized firms	7–15 years	Fixed	Tailored terms, no public rating needed	Less liquid; negotiated covenants
Lease	Asset-intensive firms	Life of asset	Implicit rate	Asset-linked financing, tax benefits	Obligation on balance sheet; residual risk
Convertible bond	Growth firms	5–10 years	Low coupon	Cheaper coupon; delayed equity	Potential dilution; complex valuation
Securitization	Firms with large receivable or loan pools	Matches pool	Tranche-specific	Funding based on assets rather than corporate credit	Complexity; reputational and legal risk

14.1.3 The cost of debt

The pre-tax cost of debt is the **yield to maturity** on the firm's debt, not the coupon on past borrowings. It comprises a risk-free rate for the maturity, a **credit spread** for expected default loss and credit risk premium, and liquidity and term premia. Because interest is deductible in most tax systems, the after-tax cost is:

```
After-tax cost of debt = Pre-tax yield × (1 − Tax rate)
```

For example, a firm that borrows at 6.5% and pays tax at 25% has an after-tax cost of 6.5% × 0.75 = 4.875%. This formula assumes the firm has taxable income against which to deduct interest. A firm with losses or large non-debt tax shields gets less or no benefit. Moreover, many jurisdictions limit deductibility of net interest to a share of earnings. The tax advantage should therefore be applied cautiously for highly leveraged borrowers.

Do not confuse the cost of debt with the cost of *capital*. Debt is cheaper than equity because it is senior, but adding debt makes the remaining equity riskier and raises its required return; a firm that substitutes cheap debt for equity does not necessarily lower its overall cost of capital until the expected costs of financial distress are considered.

14.2 Bank Loans and Covenants

14.2.1 Types of bank credit

Term loans advance a fixed sum, repaid on a schedule. **Revolving credit facilities (revolvers)** let a borrower draw, repay and redraw up to a commitment limit during the facility's life. Borrowers pay interest on drawn amounts and a **commitment fee** (often a fraction of a percent per year) on undrawn amounts. Revolvers are the primary liquidity backstop

for most corporates; their pricing often moves with the borrower's credit rating or leverage through a **pricing grid**. **Syndicated loans** are arranged by lead banks and funded by a group of lenders, allowing amounts larger than any one bank's appetite. Term Loan B-style tranches in leveraged finance are sold to institutional investors and typically carry minimal amortization and lighter maintenance covenants.

Asset-based lending lends against receivables and inventory with a **borrowing base** (for instance, 85% of eligible receivables plus 50% of eligible inventory). Capacity moves with the collateral, which makes it suitable for working-capital-heavy firms.

Project finance lends to a special-purpose project company, repaid solely from the project's cash flows with limited or no recourse to the sponsors. Lenders rely on contracts (offtake agreements, construction contracts) and the **debt service coverage ratio (DSCR)**: cash flow available for debt service divided by scheduled debt service.

14.2.2 Loan pricing

Loan pricing combines a benchmark rate plus a **margin** for credit risk, upfront and arrangement fees, commitment fees, and sometimes floors. The all-in cost should be computed as an internal rate of return on the net cash flows (proceeds net of fees against interest and principal), not read off the headline margin. A 0.75% upfront fee on a five-year loan adds roughly 0.17 percentage points to the annual cost, because the fee is received by the lender at the start while the interest is spread across the life.

14.2.3 Amortization schedules

An amortizing loan with level payments (an annuity) has payment:

```
Payment = P × r / (1 − (1 + r)^(−n))
```

where P is principal, r the periodic rate and n the number of periods. Early payments consist mostly of interest, and later payments mostly of principal.

Example 14.1 – Level-payment term loan. Kestrel borrows $10,000,000 from a bank for six years at a fixed 7.0% with equal annual payments.

Step 1: Payment = 10,000,000 × 0.07 / (1 − 1.07^(−6)) = 700,000 / (1 − 0.66634) = 700,000 / 0.33366 = $2,097,958.

Step 2: Year-by-year schedule.

Table 14.2 – Amortization schedule ($)

Year	Opening balance	Interest (7%)	Principal repaid	Payment	Closing balance
1	10,000,000	700,000	1,397,958	2,097,958	8,602,042
2	8,602,042	602,143	1,495,815	2,097,958	7,106,227
3	7,106,227	497,436	1,600,522	2,097,958	5,505,705
4	5,505,705	385,399	1,712,559	2,097,958	3,793,146
5	3,793,146	265,520	1,832,438	2,097,958	1,960,708
6	1,960,708	137,250	1,960,708	2,097,958	0

Step 3: Total interest = 6 × 2,097,958 − 10,000,000 = $2,587,748. At a 25% tax rate the first-year tax saving on interest is 25% × 700,000 = $175,000, so the first year after-tax interest cost is $525,000.

The schedule shows two features that matter to the borrower. The interest tax shield declines as the balance falls. The cash-flow burden is level, which makes forecasting easy, but the borrower must generate $2.1 million a year regardless of operating conditions, in contrast to a bullet loan, where cash flow is lighter but refinancing risk is concentrated at maturity.

14.2.4 Why covenants exist

A lender's return is capped at the contractual interest while its downside is large. After the loan is granted, shareholders have incentives to take actions that transfer value from lenders: paying large dividends, issuing senior or equally ranking debt, selling collateral, or investing in riskier projects (**asset substitution**; Jensen and Meckling). A related problem is **debt overhang** (Myers), where a heavily indebted firm passes up valuable projects because the benefits would accrue largely to creditors. Covenants are the contractual response: they restrict risky behavior, give lenders an early warning and a seat at the table before the firm is insolvent.

14.2.5 Types of covenants

- **Affirmative covenants** require the borrower to do things: provide audited statements, maintain insurance, pay taxes, comply with laws, maintain assets.
- **Negative covenants** limit actions: additional debt, liens on assets, asset disposals, mergers, restricted payments (dividends and buybacks), related-party transactions and investments.
- **Financial maintenance covenants** require ratios to be met at each test date: maximum leverage (Debt/EBITDA), minimum interest coverage (EBITDA/interest), minimum fixed-charge coverage, minimum net worth or minimum liquidity.
- **Incurrence covenants** apply only when the borrower takes a specified action, such as issuing new debt; the borrower must show that the ratio would still be met after doing so. Bonds usually carry incurrence covenants; bank loans typically use maintenance covenants.

Definitions matter greatly. "EBITDA" in a credit agreement is a defined term that may permit add-backs for restructuring costs, projected synergies or non-cash items; "Debt" may or may not include leases and letters of credit. Negotiating the definitions can be worth as much as negotiating the ratio level.

Example 14.2 – Covenant headroom. Kestrel's credit agreement requires Debt/EBITDA ≤ 4.00x and EBITDA/Interest ≥ 3.00x, tested quarterly on a rolling twelve-month basis. At the test date, Debt = $300 million, EBITDA = $80 million and interest expense = $24 million.

Step 1: Leverage = 300 / 80 = 3.75x (limit 4.00x). Coverage = 80 / 24 = 3.33x (limit 3.00x). Both tests are passed.

Step 2: EBITDA floor under the leverage test, holding debt constant: 300 / 4.00 = $75.0 million. Headroom = (80 − 75) / 80 = 6.25%.

Step 3: EBITDA floor under the coverage test, holding interest constant: 3.00 × 24 = $72.0 million. Headroom = (80 − 72) / 80 = 10.0%.

Step 4: The binding constraint is leverage, with 6.25% headroom. A drop in EBITDA of just $5 million (for instance from a wet season that halts quarrying) would trigger a breach.

Step 5: Additional borrowing capacity at constant EBITDA = 4.00 × 80 − 300 = $20 million, but borrowing it would leave zero leverage headroom, and extra interest at 8% ($1.6 million) would cut coverage headroom to 80 / 25.6 = 3.13x.

A prudent treasurer manages to an **internal target** well inside covenant levels (here, perhaps 3.0x) and models covenants under downside scenarios.

14.2.6 What happens at breach

A covenant breach is an **event of default** unless waived. In practice, lenders typically negotiate: they may grant a **waiver** or **amendment** in return for fees, a higher margin, tighter terms, additional collateral or an equity cure. The borrower's bargaining power diminishes sharply at this point, which is why covenant design is as much about control as about money. Many modern leveraged loans are "covenant-lite," with few or no maintenance tests, shifting protection toward documentation of incurrence tests and sponsors' equity cushions.

Managerial implications. (1) Negotiate covenant levels with ample headroom against a downside forecast, not the base case. (2) Seek "equity cure" rights and sensible EBITDA definitions. (3) Report covenant compliance forecasts to the board quarterly. (4) Keep relationships warm: banks are more flexible with borrowers that communicate early.

14.3 Bonds

14.3.1 Features and the issuance process

A **bond** is a tradable debt security. The issuer receives the principal at issuance and then pays a periodic **coupon** (fixed or floating) and the **face value** at maturity. Bonds may be issued domestically or in the international market (Eurobonds, issued outside the jurisdiction of the currency), and may be investment grade or high yield, based on credit ratings.

An issue involves selecting underwriters, obtaining ratings from rating agencies, drafting the offering document and the **indenture** (the contract including covenants, events of default and trustee arrangements), pricing relative to a benchmark government bond or swap rate, and distribution. Large, frequent issuers establish programs that allow rapid issuance when market windows open. Typical fees are lower than for equity, and the all-in cost includes the credit spread, underwriting fees, rating fees and legal costs.

Credit ratings summarize default likelihood and, in some cases, expected loss. Agencies rank issuers on scales from the highest-quality category down through the **investment-grade** categories to the speculative-grade, or **high-yield**, categories. The boundary matters because many institutional investors are restricted to investment-grade securities; a downgrade to high yield can force selling and sharply raise the issuer's cost of funding. Ratings are inputs, not guarantees: agencies have been late and wrong, notably in structured finance.

14.3.2 Bond pricing

A bond's price is the present value of its cash flows at the market yield for its risk and maturity:

```
Price = Σ [Coupon_t / (1 + y/m)^t] + Face / (1 + y/m)^(m×T)
```

where y is the annual yield (bond-equivalent), m the number of coupon periods per year, and T the years to maturity. When the yield equals the coupon rate, the bond trades at par; when the yield is above the coupon, it trades at a discount; when below, at a premium.

Example 14.3 – Pricing a bond and measuring interest-rate risk. Kestrel issues a five-year bond with a face value of $100, a 6.0% annual coupon paid semiannually ($3.00 every six months).

Step 1: Price at yield 7.0% (semiannual rate 3.5%, 10 periods): Annuity factor = (1 − 1.035^(−10)) / 0.035 = 8.3166. PV of coupons = 3.00 × 8.3166 = 24.95. PV of face = 100 / 1.035^10 = 100 / 1.41060 = 70.89. Price = 24.95 + 70.89 = $95.84.

Step 2: At 6.0% the price is $100.00 (par). At 5.0% (semiannual 2.5%) the price is $104.38. At 8.0% (4.0% semiannual) the price is $91.89.

Step 3: Duration. The present values of the cash flows at 7.0% weighted by time (in years) give Macaulay duration of 4.377 years. Modified duration = 4.377 / (1 + 0.035) = 4.229.

Step 4: Predicted price change for a one-percentage-point yield rise from 7.0% to 8.0%: −4.229 × 0.01 × 95.84 = −$4.05, compared with the actual change 91.89 − 95.84 = −$3.95. The slight overstatement arises because price is a convex function of yield: duration is a linear approximation.

Interpretation: a bond with a duration of about 4.2 years loses about 4.2% of its price for each one-percentage-point rise in yield. Long-dated, low-coupon bonds have higher durations and therefore are much more sensitive. For a corporate treasurer, duration matters in two ways: when estimating the price at which the firm could repurchase its bonds, and when matching asset and liability sensitivity.

Table 14.3 – Price of the 6% five-year bond at different yields

Yield	Price per $100	Premium / discount
5.0%	104.38	Premium
6.0%	100.00	Par
7.0%	95.84	Discount
8.0%	91.89	Discount

14.3.3 Credit spreads and the yield curve

The yield on a corporate bond equals a benchmark government yield for the same maturity plus the **credit spread**. If the five-year government yield is 4.5% and Kestrel's bond yields 7.0%, the spread is 2.5 percentage points, or 250 basis points. Spreads widen when default risk rises or when investors demand more compensation for risk and illiquidity; they are typically higher for longer maturities, lower ratings and subordinated claims. Spreads also have a market-wide component: in recessions, spreads rise even for firms whose fundamentals are unchanged. A firm should therefore think of its borrowing cost as a combination of three things it influences differently: the base rate (not at all), the market spread level (not at all) and its individual credit quality (substantially).

14.3.4 Bond features that matter for the issuer

- **Call provisions.** An issuer may redeem a callable bond before maturity at a stated price, usually after a non-call period. Investors demand a higher coupon in exchange. Calls give the issuer an option to refinance if rates fall, but they are not free.
- **Make-whole calls.** The issuer may redeem early only by paying the present value of remaining payments at a spread over a benchmark, so the call is only rarely economic but protects investors.
- **Sinking funds.** Scheduled retirement reduces refinancing risk.
- **Floating-rate notes and inflation-linked bonds.** These change the pattern of cash flows to match revenues.
- **Green and sustainability-linked bonds.** Proceeds or coupon step-ups are tied to environmental or social targets. They broaden the investor base, but require reporting discipline and the claims should be credible.
- **Subordination and hybrids.** Junior or hybrid debt may receive partial equity credit from agencies, preserving rating headroom, in exchange for a higher coupon.

14.3.5 Public bonds versus bank loans

Bonds offer longer tenors, fixed rates, larger sizes and fewer maintenance covenants, but the issuer faces a public market with fixed costs (ratings, documentation, disclosure), an anonymous investor base that is harder to renegotiate with in distress, and early repayment frictions. Bank loans are more flexible and easier to renegotiate, and banks provide relationship services and monitoring, but they offer shorter maturities and are exposed to the banks' own balance-sheet constraints. Most large firms use both: bonds for the permanent base of funding and revolvers as liquidity backup. Because of fixed costs, there is a minimum efficient size for a public bond; mid-sized companies commonly use private placements or bank debt.

14.4 Leases

A **lease** gives the lessee the right to use an asset in return for periodic payments to the lessor, who owns it. Leasing is a form of secured borrowing: the lessor's claim is backed by the asset, which it can repossess. It is widely used for aircraft, vehicles, real estate, medical equipment and technology.

14.4.1 Why firms lease

- **Tax.** If the lessor can use depreciation and interest deductions more efficiently than the lessee (the lessee has losses or limited taxable income), leasing transfers tax benefits that can be shared through lower lease payments.
- **Asset-specific expertise and residual risk.** Lessors that specialize in an asset class know the secondary market and can bear residual-value risk more cheaply.
- **Financing at 100% of cost.** Leases may need less upfront cash than a purchase financed by a loan with a down payment.
- **Flexibility and obsolescence.** Operating leases of short duration let the lessee return the asset when its need or technology changes.
- **Covenant and capacity considerations.** Debt capacity under bank agreements may be tied to defined indebtedness; leases may fall outside it, depending on drafting.

Some reasons offered for leasing are weak. "Off-balance-sheet" financing no longer holds under the modern standards described below, and sophisticated lenders and rating agencies adjust for leases in any case.

14.4.2 Accounting: IFRS 16

Under IFRS 16, almost all leases (with limited exemptions for short-term leases of twelve months or less and low-value assets) are recognized by the lessee as a **right-of-use asset** and a **lease liability**, measured as the present value of the lease payments discounted at the rate implicit in the lease or, if that cannot be determined, the lessee's incremental borrowing rate. The income statement shows depreciation of the right-of-use asset and interest on the liability instead of a single operating lease expense. Consequently, EBITDA rises (the lease cost is replaced by depreciation and interest, both below EBITDA), reported debt rises, and the pattern of expense is front-loaded because interest is highest early in the lease. Lessors continue to classify leases as finance or operating, depending on whether substantially all risks and rewards of ownership are transferred. U.S. GAAP (ASC 842) also brings leases onto the balance sheet but retains a distinction between finance and operating leases in income-statement presentation. For credit analysis and covenant compliance, the analyst must know whether lease liabilities are included in "Debt" and whether lease costs are in "EBITDA."

14.4.3 The lease-versus-buy decision

The economic analysis compares the cash flows of leasing with those of buying the asset financed by debt. The standard approach computes the **net advantage to leasing (NAL)**, discounting after-tax cash flows at the **after-tax cost of secured debt**, because lease payments are fixed obligations of similar risk to debt.

```
NAL = Cost of asset − PV[ after-tax lease payments ] − PV[ depreciation tax shields forgone ] − PV[ after-tax residual value forgone ]
```

If NAL > 0, leasing is cheaper than buying.

Example 14.4 – Lease versus buy. Kestrel needs a $1,000,000 crusher. The asset has a five-year life, is depreciated straight line to zero for tax, and has no residual value. The tax rate is 25%, and Kestrel's pre-tax secured borrowing rate is 8.0%. A lessor offers a five-year lease at $240,000 per year, payable at the end of each year.

Step 1: After-tax discount rate = 8.0% × (1 − 0.25) = 6.0%. Five-year annuity factor at 6.0% = (1 − 1.06^(−5)) / 0.06 = 4.2124.

Step 2: After-tax lease payment = 240,000 × (1 − 0.25) = $180,000 per year. PV = 180,000 × 4.2124 = $758,225.

Step 3: Annual depreciation = 1,000,000 / 5 = $200,000; tax shield forgone = 200,000 × 0.25 = $50,000 per year. PV = 50,000 × 4.2124 = $210,618.

Step 4: NAL = 1,000,000 − 758,225 − 210,618 = **+$31,156**. Leasing is cheaper by about $31,000 in present value.

Step 5: Break-even lease payment (NAL = 0). Solve 1,000,000 − L × 0.75 × 4.2124 − 210,618 = 0 so L = 789,382 / 3.1593 = $249,860. Kestrel would be indifferent at a lease payment of about $250,000; the offer of $240,000 therefore has a modest advantage.

Step 6: IFRS 16 treatment. The lease liability is the PV of $240,000 for five years at the incremental borrowing rate of 8.0%: annuity factor 3.9927, so liability = 240,000 × 3.9927 = $958,250. The right-of-use asset is also $958,250. In year 1, interest on the liability is 8% × 958,250 = $76,660, depreciation of the right-of-use asset (straight line) is 958,250 / 5 = $191,650, and the total expense is $268,310, compared with a cash payment of $240,000. EBITDA is higher than if the payment were an operating expense by the full $240,000 (it is removed from operating costs), and reported debt increases by $958,250.

Sensitivity matters. If Kestrel expects the crusher to have a residual value of $150,000 at year five, buying keeps that value, and the PV of the forgone after-tax residual (ignoring any tax on sale for simplicity) is 150,000 / 1.06^5 = $112,089. The NAL falls to 31,156 − 112,089 = **−$80,933**, so purchase becomes preferable. The conclusion in a lease analysis often depends more on the residual-value assumption than on the financing rate.

Common mistakes. (1) Discounting after-tax lease payments at the firm's WACC rather than after-tax debt cost; leases are debt-like. (2) Ignoring the residual value. (3) Comparing lease payments to the loan payments without accounting for tax shields on depreciation and interest. (4) Overlooking maintenance, insurance and end-of-lease conditions, which can be shifted between the parties.

14.5 Convertible Bonds

A **convertible bond** is a bond that the holder may exchange for a fixed number of the issuer's shares. It combines a straight bond with a call option on the issuer's equity. Key terms are the **conversion ratio** (shares per bond), **conversion price** (face value divided by the ratio), **conversion premium** (conversion price relative to the current share price), and call

features (issuers can often force conversion once the shares trade well above the conversion price).

14.5.1 Valuation by decomposition

```
Convertible value ≈ Straight bond value + Value of the conversion option
Conversion value (parity) = Conversion ratio × Share price
Floor value = max(Straight bond value, Conversion value)
```

The straight-bond value is the PV of coupons and principal at the yield on comparable non-convertible debt of the same issuer. The convertible always trades at or above its floor. Its price is greater when the shares are volatile, the maturity is long and the conversion price is close to the share price (the option is more valuable).

Example 14.5 – Valuing and assessing a convertible. Kestrel considers issuing $100 million of ten-year convertible notes in $1,000 denominations with a 3.0% annual coupon (paid annually for simplicity) and a conversion ratio of 20 shares per note. The share price is $40. Kestrel's straight ten-year debt yields 7.0%. Kestrel has 50 million shares outstanding; the tax rate is 25%.

Step 1: Conversion price = $1,000 / 20 = $50.00. Conversion premium = 50 / 40 − 1 = 25%.

Step 2: Conversion value (parity) = 20 × $40 = $800.

Step 3: Straight bond value = 30 × (1 − 1.07^(−10)) / 0.07 + 1,000 / 1.07^10 = 30 × 7.0236 + 508.35 = 210.71 + 508.35 = $719.06 per note. (Using $1,000 / 1.96715 = 508.35.)

Step 4: If the note is sold at par ($1,000), the implied value of the conversion option is $1,000 − $719.06 = $280.94 per note, or 28.1% of the issue price. At a parity of $800 and a premium over parity of 25% ($1,000 / $800 − 1), investors are paying for the option's time value, since it is currently out of the money (parity < 1,000).

Step 5: Interest savings. The straight-debt coupon would be 7.0% × $100M = $7.0 million per year, versus $3.0 million on the convertible; the pre-tax saving is $4.0 million and after tax $3.0 million per year.

Step 6: Dilution. There are 100,000 notes. If all convert: 100,000 × 20 = 2,000,000 new shares, which is 2.0 / 52.0 = 3.85% of the enlarged share count (4.0% of existing shares). The effective issue price per share is $50, a 25% premium to today's share price, compared with the discount that would usually apply to a seasoned equity issue.

Step 7: Outcomes at maturity. If the share price is $60, the conversion value is $1,200: holders convert, and Kestrel issues 2 million shares worth $120 million for debt of $100 million. If the share price is $30, parity is $600: holders take cash, $1,000 per note, and Kestrel has paid 3% coupons on what turned out to be debt.

The "cheap coupon" is not free money. The $4.0 million annual interest saving is the price the investors pay for a free call option on the stock. In an efficient market, the cost of the convertible equals the cost of straight debt plus the cost of the option; the saving in coupon is exactly offset by the value of the equity upside given away. Managers therefore need to compare the *all-in* cost, including expected dilution, with alternatives.

14.5.2 Why issue convertibles?

- **Delayed equity ("back-door equity").** Issuers believe their shares are undervalued today and expect to sell equity at a higher price later. A conversion premium of 25% means selling shares later at $50 instead of today at $40.
- **Reduced agency costs.** Convertible holders gain if the firm takes risk (as shareholders do), which mitigates asset-substitution concerns and makes the instrument attractive when it is hard to assess the firm's risk (Green's argument).

- **Lower information-sensitivity.** Because a convertible's value depends on both debt and equity components, it is less sensitive to disagreement about firm risk or value than straight debt or equity alone, which is why it is common in young, high-growth, high-volatility firms.
- **Lower cash interest.** Helpful when cash coverage is a constraint.

The costs include the dilution if the stock does well, the potential for forced conversion to disappoint, accounting complexity, and capital markets' perception of convertibles as a signal (Section 13.7 explains the equity announcement effect). Many investors in convertibles are arbitrage funds that buy the convertible and short the stock; issuance can therefore put pressure on the share price.

14.6 Credit Analysis

Credit analysis assesses the borrower's ability and willingness to meet its obligations. The same framework is used by lenders to underwrite and price loans, by rating agencies to assign ratings, and by CFOs who want to know how outside parties see them.

14.6.1 The classic "Cs"

Lenders traditionally assess **character** (management's record and integrity), **capacity** (cash flow to service the debt), **capital** (the equity cushion), **collateral** (the assets that can be claimed) and **conditions** (industry and economic context, and the purpose of the loan). Modern analysis adds **covenants** and **cycle position**. Capacity is the dominant consideration for unsecured lending: the first source of repayment is cash flow, the second is refinancing, and the third is collateral.

14.6.2 Core credit ratios

Ratio	Formula	Question answered
Leverage	Total debt / EBITDA	How many years of earnings to repay debt?
Net leverage	(Debt − cash) / EBITDA	Same, net of liquid assets
Interest coverage	EBITDA / interest expense	How much earnings cushion over interest?
Cash coverage	(EBITDA − capex) / interest	Coverage after maintaining the asset base
FFO / debt	Funds from operations / debt	Cash generation relative to debt
Debt / capital	Debt / (debt + equity)	Equity cushion
DSCR	CFADS / (interest + scheduled principal)	Ability to cover total debt service
Liquidity	(Cash + undrawn committed lines) / near-term obligations	Ability to survive a closed market

Ratios must be adjusted: for leases, pension deficits, off-balance-sheet obligations, non-recurring items, and seasonality. Compare to peers and through the cycle, not at a single point in time.

Table 14.4 – Credit profile comparison of two illustrative firms ($ millions)

Metric	Alpha Industrial	Beta Industrial
EBITDA	200	150
Total debt	700	900
Interest expense	49	81
Capex	90	60
Funds from operations	150	80
Debt / EBITDA	3.50x	6.00x
EBITDA / interest	4.08x	1.85x
(EBITDA − capex) / interest	2.24x	1.11x
FFO / debt	21.4%	8.9%

Alpha has moderate leverage and a reasonable cushion; Beta is highly leveraged and has barely any cash coverage after capital expenditure. A lender would likely price Beta's debt many percentage points higher, if it lends at all, and would require security, tight covenants or an equity injection.

14.6.3 Expected loss and credit spreads

Credit risk can be decomposed into three components:

```
Expected loss (EL) = Probability of default (PD) × Loss given default (LGD) × Exposure at default (EAD)
LGD = 1 − Recovery rate
```

A lender needs a spread at least large enough to cover expected loss plus its cost of funding, operating costs and a risk premium for bearing unexpected losses (default is clustered with recessions, so the risk premium is not zero even for diversified portfolios).

Example 14.6 – Expected loss and spread. A bank lends $100 million to a mid-market manufacturer. Based on the borrower's rating, PD over one year is 3.0%, expected recovery is 60% (so LGD = 40%), and the exposure at default is the full $100 million.

Step 1: EL = 0.03 × 0.40 × $100M = **$1.2 million**, or 1.2% of the exposure per year.

Step 2: The cumulative probability of default over two years, assuming independent annual PD of 3%, is $1 - (1 - 0.03)^2 = 1 - 0.9409 = 5.91\%$.

Step 3: Minimum loan margin. Suppose the bank's funding cost over the benchmark is 0.40%, operating costs are 0.30%, and it wants a risk premium and a return on regulatory capital of 0.80%. The margin required is 1.2% + 0.40% + 0.30% + 0.80% = **2.70%** over the benchmark. A borrower with a PD of 6% has an EL of 2.4%, giving a margin of 3.9% if other components are the same.

Example 14.7 – Reading a bond spread as default probability. A five-year bond of an issuer trades at a spread of 2.40% over the government curve. The market's assumed recovery rate for similar senior unsecured bonds is 40%.

Approximate relation: spread ≈ annual hazard rate × LGD, so hazard ≈ 2.40% / 0.60 = **4.0% per year**. Over five years, the implied cumulative default probability ≈ $1 - 0.96^5 = 18.5\%$. This is a "risk-neutral" probability and includes risk premia and liquidity effects, so it overstates the actual default probability, which is typically well below it. Nonetheless, the calculation is a way of translating the market's price into a risk statement: a spread of 240 basis points signals meaningful default risk, and it shows how sensitive the spread is to assumed recovery (with a 20% recovery, the hazard rate would be 3.0%).

14.6.4 Scoring models: the Altman Z-score

Statistical models combine accounting ratios to predict distress. Altman's original Z-score for public manufacturing firms is:

```
Z = 1.2 × (Working capital / Total assets) + 1.4 × (Retained earnings / Total assets) + 3.3 × (EBIT / Total assets) + 0.6 × (Market value of equity / Book value of total liabilities) + 1.0 × (Sales / Total assets)
```

Scores below about 1.81 put a firm in the "distress" zone; scores above about 2.99 in the "safe" zone; the area between is "grey." Different coefficients exist for private and non-manufacturing firms.

Example 14.8 – Computing a Z-score. A manufacturer has: working capital / total assets = 0.15; retained earnings / total assets = 0.20; EBIT / total assets = 0.10; market equity / liabilities = 1.10; sales / total assets = 0.90.

Z = 1.2(0.15) + 1.4(0.20) + 3.3(0.10) + 0.6(1.10) + 1.0(0.90) Z = 0.18 + 0.28 + 0.33 + 0.66 + 0.90 = **2.35**

The firm is in the grey zone. A decline in the market value of its equity to 0.60 of liabilities would reduce the fourth term from 0.66 to 0.36 and Z to 2.05, which indicates the sensitivity of such models to equity prices. Scoring models are screening tools and are no substitute for analysis of cash flow, competitive position and liquidity.

14.7 Debt Capacity

Debt capacity is the amount of debt a firm can carry while retaining its desired credit standing, covenant compliance and strategic flexibility. It is not a single number; it is the smallest value produced by several constraints, tested under stress.

14.7.1 Approaches

1. **Leverage constraint.** Maximum Debt/EBITDA consistent with the target rating or the covenant, adjusted for the cycle. Capital-intensive and cyclical firms need lower multiples than stable service firms.
2. **Coverage constraint.** Maximum interest such that EBITDA (or EBIT) divided by interest meets a minimum. Because interest equals debt times the rate, debt capacity under this constraint falls as interest rates rise.
3. **Cash-flow (DSCR) constraint.** Maximum debt service such that cash flow available for debt service divided by scheduled debt service is at least a target (for example, 1.5x), and the debt that this service supports given the tenor and rate.
4. **Asset-coverage constraint.** Debt not exceeding some fraction of the liquidation value of assets, relevant for secured lending.
5. **Strategic and trade-off view.** The tax shield from more debt must be weighed against expected costs of distress, loss of flexibility and the option value of unused capacity. Firms with valuable growth options or volatile cash flows deliberately hold *financial slack*: borrowing capacity held in reserve.

14.7.2 Stress testing

A firm should compute capacity on **downside cash flows**, not on the plan. Reduce EBITDA by an amount that is plausible in a recession in the industry (for example, the peak-to-trough decline in the last cycle), apply higher interest rates if the debt is floating, and check that (a) covenants hold, (b) liquidity is sufficient to meet obligations without new external funding, and (c) maturities are not clustered in a window when refinancing could be difficult.

Example 14.9 – Estimating debt capacity. Kestrel has EBITDA of $120 million and can borrow at 8.0%. Its covenants and policies are: Debt/EBITDA ≤ 4.00x; EBITDA/interest ≥ 4.00x (the internal policy for an investment-grade profile); and a minimum DSCR of 1.50x. Cash flow available for debt service (CFADS) is expected to be $90 million. New debt would be repaid in level annual payments over seven years. Existing debt is $100 million.

Step 1: Leverage constraint. Maximum debt = 4.00 × 120 = $480 million.

Step 2: Coverage constraint. Maximum interest = 120 / 4.00 = $30 million. At 8.0%, maximum debt = 30 / 0.08 = **$375 million**.

Step 3: DSCR constraint. Maximum annual debt service = 90 / 1.50 = $60 million. The present value of seven level payments of $60 million at 8.0% = 60 × [(1 − 1.08^(−7)) / 0.08] = 60 × 5.2064 = **$312.4 million**.

Step 4: Debt capacity is the minimum of the three: **$312.4 million** (the DSCR constraint binds). Incremental capacity over existing debt = 312.4 − 100 = $212.4 million, compared with a funding need of $180 million. The expansion is affordable, with roughly $32 million of slack.

Step 5: Stress test. Suppose EBITDA falls 30% to $84 million and CFADS falls 30% to $63 million. With debt of $280 million after drawing the full $180 million: leverage = 280 / 84 = 3.33x (passes 4.00x). Interest at 8.0% = $22.4 million; coverage = 84 / 22.4 = 3.75x (just below the 4.00x policy but the covenant is not breached if the legal minimum is lower). DSCR depends on debt service; if service on $280 million over seven years is 280 / 5.2064 = $53.8 million, DSCR = 63 / 53.8 = 1.17x, which is thin but above 1.0x. Kestrel could service its debt in the downturn, but with little room for further adverse events.

Step 6: Interest tax shield. Debt of $280 million at 8.0% and a 25% tax rate gives a first-year tax shield of 0.25 × 22.4 = $5.6 million, worth up to 0.25 × 280 = $70 million in present value if the debt were permanent and the rate were the discount rate. Against that, a modest further increase in debt would use up slack that has option value in a downturn.

The result frames the decision. Capacity is a ceiling, not a target. A policy such as “borrow to 70% of capacity” provides a buffer and keeps flexibility for acquisitions or downturns.

14.7.3 Managerial guidelines

- Match debt service to cash flow patterns: seasonal or cyclical firms should avoid heavy amortization and back-ended refinancing wall.
- Stagger maturities to avoid concentration: a “ladder” reduces refinancing risk.
- Maintain committed liquidity (undrawn revolver and cash) sized to cover at least twelve months of obligations in a stress scenario.
- Hedge floating-rate exposure to the extent coverage cannot absorb a rate shock.
- Link debt policy to the strategy: companies with large growth opportunities should be conservative; stable cash generators can carry more.
- Remember agency problems on both sides: managers may prefer less debt (protecting their human capital), while high-leverage structures can discipline waste. Observe the evidence in the firm’s own history rather than apply an industry rule blindly.

14.8 Securitization: An Overview

Securitization converts a pool of illiquid financial assets (loans, receivables, leases) into tradable securities backed by the pool's cash flows. The originator sells the assets to a **special purpose vehicle (SPV)**, a legally separate entity designed to be "bankruptcy remote" from the originator. The SPV funds the purchase by issuing securities, which are repaid from collections on the pool. Common underlying assets include residential mortgages (mortgage-backed securities), auto loans, credit-card receivables, trade receivables, equipment leases and corporate loans (collateralized loan obligations, or CLOs).

14.8.1 Why securitize?

- **Funding based on asset quality rather than originator credit.** A high-quality pool can be financed more cheaply than the originator's unsecured debt, if the structure is truly isolated from the originator's risk.
- **Diversification of funding sources.** Access to capital markets investors, and not only bank balance sheets.
- **Balance-sheet and capital management.** If the transfer meets the derecognition criteria under the accounting standards and regulatory rules (including a true sale and transfer of risks and rewards), assets leave the originator's balance sheet. Under IFRS 9 and IFRS 10, however, retained exposure or control may require the originator to continue to recognize the assets or consolidate the SPV.
- **Risk transfer and tailored securities.** The structure can offer investors different risk-return profiles.

14.8.2 Structure and credit enhancement

The main feature is **tranching**: the SPV issues securities of different seniority. Losses on the pool are allocated from the bottom up.

- **Equity (first-loss) tranche.** Absorbs the first losses and receives residual cash flow; often retained by the originator to align incentives ("skin in the game").
- **Mezzanine tranches.** Absorb losses after the equity tranche is exhausted.
- **Senior tranches.** Protected by the subordination below them, and therefore receive the highest credit ratings.

Other forms of **credit enhancement** include **overcollateralization** (the pool exceeds the securities issued), **excess spread** (the pool's interest income exceeds the securities' interest and fees, and the excess is trapped to absorb losses), reserve accounts, guarantees, and triggers that redirect cash flows to senior tranches if performance deteriorates.

Example 14.10 – Loss allocation in a tranched structure. An SPV holds a pool of $500 million of loans and issues three tranches: Senior $400 million (80%), Mezzanine $60 million (12%) and Equity $40 million (8%). Consider three loss scenarios for the pool (ignoring recoveries timing and excess spread).

Scenario A: Pool losses of 5% = $25 million. Equity absorbs all $25 million (62.5% of the equity tranche). Mezzanine and senior suffer no loss.

Scenario B: Pool losses of 12% = $60 million. Equity absorbs its full $40 million (100%). The remaining $20 million falls on the mezzanine (33.3% of that tranche). Senior suffers no loss.

Scenario C: Pool losses of 25% = $125 million. Equity loses $40 million, mezzanine loses $60 million (100% each), and the remaining $25 million falls on the senior tranche, a loss of 25 / 400 = 6.25%.

Table 14.5 – Tranche losses by scenario ($ millions)

Tranche	Size	Scenario A (pool loss 5%)	Scenario B (12%)	Scenario C (25%)
Senior	400	0	0	25 (6.25%)
Mezzanine	60	0	20 (33.3%)	60 (100%)
Equity	40	25 (62.5%)	40 (100%)	40 (100%)
Total loss	500	25	60	125

The senior tranche has 20% of subordination beneath it, so it is hit only if the pool's losses exceed 20%. The mezzanine tranche is exposed once losses exceed 8% and is wiped out at 20%. Whether the senior tranche is "safe" depends entirely on the distribution of pool losses and on correlation: if loans default together, the probability of extreme losses is much higher than the independent-default assumption implies.

14.8.3 Risks and lessons

Securitization is a useful tool, but its history includes significant failures. The main lessons:

- **Model risk and correlation.** Ratings depended on assumptions about default correlation and house prices that failed in the 2007-2009 crisis. Senior tranches of pools of subprime mortgages suffered losses they were rated not to suffer.
- **Misaligned incentives (originate to distribute).** When originators sell loans and keep no risk, they may lower underwriting standards. Regulators now commonly require risk retention.
- **Complexity and opacity.** Re-securitizations (securities backed by tranches of other securitizations) multiplied complexity and made valuation depend on models rather than observable cash flows.
- **Liquidity and funding runs.** Structures financed with short-term paper and liquidity backstops proved fragile when market confidence evaporated.
- **Reputational and legal risk.** Even where legal separation exists, originators may feel obliged to support their vehicles.

For a corporate CFO, securitization is most often relevant for trade receivables or equipment loans, where the pool is granular, short-dated and well understood. Disciplined use means clear true-sale analysis, realistic stress assumptions, an understanding of consolidation and covenant treatment, and the capacity to continue to fund if the market is closed.

14.8.4 Choosing among sources: a synthesis

Returning to Kestrel: the $180 million expansion has an average asset life of 20 years for the quarry and rail, and 7 to 10 years for the trucks. A sensible package might combine (i) a $100 million ten-year senior note, matching the long-lived assets and fixing most of the cost; (ii) a $60 million amortizing bank term loan with a covenant-light structure for the equipment, repayable as the assets earn; (iii) a lease for roughly $20 million of trucks if the NAL is positive after residual-value risk; and (iv) an undrawn revolver of $75 million for liquidity. A convertible might make sense only if management believes the shares are undervalued and the firm has a clear case for dilution tolerance. Securitization of receivables could be held in reserve as a contingent source of funding. The total should fit inside the lowest constraint from Example 14.9, with a buffer.

Case Study: Harbourline Foods and the Refinancing Wall

Illustrative case; Harbourline Foods and all figures are fictional.

Harbourline Foods is a mid-sized packaged-food manufacturer with sales of $900 million, EBITDA of $135 million and a loyal base of supermarket customers. Over the past four years it has grown by acquisition, funding purchases with a mix of bank debt and bonds. The new CFO, Priya Raman, joined six months ago and has just completed a review of the debt stack.

Harbourline's debt totals $540 million: a $200 million term loan A maturing in 22 months at a floating rate of the benchmark plus 2.25%; $250 million of 5.75% senior notes due in 30 months, with an incurrence covenant limiting debt/EBITDA to 4.5x for new borrowing; and a $90 million revolver, fully drawn, which matures in 22 months. The credit agreement contains maintenance covenants: debt/EBITDA no higher than 4.25x (stepping down to 4.00x in 12 months) and EBITDA/interest of at least 3.0x. Leverage is currently 4.0x. Interest expense is $39 million, and coverage is therefore 3.46x. Cash on hand is $25 million. Free cash flow after capex and dividends is about $30 million a year. Harbourline also leases its distribution centers and a fleet of delivery vehicles under operating leases with total annual payments of $22 million, now recorded on the balance sheet as lease liabilities of $140 million that the credit agreement excludes from "Debt."

The problem is clear to Raman. In 22 months Harbourline faces $290 million of maturities in a single wall, and the leverage covenant steps down in 12 months, at which point headroom will be nearly gone unless EBITDA grows or debt falls. A bad commodity year would trigger a breach. Rating agencies have placed the company on negative outlook, and a downgrade to the single-B category would raise the cost of any new bond by about 150 basis points and shut some investors out.

The CEO wants to buy a regional competitor for $150 million, which would add $25 million of EBITDA but would lift leverage to about 4.6x on a pro forma basis before synergies. The board is divided. The finance team has sketched four options: (a) issue $150 million of new senior notes and refinance the term loan with a new five-year term loan, extending maturities but increasing leverage; (b) issue $100 million of convertible notes at a 3.5% coupon with a 30% conversion premium to fund the acquisition, while keeping the existing debt in place; (c) sell and lease back two distribution centers for $120 million, using the proceeds to repay the term loan; (d) abandon the acquisition, suspend the dividend and use free cash flow to repay debt, accepting slower growth.

Raman knows that the lenders will want to see a credible plan before negotiating the refinancing. She has asked her team to prepare covenant forecasts under a downside scenario with 15% lower EBITDA, a lease-adjusted leverage computation, and a comparison of the all-in cost and dilution of each option.

Discussion questions

1. Compute Harbourline's current covenant headroom for leverage (at the current 4.25x limit and the stepped-down 4.00x limit) and for coverage. By what percentage can EBITDA fall before each covenant is breached? What does the 15% downside scenario imply?
2. Evaluate option (a) in terms of pro forma leverage, coverage and maturity profile. How would you structure maturities to avoid a new wall, and what covenant package would you negotiate?
3. Compare option (b) with a straight bond for the acquisition. What are the advantages (coupon, covenant headroom, effective issue price for equity) and disadvantages (dilution, share-price pressure, complexity)? Under what share-price paths does the convertible prove expensive?
4. Assess option (c). How do the sale-and-leaseback and the lease accounting alter reported leverage, EBITDA and the covenant tests, given that lease liabilities are excluded from "Debt" in the credit agreement but are considered by rating agencies? Which option or combination would you recommend, and what sequencing would you propose in discussions with lenders?

Key Takeaways

- Debt contracts are bundles of choices on maturity, rate basis, currency, seniority, security, amortization, optionality and covenants; the right bundle matches the funding to the asset's cash-flow profile and the firm's risk.
- The cost of debt is the current market yield on comparable debt, not the historical coupon; the tax-deductibility of interest lowers the after-tax cost only to the extent the firm can use the deduction.
- Level-payment loans front-load interest, so the tax shield and the lender's risk decline over time; bullet structures concentrate refinancing risk at maturity.
- Covenants address agency conflicts between shareholders and creditors; measure headroom against a downside scenario, and negotiate definitions (EBITDA, Debt) as carefully as the ratio levels.
- Bond prices move inversely with yields; duration summarizes interest-rate sensitivity and convexity explains why duration alone overstates the loss of a rate rise and understates the gain from a fall.
- Credit spreads reflect expected loss (PD × LGD), a risk premium and liquidity; a market spread can be translated into an implied default probability, with caution about the premium embedded in it.
- Leases are a financing choice: evaluate with the net advantage to leasing at the after-tax cost of debt, with careful attention to residual value; IFRS 16 puts most leases on the balance sheet and raises reported EBITDA and debt.
- A convertible bond is straight debt plus a call option; its lower coupon is the price paid by investors for the option, and the real cost is the combination of interest and expected dilution.
- Credit analysis combines cash-flow ratios, liquidity, collateral, management quality and industry cycle; scoring models such as the Altman Z-score are screening aids.
- Debt capacity is the minimum across leverage, coverage and cash-flow constraints, tested under stress; treat it as a ceiling, hold slack and stagger maturities.
- Securitization allocates pool losses through tranching and credit enhancement; the senior tranche's safety depends on correlation assumptions, structure and incentives, as the 2007-2009 crisis showed.

Review Questions

1. Explain the principle of matching in debt financing. Give an example of a mismatch and the risks it creates.
2. Why is the after-tax cost of debt not always equal to the pre-tax yield times one minus the tax rate? In which circumstances is the tax benefit reduced?
3. What are the main agency conflicts between shareholders and lenders that covenants are designed to address? Give one example of each type of covenant (affirmative, negative, financial maintenance and incurrence).
4. Why might a borrower prefer a bond to a bank loan, and vice versa?
5. A bond with a modified duration of 6 is held by an investor. What is the approximate percentage price change if yields rise by 50 basis points? Why is this an approximation?
6. Describe how the credit spread of a bond is related to the default probability and the recovery rate. Why does the market-implied probability usually exceed the historical default rate?
7. Under IFRS 16, how does a lease change the lessee's financial statements compared with the previous operating lease treatment? What should a lender consider when its covenants define "Debt"?
8. Why does a convertible bond carry a lower coupon than a straight bond? Who bears the cost, and when?

9. Define debt capacity. Why should a firm test it under downside scenarios, and why might a firm choose to borrow less than its capacity?
10. Describe how tranching and subordination create a senior security of higher credit quality than the underlying pool. What can cause that quality to fail?

Problems

1. **Amortizing loan.** A company borrows $5,000,000 for five years at 6.0%, repayable in equal annual installments. Compute the annual payment, the first-year interest and principal, and the balance after year 1.
2. **Bond pricing.** A 10-year bond with a face value of $1,000 pays a 5.0% annual coupon in semiannual installments. Compute the price when the yield to maturity is (a) 6.0%, (b) 5.0% and (c) 4.0%.
3. **Covenant headroom.** A borrower has debt of $360 million, EBITDA of $100 million and interest expense of $28.8 million. Its covenants require Debt/EBITDA ≤ 4.00x and EBITDA/interest ≥ 3.25x. Compute both ratios, the EBITDA at which each covenant would be breached, and the percentage headroom of each. Which binds?
4. **Lease versus buy.** A firm can buy equipment for $500,000 (four-year life, straight line to zero, no residual) or lease it for $150,000 a year, paid at year end. The tax rate is 30% and the pre-tax borrowing rate is 10%. Compute the net advantage to leasing.
5. **Convertible bond.** A $1,000 convertible bond has an 8-year maturity, a 4.0% annual coupon and a conversion ratio of 25. The share price is $30, and the issuer's straight-debt yield is 8.0%. Compute the conversion price, conversion premium over the share price, conversion value (parity), straight-bond value, and the implied value of the option if the bond sells at par.
6. **Expected loss.** A bank lends $80 million to a borrower with a one-year PD of 2.5% and LGD of 45%. Compute the expected loss in dollars and as a percentage of exposure. If funding cost is 0.35%, operating cost 0.25% and risk-premium/return requirement is 0.60%, what margin is required?
7. **Debt capacity.** A firm has EBITDA of $90 million, borrows at 9.0%, and uses these limits: Debt/EBITDA ≤ 3.5x; EBITDA/interest ≥ 3.5x; DSCR ≥ 1.4x with CFADS of $60 million and level repayment over five years. Compute the debt capacity under each test and the overall capacity.
8. **Securitization.** A pool of $300 million of receivables backs a Senior tranche of $240 million, a Mezzanine tranche of $36 million and an Equity tranche of $24 million. Pool losses are 15%. Compute the loss and percentage loss for each tranche. At what pool loss percentage does the senior tranche first suffer a loss?

Answers to Selected Problems

1. Payment = 5,000,000 × 0.06 / (1 − 1.06^(−5)) = **$1,186,982**. First-year interest = $300,000; principal = $886,982; balance after year 1 = **$4,113,018**.
2. a. $925.61; (b) $1,000.00; (c) $1,081.76.
3. Leverage = 3.60x; coverage = 100 / 28.8 = 3.47x. EBITDA floors: leverage test 360 / 4.00 = $90.0 million (headroom 10.0%); coverage test 3.25 × 28.8 = $93.6 million (headroom 6.4%). The coverage covenant binds.
4. After-tax rate = 7.0%; annuity factor (4 years) = 3.3872. PV of after-tax lease payments = 150,000 × 0.70 × 3.3872 = $355,657. PV of depreciation tax shields forgone = (500,000 / 4) × 0.30 × 3.3872 = $127,020. NAL = 500,000 − 355,657 − 127,020 = **+$17,322** (leasing is slightly cheaper).

5. Conversion price = $40.00; premium = 40 / 30 − 1 = 33.3%; parity = 25 × 30 = $750; straight value = 40 × 5.7466 + 1,000 / 1.08^8 = 229.86 + 540.27 = $770.13 (rounded); option value if sold at par = 1,000 − 770.13 = **$229.87**.
6. EL = 0.025 × 0.45 × 80 = **$0.90 million**, or 1.125% of exposure. Required margin = 1.125% + 0.35% + 0.25% + 0.60% = **2.325%**.
7. Leverage: 3.5 × 90 = $315.0 million. Coverage: maximum interest = 90 / 3.5 = $25.71 million, so debt = 25.71 / 0.09 = $285.7 million. DSCR: maximum debt service = 60 / 1.4 = $42.86 million; PV over five years at 9.0% = 42.86 × 3.8897 = **$166.7 million**. The DSCR test binds; overall capacity is about $166.7 million.
8. Loss = 0.15 × 300 = $45 million. Equity loses its full $24 million (100%); Mezzanine loses $21 million (58.3% of $36 million); Senior loses $0. The senior tranche first suffers a loss when pool losses exceed the 60 million junior tranches (24 + 36), i.e., 60 / 300 = **20%**.

Chapter 15: Capital Structure

Learning Objectives

- State and prove Modigliani-Miller Propositions I and II in a world without taxes, and explain why they hold even though debt looks “cheaper” than equity.
- Extend the propositions to corporate taxes, compute the value of the interest tax shield, and derive the cost of equity and WACC of a levered firm.
- Unlever and relever betas, and use the result to estimate the cost of capital at a proposed new leverage ratio.
- Explain the direct and indirect costs of financial distress and combine them with tax shields in the static trade-off theory to locate an optimal debt level.
- Compare the pecking order theory, signaling models, and agency-cost theories (asset substitution, debt overhang, free cash flow), and say which observable facts each explains.
- Set a practical target capital structure using coverage and leverage metrics, stress tests, debt capacity analysis, and an EBIT-EPS comparison, and recognize what the numbers cannot tell you.

Opening Vignette

Illustrative scenario; the company is fictional.

Harlow Fabrication is a mid-sized maker of precision metal components, owned by public shareholders and a founding family that holds 22 percent of the stock. For a decade the company has carried almost no debt: net debt is under 0.5 times EBITDA, the balance sheet holds a comfortable cash cushion, and the CFO, Priya Nair, is rightly proud that the firm has never missed a payment or breached a covenant.

At the annual strategy meeting, the board’s newest member, a former private-equity partner, makes a pointed remark. “You earn an operating margin of 14 percent, your cash flows are stable, and you pay a tax rate of 25 percent. Your competitor Dunmore runs at three times EBITDA of leverage. You are leaving money on the table.” Several directors nod. The family representative objects: “Dunmore nearly lost its bank lines in the last downturn. We have survived four recessions precisely because we owe nothing.”

Priya is asked to come back in one month with a recommendation. She quickly realizes that both sides are partly right and partly confused. The private-equity argument is correct that interest is tax deductible, and that more debt means a larger tax shield. But the claim that debt is “cheap capital” ignores that equity becomes riskier as leverage rises. The family’s argument is correct that distress is costly, but a company with Harlow’s cash flows could carry a good deal more debt than zero before the risk of distress becomes material. And neither side has mentioned what a debt-financed recapitalization does to management’s incentives, to the firm’s ability to fund an acquisition opportunity that may appear next year, or to the signal that investors read into the announcement.

This chapter gives Priya the tools. We begin with the benchmark result of Modigliani and Miller, which says that, in a frictionless world, the question “how much debt?” is irrelevant. We then add, one at a time, the frictions that make the question matter: taxes, distress costs, information asymmetries, and conflicts of interest. We end with a practical method for converting theory into a target ratio that a board can approve, and a case study that returns to a company much like Harlow.

15.1 The Capital Structure Question

A firm's *capital structure* is the mix of financing it uses to fund its assets: debt of various maturities and seniority, preferred stock, hybrid securities, and common equity. The *capital structure decision* is the choice of that mix, and in practice it is made through a series of discrete actions: issuing bonds, drawing a revolver, retiring debt with free cash flow, repurchasing shares, or issuing new equity.

Why would the mix matter at all? Consider the firm as a collection of assets that generate operating cash flows. Those cash flows are determined by products, customers, costs, and competition. Financing decides only how the cash flows are *divided* among the claimants. If the pie is fixed, slicing it differently should not change its size. This is the deep insight behind the Modigliani-Miller (MM) theorems, and it is worth holding on to, because every real-world reason that capital structure matters must be traced to a mechanism that changes the size of the pie (taxes paid, distress costs incurred, investments foregone) or transfers value between claimants (from lenders to shareholders, or from uninformed to informed investors).

Notation and Definitions

Throughout the chapter we use the following symbols.

```
V_U   = value of the unlevered (all-equity) firm
V_L   = value of the levered firm = D + E
D, E  = market values of debt and equity
r_U   = cost of capital of the unlevered firm (cost of equity with no debt)
r_D   = cost of debt (assumed to be the expected, riskless-promise rate unless stated)
r_E   = cost of equity of the levered firm
τ     = corporate tax rate
EBIT  = earnings before interest and taxes (assumed perpetual and constant unless stated)
```

We will measure leverage in several ways, and it pays to keep them separate. The *debt-to-equity ratio* D/E and the *debt-to-value ratio* D/V = D/(D+E) are market-value measures used in valuation theory. The *net debt to EBITDA* ratio and the *interest coverage ratio* (EBIT divided by interest expense) are accounting measures used by lenders and rating agencies. Theory speaks in market values; practice speaks in accounting ratios. A good financial manager translates between them.

Why It Is Not Obvious

A common intuition goes: "Debt costs 5 percent and equity costs 12 percent, so replacing equity with debt must lower the average cost of capital and raise firm value." This reasoning contains a subtle error. The 12 percent cost of equity is the return required by shareholders *given the current leverage*. When you add debt, the equity left behind is a smaller, more volatile claim on the same cash flows. Its required return rises. The question is by how much, and the MM result shows that, absent frictions, it rises by exactly enough to offset the benefit of substituting cheaper debt.

15.2 Modigliani-Miller Without Taxes

Franco Modigliani and Merton Miller published their capital structure irrelevance results in 1958. They set the stage for modern corporate finance by showing precisely what must be true for financing to matter. The assumptions of the no-tax version are:

1. There are no taxes of any kind.
2. There are no bankruptcy or transaction costs.

3. Individuals and firms can borrow and lend at the same rate (the rate on riskless or equally risky debt).
4. Investors and managers have the same information.
5. The firm's operating cash flows are not affected by how it is financed.

Proposition I: Firm Value Is Independent of Capital Structure

Proposition I (no taxes). The market value of a firm is determined by its operating cash flows and the risk of those assets, and not by the proportions of debt and equity used to finance them:

```
V_L = V_U
```

The proof uses arbitrage and does not need any assumptions about investor preferences. Suppose two firms, U (unlevered) and L (levered), generate identical operating cash flows. If L were worth more than U, an investor could sell L's shares, borrow on personal account to replicate L's leverage, and buy U's shares. The investor would receive the same cash flows at a lower cost and pocket the difference. Such arbitrage drives the prices together. The financing that corporations do can be undone or replicated by investors at no cost, so corporate leverage adds nothing that investors cannot create for themselves. This idea is called *homemade leverage*.

Example 15.1 – Homemade leverage and arbitrage. Two firms have identical perpetual operating income of EBIT = \$1,000 per year and an unlevered cost of capital r_U = 10%. Firm U is all equity. Firm L has \$4,000 of perpetual debt at r_D = 5%.

Step 1: Value the unlevered firm.

```
V_U = EBIT / r_U = 1,000 / 0.10 = $10,000
```

Step 2: Suppose the market values the levered firm's equity at \$7,000, so V_L = 7,000 + 4,000 = \$11,000. That is higher than V_U although the cash flows are identical. Let us check that this allows arbitrage.

Step 3: Consider an investor who holds 1% of L's equity. The investment costs 0.01 × 7,000 = \$70 and pays 1% of (EBIT − interest) = 0.01 × (1,000 − 200) = \$8 per year.

Step 4: Replicate that payoff with U. The investor sells the L shares (receiving \$70) and buys 1% of U's equity for 0.01 × 10,000 = \$100. To replicate L's leverage, the investor also borrows personally an amount equal to 1% of L's debt, which is \$40. Borrowing \$40 at 5% costs \$2 per year. The net outlay for the position is \$100 − \$40 = \$60, which is less than the \$70 received.

```
Cash flow from 1% of U      = 0.01 × 1,000 = $10
Less interest on $40 loan   = 0.05 × 40    = $2
Net payoff                  = $8   (identical to holding 1% of L)
Net outlay                  = $100 − $40   = $60
```

Step 5: Compare. The investor receives the same \$8 per year but pays \$60 instead of \$70, an arbitrage profit of \$10 per 1% position. All investors seek it, pushing L's equity down to \$6,000 and V_L to \$10,000. At that point the homemade strategy costs the same as the levered share and the arbitrage vanishes.

Conclusion. In equilibrium E_L = 6,000, D = 4,000, V_L = V_U = 10,000. The firm cannot create value by choosing leverage if investors can borrow on equal terms.

Proposition II: The Cost of Equity Rises with Leverage

If V is independent of the mix, so is the weighted average cost of capital. The cost of equity must therefore rise with leverage to balance the lower cost of debt. Starting from the identity that the return on assets equals the weighted average of returns on the claims,

```
r_U = (E/V) × r_E + (D/V) × r_D
```

and solving for r_E, we obtain:

Proposition II (no taxes).

```
r_E = r_U + (D/E) × (r_U − r_D)
```

The cost of equity is a linear function of the debt-to-equity ratio. The slope is the spread between the cost of assets and the cost of debt. Shareholders bear all the operating risk of the firm after lenders have taken their fixed share, so the more debt, the greater the *financial risk* they bear, and the higher their required return.

Example 15.2 – Cost of equity under Proposition II. Continue with Example 15.1: r_U = 10%, r_D = 5%, D = $4,000, E = $6,000.

```
r_E = 0.10 + (4,000 / 6,000) × (0.10 − 0.05)
    = 0.10 + 0.6667 × 0.05
    = 0.10 + 0.03333
    = 13.33%
```

Check by cash flows. Shareholders receive EBIT − interest = 1,000 − 200 = $800 per year. Dividing by E gives 800 / 6,000 = 13.33%, which matches. *WACC check:*

```
WACC = (6,000/10,000) × 13.33% + (4,000/10,000) × 5%
     = 8.00% + 2.00% = 10.00% = r_U
```

The WACC is unchanged at 10 percent. Each extra dollar of debt replaces a dollar of 13.33 percent equity with a 5 percent dollar, but it also raises the cost of the remaining equity. The firm’s value, EBIT/WACC, stays at $10,000.

Think of a seesaw: the cheapness of debt is real, but it is paid for by shareholders through higher risk. A manager who reports “our cost of capital has fallen because we use more debt” while ignoring the rising cost of equity is making the most common mistake in this chapter.

Risk and the Role of Debt Beta

The same logic can be expressed in betas. If debt is risky, the asset beta is the value-weighted average of the debt and equity betas:

```
β_A = (E/V) × β_E + (D/V) × β_D
```

and so

```
β_E = β_A + (D/E) × (β_A − β_D)
```

When debt is riskless (β_D = 0), this reduces to β_E = β_A × (1 + D/E). Higher leverage increases the equity beta proportionally. Notice that the equity beta increases even though nothing about the firm’s business changed. This is the origin of the distinction between *business risk* (determined by the cyclicality of operating cash flows and operating

leverage, captured by β_A) and *financial risk* (determined by D/E).

What the Irrelevance Result Does and Does Not Say

A frequent misreading is that MM proved capital structure does not matter in practice. They proved something more useful: capital structure matters *only if* one of the assumptions fails. The result is a checklist. For each real-world argument about debt or equity ("equity dilutes EPS", "debt is cheaper", "we need financial flexibility"), ask: which assumption is being violated, and is the effect large enough to matter? An argument that does not identify a friction is mere accounting. For example, issuing shares to repay debt may raise EPS if the firm's earnings yield exceeds the cost of debt, but that tells us only that shares are cheaper than the earnings yield implies; the P/E ratio and risk of EPS change as well, and the stock price should not respond to the arithmetic.

15.3 Modigliani-Miller With Corporate Taxes

The first and most important friction is the tax system. In most countries, interest paid on debt is deductible from corporate taxable income, but dividends and retained earnings (the return to equity) are not. This asymmetry means that for every dollar of interest paid, the government effectively contributes τ dollars. The deduction creates an *interest tax shield*.

The Interest Tax Shield

Consider a firm with EBIT and interest expense I. Compare the after-tax cash flows available to all investors (debtholders and shareholders) with and without leverage.

```
Unlevered: total to investors = EBIT × (1 − τ)
Levered:   total to investors = (EBIT − I)(1 − τ) + I
                              = EBIT × (1 − τ) + τ × I
```

The extra amount, τ × I, is the annual interest tax shield. Since the cash flows from the operating assets are unchanged, the extra value to investors is exactly the present value of the tax shields. If the firm maintains a permanent amount of debt D at rate r_D, interest is r_D × D every year, and the shield is τ × r_D × D. Discounting at r_D (the shields have the same risk as the interest payments, assuming the firm will always be able to use them) gives:

```
PV(tax shield) = τ × r_D × D / r_D = τ × D
```

Proposition I (with corporate taxes).

```
V_L = V_U + τ × D
```

The value of the levered firm is the value of the unlevered firm plus the present value of the interest tax shield. Under this simple model, with permanent debt, the gain from leverage is proportional to the amount of debt, and the tax rate is the constant of proportionality.

Cost of Equity and WACC with Taxes

Proposition II is modified because the return to debtholders is effectively paid with a tax subsidy. The cost of equity becomes:

Proposition II (with corporate taxes).

```
r_E = r_U + (D/E) × (1 − τ) × (r_U − r_D)
```

Notice that the cost of equity still rises with leverage, but more slowly than in the no-tax world because (1 − τ) multiplies the slope. This is why the weighted average cost of capital, now defined with an after-tax cost of debt, declines with leverage:

```
WACC = (E/V) × r_E + (D/V) × r_D × (1 − τ)
     = r_U × (1 − τ × D/V)
```

The second line is an elegant simplification: WACC is the unlevered cost of capital reduced by the fraction τ × D/V. At D/V = 40 percent and τ = 25 percent, the WACC is 10 percent lower in relative terms than r_U.

Example 15.3 – MM with taxes. Use the same firm: EBIT = $1,000 perpetual, r_U = 10%, r_D = 5%, D = $4,000, and now τ = 25%.

Step 1: Unlevered value.

```
V_U = EBIT × (1 − τ) / r_U = 1,000 × 0.75 / 0.10 = $7,500
```

Step 2: Levered value.

```
PV(tax shield) = τ × D = 0.25 × 4,000 = $1,000
V_L = 7,500 + 1,000 = $8,500
E   = V_L − D = 8,500 − 4,000 = $4,500
```

Step 3: Cost of equity.

```
r_E = 0.10 + (4,000 / 4,500) × 0.75 × (0.10 − 0.05)
    = 0.10 + 0.8889 × 0.75 × 0.05
    = 0.10 + 0.03333 = 13.33%
```

Step 4: Check against cash flows. Interest is 0.05 × 4,000 = $200. Net income = (1,000 − 200) × 0.75 = $600. Dividing by E: 600 / 4,500 = 13.33%. It matches.

Step 5: WACC.

```
WACC = (4,500/8,500) × 13.33% + (4,000/8,500) × 5% × 0.75
     = 7.059% + 1.765% = 8.82%
Shortcut: r_U × (1 − τ × D/V) = 10% × (1 − 0.25 × 4,000/8,500) = 10% × 0.88235 = 8.82%
```

Interpretation. Using debt lowers the WACC from 10.00 percent to 8.82 percent and raises firm value from $7,500 to $8,500. The gain of $1,000 is the PV of tax shields. In this textbook world the optimal capital structure is nearly 100 percent debt, a conclusion that no real firm follows and that tells us the model omits something important.

Limits of the Simple Tax Shield Formula

The value formula τ × D relies on strong assumptions: debt is permanent (rolled over indefinitely), the firm always has enough taxable income to use the deduction, and the discount rate for shields is r_D. Each assumption can be relaxed.

- *Finite or growing debt.* If the firm targets a constant D/V ratio rather than a fixed dollar amount, the debt grows with firm value and the shield is as risky as the firm's assets. The appropriate discount rate then is r_U, and the value is computed in the adjusted present value (APV) or WACC framework with leverage recomputed each period. Many practitioners find the constant-ratio assumption more realistic, particularly for growing firms.

- *Limited ability to use shields.* A firm with low or negative taxable income cannot immediately use interest deductions. Unused deductions carry forward, but their value is delayed, and some jurisdictions restrict deductibility of interest above a percentage of EBITDA. The effective τ on the margin may be much lower than the statutory rate for firms already shielded by depreciation, tax credits, or losses.
- *Non-debt tax shields.* Depreciation, depletion allowances, and loss carryforwards also reduce taxable income. A firm with ample non-debt shields faces a lower marginal benefit from debt.

Personal Taxes: Miller's Refinement

In 1977 Miller pointed out that investors also pay personal taxes, and that the tax rate on interest income typically exceeds the tax rate on equity returns (dividends and capital gains). Let τ_C be the corporate rate, τ_E the personal rate on equity income, and τ_D the personal rate on interest. The net advantage of debt per dollar of debt is:

```
G = 1 − (1 − τ_C)(1 − τ_E) / (1 − τ_D)
```

With τ_C = 25%, τ_E = 15%, and τ_D = 35%:

```
G = 1 − (0.75 × 0.85) / 0.65 = 1 − 0.98077 = 0.0192
```

The tax advantage of debt falls from 25 cents per dollar to under 2 cents, because lenders demand extra yield to compensate for their higher personal tax. In the extreme case where (1 − τ_C)(1 − τ_E) = (1 − τ_D), debt has no net advantage at the aggregate level. Miller's theory also predicts that firms in the aggregate issue debt until the tax rates of the marginal buyers equalize across debt and equity, so the overall supply of debt is determined by the market, and an individual firm gains nothing from changing its mix. While few believe the extreme conclusion, the personal-tax effect usefully reminds us that the corporate tax shield overstates the net benefit. A conservative planning figure for the net tax benefit of debt is usually well below the statutory corporate rate.

Unlevering and Relevering Betas

For practical valuation we rarely observe the unlevered cost of capital directly. We estimate it from comparable firms by "unlevering" the observed equity betas, averaging the unlevered asset betas, and relevering at the target firm's capital structure. With corporate taxes and riskless debt (a standard simplification attributed to Hamada), the relationships are:

```
β_E = β_U × [1 + (1 − τ) × D/E]
β_U = β_E / [1 + (1 − τ) × D/E]
```

Example 15.4 – Estimating the cost of equity at a new leverage ratio. Harlow Fabrication's closest comparable has an observed equity beta of 1.20, a market debt-to-equity ratio of 0.50, and faces a 25 percent tax rate. The risk-free rate is 4.0% and the equity risk premium is 5.5%. Harlow is considering a leverage of D/E = 1.00.

Step 1: Unlever the comparable.

```
β_U = 1.20 / [1 + 0.75 × 0.50] = 1.20 / 1.375 = 0.8727
```

Step 2: Relever at D/E = 1.00.

```
β_E(new) = 0.8727 × [1 + 0.75 × 1.00] = 0.8727 × 1.75 = 1.5273
```

Step 3: Cost of equity by CAPM.

```
r_E(comparable) = 4.0% + 1.20 × 5.5% = 10.60%
r_E(new)        = 4.0% + 1.5273 × 5.5% = 4.0% + 8.40% = 12.40%
```

Step 4: Interpretation. Raising leverage from 0.5 to 1.0 times equity adds 1.8 percentage points to the cost of equity. If the pre-tax cost of debt at this leverage is 6 percent, the WACC at D/E = 1.00 (D/V = 50%) is:

```
WACC = 0.50 × 12.40% + 0.50 × 6% × 0.75 = 6.20% + 2.25% = 8.45%
```

The analyst must verify that the cost of debt used corresponds to the new leverage: at higher leverage, lenders charge higher spreads, and a debt beta greater than zero would reduce the relevered equity beta somewhat. Not doing so overstates the benefit of leverage, which is a common error.

15.4 Financial Distress and the Static Trade-Off Theory

If taxes were the whole story, every profitable firm would borrow until its taxable income vanished. They do not, and the reason is that debt creates a legal obligation. When a firm cannot meet it, the consequences are costly and sometimes catastrophic.

Direct and Indirect Costs of Financial Distress

Financial distress is the condition in which a firm has difficulty meeting its obligations to creditors. It need not end in bankruptcy, but the costs begin well before a filing and they exist even for firms that eventually recover.

Direct costs are the explicit expenses of default and restructuring: legal, accounting, and advisory fees; court costs; and the management time absorbed by negotiation with creditors. Empirical research suggests that direct costs are a modest fraction of firm value for large firms, though proportionally larger for small ones, as many of the fees are fixed.

Indirect costs are generally much larger and harder to measure. They include:

- *Lost customers and suppliers.* Buyers of durable products with long-term warranties, or of complex systems requiring service for years, hesitate to deal with a firm that may not exist. Suppliers demand cash on delivery, tightening liquidity just when the firm needs it.
- *Loss of key employees.* Talented staff leave for secure employers; recruiting becomes costly.
- *Underinvestment.* Managers who see the firm's cash flowing to creditors cut positive-NPV projects (the debt overhang problem discussed in Section 15.6).
- *Fire sales.* Forced asset sales to meet obligations realize less than going-concern value, particularly when the industry itself is distressed and the natural buyers are also short of capital.
- *Management distraction.* Time is spent on creditors rather than customers.

Two features determine how large expected distress costs are. First, the *probability* of distress, which rises with leverage and with the volatility of operating cash flows. Second, the *cost conditional on distress*, which is high for firms whose value depends on intangible assets, growth options, human capital, and customer trust, and low for firms with tangible, redeployable assets such as real estate, aircraft, or standard equipment. This explains why pharmaceutical and software firms typically carry little debt while shipping companies, utilities, and real estate firms carry much more.

The Trade-Off Theory

The *static trade-off theory* says that a firm picks the debt level that balances the marginal tax benefit of debt against the marginal expected cost of distress. Algebraically:

```
V_L = V_U + PV(interest tax shield) − PV(expected costs of financial distress)
```

The tax shield term rises roughly linearly with debt, while the expected distress cost term rises at an increasing rate, because both the probability of default and the severity of consequences rise with leverage. The sum therefore has an interior maximum at which marginal benefit equals marginal cost. The theory delivers several testable predictions: firms with more tangible assets, more stable cash flows, and higher taxable income should borrow more; firms with high growth options and volatile earnings should borrow less; and firms should move toward their targets after shocks.

Example 15.5 – Locating the optimum. An unlevered firm is worth V_U = $7,500 (Example 15.3). If financial distress occurs, costs equal 25 percent of V_U, or $1,875. Management estimates the probability of distress at various levels of permanent debt, and τ = 25 percent.

Method. For each debt level compute PV(tax shield) = 0.25 × D, expected distress cost = probability × $1,875, and V_L = 7,500 + shield − expected cost.

Debt D ()|*PVtaxshield*()	Prob. of distress	Expected distress cost ()|*FirmvalueV*$_L$()		
0	0	0%	0.00	7,500.00
1,000	250	1%	18.75	7,731.25
2,000	500	3%	56.25	7,943.75
3,000	750	8%	150.00	8,100.00
4,000	1,000	16%	300.00	8,200.00
5,000	1,250	28%	525.00	8,225.00
6,000	1,500	45%	843.75	8,156.25

Reading the table. Going from $4,000 to $5,000 of debt adds $250 of shield and $225 of expected distress cost, a net gain of $25. Going from $5,000 to $6,000 adds another $250 of shield but $318.75 of expected cost, a net loss of $68.75. The maximum lies at D = $5,000, with V_L = $8,225 and D/V = 5,000 / 8,225 = 60.8%. This is a toy calculation, and a real analysis would place much more weight on how fast the probability curve steepens. Notice how flat the value function is around the optimum: moving from $4,000 to $5,000 changes value by 0.3 percent. The right conclusion is not “the optimum is exactly $5,000” but “there is a broad plateau, and the cost of being on it imprecisely is small, while the cost of being far past it is large and asymmetric.”

Evidence and Limits

Static trade-off theory explains industry differences in leverage and the fact that distressed firms have difficulty borrowing. It struggles with two stylized facts. First, many highly profitable firms with low distress risk carry little debt, whereas the theory predicts they should use the most shield. Second, within an industry, more profitable firms tend to borrow *less*, not more. These observations motivated the information-based theories below. Dynamic trade-off models, which add transaction costs of adjusting leverage and uncertainty over future earnings, can explain some of this by showing that firms rationally let leverage drift within a band and rebalance only when it crosses a boundary.

15.5 Information Asymmetry: Pecking Order and Signaling

The MM world assumes managers and investors see the same information. In reality, managers know more about the firm's prospects than outsiders do. This asymmetry changes how financing choices are interpreted and priced.

The Pecking Order Theory

Myers and Majluf showed that when managers know more than investors and act in the interest of existing shareholders, equity issuance is a bad signal and can lead firms to forego good projects. Investors reason: "Managers will issue shares when they believe the shares are overvalued." The market therefore marks down the price on announcement, and good firms, which would be selling shares cheaply, may prefer not to issue at all. Myers later organized this into the *pecking order* of financing preferences:

1. Use internally generated funds (retained earnings) first, because they carry no information cost and no issuance fees.
2. If external funds are required, issue debt next, because the value of safe debt is almost insensitive to the manager's private information.
3. Issue equity last, only when debt capacity is exhausted or the cost of distress is high.

Under this theory there is no optimal target capital structure. Leverage is the cumulative result of the firm's financing deficit over time: profitable firms accumulate cash and have low debt; fast-growing firms with large investment needs borrow more. This neatly explains why profitability and leverage are negatively related within industries.

Example 15.6 – Why the good firm refuses to issue equity. A firm has assets in place worth either \$140 (good state) or \$60 (bad state), each with probability 50 percent. Only management knows the state. A project requires an investment of \$40, financed by new equity, and has an NPV of +\$10 (value of \$50 for a \$40 outlay). Outside investors have no way to distinguish the types.

Step 1: If both types issue, the market values the firm at the average.

```
Expected assets in place = 0.5 × 140 + 0.5 × 60 = $100
Value of firm after issue (market view) = 100 + 50 = $150
Fraction demanded by new investors for $40 = 40 / 150 = 26.67%
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Step 2: What does the good firm's current shareholder get? True value of the good firm with the project is 140 + 50 = \$190. After giving away 26.67 percent, existing owners retain 73.33 percent:

```
0.7333 × 190 = $139.33
```

That is less than the \$140 they would have kept by skipping the project. So the good firm will not issue.

Step 3: The equilibrium unravels. If only bad firms issue, investors know the firm is bad and value assets at \$60. The value after issue is 60 + 50 = \$110, new investors demand 40 / 110 = 36.4 percent, and existing owners of the bad firm keep 63.6 percent × 110 = \$70, which exceeds the \$60 they would have without the project. Bad firms issue.

Conclusion. The good firm passes up an NPV-positive project because of information asymmetry, and the announcement of equity issuance reveals the firm to be the bad type. The average announcement effect of seasoned equity issues in practice is negative, and the pecking order theory takes this as support. The cost of this adverse selection is a reason to preserve *financial slack*: cash, unused debt capacity, and credit lines that allow positive-NPV investments to be funded without turning to equity.

Signaling Models

Ross (1977) proposed that managers use capital structure to *signal* private information. Because bankruptcy is costly to managers (loss of job, reputation), only managers of strong firms can credibly commit to high debt service. A rise in leverage thus signals that management expects robust cash flows, and the stock price rises on announcement. Conversely, equity issues signal weakness.

Empirical regularities are consistent with this: announcements of leveraged recapitalizations and debt-for-equity exchanges tend to be received positively, while equity offerings and equity-for-debt exchanges are received negatively. A caution is warranted, though. Announcement effects combine several things (signaling, tax benefits, agency effects, resolution of uncertainty about financing needs), and it is difficult to separate them. The managerial implication is more practical: *every financing announcement is a message.* A firm that issues shares to fund an acquisition will be read differently from one that buys back shares, and communication around the transaction (why equity, why now, what the firm's own insiders are doing) shapes the price response.

Market Timing

A related behavioral idea is *market timing*: managers issue equity when their stock price is high relative to book or past values and repurchase when it is low. Evidence that the effects of past market valuations persist in leverage for many years has been used to argue that capital structure is partly the accumulation of past timing attempts. Whether managers truly outsmart the market is doubtful, but the pattern is consistent with the lemon-avoidance logic above: issue when information costs are lowest, which is when prices are high and the stock has risen.

15.6 Agency Costs of Debt and Equity

The final family of theories focuses on conflicts of interest among the parties to the firm: shareholders, creditors, and managers. Jensen and Meckling (1976) formalized the idea that financing choices change incentives, and that the costs of these incentive problems are borne ultimately by the parties who anticipate them.

Shareholder-Creditor Conflicts

When debt is risky, equity is similar to a call option on the firm's assets with a strike equal to the face value of the debt. Option value rises with the volatility of the underlying. Shareholders therefore have an incentive to take on risk, even at the expense of firm value, because the downside is limited by limited liability and the upside is theirs.

Asset substitution. After creditors lend at a rate based on the firm's current low-risk projects, shareholders may switch to riskier ones. Creditors, anticipating this, charge higher rates or impose covenants.

Example 15.7 – Asset substitution. A firm owes \$100 (a single zero-coupon debt due next year). It can choose one of two projects, each costing the same.

- Project A: produces a certain \$120.
- Project B: produces \$190 with probability 50% or \$30 with probability 50% (expected value \$110).

Step 1: Payoffs under A. Debt receives \$100 for sure, equity receives \$20. Firm value \$120.

Step 2: Payoffs under B.

```
Good state: debt gets 100, equity gets 190 − 100 = 90
Bad state:  debt gets 30 (all that is available), equity gets 0
E[debt]   = 0.5 × 100 + 0.5 × 30 = $65
E[equity] = 0.5 × 90 + 0.5 × 0   = $45
Total     = $110
```

Step 3: Compare. Equity prefers B ($45 vs $20), a gain of $25, even though B destroys $10 of expected total value (110 vs 120). Debtholders lose $35 (100 → 65). The value transfer from creditors to shareholders more than offsets the value destroyed, so a shareholder-maximizing manager picks the inferior project. If lenders foresee this, they price loans as though B will be chosen, and the firm bears the cost through a higher interest rate or tighter covenants. This is the agency cost of debt.

Debt overhang. In the converse situation, a heavily indebted firm may decline a positive-NPV project because the benefits accrue mostly to creditors (who are more likely to be repaid) and the new investment must be funded by shareholders. Suppose a firm owes $100 and its assets are worth $80, and it has a project costing $10 that would raise asset value by $15 (NPV +$5). Even after the project the assets are worth only $95, less than the debt, so the entire $15 goes to creditors. Shareholders would pay the full $10 and receive nothing, so they rationally decline. This underinvestment is one reason distressed firms forego opportunities, and it is an important component of indirect distress costs.

Creditors protect themselves with covenants (restrictions on additional debt, dividends, asset sales), collateral, shorter maturities, and the right to renegotiate. Each solution has its own costs, in lost flexibility and monitoring.

Manager-Shareholder Conflicts and Free Cash Flow

Managers may prefer to grow the firm, retain cash, and enjoy perquisites rather than pay out cash to shareholders. Jensen's *free cash flow hypothesis* argues that debt disciplines management: required interest and principal payments reduce the cash available for wasteful investment, and the threat of default forces efficiency. The benefit is greatest in mature, cash-generative firms with few growth opportunities (typically in industries like tobacco, certain retail, and older energy), and it explains the effect of leveraged buyouts. It is least valuable, and potentially harmful, in firms with many valuable investment options, where debt may cause underinvestment.

The agency cost of equity arises from the fact that managers who own less than 100 percent of the firm bear only a fraction of the cost of perks and empire-building. The costs of monitoring, bonding, and the residual loss together make up the total agency cost. Optimal capital structure, in this view, minimizes the sum of the agency costs of debt and equity.

Comparing the Theories

Theory	Key friction	Predicts leverage will be	Main observable implication
MM (no taxes)	None	Irrelevant	Financing announcements should not change value
Static trade-off	Taxes vs. distress costs	Higher for tangible assets, stable cash flows	Target ratio; reversion after shocks
Pecking order	Information asymmetry	Result of financing deficits	Profitable firms borrow less; negative equity-issue announcements
Signaling	Information plus costly default	Higher when management is confident	Positive reaction to leverage increases
Agency (Jensen-Meckling)	Conflicts between managers, shareholders, creditors	Higher for mature, high-cash-flow firms; lower with growth options	Covenants; buyouts raise value

No single theory dominates the evidence. Managers' stated behavior suggests a mixture: they speak of target ratios and ratings (trade-off), they value financial flexibility and reluctantly issue equity (pecking order), and they take care over investor perceptions (signaling).

15.7 Setting a Practical Target Capital Structure

No manager can read the optimal debt ratio off a graph. In practice, the decision combines theory-based reasoning with market constraints (what lenders and rating agencies will accept) and strategic needs (what the firm must preserve in order to pursue its plan). A disciplined process has five steps.

1. *Establish the business risk profile.* How volatile are EBITDA and cash flow across the cycle? What fraction of costs is fixed? How liquid and redeployable are the assets? Peer leverage in the same industry gives a first anchor.
2. *Choose the credit standing the firm wishes to maintain.* Many companies target a particular credit rating (for example, a solid investment-grade rating) because it preserves access to bond markets, commercial paper, and favorable supplier terms even in downturns. The rating, in turn, depends on leverage and coverage metrics consistent with business risk.
3. *Stress test.* Compute coverage and leverage under a severe but plausible downturn, not only under the base case. The debt level should leave room to keep investing and paying obligations even in the stress scenario.
4. *Test the value case.* Estimate the tax benefit (using a realistic marginal rate, net of personal tax effects) and compare it with the loss of flexibility and expected distress costs, as in Example 15.5.
5. *Express the target as a range, with a plan to correct deviations.* A band (for example, net debt/EBITDA of 1.5 to 2.5 times) allows tactical flexibility and avoids constant small transactions.

Key Metrics

Metric	Formula	Typical use	Illustrative comfort level for a stable manufacturer
Net debt / EBITDA	(Debt − cash) / EBITDA	Lender covenants, rating analysis	below 2.5x
Interest coverage	EBIT / interest expense	Ability to service debt	above 4x base, above 3x stressed
FFO / debt	Funds from operations / total debt	Rating agency cash-flow measure	above 30%
Debt / capital (market)	D / (D + E)	Cost-of-capital theory	25% to 40%
Fixed-charge coverage	(EBITDA + rent) / (interest + rent)	Firms with large leases	above 2.5x

These comfort levels are *illustrative only*. Appropriate levels depend on industry, size, cash-flow volatility, and prevailing market conditions. Leases, pension deficits, and other debt-like obligations should be included in the debt measure that lenders actually use. Practitioners often use market-value D/V for valuation and book-based or cash-flow ratios for covenants, and a good target is expressed in both.

EBIT-EPS Analysis

A tool commonly used in boardrooms is the EBIT-EPS comparison. It plots earnings per share under alternative financing plans across a range of EBIT values. The *indifference point* is the EBIT at which the plans give the same EPS. Above it, leverage raises EPS (positive financial leverage); below it, leverage lowers EPS.

Example 15.8 – EBIT-EPS indifference. Lumen Instruments has 10 million shares priced at $40 and needs $100 million for an expansion. It can (A) issue 2.5 million new shares at $40, or (B) issue $100 million of 6% debt. Existing debt is zero and the tax rate is 25%.

Step 1: Share counts and interest. Plan A: shares = 10 + 2.5 = 12.5 million, interest = 0. Plan B: shares = 10 million, interest = 0.06 × $100m = $6 million.

Step 2: Set EPS equal.

```
EBIT × (1 − 0.25) / 12.5  =  (EBIT − 6) × (1 − 0.25) / 10
EBIT / 12.5 = (EBIT − 6) / 10
10 × EBIT = 12.5 × EBIT − 75
EBIT = 75 / 2.5 = $30 million
```

Step 3: Verify.

```
Plan A at EBIT = 30: 30 × 0.75 / 12.5 = $1.80
Plan B at EBIT = 30: 24 × 0.75 / 10   = $1.80
```

Step 4: Compare at other EBIT levels. At EBIT = $50 million: Plan A gives 50 × 0.75 / 12.5 = $3.00; Plan B gives 44 × 0.75 / 10 = $3.30. At EBIT = $20 million: Plan A gives $1.20; Plan B gives 14 × 0.75 / 10 = $1.05.

Interpretation and warning. Debt raises EPS when EBIT exceeds $30 million and reduces it below. But EPS is not value. EBIT-EPS analysis ignores the higher risk of EPS under Plan B (the spread between $1.05 and $3.30 is wider than between $1.20 and $3.00) and the change in the P/E ratio that the market should apply. It is useful to see the *sensitivity* of earnings to

operating results, and in particular to see how far EBIT must fall before the debt plan produces a loss relative to equity, but it should never be the criterion that determines the capital structure.

Debt Capacity from Stress Tests

Example 15.9 – Sizing debt with a stressed coverage constraint. Meridian Components has EBITDA of $400 million, depreciation and amortization of $100 million, and a marginal borrowing cost of 6%. The board wants to keep EBIT interest coverage at or above 3.0 times even if EBITDA falls 35% (the severe downturn scenario, with depreciation unchanged).

Step 1: Stressed EBITDA and EBIT.

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Stressed EBITDA = 400 × (1 − 0.35) = $260 million
Stressed EBIT   = 260 − 100        = $160 million
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Step 2: Maximum interest at 3.0x coverage.

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Max interest = 160 / 3.0 = $53.3 million
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Step 3: Maximum debt.

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Max debt = 53.3 / 0.06 = $888.9 million
```

Step 4: Cross-check with other metrics. Base-case coverage at this debt: base EBIT is $300 million, interest $53.3 million, so coverage is 5.6x. Leverage is 888.9 / 400 = 2.2x EBITDA in the base case, but 888.9 / 260 = 3.4x in the stress case. If current debt is $1,200 million (3.0x EBITDA, interest $72 million, base coverage 300 / 72 = 4.17x), then stressed coverage is 160 / 72 = 2.22x, and stressed leverage is 1,200 / 260 = 4.6x. The existing capital structure fails the test, even though base-case metrics look acceptable.

Interpretation. The stress-constrained debt capacity is about $890 million. The company should treat that as a ceiling for sustained debt, not as a target, and leave a margin of unused capacity for acquisitions or shocks. This is exactly the sort of reasoning that distinguishes a defensible target from a simple copy of a competitor's ratio.

Practical Pitfalls

- *Copying peers.* Peer averages embed different business mixes, tax positions, and histories. Use them as a reference, not a justification.
- *Optimizing to the day's cheapest funding.* Issuing debt because rates are low, or equity because the stock price is high, is market timing; it should be consistent with a long-run target.
- *Ignoring non-financial claims.* Operating leases, pension deficits, guarantees, and contingent liabilities add to effective leverage.
- *Ignoring maturity structure.* A debt stack with a large refinancing wall in a single year is far riskier than the same total spread over many years. Choosing maturities, fixed versus floating, and currency are part of the capital structure decision.
- *Forgetting the equity side.* A buyback to reach the target leverage reduces shareholders' equity and book value, which can affect covenants and ratings; and shareholders' needs for dividends or growth signal may constrain the choice.
- *Treating the target as static.* Business risk changes with strategy, so the target should be reviewed at least annually and after major transactions.

Case Study: Brightwater Packaging’s Recapitalization Debate

Illustrative case; the company and all figures are fictional.

Brightwater Packaging manufactures corrugated and specialty packaging for consumer goods customers. It generates EBITDA of $240 million on revenue of $1.6 billion, with depreciation of $60 million, and pays a 25 percent tax rate. The firm has $360 million of debt at an average 5.5 percent cost, a market capitalization of $1.8 billion, and $90 million of cash. Net debt to EBITDA is 1.1 times. The company’s stock trades at 9 times EBITDA, in line with peers. Peers have net debt to EBITDA between 2.0 and 3.0 times.

A new CEO, Marcus Delacroix, wants to launch a $450 million leveraged recapitalization: Brightwater would issue $450 million of new debt at 6.5 percent, and use the proceeds for a special repurchase of shares. He argues that the tax shield would add about $112 million of value (25 percent of $450 million), that EPS would increase by double digits, and that the discipline of debt would end the habit of “lazy cash” that led to two marginal acquisitions in the past three years. He also notes that the stock has underperformed its peers, and thinks buying shares now signals confidence.

The CFO, Anjali Rao, is not so sure. Brightwater’s business is stable but not immune to cycles: during the last recession, EBITDA fell 30 percent from peak. About 55 percent of revenue comes from three large customers, who have used their purchasing power to renegotiate prices in tough times. Moreover, the company has identified a promising $300 million investment in a new line of recyclable fiber-based packaging, expected to earn returns well above its cost of capital, which will need funding in 18 months. The new debt would take net debt to EBITDA from 1.1 to 3.0 times ((360 + 450 − 90) / 240 = 3.0) and push stressed coverage uncomfortably low.

She builds a simple stress test. In a 30 percent downturn, EBITDA would be $168 million and EBIT $108 million. Interest on the enlarged debt of $810 million, assuming the whole stack reprices at 6.5 percent, is about $52.7 million, so stressed coverage would be 2.05 times, down from the current stressed coverage of 108 / (0.055 × 360 = 19.8) = 5.45 times. The recapitalization would leave almost no capacity for the $300 million expansion without issuing equity, which, given the information asymmetries between management and the market, might be costly.

On the other side, Anjali accepts that Brightwater is under-levered relative to its stable cash flows and tangible, redeployable assets, and that the board has been too relaxed about the use of free cash flow. She proposes a middle path: a target range of 1.5 to 2.0 times net debt to EBITDA, a moderate $250 million debt-financed repurchase over twelve months, a committed revolving credit facility sized for the expansion, and a policy requiring that all acquisitions exceeding $100 million be justified with explicit hurdle rates and post-audited.

The board must choose.

Discussion Questions

1. Using Proposition I with taxes, evaluate the CEO’s claim of a $112 million gain. What assumptions underlie it, and how might personal taxes, distress costs, and the loss of flexibility reduce the net benefit?
2. Compute net debt/EBITDA and stressed coverage under the CEO’s proposal and under the CFO’s compromise ($250 million of new debt at 6.5 percent on the incremental amount only). Which plan better satisfies a 3.0x stressed coverage criterion?
3. Explain the CEO’s “free cash flow” argument in terms of Jensen’s hypothesis. Is Brightwater a good candidate for it? What evidence from the case supports or contradicts the argument?
4. Discuss how the planned $300 million expansion and the information asymmetry between management and investors affect the choice. Which theory (trade-off, pecking order, agency) is most relevant, and why?

Key Takeaways

- Modigliani-Miller Proposition I says that, in a frictionless market, the mix of debt and equity does not change firm value, because investors can replicate or undo corporate leverage on their own account.
- Proposition II says the cost of equity rises linearly with the debt-to-equity ratio, so cheaper debt is exactly offset by riskier equity and the WACC is unchanged.
- With corporate taxes, the interest tax shield adds $\tau \times D$ to firm value for permanent debt; the cost of equity still rises with leverage, but more slowly, so WACC declines.
- Personal taxes reduce the net advantage of debt: the effective gain per dollar of debt is $1 - (1 - \tau_C)(1 - \tau_E)/(1 - \tau_D)$, often far below the corporate rate.
- Unlever comparable betas and relever at the target capital structure; always check that the cost of debt used matches the leverage assumed.
- Financial distress imposes direct costs (fees) and larger indirect costs (lost customers, employees, underinvestment); the trade-off theory sets debt where the marginal tax shield equals the marginal expected distress cost, and the value curve is flat near the optimum.
- Under asymmetric information, equity issuance conveys bad news; the pecking order (retained earnings, debt, equity) and the value of financial slack follow from this.
- Debt creates agency conflicts (asset substitution, debt overhang) but also disciplines managers with excess free cash flow; the balance depends on growth opportunities and asset tangibility.
- EBIT-EPS analysis shows earnings sensitivity to financing but is not a value criterion; the P/E and risk change with leverage.
- A practical target is a range based on business risk, stressed coverage, rating objectives, and strategic flexibility, reviewed regularly, and not a copy of peers.

Review Questions

1. State the assumptions underlying the no-tax MM propositions. Which are the least realistic, and what real-world factors correspond to each failure?
2. Explain homemade leverage. Why does it imply that a firm cannot create value simply by changing its debt-equity mix?
3. Why does the cost of equity rise with leverage, even though debt is cheaper? What is the relationship to the beta of equity?
4. Why is the WACC lower for a levered firm when corporate taxes exist? Derive the expression $WACC = r_U \times (1 - \tau \times D/V)$ under permanent debt.
5. What assumptions are required for the value of the tax shield to equal $\tau \times D$, and how would the value differ if the firm maintained a constant debt-to-value ratio?
6. Distinguish between direct and indirect costs of financial distress. Which industries would you expect to have the highest indirect costs, and why?
7. Why does the trade-off theory predict an interior optimum? What empirical regularity (about profitability and leverage) is hard for it to explain?
8. Describe the pecking order. Why might a firm with a profitable investment opportunity decline to issue equity?

9. Define asset substitution and debt overhang. How do covenants address each, and what are their costs?
10. A CFO argues that issuing debt to repurchase shares will raise EPS and therefore the stock price. Evaluate the argument using the MM framework and the EBIT-EPS analysis.

Problems

1. A firm has perpetual EBIT of $2,400 per year, an unlevered cost of capital of 12%, and no taxes. It borrows $6,000 permanently at 6%. Compute the firm value, equity value, and cost of equity. Verify the cost of equity from the cash flows.
2. Repeat Problem 1 with a 30% corporate tax rate. Compute V_U, V_L, E, r_E, and the WACC.
3. A comparable firm has an equity beta of 1.40, D/E of 0.80, and a 21% tax rate. Estimate the unlevered beta, and the relevered equity beta for a target D/E of 0.25. With a risk-free rate of 4% and a market risk premium of 5.5%, compute the cost of equity at the target.
4. V_U = $5,000, τ = 30%, and distress costs equal 20% of V_U ($1,000). Probabilities of distress at debt of $0, 500, 1,000, 1,500, 2,000, 2,500, 3,000, and 3,500 are 0%, 1%, 3%, 7%, 14%, 25%, 45%, and 65%. Compute V_L at each level and identify the optimal debt.
5. A company with 20 million shares at $25 needs $150 million. Plan A issues new shares at $25; Plan B issues 7% debt. The tax rate is 25% and there is no other debt. Find the indifference EBIT and the EPS at that point.
6. A firm owes $80 due next year. Project A yields a certain $100. Project B yields $160 or $20 with equal probability. Compute the expected payoffs to debt and equity under each project. Which project do shareholders prefer, and which maximizes firm value?
7. A firm has EBITDA of $500 million, D&A of $120 million, a borrowing cost of 7%, a minimum stressed coverage (EBIT/interest) of 3.0x, and a stress scenario with EBITDA down 30%. Find the maximum debt supported by the coverage criterion, and compare with a ceiling of 2.5 times base EBITDA.
8. With τ_C = 21%, τ_E = 12%, and τ_D = 32%, compute Miller's net tax advantage of debt per dollar and for $1,000 of debt. Interpret.

Answers to Selected Problems

1. V = 2,400 / 0.12 = $20,000; E = 14,000; r_E = 12% + (6,000 / 14,000)(12% − 6%) = 14.57%; check: (2,400 − 360) / 14,000 = 14.57%.
2. V_U = 2,400 × 0.7 / 0.12 = $14,000; V_L = 14,000 + 0.3 × 6,000 = $15,800; E = $9,800; r_E = 12% + (6,000/9,800)(0.7)(6%) = 14.57%; WACC = 12% × (1 − 0.3 × 6,000/15,800) = 10.63%.
3. β_U = 1.40 / (1 + 0.79 × 0.80) = 0.858; β_E(target) = 0.858 × (1 + 0.79 × 0.25) = 1.027; r_E = 4% + 1.027 × 5.5% = 9.65%.
4. V_L: $5,000, 5,140, 5,270, 5,380, 5,460, 5,500, 5,450, 5,400 for D = 0 through 3,500. The optimum is D = $2,500 with V_L = $5,500.
5. New shares under Plan A: 6 million (total 26 million); interest under Plan B: $10.5 million. Indifference EBIT = $45.5 million; EPS = $1.3125 under both.
6. Under A: debt $80, equity $20. Under B: debt = 0.5 × 80 + 0.5 × 20 = $50; equity = 0.5 × 80 + 0 = $40; total $90. Shareholders prefer B ($40 > $20) although A has greater firm value ($100 vs $90).

7. Stressed EBITDA = $350 million; stressed EBIT = $230 million; max interest = $76.7 million; max debt = $1,095 million. The 2.5x ceiling gives $1,250 million, so the coverage criterion is binding.
8. G = 1 − (0.79 × 0.88)/0.68 = −0.0224, or −2.2 cents per dollar; for $1,000 of debt, −$22.35. The personal tax cost on interest outweighs the corporate shield, so debt has a net disadvantage at these rates.

Chapter 16: Payout Policy

Learning Objectives

- Describe the mechanics, timeline, and legal constraints of cash dividends, share repurchases, special dividends, and stock dividends.
- Prove the Modigliani-Miller dividend irrelevance proposition using homemade dividends, and identify the assumptions whose failure makes payout policy matter.
- Compare dividends and repurchases in terms of flexibility, taxation, EPS effects, and value transfers between selling and continuing shareholders.
- Explain tax-based clienteles, the ex-dividend price drop, and the signaling and agency theories of payout, including Lintner's partial-adjustment model.
- Distinguish stock splits and stock dividends from cash payouts and analyze their accounting and valuation effects.
- Design a payout policy in practice by combining the residual approach, sustainable growth, target payout ranges, and a plan for returning surplus cash.

Opening Vignette

Illustrative scenario; the company is fictional.

Calder Semiconductor Equipment designs and manufactures specialized tools for chip fabrication. After a strong three-year expansion, the company has $1.2 billion of cash on its balance sheet, generates about $600 million of free cash flow a year, and expects to need roughly $350 million annually for capital spending and research. The company has never paid a dividend. Its CEO, Tomas Reyes, believes the stock is undervalued after a sector sell-off. Its CFO, Elena Park, is under pressure from three directions at once.

An activist fund holding 4 percent of the shares has published a letter demanding that Calder "return excess capital" through a $1 billion buyback and a regular dividend. A long-term pension fund, one of the largest holders, has written privately that it prefers a dividend because its mandate requires current income. And Calder's head of strategy reminds the board that the semiconductor equipment industry is deeply cyclical: orders fell 40 percent in the last downturn, and the firm's rivals that had distributed cash aggressively were forced to cut payouts and borrow at punitive rates when business slowed.

Elena has to answer a set of questions that sound simple but are not. Should Calder pay cash at all, or keep funding growth? If it pays, should it be a dividend, which investors read as a commitment, or a repurchase, which is flexible but may be bought at the wrong price? Does the choice matter for the share price, or merely for who is happy? How does the answer depend on taxes, on what investors infer from the announcement, and on how management would otherwise spend the cash?

This chapter develops a framework for those questions. We begin with the mechanics, then present the benchmark irrelevance result of Modigliani and Miller, and then work through the frictions (taxes, clienteles, information, agency problems) that make payout policy a real management decision. We close with practical guidance for setting a policy and a case in which a board must choose.

16.1 The Payout Decision and Its Mechanics

A firm that earns more cash than it can profitably reinvest must eventually return it to its owners. *Payout policy* is the set of decisions about how much to distribute, in what form, and on what schedule. Along with investment and financing decisions, it is one of the three core decisions of corporate finance. The link is the cash flow identity: free cash flow to equity equals dividends plus repurchases minus net share issuance. Given a firm's investment and borrowing plans, the payout is fixed; to pay more, the firm must invest less, borrow more, or issue shares. This is why payout cannot be treated in isolation.

Forms of Payout

Form	How it works	Typical features
Regular cash dividend	Declared by the board, paid per share each quarter (or half-year)	Sticky; cuts are rare and read as bad news
Special (extra) dividend	One-time distribution declared separately	No expectation of repetition
Open-market repurchase	Firm buys its own shares on the exchange over time	Flexible; no obligation to complete
Tender offer / accelerated repurchase	Firm offers to buy a stated number of shares at a stated price or range	Faster, more visible, usually a premium
Stock dividend / stock split	Additional shares distributed pro rata	No cash leaves the firm
Dividend in kind / spin-off	Distribution of assets or shares of a subsidiary	Restructures the business

Many firms also operate a *dividend reinvestment plan* (DRIP), which lets shareholders use dividends to buy additional shares, sometimes at a discount.

The Dividend Timeline

Four dates matter for a cash dividend.

- *Declaration date.* The board announces the amount, the record date, and the payment date. From this moment the dividend is a legal liability of the firm.
- *Ex-dividend date.* Shares trade without the right to the dividend from this date. In markets with standard settlement cycles, the ex-date is set shortly before the record date.
- *Record date.* Holders on the company's books receive the dividend.
- *Payment date.* The cash is actually paid.

The price of a share should fall around the ex-dividend date by approximately the value of the dividend to the marginal investor (Section 16.4), since a buyer on or after that date no longer receives it.

Legal and Contractual Constraints

Corporate law places limits on distributions, intended to protect creditors. Typically, dividends and buybacks cannot be paid from capital that impairs the firm's ability to pay its debts as they fall due, and some jurisdictions restrict payouts to the extent of retained earnings or distributable reserves. Loan covenants often limit dividends and share repurchases explicitly: for instance, by tying them to a percentage of net income or a leverage threshold. A manager must check the legal capacity, the covenants, and the liquidity position before announcing any payout.

Trends in Payout Form

In many developed markets, repurchases have grown considerably relative to dividends over recent decades. Two reasons are commonly cited: repurchases offer flexibility (they can be scaled back without the negative signal of a dividend cut), and in many tax systems they receive better tax treatment than dividends for some shareholders. Dividends nevertheless remain the foundation of payout for mature, stable firms, and many firms use a combination: a regular dividend sized to be sustainable through a downturn, supplemented by repurchases of variable size.

16.2 Dividend Irrelevance: Modigliani and Miller

In 1961 Miller and Modigliani showed that, in a perfect capital market, the value of a firm is determined by its investment policy (the cash flows from its assets) and not by how it splits earnings between dividends and retained earnings. The argument is closely related to the capital structure irrelevance of Chapter 15.

The Assumptions

The result holds when there are no taxes, no transaction or issuance costs, no information asymmetries, and no agency problems, and when the firm's investment policy is fixed and independent of its payout. Fixing the investment policy is crucial. The proposition does not say that retaining cash and then wasting it on bad projects is harmless; it compares payout alternatives for *given* investments.

The Logic

If a firm decides to pay a larger dividend than its free cash flow, it must raise the shortfall by issuing new shares (or debt), whose value dilutes the existing shareholders by exactly the amount of the extra dividend received. If it pays less, the retained cash adds to firm value and so to the share price in the same measure. The shareholder receives either cash or share price appreciation, and in a perfect market these are worth exactly the same. Moreover, shareholders can create any pattern of cash flows they want by selling shares or reinvesting dividends, which is the *homemade dividend* argument.

Example 16.1 – Homemade dividends. A firm with 1,000,000 shares has equity worth $50,000,000 ($50 per share) and a planned $5.00 per share cash payout ($5,000,000 in total). Alternatively, it could pay no dividend. Consider investors with different preferences, and no taxes or transaction costs.

Case A: An investor who wants cash income but the firm pays nothing. The investor owns 1 share at $50. To raise $5.00, the investor sells 5/50 = 0.10 share. Remaining holding: 0.90 share × $50 = $45, plus $5.00 cash. Total wealth: $50.

Case B: An investor who wants no cash but the firm pays the $5.00 dividend. After the dividend the share trades ex-dividend at $50 − $5 = $45. The investor reinvests the $5 by buying 5/45 = 0.1111 additional share. New holding: 1.1111 shares × $45 = $50. Total wealth: $50.

Conclusion. Whether the firm pays $5.00 or nothing, an investor can get the same position (cash plus shares worth $50) by trading on their own account. Because a shareholder can achieve any preferred cash pattern without cost, the firm's choice is irrelevant to value. If trading costs, taxes, or information frictions made the homemade strategy costly, the firm's policy could matter; that is the topic of the rest of the chapter.

The Dividend Is Not Free Cash in the Valuation Sense

A common source of confusion is the dividend discount model, which values a share as the present value of all future dividends, and which seems to imply that higher dividends mean higher value. The resolution is that the model discounts *all* future dividends, including those funded by retained cash. Raising dividends today by issuing shares reduces future

dividends per share exactly enough to leave the present value unchanged. Consider a firm that can only fund a larger dividend by issuing new shares; the existing owners receive cash today but own a smaller fraction of future dividends. The model confirms irrelevance, rather than contradicting it.

What Matters if Not Dividends

If payout form is irrelevant in the perfect market, then what drives value? The answer is the quality of the investment decisions. Returning cash to shareholders when the firm lacks positive-NPV projects is value-increasing relative to retaining it and misusing it. The payout question is therefore inseparable from the question: *what is the opportunity cost of retained cash?* The relevant comparison is the return the firm can earn on the cash against the return shareholders could earn on equivalently risky investments elsewhere. A firm that cannot earn its cost of capital on retained earnings should pay them out.

16.3 Dividends versus Repurchases

In the MM world, a repurchase is equivalent to a dividend: a shareholder who wanted cash would sell shares in the buyback; one who did not could simply keep holding, and the price per share would be unaffected. Cash leaves the firm, the number of shares falls, and the value of each remaining share is unchanged. Differences between the two arise from frictions, but understanding the neutral case is essential to avoid the most common fallacies about buybacks.

The EPS Fallacy

Repurchases reduce the share count, so EPS rises mechanically, and many managers and commentators describe this as a benefit. It is not necessarily so. The money used to repurchase shares was earning something (interest on cash) or could have been used to retire debt. The reduction in earnings is smaller than the proportional reduction in shares only if the earnings yield on the stock exceeds the after-tax return on the cash spent. Even then, the P/E multiple adjusts to reflect the changed risk and growth; the value per share is unaffected unless something else changes.

Example 16.2 – Dividend versus repurchase at a fair price. Halden Instruments has 100 million shares trading at $40, net income of $200 million (EPS $2.00, P/E 20), and equity value $4,000 million. It will distribute $100 million to shareholders. Cash earns 4 percent before tax; the tax rate is 25 percent. Assume the stock is fairly priced.

Option 1: Dividend. Dividend per share = 100 / 100 = $1.00. After the ex-date the equity value is 4,000 − 100 = $3,900 million, so the price is 3,900 / 100 = $39.00. Shareholders hold a share worth $39 plus $1.00 cash = $40. If the cash was earning interest, the foregone after-tax interest is 100 × 0.04 × 0.75 = $3 million, so net income falls to $197 million and EPS to $1.97.

Option 2: Repurchase. Shares bought = 100 / 40 = 2.5 million. Remaining shares = 97.5 million. Equity value after = 4,000 − 100 = $3,900 million, so the price is 3,900 / 97.5 = $40.00. The price is unchanged, as MM predicts, and a seller receives $40 per share while continuing holders keep $40.

EPS comparison. Ignoring foregone interest, EPS after the buyback = 200 / 97.5 = $2.051 (up 2.6 percent). Accounting for the $3 million of lost after-tax interest, net income is $197 million and EPS = 197 / 97.5 = $2.0205 (up 1.0 percent). The P/E ratio falls from 20.0 to 40 / 2.0205 = 19.8, and the value per share is the same $40. A buyback “raises EPS” without increasing value.

Interpretation. The apparent EPS accretion is an arithmetic artifact. What matters is whether the buyback is done at a price below intrinsic value (a real benefit to continuing shareholders) or above it (a transfer to sellers), as the next example shows.

Value Transfers When Shares Are Mispriced

If the firm's managers believe the market price differs from intrinsic value, a repurchase transfers wealth between those who sell and those who stay. Buying below intrinsic value benefits continuing shareholders at the expense of sellers; buying above intrinsic value does the opposite. Managers who maximize the wealth of long-term holders should therefore repurchase when they think shares are undervalued and avoid it when they are overvalued, subject to two cautions. The first is that managers' views of undervaluation are not always correct, and the second is that selling shareholders who are harmed are also shareholders (some of whom are long-term holders who sell for liquidity reasons).

Example 16.3 – Buying back undervalued and overvalued shares. Halden's intrinsic value, estimated by management, is \$50 per share (equity value \$5,000 million) before a \$100 million repurchase.

(a) Market price \$40 (undervalued). Shares retired = 100 / 40 = 2.5 million, leaving 97.5 million. True value after = 5,000 − 100 = \$4,900 million. Intrinsic value per remaining share = 4,900 / 97.5 = \$50.26. Continuing shareholders gain \$0.26 per share (0.5 percent); sellers received \$40 for shares worth \$50, losing \$10 each.

(b) Market price \$60 (overvalued). Shares retired = 100 / 60 = 1.667 million, leaving 98.333 million. Intrinsic value per remaining share = 4,900 / 98.333 = \$49.83. Continuing shareholders lose \$0.17 per share; sellers gained \$10 each.

Interpretation. The gain to continuing shareholders is small relative to the gain or loss to sellers, because only a small fraction of shares is retired. The real consideration for a buyback is rarely clever timing; it is whether the firm has surplus capital, and whether the return of that capital is preferable to other uses.

Flexibility, Signals, and Commitment

Feature	Regular dividend	Open-market repurchase
Commitment	High; cuts are penalized by markets	Low; programs can be slowed or stopped
Signal content	Confidence in stable future earnings	Possible undervaluation; weaker as a commitment
Investor tax treatment	Taxed on receipt (rate varies by jurisdiction)	Taxed on gain only for selling holders, timing at holder's choice
Suitable for	Stable, mature cash flows	Variable cash flows; one-off surpluses
Effect on share count	None	Reduces shares
Risk of management misuse	Low	Buying high, funding by debt at the top of a cycle; offsetting option dilution
Effect on insiders	Neutral	Can raise value of options and EPS-linked bonuses

A significant share of repurchases are used merely to offset the dilution from employee stock options and grants. In such cases the net reduction in share count is zero or small, and the cash outflow is, in substance, a labor cost. Analysts should compute *net* repurchases (gross buybacks less issuance) and the share count trend rather than headline buyback announcements.

16.4 Taxes and Clienteles

Taxes are the most commonly cited friction in payout policy. Their effect depends on the specific tax system, but several general principles apply.

The Tax Disadvantage of Dividends

Where dividends are taxed at a higher rate than capital gains, or where gains are taxed only on realization, an extra dollar of dividends is worth less after tax than an extra dollar of capital gain. A firm paying a dividend pushes shareholders to realize taxable income immediately, while buybacks allow investors to choose when and whether to realize gains (and to postpone the tax indefinitely by never selling). Even if rates are equal, deferral has value. Tax-exempt institutions and some corporate shareholders in some jurisdictions are indifferent or even favor dividends. This predicts that high-tax investors prefer low payouts and repurchases.

The Ex-Dividend Price Drop

If investors' marginal tax rate on dividends is τ_D and on capital gains is τ_G, then in equilibrium, the price should fall on the ex-dividend date by less than the full dividend if dividends are taxed more heavily. The marginal investor is indifferent between holding the stock through the ex-date and selling it just before:

```
(P_cum − P_ex) × (1 − τ_G) = D × (1 − τ_D)
Price drop = D × (1 − τ_D) / (1 − τ_G)
```

Example 16.4 – The ex-dividend drop. A stock trades cum-dividend at \$50, and pays a dividend of \$2.00 per share. The marginal investor faces a 30 percent tax on dividends and a 15 percent tax on capital gains.

Step 1: After-tax dividend. 2.00 × (1 − 0.30) = \$1.40.

Step 2: Equivalent pre-tax capital loss. The investor would be indifferent if the after-tax value of the lost price equals the after-tax dividend: drop × (1 − 0.15) = 1.40, so drop = 1.40 / 0.85 = \$1.647.

Step 3: Ex-dividend price. 50 − 1.647 = \$48.35.

Interpretation. The price falls by 82 percent of the dividend, not 100 percent. If the tax rates were equal, the drop would be \$2.00. A drop that is smaller than the dividend is consistent with dividend-taxed investors being the marginal price setters, but the empirical evidence is mixed because transaction costs, short-term traders, and tax-exempt arbitrageurs also influence prices around the ex-date. The key lesson is that the ex-dividend price drop and the after-tax value of the dividend are linked through the tax rates of the marginal investor, so the drop is a way of reading what that investor thinks of dividends.

Clienteles

If investors differ in their tax rates and preferences, they will sort into groups, or *clienteles*, each choosing firms whose payout policy suits them. High-tax individuals favor low-payout stocks; pension funds, endowments, and retirees needing income favor high-payout stocks. In equilibrium, supply of payout levels from firms matches the demand from clienteles, and a firm that changes policy simply swaps one clientele for another, at some transaction cost. This is a variation of the MM irrelevance: if clienteles are fully accommodated, no firm can create value by changing its payout. However, the changeover has costs: when a firm introduces or cuts a dividend, it may trigger selling by dissatisfied investors and temporary price pressure.

Investor type	Likely preference	Reason
High-income individual in a jurisdiction taxing dividends heavily	Low payout, repurchases	Defers tax on gains
Retiree needing cash flow	Dividends	Avoids selling costs; psychological preference for spending income not principal
Tax-exempt pension fund	Indifferent on tax; may prefer dividends for liability matching	No tax disadvantage
Corporate investor with dividend exclusion	Dividends	Partial tax exemption on dividends received
Growth fund	Low payout	Wants reinvestment of earnings
Trust/endowment with spending rule	Dividends or total return	Spending rules often tied to income

Many retirees say they prefer "living off the dividends." In the perfect market this is a psychological preference rather than an economic one, since selling a bit of stock produces the same cash. Behavioral explanations (self-control, mental accounting) and institutional rules (trusts that may spend income but not principal) give the preference some force in practice.

16.5 Signaling, Agency Costs, and Dividend Smoothing

Taxes explain why payout might be undesirable. Other theories explain why firms pay anyway, and why they pay in the particular patterns we observe.

Signaling

Managers know more about the firm's prospects than outsiders do. A dividend increase is a costly commitment: if future earnings disappoint, the firm must either cut the dividend (which is punished) or find external funds. Only firms confident in future cash flow can afford to raise the dividend, so an increase *signals* good news, and the stock tends to rise at announcement. Cuts and omissions generally signal the opposite and are accompanied by price declines. The classic models of Bhattacharya, and of Miller and Rock, formalize this idea. The implication for managers is that the signal depends on the *surprise* relative to expectations, and that dividend policy has an informational role that extends beyond the cash itself.

Repurchases also carry information. An open-market program is a weaker commitment than a dividend and so a weaker signal. Tender offers at a premium, which require the firm to buy a fixed number of shares, are stronger signals of management's belief that the stock is undervalued. Because signaling requires cost to be credible, there is also a limit: if every firm announced buybacks as signals without following through, the signal would lose meaning, and investors have learned to look at actual completion rates.

Agency Costs of Free Cash Flow

Jensen's free cash flow argument, introduced in Chapter 15, applies equally to payout. When managers retain cash that exceeds profitable investment opportunities, they may spend it on pet projects, empire-building acquisitions, or perquisites. Paying cash out reduces the resources under management's control and forces the firm to return to capital markets when it needs funds, subjecting its plans to external scrutiny. Seen this way, a commitment to pay dividends is a bonding device, and the predicted gains are largest for mature firms with few growth options and substantial surplus cash. Evidence suggests that the market responds more positively to payout increases by firms with poor investment opportunities, consistent with this theory, and that firms with weak governance hold cash that is valued at a discount by investors.

There is a conflict on the other side as well. Shareholders can use high payouts to transfer wealth from creditors: paying a large dividend reduces the assets backing the debt. Bondholders protect themselves with covenants, and aggressive payouts by highly leveraged firms can raise borrowing costs. Managers therefore balance the demands of shareholders for cash against the claims of lenders.

Lintner's Model of Dividend Smoothing

Lintner's 1956 survey of managers found that:

- Firms think about dividends in terms of a long-run target payout ratio.
- Managers are reluctant to cut dividends and raise them only when they believe the increase is sustainable.
- As a consequence, dividends adjust only partially toward the target, and are much smoother than earnings.

This is summarized by a partial adjustment model:

```
D_t = D_{t−1} + s × (p × E_t − D_{t−1})
```

where D_t is the dividend this period, E_t is earnings (or EPS), p is the target payout ratio, and s is the speed of adjustment ($0 < s \le 1$). A smaller s implies a smoother policy. Despite decades of structural change in how firms pay out, smoothing remains a robust feature of dividend behavior, particularly among mature firms.

Example 16.5 – Lintner smoothing. A firm has a target payout ratio of 40 percent, an adjustment speed s = 0.30, and last year's dividend of \$1.60 per share, which corresponded to EPS of \$4.00 (a 40 percent payout). This year EPS jumps to \$5.00; next year it rises to \$5.50 and then \$6.00.

Year 1. Target dividend = 0.40 × 5.00 = \$2.00. Adjustment = 0.30 × (2.00 − 1.60) = \$0.12. D_1 = 1.60 + 0.12 = \$1.72. Actual payout = 1.72 / 5.00 = 34.4%.

Year 2. Target = 0.40 × 5.50 = \$2.20. Adjustment = 0.30 × (2.20 − 1.72) = 0.144. D_2 = 1.72 + 0.144 = \$1.864. Payout = 1.864 / 5.50 = 33.9%.

Year 3. Target = 0.40 × 6.00 = \$2.40. Adjustment = 0.30 × (2.40 − 1.864) = 0.1608. D_3 = 1.864 + 0.1608 = \$2.0248. Payout = 33.7%.

Interpretation. Over three years, EPS rises 50 percent (from \$4.00 to \$6.00) while the dividend rises 26.5 percent (from \$1.60 to \$2.02). The payout ratio falls below the target of 40 percent because the dividend lags the earnings growth. Conversely, if earnings fell, the dividend would stay high and the payout ratio would exceed the target temporarily. The smoothing explains why payout ratios swing more than dividends and why a dividend cut is so informative: management would only make it when the earnings shortfall appears permanent.

16.6 Stock Splits and Stock Dividends

Stock splits and *stock dividends* distribute additional shares to existing holders in proportion to their holdings. No cash leaves the firm, and the company's assets and earnings power are unchanged, so in a frictionless market the total value of a shareholder's position does not change.

Mechanics and Accounting

In a *stock split* (for example, 2-for-1), the number of shares doubles and the par value (if any) per share halves; the equity accounts are unchanged. In a *stock dividend* (for example, 10 percent), the firm issues new shares worth 10 percent of the outstanding amount, and under conventional accounting treatment for small stock dividends (usually below about 20 to 25 percent of shares outstanding) transfers the market value of the new shares from retained earnings to common stock and paid-in capital. Larger distributions are treated like splits and only change the par value and share count. Local accounting rules vary, and managers should check the applicable standard.

Example 16.6 – Split and stock dividend. Pendleton Holdings has 100 million shares trading at $80 (market capitalization $8,000 million), common stock at $1 par.

(a) 2-for-1 split. Shares become 200 million, par becomes $0.50, and the price should fall to 8,000 / 200 = $40. A holder of 1,000 shares ($80,000) now owns 2,000 shares worth $40 each, still $80,000. Total equity on the balance sheet is unchanged.

(b) 10% stock dividend (instead of the split). New shares = 10 million. At a market price of $80, the accounting transfer is 10 million × $80 = $800 million from retained earnings: $10 million to common stock (par $1) and $790 million to additional paid-in capital. Shares become 110 million, and the price should be about 8,000 / 110 = $72.73. A holder of 1,000 shares now has 1,100 shares at $72.73 = $80,000.

(c) Reverse split. A 1-for-4 reverse split on a $4 stock reduces the share count by four and raises the price to $16, with no change in total value.

Why Do Firms Split Shares?

Since splits create no cash and change no cash flow, they should not matter in theory. Three explanations are offered for why firms do them:

- *Trading range.* Managers believe there is an optimal price range for the shares, which makes them accessible to small investors and increases liquidity. Evidence on liquidity is mixed: splits often reduce price but increase the proportional bid-ask spread, in some markets.
- *Signaling.* Splits are costly in terms of fees and typically are done by firms whose stock has risen strongly, so they are taken to signal confidence about future earnings. Announcement returns are positive on average, but much of this reflects the earnings and dividend information accompanying the split.
- *Dividend commitments.* A split often accompanies a promise to maintain the per-share dividend, which implicitly increases the total dividend.

Stock dividends and splits are not cash payouts and are not substitutes for them. A stock dividend leaves shareholders with the same economic stake, whereas a cash dividend or repurchase returns value. Managers sometimes announce stock dividends as a way of “rewarding shareholders” without cash; the right response is to ask what they actually accomplish.

16.7 Payout Decisions in Practice

Surveys of Managers

Surveys of financial executives consistently report a set of priorities. Managers say that maintaining the existing dividend per share is a very high priority, that they are reluctant to cut it, and that they would forego positive-NPV investments before cutting it in extreme cases. They say repurchase decisions are made after investment needs are met and are treated as

more flexible, and that the level of payout is influenced by the stability of earnings, the firm's desire to preserve financial flexibility, and market conditions. Few managers cite tax considerations as the dominant factor, which is notable given how large they loom in theory. The practical consequence is that dividends are treated like fixed charges and repurchases like a residual.

The Residual Dividend Approach

The *residual dividend policy* treats dividends as what is left after the firm has funded all positive-NPV investments while maintaining its target capital structure. The steps:

1. Determine the capital budget (the set of positive-NPV projects).
2. Choose the equity portion of that budget using the target capital structure.
3. Fund that equity portion with retained earnings.
4. Pay out any remaining earnings as dividends (or repurchases).

Example 16.7 – Residual dividend. Marlowe Industries has a target capital structure of one-third debt and two-thirds equity (D/V = 1/3). In Year 1, planned capital spending is $300 million and net income is $260 million.

Year 1. Equity needed = 300 × (2/3) = $200 million. Dividend = 260 − 200 = $60 million (payout ratio 60 / 260 = 23.1%). The remaining $100 million of investment is financed with new debt.

Year 2. Capital spending rises to $450 million with net income still $260 million. Equity needed = 450 × (2/3) = $300 million. Retained earnings of $260 million fall short by $40 million. Dividend = 0, and the firm must either issue $40 million of new equity or let leverage rise temporarily above target (with more borrowing).

Interpretation. A strict residual policy produces dividends that vary wildly from year to year (60 and then 0), which conflicts with managers' strong preference for smoothness and with the signaling consequences of cuts. In practice, firms use the residual idea over a multi-year horizon: they plan average investment and average earnings and set a *sustainable* dividend that they can maintain through the cycle, with repurchases or special dividends absorbing the surplus in good years. Repurchases fit the residual role well.

Payout and Growth: Sustainable Growth and Value

Payout policy also determines growth, because retained earnings fund reinvestment. With ROE constant and no external equity, the *sustainable growth rate* is:

```
g = ROE × (1 − payout ratio) = ROE × b
```

where b is the retention ratio. Value depends on whether the reinvested funds earn more than the cost of equity.

Example 16.8 – When retention creates value. A firm expects next-year EPS of $4.00 and has a cost of equity of 11 percent. Compare two payout ratios under two different returns on equity, using the constant-growth model P = D_1 / (r − g).

Case 1: ROE = 15%.

```
Payout 40%: g = 0.15 × 0.60 = 9.0%;  D_1 = 1.60;  P = 1.60 / (0.11 − 0.09) = $80.00
Payout 70%: g = 0.15 × 0.30 = 4.5%;  D_1 = 2.80;  P = 2.80 / (0.11 − 0.045) = $43.08
```

Case 2: ROE = 11% (equal to the cost of equity).

```
Payout 40%: g = 0.11 × 0.60 = 6.6%;  D_1 = 1.60;  P = 1.60 / (0.11 − 0.066) = $36.36
Payout 70%: g = 0.11 × 0.30 = 3.3%;  D_1 = 2.80;  P = 2.80 / (0.11 − 0.033) = $36.36
```

Interpretation. When the firm earns more than the cost of equity on retained funds (Case 1), retention is worth more to shareholders than payout: price falls from $80 to $43 when payout is raised, because the firm would be forced to give up growth that creates value. When ROE equals the cost of equity (Case 2), payout is irrelevant, exactly as MM predicts: both policies give $36.36. If ROE were below 11 percent, higher payout would raise value. The message: *the payout decision is an investment decision in disguise.* Whether to pay out depends on the firm's marginal return on reinvestment, not on how investors feel about dividends.

A Life-Cycle View

Stage	Investment opportunities	Cash generation	Typical payout approach
Start-up / early growth	Abundant	Negative or low	No payout; external financing
High growth	Strong	Moderate	Little or no dividend; retain for expansion
Maturity	Moderate	Strong	Regular dividend, plus repurchases
Decline	Scarce	Strong but shrinking	High payout, special dividends, liquidation of assets

Calder, in the opening vignette, sits between the high-growth and maturity stages, with a cyclical business and large cash balance. Such firms often choose a moderate regular dividend, covered comfortably even in a downturn, and a flexible repurchase program.

Measuring Total Shareholder Yield

Because dividends and repurchases are substitutes, analysis should use total payout. The *total shareholder yield* adds the dividend yield to the net buyback yield:

```
Total yield = (Dividends + Net repurchases) / Market capitalization
```

A firm with a market capitalization of $4,000 million, dividends of $60 million, and net repurchases of $140 million has a total yield of (60 + 140) / 4,000 = 5.0%. The dividend yield alone, 1.5 percent, understates the cash returned to shareholders. A related measure, the *payout ratio relative to free cash flow*, indicates sustainability: a company distributing more than its free cash flow for several years is borrowing, depleting cash, or issuing shares to pay.

A Practical Framework for Setting Policy

1. *Identify surplus capital.* After the capital budget, the strategic reserve (for acquisitions, a downturn buffer), and the target capital structure, how much cash is truly surplus, both now and on average over the cycle?
2. *Decide the base dividend.* Set a regular dividend that can be maintained through a severe downturn from internally generated cash. Express as a payout ratio range and per-share growth aspiration, and treat as a commitment.
3. *Use repurchases for the variable component.* Authorize programs sized to surplus cash, with discretion on timing. Net out issuance, and publish the logic.
4. *Use special dividends for one-time windfalls* (an asset sale, a settlement) that are not expected to recur.

5. *Check the constraints.* Covenants, legal distributable reserves, rating agency tolerance, and the liquidity position under stress.
6. *Communicate.* State the principles (priorities for cash, the dividend's coverage, how buybacks are decided) so that investors do not over-read individual announcements.
7. *Review annually and after major events.*

Common Mistakes

- Funding buybacks with debt at the top of the cycle and then cutting them when credit conditions tighten, effectively buying high and selling low through issuance.
- Setting a dividend based on peak earnings and then cutting it in the downturn, with a double penalty of lower payout and negative signal.
- Using repurchases only to offset dilution and calling it a return of capital.
- Treating EPS accretion as the justification for a buyback.
- Paying out cash while ignoring positive-NPV opportunities; or hoarding cash with no plan, which erodes value through agency costs and invites activist pressure.
- Announcing large authorizations that are never executed, thereby damaging credibility.

Case Study: Calder Semiconductor Equipment Sets a Payout Policy

Illustrative case; the company and all figures are fictional.

Calder Semiconductor Equipment has 200 million shares trading at $60 (market capitalization $12 billion), $1.2 billion of cash, $300 million of debt, and annual free cash flow before payouts that has averaged $600 million over the cycle but ranged from $150 million in the last trough to $900 million at the peak. The firm has never paid a dividend, and earlier in its life it funded growth by retaining all earnings. Its investment plan calls for around $350 million per year in capital and research (already counted in free cash flow) plus $200 million of possible bolt-on acquisitions that management considers attractive but uncertain.

The board has before it three proposals.

Proposal A (the activist). A $1 billion accelerated share repurchase this year, financed from cash and $400 million of new debt, followed by a regular dividend of $0.50 per share ($100 million per year). The activist argues that the cash is idle, that the stock trades at a discount to peers (a P/E of 14 compared with 18), and that management's earlier acquisitions have destroyed value. An accelerated repurchase at $60 would retire about 16.7 million shares, 8.3 percent of the total.

Proposal B (the pension fund). A regular dividend equal to 40 percent of average-cycle earnings, about $240 million a year or $1.20 per share, with no buybacks. The fund argues that a dividend is a firm commitment and provides discipline against empire-building.

Proposal C (management). A modest regular dividend of $0.25 per share ($50 million a year), covered more than three times by even the trough free cash flow of $150 million, together with a flexible repurchase program of up to $300 million a year, executed when the stock trades below management's estimate of intrinsic value and when cash exceeds a minimum liquidity reserve of $800 million. A further special dividend would be considered following major asset sales.

Elena Park, the CFO, notes several facts. First, in the last downturn, two competitors that had committed to large dividends were forced to cut them within a year and saw their share prices fall by more than their peers. Second, repurchase programs in the industry have been poorly timed: the sector's largest buybacks took place near the peak of the cycle. Third, her tax advisors say that a large fraction of Calder's shareholders (taxable individuals) are subject to a higher rate on dividends than on long-term capital gains, while the pension holders are untaxed. Fourth, Calder's research budget gets cut during downturns, which costs the firm its technological lead; a liquidity buffer therefore has strategic value beyond covering fixed charges. Finally, an internal analysis suggests that the average return on the recent acquisitions has been below the firm's cost of capital, supporting the activist's critique.

Elena believes that Proposal A takes on debt at the wrong point in the cycle and that the dividend of $100 million combined with a large buyback leaves too little in a trough: free cash flow of $150 million in the trough would barely cover the dividend and interest on $700 million of debt. Proposal B places too much weight on the dividend. But she sees merit in the activist's governance argument and wants some mechanism of accountability for acquisitions and a public statement of how the buyback will be sized.

Discussion Questions

1. Using MM dividend irrelevance, what assumptions would have to hold for Calder's choice among Proposals A, B, and C to not affect firm value? Which of those assumptions fail in this case?
2. Compute the dividend coverage under the trough free cash flow of $150 million for Proposals B and C. What does this imply about their relative risk of a future cut, and how would a cut be interpreted by the market?
3. A taxable shareholder faces a 30 percent tax on dividends and a 15 percent tax on gains. A pension fund pays no tax. Compare the after-tax value to each of $100 of dividends and $100 of repurchase proceeds that is 60 percent gain, and discuss how clienteles influence the choice.
4. Evaluate management's rule of buying back only when the stock is below intrinsic value. What are the risks of this rule, and what governance devices could make it credible to investors?

Key Takeaways

- In a perfect market, the value of the firm depends on its investment policy, not its payout policy; shareholders can create any payout pattern by trading (homemade dividends).
- Payout is tied to the cash flow identity: paying more means investing less, borrowing more, or issuing shares, so the right question is the return on retained cash compared with shareholders' alternatives.
- Dividends and repurchases are close substitutes; a buyback raises EPS mechanically, but value per share changes only through mispricing or taxes, so EPS accretion is not a justification.
- Repurchases transfer value between sellers and continuing holders when the price differs from intrinsic value; the amounts are modest for small programs.
- Where dividends are taxed more heavily than gains, the ex-dividend drop is smaller than the dividend: price drop = $D(1 - \tau_D)/(1 - \tau_G)$; clienteles sort investors into firms whose policies suit their tax status.
- Dividend increases signal confidence and cuts signal trouble; Lintner's model shows managers smooth dividends and adjust only partially to earnings changes.
- Payout can discipline managers by reducing free cash flow they might misuse, with the greatest value in mature firms with few opportunities.

- Splits and stock dividends redistribute no value; they change the share count and the price, with accounting differences between splits and small stock dividends.
- A residual policy applied strictly yields volatile dividends, so firms smooth: a sustainable base dividend plus flexible repurchases and occasional specials.
- Total shareholder yield, free cash flow coverage, and the liquidity position under stress are the measures that matter more than headline announcements.

Review Questions

1. State the Miller-Modigliani dividend irrelevance proposition and the conditions under which it holds. Why must the investment policy be held fixed?
2. Explain homemade dividends with an example of an investor who wants cash when the firm pays none, and one who wants no cash when the firm pays a dividend.
3. Why does a share repurchase increase EPS? Under what condition would it also increase the share price?
4. Describe how a repurchase at a price below intrinsic value transfers wealth among shareholders. Why is the effect small for a modest program?
5. What is the relationship between the ex-dividend price drop and shareholders' tax rates? What does a drop of less than the dividend suggest?
6. What is a dividend clientele? Does the existence of clienteles imply that a firm should never change its payout policy?
7. Explain the signaling interpretation of dividend increases and cuts. Why might repurchases be a weaker signal?
8. How does the free cash flow hypothesis relate to payout policy? In what kind of firm is it most relevant?
9. Summarize Lintner's model. If earnings fall temporarily, what happens to the dividend and the payout ratio?
10. Why would a firm announce a stock split? Does a split create value? What other effects might make it appear to do so?

Problems

1. A firm has 2 million shares at $30 and will pay a $3.00 dividend. An investor owning 1,000 shares wants no cash. After the dividend, how many shares must the investor buy with the proceeds, and what is the investor's total position value? Assume no taxes or costs.
2. A firm has 60 million shares at $36, net income of $120 million, and distributes $90 million. (a) Compute dividend per share if paid as a dividend. (b) Compute shares retired and EPS after a repurchase at $36, ignoring foregone interest. (c) Recompute EPS if the $90 million was earning 4% pre-tax, with a 25% tax rate. (d) Compute the P/E after the buyback for part (c) at a price of $36.
3. The firm in Problem 2 has intrinsic value of $45 per share (total $2,700 million) while the market price is $36. It repurchases $90 million. Compute intrinsic value per remaining share. Then suppose instead intrinsic value is $30 per share ($1,800 million) with the market price still $36 and compute the new intrinsic value per remaining share.
4. A stock trades at $60 cum-dividend and pays a $3.00 dividend. The marginal investor's tax rates are 35% on dividends and 15% on capital gains. Compute the ex-dividend price.

5. A firm's last dividend was $1.00 per share, its target payout is 50%, and its adjustment speed is 0.25. EPS this year is $3.00 and next year it is $3.60. Compute the dividend in each year under Lintner's model.
6. A firm with 40 million shares at $90 announces a 3-for-2 split. (a) Compute the new share count and price. (b) Instead of the split, it declares a 5% stock dividend when the price is $60 and 60 million shares are outstanding. Compute new shares, the amount transferred from retained earnings, and the expected ex-price.
7. A firm targets a capital structure with 40 percent debt (D/V = 0.40) and net income of $400 million. (a) Under a residual policy with capital spending of $540 million, compute the dividend and payout ratio. (b) If capital spending rises to $700 million, compute the dividend or the external equity needed.
8. A firm expects EPS of $4.00 next year, ROE of 12%, and has a cost of equity of 11%. Using the constant-growth model, compute the price at payout ratios of 25% and 60%. Interpret the result.

Answers to Selected Problems

1. Ex-dividend price = $27. Dividends received = 1,000 × $3 = $3,000; shares purchased = 3,000 / 27 = 111.11. Total shares = 1,111.11; value = 1,111.11 × $27 = $30,000, the same as before.
2. a. $90m / 60m = $1.50 per share. (b) Shares retired = 90 / 36 = 2.5 million; remaining 57.5 million; EPS = 120 / 57.5 = $2.087 (from $2.00). (c) Lost after-tax interest = 90 × 0.04 × 0.75 = $2.7 million; net income = $117.3 million; EPS = 117.3 / 57.5 = $2.04. (d) P/E = 36 / 2.04 = 17.65 (versus 18.0 initially).
3. At intrinsic $45: (2,700 − 90) / 57.5 = $45.39, a gain of $0.39 per continuing share. At intrinsic $30: (1,800 − 90) / 57.5 = $29.74, a loss of $0.26.
4. Price drop = 3.00 × 0.65 / 0.85 = $2.294; ex-dividend price = 60 − 2.294 = $57.71.
5. Year 1 target = 0.5 × 3.00 = 1.50; D_1 = 1.00 + 0.25 × (1.50 − 1.00) = $1.125. Year 2 target = 0.5 × 3.60 = 1.80; D_2 = 1.125 + 0.25 × (1.80 − 1.125) = $1.29375, about $1.29.
6. a. New shares = 60 million; price = 90 × 40 / 60 = $60. (b) New shares = 3 million; transfer = 3 × $60 = $180 million; ex-price ≈ 60 × 60 / 63 = $57.14.
7. a. Equity needed = 540 × 0.60 = $324 million; dividend = 400 − 324 = $76 million; payout = 19%. (b) Equity needed = 700 × 0.60 = $420 million; shortfall = $20 million, so the dividend is zero and the firm needs $20 million of external equity (or more debt).
8. Payout 25%: g = 0.12 × 0.75 = 9%; D_1 = 1.00; P = 1.00 / (0.11 − 0.09) = $50.00. Payout 60%: g = 0.12 × 0.40 = 4.8%; D_1 = 2.40; P = 2.40 / (0.11 − 0.048) = $38.71. Because ROE (12%) exceeds the cost of equity (11%), retention creates value and the lower payout gives the higher price.

Chapter 17: Working Capital and Short-Term Financial Planning

Learning Objectives

After studying this chapter you should be able to:

- Define net working capital and the operating and cash conversion cycles, compute DSO, DIO, DPO, and the cash conversion cycle from financial statements, and translate days of improvement into cash released.
- Evaluate receivables policy (credit terms, credit standards, collection effort) with an incremental-profit framework, and calculate the implicit cost of trade-credit discounts.
- Apply inventory models (economic order quantity, reorder point, safety stock) and explain the trade-off between carrying costs, ordering costs, and stock-out costs.
- Design a cash management policy using the Baumol and Miller-Orr models, and describe the main collection, concentration, and disbursement tools.
- Compare short-term financing sources (bank lines, term loans, commercial paper, factoring, asset-based lending) and compute their effective annual costs including fees and compensating balances.
- Build a monthly cash budget, identify the peak financing requirement, and evaluate supply chain finance programs from the viewpoints of buyer, supplier, and financier.

Opening Vignette

Illustrative scenario: Northgate Cycle Components is a fictional company.

Northgate Cycle Components makes aluminum frames and drivetrain parts for mid-market bicycle brands. For three years its income statement looked healthy: sales grew from $410 million to $600 million, and operating margins held near 9 percent. Yet in the autumn of the third year the treasurer, Amara Okafor, found herself on a call with her bank asking for an emergency $40 million increase to the revolving credit facility.

The board could not understand it. “We are profitable and growing. Why are we short of cash?” Okafor put three numbers on the whiteboard. Receivables had risen from 41 days of sales to 50 days, because the sales team had granted 60-day terms to win two large retail chains. Inventory had risen from 48 days of cost of goods sold to 61 days, because the plant manager had bought aluminum in bulk to hedge price increases and had added a dozen new SKUs. And payables had fallen from 47 days to 40 days, because the purchasing group, anxious about supply reliability, had begun paying suppliers early to secure priority allocation.

Each decision was individually defensible. Together they lengthened the cash conversion cycle from about 42 days to about 71 days. With $600 million of sales and about $420 million of cost of goods sold, the extra nine days of receivables, thirteen days of inventory, and seven fewer days of payables tied up roughly $38 million of additional cash in the working capital base. That is cash the company had not earned yet in any usable form: it was sitting in unsold frames, unpaid invoices, and prepayments to suppliers. Growth was consuming cash faster than profits produced it.

This chapter is about how to avoid Northgate's situation. Working capital management is not glamorous, but it is where a large share of a firm's liquidity is won or lost. Firms with excellent strategy and mediocre working capital discipline fail more often than the reverse. We will learn to measure the cycle, to price each policy lever, to plan cash on a month-by-month basis, and to use modern tools such as supply chain finance without mistaking them for free money.

17.1 The Working Capital Cycle

17.1.1 Definitions

Working capital has two common meanings. *Gross working capital* is the total of current assets: cash and marketable securities, receivables, inventories, and prepaid expenses. *Net working capital* (NWC) is current assets minus current liabilities. For managerial purposes it is usually more useful to work with **operating working capital**, which strips out financing items:

```
Operating NWC = (Receivables + Inventory + Prepaids) − (Payables + Accruals)
```

Cash and short-term debt are excluded because they are the *result* of working capital and financing decisions, not part of the operating process itself. A firm that sells goods on credit, holds inventory, and buys on credit from suppliers will always have an operating NWC investment that must be financed by someone: shareholders, long-term lenders, or short-term lenders.

Why does this matter to valuation? Increases in operating NWC are a use of cash in the free-cash-flow calculation. A firm whose operating NWC equals 20 percent of sales faces $\Delta \text{NWC} = 0.20 \times \Delta \text{Sales}$, so each \$100 of added sales requires \$20 of added financing for working capital, in addition to fixed capital. Growth is cash-hungry.

17.1.2 The operating cycle and the cash conversion cycle

Consider the life of a dollar in a manufacturing business. The firm buys raw materials (often on credit), converts them to finished goods, sells the goods (often on credit), and finally collects cash. The time line has three components:

- **Days inventory outstanding (DIO)**: the average time inventory sits before being sold.
- **Days sales outstanding (DSO)**: the average time between a credit sale and collection.
- **Days payables outstanding (DPO)**: the average time the firm takes to pay its suppliers.

The **operating cycle** is DIO + DSO: the days from the arrival of inventory until the cash from its sale is collected. The **cash conversion cycle (CCC)** subtracts the period during which suppliers are financing the firm:

```
CCC = DIO + DSO − DPO

DIO = (Average inventory / Cost of goods sold) × 365
DSO = (Average receivables / Credit sales) × 365
DPO = (Average payables / Cost of goods sold) × 365
```

The CCC measures the number of days the firm must finance its operations from non-spontaneous sources. A short or negative CCC means customers and suppliers fund the business. Retailers such as large supermarkets and some online platforms often have negative cycles: they sell for cash within days and pay suppliers in 45 or 60 days. A capital-goods manufacturer may have a CCC of 150 days or more.

A few technical cautions apply. First, use consistent denominators. DPO conventionally uses cost of goods sold, but strictly it should use purchases; where purchases are not disclosed, COGS is the common proxy. Second, use *average* balances when seasonality is present, because a year-end balance may be unrepresentative. Third, when sales are growing quickly, balances at year-end divided by full-year sales understate the days on a current-run-rate basis; some analysts annualize the last quarter instead.

Example 17.1 – Computing and improving the cash conversion cycle. Northgate-style data for a fictional firm, Harbor Industrial: annual sales (all on credit) are $600 million; cost of goods sold $420 million; average receivables $82 million; average inventory $70 million; average payables $46 million. Use a 365-day year.

Step 1: DSO. 82 / 600 × 365 = 0.136667 × 365 = 49.9 days.

Step 2: DIO. 70 / 420 × 365 = 0.166667 × 365 = 60.8 days.

Step 3: DPO. 46 / 420 × 365 = 0.109524 × 365 = 40.0 days.

Step 4: CCC. 49.9 + 60.8 − 40.0 = 70.7 days.

Now management sets targets: collections tightened by 5 days of sales, inventory reduced by 6 days of COGS, and payables lengthened by 4 days of COGS (without losing discounts).

Cash released from receivables: 5 × (600 / 365) = 5 × 1.6438 = $8.22 million. *Cash released from inventory:* 6 × (420 / 365) = 6 × 1.1507 = $6.90 million. *Cash released from payables:* 4 × 1.1507 = $4.60 million. *Total:* 8.22 + 6.90 + 4.60 = $19.73 million. The CCC falls by 15 days to 55.7 days.

If the firm's marginal short-term borrowing cost is 6 percent, the annual saving is 0.06 × 19.73 = $1.18 million, which flows straight to pre-tax profit. Note the asymmetry: a receivable day is worth sales per day ($1.644 million), while inventory and payables days are worth COGS per day ($1.151 million). Receivables are measured at selling price and so carry the margin.

17.1.3 Working capital as a source of value

Because it is a perpetual investment, working capital deserves the same discipline as capital budgeting. The present value of a one-time release of $19.73 million is the cash itself (it is an immediate inflow); but the *ongoing* benefit is the interest or return avoided on that sum every year. Capitalizing a perpetual saving of $1.18 million at a 9 percent required return gives roughly $13 million of value only if the cash would otherwise be financed by long-term capital; if instead the cash is used to retire revolver borrowings at 6 percent, the benefit is simply the 6 percent saving. Be careful not to double count: the one-time release of cash and the perpetuity of interest savings are two descriptions of the *same* benefit.

17.1.4 Why cycles differ across firms and over time

Industry structure explains most cross-sectional variation. A firm with long production processes (shipbuilding, aerospace, aged spirits, tobacco) holds inventory for months by physical necessity. Firms selling to governments or large industrial customers face long payment terms, because the customer has bargaining power. Firms selling to consumers by card get paid in two to three days. Within an industry, the leaders typically have the shortest cycles; a persistent gap relative to peers is a management signal worth investigating.

Over the business cycle, working capital behaves procyclically in the upswing (inventories and receivables expand with sales) and sometimes sharply in a downturn (customers stretch payments, inventories pile up unsold). Treasury planning must therefore distinguish between *permanent* working capital, the minimum level held through the cycle, and *temporary* or seasonal working capital that fluctuates around it.

17.2 Financing Policy: Matching, Aggressive, and Conservative

17.2.1 Permanent and temporary current assets

A firm's investment in current assets has a floor. Even in its slowest month, Northgate needs a base level of receivables, inventory, and cash; this is **permanent current assets**. Above the floor sit seasonal peaks, which are **temporary current assets**. Fixed assets are permanent as well. A finance policy can then be described by how the firm funds each layer.

- **Maturity matching (hedging) policy.** Fund permanent assets (fixed plus permanent current) with long-term capital: equity and long-term debt. Fund temporary current assets with short-term borrowing. The principle is that the maturity of financing should match the life of the asset financed. Seasonal borrowing is repaid out of the asset conversion that created the need.
- **Aggressive policy.** Fund some permanent assets with short-term debt. Because the short-term yield curve is normally lower than the long-term, expected interest cost falls. But the firm must continually *roll over* its borrowings, exposing it to refinancing risk, interest-rate risk, and the risk that credit markets close just when it needs them.
- **Conservative policy.** Fund even some temporary needs with long-term capital and hold surplus cash or marketable securities in the troughs. Risk is low, but the carrying cost of idle long-term money is high.

17.2.2 The cost-risk trade-off

The textbook argument for aggressive financing is a lower expected interest cost; the practical argument against is the 2008-2009 lesson that short-term funding can evaporate. Firms with volatile cash flows, lumpy capital needs, or poor credit access should lean conservative. Firms with predictable cash flows and committed backup lines can lean more aggressive. A mature rule of thumb: the *committed* liquidity (cash plus undrawn committed facilities) should cover the plausible worst-case cumulative cash shortfall over the planning horizon, plus a margin.

Common mistake: treating a "revolving line" as permanent capital. Most are committed for 364 days or three to five years and carry covenants. They can be reduced, repriced, or not renewed. Lines are a backup for seasonality, not a substitute for equity.

Table 17.1 – Financing policies compared.

Feature	Conservative	Matching	Aggressive
Short-term debt funds	Little or none; surplus cash held	Temporary current assets only	Temporary plus part of permanent assets
Expected interest cost	Highest	Moderate	Lowest (if curve upward-sloping)
Refinancing / rate risk	Lowest	Moderate	Highest
Liquidity buffer	Large	Moderate	Thin
Typical user	Cyclical, high-distress-cost firms	Most operating firms	Stable, strongly rated firms

17.3 Receivables and Credit Policy

17.3.1 Why firms extend credit

Trade credit is a financing product sold alongside the goods. Suppliers extend it for several reasons: it is a competitive necessity in many industries; it lets customers verify quality before paying; it is a form of price discrimination (customers who value the delay pay an effective higher price, while those who do not can take a discount); and suppliers sometimes

have an information or monitoring advantage over banks, since they see order patterns and can repossess or resell goods. Credit policy therefore has four levers.

1. **Credit standards**: which customers qualify, determined by credit scoring, financial analysis, references, and payment history. The classic "five Cs" are character, capacity, capital, collateral, and conditions.
2. **Credit terms**: the length of the credit period and any early-payment discount, written in the shorthand "2/10 net 30" (2 percent discount if paid within 10 days; otherwise the full amount is due in 30).
3. **Collection policy**: reminders, dunning letters, calls, credit holds, third-party collection, legal action.
4. **Credit limits and monitoring**: exposure caps by customer, credit insurance, and aging reports.

17.3.2 The incremental analysis framework

Any change in policy should be assessed by comparing incremental cash flows to the incremental financing cost of the added receivables. The relevant items are:

```
Incremental contribution margin on added sales
− Additional bad-debt losses (on all sales if the rate rises)
− Carrying cost on the added investment in receivables
− Added collection or administrative cost
+ Reduction in discounts lost / minus discounts granted
= Net incremental profit
```

Two points of rigor deserve emphasis. First, the **investment in receivables** is usually measured at *variable cost* rather than at sales price: what the firm has actually laid out while it waits is the cost of the goods, not the profit it has yet to collect. (Some analysts use full sales value on the grounds that the whole receivable is outstanding; that is more conservative, and the choice should be applied consistently.) Second, a lengthened period raises bad debts and collections costs on the *whole* sales base if the new terms apply to everyone, not only on the incremental sales.

Example 17.2 – Loosening credit terms. A fictional distributor, Pacific Fastener, has annual credit sales of $24.0 million, variable cost 70 percent of sales (contribution margin 30 percent), average collection period 30 days, bad debts 1 percent of sales, and a required return on receivables investment of 9 percent. Management proposes extending terms so that the average collection period lengthens to 45 days; sales are expected to rise 8 percent, and bad debts to rise to 2 percent of (all) sales.

Step 1: new sales. 24.0 × 1.08 = $25.92 million; incremental sales $1.92 million.

Step 2: incremental contribution. 1.92 × 0.30 = $0.576 million = $576,000.

Step 3: receivables investment at variable cost. Current: (24,000,000 / 365) × 30 × 0.70 = 65,753.42 × 30 × 0.70 = $1,380,822. Proposed: (25,920,000 / 365) × 45 × 0.70 = 71,013.70 × 45 × 0.70 = $2,236,932. Increase: $856,110.

Step 4: carrying cost. 856,110 × 0.09 = $77,050.

Step 5: incremental bad debts. Proposed: 25,920,000 × 0.02 = $518,400. Current: 24,000,000 × 0.01 = $240,000. Increase: $278,400.

Step 6: net. 576,000 − 77,050 − 278,400 = **$220,550 per year**. The policy adds value under these assumptions.

Sensitivity. The result is dominated by the bad-debt assumption and the sales response. If sales rose only 4 percent (incremental contribution 0.04 × 24,000,000 × 0.30 = $288,000), bad debts at 2 percent of 24,960,000 would be $499,200, an increase of $259,200; the receivables investment would be (24,960,000 / 365) × 45 × 0.70 = $2,154,082, an increase of

$773,260 and a carrying cost of $69,593. The net would be 288,000 − 259,200 − 69,593 = −$40,793: negative. Managers should therefore ask, before approving looser terms, *which* customers will be attracted, what their default experience is, and whether competitors will simply match.

17.3.3 Cash discounts and the cost of trade credit

When a supplier offers “2/10 net 30”, the buyer who skips the discount is, in effect, borrowing the invoice amount less the discount for 20 extra days (day 10 to day 30) and paying 2 percent of the discounted price for the privilege. For every $98 of effective borrowing, the buyer pays $2 after 20 days. The periodic rate is 2/98 = 2.04 percent. There are 365/20 = 18.25 such periods in a year.

```
Simple annualized cost = [d / (1 − d)] × [365 / (credit period − discount period)]
Effective annual rate   = (1 + d / (1 − d))^(365 / (credit period − discount period)) − 1
```

where d is the discount fraction.

Example 17.3 – The price of forgoing a discount. Terms are 2/10 net 30.

Step 1: periodic rate. 0.02 / 0.98 = 0.020408. *Step 2: periods per year.* 365 / (30 − 10) = 18.25. *Step 3: simple annual cost.* 0.020408 × 18.25 = 37.2 percent. *Step 4: effective annual rate.* (1.020408)^18.25 − 1 = e^(18.25 × ln 1.020408) − 1 = e^(18.25 × 0.020203) − 1 = e^0.36871 − 1 = 0.4459, or **44.6 percent**.

A buyer who can borrow from a bank at 8 percent should therefore pay on day 10 and take the discount. Stretching payment to day 60 changes the arithmetic: the cost becomes (0.020408 × 365/50) = 14.9 percent simple (15.9 percent effective), still above bank debt but far lower. However, stretching payables (stacking) beyond stated terms may incur late fees, loss of supplier goodwill, credit-rating effects, and supply interruptions; those hidden costs are not in the formula.

Table 17.2 – Implicit annual cost of forgoing cash discounts.

Terms	Discount period (days)	Net period (days)	Simple annual cost	Effective annual rate
1/10 net 30	10	30	18.4%	20.1%
2/10 net 30	10	30	37.2%	44.6%
2/10 net 45	10	45	21.3%	23.5%
3/10 net 60	10	60	22.6%	24.9%
1/15 net 30	15	30	24.6%	27.7%

The pattern is that a large discount over a short extra period produces an extreme rate. A seller offering such a discount is essentially paying a very high price for early cash; it should compare that to its own cost of capital.

17.3.4 Monitoring: aging, DSO, and the weighted average

DSO is a blunt instrument because it mixes growth effects and payment behavior. An **aging schedule** classifies receivables by days outstanding (current, 1-30 past due, 31-60, and so on). Two portfolios with the same DSO can have very different risk: one with all customers paying at day 45, the other with most paying at day 30 and a tail beyond 120 days. The tail is where losses live. Companies should also track the **collection effectiveness index**, the percentage of collectible receivables actually collected in a period, and **average days delinquent**, the gap between the average collection period and the average terms. Provisioning (expected-loss allowances) should respond to aging buckets and historical loss rates by bucket.

17.3.5 Tools for managing receivables

- *Credit insurance* transfers customer-default risk to an insurer in exchange for a premium, often 0.1 to 0.5 percent of insured sales, and can allow cheaper borrowing against the receivables.
- *Factoring* is the sale of receivables to a finance company. In *non-recourse* factoring the factor bears default risk; in *recourse* factoring the seller retains it. The factor typically advances 70 to 90 percent immediately and pays the remainder, less fees, on collection.
- *Securitization* pools receivables in a special-purpose vehicle that issues notes; it is available only to firms with large, homogeneous, well-documented receivable books.
- *Electronic invoicing and dynamic discounting* shorten the lag between invoice and payment and create the option of discounts at variable rates tied to how early the buyer pays.

17.4 Inventory Management

17.4.1 The role and costs of inventory

Inventory serves transaction, precautionary, and speculative purposes: it decouples production from sales, buffers against supply or demand variability, and sometimes captures expected price rises or volume discounts. Its costs fall in three groups.

- **Carrying (holding) costs**: capital cost on funds invested, storage, insurance, handling, spoilage, obsolescence, and shrinkage. In practice, 15 to 30 percent of inventory value per year is common, much of which is not the interest rate but obsolescence and storage.
- **Ordering (setup) costs**: placing orders, receiving, inspection, or production changeover costs; fixed per order and independent of order size.
- **Stock-out costs**: lost sales, expediting, line stoppages, and customer attrition.

The art of inventory management is balancing these. Carrying cost rises with order size; ordering cost per year falls with order size.

17.4.2 The economic order quantity

Suppose annual demand D is steady, each order costs S, and holding one unit for a year costs H. If the firm orders Q units at a time and consumption is smooth, the average inventory is Q/2. Total annual cost is

```
TC(Q) = (D / Q) × S + (Q / 2) × H
```

Minimizing with respect to Q (the first derivative $-DS/Q^2 + H/2 = 0$) gives the **EOQ** model due to Harris and later Wilson:

```
Q* = √(2 D S / H)
```

At the optimum the ordering cost equals the carrying cost, and the total is TC* = √(2 D S H). The curve is flat near the optimum: ordering 30 percent more or less than Q* raises cost by only about 4 to 5 percent, so approximate answers are acceptable.

Example 17.4 – EOQ, reorder point, and safety stock. A fictional retailer, Lakeview Hardware, sells 48,000 units of a fastener kit per year. Each order costs $150 to place and receive. Holding a unit costs $6 per year. Lead time is 6 days; average daily demand is 48,000 / 365 = 131.5 units; daily demand has a standard deviation of 40 units; the firm wants a 95

percent cycle service level (z = 1.65).

Step 1: EOQ. Q* = √(2 × 48,000 × 150 / 6) = √(2,400,000) = 1,549 units.

Step 2: orders per year. 48,000 / 1,549 = 31.0 orders.

Step 3: costs at the optimum. Ordering: 31.0 × 150 = $4,648. Carrying: (1,549 / 2) × 6 = $4,648. Total: **$9,295**.

Step 4: compare with ordering 2,000 at a time. Ordering: (48,000 / 2,000) × 150 = $3,600; carrying: 1,000 × 6 = $6,000; total $9,600, which is $305 (3.3 percent) above the optimum.

Step 5: reorder point without safety stock. Demand during lead time = 131.5 × 6 = 789 units.

Step 6: safety stock. Standard deviation of lead-time demand = 40 × √6 = 98.0; safety stock = 1.65 × 98.0 = 162 units. (The product 1.65 × 40 × √6 = 161.7.)

Step 7: reorder point. 789 + 162 = **951 units**.

Table 17.3 – Total annual cost at different order sizes (Example 17.4).

Order quantity	Orders per year	Ordering cost	Carrying cost	Total cost
500	96.0	$14,400	$1,500	$15,900
1,000	48.0	$7,200	$3,000	$10,200
1,549 (EOQ)	31.0	$4,648	$4,647	$9,295
2,000	24.0	$3,600	$6,000	$9,600
3,000	16.0	$2,400	$9,000	$11,400

(Small rounding differences arise because Q = 1,549 is rounded from 1,549.19.)

17.4.3 Beyond the EOQ

The EOQ's assumptions of constant demand, instantaneous replenishment, and constant costs are rarely true, but the model's logic is robust. Extensions include quantity discounts (compare total cost including purchase price at each discount break), finite production rates (economic production quantity), and stochastic demand (the safety stock term above, or periodic-review base-stock policies). Modern practice adds several ideas.

- **ABC analysis.** Classify items by annual value: perhaps 20 percent of items account for 80 percent of dollar usage. Apply tight control and frequent review to class A, simple rules to class C.
- **Just-in-time (JIT) and lean.** Reduce setup costs S toward zero and the EOQ shrinks toward continuous flow. JIT is thus not a repudiation of EOQ logic but a campaign to change its parameters. The cost is vulnerability: low buffers magnify the effect of supplier or logistics disruption, so firms increasingly hold "just-in-case" strategic reserves for critical components.
- **Vendor-managed inventory and consignment.** The supplier holds the stock at the customer's site and is paid on usage; inventory financing shifts to the supplier, which will price accordingly.
- **Postponement and demand forecasting.** Delaying product differentiation until orders are known (generic frames painted at the end) lets safety stock be held on fewer items and reduces the total inventory needed for a given service level, a result linked to risk pooling: the standard deviation of aggregated demand grows more slowly than the sum of individual deviations.

17.4.4 Inventory and financial reporting

Costing method (FIFO, weighted average; LIFO is not allowed under IFRS) affects reported margins and, in inflationary periods, the apparent turnover. Inventories are carried at the lower of cost and net realizable value, so write-downs recognize obsolescence only after it has occurred; managers should therefore watch physical indicators (slow-moving stock, aging by receipt date) before the accounting catches up. Inventory turnover (COGS / average inventory) is the standard ratio, but when comparing across firms ensure that cost definitions match.

17.5 Cash Management

17.5.1 Why firms hold cash

Keynes identified the transactions, precautionary, and speculative motives for holding money. For a firm, a fourth reason is compensating banks for services. The cost of holding cash is the return forgone on the best alternative use, either the yield on short-term securities or, for a borrowing firm, the interest rate on the revolver. Against this, the benefits are the avoidance of transaction costs of converting securities, the ability to meet payments on time, and resilience to shocks. Large corporate cash balances at some firms also reflect tax and repatriation frictions and the option value of being able to invest when others cannot; in this chapter we focus on operational cash.

17.5.2 Float and the collection-disbursement system

Float is the time cash is in transit between payer and payee. *Collection float* is the delay between a customer's payment and the firm's use of the funds (mail, processing, and clearing). *Disbursement float* is the delay between a firm's issuing a payment and its funds leaving the bank account. Firms want to shrink collection float and, within ethical and contractual limits, preserve disbursement float. Float is worth money: if a firm receives $2 million a day and can accelerate availability by 2 days, it frees $4 million, which at 5 percent is worth $200,000 per year.

Practical tools include:

- **Lockboxes**: customers pay to a post-office box near them, and the bank empties it several times a day. With the shift to electronic payments, "virtual lockboxes" and remittance-data capture serve the same purpose.
- **Electronic funds transfer, direct debit, and card acquiring**: collection in a day or less, with fees to compare against float savings.
- **Concentration banking and cash pooling**: balances in many local accounts are swept to a central account daily. *Physical pooling* moves cash; *notional pooling* offsets balances for interest purposes without moving them, subject to local rules. Pooling reduces the total precautionary balance, because shortages and surpluses in different units offset one another.
- **Controlled disbursement and zero-balance accounts**: the bank tells the firm each morning how much will be presented, and the firm funds only that amount; subsidiary payments draw on a master account.
- **Payables automation**: scheduling payments to land on the due date, not before.

A principal governance rule: separate the duties of authorizing, executing, recording, and reconciling payments; require dual control on payment instructions; and verify any change of beneficiary account details by an independent call-back, because payment-redirection fraud is among the most common corporate losses.

17.5.3 The Baumol model

William Baumol (1952) borrowed the inventory logic of the EOQ for cash. Imagine a firm with steady net cash needs T per period. It holds cash C, and when it runs out, sells C of securities at a fixed cost F per transaction. Securities yield i per period. Average cash is C/2 and number of transactions is T/C:

```
Total cost = (T / C) × F + (C / 2) × i
Optimal C* = √(2 F T / i)
```

Example 17.5 – Baumol and Miller-Orr targets. A fictional company, Redwood Packaging, uses $3.0 million of cash per year in a smooth pattern. Each securities sale costs $50. The yield on securities is 5 percent per year.

Baumol. C* = √(2 × 50 × 3,000,000 / 0.05) = √(6,000,000,000) = **$77,460** per replenishment. Transactions per year = 3,000,000 / 77,460 = 38.7. Transaction cost = 38.7 × 50 = $1,936; opportunity cost = (77,460 / 2) × 0.05 = $1,936. Total **$3,873** per year. If the firm instead replenished with $100,000: transaction cost 30 × 50 = $1,500; opportunity cost 50,000 × 0.05 = $2,500; total $4,000, a mere 3 percent more: again the flat bottom of the cost curve.

The Baumol model assumes predictable, steady outflows; cash flows are usually erratic.

17.5.4 The Miller-Orr model

Merton Miller and Daniel Orr (1966) treated daily net cash flows as a random walk with standard deviation σ per day, and proposed a **control-limit policy**. Set a lower limit L (determined by management as the minimum safety balance), a return point, and an upper limit. While the balance wanders between the limits do nothing. When it touches the upper limit, buy securities to bring the balance back to the return point; when it touches L, sell securities to return to the return point. The optimum spread is

```
z = [ 3 F σ² / (4 i) ]^(1/3)        (spread from L to return point)
Return point R = L + z
Upper limit  U = L + 3z
Average balance ≈ L + 4z / 3
```

where i is the *daily* opportunity rate. Intuition: the spread widens with transaction cost F and volatility σ^2, and narrows with the interest rate i. Notice that the return point is one-third of the way up the band: the upper limit is 2z above the return point but the lower limit only z below it. The asymmetry arises because idle cash earns nothing while a shortage is handled by a single, costly sale of securities, so the optimal policy sits closer to the floor and tolerates more room above.

Example 17.5 (continued) – Miller-Orr. Redwood also keeps a separate operating account. The standard deviation of daily net flows is $2,000; transaction cost F = $50; annual securities yield is 6 percent so i = 0.06 / 365 = 0.00016438 per day; management sets L = $10,000.

Step 1. 3 F σ^2 = 3 × 50 × $2{,}000^2$ = 3 × 50 × 4,000,000 = 600,000,000. *Step 2.* 4 i = 4 × 0.00016438 = 0.00065753. *Step 3.* Ratio = 600,000,000 / 0.00065753 = 9.1250×10^{11}, and its cube root is z ≈ **$9,699**. *Step 4.* Return point = 10,000 + 9,699 = **$19,699**; upper limit = 10,000 + 3 × 9,699 = **$39,098**. *Step 5.* Average balance = 10,000 + (4 × 9,699)/3 = $22,933.

When the balance rises to $39,098, buy $19,399 of securities (to bring it back to $19,699); when it falls to $10,000, sell $9,699 of securities.

Both models show the right structure of the problem even if the numbers are rough. Real treasuries use forecasts, sweep accounts and automatic overnight investing, which lowers F nearly to zero and the optimum balances with it. Today the relevant frictions are credit-line fees, minimum operating balances, and concentration limits, not brokerage commissions.

17.5.5 Short-term investment of surplus cash

Surplus cash should be invested under a written policy that ranks **safety, liquidity, and yield**, in that order. Eligible instruments typically include government bills, bank time deposits, high-quality commercial paper, repurchase agreements, and money-market funds. Key risks are credit (a money-market instrument can default), liquidity (a fund may gate redemptions in stress), and reinvestment. Maturity should be matched to forecasted needs, with concentration limits by issuer and counterparty rating triggers. Counterparty exposure to a single bank deserves particular attention in a crisis.

17.6 Short-Term Financing

17.6.1 Sources

Trade credit (already discussed) is the largest source of short-term financing for many firms, but is "free" only when the discount is taken or the terms are honored without penalty.

Accruals (wages, taxes, and interest owed but unpaid) are a spontaneous, no-explicit-cost source but cannot be managed beyond their natural cycle.

Bank lines of credit. An *uncommitted* line is an informal arrangement that the bank may withdraw. A *committed* revolving credit facility obliges the bank to lend up to the limit for a term (commonly one to five years), in exchange for a *commitment fee* on the unused portion and financial covenants (leverage, interest cover). Pricing is usually a floating reference rate plus a margin that may depend on the firm's credit rating or leverage ratio. A **clean-up** requirement obliges the firm to be out of debt on the line for a specified period each year, ensuring that it is used for seasonal rather than permanent needs.

Term loans have a fixed amortization schedule and maturity; they suit financing of permanent working capital or capital assets.

Commercial paper (CP) is unsecured short-term (up to 270 days in the United States, or up to a year in other markets) debt issued by large, highly rated borrowers at a discount. CP is generally cheaper than bank borrowing, but needs a backup liquidity line since the market can close, and so the bank line's commitment fee is part of CP's true cost.

Secured financing and asset-based lending. Receivables or inventory are pledged as collateral; the lender advances a percentage (the *advance rate*, typically 70 to 85 percent of eligible receivables and a lower fraction of inventory) and monitors collateral through field exams and borrowing-base certificates. Costs are higher than for unsecured lending, but availability is greater for weaker credits.

Factoring and invoice discounting. Discussed in section 17.3: the receivable is sold or financed, with or without notice to customers.

17.6.2 Computing effective borrowing costs

The stated rate seldom equals the real cost. Four adjustments recur.

1. **Discount interest**: interest is deducted in advance, so the borrower gets less than the face amount.
2. **Compensating balances**: the bank requires a share of the loan to be kept on deposit at low or zero interest, so the usable amount is smaller.
3. **Commitment and facility fees**: charged on unused or total commitment.
4. **Compounding**: more frequent compounding raises the effective annual rate.

Example 17.6 – The true cost of bank financing. Three alternatives are available to Redwood Packaging, which needs $400,000 of usable cash for one year (all amounts are illustrative).

Alternative A, discount loan at 8%. Interest at 8 percent is deducted in advance. Take a $1,000,000 face amount as a reference. Proceeds = 1,000,000 × (1 − 0.08) = $920,000; interest $80,000. Effective rate = 80,000 / 920,000 = 8.70 percent (equivalently 0.08 / 0.92). Scaling for the $400,000 need: face = 400,000 / 0.92 = $434,783.

Alternative B, 7% loan with 10% compensating balance. The firm must borrow B such that B × (1 − 0.10) = $400,000, so B = $444,444. Interest = 0.07 × 444,444 = $31,111. Effective rate = 31,111 / 400,000 = 7.78 percent (equivalently 0.07 / 0.90).

Alternative C, committed line of $5 million at 6.5 percent with a 0.40% fee on the unused portion; the firm expects an average drawn balance of $2.0 million. Interest on drawn: 2,000,000 × 0.065 = $130,000. Fee on unused: 3,000,000 × 0.004 = $12,000. Total cost = $142,000 on an average usage of $2.0 million = **7.10 percent**. If the firm were to use only $1.0 million on average, the cost would be (65,000 + 16,000)/1,000,000 = 8.10 percent: the fee is a fixed price of liquidity insurance, so the cost per dollar borrowed rises as utilization falls.

Ranking by cost: C (7.10 percent, at expected usage) is cheapest for borrowing, B (7.78) is next, A (8.70) is highest; but the line gives flexibility and does not require a full year of borrowing, and a short-term loan must be priced against exact days needed. Where interest is quoted on a 360-day basis for a 365-day year the effective rate is further increased by 365/360, roughly 1.4 percent of the rate itself.

Common mistake. Comparing quoted rates without adjusting for fees, balances, day counts, and the timing of cash. The correct comparison is always the effective annual cost (or the IRR of the cash flows) per dollar of *usable* funds.

17.6.3 Choosing among sources

Financial managers weigh: (1) cost; (2) availability, including backup if one source closes; (3) flexibility, i.e., the ability to prepay or reborrow; (4) covenants and security demanded; (5) timing and administration; (6) relationship value, since banks that provide credit expect ancillary business. Unused capacity under one facility reduces the risk of drawing another in a bad month. A diversified funding stack with staggered maturities, a mix of bank and market sources, and committed backup beyond the expected peak is the hallmark of mature treasury practice.

17.7 Cash Budgeting and Short-Term Financial Planning

17.7.1 Purpose and structure

The **cash budget** is a forecast of cash receipts and disbursements, by period, that reveals when the firm will have surpluses to invest and shortfalls to finance. It is distinct from the income statement because it follows cash timing, not accrual. Its steps are:

1. Forecast sales (the driver) and convert them to **cash collections** using the historical collection pattern.
2. Forecast **disbursements**: purchases (with payment lag), wages, overheads, interest, taxes, dividends, and capital expenditures.
3. Compute **net cash flow** for each period and add it to opening cash to get cash before financing.
4. Compare with the **minimum desired cash balance**; the difference is the borrowing needed (or surplus available).
5. Roll the loan balance and interest forward, since borrowing in one month creates interest in the next.

The planning horizon depends on use: 13 weeks for tight liquidity control, 12 months for seasonal planning, and a three-to-five-year financial plan for strategy. Cash budgets should be rolling: update monthly (or weekly) and extend the horizon by one period.

17.7.2 A worked cash budget

Example 17.7 – Six-month cash budget with a revolving line. A fictional seasonal company, Summit Garden Supply, has the following sales forecast (in $ thousands): November (actual) 400; December (actual) 450; January 300; February 350; March 500; April 600; May 700; June 650.

Collections pattern: 20 percent of sales are collected in the month of sale, 50 percent in the next month, 28 percent in the second month after sale, and 2 percent are never collected (bad debts).

Disbursements: purchases each month equal 60 percent of that month's sales and are paid in the same month (a simplification of a one-month lead and lag); wages 12 percent of sales; fixed overhead $40 thousand per month; interest on term debt $6 thousand per month; a tax payment of $60 thousand in March; capital expenditure of $150 thousand in April; and a dividend of $40 thousand in June.

Financing: opening cash on 1 January is $80 thousand; the minimum cash balance is $50 thousand; shortfalls are met by drawing on a revolving line charging 8 percent per year (0.667 percent per month), with interest paid on the opening line balance each month and surplus cash used to repay the line first.

Step 1: collections. - January: 0.20 × 300 + 0.50 × 450 + 0.28 × 400 = 60 + 225 + 112 = 397. - February: 0.20 × 350 + 0.50 × 300 + 0.28 × 450 = 70 + 150 + 126 = 346. - March: 0.20 × 500 + 0.50 × 350 + 0.28 × 300 = 100 + 175 + 84 = 359. - April: 0.20 × 600 + 0.50 × 500 + 0.28 × 350 = 120 + 250 + 98 = 468. - May: 0.20 × 700 + 0.50 × 600 + 0.28 × 500 = 140 + 300 + 140 = 580. - June: 0.20 × 650 + 0.50 × 700 + 0.28 × 600 = 130 + 350 + 168 = 648.

Step 2: disbursements. January purchases 0.60 × 300 = 180, wages 0.12 × 300 = 36, overhead 40, term interest 6, total 262, and likewise for each month, with the one-off items added.

Step 3: the budget (in $ thousands).

	Jan	Feb	Mar	Apr	May	Jun
Sales	300	350	500	600	700	650
Collections	397.0	346.0	359.0	468.0	580.0	648.0
Purchases	180.0	210.0	300.0	360.0	420.0	390.0
Wages	36.0	42.0	60.0	72.0	84.0	78.0
Overhead + term interest	46.0	46.0	46.0	46.0	46.0	46.0
Tax / capex / dividend	0	0	60.0	150.0	0	40.0
Interest on line	0	0	0	0	0.36	0.16
Total disbursements	262.0	298.0	466.0	628.0	550.36	554.16
Net cash flow	135.0	48.0	−107.0	−160.0	29.64	93.84
Opening cash	80.0	215.0	263.0	156.0	50.0	50.0
Cash before financing	215.0	263.0	156.0	−4.0	79.64	143.84
Line draw / (repayment)	0	0	0	54.0	(29.64)	(24.36)
Line balance (end)	0	0	0	54.0	24.36	0
Ending cash	215.0	263.0	156.0	50.0	50.0	119.48

Step 4: reading the result. Cash is ample in the winter because the large December collections arrive while spending is low. The squeeze comes in April when the capital expenditure and the pre-season purchasing build coincide with collections that lag the sales surge: cash before financing is −$4 thousand against a $50 thousand minimum, so the firm must draw **$54 thousand**, the peak borrowing. Interest in May is 54 × 0.08 / 12 = $0.36 thousand, and in June 24.36 × 0.08 / 12 = $0.16 thousand. By June the line is repaid. Ending cash of $119.48 thousand could fund the next dividend, an early repayment of term debt, or inventory prebuying.

Step 5: managerial uses. The peak requirement of $54 thousand is small here, but the firm should arrange a committed line of perhaps $100 to $120 thousand to cover forecast error. A simple stress test: suppose sales are 15 percent below forecast or the 50 percent collection bucket slips by a month; recompute and see whether the line limit and covenants hold. A cash budget is only as good as the collection assumptions; monitor actual versus budget and revise the percentages.

17.7.3 The percentage-of-sales and the sustainable-growth links

For longer horizons the cash budget gives way to pro forma statements. The simplest approach scales items that vary with sales (receivables, inventory, payables) as a percentage of sales and treats the rest as fixed, with external financing needed (EFN) as the balancing item: EFN = Required assets increase − spontaneous liability increase − retained earnings. A firm may be profitable but grow faster than its **sustainable growth rate** (ROE × retention, in simple form) and so require outside funds. The CCC tells why: each additional dollar of sales needs about (CCC/365) of financing per dollar of daily cost base. Reducing the cycle therefore raises the growth that internal funds can support.

17.7.4 Short-term forecasting and risk

Forecasts should carry ranges, not points. Techniques include scenario analysis, simulation of collection delays, and variance analysis of actual versus forecast flows. For firms with foreign-currency flows, add a currency layer to the budget and decide whether to hedge forecast exposures. Key controls are a single source of truth for the forecast, assigned accountability for each cash flow line, and a monthly post-mortem of forecast errors, which tend to be systematically biased (usually optimistic on collections and conservative on payments).

17.8 Supply Chain Finance

17.8.1 What it is

Supply chain finance (SCF), also called reverse factoring or approved-payables financing, is a buyer-led program in which a bank or fintech platform pays suppliers' approved invoices early, at a discount priced off the *buyer's* credit risk, and the buyer pays the financier on the original (or an extended) due date. The mechanics are:

1. The supplier delivers and invoices; the buyer approves the invoice, which is thereby an irrevocable payment obligation due on day T.
2. The platform offers the supplier early payment at day t < T, at a discount: Invoice × (1 − r × (T − t)/360), where r is a reference rate plus a margin reflecting the buyer's credit.
3. The supplier chooses invoice by invoice whether to accept early payment.
4. On day T, the buyer pays the financier the full invoice.

Because the buyer is typically a larger and better-rated entity than the supplier, r is lower than the supplier could obtain on its own, especially for small suppliers in emerging markets. A related product, **dynamic discounting**, has the buyer itself use its surplus cash to pay early for a sliding-scale discount.

17.8.2 Who gains

- **Supplier**: cheaper liquidity, earlier cash, reduced collection risk, and easier forecasting.
- **Buyer**: the supplier's relationship is stabilized; and, importantly, the buyer can use the program to *negotiate longer payment terms* (say from 45 to 75 days) without suppliers bearing the full cost, because suppliers can still obtain early cash from the platform. The extension of DPO is a working capital release.
- **Financier**: fee and interest income on a short-term, low-risk asset (exposure to the buyer's credit), with potential data and cross-selling benefits.

Example 17.8 – Extending payment terms with supply chain finance. A fictional manufacturer, Brightwave Appliances, purchases $360 million of materials a year. It currently pays in 45 days. Its sourcing team announces new terms of 75 days, accompanied by an SCF program offered by its bank at a rate of 5.5 percent per year (360-day basis) to its suppliers.

Buyer's benefit. Payables rise by 30 days of purchases: 360,000,000 / 365 × 30 = 986,301 × 30 = **$29.59 million** of cash released. If Brightwave's alternative cost of short-term funds is 7 percent, the annual value is 0.07 × 29.59 = **$2.07 million**. (If the funds retire revolver borrowings, they reduce interest paid; if held, they increase cash balances. Either way, liquidity improves.)

Supplier's economics, one invoice. A supplier has a $1,000,000 invoice due on day 75. If it waits, it must carry the receivable for 75 days and fund itself at 11 percent per year (its own bank borrowing cost). Under SCF, it requests payment on day 15, 60 days early.

Discount at 5.5 percent: 1,000,000 × 0.055 × 60 / 360 = $9,166.67. Proceeds = $990,833.33. Supplier's own financing cost for the same 60 days: 1,000,000 × 0.11 × 60 / 360 = $18,333.33. Saving to the supplier relative to its own borrowing = **$9,166.67** per invoice.

Under the old terms the supplier was paid on day 45. Under the new terms it receives $990,833 on day 15, which is 30 days *sooner* than before, at an annual cost of 5.5 percent. A supplier that previously financed itself at 11 percent is therefore better off, not worse, even though the buyer's contractual date moved from 45 to 75 days. The program transfers the buyer's cheaper credit to the supplier, while the buyer captures the 30-day extension.

What to check. The supplier is better off only if (a) the SCF rate is below its own borrowing cost, (b) it is not forced to accept long terms with no real alternative, and (c) the program's availability is reliable. If the buyer's credit deteriorates, the facility may be withdrawn, and the supplier suddenly holds a 75-day receivable.

17.8.3 Risks, accounting, and disclosure

SCF is not free money, and several high-profile corporate failures in recent years involved opaque payables financing. The issues are:

- **Classification.** If a payable extended through a program loses its trade character, it may look like bank debt, and rating agencies may reclassify it as financial debt. Accounting standards now require disclosure of supplier finance arrangements (their terms, the carrying amount of liabilities in them, and the range of payment due dates versus comparable payables), so investors can see the dependence. IFRS requires such disclosure under amendments to IAS 7 and IFRS 7.
- **Concentration and refinancing risk.** The buyer's liquidity may rely on suppliers' acceptance of terms that depend on the program's continuing; a withdrawal creates a sudden demand for cash.
- **Supplier dependence.** If the supplier uses SCF as its main financing, it faces a cliff if the buyer's credit or the program changes.
- **Ethics and legal.** Payment-term extension should not squeeze small suppliers; several jurisdictions limit payment terms or require reporting of payment practices.

A sound policy is to use SCF to improve supply-chain stability, to evaluate extended terms as a price negotiation (the supplier will often reflect any financing cost in the invoice price), and to disclose the program fully.

17.8.4 Other supply-chain and trade finance tools

Inventory finance (warehouse receipts), *letters of credit* and documentary collections for cross-border sales, *forfaiting* of export receivables, and *pre-shipment finance* for suppliers. They shift risk from one party to another and are priced according to the party who bears the risk. The central lesson: **a working capital improvement is genuine only if it creates value (lower total system cost of capital or friction) rather than merely shifting funding costs to a weaker party who will price them back.**

Case Study

Illustrative case: Alder Valley Foods is a fictional company. [328]

Alder Valley Foods is a regional manufacturer of frozen vegetable products with annual sales of $280 million, cost of goods sold of $196 million, and a loyal customer base among grocery chains. Over the last two years the company has grown at 14 percent annually after winning private-label contracts from three supermarket groups. The chief financial officer, Daniel Reyes, has just received the third-quarter treasury report with two uncomfortable findings: the revolving credit facility, a committed $45 million line, was 92 percent drawn at the seasonal peak in September, and the bank has indicated that its covenant on net debt to EBITDA, currently at 3.1 times against a limit of 3.5 times, will be tested more strictly at the next renewal.

Where the cash went. Reyes asks the controller, Mei Lin Zhou, for a working capital diagnostic. She finds that receivables have risen from 36 days of sales to 47 days: the private-label contracts carry 60-day terms, and the three groups now account for 38 percent of sales. Inventory has risen from 52 to 64 days of cost of goods sold, in part because the company holds finished stock for peak summer demand and in part because it now manufactures 140 SKUs against 85 two years ago, including many low-volume private-label variants. Payables have shrunk slightly from 38 to 35 days, since the purchasing manager, Kofi Mensah, has been paying small growers early to protect supply of the harvest crop. Zhou estimates that, relative to two years ago, working capital absorbs about $16.5 million more cash than it would have at the earlier days ratios.

Options on the table. The sales director, Priya Nair, argues that the private-label business is strategic and its terms cannot be reopened: the supermarket groups might simply move to a competitor. She suggests instead that the company factor the receivables from those three customers, which would cost about 1.2 percent of invoice value for a 60-day average advance. The operations head, Tomasz Wrona, proposes SKU rationalization, which he estimates would cut inventory by eight days but reduce sales by about $6 million with contribution margin of 28 percent. Mensah suggests that the company could join the largest customer’s supply chain finance program: the customer is rated investment grade and has just announced that it will offer suppliers early payment at a spread of 90 basis points over the reference rate, which is currently 4.5 percent. The program would not change Alder Valley’s terms but would give it cash on day 5 after invoice approval instead of day 60.

Constraints. The grower payment issue is delicate: growers are small, depend on prompt payment, and are few in number in the region. Mensah says that at least a dozen growers would walk away if payment extended past 30 days. The covenant leaves little room for additional drawn debt, but factored receivables, if treated as true sales, would not count as debt. Reyes recalls that rating analysts and auditors might look through arrangements designed solely to circumvent the covenant. He also notes that the company’s equity owners want a $12 million expansion of the cold-storage facility, which would require financing from the same line.

The decision. Reyes has asked for a recommendation to the board that addresses the next twelve months. He wants an estimate of the cash that each lever could release, its annual cost, its effect on liquidity and covenants, the operational risks, and an explanation of how to track progress. He also wants a proposed target for the cash conversion cycle and a plan for monitoring it, because he does not want to find himself in the same position next year.

Discussion questions

1. Calculate Alder Valley’s current DSO, DIO, DPO, and cash conversion cycle using the data above (assuming the stated days are current), and the cycle two years ago. Verify Zhou’s estimate of $16.5 million of added working capital to within a reasonable approximation, noting any differences in how receivables and inventory are valued.
2. Compare the annualized cost of factoring (1.2 percent for 60 days) with the supply chain finance offer (reference rate plus 90 basis points). For which customers could each be used, and what does the comparison imply for how the company should negotiate its terms?

3. Evaluate Wrona's SKU rationalization: estimate the cash released from eight days of inventory, the lost contribution from a $6 million sales reduction, and the break-even carrying cost. What non-financial considerations might alter the decision?
4. Prepare the recommendation: which combination of levers would you present to the board, how would you treat the covenant and the disclosure obligations, how would you handle the growers, and what metrics and targets would you set to prevent a recurrence?

Key Takeaways

- Profitable growth can still consume cash; the cash conversion cycle (DIO + DSO − DPO) measures the number of days of operations the firm must finance, and each day of improvement releases a calculable sum of cash.
- Receivable days are worth sales per day; inventory and payable days are worth cost of goods sold per day. Specify the base before converting days into dollars.
- Match the maturity of financing to the life of the assets: permanent working capital belongs on long-term funding, and seasonal peaks on committed short-term lines. Aggressive funding lowers expected cost but raises refinancing risk.
- Credit policy decisions are incremental-profit problems: added contribution margin, minus carrying cost on the added receivables investment, minus incremental bad debts on the whole affected sales base.
- Forgoing a cash discount is expensive borrowing: 2/10 net 30 costs about 37 percent simple and 44.6 percent effective annually, so the discount should almost always be taken unless the firm is truly credit-constrained.
- The EOQ, $Q^* = \sqrt{(2DS/H)}$, balances ordering and carrying cost; the cost curve is flat near the optimum, so precise inputs are less important than recognizing the trade-off. Safety stock addresses variability during lead time.
- The Baumol and Miller-Orr models give the structure of the cash-versus-securities decision; the Miller-Orr spread increases with transaction cost and volatility and decreases with the interest rate.
- Quoted loan rates mislead; discount interest, compensating balances, fees, and day-count conventions all raise the effective cost per dollar of usable funds, and commitment fees make a line's per-dollar cost rise as utilization falls.
- A cash budget links sales forecasts and collection patterns to the timing of financing needs. The peak cumulative shortfall, not the average, sizes the committed facilities; stress-test the collection assumptions.
- Supply chain finance can lower system financing costs by lending on the buyer's credit, but it can also hide leverage and concentrate refinancing risk; disclose and size it prudently.

Review Questions

1. Distinguish between gross working capital, net working capital, and operating working capital. Why is the operating measure preferred in valuation and planning?
2. Explain the cash conversion cycle. Why does a negative cycle favor a firm, and what business models typically produce it?
3. Why is it incorrect to measure all three components of the cycle on the same base (sales)? What happens to the conversion of days into cash if you do so?
4. Describe the matching, aggressive, and conservative financing policies and the risks of each. In what circumstances might an aggressive policy be justified?

5. Describe the four levers of credit policy. How would you decide whether a firm should relax its credit standards?
6. Derive the cost of forgoing a 2/10 net 30 discount, and explain why the effective annual rate exceeds the simple annualized rate. What other considerations should limit stretching payables?
7. State the assumptions of the EOQ model and explain how just-in-time practices can be seen as an attack on its inputs rather than its logic.
8. What are the main sources of float and the tools used to manage it? Why does pooling reduce the total cash needed?
9. Compare the Baumol and Miller-Orr models in terms of assumptions and use. Why is the return point in Miller-Orr below the midpoint of the control band?
10. What is supply chain finance? Discuss how the benefits can be real and under what conditions it becomes a concern for investors and rating agencies.

Problems

1. **Cash conversion cycle.** A firm reports annual credit sales of $900 million, cost of goods sold of $630 million, average receivables of $105 million, average inventory of $98 million, and average payables of $63 million. Compute DSO, DIO, DPO, and the cash conversion cycle (365-day year).
2. **Cost of trade credit.** Suppliers offer terms of 3/10 net 45. Compute the simple annualized cost and the effective annual rate of forgoing the discount. Should a firm that can borrow at 12 percent take the discount?
3. **EOQ.** Annual demand is 36,000 units, ordering cost is $80 per order, and holding cost is $4 per unit per year. Compute the EOQ, number of orders per year, and total annual ordering plus carrying cost.
4. **Baumol.** A company needs $1.2 million of cash per year at a steady rate, securities transactions cost $40 each, and the interest rate is 6 percent. Find the optimal transfer size and the number of transfers per year.
5. **Miller-Orr.** The standard deviation of daily net cash flow is $1,500, the transaction cost is $40, the annual interest rate is 7.3 percent (daily rate 0.02 percent), and the lower limit is $5,000. Compute the spread, the return point, and the upper limit.
6. **Discount loan.** A firm needs $400,000 of usable funds for one year and is offered a discount loan (interest deducted in advance) at 9 percent. Find the face amount, the dollar interest, and the effective annual rate.
7. **Credit tightening.** A firm has credit sales of $18.0 million, variable cost of 65 percent of sales, DSO of 60 days, and bad debts of 3 percent of sales. A tighter policy would reduce sales by 5 percent, DSO to 40 days, and bad debts to 1.5 percent of the new sales level. The required return on receivables is 10 percent. Compute the net annual benefit (investment in receivables measured at variable cost).
8. **Supply chain finance.** A buyer with $240 million of annual purchases extends its payment terms from 40 to 70 days. If its cost of short-term funds is 6 percent, compute the cash released and the annual saving.

Answers to Selected Problems

1. DSO = 105 / 900 × 365 = 42.6 days; DIO = 98 / 630 × 365 = 56.8 days; DPO = 63 / 630 × 365 = 36.5 days; CCC = 42.6 + 56.8 − 36.5 = 62.9 days.
2. Periodic rate 3/97 = 3.093 percent; periods per year 365 / 35 = 10.43; simple cost = 32.3 percent; effective = (1.03093)^10.43 − 1 = 37.4 percent. Yes, take the discount (and borrow at 12 percent to do so).
3. EOQ = √(2 × 36,000 × 80 / 4) = √1,440,000 = 1,200 units; 30 orders per year; ordering 30 × 80 = $2,400; carrying 600 × 4 = $2,400; total $4,800.

4. C* = √(2 × 40 × 1,200,000 / 0.06) = √1,600,000,000 = $40,000; transfers per year = 1,200,000 / 40,000 = 30.

5. z = [3 × 40 × 1,500² / (4 × 0.0002)]^(1/3) = [270,000,000 / 0.0008]^(1/3) = (3.375 × 10¹¹)^(1/3) = $6,962 (the daily rate 7.3% / 365 is exactly 0.0002). Return point = 5,000 + 6,962 = $11,962; upper limit = 5,000 + 3 × 6,962 = $25,887.

6. Face = 400,000 / 0.91 = $439,560; interest = $39,560; effective rate = 0.09 / 0.91 = 9.89 percent.

7. New sales = $17.1 million. Receivables investment: before (18.0m / 365) × 60 × 0.65 = $1,923,288; after (17.1m / 365) × 40 × 0.65 = $1,218,082; reduction $705,205, saving 10 percent = $70,521. Bad debts: before 0.03 × 18.0m = $540,000; after 0.015 × 17.1m = $256,500; saving $283,500. Lost contribution: 0.9m × 0.35 = $315,000. Net = 70,521 + 283,500 − 315,000 = **+$39,021** per year; the tighter policy is marginally favorable.

8. Cash released = 240m / 365 × 30 = $19.73 million; annual saving = 0.06 × 19.73 = $1.18 million.

Part V — Derivatives and Risk Management

Chapter 18: Options

Learning Objectives

After studying this chapter you should be able to:

- Describe call and put options, compute payoffs and profits at expiration for single options and common combinations (protective put, covered call, straddle), and identify breakeven points.
- State the no-arbitrage bounds on option prices and use put-call parity to price one option from the others or to construct an arbitrage.
- Value European and American options with the binomial model, using both replicating portfolios and risk-neutral probabilities.
- Apply the Black-Scholes-Merton formula, including dividends, and interpret its inputs and its assumptions.
- Compute and use delta, gamma, vega, theta, and rho to understand and hedge option risk.
- Value employee stock options, and analyze equity, debt, and convertible securities as options on the firm's assets.

Opening Vignette

Illustrative scenario: Meridian Biosystems is a fictional company.

Meridian Biosystems, a mid-sized diagnostics firm with a share price of $40, is preparing its annual equity compensation grant. The compensation committee wants to give senior managers one million options struck at the current price, expiring in ten years and vesting over four. Chief Financial Officer Lucia Ferrante has been asked two questions the committee cannot answer from intuition. First: what is the grant worth to the executives, and what must the company record as an expense? Second: is the cost to shareholders the same as the value to the executives?

Ferrante's analyst brings a simple spreadsheet. The share price is $40 and the exercise price is $40, so the options are "at the money" and have no intrinsic value today. Is the value therefore zero? Of course not. If the stock rises to $70 in five years, the manager earns $30 per option. If it falls to $20, the manager simply does not exercise and loses nothing but the opportunity. The asymmetry, unlimited upside with limited downside, is worth real money. The analyst's spreadsheet, using an option-pricing model with a 30 percent volatility, a 4 percent interest rate, and an expected life of five years, values each option at about $12, so the grant is worth roughly $12 million before forfeitures.

A second surprise follows. The treasurer wants to hedge the share-count dilution, and a banker proposes that Meridian buy call options on its own shares. The banker quotes a price, and Ferrante wants to know why it differs from the model value and what could be learned from the market's *implied volatility*. Meanwhile, the firm's own convertible bond, issued two years earlier, has fallen in value even though the firm's credit is unchanged. The reason is the same: a convertible is a bond plus a call option, and the call option depends on the stock price and on volatility.

Options are everywhere in corporate finance. They are traded on exchanges, embedded in compensation, hidden in financing contracts, and implied in the capital structure itself. This chapter builds the toolkit: payoffs, pricing logic, the binomial and Black-Scholes models, the Greeks, and the application to employee options and corporate securities.

18.1 Options and Their Payoffs

18.1.1 Definitions

An **option** is a contract giving its holder the *right, but not the obligation*, to buy or sell an underlying asset at a fixed price on or before a specified date. The seller (writer) bears the obligation if the holder exercises. The terms are:

- **Call option**: right to *buy* the underlying at the **exercise (strike) price** K.
- **Put option**: right to *sell* the underlying at K.
- **European option**: exercisable only at **expiration** T. **American option**: exercisable at any time up to T.
- **Premium**: the price paid by the buyer to the writer when the contract is struck.
- **In the money / at the money / out of the money**: a call is in the money if S > K (the underlying price exceeds the strike), a put if S < K. Moneyness refers to the immediate exercise value.

Listed equity options typically cover 100 shares per contract; the premium is quoted per share. Options can also be written on indices, currencies, interest rates, commodities, and futures. Unlike forwards, which impose an obligation on both sides at no initial cost, an option has an up-front price because its holder cannot lose beyond the premium.

18.1.2 Payoffs at expiration

Let S_T denote the underlying price at expiration. The payoff to the **holder** of a European call and put are

```
Call payoff = max(S_T − K, 0)
Put payoff  = max(K − S_T, 0)
```

The payoff to the **writer** is the negative of the holder's. **Profit** subtracts the premium paid (or adds the premium received), ignoring interest on the premium. The key properties: a long call has limited downside (the premium) and unlimited upside; a long put gains as the asset falls, limited only by the fact that the price cannot go below zero (maximum payoff K); a short call has unlimited potential loss; a short put's loss is limited to K minus the premium.

Intrinsic value is the immediate exercise payoff, max(S − K, 0) for a call. **Time value** is the premium minus the intrinsic value. Time value reflects the chance that the option will become more valuable before expiry and is largest for at-the-money options with long maturities and high volatility.

18.1.3 Combinations

Options can be combined with each other and with the underlying to shape a risk profile.

- **Protective put**: own the asset and a put. The put sets a floor under the portfolio's value at K, like insurance, at the cost of the premium (the "deductible" is S − K).
- **Covered call**: own the asset and write a call. The premium cushions small losses and adds income, but upside is capped at K. This is common among investors seeking yield in flat markets.
- **Straddle**: buy a call and a put with the same strike and expiry. It profits from a large move in either direction, and is a pure bet on volatility.
- **Bull spread**: buy a call at K_1, sell a call at $K_2 > K_1$. Limited profit and loss.
- **Collar**: own the asset, buy a put at a low strike, and sell a call at a high strike; frequently structured at zero net premium to protect a concentrated holding.

Example 18.1 – Payoffs and profits of option strategies. The underlying stock trades at $50. A six-month call and put, both with strike $50, cost $4 and $3 respectively. An investor considers four strategies, each per share; the stock is bought at $50 where relevant. We compute profit (payoff minus cost, ignoring interest) at five expiry prices.

Long call: profit = max(S_T − 50, 0) − 4. *Long put:* profit = max(50 − S_T, 0) − 3. *Straddle:* sum of the two; cost $7. *Protective put:* (S_T − 50) + max(50 − S_T, 0) − 3. *Covered call:* (S_T − 50) − max(S_T − 50, 0) + 4.

For S_T = 35: call: 0 − 4 = −4; put: 15 − 3 = 12; straddle: 8; protective put: −15 + 15 − 3 = −3; covered call: −15 − 0 + 4 = −11. For S_T = 45: call −4; put 5 − 3 = 2; straddle −2; protective put: −5 + 5 − 3 = −3; covered call: −5 + 4 = −1. For S_T = 50: call −4; put −3; straddle −7; protective put: −3; covered call: 0 − 0 + 4 = 4. For S_T = 55: call 5 − 4 = 1; put −3; straddle −2; protective put: 5 + 0 − 3 = 2; covered call: 5 − 5 + 4 = 4. For S_T = 65: call 15 − 4 = 11; put −3; straddle 8; protective put: 15 − 3 = 12; covered call: 15 − 15 + 4 = 4.

Table 18.1 – Profit per share at expiration (Example 18.1).

Final stock price	Long call	Long put	Straddle	Protective put	Covered call
$35	−4	12	8	−3	−11
$45	−4	2	−2	−3	−1
$50	−4	−3	−7	−3	4
$55	1	−3	−2	2	4
$65	11	−3	8	12	4

Breakeven points. The long call breaks even at 50 + 4 = $54; the long put at 50 − 3 = $47. The straddle breaks even at 50 − 7 = $43 and 50 + 7 = $57, so the stock must move more than 14 percent either way for the buyer to profit. The protective put's worst loss is $3 (the premium), at any price at or below $50. The covered call is the mirror: it earns $4 at any price at or above $50 but loses $11 if the stock falls to $35. It can be seen that the protective put and the covered call are not symmetric: one buys insurance, the other sells it.

18.1.4 Why options matter to managers

Options matter in corporate finance for four reasons. First, managers use them to hedge exposures at a known cost (currency and commodity options). Second, they are an element of pay design. Third, many corporate securities, such as convertibles, warrants, callable bonds, and even common equity in a levered firm, are packages that contain options. Fourth, real investment decisions embed options: the right to defer, expand, abandon, or switch is worth something, and ignoring it understates the value of flexible projects (the pricing logic of this chapter carries over to such real options).

18.2 Arbitrage Bounds and Put-Call Parity

18.2.1 Bounds on option values

Without any model of the stock price process, no-arbitrage restrictions place bounds on option prices. For a European call on a non-dividend-paying stock with strike K, maturity T, risk-free continuously compounded rate r:

```
max(0, S − K e^(−rT)) ≤ C ≤ S
max(0, K e^(−rT) − S) ≤ P ≤ K e^(−rT)
```

Reasoning for the lower bound on the call: a portfolio of one call plus Ke^(−rT) in cash is worth at least S_T at expiry (if S_T > K, it is worth S_T; otherwise K > S_T), so it must be worth at least S today. Hence C ≥ S − Ke^(−rT). The call cannot be worth more than the stock itself, since the stock delivers everything the call delivers.

Other general results: American options are worth at least as much as otherwise identical European options; options with longer maturities (for American) are worth at least as much; an option's value is convex in the strike. A call on a stock that pays no dividend should never be exercised early, because exercising forfeits the time value and the interest on the strike; its American value equals its European value. American puts, and American calls on dividend-paying stocks, may be optimally exercised early.

18.2.2 Put-call parity

Consider two portfolios at time 0 for European options on a non-dividend-paying stock with the same K and T:

- Portfolio A: one call plus cash Ke^(−rT) invested at the risk-free rate.
- Portfolio B: one put plus one share.

At expiry, A is worth max(S_T − K, 0) + K = max(S_T, K). B is worth max(K − S_T, 0) + S_T = max(K, S_T). They are identical in every state, so they must cost the same today:

```
C + K e^(−rT) = P + S
```

or equivalently C − P = S − Ke^(−rT). With a continuous dividend yield q the stock term becomes Se^(−qT); with discrete dividends, subtract their present value from S. The relationship is model-independent: it needs only no-arbitrage. It implies that a call can be synthesized from a put, the stock, and borrowing (C = P + S − Ke^(−rT)), that a long call plus short put equals a forward purchase at K, and that put prices are determined by call prices.

For American options, parity holds only as an inequality: S − K ≤ C − P ≤ S − Ke^(−rT).

Example 18.2 – A parity arbitrage. A non-dividend-paying stock trades at $100. A six-month European call and put, both struck at $100, are quoted at $9.50 and $6.50. The continuously compounded risk-free rate is 5 percent.

Step 1: PV of the strike. 100 × e^(−0.05 × 0.5) = 100 × e^(−0.025) = 100 × 0.975310 = $97.53.

Step 2: parity check. C − P = 9.50 − 6.50 = $3.00. S − Ke^(−rT) = 100 − 97.53 = $2.47. The call is $0.53 too expensive relative to the put.

Step 3: arbitrage. Sell the call (+9.50), buy the put (−6.50), buy the stock (−100), and borrow $97.53 (+97.53). Net cash today: 9.50 − 6.50 − 100 + 97.53 = **+$0.53**.

Step 4: expiry. The short call, long put, and the stock combine to deliver exactly K = $100 in every state: if S_T > 100 the call is exercised against us (−(S_T − 100)) and we hold the stock worth S_T, net 100; if S_T < 100 the put pays 100 − S_T, and with the stock again 100. Repaying the loan costs 97.53 × e^(0.025) = $100. The net at expiry is zero in every state. The profit is the $0.53 of cash received today.

If the parity gap were closed by market forces, the call would fall or the put rise until C − P = $2.47. In practice, bid-ask spreads, borrowing costs, dividends, and early-exercise premiums narrow the gap before costs eliminate it.

18.2.3 Common mistakes

Two errors are frequent. The first is to apply parity to American options without adjusting for early exercise. The second is to forget dividends: a stock paying a known dividend before expiration reduces the call and raises the put, and parity must use the stock price net of the present value of those dividends.

18.3 The Binomial Option Pricing Model

18.3.1 The one-period model: replication

The binomial model of Cox, Ross, and Rubinstein (building on Sharpe) values an option by building a portfolio of the underlying stock and risk-free borrowing that has *exactly* the same payoffs as the option. If it does, the option must cost what the portfolio costs. Suppose that over one period the stock price S can move to either uS (up) or dS (down), where u > 1 > d, and that the gross risk-free return per period is R = 1 + r_f, with d < R < u (otherwise arbitrage exists). The option pays C_u in the up state and C_d in the down state.

Form a portfolio of Δ shares and B dollars of risk-free lending (negative for borrowing). To match payoffs:

```
Δ·uS + B·R = C_u
Δ·dS + B·R = C_d
```

Subtracting gives the **hedge ratio** (delta):

```
Δ = (C_u − C_d) / (S (u − d))
B = (C_u·d − C_d·u) / ((u − d) R)    [the cash holding; negative means borrowing]
```

The option's value is C = ΔS + B. This logic is strikingly general: no assumption is made about investors' risk preferences or about the actual probabilities of the up and down moves. The reason is that the replicating portfolio *removes* the risk, so the option need only be priced relative to the stock.

18.3.2 Risk-neutral valuation

The same value can be written as a discounted expectation under artificial **risk-neutral probabilities**:

```
p = (R − d) / (u − d)
C = [ p · C_u + (1 − p) · C_d ] / R
```

Here p is not the true probability of an up-move. It is the probability that would make the stock's expected return equal to the risk-free rate, so that discounting at the risk-free rate is valid. The result extends multi-period by rolling back through the tree, applying this formula at each node. This is one of the most important ideas in modern finance: pricing derivatives by expectation under a measure in which assets earn the risk-free rate, which works because the replication argument is available at every node.

Example 18.3 – One-period call by replication and risk-neutral valuation. A stock is at \$100. Over one year it will either rise to \$120 (u = 1.20) or fall to \$90 (d = 0.90). The risk-free rate is 5 percent (R = 1.05). A European call has strike K = \$105.

Step 1: payoffs. C_u = max(120 − 105, 0) = 15; C_d = max(90 − 105, 0) = 0.

Step 2: delta. Δ = (15 − 0) / (100 × (1.20 − 0.90)) = 15 / 30 = 0.5 share.

Step 3: borrowing. If the stock falls to \$90, 0.5 shares are worth \$45 and the call pays 0, so the loan must be repaid with \$45: B = −45 / 1.05 = −\$42.857. Check the up state: 0.5 × 120 = 60; repay 42.857 × 1.05 = 45; net 15 = C_u ✓.

Step 4: call value. C = 0.5 × 100 − 42.857 = **\$7.143**.

Step 5: risk-neutral check. p = (1.05 − 0.90) / 0.30 = 0.50. C = (0.5 × 15 + 0.5 × 0) / 1.05 = 7.5 / 1.05 = \$7.143 ✓.

Interpretation. Suppose someone offered the call at \$8.00. We would sell it and buy the replicating portfolio for \$7.143, pocketing \$0.857 with no risk. Suppose the true probability of an up-move were 80 percent: the option's *expected* payoff would be 12, but the option's price would remain \$7.143, because the stock's own price already incorporates the true probabilities and investors' required returns.

18.3.3 Multi-period trees and American options

Divide the life of the option into n steps. The stock moves up by u or down by d each step, so after n steps there are n + 1 possible prices. Starting at the end of the tree, value the option at each terminal node by its payoff, then step backward: at each earlier node, the value is the discounted risk-neutral expectation of its two successors. For an **American** option, at each node compare this "continuation value" with the immediate exercise value, and take the larger.

To make the tree consistent with a volatility σ (annualized) over steps of length Δt, Cox, Ross, and Rubinstein set

```
u = e^(σ√Δt),   d = 1/u,   p = (e^(rΔt) − d) / (u − d)
```

As the number of steps grows large, the binomial price converges to the Black-Scholes price for European options. The tree remains valuable because it handles early exercise, discrete dividends, and path-dependent or state-dependent features that closed-form formulas cannot.

Example 18.4 – A two-period American put. A stock is at \$100. Each period (six months), it either rises by 10 percent (u = 1.10) or falls by 10 percent (d = 0.90). The risk-free rate is 3 percent per period (R = 1.03). Value a put with strike K = \$100 expiring after two periods, both European and American.

Step 1: risk-neutral probability. p = (1.03 − 0.90) / (1.10 − 0.90) = 0.13 / 0.20 = 0.65; 1 − p = 0.35.

Step 2: terminal stock prices. uu: 121; ud = du: 99; dd: 81. Terminal put payoffs: 0, 1, 19.

Step 3: European put values at period 1. Up node (S = 110): P_u = (0.65 × 0 + 0.35 × 1) / 1.03 = 0.35 / 1.03 = 0.3398. Down node (S = 90): P_d = (0.65 × 1 + 0.35 × 19) / 1.03 = (0.65 + 6.65) / 1.03 = 7.30 / 1.03 = 7.0874.

Step 4: European put at time 0. P = (0.65 × 0.3398 + 0.35 × 7.0874) / 1.03 = (0.2209 + 2.4806) / 1.03 = 2.7015 / 1.03 = **\$2.623**.

Step 5: American put. At the up node, exercise value = 100 − 110 < 0, so hold: 0.3398. At the down node (S = 90), exercise value = 100 − 90 = \$10.00, which exceeds the continuation value of 7.0874. So exercise early: the node value is 10.00. At time 0: P = (0.65 × 0.3398 + 0.35 × 10.00) / 1.03 = (0.2209 + 3.50) / 1.03 = 3.7209 / 1.03 = **\$3.612**. (Immediate exercise at time 0 would pay 0.)

Interpretation. The right to exercise early adds \$0.99 (3.612 − 2.623), about 38 percent to the value. In the down state the holder does better to collect \$10 now, and invest it at 3 percent, than to wait for the chance of a further decline. Deep in-the-money puts, high interest rates, and low volatility make early exercise more attractive.

Table 18.2 – Option replication and valuation summary (Examples 18.3 and 18.4).

Item	One-period call	Two-period European put	Two-period American put
Spot	100	100	100
Strike	105	100	100
u, d	1.20, 0.90	1.10, 0.90	1.10, 0.90
Risk-free gross rate per period	1.05	1.03	1.03
Risk-neutral p	0.500	0.650	0.650
Value	7.143	2.623	3.612

18.3.4 Practical notes on the binomial approach

Calibration is the real work: choosing u, d consistent with volatility estimates and the time grid. Convergence is not monotone because of the position of the strike relative to the nodes, so practitioners use a few hundred steps or averaging. The tree also supports numerical valuation of options on assets with dividend yields (set p = (e^((r−q)Δt) − d)/(u − d)), currencies (q = foreign rate), and futures (q = r).

18.4 The Black-Scholes-Merton Model

18.4.1 The formula

Fischer Black and Myron Scholes (1973), and independently Robert Merton, derived a closed-form price for European options by the same replication logic as the binomial model, extended to continuous trading. They assumed that the stock price follows a geometric Brownian motion (constant volatility σ, log returns normally distributed), that interest rates are constant, that there are no transaction costs or taxes, that the asset can be traded continuously and shorted, and that no dividends are paid (an extension handles dividends). The price of a European call is

```
C = S N(d₁) − K e^(−rT) N(d₂)
P = K e^(−rT) N(−d₂) − S N(−d₁)

d₁ = [ ln(S/K) + (r + σ²/2) T ] / (σ √T)
d₂ = d₁ − σ √T
```

where N(·) is the standard normal cumulative distribution function. With a continuous dividend yield q, replace S with Se^(−qT) and use (r − q + σ²/2) in d_1.

The formula has a financial interpretation: $N(d_2)$ is the risk-neutral probability that the option finishes in the money, so K e^(−rT) $N(d_2)$ is the present value of the expected payment of the strike. S $N(d_1)$ is the present value of the expected stock received, conditional on exercise; $N(d_1)$ is also the delta. The call is thus a leveraged position in the stock: it is a replicating portfolio of $N(d_1)$ shares financed by borrowing Ke^(−rT)$N(d_2)$.

18.4.2 The inputs

- **S**, the current price of the underlying, and **K**, the strike: observable.
- **T**, time to expiry in years.
- **r**, the continuously compounded risk-free rate for maturity T (a government rate or OIS rate).
- **σ**, the annualized volatility of the underlying's log returns: the one unobservable input. It can be estimated from historical returns (the standard deviation of daily log returns multiplied by √252) or inferred from market prices as **implied volatility**, the σ that makes the model price equal the observed price. Implied volatility is a quotation

convention: traders talk of volatility, not price. Across strikes and maturities, implied volatilities typically are not flat: the "smile" or "skew," in which low-strike options carry higher implied volatility in equity markets, is evidence that the constant-volatility assumption fails, particularly in the tail.

- **q**, the dividend yield or the present value of discrete dividends.

An important feature is that the **expected return on the stock does not appear**. This follows directly from the replication argument. Two investors with very different views about the stock's growth should agree on the call's price if they agree on volatility.

18.4.3 Worked pricing

Example 18.5 – Pricing a European call and put. A non-dividend-paying stock trades at S = \$100. A European option with strike K = \$105 expires in six months (T = 0.5). The risk-free rate is r = 4 percent (continuous) and the volatility is σ = 25 percent.

Step 1: d_1. ln(100 / 105) = ln(0.952381) = −0.048790. (r + σ²/2) T = (0.04 + 0.03125) × 0.5 = 0.035625. Numerator = −0.048790 + 0.035625 = −0.013165. Denominator σ√T = 0.25 × 0.707107 = 0.176777. **$d_1 = -0.0745$.**

Step 2: d_2. $d_2 = -0.0745 - 0.1768 =$ **-0.2513.**

Step 3: normal probabilities. $N(d_1) = N(-0.0745) = 0.4703$; $N(d_2) = N(-0.2513) = 0.4008$. Then $N(-d_1) = 0.5297$ and $N(-d_2) = 0.5992$.

Step 4: PV of the strike. 105 × e^(−0.02) = 105 × 0.980199 = \$102.921.

Step 5: call. C = 100 × 0.4703 − 102.921 × 0.4008 = 47.03 − 41.25 = **\$5.78.**

Step 6: put. P = 102.921 × 0.5992 − 100 × 0.5297 = 61.67 − 52.97 = **\$8.70.**

Step 7: parity check. C − P = 5.78 − 8.70 = −2.92. S − Ke^(−rT) = 100 − 102.92 = −2.92 ✓.

Interpretation. The call is out of the money (S < K) yet worth \$5.78, entirely time value. Its delta of 0.47 means it behaves like 0.47 shares for small moves. If volatility were 30 percent instead of 25 percent, the call would be worth \$7.19, an increase of 24 percent; options are volatility bets.

Table 18.3 – At-the-money call value as a function of volatility (S = K = \$100, T = 0.5, r = 5%).

Volatility	Call value	Increase over 15% case
15%	5.53	–
20%	6.89	24.6%
25%	8.26	49.4%
30%	9.63	74.3%
40%	12.39	124.1%

Value is almost linear in volatility for at-the-money options, which is why a rough rule is C ≈ 0.4 S σ √T when r is small: for σ = 30 percent, that gives 0.4 × 100 × 0.3 × 0.7071 = 8.49, near the true 9.63 (the interest component adds some value).

18.4.4 Limitations

The model's flaws are well understood: volatility is not constant and jumps occur; markets have transaction costs and cannot trade continuously; interest rates vary; American features and dividends complicate matters. Despite this, the model remains the benchmark language of derivatives markets, mainly because deviations are managed by quoting a *different*

implied volatility for each strike. In corporate settings, where options are valued for accounting or pay design, its inputs (especially the volatility over long horizons) deserve judgment, and the sensitivity of the result to them should be tested.

18.5 The Greeks and Hedging

18.5.1 Sensitivities

The **Greeks** are partial derivatives of the option price with respect to its inputs. They tell a trader or risk manager how the value will change when the market moves, and so how to hedge. For a European call on a stock with dividend yield q (set q = 0 if none), with φ(x) denoting the standard normal density:

Greek	Meaning	Call	Put
Delta Δ	$\partial V/\partial S$	$e^{-qT} N(d_1)$	$e^{-qT} [N(d_1) - 1]$
Gamma Γ	$\partial^2 V/\partial S^2$	$e^{-qT} \varphi(d_1) / (S \sigma \sqrt{T})$	same as call
Vega ν	$\partial V/\partial \sigma$	$S e^{-qT} \varphi(d_1) \sqrt{T}$	same as call
Theta Θ	$\partial V/\partial t$	$-S e^{-qT} \varphi(d_1) \sigma / (2\sqrt{T}) - r K e^{-rT} N(d_2) + q S e^{-qT} N(d_1)$	$-S e^{-qT} \varphi(d_1) \sigma / (2\sqrt{T}) + r K e^{-rT} N(-d_2) - q S e^{-qT} N(-d_1)$
Rho ρ	$\partial V/\partial r$	$K T e^{-rT} N(d_2)$	$-K T e^{-rT} N(-d_2)$

Delta is the hedge ratio and the approximate probability (in the risk-neutral world) of finishing in the money. Call deltas run from 0 to 1 and put deltas from −1 to 0. *Gamma* measures the curvature: how fast delta changes. It is highest for at-the-money options close to expiry. *Vega* is sensitivity to volatility (conventionally quoted per one percentage point). *Theta* is time decay: the option loses time value as expiry approaches, and long option holders "pay" theta daily while writers collect it. Positive gamma and negative theta go together: the holder benefits from large moves and pays for the privilege through time decay. *Rho* is interest-rate sensitivity and is usually the least important for short-dated equity options.

A useful relationship (for a delta-hedged portfolio, derived from the Black-Scholes equation) is $\Theta + r S \Delta + \tfrac{1}{2} \sigma^2 S^2 \Gamma = r V$. For a delta-neutral position this reduces to $\Theta \approx -\tfrac{1}{2} \sigma^2 S^2 \Gamma$: the daily time decay received by the seller of options is compensation for the gamma risk borne.

18.5.2 Delta hedging

A market maker who sells a call can hedge by buying Δ shares per option. The position's value is then insensitive to small moves in S. But as S moves, Δ changes (gamma), so the hedge must be *rebalanced*. Profit or loss arises from the gap between realized volatility and the volatility priced into the option: the hedged option seller who sold at 30 percent implied volatility makes money if the stock subsequently realizes less than 30 percent. In practice, rebalancing is discrete, transaction costs matter, and volatility itself moves (vega risk), so the hedge is imperfect. Large books are hedged in aggregate: net delta, gamma, and vega limits are set by risk managers.

Example 18.6 – Greeks of an at-the-money option and delta-gamma hedging. An at-the-money call has S = K = \$100, T = 0.5, r = 5 percent, σ = 30 percent, no dividends.

Step 1: d_1 and d_2. $d_1 = [0 + (0.05 + 0.045) \times 0.5] / (0.30 \times 0.707107) = 0.0475 / 0.212132 = 0.2239$; $d_2 = 0.2239 - 0.2121 = 0.0118$.

Step 2: price. $N(d_1) = 0.5886$, $N(d_2) = 0.5047$. $Ke^{-rT} = 97.531$. $C = 100 \times 0.5886 - 97.531 \times 0.5047 = 58.86 - 49.22 =$ **\$9.63**.

Step 3: Greeks. φ(0.2239) = 0.3891 (the standard normal density at 0.2239). - Delta = $N(d_1)$ = **0.5886**. - Gamma = 0.3891 / (100 × 0.30 × 0.7071) = 0.3891 / 21.213 = **0.01834** per $1. - Vega = 100 × 0.3891 × 0.7071 = **27.51** per 1.00 of volatility, or $0.275 per volatility point. - Theta ≈ **−$10.71 per year**, or about −$0.0294 per calendar day. - Rho = K T e^(−rT) $N(d_2)$ = 100 × 0.5 × 0.9753 × 0.5047 = **24.61** per 1.00, or $0.246 per percentage point of rate.

Step 4: hedging a short position. Treat each call as covering one share. A bank sells 10,000 calls and buys 10,000 × 0.5886 = **5,886 shares** to be delta neutral.

Step 5: a $2 move. Suppose the stock rises to $102. The call reprices to $10.848 (a rise of $1.213 per call), so the loss on 10,000 short calls is $12,133. The 5,886 shares gain 5,886 × 2 = $11,772. Net loss: **$361**. A delta-only estimate of the call change is 0.5886 × 2 = 1.177. The delta-gamma estimate adds ½ × 0.01834 × 2^2 = 0.0367, giving 1.214, close to the actual 1.213. The residual loss of $361 is gamma: a short-gamma position loses whichever way the stock moves.

Step 6: rebalancing. New delta ≈ 0.5886 + 0.01834 × 2 = 0.6253, so the bank needs 6,253 shares: buy about **367 more** at $102 (buying after the rise, selling after a fall, "buy high, sell low" being the cost of being short gamma).

Step 7: other exposures. A one-point rise in implied volatility raises each call by $0.275, a loss of about $2,751 for the bank. One day of decay earns the bank 10,000 × 0.0294 = about **$294**, which is the compensation for bearing gamma and vega risk.

18.5.3 Using Greeks in corporate risk management

Corporate treasurers rarely run option books, but they use the Greeks conceptually. The cost of a currency collar can be set so that the delta of the sold option and bought option nearly offset in premium terms. The effective hedge ratio of a protective put is its delta, not 1. Sizing a hedge program with options requires deciding whether the exposure should be protected against rate or price moves *at expiry* (use the option's payoff) or *during the life* (account for delta and mark-to-market). And whenever a firm sells options (callable debt, embedded customer options), it should recognize that it is short gamma and vega and incurs valuation volatility under fair-value accounting.

18.6 Employee Stock Options

18.6.1 Features and incentive logic

An **employee stock option** (ESO) gives an employee the right to buy company shares at a fixed exercise price, usually set at the market price on the grant date, for a period of up to ten years. Typical features:

- **Vesting**: the option cannot be exercised for a period (cliff or graded, e.g., four years). Employees who leave before vesting forfeit unvested options.
- **Non-transferability**: ESOs cannot be sold or hedged freely, unlike traded options. A risk-averse, under-diversified employee will usually value them less than a diversified investor would, and will exercise early to lock in gains.
- **Dilution**: exercise creates new shares (or uses treasury shares), so the proceeds are the strike, but the value given up is the market price. The existing shareholders bear the dilution.

The incentive argument is that a manager with options benefits from share price appreciation and, because of the convex payoff, from *volatility* as well. This can align interests with those of shareholders, but it can also encourage risk taking (a convex payoff gives a manager an incentive to raise risk) and short-term focus. Design devices to counter this include longer vesting, holding requirements, performance-conditioned vesting, and strike prices indexed to peers or set above market.

18.6.2 Valuation and expensing

Under IFRS 2 and the corresponding US standard (ASC 718), equity-settled share-based payments are measured at the **grant-date fair value** and recognized as an expense over the vesting period, adjusted for the number of options expected to vest. Because employee options are not traded, firms use an option pricing model (Black-Scholes-Merton or a lattice) with these adaptations:

1. Use the **expected life** of the option, not the contractual term, to reflect early exercise, which is typically triggered at some multiple of the strike or at departure.
2. Adjust the expected number of options for **forfeitures** during vesting.
3. Use an **expected volatility** that considers historical volatility and implied volatility of traded options, if available, and the term structure.
4. Include **expected dividends** (as a yield q).

Strictly, the Black-Scholes value ignores dilution. If the firm has N shares and issues M options, an exercised option increases the share count, so the holder shares the gain with others. A simple adjustment multiplies the standard value by N / (N + M) (the dilution factor) when the market has already priced the options.

The question asked by the compensation committee is whether the cost to shareholders equals the value to the employee. In general, it does not: the cost to the firm (fair value) is typically higher than the employee's subjective value because of non-diversification and non-tradability. This wedge is a reason that firms sometimes prefer restricted stock or performance shares for senior staff and options for broad-based start-up grants where cash is scarce.

Example 18.7 – Valuing and expensing an executive option grant. Meridian Biosystems grants 1,000,000 options at-the-money: S = K = \$40. Expected life is 5 years (the contractual term is 10); expected volatility is 30 percent; the risk-free rate is 4 percent; the dividend yield is 1 percent. Vesting is graded straight-line over 4 years, and forfeitures are expected to remove 8 percent of the options. There are 50 million shares outstanding.

Step 1: d_1. $\ln(40 / 40) = 0$. $(r - q + \sigma^2/2)\,T = (0.04 - 0.01 + 0.045) \times 5 = 0.075 \times 5 = 0.375$. $\sigma\sqrt{T} = 0.30 \times 2.23607 = 0.670820$. $d_1 = 0.375 / 0.670820 =$ **0.5590**.

Step 2: d_2. $0.5590 - 0.6708 =$ **−0.1118**.

Step 3: normal probabilities. $N(d_1) = 0.7119$, $N(d_2) = 0.4555$.

Step 4: discount factors. e^(−qT) = e^(−0.05) = 0.951229; e^(−rT) = e^(−0.20) = 0.818731.

Step 5: value per option. $C = 40 \times 0.951229 \times 0.7119 - 40 \times 0.818731 \times 0.4555 = 27.088 - 14.918 =$ **\$12.17**.

Step 6: total fair value. Expected vesting options = 1,000,000 × 0.92 = 920,000. Total = 920,000 × 12.17 = **\$11.20 million**.

Step 7: annual expense. Straight-line over 4 years: 11.20 / 4 = **\$2.80 million** per year in the income statement (a non-cash charge; the offset is credited to equity). True-up adjustments occur if actual forfeitures differ.

Step 8: dilution adjustment (informational). 12.17 × 50 / 51 = \$11.93 per option.

Step 9: sensitivities. The result is quite sensitive to the inputs, as the table shows.

Table 18.4 – Value per option for the Meridian grant under alternative assumptions (S = K = \$40, r = 4%, q = 1%).

Volatility	Expected life	Value per option
20%	5 years	$9.26
25%	5 years	$10.71
30%	3 years	$9.40
30%	4 years	$10.89
30%	5 years	$12.17
30%	6 years	$13.30
35%	5 years	$13.61
40%	5 years	$15.03

An error of ten volatility points (20 versus 30 percent) shifts the value by about 30 percent, so documentation of how volatility and life were chosen is part of good governance and audit readiness. Note also that the expense is fixed at grant date: if the stock later falls and the options go underwater, the expense is not reversed (unless the performance or service conditions fail), which has led some boards to reprice options, a controversial act because it rewards failure.

18.6.3 Design issues and the tax and governance context

Tax treatment varies by jurisdiction (incentive stock options, qualified plans, and so on) and affects the form and timing of grants. Governance issues include "spring-loading" (grants made before good news), backdating (illegal when undisclosed), and overhang (options outstanding as a percentage of shares). Shareholder-approved plans, independent compensation committees, and transparent disclosure of valuation assumptions reduce these concerns.

18.7 Corporate Securities as Options

18.7.1 Equity as a call on the firm's assets

Merton (1974) observed that the capital structure of a firm with a single zero-coupon debt issue can be described in option terms. Let V be the value of the firm's assets and D the face value of debt due at date T. At T:

- If $V_T > D$, creditors are paid in full, and shareholders keep $V_T - D$.
- If $V_T < D$, shareholders walk away (limited liability); creditors receive V_T.

So the equity payoff is $\max(V_T - D, 0)$: a **call option on the firm's assets with strike D**. The debt payoff is $D - \max(D - V_T, 0)$: a risk-free bond *minus* a put option on the assets. Thus

```
E = Call(V, D, T, r, σ_V)
Debt = V − E = D e^(−rT) − Put(V, D, T, r, σ_V)
```

These relationships are put-call parity applied to the firm. They produce several managerial insights.

1. **Risk shifting.** Equity is a call; calls rise in value with volatility. Shareholders therefore gain, at the expense of creditors, when the firm shifts into riskier assets, an incentive that grows as leverage rises. This is the "asset substitution" problem, and explains covenants.
2. **The credit spread is the price of a put.** The yield on risky debt exceeds the risk-free rate by an amount that compensates creditors for being short a put on the firm's assets. Credit models used by banks and rating agencies are built on this framework.

3. **Shareholders hold an option value even in distress.** A firm with V < D can still have positive equity value, as long as there is time and volatility. This explains why equity does not fall to zero at the point of technical insolvency, and why equity holders resist early liquidation.
4. **Equity beta rises with leverage** because the delta of a call is less than one but its elasticity (percentage change in E per percentage change in V) exceeds one: $\beta_E = \beta_V \times \Delta \times (V/E)$.

Example 18.8 – Equity and debt valued with the Merton model. A firm has asset value V = $120 million, a single zero-coupon bond with face value D = $80 million due in T = 2 years, asset volatility σ = 30 percent, and risk-free rate r = 4 percent.

Step 1: d_1. ln(120 / 80) = ln(1.5) = 0.405465. $(r + \sigma^2/2)T = (0.04 + 0.045) \times 2 = 0.17$. Numerator 0.575465. $\sigma\sqrt{T} = 0.30 \times 1.414214 = 0.424264$. d_1 = **1.3564**.

Step 2: d_2. 1.3564 − 0.4243 = **0.9321**.

Step 3: probabilities. $N(d_1) = 0.9125$; $N(d_2) = 0.8244$.

Step 4: PV of debt face. 80 × e^(−0.08) = 80 × 0.923116 = $73.849 million.

Step 5: equity. E = 120 × 0.9125 − 73.849 × 0.8244 = 109.50 − 60.88 = **$48.62 million**.

Step 6: debt. D_0 = 120 − 48.62 = **$71.38 million**.

Step 7: credit yield. y = −ln(71.38 / 80) / 2 = −ln(0.89225) / 2 = 0.11400 / 2 = **5.70 percent**, a spread of **1.70 percent** over the risk-free rate. The risk-neutral probability of default is $1 - N(d_2) = 17.6$ percent.

Step 8: parity check. The embedded put value is 73.849 − 71.38 = **$2.47 million**, so the creditors' payoff is a risk-free bond worth $73.85 million less a $2.47 million put.

Step 9: risk shifting. The firm replaces assets with riskier assets of the same value, raising σ to 45 percent. Recomputing gives E = $53.61 million and debt value $66.39 million, with a yield of 9.33 percent. Shareholders gain $4.99 million and creditors lose exactly that amount, even though firm value V did not change. This is the transfer of wealth that creditors guard against with covenants and security.

18.7.2 Convertible bonds and warrants

A **convertible bond** is a straight bond plus a call option on the issuer's shares, exercisable by giving up the bond. A **warrant** is a call option issued by the company itself, with dilution when exercised. Valuation separates the pieces:

```
Convertible value ≈ Straight-bond value + (Conversion ratio × Option value) [adjusted for call provisions, credit risk, and dilution]
```

The conversion option lets issuers pay a lower coupon (the buyer is paid partly in an option) and reduces the agency costs of debt, because the option offsets the risk-shifting incentive described above: if managers take extra risks, the convertible holder participates through the option.

Example 18.9 – Decomposing a convertible bond. A fictional issuer sells a five-year convertible with face value $1,000 and a 3 percent annual coupon. Each bond converts into 20 shares (conversion price $50). The stock trades at $40, volatility is 30 percent, the risk-free rate is 4 percent, no dividends are paid. A comparable straight bond of the same issuer would yield 6 percent.

Step 1: straight bond value. PV of coupons: 30 × [1 − 1.06^(−5)] / 0.06 = 30 × 4.21236 = \$126.37. PV of face: 1,000 / 1.06^5 = 1,000 / 1.338226 = \$747.26. Total **\$873.63**.

Step 2: option per share. S = 40, K = 50, T = 5, r = 4 percent, σ = 30 percent. ln(40 / 50) = −0.223144. (0.04 + 0.045) × 5 = 0.425. Numerator 0.201856. σ√T = 0.670820. d_1 = 0.3009; d_2 = −0.3699. $N(d_1)$ = 0.6182; $N(d_2)$ = 0.3557. PV of strike = 50 × 0.818731 = 40.937. C = 40 × 0.6182 − 40.937 × 0.3557 = 24.73 − 14.56 = **\$10.17**.

Step 3: option per bond. 20 × 10.17 = **\$203.36**.

Step 4: convertible value. 873.63 + 203.36 = **\$1,076.99**, about 107.7 percent of par.

Interpretation. Investors accept a 3 percent coupon instead of 6 percent because they receive a conversion option. Relative to a 6 percent coupon, the 3 percent coupon is worth \$126.37 less in present value (30 × 4.21236); the option is worth \$203.36. Because the package is worth \$1,077 against a \$1,000 issue price, the issuer is underpricing it: in practice it would raise the conversion price, cut the conversion ratio, or lower the coupon until the value equals par. The lesson for issuers is that convertible financing is not cheap debt; its low coupon is paid for with an option on the upside, and the true cost of the bond includes the option's value.

18.7.3 Other embedded options

Many corporate instruments embed options: **callable bonds** (the issuer holds a call, so the bond is worth less than a straight bond: callable = straight − call), **putable bonds**, prepayment options in mortgages, **loan commitments** (the lender writes a put and a call), guarantees (a put on the guaranteed asset), and **rights issues** (short-dated calls). The pricing methods of this chapter apply with modification.

18.8 Practical Guidance for Managers

1. **Distinguish value from payoff.** An option's market value includes time value; do not equate it with the intrinsic value at today's price. An at-the-money option is not worthless.
2. **Volatility is the key assumption.** Compare historical, implied, and peer volatility, and run sensitivities. When valuing long-dated corporate options, the term structure of volatility matters.
3. **Use parity as a sanity check.** If a quoted call, put, stock price, and strike do not satisfy parity, there is an error or a market friction (borrow cost, dividends, early exercise).
4. **Recognize options in contracts.** Whenever a contract gives one party a right without an obligation (an extension, a termination, a price cap, a conversion), it has an option value that should be priced, not ignored.
5. **Hedge with understanding.** Know the delta, gamma, and vega of any option position, and the accounting implications (hedge accounting, fair-value changes, and disclosure requirements).
6. **Do not mistake the model for the market.** Black-Scholes is a quoting convention and an approximation; model risk is real, particularly for long maturities, thinly traded underlyings, and exotic features.

Case Study

Illustrative case: Halden Marine Technologies is a fictional company.

Halden Marine Technologies designs and builds autonomous survey vessels for offshore wind developers and port authorities. Listed on a mid-cap exchange for four years, it has 80 million shares outstanding trading at \$25, a market capitalization of \$2.0 billion, and net debt of \$400 million, of which \$300 million is a single five-year bond with a 4.5

percent coupon issued three years ago. The chief financial officer, Ingrid Solberg, is preparing three decisions for the March board meeting, each of which turns out to be about options.

Decision one: the management incentive plan. The compensation committee, chaired by independent director Hassan Idris, wants to replace the existing restricted-share awards with a grant of 2.4 million at-the-money options (3 percent of shares) to 40 senior employees, with a 10-year term and cliff vesting after three years. The human resources director, Camille Dubois, argues that options are “free” because no cash leaves the company. A shareholder representative has written to the board objecting that the grants reward managers for volatility rather than performance. The company’s historical annualized volatility is 38 percent, but the implied volatility on listed options with a one-year maturity is 31 percent, and the average of peers’ five-year volatilities is 34 percent. The risk-free rate is 4 percent and Halden pays no dividend. Employee surveys suggest that executives typically exercise about five years after grant, if vested, unless the stock has doubled, in which case they exercise sooner. Solberg estimates the expected forfeiture rate at 10 percent.

Decision two: the convertible. Halden needs $250 million to fund a new shipyard. Bankers propose a seven-year convertible bond with a 2 percent coupon and a 40 percent conversion premium (a conversion price of $35). A straight seven-year bond would cost Halden about 6.5 percent. The treasury director, Pieter Van Dijk, notes that the convertible’s coupon saving is $11.25 million a year and that the shares reserved for conversion would amount to 7.1 million (about 8.9 percent of the current share count). Board member Elena Rossi wants to know if the company is being given “something for nothing” and whether a lower conversion premium would change the economics. The rating agency has said that the equity component of a convertible does not receive full equity credit.

Decision three: the hedge and the bank’s request. Halden has a euro-denominated contract, due for payment in nine months, which will deliver EUR 60 million. Solberg could sell the euros forward, but she dislikes locking in a rate if the euro rallies. She is considering a put option on the euro (the right to sell euros at a floor rate), which costs a premium of about 2.1 percent of the notional at the current forward rate. The bank has also asked whether Halden would accept a *zero-cost collar* in which it buys the put and sells a call on the euro at a higher strike. The bank’s hedging desk will delta hedge the position and has warned that during the contract period, gamma risk could cause the bank to widen the price. Solberg must decide whether the collar’s loss of upside is worth saving the premium.

Other considerations. The stock has recently been volatile because of a contested bid for one of Halden’s competitors. A share buyback is on the agenda as well, funded from the proceeds of the convertible; Dubois remarks that buybacks might be used to offset the dilution from the option grant. Meanwhile, the auditors have asked for documentation of how the volatility and expected life assumptions will be selected, noting that they will test the valuation for reasonableness and consistency with prior grants.

Solberg realizes that she needs a consistent framework: a way to value the incentive grant and understand its cost to shareholders, a way to unbundle the convertible into bond and option, and a way to compare the put with the collar. She also needs to explain to the board that “free” is the wrong word for any of these instruments.

Discussion questions

1. For decision one, explain which volatility and expected life you would use to value the option grant and why. Describe qualitatively how the value per option and the accounting expense change if the committee moves to a four-year vesting cliff or an exercise price 10 percent above market. Is Dubois correct that options are free?
2. For decision two, decompose the convertible into a straight bond and a call option. Using the information in the case, estimate whether the convertible is likely to be priced fairly and discuss how a lower conversion premium would change the coupon needed. What are the costs of the convertible that the coupon saving fails to capture?

3. For decision three, compare a forward sale, a purchased put, and a zero-cost collar in terms of payoffs, cost, and the retained upside. When would the collar be preferable? How does put-call parity relate the put and call prices in structuring the zero-cost collar?
4. The shareholder representative argues that options encourage excessive risk taking. Use the idea of equity as a call option on firm assets to explain the argument, and propose two design features that reduce the problem.

Key Takeaways

- An option gives a right without an obligation; its payoff is max(S − K, 0) for a call and max(K − S, 0) for a put, and the asymmetry makes options valuable even when at the money.
- Combinations of options and the underlying (protective puts, covered calls, straddles, collars) shape the distribution of outcomes; each has a clear cost, breakeven, and set of bets embedded within it.
- Put-call parity, C + Ke^(−rT) = P + S, follows from no-arbitrage alone and holds regardless of any model of the stock price; violations signal an arbitrage or an error in the inputs.
- In the binomial model an option is priced by building a replicating portfolio of stock and borrowing, equivalent to discounting expected payoffs under risk-neutral probabilities at the risk-free rate; true probabilities and risk preferences are irrelevant.
- American options can be valued on a tree by comparing at each node the continuation value with exercise value; early exercise is never optimal for American calls on non-dividend-paying stocks but can be for puts.
- The Black-Scholes-Merton formula depends on five inputs; volatility is the only unobservable, and the expected return on the stock does not appear.
- Delta is the hedge ratio, gamma the curvature, vega the volatility sensitivity, theta the time decay, and rho the rate sensitivity; a delta-hedged option seller earns theta in return for bearing gamma and vega risk.
- Employee stock options are valued at grant date using an option model with expected life, expected volatility, and forfeiture adjustments; the expense is a real cost to shareholders, even though no cash is paid at grant.
- Equity in a levered firm is a call option on the firm’s assets with strike equal to the debt; risky debt is risk-free debt minus a put, which explains risk shifting, credit spreads, and the equity value of distressed firms.
- Convertibles, callable bonds, warrants, and many contract clauses embed options; unbundling a hybrid security into its components is the first step in pricing it correctly.

Review Questions

1. Define in-the-money, at-the-money, and out-of-the-money for calls and puts. Why does an at-the-money option still have value?
2. Compare the profit profiles of a protective put and a covered call. In what market views and risk positions would an investor choose each?
3. Derive put-call parity using two portfolios. How does the relationship change if the stock pays a dividend, or if the options are American?
4. Explain why the value of a European call on a non-dividend-paying stock is bounded below by max(0, S − Ke^(−rT)). Why should such a call not be exercised early?

5. In the binomial model, why do the true probabilities of an up- or down-move not appear in the option price? What do the risk-neutral probabilities represent?
6. What makes early exercise attractive for an American put? What features of the parameters increase the early exercise premium?
7. List the inputs of the Black-Scholes-Merton formula and explain how each affects a call's value. Why does the expected return on the stock not appear?
8. What is implied volatility, and what does a volatility smile tell us about the limitations of the model?
9. Describe delta, gamma, vega, and theta. Explain why a delta-hedged option seller has positive theta and negative gamma.
10. Explain, using the option view of the firm, why shareholders of a highly levered company may favor risky projects even if they have negative NPV. What contractual devices mitigate the problem?

Problems

1. **Protective put.** An investor owns a stock bought at \$42 and buys a put with strike \$40 for a premium of \$2.50. Compute the profit at expiry if the stock finishes at \$30, \$40, and \$50, and find the maximum loss and the breakeven stock price.
2. **Put-call parity.** A non-dividend stock trades at \$50. A one-year European call with strike \$50 costs \$6.20. The continuously compounded risk-free rate is 4 percent. What is the no-arbitrage price of the put with the same strike and maturity?
3. **One-period binomial.** A stock is at \$80 and will be either \$100 (u = 1.25) or \$64 (d = 0.80) after one year. The gross risk-free rate is 1.03. Find the risk-neutral probability and the value of a European call with strike \$85.
4. **Black-Scholes.** A non-dividend stock trades at \$50, strike \$50, one year to expiry, r = 5 percent, σ = 20 percent. Compute d_1, d_2, the call and put values, and verify parity.
5. **Delta hedge.** A trader has sold 5,000 calls (each on one share) with delta 0.55 and gamma 0.04. How many shares should be held to be delta neutral? If the stock then rises by \$1.50, estimate the new delta and the share purchase needed to rebalance.
6. **Employee options.** A company grants 500,000 at-the-money options on stock worth \$25 with an expected life of 4 years, expected volatility 40 percent, risk-free rate 3.5 percent, no dividends. Compute the value per option and, with 10 percent expected forfeitures and three-year straight-line vesting, the annual expense.
7. **Option view of equity.** A firm has assets of \$200 million, zero-coupon debt with a face value of \$150 million due in one year, asset volatility of 25 percent, and risk-free rate 3 percent. Compute the equity value, the value of the debt, and the debt's yield and credit spread.
8. **Straddle.** A trader buys a straddle on a stock with both strike \$60, paying \$3.50 for the call and \$2.80 for the put. Find the breakeven prices and the profit if the stock finishes at \$72 and \$58.

Answers to Selected Problems

1. Profit = (S_T − 42) + max(40 − S_T, 0) − 2.50. At \$30: −12 + 10 − 2.50 = −4.50. At \$40: −2 + 0 − 2.50 = −4.50. At \$50: 8 − 2.50 = +5.50. Maximum loss \$4.50 (for any S_T ≤ \$40); breakeven S_T = 42 + 2.50 = \$44.50.
2. Ke^(−rT) = 50 × e^(−0.04) = 50 × 0.960789 = 48.039. P = C − S + Ke^(−rT) = 6.20 − 50 + 48.039 = **\$4.24**.

3. p = (1.03 − 0.80) / (1.25 − 0.80) = 0.23 / 0.45 = 0.5111. C_u = 100 − 85 = 15, C_d = 0. C = 0.5111 × 15 / 1.03 = **$7.44**. (Equivalent hedge ratio: Δ = 15 / (100 − 64) = 0.4167.)

4. d_1 = [0 + (0.05 + 0.02) × 1] / 0.20 = **0.35**; d_2 = **0.15**. N(0.35) = 0.6368, N(0.15) = 0.5596. PV of strike = 50 × e^(−0.05) = 47.561. C = 50 × 0.6368 − 47.561 × 0.5596 = 31.84 − 26.62 = **$5.23**. P = 47.561 × (1 − 0.5596) − 50 × (1 − 0.6368) = 20.95 − 18.16 = **$2.79**. Parity: C − P = 2.44 = 50 − 47.56 = 2.44 ✓.

5. Shares = 5,000 × 0.55 = 2,750. New delta ≈ 0.55 + 0.04 × 1.50 = 0.61; shares needed 5,000 × 0.61 = 3,050; buy **300** more.

6. d_1 = (0.035 + 0.08) × 4 / (0.40 × 2) = 0.46 / 0.80 = 0.575; d_2 = −0.225. N(d_1) = 0.7173, N(d_2) = 0.4110. PV of strike = 25 × e^(−0.14) = 21.735. C = 25 × 0.7173 − 21.735 × 0.4110 = 17.93 − 8.93 = **about $9.00**. Vesting options 450,000 × 9.00 = $4.05 million; expense **$1.35 million per year** for three years.

7. d_1 = [ln(200/150) + (0.03 + 0.03125)] / 0.25 = (0.287682 + 0.06125) / 0.25 = 1.3957; d_2 = 1.1457. N(d_1) = 0.9186, N(d_2) = 0.8741. PV of debt face = 150 × e^(−0.03) = 145.57. E = 200 × 0.9186 − 145.57 × 0.8741 = 183.72 − 127.24 = **about $56.5 million**. Debt value = **$143.5 million**; yield = ln(150 / 143.5) = 4.42 percent; spread ≈ **1.42 percent**.

8. Cost = 6.30, so breakeven prices are 60 − 6.30 = **$53.70** and 60 + 6.30 = **$66.30**. At $72: call 12 − 3.50 = 8.50, put −2.80, profit **$5.70**. At $58: put pays 2, call expires worthless, total 2 − 6.30 = **−$4.30** (a loss).

Chapter 19: Forwards, Futures, Swaps and Hedging

Learning Objectives

After studying this chapter you should be able to:

- Explain how forwards, futures and swaps transfer risk, and distinguish exchange-traded from over-the-counter contracts in terms of credit risk, flexibility and liquidity.
- Derive and apply the cost-of-carry relationship to price forwards and futures on financial assets and commodities, and identify arbitrage when quoted prices deviate from fair value.
- Compute a minimum-variance hedge ratio and the number of futures contracts needed, and interpret hedge effectiveness and basis risk.
- Value a plain-vanilla interest rate swap, find its par fixed rate from a discount curve, and explain how a swap converts floating-rate debt into fixed-rate debt.
- Describe the cash flows of a currency swap and explain why principal is exchanged at inception and maturity.
- Summarize the concept of hedge accounting under IFRS 9 (fair value hedges, cash flow hedges, net investment hedges) and explain why it exists, what qualifies, and how effectiveness and ineffectiveness are treated.

Opening Vignette

(Illustrative scenario; the company is fictional.)

Calder Mills is a mid-sized food manufacturer that buys about 2 million gallons of diesel fuel a year to run its delivery fleet and finances its plants with a floating-rate bank term loan of 50 million. In a good year the company earns a 9 percent operating margin, and its treasurer, Imani Okafor, knows that a sustained 30 percent rise in fuel prices would erase roughly a third of that margin. At the same time every 1 percentage point rise in the base lending rate costs the company 500,000 in additional annual interest.

At the budget meeting the chief executive asks a blunt question: “Why are we exposed to oil prices and central bank decisions? We make bread and pasta.” Imani’s answer is that exposure is a fact, not a choice; the choice is whether to carry it unmanaged. She proposes three actions. First, buy diesel forward through the company’s supplier for roughly half of next year’s expected volume. Second, where the supplier cannot offer fixed prices, use exchange-traded heating-oil futures as a proxy. Third, enter an interest rate swap that converts the floating-rate loan into a fixed-rate obligation for three years.

The chief financial officer raises objections. If fuel prices fall, will the company look foolish for having locked in a higher price? Will the swap appear as a loss in earnings every time rates dip, creating noise in reported profit? What happens to the heating-oil hedge if the price of heating oil and the price of diesel at the company’s depots stop moving together? And what are the margin calls if futures prices drop?

This chapter answers those questions. It builds from the pricing logic of forwards and futures, through hedge ratios and basis risk, to interest rate and currency swaps, and ends with the accounting concept that keeps hedging from distorting reported results. By the end you should be able to evaluate Imani’s proposal as a leader who understands what is being bought, what can go wrong, and how it will show up in the financial statements.

19.1 Derivatives as Risk-Transfer Instruments

A derivative is a contract whose value depends on the price of an underlying asset, rate or index. Corporate treasurers use derivatives not because they create value out of thin air but because they let a firm separate the *ownership of an asset or operating position* from the *price risk attached to it*. A bakery can keep buying flour from a local mill and still have fixed its flour price; a manufacturer can keep a floating-rate loan with its relationship bank and still pay a fixed rate.

The four building blocks

Almost every corporate derivative is a combination of four elementary contracts:

1. **Forward contract.** A customized agreement between two parties to buy or sell an asset at a fixed price (the delivery price, K) on a future date. No cash changes hands at inception; the forward price is set so the contract has zero value at the start. It is traded over the counter (OTC).
2. **Futures contract.** A standardized forward traded on an exchange and settled daily through a clearing house. Daily settlement ("marking to market") largely removes counterparty credit risk but creates interim cash flows.
3. **Swap.** An agreement to exchange streams of cash flows over several periods. A swap can be viewed as a portfolio of forward contracts bundled into one agreement.
4. **Option.** A contract giving the holder the right, but not the obligation, to buy (call) or sell (put) at a strike price. Options are covered in earlier chapters and return in Chapter 20 for currency hedging; here we concentrate on the linear instruments (forwards, futures, swaps), whose payoffs are symmetric.

Hedging versus speculating

The same contract can hedge or speculate; what differs is the underlying exposure. A farmer who sells wheat futures against a growing crop reduces risk. A trader with no wheat who sells the same futures adds risk. For a corporation the test is whether the derivative *offsets* an existing or highly probable exposure arising from operations or financing. Sound governance requires that this test be written into policy (see Chapter 20) and monitored independently of the people who trade.

The cost of hedging is not the premium

A forward or swap usually costs nothing to enter, yet hedging is never free. It has an *opportunity cost*: if the market moves in your favor you do not benefit. Judging a hedge by whether it "made money" is a classic managerial mistake. A hedge is insurance against variance, and insurance that was not claimed upon was not wasted. What can be judged is whether the hedge achieved its purpose: stable cash flows, protection of covenants, or predictable pricing for customers.

Forwards versus futures

Table 19.1 summarizes the practical differences.

Feature	Forward	Futures
Market	Over the counter, bilateral	Organized exchange
Contract terms	Customized (amount, date, grade)	Standardized size, delivery months, grades
Credit risk	Direct counterparty exposure unless collateralized	Clearing house guarantee; margin posted daily
Cash flows before maturity	None (unless collateral agreements require)	Daily settlement of gains and losses
Liquidity and exit	Reversal needs a second contract or termination with the dealer	Easily closed out by an offsetting trade
Typical use	Tailored hedges of a known amount and date	Hedges of standard exposures, proxy hedges, tactical positions
Transparency	Prices known only from dealer quotes	Public prices and volumes

Table 19.1 – Forwards compared with futures.

The trade-off is between *precision* and *liquidity*. A forward can match an exposure exactly, down to the day and quantity. A futures contract rarely matches the exposure exactly, creating basis risk, but can be entered and exited cheaply and with little credit concern.

19.2 Forward Pricing and the Cost of Carry

19.2.1 The no-arbitrage logic

Suppose you wish to own an asset at time T. You have two ways to get there. You can enter a forward and pay K at T. Or you can buy the asset now with borrowed money and hold it. If the asset can be stored and financed, these two routes must cost the same at T; otherwise a riskless profit exists. Equating them yields the **cost-of-carry model**, in which the forward price equals the spot price compounded forward at the *net cost of carrying* the asset: financing cost plus storage, minus any income the asset yields.

For an asset providing no income, with continuously compounded risk-free rate r and maturity T years:

```
F₀ = S₀ · e^(rT)
```

Equivalently, with annual compounding, `F₀ = S₀ (1 + r)^T`. We use continuous compounding for convenience because exponents add: carrying costs expressed as annual rates simply sum in the exponent.

The general form is:

```
F₀ = S₀ · e^((r + u − y) T)
```

where `u` is storage cost as an annual percentage of the asset price (insurance, warehousing, spoilage), and `y` is the yield the holder of the physical asset receives, which includes dividends on shares, coupons on bonds, the foreign interest rate for currency, or the “convenience yield” for commodities.

19.2.2 Components of carry

- **Financing (r).** Owning the asset ties up capital, whose cost is the risk-free rate in the arbitrage argument because the arbitrageur’s position is riskless.
- **Storage (u).** Physical commodities require warehousing and insurance. Financial assets have negligible storage costs.

- **Income (q or y).** A share paying a 2 percent dividend yield rewards the owner of the share but not the holder of the forward; therefore the forward price is lower.
- **Convenience yield.** Holders of physical commodities may benefit from having inventory on hand to avoid stock-outs or keep a plant running. This benefit accrues to owners of the physical, not holders of the futures, so it lowers the forward price relative to the pure financing-plus-storage calculation. When convenience yield is high, futures prices can be below spot (*backwardation*).

Example 19.1 – Forward price of a non-dividend-paying share

A share trades at 80. The continuously compounded risk-free rate is 5 percent per year. Find the six-month forward price.

```
F₀ = S₀ · e^(rT)
   = 80 · e^(0.05 × 0.5)
   = 80 · e^0.025
   = 80 · 1.025315
   = 82.025
```

The forward price is 82.03, a premium of 2.03 over spot (2.5 percent over six months) that exactly compensates a seller for financing the share. A buyer pays nothing today and agrees to pay 82.03 in six months.

Example 19.2 – Commodity and index forwards with carry components

(a) Crude oil trades at 70 per barrel. The risk-free rate is 4 percent, storage and insurance cost 3 percent of value per year, and the convenience yield is 2 percent. Find the one-year forward price.

```
F₀ = S₀ · e^((r + u − y) T)
   = 70 · e^((0.04 + 0.03 − 0.02) × 1)
   = 70 · e^0.05
   = 70 · 1.051271
   = 73.589
```

The one-year forward is 73.59, a contango of 3.59.

(b) A stock index stands at 2,500 and pays a continuous dividend yield of 1.8 percent. The risk-free rate is 4 percent. Find the nine-month forward.

```
F₀ = 2,500 · e^((0.04 − 0.018) × 0.75)
   = 2,500 · e^0.0165
   = 2,500 · 1.016637
   = 2,541.59
```

The index forward is above spot because the financing rate exceeds the dividend yield. If the dividend yield were above the risk-free rate, the forward would be below spot.

Interpreting contango and backwardation

When the forward curve slopes upward (forward above spot) the market is in **contango**; when it slopes downward it is in **backwardation**. For financial assets, the slope is determined entirely by interest differentials and yields. For commodities, supply conditions and convenience yield move the slope. A hedger who repeatedly rolls short-dated futures forward into later contracts earns or loses the *roll yield*: in contango a long hedger buys each new contract at a price above the expiring one and bears a cost; in backwardation the roll adds a gain. A treasurer should know the shape of the curve before choosing between a long-dated forward and a rolling program.

19.2.3 Arbitrage when prices deviate

If the market forward price is above the fair value $S_0 e^{rT}$, an arbitrageur performs a **cash-and-carry**: borrow to buy the asset, sell the forward, deliver at maturity, repay the loan, and keep the difference. If the market price is below fair value, the arbitrageur performs a **reverse cash-and-carry**: short the asset (where possible), invest the proceeds, and buy the forward. Real-world frictions (transaction costs, borrowing and lending spreads, restrictions on short selling) create a *no-arbitrage band* around fair value rather than a single price.

Example 19.3 – Cash-and-carry arbitrage

Using the share from Example 19.1 (S_0 = 80, r = 5 percent, T = 0.5), the dealer quotes a six-month forward at 84.00 for 100,000 shares. Show the arbitrage.

Fair value is 82.025. The quoted forward is 84.00, which is too high.

1. Borrow 8,000,000 (80 × 100,000) at 5 percent continuous for six months and buy 100,000 shares.
2. Enter a forward to sell 100,000 shares at 84.00 in six months.
3. At maturity, deliver the shares and receive 84.00 × 100,000 = 8,400,000.
4. Repay the loan: 8,000,000 × e^0.025 = 8,202,521.
5. Profit = 8,400,000 − 8,202,521 = 197,479.

Per share the profit is 84.00 − 82.025 = 1.975. The profit is certain and requires no investment, so in an efficient market arbitrageurs would sell forwards and buy shares until the forward falls back toward 82.03. The managerial message: for liquid financial assets, a dealer's forward quote should be explainable by the carry formula, and a wide divergence is a sign to question the quote or the credit terms behind it.

19.2.4 Value of an existing forward

At inception a forward has zero value. Later, its value changes as the spot price and the time remaining change. For a long forward with delivery price K, current spot S, remaining life τ and no income:

```
f = S − K · e^(−rτ)
```

The long forward is worth the spot price less the present value of the delivery price.

Example 19.4 – Valuing a forward after three months

The company in Example 19.3 holds a long forward to buy 100,000 shares at K = 82.025, entered when spot was 80. Three months later spot is 84 and the risk-free rate is still 5 percent. What is the value of the forward?

```
τ = 0.25
f = 84 − 82.025 · e^(−0.05 × 0.25)
  = 84 − 82.025 · 0.987578
  = 84 − 81.006
  = 2.994 per share
```

For 100,000 shares, the value is about 299,400 in favor of the long. The current forward price for the remaining three months is 84 · e^0.0125 = 85.06, and the present value of the difference between that and the contract price (85.06 − 82.025 = 3.03) is 3.03 × 0.987578 = 2.99, confirming the result. This is the amount the long party would receive if the contract

were terminated today, and it is the amount the counterparty owes: a credit exposure for the holder, and a liability for the other party. Accounting standards require such contracts to be recognized at fair value, which is why the valuation mechanics matter for the financial statements as well as for risk management.

19.3 Futures Contracts and Daily Settlement

19.3.1 How a futures position works

A futures contract is a forward standardized in size, grade and delivery date and cleared through a central clearing house, which becomes the buyer to every seller and the seller to every buyer. Participants post **initial margin** (a performance bond, typically a small percentage of the contract's notional value) in a margin account held with their broker. Each day, the clearing house recalculates the settlement price. The loser's account is debited and the winner's account is credited by the change in value; this is **marking to market**. If the balance in an account falls below the **maintenance margin**, the holder receives a **margin call** and must restore the account to the initial level (variation margin) by the next morning, or the position will be closed.

Daily settlement has three consequences that matter for corporate users:

- **Credit risk is almost eliminated.** Losses are collected as they occur, so no large unpaid obligation accumulates. This is why futures can be traded anonymously among thousands of participants.
- **Liquidity risk is created.** A hedger whose futures lose money must find cash immediately, even though the offsetting gain on the physical position will materialize only when the commodity is eventually purchased or sold. Many corporate hedging crises have been liquidity crises in which an economically sound hedge failed because the firm could not fund the margin calls.
- **Convergence.** As delivery approaches, the futures price converges to the spot price of the deliverable asset. Most contracts are closed out before delivery by an offsetting trade, so physical delivery is the exception.

19.3.2 Forward versus futures prices

When interest rates are constant and known, forward and futures prices for the same maturity are identical. When rates are stochastic, the two can differ slightly because the daily reinvestment of margin flows correlates with the futures price. For most corporate purposes, the difference is negligible compared with basis risk, and the cost-of-carry formulas of Section 19.2 are used for both.

Example 19.5 – Margin account for a long fuel hedge

Calder Mills buys 10 heating oil futures contracts to hedge diesel purchases. Each contract covers 42,000 gallons, so the position is 420,000 gallons. The initial margin is 60,000 in total and the maintenance margin is 45,000. The futures settlement prices (per gallon) over the first four days after the purchase at 2.50 are 2.46, 2.39, 2.43 and 2.52. Trace the margin account.

Daily gain or loss for a long position = (price change) × 420,000 gallons.

Day	Settle price	Price change	Gain / (loss)	Account balance before call	Margin call	Balance after
0	2.50	–	–	60,000	–	60,000
1	2.46	−0.04	(16,800)	43,200	16,800	60,000
2	2.39	−0.07	(29,400)	30,600	29,400	60,000
3	2.43	+0.04	16,800	76,800	–	76,800
4	2.52	+0.09	37,800	114,600	–	114,600

Table 19.2 – Margin account of the long futures position.

Calculations: Day 1: 0.04 × 420,000 = 16,800 loss, balance 43,200 falls below 45,000, so a call of 60,000 − 43,200 = 16,800 is made. Day 2: 0.07 × 420,000 = 29,400 loss, balance 30,600, call of 29,400. Day 3: 0.04 × 420,000 = 16,800 gain, balance 60,000 + 16,800 = 76,800. Day 4: 0.09 × 420,000 = 37,800 gain, balance 114,600, of which 54,600 above the initial margin may be withdrawn.

Net result over four days: (2.52 − 2.50) × 420,000 = +8,400, which equals −16,800 − 29,400 + 16,800 + 37,800. But the company had to wire 46,200 on days 1 and 2, while its gain arrived later. The profit and loss is the same as a forward would produce; the *timing of cash* is not. A treasury that hedges 50 million gallons and sees a 10 percent price drop must be prepared for margin calls in the millions, and should arrange a committed credit line for variation margin before the hedging program is launched.

19.3.3 Tailing the hedge

Because gains and losses on futures are received or paid daily, a position held to maturity is slightly different from a forward. Hedgers sometimes “tail” the hedge by scaling the number of contracts by a discount factor (approximately `e^(−rτ)`) to reflect interest earned or paid on margin flows. For short-dated hedges the adjustment is small and usually ignored; for multi-year strips of futures, it can be material.

19.3.4 Rolling hedges and the “stack and roll” problem

Liquid futures exist only for relatively near maturities. A company that wants to hedge fuel needs for three years may buy a “stack” of nearby contracts and roll them forward as they near expiry. This creates (a) roll cost or gain depending on the shape of the curve, (b) margin funding risk if prices fall while the real purchases lie years in the future, and (c) basis risk that repeats at each roll. A well-known category of corporate losses came from long-dated fixed-price sales to customers hedged with short-dated futures; when prices fell sharply the futures losses required cash immediately while the offsetting profit on the fixed-price contracts accrued over years. The lesson is to match the *maturity* of the hedge to the maturity of the exposure whenever possible, and otherwise to size the liquidity buffer for the mismatch.

19.4 Hedge Ratios and Basis Risk

19.4.1 Why a one-for-one hedge is not always optimal

The simplest hedge sells (or buys) futures equal in quantity to the exposure: a hedge ratio of 1. This works if the futures price moves exactly one-for-one with the price of the exposure. In practice the asset hedged and the contract used differ in grade, location or timing. Imani’s diesel at the company’s depots is not the same product as heating-oil futures at the exchange’s delivery point. The hedger therefore chooses the ratio that minimizes the variance of the combined position.

19.4.2 The minimum-variance hedge ratio

Let ΔS be the change in the spot price (or value) of the exposure over the hedge horizon, and ΔF the change in the futures price. Let σS and σF be their standard deviations and ρ their correlation. A hedger who takes h units of futures per unit of exposure sees the change in the hedged position as $\Delta S - h\Delta F$. The variance is minimized at:

```
h* = ρ · σS / σF
```

This is the slope of a regression of ΔS on ΔF, and it equals the covariance of the two changes divided by the variance of the futures change. Three properties matter:

1. If $\rho = 1$ and $\sigma S = \sigma F$, then $h^* = 1$.
2. If the exposure is more volatile than the futures ($\sigma S > \sigma F$), the hedge ratio exceeds ρ and can exceed 1.
3. Low correlation lowers the optimal ratio and, more importantly, limits how much risk the hedge can remove.

The effectiveness of the optimal hedge is measured by the R^2 of the regression, which equals ρ^2. The share of variance eliminated is ρ^2, and the remaining standard deviation of the hedged position is $\sigma S \sqrt{1 - \rho^2}$.

The number of futures contracts is:

```
N* = h* · Q_A / Q_F
```

where Q_A is the size of the exposure and Q_F the size of one futures contract.

Example 19.6 – Minimum-variance hedge for a diesel exposure

Calder Mills expects to buy 2,000,000 gallons of diesel over the next quarter. The standard deviation of weekly percentage changes in the diesel price is 2.0 percent; for heating oil futures it is 1.6 percent. The correlation is 0.85. Each futures contract covers 42,000 gallons. Find the optimal hedge ratio, the number of contracts and the hedge effectiveness.

Step 1: hedge ratio

```
h* = ρ · σS / σF = 0.85 × 2.0 / 1.6 = 1.0625
```

Step 2: number of contracts

```
N* = h* × Q_A / Q_F = 1.0625 × 2,000,000 / 42,000
   = 2,125,000 / 42,000
   = 50.6
```

The company should buy 51 contracts (rounding to the nearest whole contract; 50 would also be defensible).

Step 3: effectiveness

```
R² = ρ² = 0.85² = 0.7225
```

The hedge removes 72.25 percent of the variance of weekly price changes. The residual standard deviation is

```
σS · √(1 − ρ²) = 2.0% × √(0.2775) = 2.0% × 0.5268 = 1.054%
```

Observe that a correlation as high as 0.85 still leaves more than half of the original *standard deviation* (1.054 versus 2.0 percent, about 53 percent). Standard deviation falls much more slowly than variance, and a manager who hears “72 percent effective” should translate it correctly: the hedge reduces volatility by about 47 percent, not 72 percent. If the treasurer

wants more protection, a closer-matching instrument (an OTC diesel swap with the supplier's pricing index, for example) is needed.

19.4.3 Basis risk

The **basis** is the spot price of the hedged asset minus the futures price of the contract used:

```
b = S − F
```

A perfect hedge exists only if the basis is known at the time the hedge is lifted. Otherwise the hedger has exchanged *price risk* for *basis risk*. For a hedger who is long the physical (will sell it) and short futures, the effective sale price is $F_1 + b_2$; for a hedger who will buy the physical and is long futures, the effective purchase price is likewise $F_1 + b_2$, where F_1 is the futures price when the hedge is placed and b_2 the basis when it is lifted. The derivation is simple: the hedger pays S_2 in the market and gains $F_2 - F_1$ on futures, so net cost is $S_2 - (F_2 - F_1) = F_1 + (S_2 - F_2) = F_1 + b_2$.

Basis risk arises from four sources:

- **Product mismatch** (cross-hedging a different grade or commodity).
- **Location mismatch** (delivery point differs from the exposure's location).
- **Timing mismatch** (hedge lifted before the futures expire, so convergence is incomplete).
- **Carry changes** (changes in interest rates, storage costs or convenience yield alter the spread between spot and futures).

Example 19.7 – Basis risk in a long hedge

Three months ahead of a purchase of 420,000 gallons, Calder buys 10 heating oil futures at 2.55. At that time the local diesel price is 2.50, so the initial basis is −0.05. At the purchase date, local diesel costs 2.80 and the futures are at 2.88. Compute the effective purchase price and the effect of basis.

Futures gain: 2.88 − 2.55 = 0.33 per gallon, or 0.33 × 420,000 = 138,600. Cost in the physical market: 2.80 × 420,000 = 1,176,000. Net cost: 1,176,000 − 138,600 = 1,037,400, an effective price of 2.47 per gallon.

Check with the formula: $F_1 + b_2 = 2.55 + (2.80 - 2.88) = 2.55 - 0.08 = 2.47$.

If the basis had stayed at −0.05, the price would have been 2.55 − 0.05 = 2.50. The basis moved from −0.05 to −0.08, which *favored* the buyer by 0.03 per gallon (12,600 in total). Had the basis strengthened to +0.02 the effective price would have been 2.57, a loss of 0.07 per gallon versus the 2.50 expected (29,400). The hedge fixed the *futures* price at 2.55 but left the company exposed to the *change in basis*. Importantly, the volatility of the basis is typically much smaller than the volatility of the outright price, which is why imperfect hedges usually still reduce risk substantially.

19.4.4 Managerial rules for choosing the hedge

Question	Guidance
Is a closer instrument available?	Prefer OTC contracts referencing the exact index on which the company pays or receives.
Is the hedge horizon short or long?	Short horizons have lower basis volatility but may create rolling and liquidity issues.
How much correlation is acceptable?	Hedge only if the regression is stable and economically explainable; a high historical correlation without a causal link can vanish.
What if margin funding is a constraint?	Use OTC forwards with collateral thresholds, swaps with the bank's credit line, or options (premium paid once).
What proportion to hedge?	Matching exposure fully gives certainty but removes upside; policies typically hedge a declining percentage as the horizon lengthens (see Chapter 20).

Table 19.3 – Practical decisions in designing a commodity hedge.

19.5 Interest Rate Swaps

19.5.1 What a plain-vanilla swap does

In a plain-vanilla interest rate swap, one party pays a fixed rate on a notional principal and receives a floating rate (such as a benchmark overnight-based or term reference rate) on the same notional; the counterparty does the opposite. Only the *net* difference is paid on each settlement date, and the notional is never exchanged. The party that pays fixed is called the **payer** (or "long" the swap); it benefits when floating rates rise.

A company with a floating-rate loan that enters a pay-fixed swap has a **synthetic fixed-rate loan**. It pays the bank floating on the loan and receives floating from the swap, with the two floating legs cancelling approximately (if the reference rates match), leaving the swap's fixed rate plus the loan margin.

Why not simply borrow at a fixed rate? There are several reasons. The company may have an existing floating loan whose refinancing would incur fees and a prepayment penalty. Its relationship bank may offer floating loans on better terms. The company may want to change the mix of fixed and floating liabilities as its view of the interest rate environment or its balance sheet changes, which a swap can do without touching the underlying debt. A comparative-advantage argument (borrow where relatively cheap, swap into the desired form) is often cited, but in efficient markets it mainly reflects differences in credit spreads at different maturities and should be examined skeptically: the swap exposes the company to the counterparty's credit risk, which is part of the supposed saving.

19.5.2 Valuation: a swap as two bonds

At inception, a swap is set so that the present value of the fixed leg equals the present value of the floating leg; hence its value is zero. The **par swap rate** `c` for annual payments over n years, given discount factors `DF_t`, is:

```
c = (1 − DF_n) / Σ DF_t      (t = 1 … n)
```

The numerator is the value of a floating-rate bond per unit of principal less its final discounted principal; the floating leg is worth 1 − DF_n per unit of notional because a floating-rate note resets to par at each payment date. The denominator is the present value of an annuity of 1 per year.

After inception the swap's value is the difference between the PV of the leg received and the PV of the leg paid. For a payer swap with contract fixed rate K when the current par rate is c:

```
Value (payer) = Notional × (c − K) × Σ DF_t
```

For the receiver, the sign is reversed. The formula shows that a swap behaves like an annuity of the *rate difference*, and that its sensitivity to rates rises with the remaining life and the notional.

Example 19.8 – Finding the par swap rate

A three-year swap with annual payments is priced from the zero-coupon curve: 1-year 3.0 percent, 2-year 3.4 percent, 3-year 3.7 percent (annual compounding). Find the discount factors, the implied forward rates and the par swap rate. Notional is 50 million.

Discount factors:

```
DF₁ = 1 / 1.030        = 0.970874
DF₂ = 1 / 1.034²       = 0.935317
DF₃ = 1 / 1.037³       = 0.896734
Σ DF = 2.802925
```

Implied one-year forward rates (rate for each year, as seen today):

```
f(0,1) = 3.000%
f(1,2) = DF₁/DF₂ − 1 = 0.970874/0.935317 − 1 = 3.802%
f(2,3) = DF₂/DF₃ − 1 = 0.935317/0.896734 − 1 = 4.303%
```

Par swap rate:

```
c = (1 − DF₃) / Σ DF = (1 − 0.896734) / 2.802925 = 0.103266 / 2.802925 = 3.6842%
```

The fixed payment each year is 50,000,000 × 3.6842% = 1,842,108. As a check, the expected floating payments are 1,500,000, 1,900,777 and 2,151,307 with present values 1,456,311, 1,777,829 and 1,929,150, summing to 5,163,290. The fixed payments of 1,842,108 discounted at the same factors total 1,842,108 × 2.802925 = 5,163,290. The two legs are equal, so the swap has zero value at inception.

Notice that the par rate (3.684 percent) lies between the early and late forward rates: the fixed rate is a weighted average of the forwards, weighted by the discount factors. The payer therefore overpays in year 1 (fixed 3.684 versus expected floating 3.000), and underpays in year 3 (3.684 versus 4.303). That front-loaded overpayment is a real cash cost in the early years, which the CFO should understand before locking in.

Example 19.9 – Value of an existing swap

A year ago Calder entered a four-year pay-fixed swap at 3.50 percent on 50 million, and the first payment has just been made. Three years remain, and we assume the curve of Example 19.8 still applies, so that the par rate for the remaining term is 3.6842 percent. What is the swap worth to Calder?

```
Value = Notional × (c − K) × Σ DF
      = 50,000,000 × (0.036842 − 0.035000) × 2.802925
      = 50,000,000 × 0.001842 × 2.802925
      = 258,171
```

Calder's swap is an asset of about 258,000: it pays 3.50 percent when market-equivalent swaps would require 3.684 percent. This is the mark-to-market amount used for the balance sheet and the amount the bank would pay to terminate it. If rates had fallen so that the par rate was 3.0 percent, the value would be 50,000,000 × (0.030 − 0.035) × 2.802925, or roughly −700,700, a liability. A manager must therefore expect a pay-fixed swap to show a loss when rates fall; the compensation is that the underlying floating-rate loan costs less. This is the reason accounting treatment matters: without hedge accounting, the loss on the swap would appear in profit or loss while the matching saving on the loan emerges only gradually.

Example 19.10 – Converting a floating loan into fixed

Calder's 50 million term loan pays the reference rate plus a 1.20 percent margin annually. It enters the swap of Example 19.8 as fixed-rate payer at 3.6842 percent. What is the effective cost, and what happens if rates rise by 1 percentage point?

Effective fixed cost = swap fixed rate + loan margin = 3.6842% + 1.20% = 4.8842%. Annual interest = 50,000,000 × 4.8842% = 2,442,108.

If the reference rate rises 1 percentage point above the forward in a given year, the loan interest increases by 500,000, but the swap receives 1 percentage point more floating, an additional 500,000. The net cost stays at 2,442,108. The company's exposure to rate changes on this debt has been eliminated for the three-year term, at the cost of giving up gains if rates fall. Remaining risks are (1) counterparty default by the swap bank (the exposure is the positive mark-to-market value, not the notional), (2) basis risk if the loan reference rate and the swap floating rate differ, and (3) the opportunity cost of being locked in when rates move below the swap rate.

19.5.3 Credit risk and clearing

Swap counterparty exposure is measured by replacement cost: if the counterparty defaults, the non-defaulting party must replace the swap at current market rates. This is a function of mark-to-market value, not notional. After the financial crisis of 2007 to 2009, regulators in many jurisdictions required standardized OTC derivatives to be cleared through central counterparties and imposed margin requirements for non-cleared trades. Corporate users meeting certain criteria can often be exempt from clearing, but must manage collateral under credit support annexes to their master agreements. The practical implication is that the swap, though it has no up-front cost, requires credit lines or collateral capacity.

19.6 Currency Swaps

19.6.1 Structure

A **currency swap** exchanges principal and interest payments in one currency for principal and interest payments in another. Unlike an interest rate swap, the principal amounts are usually exchanged at the start (at the spot exchange rate) and re-exchanged at maturity at the *same* rate. Interest on each principal is paid in its own currency, at fixed or floating rates. Typical uses:

- **Synthetic foreign-currency borrowing.** A company can raise funds in the market where it has the strongest access (say, domestic bonds), and swap into the currency where it has revenues or assets.
- **Hedging a foreign-currency asset or net investment.** A parent funding a foreign subsidiary can borrow in the subsidiary's currency through a swap.
- **Managing long-dated exposures.** Forward markets thin out beyond a year or two; currency swaps provide hedging for five to ten years.

The re-exchange at the original rate is the essential feature. It fixes the amount of foreign currency that will be needed to repay the home-currency principal, removing the exchange rate risk on the principal. The interest payments are also fixed in foreign currency terms for the fixed-for-fixed case.

Example 19.11 – A fixed-for-fixed currency swap

Calder Mills wants to finance a 9.09 million euro plant in the euro area. It raises 10 million US dollars through a three-year bond with a 5.0 percent annual coupon, then enters a currency swap with a bank. The spot rate is 1.10 USD per EUR. In the swap, Calder pays EUR 3.5 percent fixed and receives USD 5.0 percent fixed. Describe the cash flows, and show the effect of a euro depreciation to 0.95 USD per EUR if Calder had not swapped.

Principal in euros = 10,000,000 / 1.10 = EUR 9,090,909.

Date	Calder pays (EUR)	Calder receives (USD)	Passes to bondholders (USD)
Year 0	–	–	Calder gives bank USD 10,000,000 and receives EUR 9,090,909
Year 1	318,182	500,000	500,000 coupon
Year 2	318,182	500,000	500,000 coupon
Year 3	9,409,091 (9,090,909 + 318,182)	10,500,000	10,500,000 (principal + coupon)

Table 19.4 – Cash flows of the currency swap.

The euro interest is 9,090,909 × 3.5% = 318,182. The dollar interest received (USD 500,000) funds the bond coupon exactly, and the principal received (USD 10,500,000 at Year 3) repays the bond. Calder's net position is a euro liability of 9,090,909 at 3.5 percent, a synthetic euro bond.

Without the swap: Calder would owe USD 10,000,000 at maturity. If the euro had depreciated to 0.95 USD per EUR, repaying the principal would need EUR 10,000,000 / 0.95 = 10,526,316, compared with EUR 9,090,909 at the original rate, an additional euro cost of 1,435,407 (about 15.8 percent). With the swap, Calder pays only EUR 9,090,909 for the principal. Of course, had the euro appreciated, the unhedged firm would have gained; the swap eliminates that benefit as well as the risk.

At inception the swap has zero value if the fixed rates reflect the two yield curves. Over time its value fluctuates with exchange rates and both interest rate curves; because principals are exchanged at maturity, a currency swap has more credit exposure than an interest rate swap of the same size, and exposures tend to grow with maturity.

19.7 Hedge Accounting: The Concept

19.7.1 The accounting mismatch problem

Derivatives are measured at fair value, with changes recognized in profit or loss (IFRS 9 classifies derivatives that are not designated as hedging instruments as held for trading). The items they hedge, however, are usually measured differently: forecast purchases are not recognized at all until they occur, loans are typically at amortized cost, inventory at cost. If the derivative is marked to market through earnings while the hedged item is not, reported profit will swing even though the economic position is stable. In Example 19.9, the swap may show a profit of 258,000 in one quarter and a loss in the next, while the company's real cash cost of borrowing has been fixed at 4.88 percent.

Hedge accounting is an optional set of rules that allows the accounting for the hedging instrument and the hedged item to be aligned, so that the effects of the hedge appear in the same period as the effects of the hedged item. It does not change cash flows or economics; it changes timing and location of recognition. Because it is a departure from the default rules, it is permitted only when strict conditions are met.

19.7.2 Qualifying conditions (IFRS 9)

For a hedging relationship to qualify, all of the following must hold:

1. **Eligible items.** The hedging instrument is typically a derivative (or, for currency risk, a non-derivative financial asset or liability); the hedged item is a recognized asset or liability, an unrecognized firm commitment, a highly probable forecast transaction or a net investment in a foreign operation. Components of items, such as a benchmark interest rate component of a loan or the currency component of a purchase, can be designated.
2. **Formal designation and documentation** at inception: the risk management objective and strategy, the hedged item, the hedging instrument, the nature of the risk being hedged, and how effectiveness will be assessed (including sources of ineffectiveness and how the hedge ratio is determined).
3. **Effectiveness requirements.** There must be an *economic relationship* between hedged item and hedging instrument; the effect of *credit risk* must not dominate the value changes; and the *hedge ratio* of the relationship must be the same as that actually used for risk management. IFRS 9 replaced the older rule-based 80 to 125 percent test with this more principles-based and forward-looking assessment, though ineffectiveness must still be measured and recognized.

If the risk management objective is unchanged but the relationship no longer meets the ratio requirement, the entity *rebalances* by adjusting the quantities rather than discontinuing the hedge. Discontinuation is voluntary only in the sense that it follows from a change of objective or from the criteria ceasing to be met.

19.7.3 The three types of hedge

Type	Hedged exposure	Treatment of the instrument	Treatment of the hedged item
Fair value hedge	Changes in fair value of a recognized asset or liability or firm commitment (e.g., a fixed-rate bond exposed to interest rates)	Gain or loss in profit or loss	Carrying amount adjusted for the hedged risk, with the adjustment in profit or loss, so the two offset
Cash flow hedge	Variability in future cash flows (e.g., floating-rate interest, forecast sales in foreign currency, forecast fuel purchases)	Effective portion in other comprehensive income (cash flow hedge reserve); ineffective portion in profit or loss	Not remeasured; when the hedged cash flows affect profit or loss, the reserve is reclassified to profit or loss in the same period. For a forecast purchase of a non-financial item, the reserve is removed and included in the initial cost of the asset (basis adjustment)
Net investment hedge	Foreign currency exposure in the net assets of a foreign operation	Effective portion in OCI (foreign currency translation reserve); ineffective portion in profit or loss	Translation differences on the foreign operation are already in OCI; amounts are recycled to profit or loss on disposal

Table 19.5 – Types of hedge relationships under IFRS 9.

19.7.4 Measuring the effective portion in a cash flow hedge

In a cash flow hedge, the amount recognized in OCI is the *lower* (in absolute terms) of:

- the cumulative gain or loss on the hedging instrument since inception, and
- the cumulative change in fair value (present value) of the expected future cash flows of the hedged item since inception, usually measured using a *hypothetical derivative* that perfectly matches the hedged risk.

The remainder is ineffectiveness recognized in profit or loss. If the hedging instrument has gained *less* than the hedged item has lost, the shortfall simply remains unrecognized in earnings until it affects the item itself (this is the effect of the "lower of" rule).

Example 19.12 – Cash flow hedge accounting for the swap

At the reporting date Calder's pay-fixed swap (hedging the variable interest on the 50 million loan) has a cumulative fair value gain of 260,000. A hypothetical derivative with perfect terms matching the loan's cash flows shows a cumulative gain of 250,000. The tax rate is 25 percent. Show the accounting for the period.

Cumulative gain on the instrument: 260,000. Cumulative change in the hedged item (hypothetical derivative): 250,000. Lower of the two: 250,000, which is recognized in OCI (cash flow hedge reserve). Ineffectiveness: 260,000 − 250,000 = 10,000, recognized in profit or loss immediately.

Journal entries (before tax):

```
Dr  Derivative asset (swap)            260,000
    Cr  Cash flow hedge reserve (OCI)           250,000
    Cr  Profit or loss (ineffectiveness)         10,000
```

Deferred tax at 25 percent: on the OCI portion 250,000 × 25% = 62,500, charged against OCI, giving a net reserve of 187,500; on the 10,000 in profit or loss, tax of 2,500.

In each later period when the loan interest is accrued, the cumulative amount in the reserve is reclassified to profit or loss in line with the hedged cash flows. The hedging gain lowers interest expense (or recoups earlier losses) in the same periods as the loan interest it hedges, so the reported interest cost reflects the fixed 4.88 percent economic rate instead of oscillating with the swap's mark-to-market. Ineffectiveness arises here from the 10,000 difference, which might reflect differences in reference rates, credit valuation adjustments or payment dates.

19.7.5 Practical issues

- **Documentation discipline.** Hedge accounting is prospective. A hedge undocumented on day one cannot be retrospectively designated; the derivative's marks go through profit or loss.
- **Cost of hedging.** IFRS 9 allows the time value of an option or the forward points of a forward contract to be accounted for separately and deferred in OCI, reducing earnings volatility from those components.
- **Disclosure.** IFRS 7 requires disclosure of risk management strategy, the amount, timing and uncertainty of future cash flows from hedging instruments, and the effects on the statement of financial position, profit or loss and equity.
- **Macro hedging.** Banks and others managing dynamic portfolios continue to rely on the separate IAS 39 option for portfolio fair value hedges of interest rate risk while a dedicated macro hedging project is developed.

- **US GAAP.** Under US GAAP, hedge accounting rules differ in detail, but the objective is similar; this chapter describes the IFRS concept.
- **Hedge accounting is not hedging.** A firm may have an effective economic hedge and still report volatility because it did not qualify or elect hedge accounting. Executives reading earnings should distinguish economic hedging results from accounting results.

Case Study: Tidewater Freight Lines and the Fuel Hedge

(Illustrative case; the company and all figures are fictional.)

Tidewater Freight Lines operates a fleet of coastal cargo vessels and a network of trucking depots. Annual revenue is 900 million, of which fuel accounts for roughly 18 percent of operating costs in a normal year. Roughly two-thirds of its revenue comes from contracts with fixed fuel surcharge tables that lag fuel price changes by about three months; the remaining third comes from spot shipments priced at prevailing market rates. The company is financed with 400 million of debt: 250 million of floating-rate bank loans due in five years and 150 million of fixed-rate notes.

Three years ago, with fuel prices at a cyclical peak and the company's equity cushion thin after an acquisition, the board adopted a fuel-hedging policy proposed by the treasurer, Daniel Whitcombe. The policy required hedging at least 70 percent of the next 12 months of expected consumption and 50 percent of months 13 to 24. Daniel executed the program using marine gas oil futures, which he felt were the most liquid contracts, to hedge the company's actual purchases of bunker fuel and truck diesel at ports. He bought contracts out to 24 months and rolled the nearest months as they expired. No analysis of the correlation between marine gas oil futures and the company's delivered prices was documented, and the hedges were not designated for hedge accounting because the finance team judged the paperwork "not worth the effort." The company's bank credit agreement had a leverage covenant of net debt to EBITDA below 3.5 times, and an interest coverage covenant above 4.0 times.

In the first nine months the fuel price rose, and the hedge produced gains of 31 million that were recognized as other income. The operating results showed a corresponding rise in fuel cost. The chief executive praised the hedging program at the annual meeting. In the following year, however, global demand weakened and fuel prices fell by about 40 percent over four months. The futures position, which covered about 70 percent of the next year's consumption and 50 percent of the following year's, generated mark-to-market losses of 58 million, all recognized in profit or loss as the contracts were not in a hedge relationship. Variation margin calls totaled 47 million over that period, and the treasury had to draw on its revolving credit facility to fund them. Meanwhile, the fuel surcharge tables adjusted downward with a three-month lag, so Tidewater's revenue fell before the physical fuel purchases at lower prices fully fed through. Because the physical savings accrue only as fuel is consumed over the next two years, reported EBITDA in the period showed the hedge losses but not the offsetting lower fuel costs. Reported EBITDA fell 22 percent, and net debt to EBITDA rose to 3.3 times, near the covenant limit.

The new chief financial officer, Priya Ramanathan, was asked by the board to review the program. Her preliminary findings were:

1. The regression of the company's delivered fuel prices on marine gas oil futures showed a correlation of 0.78 and a slope of 0.9, indicating a one-for-one hedge ratio overstated the appropriate ratio by about 10 percent. The correlation had been lower in two of the last six quarters because of port-specific supply disruptions.

2. Because two-thirds of the revenue base carried fuel surcharges that pass through fuel price changes after a lag, the company's net exposure to fuel prices was much smaller than the gross consumption suggested. Hedging 70 percent of gross consumption therefore left the company *net long* fuel price in some months: when fuel fell, it lost on the hedge and also lost surcharge revenue.
3. No liquidity plan or committed margin facility had been established; the margin calls were met through borrowing that had been intended for fleet maintenance.
4. The floating-rate loans were unhedged. The board had assumed the fuel hedges were the only market risk worth managing, yet a 2 percentage point rise in rates would increase interest cost by 5 million per year.

Priya proposes to (a) re-estimate the net fuel exposure after surcharge pass-through, (b) reduce the hedge to a policy level of 40 to 50 percent of *net* exposure with a declining profile across the horizon, (c) document hedge relationships and adopt hedge accounting for the retained program, (d) secure a committed margin facility sized with stress scenarios for a 40 percent price fall, and (e) evaluate a pay-fixed swap for about half of the floating loans. Daniel argues that the program "worked as intended" because it protected the company when prices rose and that unwinding now would crystallize the losses.

Discussion Questions

1. Using the concepts of net exposure and pass-through, explain why Tidewater's economic hedge position was larger than its actual risk. How would you estimate the net exposure quantitatively?
2. Distinguish between the economic outcome and the reported outcome of the fuel hedge during the price decline. How would hedge accounting (cash flow hedge) have changed reported profit and EBITDA, and what conditions would Tidewater have needed to satisfy?
3. Evaluate Daniel's claim that the program "worked as intended" and that unwinding would "crystallize" the losses. Is a mark-to-market loss on a hedge a real loss? What does the analysis of basis risk and hedge ratio add?
4. Design the liquidity and governance elements Tidewater should adopt before re-launching the program, including margin funding, limits, reporting to the board, and the role of the covenant package.

Key Takeaways

- Derivatives separate the ownership of an asset or position from its price risk; a hedge is judged by whether it reduced the variability the firm cares about, not by whether it made money.
- Forwards are customized OTC contracts with credit exposure; futures are standardized, exchange-traded and marked to market daily, which swaps credit risk for liquidity risk.
- The cost-of-carry formula $F_0 = S_0\, e^{(r + u - y)T}$ prices forwards and futures; deviations beyond transaction costs imply arbitrage, and the shape of the curve (contango or backwardation) determines roll costs.
- The value of an existing forward is $S - K\, e^{-r\tau}$; this fair value is what appears in the balance sheet and what the counterparty owes or is owed.
- The minimum-variance hedge ratio $h^* = \rho\sigma_S/\sigma_F$ and the number of contracts $N^* = h^* \cdot Q_A/Q_F$ determine the optimal position; hedge effectiveness is ρ^2 for variance, but the reduction in standard deviation is smaller.
- Basis risk replaces price risk: the effective price of a hedged transaction is $F_1 + b_2$, so the hedger bears uncertainty about the basis change, not the price level.

- An interest rate swap is equivalent to a portfolio of forward-rate agreements; the par rate is `(1 − DF_n) / Σ DF_t`, and a swap's value is `Notional × (c − K) × Σ DF`.
- A swap converts floating debt into fixed (or vice versa) without refinancing; its credit exposure equals positive mark-to-market value, not notional.
- Currency swaps exchange principal at the same rate at start and maturity, removing exchange rate risk on principal and fixing interest in foreign currency terms for long horizons.
- Hedge accounting (IFRS 9) aligns the timing of derivative gains and losses with the hedged item, requires documentation at inception, an economic relationship and a sensible hedge ratio, and puts the effective portion of a cash flow hedge in OCI while ineffectiveness goes to profit or loss.

Review Questions

1. Explain why a forward contract has zero value at inception, and why its forward price is not an unbiased prediction of the future spot price.
2. Describe the cash-and-carry arbitrage, and explain what transaction costs and short-selling restrictions do to the no-arbitrage band.
3. What is convenience yield, and why can it cause futures prices of a commodity to be below its spot price?
4. Why do margin calls on a futures hedge create a funding risk even when the hedge is economically sound? Suggest three ways a corporate treasury can manage it.
5. Why can the optimal hedge ratio differ from 1? Provide a business situation in which h* is greater than 1.
6. A hedge has an R^2 of 0.64. By what proportion is the standard deviation of the hedged position reduced relative to unhedged? Explain why this differs from 64 percent.
7. Define basis and basis risk. Why might a company prefer a slightly imperfect futures hedge to no hedge at all?
8. Describe how a pay-fixed interest rate swap transforms floating-rate debt. Which party bears the risk if rates fall, and where does the swap's counterparty credit exposure arise?
9. Why does a currency swap exchange principal, and why does it have greater credit exposure than a single-currency interest rate swap of the same size?
10. Explain the "lower of" rule in a cash flow hedge under IFRS 9, and why a company with an effective economic hedge might still report volatile earnings if it does not apply hedge accounting.

Problems

1. **Forward price with dividends.** A share trades at 120. The risk-free rate is 6 percent and the dividend yield is 2 percent (both continuously compounded). Find the nine-month forward price.
2. **Commodity forward.** Gold is at 1,900 per ounce. The risk-free rate is 4 percent per year, storage and insurance cost 0.6 percent per year, and there is no convenience yield. Find the two-year forward price (continuous compounding).
3. **Arbitrage.** A non-dividend-paying share is at 50, the one-year risk-free rate is 5 percent (continuous) and the one-year forward is quoted at 54. Identify the arbitrage and compute the profit per share at maturity.
4. **Hedge ratio.** A company will buy 1,500,000 gallons of fuel. The standard deviation of the fuel price change is 3.0 percent and that of the futures is 2.4 percent, with a correlation of 0.90. Each contract is 42,000 gallons. Find h*, the number of contracts, the variance reduction and the residual standard deviation of the hedged position.

5. **Margin account.** A farmer sells five wheat futures contracts (5,000 bushels each) at 6.00 per bushel. The initial margin is 7,500 in total, the maintenance margin is 5,600. Settlement prices on the next three days are 6.10, 5.95 and 5.80. Show the margin account, any margin calls and the total gain.
6. **Swap rate and value.** The 1-year zero rate is 2.5 percent and the 2-year zero rate is 3.0 percent (annual compounding). Find the par two-year swap rate (annual payments), and the value to a payer of a 20 million swap with a fixed rate of 2.8 percent.
7. **Currency swap.** A US firm with revenues in sterling needs to fund a 20 million US dollar investment. The spot rate is 1.25 USD per GBP. It raises USD 20 million at 4.5 percent and enters a swap paying GBP 3.8 percent on the sterling equivalent and receiving USD 4.5 percent. Find the sterling principal and the annual sterling interest, and compute the extra sterling the firm would have needed at maturity to repay the loan without a swap if spot becomes 1.10 USD per GBP.
8. **Hedge accounting.** A cash flow hedge has a cumulative instrument gain of 90,000 and a cumulative hypothetical derivative gain of 100,000 at the end of period 1. At the end of period 2 the cumulative figures are 210,000 and 195,000. Compute the amounts in OCI and profit or loss for each period.

Answers to Selected Problems

1. F = 120 × e^((0.06 − 0.02) × 0.75) = 120 × e^0.03 = 120 × 1.030455 = 123.65.
2. F = 1,900 × e^((0.04 + 0.006) × 2) = 1,900 × e^0.092 = 1,900 × 1.096368 = 2,083.09 per ounce.
3. Fair forward = 50 × e^0.05 = 52.56. The quoted 54 is higher, so borrow 50 and buy the share, sell the forward at 54, deliver at maturity and repay 52.56. Profit = 54 − 52.56 = 1.44 per share.
4. h* = 0.90 × 3.0 / 2.4 = 1.125. N* = 1.125 × 1,500,000 / 42,000 = 40.18, so 40 contracts. Variance reduction = 0.81 (81 percent). Residual standard deviation = 3.0% × √0.19 = 1.31%.
5. The seller loses when prices rise. Day 1: −0.10 × 25,000 = −2,500, balance 5,000 < 5,600, so call 2,500 to restore 7,500. Day 2 (5.95): +0.15 × 25,000 = +3,750, balance 11,250. Day 3 (5.80): +3,750, balance 15,000. Total gain = (6.00 − 5.80) × 25,000 = 5,000 (−2,500 + 3,750 + 3,750). One margin call of 2,500.
6. DF_1 = 0.975610; DF_2 = 1/1.03² = 0.942596; Σ DF = 1.918206. Par rate = (1 − 0.942596)/1.918206 = 2.9926%. Value = 20,000,000 × (0.029926 − 0.028) × 1.918206 = about 73,900 (positive to the payer).
7. Sterling principal = 20,000,000 / 1.25 = GBP 16,000,000. Annual interest = 16,000,000 × 3.8% = GBP 608,000 (receiving USD 900,000 to pay the dollar loan coupon). Without a swap at spot 1.10: 20,000,000 / 1.10 = GBP 18,181,818, so the extra sterling needed is 2,181,818.
8. Period 1: the lower of 90,000 and 100,000 is 90,000, all to OCI; ineffectiveness in profit or loss is 0. Period 2: the lower of 210,000 and 195,000 is 195,000, so the cumulative reserve is 195,000 and the OCI movement in the period is 105,000; cumulative ineffectiveness is 210,000 − 195,000 = 15,000, recognized in profit or loss in period 2.

Chapter 20: Enterprise Risk Management and Foreign Exchange

Learning Objectives

After studying this chapter you should be able to:

- Classify the main categories of corporate risk (market, credit, liquidity, operational, strategic and others) and explain how enterprise risk management integrates them into a single framework tied to strategy and risk appetite.
- Explain the concept of Value at Risk (VaR), compute parametric VaR for single assets and simple portfolios, scale it across horizons, and describe its limitations along with the role of expected shortfall and stress testing.
- Design the elements of a corporate hedging policy: objectives, hedge ratios and horizons, permitted instruments, limits, counterparty rules and governance.
- Distinguish transaction, translation and economic exposure to exchange rates, and measure each with simple tools.
- Apply covered interest rate parity, purchasing power parity and related relationships to price forward rates, detect arbitrage and form disciplined expectations about long-run currency movements.
- Compare hedging a currency exposure with forwards, money market positions, options and collars, and choose among them using payoff analysis and managerial objectives.

Opening Vignette

(Illustrative scenario; the company is fictional.)

Verdant Bay Instruments designs and manufactures laboratory analyzers at a plant in the United States. Last year it sold equipment worth 480 million: 55 percent in the United States, 30 percent in the euro area and 15 percent in the United Kingdom. Components are bought in dollars, euros and Swiss francs; the largest subsidiary, in Germany, holds net assets of 25 million euros. The company’s debt is 220 million, of which about a third is floating-rate.

In the last three years the chief financial officer, Marcus Lindqvist, has seen quarterly earnings move in ways that operating performance alone does not explain. In one quarter a strengthening dollar reduced reported euro revenue by 14 million. In another, a customer in the euro area paid an invoice for 5 million euros six months after it was issued, and the treasury received 220,000 dollars less than the amount recorded in the budget. In a third, the board noticed that the German subsidiary’s net assets, translated into dollars, shrank by almost 2 million, lowering reported equity although nothing had happened to the subsidiary’s operations.

The board’s audit and risk committee asks Marcus three questions. What are all the risks the company runs, and which ones are we paid to take? How bad could a bad month be, in a single number the committee can track? And what is the policy for currency exposure; who decides how much to hedge, with what instruments, and how will we know if it is working?

Marcus recognizes that these questions cannot be answered by the treasury alone. Risks arise in procurement, sales, financing and technology; they interact; and the choice of which to retain is a strategic one. This chapter gives him the framework: a taxonomy of risks, a quantitative measure of market risk (VaR), the elements of a hedging policy, and the economics of foreign exchange, from the parity relationships that discipline exchange rates to the instruments (forwards, money market hedges, options) that manage them.

20.1 A Taxonomy of Corporate Risk

20.1.1 Why risk management matters to shareholders

In the frictionless world of Modigliani and Miller, hedging cannot create value: shareholders can diversify and hedge on their own, and a firm's value depends on the cash flows of its assets. Risk management matters in practice because of *frictions*:

- **Costs of financial distress.** Volatile cash flows raise the probability that the firm cannot meet its obligations; distress involves lost customers, strained suppliers, forced asset sales and legal costs.
- **Convex taxes.** If tax is progressive or losses cannot be fully carried back, smoother income lowers expected tax.
- **Underinvestment.** When internal cash flows fall unexpectedly, external financing is costly, and profitable projects are cut. Hedging helps match internal funds to investment needs.
- **Managerial risk aversion and incentives.** Managers hold undiversified human and financial capital in the firm.
- **Contracting and covenants.** Debt covenants based on accounting ratios can be breached by volatility rather than by poor performance.

These frictions imply that firms should hedge risks that are *not core to their competitive advantage*, are *large relative to the firm's capacity to absorb them*, and are *cheaply transferable* in liquid markets. They should retain risks they are better placed than the market to understand and manage.

20.1.2 The main categories

Risk category	Definition	Typical sources	Typical tools
Market risk	Loss from changes in prices and rates	Exchange rates, interest rates, commodity prices, equity prices	Forwards, futures, swaps, options; natural hedges
Credit risk	Loss from a counterparty failing to perform	Customer receivables, derivative counterparties, bank deposits, suppliers	Credit limits, insurance, collateral, diversification
Liquidity risk	Inability to meet cash obligations when due, or to trade without large price impact	Margin calls, maturing debt, covenant breaches, market disruptions	Committed credit lines, cash buffers, staggered maturities
Operational risk	Loss from failed processes, people, systems or external events	Fraud, errors, cyber events, supply chain failure, natural disasters	Controls, business continuity, insurance
Strategic and business risk	Loss from poor strategic decisions or changes in competitive conditions	Technology disruption, shifts in demand, acquisitions, pricing	Scenario planning, portfolio diversification, real options
Legal, regulatory and compliance risk	Loss from violation of laws or changes in regulation	Sanctions, privacy, tax, product liability	Compliance programs, legal review
Reputational and ESG risk	Loss of stakeholder trust or exposure to environmental and social factors	Product safety, labor practices, climate-related physical and transition effects	Policies, disclosure, monitoring

Table 20.1 – A taxonomy of corporate risks.

Two cautions apply. First, categories overlap: a margin call is a market-risk event that becomes a liquidity-risk event. A supplier's failure may be a credit, operational and strategic problem at once. Second, risk is not one-sided; the same uncertainty that produces losses produces gains. A firm that eliminates all risk eliminates return.

20.1.3 Enterprise risk management

Traditional risk management was siloed: treasury handled currencies, insurance managers bought policies, internal audit tested controls. **Enterprise risk management (ERM)** is an integrated, firm-wide process, overseen by the board and executed by management, that identifies the events that might affect the achievement of objectives, assesses them consistently, and responds within an agreed appetite for risk. Widely used frameworks, such as those published by COSO and ISO 31000, share the same logic:

1. **Governance and culture.** The board sets risk appetite and holds management accountable; managers have clear ownership of risks.
2. **Strategy and objective-setting.** Risk is considered when strategy is chosen, not afterwards. The risk appetite (how much loss the firm is willing to bear in pursuit of its objectives) and risk tolerance (acceptable variation around specific objectives) are made explicit.
3. **Identification and assessment.** Risks are catalogued in a risk register and assessed by likelihood and impact, ideally in quantified terms (cash-flow-at-risk, earnings-at-risk, VaR).
4. **Response.** For each risk the options are to avoid, reduce (mitigate), transfer (hedge or insure) or accept.
5. **Monitoring and reporting.** Key risk indicators, limit utilization and incidents are reported to management and the board, with independent challenge from a risk function.

The managerial gain from ERM is *aggregation*. A firm may discover that its currency exposure and its commodity exposure offset each other, that two apparently different risks share a common driver (for example, a recession lowering demand, raising credit losses and tightening liquidity simultaneously), or that its risk appetite is being used up by exposures it did not intend to take. Aggregation also reveals natural hedges, which are cheaper than financial ones.

20.1.4 Risk appetite and the three lines

Risk appetite is expressed in limits that connect to what the firm cannot afford to lose. Examples: "earnings before interest and tax should not fall by more than 10 percent from currency movements in a one-in-twenty year event"; "liquidity headroom must exceed the largest 30-day cash outflow in the stress scenario"; "net debt to EBITDA must remain at least 0.5 turns inside covenant levels under the stress scenario." Responsibility is commonly allocated through the **three lines** model: business units own and manage risks (first line); an independent risk and compliance function sets standards and monitors (second line); internal audit gives independent assurance to the board (third line).

20.2 Measuring Market Risk: Value at Risk

20.2.1 The concept

Value at Risk (VaR) answers the question: "What is the maximum loss that will not be exceeded with a given probability (the confidence level) over a given horizon?" A one-day 99 percent VaR of 2.8 million means that, under normal market conditions, the portfolio should lose more than 2.8 million on only about 1 trading day in 100 (about two or three days a year). VaR has three parameters: the **horizon** (one day, ten days, one month), the **confidence level** (95 or 99 percent), and the **portfolio** with its risk factors. It does not say how much is lost on the bad days; it is a threshold, not a worst case.

Three methods of estimating VaR are common:

- **Parametric (variance-covariance).** Assume returns are normally distributed with a volatility estimated from history. VaR is a multiple of the portfolio standard deviation. Fast and transparent, but sensitive to the normality assumption and poor for options.
- **Historical simulation.** Revalue the current portfolio using actual historical changes in risk factors; read off the percentile. Captures fat tails present in the data but assumes the past sample is representative.
- **Monte Carlo simulation.** Simulate thousands of scenarios from a specified model of the risk factors and revalue the portfolio in each. Flexible, handles options and nonlinearities, but model-dependent and computationally heavier.

20.2.2 Parametric VaR

For a portfolio of value V with daily standard deviation of returns σ and a confidence level c with normal critical value z_c (z = 1.645 for 95 percent, 2.326 for 99 percent), assuming a zero mean daily return:

```
VaR (1 day) = z_c · σ · V
```

Scaling to a longer horizon of T days under independent daily returns:

```
VaR (T days) = VaR (1 day) · √T
```

The square-root-of-time rule follows from the variance of independent returns adding over time. It overstates risk where returns mean-revert and understates it where returns are autocorrelated or volatility clusters.

Example 20.1 – Single-position VaR and scaling

A treasury holds a 100 million portfolio of foreign-currency-denominated investments. Daily return volatility is 1.2 percent. Compute the one-day 99 percent VaR, the ten-day 99 percent VaR and the one-day 95 percent VaR. Also compute the expected shortfall at 99 percent.

```
1-day 99% VaR  = 2.326 × 0.012 × 100,000,000 = 2,791,617
10-day 99% VaR = 2,791,617 × √10 = 2,791,617 × 3.1623 = 8,827,869
1-day 95% VaR  = 1.645 × 0.012 × 100,000,000 = 1,973,824
```

(Using unrounded critical values 2.32635 and 1.64485.) Interpretation: on 99 days in 100 the one-day loss should not exceed about 2.79 million; over ten days, the loss is expected to stay below 8.83 million with 99 percent probability. The ten-day figure is about 3.16 times the one-day figure, not 10 times.

Expected shortfall (ES), also called conditional VaR, is the average loss *given that* the loss exceeds VaR. For a normal distribution with zero mean:

```
ES = σ · V · φ(z) / (1 − c)
```

where φ is the standard normal density. At 99 percent, φ(2.326) = 0.02665, so

```
ES = 0.012 × 100,000,000 × 0.02665 / 0.01 = 3,198,257
```

The average loss on the worst 1 percent of days is about 3.20 million, 14.6 percent higher than the VaR threshold. With fat-tailed real data the gap is larger, which is why regulators and risk committees increasingly look at ES alongside VaR.

20.2.3 Portfolio VaR and diversification

For a portfolio of positions with standard deviations (in money terms) $a = w_1\sigma_1$ and $b = w_2\sigma_2$, and correlation ρ, the portfolio's money standard deviation is

$$\sigma_P = \sqrt{a^2 + b^2 + 2\rho ab}$$

and VaR = $z_c \cdot \sigma_P$. Because the correlation is less than 1, portfolio VaR is lower than the sum of the stand-alone VaRs: the difference is the **diversification benefit**.

Example 20.2 – Two-asset VaR with diversification

A firm has 60 million of exposure to asset 1 (daily volatility 1.5 percent) and 40 million to asset 2 (daily volatility 1.0 percent), correlation 0.3. Find the one-day 99 percent VaR of the portfolio, and the diversification benefit.

$$a = 60{,}000{,}000 \times 0.015 = 900{,}000$$
$$b = 40{,}000{,}000 \times 0.010 = 400{,}000$$

$$\sigma_P = \sqrt{900{,}000^2 + 400{,}000^2 + 2 \times 0.3 \times 900{,}000 \times 400{,}000}$$
$$= \sqrt{810{,}000{,}000{,}000 + 160{,}000{,}000{,}000 + 216{,}000{,}000{,}000}$$
$$= \sqrt{1{,}186{,}000{,}000{,}000}$$
$$= 1{,}089{,}036$$

$$\text{Portfolio VaR} = 2.326 \times 1{,}089{,}036 = 2{,}533{,}477$$
$$\text{Sum of stand-alone VaRs} = 2.326 \times (900{,}000 + 400{,}000) = 2.326 \times 1{,}300{,}000 = 3{,}024{,}252$$
$$\text{Diversification benefit} = 3{,}024{,}252 - 2{,}533{,}477 = 490{,}775$$

(The sums are computed with unrounded z.) Diversification reduces VaR by about 16 percent. In a crisis correlations tend to rise toward 1, eroding the benefit, which is why stress testing with elevated correlations matters.

20.2.4 Limitations of VaR

- **Not a worst case.** Losses beyond VaR can be far larger.
- **Distribution assumptions.** Financial returns have fat tails and volatility clusters; normal-based VaR understates extreme moves.
- **Not subadditive in general.** VaR can, in unusual cases, increase when portfolios are combined, violating the intuition that diversification reduces risk. Expected shortfall does not have this flaw.
- **Backward-looking.** Calibration windows that end in calm periods produce low VaR just before volatility rises.
- **Liquidity ignored.** VaR assumes positions can be exited at current prices within the horizon.
- **Gaming and false comfort.** A single number can mask concentrated, illiquid or complex positions.

For these reasons VaR is complemented by **stress testing** (revaluing the portfolio under severe but plausible scenarios, such as a 20 percent currency move or a 300 basis point rate shock), **scenario analysis** (historical episodes replayed), and **reverse stress testing** (asking what event would breach covenants or exhaust liquidity). Nonfinancial corporations often prefer **cash-flow-at-risk** or **earnings-at-risk**, which apply the same statistical idea to the firm's operating cash flows or EBIT over the planning horizon, because their exposures are not marketable securities and what matters is the ability to fund operations and investment.

20.3 Designing a Hedging Policy

A hedging policy turns the risk appetite of Section 20.1 into rules that treasury staff can apply and the board can monitor. Its purpose is to prevent two opposite failures: leaving material exposures unmanaged, and allowing the treasury to become a profit center that takes unauthorized speculative positions.

20.3.1 Elements of a sound policy

1. **Objectives.** State what the hedging is for: protect budgeted margins, stabilize cash flows available for investment, secure covenant compliance, or lock in prices for customers. An objective such as “beat the market” is not a hedging objective. Specify the metric that will be protected (EBIT, cash flow, a specific margin, net debt to EBITDA).
2. **Scope of exposures.** List which exposures are hedged: recognized balances (receivables, payables, loans), firm commitments, forecast transactions, and net investments. Define how forecast exposures are identified, who certifies the forecast, and how it is updated.
3. **Hedge ratios and horizon.** Specify minimum and maximum percentages by time bucket. A typical pattern is higher ratios for near-term, more certain exposures and lower ratios further out.
4. **Permitted instruments.** Forwards, swaps and purchased options are usually permitted; sold options, leveraged or exotic instruments are typically prohibited or require approval, because they can create unlimited losses or obscure risk.
5. **Counterparty and credit rules.** Minimum credit rating, maximum exposure per bank (measured by mark-to-market plus an allowance for potential movement), netting and collateral agreements.
6. **Limits and authorities.** Notional limits per instrument and per currency, VaR or cash-flow-at-risk limits, stop-loss escalation, authority levels for dealers.
7. **Segregation of duties.** The front office executes, the middle office monitors risk and valuation independently, the back office confirms and settles, and accounting records. No one person can execute, confirm and record a trade.
8. **Liquidity planning.** Margin and collateral requirements under stress; committed facilities sized to the stressed outflow.
9. **Accounting and reporting.** Rules on documentation for hedge accounting (Chapter 19), the frequency and content of reports, and escalation of breaches to the board.
10. **Review.** Periodic review of the policy and a post-mortem of hedge performance, benchmarked against pre-agreed alternatives (such as no hedge, fully hedged, or a static 50 percent hedge), so that results are judged by process rather than hindsight.

20.3.2 Natural hedges and operational hedges first

Financial hedging is the last resort, not the first. Cheaper tools are:

- **Natural hedging.** Match currency of costs and revenues. A firm with euro revenues can buy components or locate production in the euro area, or borrow in euros so that debt service is paid from euro income.
- **Netting.** Combine receivables and payables in the same currency across subsidiaries and hedge only the net amount; multilateral netting centers reduce transaction volumes and fees.
- **Invoicing in home currency.** Shifts the exposure to the customer or supplier, but may reduce competitiveness or simply move the risk into the price negotiation.

- **Leading and lagging.** Accelerating or delaying payments in anticipation of exchange rate moves; this is partly speculative and often constrained by transfer pricing and tax rules.
- **Pricing clauses and surcharges.** Contractual adjustment of price for exchange rate movements beyond a band.
- **Diversifying operations and suppliers** across currency areas.

20.3.3 How much to hedge

The correct hedge ratio is neither zero nor one by default. Considerations: the certainty of the forecast (hedging a forecast that does not materialize creates a speculative position), the firm's capacity to absorb adverse moves, the cost of the hedge, the firm's view of whether competitors are hedged (an unhedged competitor with lower costs after a currency move may undercut prices), and the relationship between the exposure and other parts of the business. Many firms use a **layered (rolling) hedge**: they hedge a high proportion of exposures in the next quarter and progressively less for later periods, and add layers each period. The average hedged rate is then a moving average of forward rates, smoothing the transition when the currency permanently shifts and giving the firm time to adjust prices and costs.

Example 20.3 – A layered hedging program

Verdant Bay forecasts euro-denominated net inflows of EUR 12 million in each of the next four quarters. Policy requires hedging 90 percent of the first quarter, 70 percent of the second, 50 percent of the third and 30 percent of the fourth. The spot rate is 1.08 USD per EUR. (a) How much is hedged? (b) What is the dollar effect of an immediate, permanent 5 percent fall in the euro with and without the program?

a. Hedged amounts:

```
Q1: 12 × 0.90 = 10.8
Q2: 12 × 0.70 =  8.4
Q3: 12 × 0.50 =  6.0
Q4: 12 × 0.30 =  3.6
Total hedged   = 28.8 million EUR out of 48 million  (60% overall)
Unhedged       = 48 − 28.8 = 19.2 million EUR
```

b. A 5 percent fall in the euro lowers the rate by 1.08 × 0.05 = 0.054 USD per EUR (to 1.026).

```
Without hedges: loss = 48,000,000 × 0.054 = 2,592,000 USD
With the program: unhedged loss = 19,200,000 × 0.054 = 1,036,800 USD
Protected amount = 2,592,000 − 1,036,800 = 1,555,200 USD
```

(Ignoring forward points.) The firm has protected 60 percent of the dollar value of the shock. The protection is heavily weighted toward the near term, so the firm retains pressure to adapt its prices and cost base for the later quarters, where the permanent currency change will bite. Conversely, if the euro rises 5 percent, the firm gives up the matching gain on the hedged part: that is the cost of certainty. The next quarter's layer would be added at the new forward rate, so the average hedge rate gradually adjusts.

20.3.4 Governance lessons from hedging failures

Published cases of large derivative losses at corporations and public bodies in past decades tend to share features: positions much larger or more complex than the underlying exposure; leverage embedded in structured products; inadequate valuation and independent reporting; board members who did not understand the instruments; and the reporting of

derivatives as profit centers. The remedies are exactly the ones listed in 20.3.1: clear objectives, restricted instruments, independent monitoring, and board-level reporting in plain language.

20.4 Foreign Exchange Exposure

Exchange rate risk is the effect of unexpected exchange rate changes on the value of the firm. It is usually divided into three types, which differ in time horizon, measurement and hedge.

Dimension	Transaction exposure	Translation exposure	Economic (operating) exposure
What is affected	Cash flows from contracted foreign-currency receipts and payments	Reported values of foreign subsidiaries' assets, liabilities, income when consolidated	Present value of future operating cash flows and competitive position
Time horizon	Short term, contractually fixed	Balance sheet dates, reporting periods	Long term, uncertain
Measurement	Net contracted foreign-currency amounts by date	Net assets (and income) by functional currency	Regression, scenario or simulation of cash flows against exchange rates
Cash effect	Real cash gain or loss at settlement	No direct cash effect (accounting), though covenants and capital ratios may be affected	Real, via sales volume, prices, costs
Hedge tools	Forwards, money market hedges, options, natural hedges	Foreign-currency debt or derivatives designated as net investment hedges; or accept it	Operational: diversify production, sourcing and markets; pricing strategy; foreign-currency financing

Table 20.2 – The three exposures.

20.4.1 Transaction exposure

Transaction exposure arises between the moment a price in foreign currency is agreed and the moment it is settled. A US exporter invoices EUR 5 million payable in six months; if the euro falls before payment, the dollars received fall. The exposure also extends to *contingent* exposures, such as a bid in foreign currency for a contract that may or may not be won. Because the amount and date are known, transaction exposure is the easiest to hedge, and the hedging choices (forwards, money market, options) are covered in Section 20.6.

20.4.2 Translation exposure

When a parent consolidates a foreign subsidiary, it translates the subsidiary's financial statements from its functional currency into the parent's presentation currency. Under IAS 21, assets and liabilities are translated at the closing rate, and income and expenses at the transaction-date rates (or an average approximation); the resulting exchange differences accumulate in other comprehensive income in a foreign currency translation reserve and are recycled to profit or loss when the foreign operation is disposed of. Translation effects are therefore not usually in earnings, but they change equity, leverage ratios and book value, and may be disclosed prominently.

Hedging translation exposure means creating a liability or derivative in the foreign currency of equal size to the net assets, designated as a *net investment hedge* so that offsetting gains and losses sit together in OCI (Chapter 19). Many firms choose not to hedge translation exposure because it is non-cash and shareholders can diversify; others hedge to protect leverage

covenants or capital ratios.

Example 20.4 – Translation exposure and a net investment hedge

Verdant Bay's German subsidiary has net assets of EUR 25 million. The rate is 1.10 USD per EUR at the start of the year and falls to 1.04 at year-end. (a) Compute the translation loss. (b) Suppose the parent had sold EUR 25 million forward at 1.10 (ignoring forward points) designated as a net investment hedge. What is the net effect on OCI?

a. Net assets in dollars:

```
Start: 25,000,000 × 1.10 = 27,500,000
End:   25,000,000 × 1.04 = 26,000,000
Translation loss = 1,500,000 (recognized in OCI)
```

b. The forward sells EUR at 1.10 while the market rate is 1.04: gain per EUR of 0.06, or 25,000,000 × 0.06 = 1,500,000. The gain on the hedge offsets the loss on net assets in OCI, leaving the reserve unchanged. The hedge generates cash settlement of 1.5 million at maturity (an actual cash flow), while the translation loss is non-cash; so the program creates a liquidity need if the currency moves the other way. This mismatch is why some firms hedge translation exposure partially, or hedge with euro borrowings (which require no cash settlement until repaid).

20.4.3 Economic exposure

Economic exposure is the sensitivity of the firm's *future* operating cash flows, and therefore its value, to exchange rate changes. It is not an accounting concept and cannot be read off the balance sheet. A currency appreciation can hurt a firm that has no foreign-currency contracts: an exporter loses price competitiveness at home and abroad; a domestic producer facing imports from a country whose currency depreciates loses sales or margin. Conversely, a firm importing inputs benefits from a stronger home currency. Exposure depends on:

- the currency of costs and revenues (the *net* position);
- pricing power and the degree of pass-through to customers;
- elasticity of demand, and the currency of competitors' costs;
- the speed at which the firm can change sourcing and production;
- hidden dependencies, such as suppliers whose costs are in a foreign currency, even if the firm pays them in domestic currency.

Measurement typically uses regression: the dependent variable is the firm's cash flow (or stock return) and the independent variable the exchange rate or an index; the slope is the *exposure coefficient*. Scenario analysis complements it by modeling cash flows under alternative rate paths.

Example 20.5 – Estimating economic exposure with a regression

A division's quarterly operating cash flow (USD millions) and the average USD per EUR rate over six quarters are as follows. Estimate the exposure.

Quarter	USD per EUR (x)	Cash flow, USD millions (y)
1	1.00	118
2	1.04	121
3	1.08	125
4	1.12	126
5	1.16	131
6	1.20	133

Table 20.3 – Illustrative data for an exposure regression.

Means: $\bar{x} = (1.00 + 1.04 + 1.08 + 1.12 + 1.16 + 1.20) / 6 = 6.60 / 6 = 1.10$; $\bar{y} = (118 + 121 + 125 + 126 + 131 + 133) / 6 = 754 / 6 = 125.667$.

Deviations and products:

```
x − x̄:  −0.10, −0.06, −0.02, +0.02, +0.06, +0.10
y − ȳ:  −7.667, −4.667, −0.667, +0.333, +5.333, +7.333
Σ(x−x̄)(y−ȳ) = 0.7667 + 0.2800 + 0.0133 + 0.0067 + 0.3200 + 0.7333 = 2.1200
Σ(x−x̄)²     = 0.01 + 0.0036 + 0.0004 + 0.0004 + 0.0036 + 0.01 = 0.0280

slope b = 2.1200 / 0.0280 = 75.71
intercept a = 125.667 − 75.71 × 1.10 = 42.38
```

So `Cash flow ≈ 42.4 + 75.7 × (USD per EUR)`, with R^2 of about 0.98 in this illustrative sample. Interpretation: each 0.01 rise in the euro adds about 0.76 million to quarterly cash flow; a 0.05 fall in the euro (4.5 percent from 1.10) reduces it by $75.71 \times 0.05 = 3.79$ million, about 3 percent of the cash flow. The division behaves as if it were *long* about EUR 75.7 million per quarter. This is a measure of economic exposure and may be much greater than its contractual transaction exposure. A hedge against a currency fall would sell euros forward in proportion, but a regression based on six points is fragile: the manager should confirm with scenario analysis and ask why the relationship exists (for example, euro-denominated sales with dollar costs).

The practical implication is that economic exposure is managed mainly through *strategy*: locating plants in the currency areas where sales occur, choosing suppliers with costs in the same currency as revenues, financing foreign operations in local currency, and adjusting pricing. Financial hedges can buy time (as in the layered program) but cannot permanently protect a firm whose cost structure is structurally misaligned with its revenue base.

20.5 Exchange Rate Relationships

Exchange rates are determined by trade flows, capital flows, expectations and policy. Over short horizons they are very hard to predict; over longer horizons and in terms of arbitrage, a set of *parity relationships* provide discipline. We quote exchange rates as the domestic-currency price of one unit of foreign currency (USD per EUR). With that convention, a rising rate means the foreign currency appreciates.

20.5.1 Covered interest rate parity (CIP)

Covered interest rate parity states that the forward exchange rate is determined by the spot rate and the interest rate differential. An investor can invest domestically, or convert to foreign currency, invest abroad, and lock in the conversion back with a forward contract. If both routes are riskless, they must give the same return:

```
(1 + i_d · t) = (S / F) · (1 + i_f · t)      ⇒      F = S · (1 + i_d · t) / (1 + i_f · t)
```

where i_d is the domestic interest rate, i_f the foreign rate, t the period as a fraction of a year (using simple interest for periods up to a year; for longer periods use compounding), S and F quoted as domestic per foreign. The currency with the higher interest rate trades at a **forward discount**; the currency with the lower interest rate trades at a forward premium. The forward rate is *not* a forecast of the future spot rate; it is a no-arbitrage price based on interest rate differentials, and CIP holds closely (apart from stresses in funding markets, when cross-currency basis spreads widen) because deviations can be arbitraged with very low risk.

Example 20.6 – Pricing a forward rate with CIP

The spot rate is 1.0800 USD per EUR. The six-month USD interest rate is 5.0 percent per year and the six-month EUR rate is 3.0 percent per year (simple interest). Find the six-month forward rate.

```
F = 1.0800 × (1 + 0.05 × 0.5) / (1 + 0.03 × 0.5)
  = 1.0800 × 1.025 / 1.015
  = 1.0800 × 1.009852
  = 1.09064
```

The euro trades at a forward premium of 1.09064 − 1.0800 = 0.01064 USD per EUR, about 0.985 percent over six months (1.97 percent annualized), approximately the interest differential of 2 percentage points per year. These “forward points” (106 points in the quotation convention where one point is 0.0001) are the price of carrying one currency instead of the other, not a prediction that the euro will rise.

Example 20.7 – Covered interest arbitrage

Using the data in Example 20.6, a bank quotes the six-month forward at 1.0950, above the parity value of 1.09064. Show how to exploit this with a 1,000,000 dollar borrowing.

The forward price of euros (1.0950) is too high relative to parity, so sell euros forward and finance by borrowing dollars:

1. Borrow USD 1,000,000 for six months at 5 percent per year; at maturity the repayment is 1,000,000 × 1.025 = USD 1,025,000.
2. Convert at spot: 1,000,000 / 1.0800 = EUR 925,926.
3. Invest the euros at 3 percent per year for six months: 925,926 × 1.015 = EUR 939,815.
4. Sell EUR 939,815 forward at 1.0950: receive USD 939,815 × 1.0950 = USD 1,029,097.
5. Repay the loan: USD 1,025,000. Profit = 1,029,097 − 1,025,000 = **USD 4,097** per million borrowed, with no exposure to the exchange rate and no investment of own funds.

Arbitrageurs doing this would push spot up, forward down, euro interest down, and dollar interest up until the profit disappears. Therefore corporate treasurers should expect a bank’s forward quotes to follow CIP closely, and any difference beyond a reasonable dealer margin is a negotiation point.

20.5.2 Purchasing power parity (PPP)

The **law of one price** says that identical tradable goods should cost the same in different countries when prices are converted at the market exchange rate, otherwise arbitrageurs would buy where cheap and sell where dear. **Absolute PPP** extends this to price levels: S = P_d / P_f. **Relative PPP** says that the percentage change in the exchange rate equals the inflation differential:

```
S_t = S₀ · [(1 + π_d) / (1 + π_f)]^t
```

where π_d and π_f are domestic and foreign annual inflation rates. A country with higher inflation should see its currency depreciate.

The empirical record is that PPP is a weak guide in the short run (deviations last for years because of nontradable goods, transport costs, tariffs, pricing-to-market and capital flows), but a better guide over long horizons, especially for large inflation differentials. It is useful to a manager in three ways: as a *benchmark of valuation* (is a currency far from its PPP level?), as a *basis for long-horizon planning* when forward markets do not exist, and as a *warning* that a high-inflation location will probably experience currency depreciation that offsets any apparent cost advantage.

Example 20.8 – Relative PPP and real exchange rate

Spot is 1.08 USD per EUR. Expected annual inflation is 3 percent in the United States and 2 percent in the euro area. (a) What does relative PPP predict for the rate after one year and after three years? (b) A basket of goods costs USD 6.00 in the US and EUR 5.20 in the euro area. What is the absolute PPP rate and the real exchange rate?

a. One year:

```
S₁ = 1.08 × (1.03 / 1.02) = 1.08 × 1.009804 = 1.0906
```

The euro is expected to appreciate by about 0.98 percent per year against the dollar.

Three years:

```
S₃ = 1.08 × (1.03 / 1.02)³ = 1.08 × 1.029707 = 1.1121
```

b. The absolute PPP rate = 6.00 / 5.20 = 1.1538 USD per EUR. The actual rate of 1.08 is lower than 1.1538, so the euro is undervalued relative to PPP by 1.08/1.1538 − 1 = −6.4 percent. The real exchange rate is

```
q = S · P_f / P_d = 1.08 × 5.20 / 6.00 = 0.936
```

A real rate below 1 means that euro-area goods are about 6.4 percent cheaper than US goods when measured in a common currency, a source of competitive advantage for euro-area producers *if* the basket is representative. The interest differential (5 percent versus 3 percent) in Example 20.6 and the inflation differential (3 percent versus 2 percent) here are both roughly consistent with the dollar being the weaker currency in the long run, though the forward rate is by itself no forecast.

20.5.3 The international Fisher effect and uncovered parity

The **Fisher effect** states that nominal interest rates equal the real rate plus expected inflation. If real rates are equal across countries, interest differentials reflect inflation differentials: `(1 + i_d) / (1 + i_f) = (1 + π_d) / (1 + π_f)`. Together with relative PPP, this gives the **international Fisher effect** or **uncovered interest parity** (UIP): the expected change in the spot rate equals the interest differential. In its pure form the expected future spot rate would equal the forward rate. The empirical evidence has shown otherwise: high-interest-rate currencies have often not depreciated as UIP predicts, which is why the “carry trade” (borrowing in low-rate currencies to invest in high-rate ones) has delivered positive average returns punctuated by sharp crashes. For corporate managers, the lesson is that forward rates are not unbiased predictors, that a hedge at the forward rate is a fair-value transaction rather than an advantageous or disadvantageous one, and that policy should not depend on forecasting currencies.

Relationship	Statement	Basis	Reliability	Use for the manager
Covered interest parity	F/S = (1 + i_d)/(1 + i_f)	Riskless arbitrage	Very high (except in funding stress)	Price and sanity-check forwards; money market hedges equal forwards
Relative PPP	ΔS ≈ π_d − π_f	Goods arbitrage	Weak short run, better long run	Long-horizon planning, valuation benchmarks
Fisher effect	i ≈ r + π	Real-rate equalization	Moderate	Interpret interest differentials as inflation expectations
Uncovered interest parity	E[ΔS] ≈ i_d − i_f	Expectations; risk-neutral investors	Poor over short horizons	Do not assume forwards forecast spot

Table 20.4 – Parity relationships.

20.6 Hedging Foreign Exchange Exposure

20.6.1 The set of instruments for transaction exposure

Suppose Verdant Bay will receive EUR 5 million in six months from a euro-area customer. The treasurer has three main alternatives for locking in or bounding the dollar value:

1. **Forward contract.** Sell EUR 5 million forward to a bank at the six-month forward rate.
2. **Money market hedge.** Borrow the present value of the euros today, convert at spot, invest the dollars; use the euro receipt to repay the loan.
3. **Currency option.** Buy a put option on euros, which guarantees a minimum dollar value while preserving the upside.

A fourth alternative is to stay unhedged, which is a conscious decision to bear the risk.

Example 20.9 – Forward and money market hedge compared

Verdant Bay expects EUR 5,000,000 in six months. Spot is 1.0800; the six-month USD rate is 5 percent and the EUR rate is 3 percent per year (simple). Compute the dollar proceeds with (a) the forward hedge and (b) the money market hedge. (c) Compare with the unhedged conversion at today's spot.

a. Forward rate from Example 20.6: 1.09064. Proceeds = 5,000,000 × 1.09064 = USD 5,453,202.

b. Money market hedge:

```
Step 1: Borrow euros whose repayment at six months equals the receipt:
        PV = 5,000,000 / (1 + 0.03 × 0.5) = 5,000,000 / 1.015 = EUR 4,926,108
Step 2: Convert to dollars at spot:
        4,926,108 × 1.0800 = USD 5,320,197
Step 3: Invest dollars at 5% for six months:
        5,320,197 × 1.025 = USD 5,453,202
Step 4: At month 6, the EUR 5,000,000 receipt repays the euro loan.
```

The result, USD 5,453,202, is identical to the forward hedge, as covered interest parity requires. In practice the two differ slightly because of bid-ask spreads, differences between deposit and borrowing rates, and credit lines used.

c. Selling the euros at today's spot would give 5,000,000 × 1.08 = 5,400,000 but is not available (the cash arrives in six months). The extra 53,202 from the hedges (5,453,202 − 5,400,000) is the forward premium, which compensates for the higher dollar interest rate: it is not a free gain. If the spot rate in six months were 1.10, the unhedged outcome would be 5,500,000; if 1.05, it would be 5,250,000. The hedge removes this variability: the outcome is 5,453,202 whatever the rate.

20.6.2 Options

A **currency put option** on euros gives the holder the right to sell euros at the strike price. For an exporter with euro receivables, buying a put sets a floor for the dollar value while retaining the benefit of a stronger euro, at the price of the premium. The same economics applies to importers with euro payables, who buy calls.

Example 20.10 – Hedging with a put option

Verdant Bay buys a six-month put option on EUR 5,000,000 with a strike of 1.0800 for a premium of 0.0200 USD per EUR. The USD interest rate is 5 percent per year (simple), so the premium is financed for six months. Compute the net proceeds for several spot rates at maturity, compare with the forward hedge of Example 20.9 and find the break-even rate.

Premium paid today = 0.0200 × 5,000,000 = USD 100,000. Future value at six months = 100,000 × 1.025 = USD 102,500.

Net proceeds = 5,000,000 × max(S_T, 1.0800) − 102,500.

Spot in six months	Unhedged (USD)	Forward hedge (USD)	Put option, net of premium (USD)
1.00	5,000,000	5,453,202	5,297,500
1.05	5,250,000	5,453,202	5,297,500
1.08	5,400,000	5,453,202	5,297,500
1.11	5,550,000	5,453,202	5,447,500
1.15	5,750,000	5,453,202	5,647,500
1.20	6,000,000	5,453,202	5,897,500

Table 20.5 – Outcomes of alternative hedging strategies for a EUR 5 million receivable.

Calculation checks: at 1.00 and 1.05 and 1.08 the option's floor of 1.0800 applies: 5,400,000 − 102,500 = 5,297,500. At 1.11: 5,550,000 − 102,500 = 5,447,500. At 1.15: 5,750,000 − 102,500 = 5,647,500. At 1.20: 6,000,000 − 102,500 = 5,897,500.

The option guarantees at least 5,297,500, which is 155,702 less than the forward (5,453,202 − 5,297,500). In return it offers unlimited upside. The break-even spot rate at which the option outcome equals the forward outcome solves

```
5,000,000 × S − 102,500 = 5,453,202   ⇒   S = 5,555,702 / 5,000,000 = 1.1111
```

If the euro ends above 1.1111 the option beats the forward; below it, the forward is better. The choice depends on the treasurer's view of the distribution of outcomes and, more importantly, on the objective. Options are valuable when the exposure itself is uncertain (a bid that may not be won: with a forward, failure to win the contract leaves an open position that could lose, while the option's loss is limited to the premium) or when the firm wants to avoid regret.

20.6.3 Collars and other structures

A **collar** (or range forward) combines a purchased put with a sold call, so the premium received on the call finances the put. The exporter has a floor and a cap on the effective rate; between them the firm participates in the market.

Example 20.11 – A zero-cost collar

Verdant Bay buys a six-month put with strike 1.0600 and sells a call with strike 1.1200 on EUR 5,000,000. The premiums are equal, so the collar has zero cost. Find the net proceeds for the spot rates of Table 20.5.

Net proceeds = 5,000,000 × min(max(S_T, 1.06), 1.12).

```
S_T = 1.00:  5,000,000 × 1.06 = 5,300,000   (put exercised)
S_T = 1.05:  5,000,000 × 1.06 = 5,300,000
S_T = 1.08:  5,000,000 × 1.08 = 5,400,000   (neither exercised)
S_T = 1.11:  5,000,000 × 1.11 = 5,550,000
S_T = 1.15:  5,000,000 × 1.12 = 5,600,000   (call exercised against the firm)
S_T = 1.20:  5,000,000 × 1.12 = 5,600,000
```

Proceeds are constrained to the band 5,300,000 to 5,600,000, a range of 300,000 compared with 1,000,000 for the unhedged position over the same scenarios. The collar costs nothing up front, and gives a floor (5,300,000) slightly above the net floor of the single put (5,297,500), but surrenders gains above 1.12. The sold call is an obligation that could lose money for the firm in a rising euro scenario, though it is offset by the gain on the underlying receivable. Because sold options can be considered speculative, policy must permit them only when matched against an underlying exposure.

20.6.4 Choosing among hedges

Instrument	Cost	Protection	Upside retained	Best used when
Forward	Zero premium; embedded forward points	Complete at a fixed rate	None	Exposure certain in amount and date; objective is certainty
Money market hedge	Borrowing and lending spreads	Complete	None	No forward market available for the currency or maturity; credit lines in place
Purchased option	Upfront premium	Floor at strike	Full	Exposure uncertain (bids, forecasts); desire to retain upside
Collar	Zero or low	Floor and cap	Partial	Wants to limit cost and keep some participation
Natural or operational hedge	Often low	Partial, long-lasting	Depends	Structural exposures; first line of defense
No hedge	None	None	Full	Exposure immaterial, or offset elsewhere, or deliberate risk-taking within policy

Table 20.6 – Comparison of FX hedging approaches.

20.6.5 Practical issues in FX hedging

- **Forecast risk.** A hedge of a forecast that fails to materialize leaves a speculative position. Use lower ratios for uncertain forecasts or use options.
- **Opportunity cost and regret.** If the hedge shows a "loss" relative to spot at settlement, the underlying receivable shows a matching gain. Report the combined position.
- **Currency of competitors.** Hedging transaction exposure for a year does not protect market share if rivals' costs are in a currency that has weakened. The hedge buys time.
- **Cross-hedging.** Where forward markets do not exist for a currency, a related currency (a proxy) may be used, creating basis risk; the minimum-variance approach of Chapter 19 applies.
- **Cash and credit.** Forwards use credit lines at banks. Option premiums are paid in cash up front. Margin and collateral requirements should be budgeted.
- **Accounting.** Forwards and options hedging forecast foreign-currency cash flows can be designated as cash flow hedges, with the forward points or option time value accounted for as costs of hedging, as described in Chapter 19.
- **Tax.** The tax treatment of hedging gains and losses may differ across jurisdictions; check before finalizing.

20.7 Integrating Currency, Interest Rate and Liquidity Risk

A multinational should treat its exposures as a portfolio. Currency exposures from sales, costs, intercompany loans and foreign debt may partially offset; interest rate exposures on foreign-currency debt are linked to currency exposure through the interest rate differentials that drive forward points; and liquidity risk links them all, because hedges that are economically perfect can still demand cash on a bad day.

An integrated approach has four steps. First, build a *map of exposures* by currency, tenor and type, netting where legal and operationally feasible. Second, estimate a *measure of risk* of the net portfolio (VaR for treasury positions; cash-flow-at-risk or earnings-at-risk for operating exposures), including stress scenarios that move currencies, rates and credit spreads together. Third, *compare the risk to appetite*, expressed in terms of covenants, ratings thresholds and investment capacity. Fourth, *choose the cheapest effective responses*, starting with operational and natural hedges, followed by financing in matching currencies, and finally financial derivatives sized according to the hedging policy.

Reporting should answer simple questions: What is our net exposure by currency this quarter and next year? What portion is hedged and at what average rate? What would a 10 percent move in each currency do to EBIT and to net debt to EBITDA? How much cash might we need to post if rates move adversely by a given stress amount? And how does our actual hedged result compare with the benchmarks decided in advance? When Marcus Lindqvist presents answers to these questions to the audit and risk committee, risk management becomes a management discipline rather than a collection of trades.

Case Study: Northfield Agricultural Machinery

(Illustrative case; the company and all figures are fictional.)

Northfield Agricultural Machinery builds tractors and harvesting equipment at two plants in the Midwest of the United States and one plant in Poland. Annual revenue is 1.2 billion dollars: 50 percent from North America, 28 percent from Europe (invoiced mainly in euros) and 22 percent from Brazil and Argentina (invoiced in local currencies). Major cost lines

are steel and components bought in dollars, hydraulic systems from a Japanese supplier invoiced in yen, and labor in dollars, euros and Polish zloty. Net debt is 480 million, 60 percent floating-rate. The covenant package requires net debt to EBITDA below 3.5 times.

For many years Northfield had no formal risk policy. Regional finance directors hedged invoices on request, sometimes with forwards and sometimes with sold options sold to generate premium income; the group treasurer in Chicago recorded the results at year-end. After a currency shock in which the real and the peso fell roughly 25 percent in one year while the euro weakened modestly, group EBITDA fell 18 percent. Reported net debt to EBITDA rose from 2.6 to 3.4 times, just inside the covenant. In the same year the Brazilian subsidiary received a large government-linked receivable in reais that had been hedged with a sold-option structure; the structure paid out a loss of 14 million when the real appreciated sharply in the following quarter, which the Brazilian director described as “lost premium income.”

The board asked the new chief risk officer, Adaeze Mbeki, to produce an enterprise-wide analysis. Her team found the following.

First, the largest exposures were not in the hedged transactions. Economic exposure to the euro was significant: the Polish plant produced in zloty and sold in euros, while competitors’ costs in Turkey and Italy moved in other directions. Brazil and Argentina were a large unhedged exposure in revenues, with local-currency sales and dollar-denominated component costs. The yen exposure was small and partially hedged, but a single supplier relationship made the company dependent on that currency in the medium term.

Second, correlations among exposures were not stable. In the stress period the real, peso and euro all fell against the dollar, and commodity prices declined at the same time, depressing farm incomes and equipment demand. The firm’s market risk, credit risk (dealer receivables) and liquidity risk (dealer financing lines) all worsened together. A parametric VaR of the treasury’s derivative book was 3.1 million at 99 percent, but the earnings-at-risk from the operating exposures, estimated by scenario analysis, was about 95 million for a one-in-twenty year event.

Third, instrument use violated no formal rule because none existed. The sold-option structures were used in several subsidiaries to reduce the cost of hedging. None was designated for hedge accounting, so mark-to-market swings flowed through profit and loss, obscuring operating performance.

Adaeze proposed a policy with the following features. Exposures will be netted by currency at group level and hedged centrally by the treasury using forwards and purchased options; sold options are allowed only in collars matched with an underlying exposure and approved by the risk committee. Forecast sales in euros will be hedged using a layered program of 80 percent for months 1 to 3, 60 percent for months 4 to 6, 40 percent for months 7 to 12 and 20 percent for months 13 to 18. Local-currency sales in Brazil and Argentina will be partially hedged where deliverable markets exist; elsewhere the company will adjust pricing clauses and shift more component sourcing to local suppliers. The Polish plant will gradually source in euros. A committed 150-million liquidity facility will be arranged against stress margin outflows. The group will adopt hedge accounting for forecast euro sales and for the net investment in the Polish subsidiary. A quarterly risk dashboard reports exposures, hedge ratios, average hedge rates, VaR, earnings-at-risk and liquidity headroom, and compares results with two benchmarks set in advance: the unhedged position and a static 50 percent hedge.

The sales director objected that hedging reduces the ability of regional managers to price flexibly, and that centralization will slow responses to customers. The Brazilian director argued that premium income from selling options was a legitimate source of financial return. The chief financial officer, while supportive, worried about the cost of purchased options and the accounting resources required.

Discussion Questions

1. Classify Northfield's exposures into transaction, translation and economic exposure. Which are the most material to the firm's value and covenant position, and what evidence in the case supports your view?
2. Evaluate the argument that selling options to earn premium income is a legitimate source of return for a corporate treasury. Use payoffs to explain what risk the Brazilian structure created.
3. Explain why a VaR of 3.1 million and an earnings-at-risk of 95 million can both be correct. Which better informs the decision about the liquidity facility and the hedging ratios?
4. Critique the proposed layered hedge ratios and the centralization of hedging. What operational hedges would you recommend alongside, and how should hedge outcomes be judged against the benchmarks?

Key Takeaways

- Hedging creates value only through frictions such as financial distress costs, convex taxes, underinvestment and covenants; firms should hedge large, non-core, cheaply transferable risks and retain those where they hold an advantage.
- Corporate risks span market, credit, liquidity, operational, strategic, legal and reputational categories; they interact, and a margin call turns a market-risk event into a liquidity event.
- Enterprise risk management integrates risks into one framework tied to strategy and risk appetite, using governance, assessment, response and monitoring, with three lines of accountability.
- VaR states the loss not exceeded with a given confidence over a horizon: `VaR = z · σ · V`, scaled by √T for longer horizons; expected shortfall and stress tests are essential complements because VaR ignores the tail beyond the threshold.
- Portfolio VaR reflects correlations; diversification benefits shrink in crises when correlations rise.
- A hedging policy defines objectives, scope, hedge ratios by horizon, permitted instruments, limits, counterparty rules, segregation of duties, liquidity planning and review, and judges outcomes against benchmarks set in advance.
- Transaction, translation and economic exposure differ in horizon, measurement and cash effect; economic exposure is the most important and the hardest to hedge, and is managed mainly by operational and strategic means.
- Covered interest parity `F = S(1 + i_d t)/(1 + i_f t)` prices forwards, makes the money market hedge equal to the forward hedge, and the forward rate is not a forecast of the spot rate.
- Relative PPP is a long-run benchmark rather than a short-run predictor; uncovered interest parity performs poorly, so hedging policy should not rely on currency forecasts.
- Forwards give certainty, options retain upside at the price of a premium, collars trade upside for lower cost; the instrument should match the certainty of the exposure and the objective of the firm.

Review Questions

1. Explain, using Modigliani and Miller's irrelevance argument, why risk management might still create value for a corporation. Name three frictions and how each justifies hedging.
2. Give two examples showing how a risk in one category (market, credit, liquidity, operational) can become a risk in another category.

3. What are risk appetite and risk tolerance? Provide an example of each for a manufacturing company with foreign sales.
4. Define VaR in words and state its three parameters. Why is the ten-day VaR not ten times the one-day VaR?
5. Why is expected shortfall considered superior to VaR in describing tail risk? What do stress tests add?
6. List five elements that a corporate hedging policy should contain and explain why segregation of duties is critical.
7. Distinguish transaction, translation and economic exposure with an example of each for a US exporter with a German subsidiary.
8. Why is it usually better to address economic exposure by operational means than by financial derivatives?
9. State covered interest parity. Why does the forward rate not forecast the future spot rate? Why is the money market hedge equivalent to the forward hedge?
10. When would a firm prefer a purchased option to a forward contract for a foreign-currency exposure? What is the trade-off of a zero-cost collar?

Problems

1. **VaR scaling.** A portfolio worth 250 million has daily volatility of 0.8 percent. Compute the one-day 99 percent VaR, the ten-day 99 percent VaR and the one-day 95 percent VaR (z = 2.326 and 1.645).
2. **Two-asset VaR.** A firm has exposures of 30 million (daily volatility 2.0 percent) and 70 million (daily volatility 1.0 percent), with correlation −0.2. Compute the one-day 95 percent VaR, the sum of stand-alone VaRs and the diversification benefit.
3. **Forward rate.** The spot is 1.2500 USD per GBP. One-year interest rates are 4.5 percent in USD and 3.0 percent in GBP (annual compounding). Compute the one-year forward rate and state whether sterling is at a premium or discount.
4. **PPP.** The USD/JPY spot is 150 yen per dollar. Expected annual inflation is 2.5 percent in the US and 0.5 percent in Japan. Use relative PPP to find the expected rate after four years.
5. **Money market hedge.** A US importer owes GBP 2,000,000 in six months. Spot is 1.2500 USD per GBP; the six-month USD rate is 4 percent per year and the GBP rate is 3 percent per year (simple interest). Describe a money market hedge and compute the dollar cost at maturity. Verify that it equals the forward cost.
6. **Option hedge.** For the importer in Problem 5, a six-month call on GBP 2,000,000 with strike 1.2600 costs 0.0150 USD per GBP. Financing at 4 percent per year (simple) for six months, compute the total dollar cost at maturity if the spot rate is 1.20 and if it is 1.35, and the maximum cost per pound.
7. **Translation exposure.** A subsidiary has net assets of EUR 25 million. The euro falls from 1.10 to 1.04 USD per EUR. Compute the translation loss, and the gain on a EUR 25 million forward sale at 1.10 designated as a net investment hedge.
8. **Economic exposure.** A regression gives quarterly cash flow (USD millions) = 30 + 55 × S, where S is USD per GBP. Compute the predicted change in cash flow if S rises from 1.20 to 1.32, and the GBP notional per quarter that would hedge the exposure.

Answers to Selected Problems

1. 1-day 99%: 2.326 × 0.008 × 250,000,000 = 4.65 million (4,652,696 using unrounded z). 10-day: 4.65 × √10 = 14.71 million. 1-day 95%: 1.645 × 0.008 × 250,000,000 = 3.29 million.

2. a = 30,000,000 × 0.02 = 600,000; b = 70,000,000 × 0.01 = 700,000. $\sigma P = \sqrt{(600{,}000^2 + 700{,}000^2 + 2 \times (-0.2) \times 600{,}000 \times 700{,}000)} = \sqrt{(360{,}000{,}000{,}000 + 490{,}000{,}000{,}000 - 168{,}000{,}000{,}000)} = \sqrt{682{,}000{,}000{,}000} = 825{,}833$. VaR(95%) = 1.645 × 825,833 = 1.358 million. Sum of stand-alone = 1.645 × 1,300,000 = 2.138 million. Diversification benefit about 0.78 million.
3. F = 1.2500 × 1.045 / 1.03 = 1.2682. Sterling is at a forward premium (USD rates are higher).
4. $S_4 = 150 \times (1.005 / 1.025)^4 = 150 \times 0.924 = 138.6$ yen per dollar (the dollar is expected to depreciate against the yen, since US inflation is higher).
5. Buy the PV of GBP 2,000,000 at 3 percent for six months: 2,000,000 / 1.015 = GBP 1,970,443; its dollar cost today is 1,970,443 × 1.25 = USD 2,463,054, financed by borrowing at 4 percent for six months, so the repayment is 2,463,054 × 1.02 = USD 2,512,315. The forward rate is 1.25 × 1.02 / 1.015 = 1.25616, giving 2,000,000 × 1.25616 = USD 2,512,315, which is identical.
6. Premium 0.0150 × 1.02 = 0.0153 per GBP. If S = 1.20 the call is not exercised: cost = 2,000,000 × (1.20 + 0.0153) = USD 2,430,600. If S = 1.35 the call is exercised at 1.26: cost = 2,000,000 × (1.26 + 0.0153) = USD 2,550,600. Maximum cost per pound = 1.2753.
7. Loss: 25,000,000 × (1.10 − 1.04) = USD 1,500,000 (translation reserve). The forward gain is also 25,000,000 × (1.10 − 1.04) = USD 1,500,000, fully offsetting in OCI.
8. Change = 55 × (1.32 − 1.20) = 55 × 0.12 = USD 6.6 million per quarter. The cash flow behaves as if long GBP 55 million per quarter, so that is the hedge notional (before considering the appropriate hedge ratio).

Part VI — Corporate Transactions and Strategy

Chapter 21: Enterprise Valuation: DCF and Multiples

Learning Objectives

After studying this chapter, you should be able to:

- Distinguish enterprise value from equity value, and choose correctly between free cash flow to the firm (FCFF) and free cash flow to equity (FCFE) as the object of discounting.
- Build an internally consistent multi-year forecast of FCFF from operating drivers (growth, margins, reinvestment) rather than from extrapolated line items.
- Estimate a WACC with a relevered beta and target capital structure, and explain when adjusted present value (APV) is the better tool.
- Compute and stress-test a terminal value using the Gordon growth model, the value-driver formula, and an exit multiple, and recognise when each is internally inconsistent.
- Value a company with trading comparables and precedent transactions, interpret a football field, and explain why the ranges rarely agree.
- Apply the venture capital method and scenario analysis to start-ups, and diagnose the most common errors made in practical valuation work.

Opening Vignette

Illustrative scenario. Halden Precision Components is a fictional manufacturer of machined parts for medical devices and industrial automation.

Marguerite Osei, the CFO of Halden, is preparing for a board meeting that will decide whether to accept an unsolicited indicative offer of 28 per share from a private equity sponsor. Halden has 50 million shares, revenue of 800 million, EBIT margin of 18 percent, 250 million of debt and 40 million of cash. The shares have traded between 26 and 32 over the past year.

Her investment bank has delivered three numbers. The discounted cash flow (DCF) model says 31 per share. The trading comparables say roughly 26. The precedent transactions say 27 to 36. Several directors want to know which number is “right,” and one asks whether the DCF is worthless because Marguerite’s own team can move it by 20 percent simply by changing the discount rate by one percentage point.

Marguerite’s response is a lesson in what valuation is for. She explains that no single number is right, because each method answers a different question. The comparables show what the market currently pays for similar earnings streams; the precedents show what buyers have paid for control; the DCF shows what Halden is worth if the board’s own operating plan is delivered and the cost of capital is what the team believes it is. The disagreement among the three is information: the DCF is above the comparables because the plan assumes margins and growth that peers do not currently deliver. The board’s real decision is whether it believes the plan more than the market does.

She then shows the directors something unusual: a table decomposing the DCF value into the first five years of cash flow and a terminal value, which accounts for three-quarters of the total. “If you doubt anything,” she says, “doubt what we assume happens after year five.” In this chapter, we learn how to build every number in that exchange, and how to challenge it.

21.1 The Logic of Intrinsic and Relative Valuation

21.1.1 Two families of methods

Every valuation method falls into one of two families. **Intrinsic valuation** (discounted cash flow) estimates the value of an asset from its own forecast cash flows and the return investors require for bearing their risk. **Relative valuation** (multiples, comparables) estimates value from how the market prices similar assets. The two are not rivals: a multiple is simply a compressed DCF in which growth, risk, and reinvestment have been folded into a single ratio. A company that trades at 12 times earnings is being valued as if its growth, reinvestment needs, and discount rate lie within a particular range. When you use a peer’s multiple, you are implicitly importing the peer’s DCF assumptions.

The practical consequences differ. A DCF forces explicit assumptions and so makes disagreement discussable, but it is highly sensitive to a handful of inputs. A multiple is quick, reflects market sentiment, and is hard to manipulate, but it can be wrong in the same direction as the whole market. Good practice uses both and treats the gap between them as a diagnostic.

21.1.2 Enterprise value versus equity value

The **enterprise value** (EV) of a business is the value of its operating assets, which belongs collectively to all providers of capital. The **equity value** is what remains for shareholders after the claims of debt holders and other senior claimants are settled:

```
Equity value = Enterprise value − Debt − Preferred stock − Non-controlling interests
               − Pension and other debt-like deficits + Cash and non-operating assets
```

The distinction matters because the two are discounted at different rates and measured with different cash flows. Operating cash flows are available to everyone and should be discounted at a rate reflecting the risk of the operating assets, blended across the financing mix. Cash flows to shareholders, after interest and debt repayment, are riskier when leverage is higher and need a higher discount rate.

21.1.3 FCFF versus FCFE

Free cash flow to the firm is the cash generated by operations after taxes on operating income and after the investment needed to sustain growth, before any payment to financiers:

```
FCFF = EBIT × (1 − t) + D&A − Capex − ΔNWC
     = NOPAT − Net reinvestment
```

Free cash flow to equity is what is left for shareholders after servicing debt:

```
FCFE = Net income + D&A − Capex − ΔNWC + Net borrowing
     = FCFF − Interest × (1 − t) + Net borrowing
```

Each pairs with a discount rate and a bridge:

Approach	Cash flow	Discount rate	Output	Bridge to equity value
WACC / FCFF	Operating, pre-financing	WACC	Enterprise value	Subtract net debt and other claims
FCFE	After interest and net borrowing	Cost of equity	Equity value directly	None
APV	Operating, pre-financing	Unlevered cost of capital (then add financing effects)	Enterprise value	Subtract net debt and other claims

Consistency is the cardinal rule: FCFF with WACC, FCFE with the cost of equity, never a mix. In theory, if the capital structure assumptions are the same, all three give the same equity value; in practice, they diverge because of the difficulty of modelling financing.

When should you prefer FCFE? For financial institutions, where debt is a raw material rather than a financing choice and "operating" cash flow cannot be separated from financing, equity-based methods (dividend discount, residual income, FCFE) are standard. For industrial companies with stable or targeted leverage, FCFF is usually easier because forecasting net borrowing requires a financing plan that may not exist.

21.1.4 The architecture of a DCF

A complete DCF has six steps: (1) understand the business and set a scenario; (2) forecast operating drivers for an explicit period; (3) translate the drivers into FCFF; (4) estimate the discount rate; (5) estimate a terminal value for the period beyond the forecast horizon; and (6) discount, sum, and bridge to equity value, then test the answer against other methods. We take these in turn, beginning with the forecast.

21.2 Forecasting Free Cash Flows

21.2.1 Forecast operating drivers, not line items

The most common weakness in practical models is that each line of the income statement is extrapolated separately: revenue grows 8 percent, costs grow 6 percent, capex grows 5 percent, and no one checks whether the implied economics make sense. A better discipline builds the forecast from a small set of **value drivers**:

1. **Revenue growth**, ideally decomposed into volume, price, and mix, or into market growth and market share.
2. **Operating margin** (EBIT or EBITDA), reflecting scale economies, input costs, and competitive pressure.
3. **Tax rate on operating income**, normally the marginal or long-run effective cash rate, not a temporarily low reported rate.
4. **Reinvestment**: capital expenditure net of depreciation, plus investment in net working capital, expressed relative to growth (a sales-to-capital ratio is convenient).

Then check outputs for plausibility: return on invested capital (ROIC), the implied sales-to-capital ratio, and the margin trajectory against peers. A forecast in which sales double with no net investment, or in which margins rise forever, fails the test.

21.2.2 The length of the explicit period

The explicit forecast should be long enough for the firm to reach a **steady state**: growth converging toward the long-run rate of the economy, margins stable, and returns on new investment settling at a sustainable level. For a mature industrial company, five years may suffice. For a high-growth firm, ten years or more may be needed; stopping at year five and capitalising a still-high growth rate is a classic way to overvalue.

21.2.3 Operating items and their traps

Taxes. Use taxes on EBIT, not taxes actually paid after the interest deduction: the interest tax shield is credited to the financing side, through the after-tax cost of debt in the WACC. Counting it in both the cash flows and the discount rate is double counting.

Depreciation and capex. In the terminal year, capex must be at least roughly equal to depreciation for a firm that is growing; otherwise the model assumes the asset base shrinks while revenue grows.

Working capital. Model operating working capital (receivables, inventory, less payables), excluding cash and short-term debt. Growth consumes working capital; ignoring it flatters early cash flows.

Leases and other debt-like items. Under IFRS 16 and similar standards, lease liabilities appear as debt and lease costs are split into depreciation and interest, which raises EBITDA. If you use post-lease EBITDA in the cash flows, treat lease liabilities as debt in the bridge and make the multiples consistent. Mixing pre-lease EBITDA with post-lease debt is a common error.

Stock-based compensation. It is a real cost. Excluding it from operating expense without adjusting the share count overstates value.

Example 21.1 – Forecasting FCFF for Halden Precision Components. Halden has current revenue of 800 (all figures in millions). Management plans revenue growth of 10%, 9%, 8%, 7%, and 6% over the next five years. EBIT margin is 18% of revenue, the tax rate is 25%, depreciation and amortisation (D&A) is 4% of revenue, capex is 6% of revenue, and net working capital absorbs 10% of each year’s incremental revenue.

Step 1 – Revenue. Year 1: 800 × 1.10 = 880.0. Year 2: 880.0 × 1.09 = 959.2. Year 3: 959.2 × 1.08 = 1,035.9. Year 4: 1,035.9 × 1.07 = 1,108.5. Year 5: 1,108.5 × 1.06 = 1,175.0.

Step 2 – NOPAT. EBIT = 18% of revenue; NOPAT = EBIT × 0.75. Year 1: EBIT = 158.4, NOPAT = 118.8. Year 2: EBIT = 172.7, NOPAT = 129.5. Year 3: EBIT = 186.5, NOPAT = 139.9. Year 4: EBIT = 199.5, NOPAT = 149.6. Year 5: EBIT = 211.5, NOPAT = 158.6.

Step 3 – Reinvestment items. D&A = 4% of revenue; capex = 6%; ΔNWC = 10% × (revenue change). Year 1: D&A 35.2, capex 52.8, ΔNWC = 0.10 × 80.0 = 8.0. Year 2: D&A 38.4, capex 57.6, ΔNWC = 0.10 × 79.2 = 7.9. Year 3: D&A 41.4, capex 62.2, ΔNWC = 0.10 × 76.7 = 7.7. Year 4: D&A 44.3, capex 66.5, ΔNWC = 0.10 × 72.6 = 7.3. Year 5: D&A 47.0, capex 70.5, ΔNWC = 0.10 × 66.5 = 6.7.

Step 4 – FCFF = NOPAT + D&A − Capex − ΔNWC. Year 1: 118.8 + 35.2 − 52.8 − 8.0 = 93.2. Year 2: 129.5 + 38.4 − 57.6 − 7.9 = 102.4. Year 3: 139.9 + 41.4 − 62.2 − 7.7 = 111.5. Year 4: 149.6 + 44.3 − 66.5 − 7.3 = 120.2. Year 5: 158.6 + 47.0 − 70.5 − 6.7 = 128.5.

Year	Revenue	EBIT	NOPAT	D&A	Capex	ΔNWC	FCFF
1	880.0	158.4	118.8	35.2	52.8	8.0	93.2
2	959.2	172.7	129.5	38.4	57.6	7.9	102.4
3	1,035.9	186.5	139.9	41.4	62.2	7.7	111.5
4	1,108.5	199.5	149.6	44.3	66.5	7.3	120.2
5	1,175.0	211.5	158.6	47.0	70.5	6.7	128.5

Sanity check. Net reinvestment in year 5 is capex − D&A + ΔNWC = 70.5 − 47.0 + 6.7 = 30.2, a reinvestment rate of 30.2 / 158.6 = 19.0% of NOPAT while revenue is still growing 6%. We will use that observation in section 21.4.

Notice that FCFF grows faster than revenue in this plan, partly because the working capital drag falls as growth slows. A reviewer should ask whether the asset base is growing fast enough to support the sales plan: cumulative net capex (capex minus D&A) over five years is only about 103, plus about 38 of working capital, to support an increase in revenue of 375.

21.2.4 Scenarios and the role of judgement

A single point forecast conveys false precision. Practitioners therefore build a base case and two or three scenarios (upside, downside, and sometimes a “stress” case), attach rough probabilities, and examine how value depends on the principal uncertain driver, whether it is a commodity price, regulation, or customer concentration. The point is not to produce the average of the scenarios, though that can be useful, but to find which assumptions the valuation depends on, so management can target diligence at them.

21.3 The Discount Rate: WACC Done Properly

21.3.1 Definition

The weighted average cost of capital blends the required returns of the providers of capital at market-value weights:

```
WACC = (E/V) × Ke + (D/V) × Kd × (1 − t)
```

where E and D are the market values of equity and debt, V = E + D, Ke is the cost of equity, Kd is the pre-tax cost of debt, and t is the tax rate at which interest is deductible. The cost of equity is typically estimated with the capital asset pricing model (Sharpe, Lintner):

```
Ke = Rf + βL × ERP
```

where Rf is the risk-free rate, ERP the equity risk premium, and βL the levered equity beta.

21.3.2 Choices that matter

Risk-free rate. Match the currency and the duration of the cash flows. For a long-lived business valued in a given currency, use a long-term government yield in that currency. A rate that is temporarily depressed by monetary policy needs a considered decision: either normalise it and the ERP together, or retain the market rate and a market-consistent ERP.

Equity risk premium. There is no single correct number; providers and academics publish estimates that differ by a percentage point or more. Whatever you choose, apply it consistently across all valuations in your firm, and report the sensitivity.

Beta. A single regression beta for the company is noisy. The recommended approach is a **bottom-up beta**: collect the betas of comparable firms, unlever each to remove the effect of its capital structure, take the median, and relever at the target capital structure of the firm being valued. With debt beta approximately zero and a tax shield on debt, the standard formulae are:

```
βU = βL / [1 + (1 − t) × D/E]
βL = βU × [1 + (1 − t) × D/E]
```

Weights. Use the **target** capital structure measured in market values, not the book values on the balance sheet, and not today's actual structure if the firm is expected to move toward a different one. Using book weights when the market value of equity is far above book understates the cost of capital.

Cost of debt. Use the current marginal yield the firm would pay on new long-term borrowing, not the historical coupon on existing debt.

Example 21.2 – Estimating Halden's WACC. Inputs: risk-free rate 4.0%, equity risk premium 5.5%, industry unlevered beta 0.95 (median of unlevered peer betas), target debt-to-capital 25% (so D/E = 25/75 = 0.3333), pre-tax cost of debt 6.0%, tax rate 25%.

Step 1 – Relever the beta. βL = 0.95 × [1 + 0.75 × 0.3333] = 0.95 × 1.25 = 1.1875.

Step 2 – Cost of equity. Ke = 4.0% + 1.1875 × 5.5% = 4.0% + 6.531% = 10.531%.

Step 3 – After-tax cost of debt. 6.0% × (1 − 0.25) = 4.5%.

Step 4 – WACC. 0.75 × 10.531% + 0.25 × 4.5% = 7.898% + 1.125% = 9.02%, which we round to **9.0%** for the valuation.

Contrast – a common error. If an analyst used the unlevered cost of equity (4.0% + 0.95 × 5.5% = 9.225%) with the 75/25 weights, the result would be 0.75 × 9.225% + 0.25 × 4.5% = 6.919% + 1.125% = 8.04%. A full percentage point of understatement arises from forgetting that equity becomes riskier when the company borrows. As the next section shows, one percentage point of WACC moves Halden's value by roughly 14 to 20 percent.

21.3.3 The WACC is a rate for a given risk, not for the company's mood

The WACC applies to cash flows with the same risk as the company's typical operations. A division with different risk needs its own rate; an acquisition into a different industry needs a rate for the target's business, not the buyer's WACC. Using a single corporate WACC for all projects systematically overinvests in risky divisions and underinvests in safe ones.

21.3.4 Constant capital structure and the circularity problem

WACC-based valuation has a subtle circularity: the weights depend on value, and value depends on the WACC. In practice, one assumes a target ratio of debt to value and iterates, or uses the target ratio directly. If the capital structure is expected to change considerably over the forecast, a single WACC is inadequate and the adjusted present value approach (section 21.5.4) is preferable.

21.4 Terminal Value

21.4.1 Why terminal value dominates

A company is not expected to stop operating after the explicit forecast, so we need a **terminal value** (TV): the value, at the end of the forecast, of all cash flows thereafter. For most going concerns, the terminal value accounts for 60 to 85 percent of enterprise value. That is not a defect of DCF; it reflects that most of a firm's value lies in the far future. But it means that

terminal assumptions deserve more scrutiny than the year-by-year forecast, and that a valuation presentation which hides the terminal value share is incomplete.

21.4.2 The Gordon growth formula

If FCFF grows at a constant rate g forever, starting from the first terminal year (year N+1):

```
TV_N = FCFF_(N+1) / (WACC − g)
```

The growth rate g must be less than the WACC and, for a perpetual rate, not greater than the long-run nominal growth of the economy in which the firm operates; otherwise the firm would eventually become larger than the economy. A common range for developed-market nominal growth is the rate of expected long-run inflation plus a small real component. Using a perpetual g of 5 or 6 percent in a developed economy is almost always a mistake, however strong the business.

21.4.3 The terminal cash flow must be consistent with the growth rate

The most subtle error is to take the final forecast-year FCFF and multiply by (1 + g) without asking whether that cash flow is consistent with growth of g. Growth requires reinvestment. If the final forecast year still reflects the reinvestment pattern of a faster-growth phase (or conversely, the capex of a firm in decline), the terminal FCFF is wrong. The **value-driver formula** repairs this by linking reinvestment to growth and returns:

```
Reinvestment rate = g / ROIC
FCFF_(N+1) = NOPAT_(N+1) × (1 − g / ROIC)
TV_N = NOPAT_(N+1) × (1 − g / ROIC) / (WACC − g)
```

where ROIC is the return on new invested capital in perpetuity. Two insights follow. First, growth creates value only if ROIC exceeds WACC; if ROIC = WACC, the terminal value equals NOPAT/WACC regardless of g (growth is value-neutral). Second, forecasting a high perpetual growth rate together with high perpetual ROIC assumes the firm keeps finding opportunities with excess returns forever, which in competitive markets competition tends to erode. A conservative terminal assumption sets ROIC on new investment close to WACC, while allowing the existing business to keep earning its return.

21.4.4 Exit multiple method

An alternative is to apply a multiple, typically EV/EBITDA, to the final-year metric:

```
TV_N = Multiple × EBITDA_N
```

The attraction is that it is anchored in market observation. The weakness is that today's multiple embeds today's growth and interest-rate environment, and it may be unrepresentative of a steady-state firm five years from now. The exit multiple also sneaks market sentiment into an "intrinsic" model. Every exit multiple should therefore be converted into an **implied perpetual growth rate**, and every Gordon terminal value into an **implied multiple**, and the pair inspected for reasonableness.

21.4.5 Mid-year convention

If cash flows arrive evenly through the year rather than all at year-end, discounting each year's FCFF by (1 + WACC)^(t − 0.5) is more accurate. Under the Gordon approach, the terminal value also shifts by half a year. Ignoring this understates value by roughly half a year of the discount rate, a few percent. Whichever convention you use, apply it consistently, and state it.

Example 21.3 – Discounting Halden's cash flows and terminal value. Use the FCFF from Example 21.1, a WACC of 9.0% (Example 21.2), perpetual growth g = 3.0%, and a long-run ROIC on new investment of 12%.

Step 1 – Discount factors at 9%. Year 1: 1/1.09 = 0.9174. Year 2: 0.8417. Year 3: 0.7722. Year 4: 0.7084. Year 5: 0.6499.

Step 2 – Present value of explicit FCFF. 93.2 × 0.9174 = 85.5; 102.4 × 0.8417 = 86.2; 111.5 × 0.7722 = 86.1; 120.2 × 0.7084 = 85.2; 128.5 × 0.6499 = 83.5. Sum = **426.4**.

Step 3 – Terminal NOPAT. Year-6 revenue = 1,175.0 × 1.03 = 1,210.2. NOPAT = 1,210.2 × 0.18 × 0.75 = 163.4.

Step 4 – Terminal reinvestment and FCFF. Reinvestment rate = 3% / 12% = 25.0%. $FCFF_6$ = 163.4 × (1 − 0.25) = 122.5.

Step 5 – Terminal value at end of year 5. TV = 122.5 / (0.09 − 0.03) = 122.5 / 0.06 = **2,042.2**.

Step 6 – Discount the terminal value. PV(TV) = 2,042.2 × 0.6499 = **1,327.3**.

Step 7 – Enterprise value. EV = 426.4 + 1,327.3 = **1,753.7**. The terminal value is 1,327.3 / 1,753.7 = **75.7%** of EV.

Step 8 – Implied multiples. Year-5 EBITDA = 211.5 + 47.0 = 258.5, so the Gordon TV equals 2,042.2 / 258.5 = **7.9×** EBITDA. The EV relative to current (LTM) EBITDA of 800 × (0.18 + 0.04) = 176 is 1,753.7 / 176 = 10.0×.

Step 9 – Mid-year convention. Discounting each FCFF by (1.09)^(t − 0.5) gives a present value of explicit cash flows of 445.2, and a PV of the terminal value of 2,042.2 / 1.09^4.5 = 1,385.7; EV = **1,830.9**, about 4.4% higher.

Sensitivity of EV to WACC and g.

WACC g	2.0%	3.0%	4.0%
8.0%	1,967.7	2,106.2	2,309.6
9.0%	1,678.2	1,753.7	1,856.0
10.0%	1,461.4	1,501.9	1,553.1

(Each cell recomputes the terminal value with the value-driver formula, so the reinvestment rate moves with g.) The table shows the asymmetry typical of DCF: lowering the WACC by one point adds 352 (20%), raising it subtracts 252 (14%), and the value is more sensitive to WACC than to g in this range because higher g also requires more reinvestment.

Example 21.4 – What growth is worth: the value-driver formula. A mature business has next-year NOPAT of 100 and a WACC of 9%. Compare perpetual growth of 3% under different returns on new capital.

No growth. TV = 100 / 0.09 = 1,111.1 (all NOPAT is paid out).

ROIC = 9% (equal to WACC). Reinvestment = 3/9 = 33.3%; FCFF = 66.7; TV = 66.7 / 0.06 = 1,111.1. Growth adds nothing.

ROIC = 12%. Reinvestment = 25%; FCFF = 75; TV = 75 / 0.06 = 1,250.0, an increase of 12.5% over no growth.

ROIC = 20%. Reinvestment = 15%; FCFF = 85; TV = 85 / 0.06 = 1,416.7, an increase of 27.5%.

ROIC = 6% (below WACC). Reinvestment = 50%; FCFF = 50; TV = 50 / 0.06 = 833.3, a 25% **loss** relative to no growth: growing destroys value when the returns on new capital are below the cost of capital.

ROIC on new capital	Reinvestment rate	FCFF (year 1)	Terminal value	vs. no growth
No growth	0%	100.0	1,111.1	n/a
6%	50.0%	50.0	833.3	−25.0%
9%	33.3%	66.7	1,111.1	0.0%
12%	25.0%	75.0	1,250.0	+12.5%
20%	15.0%	85.0	1,416.7	+27.5%

The managerial lesson is that growth in itself is not a source of value. A budget presentation that boasts of growth must be accompanied by evidence of returns above the cost of capital.

Example 21.5 – Checking an exit multiple against the Gordon model. Suppose the bank prefers an exit multiple of 9.0× year-5 EBITDA instead of the Gordon terminal value.

Step 1 – Terminal value. TV = 9.0 × 258.5 = 2,326.4.

Step 2 – Present value and EV. PV(TV) = 2,326.4 × 0.6499 = 1,512.0; EV = 426.4 + 1,512.0 = **1,938.4**, which is 184.7 (10.5%) above the Gordon valuation.

Step 3 – Implied perpetual growth, simple version. Solve 128.5 × (1 + g) / (0.09 − g) = 2,326.4. This gives g = 3.3%, which looks acceptable.

Step 4 – Implied growth, value-driver version. Solve 158.6 × (1 + g) × (1 − g/0.12) / (0.09 − g) = 2,326.4. The solution is g ≈ 4.6%, requiring a reinvestment rate of 4.6% / 12% = 38.4%. This is above the 3% long-run growth and well above the 19% reinvestment of year 5.

Interpretation. The simple check was misleading because year-5 FCFF reflects a low reinvestment rate. Once reinvestment is tied to growth, the 9.0× exit multiple requires growth of about 4.6% forever, hardly credible. A multiple of 7.9× is consistent with 3% growth and 12% returns on new capital. The right course is to challenge the 9.0× multiple, not to accept it because peers trade there today.

21.5 From Enterprise Value to Equity Value; FCFE; and APV

21.5.1 The equity bridge

Once the enterprise value is known, the equity value is obtained by subtracting claims senior to the shareholders and adding assets whose cash flows were not in FCFF:

- **Debt** at market value where available (book value is typically a reasonable proxy for floating-rate or recently issued debt; for fixed-rate debt issued long ago, the difference can be material).
- **Lease liabilities**, if the cash flows exclude lease payments.
- **Preferred stock and non-controlling interests** at market value or an earnings-multiple estimate.
- **Unfunded pension deficits and other debt-like obligations**, net of tax shields.
- **Cash and marketable securities in excess of operating needs**, **equity investments in associates** (valued separately because their income is excluded from EBIT), and other non-operating assets.

Be careful about what “cash” is. Cash that the business needs to operate (minimum cash balances, cash trapped in foreign jurisdictions with repatriation taxes) should not be added at face value.

21.5.2 Per-share value and dilution

Divide equity value by the **diluted** share count. Options and warrants are dealt with using the treasury stock method (TSM) as an approximation, or better, by valuing them with an option model and subtracting their value from equity before dividing by basic shares. The TSM is circular, because the in-the-money status depends on the share price you are trying to determine, so it is solved by iteration.

Example 21.6 – Bridge to a per-share value. From Example 21.3, EV = 1,753.7. Halden has debt of 250 and cash of 40, and 50.0 million basic shares. There are also 4.0 million options with a strike of 20.

Step 1 – Equity value. 1,753.7 − 250 + 40 = **1,543.7**.

Step 2 – Basic per-share value. 1,543.7 / 50.0 = 30.87.

Step 3 – TSM dilution by iteration. Start with price P = 30.87: net new shares = 4.0 × (1 − 20/30.87) = 1.41; diluted shares = 51.41; P = 1,543.7 / 51.41 = 30.03. Second pass: net new shares = 4.0 × (1 − 20/30.03) = 1.34; diluted shares = 51.34; P = 1,543.7 / 51.34 = 30.07. A third pass changes nothing material. The value per diluted share is **30.07**.

Interpretation. The options reduce value per share by 0.80 (2.6%), about 40 in equity value transferred to optionholders. Ignoring them would have overstated the per-share value, and the offer of 28 would have looked more clearly inadequate than it actually is.

21.5.3 FCFE valuation

Under the FCFE approach, we discount the cash flows available to shareholders at the cost of equity. It is natural when the firm's debt policy is a stated target, when leverage is expected to change, or for financial firms. Two points deserve emphasis: (1) the cost of equity should reflect the leverage assumed in the cash flows, and (2) the terminal FCFE must reflect the reinvestment needed to support growth, financed partly by borrowing at a consistent debt ratio.

Example 21.7 – FCFE valuation of a small leveraged firm. Forecasts for a company (millions) are in the table below. The cost of equity is 11%, terminal growth is 3%, and return on equity on new investment is 14%.

Year	Net income	D&A	Capex	ΔNWC	Net borrowing	FCFE
1	60	20	30	5	10	55
2	66	21	31	5	10	61
3	72	22	32	5	8	65
4	77	23	33	4	6	69
5	82	24	34	4	5	73

Step 1 – FCFE. Year 1: 60 + 20 − 30 − 5 + 10 = 55. Year 2: 66 + 21 − 31 − 5 + 10 = 61. Year 3: 72 + 22 − 32 − 5 + 8 = 65. Year 4: 77 + 23 − 33 − 4 + 6 = 69. Year 5: 82 + 24 − 34 − 4 + 5 = 73.

Step 2 – PV at 11%. 55/1.11 = 49.55; 61/1.11^2 = 49.51; 65/1.11^3 = 47.53; 69/1.11^4 = 45.45; 73/1.11^5 = 43.32. Sum = **235.4**.

Step 3 – Terminal FCFE. Net income in year 6 = 82 × 1.03 = 84.5. Equity reinvestment rate = g/ROE = 3% / 14% = 21.4%. $FCFE_6$ = 84.5 × (1 − 0.2143) = 66.4.

Step 4 – Terminal value. TV = 66.4 / (0.11 − 0.03) = 829.5. PV = 829.5 / 1.11^5 = 829.5 / 1.6851 = 492.3.

Step 5 – Equity value. 235.4 + 492.3 = **727.6** (million). The terminal value is 67.7% of the total.

Note that the equity reinvestment rate combines retained earnings and net borrowing; applying the formula to FCFE implicitly assumes a stable debt ratio.

21.5.4 Adjusted present value (APV)

APV, developed from the Modigliani-Miller propositions, separates the operating value from the value of financing side effects:

```
APV = Value of unlevered firm (FCFF discounted at the unlevered cost of capital, Ku)
    + PV of interest tax shields
    − PV of expected financial distress and agency costs
```

It is preferable when (a) capital structure will change materially, as in an LBO or a heavily leveraged acquisition that delevers over time; (b) the debt policy is a fixed schedule rather than a target ratio; (c) there are other financing side effects such as subsidised loans; or (d) one wants to see explicitly how much value comes from the tax shield. The central choice is the discount rate for the tax shields. If debt is fixed in amount and thought to be as safe as the debt itself, discount at the cost of debt (Modigliani-Miller); if debt is rebalanced to a constant proportion of value, shields have the risk of operating assets and are discounted at Ku (Miles-Ezzell, Harris-Pringle). The correct choice depends on the financing policy, and the resulting difference in value is often modest compared with the uncertainty in cash flows.

Example 21.8 – APV versus WACC for a stable business. A company has next-year FCFF of 100 growing at 2% forever, an unlevered cost of capital Ku = 9%, a pre-tax cost of debt of 6%, and a tax rate of 25%.

Step 1 – Unlevered value. VU = 100 / (0.09 − 0.02) = **1,428.6**.

Step 2 – Case A: permanent fixed debt of 500. The tax shield is perpetual, so its present value (discounted at the cost of debt) is t × D = 0.25 × 500 = **125.0**. Suppose also that the expected present value of distress costs equals a 4% probability of losing 20% of unlevered value: 0.04 × 0.20 × 1,428.6 = **11.4**. APV = 1,428.6 + 125.0 − 11.4 = **1,542.1**.

Step 3 – Case B: debt rebalanced to 35% of value. Under the Harris-Pringle assumption, WACC = Ku − t × Kd × D/V = 9% − 0.25 × 6% × 0.35 = 9% − 0.525% = 8.475%. Value = 100 / (0.08475 − 0.02) = 100 / 0.06475 = **1,544.4**. The implied debt is 0.35 × 1,544.4 = 540.5 and the value of the tax shield is 1,544.4 − 1,428.6 = 115.8.

Step 4 – Check the implied cost of equity. Equity value = 1,544.4 − 540.5 = 1,003.9. With debt at 6% and the Harris-Pringle formula: Ke = Ku + (Ku − Kd) × D/E = 9% + 3% × 540.5/1,003.9 = 10.62%. Then WACC = (1,003.9/1,544.4) × 10.62% + (540.5/1,544.4) × 4.5% = 6.900% + 1.575% = 8.475%, which reconciles exactly to the WACC used above.

Interpretation. The two cases give values within 0.2% of each other (1,542.1 vs 1,544.4), but they embody different assumptions: in Case A the benefit of leverage is certain but costly if distress occurs, whereas in Case B the tax shield scales with firm value. When a company intends to hold constant leverage, the WACC method and APV agree; when the debt path is irregular, APV is cleaner and easier to audit.

21.6 Relative Valuation and the Football Field

21.6.1 Trading comparables

The method has four steps: choose peers, compute their multiples, select a representative figure, and apply it to the target's corresponding metric. The most useful **enterprise-value multiples** are EV/EBITDA, EV/EBIT, and EV/Sales (for firms with temporarily depressed margins). The **equity multiples** are P/E and price-to-book (most useful for financial institutions). Rules for sound practice:

- **Match numerator and denominator.** EV multiples go with metrics before interest (EBITDA, EBIT, sales); equity multiples go with metrics after interest (net income, book equity). Dividing enterprise value by net income is

meaningless.

- **Clean the metrics.** Use recurring earnings, adjust for one-time items, align fiscal year-ends, and ensure consistent accounting for leases and stock compensation.
- **Choose peers on fundamentals**, not on industry code alone: growth, margins, returns on capital, risk, and size. Five good peers beat twenty poor ones.
- **Use the median or the harmonic mean**, because multiples are right-skewed and a mean is dragged up by outliers. Harmonic means are appropriate because they average the earnings yield (the reciprocal), which is the economically additive quantity.
- **Explain the differences.** A target that trades at a discount to peers should be evaluated for lower growth, higher risk, or worse governance, not mechanically revalued up.

21.6.2 Precedent transactions

Transaction multiples reflect prices paid for control, including a premium for synergies and for the right to direct the business. They are useful in an M&A context and typically exceed trading multiples. But deals cluster in booms, involve different financing conditions, and often have idiosyncratic strategic motives. Use a longer window, adjust for market conditions, and treat the range as indicative.

21.6.3 The regression approach

A more informative alternative to averaging is to regress a multiple (for instance EV/EBITDA) on its fundamental drivers (growth, margin, ROIC) across the peer set, then predict the target's multiple from its own drivers. This forces the question of why two firms trade differently, and exposes outliers.

Example 21.9 – Valuing Brindle Tools with trading comparables. Five peers provide the following data (millions): enterprise value (EV), EBITDA, net income (NI), and market capitalisation (Mkt cap).

Peer	EV	EBITDA	EV/EBITDA	Mkt cap	NI	P/E
A	2,400	300	8.00×	1,500	150	10.0×
B	3,100	360	8.61×	2,400	200	12.0×
C	1,500	150	10.00×	900	60	15.0×
D	5,200	560	9.29×	4,200	330	12.7×
E	1,800	240	7.50×	1,300	110	11.8×
Median			8.61×			12.0×
Mean			8.68×			12.3×

Brindle Tools has EBITDA of 220, net income of 130, debt of 300, and cash of 50 (net debt of 250).

Step 1 – EV/EBITDA. Low (7.5×): EV = 7.5 × 220 = 1,650; equity = 1,650 − 250 = 1,400. Median (8.61×): EV = 1,894.4 (8.6111 × 220); equity = 1,644.4. High (10.0×): EV = 2,200; equity = 1,950.

Step 2 – P/E. Low (10.0×): equity = 10.0 × 130 = 1,300. Median (12.0×): 1,560. High (15.0×): 1,950.

Step 3 – Reconcile. The median EV/EBITDA gives equity of 1,644, about 5% above the P/E-based 1,560. The gap reflects differences in leverage, depreciation, and tax between Brindle and its peers. A careful analyst would investigate rather than average the two blindly.

Step 4 – Harmonic mean check. The harmonic mean of the five EV/EBITDA multiples is 5 / (1/8.00 + 1/8.61 + 1/10.00 + 1/9.29 + 1/7.50) = 5 / 0.5822 = 8.59×, nearly identical to the median, so outliers are not distorting the result.

Conclusion. A fair equity range for Brindle on trading multiples is roughly 1,560 to 1,650, centered on about 1,600, with the overall peer range being much wider (1,300 to 1,950).

21.6.4 The football field

A **football field** chart summarises each method as a horizontal bar showing the low-to-high range of value per share (or EV), with a vertical line at the current price or offer. It is the standard tool in fairness opinions and board presentations. It conveys that valuation is a range, shows which methods are consistent, and highlights where an offer lies.

Example 21.10 – Football field for Halden. For Halden (net debt 210, 50 million shares), convert each method's EV range to a per-share range with (EV − 210) / 50. LTM EBITDA is 176.

Method	EV low	EV high	Per share low	Per share high
52-week trading range	1,510	1,810	26.0	32.0
Trading comps, 7.5×–10.0× LTM EBITDA	1,320	1,760	22.2	31.0
Precedents, 9.0×–11.5× LTM EBITDA	1,584	2,024	27.5	36.3
DCF, WACC 10%–8%, g = 3%	1,501.9	2,106.2	25.8	37.9
DCF, exit multiple 7.5×–9.5×	1,686.4	2,022.4	29.5	36.3

Check on one row. Comps: (1,320 − 210) / 50 = 22.2; (1,760 − 210) / 50 = 31.0. The median multiple of 8.61× gives EV = 1,515.6 and a per-share value of (1,515.6 − 210) / 50 = 26.1.

Reading the chart. The 28 offer lies inside every range except the exit-multiple DCF (low end 29.5), and it sits in the upper half of the trading comps and the lower third of the precedents and DCFs. The interpretation for the board: the offer is not a knockout, but it also offers a modest premium to the 26 median comps value and falls short of what control transactions have paid. A negotiating counter near the mid-point of the precedent range (about 32) is defensible on this evidence.

21.7 Valuing Start-ups and Young Companies

21.7.1 Why standard DCF struggles

Young firms often have negative earnings, no meaningful history, rapidly changing business models, and a high probability of failure. Standard DCF fails in three ways: (1) the cash flows in the explicit period are negative or tiny, so almost all the value is terminal; (2) the discount rate on a diversified-investor CAPM basis is far below the return venture investors need, as it ignores failure; and (3) there are valuable options (to expand, pivot, or abandon) that a single scenario ignores.

21.7.2 Approaches

Venture capital method. Estimate the exit value in a future year (using a multiple of revenue or earnings at that point), discount it at a high **target IRR** (often 30 to 50 percent for early-stage investments) to obtain the post-money value today, and back out the required ownership. The high rate is not a risk premium in the CAPM sense; it is a blend of diversifiable failure risk, illiquidity, the value added by the investor, and the optimism embedded in founders' projections.

Scenario or probability-weighted DCF. Specify distinct futures (failure, modest success, breakout), assign probabilities, value each, and take the expected value, discounting at a rate reflecting systematic risk only. This treats failure as a cash flow rather than burying it in the discount rate. It is conceptually cleaner and makes assumptions explicit.

Market-based methods. Multiples of revenue, users, or recurring revenue from comparable listed or privately financed companies. Recent financing rounds provide a reference but not an independent valuation, because they reflect negotiated terms (liquidation preferences, anti-dilution protections) that make the headline price different from the economic value of common shares.

Real options. Where the main value comes from the choice to scale up if the product proves itself, an options view is useful (see the chapter on real options), though in practice it often supplements rather than replaces the methods above.

21.7.3 Dilution and the preference stack

Founders and investors should translate a valuation into ownership only after allowing for future funding rounds and the option pool. Preferred shares carry liquidation preferences that shift the proceeds in downside outcomes, so the value of a common share is lower than the post-money valuation divided by total shares.

Example 21.11 – The venture capital method. A start-up seeks 8 million. Management projects year-5 revenue of 60 million; comparable companies are acquired at 4× revenue. The investor requires a 40% IRR, and expects future rounds to dilute existing holders by 25% before exit.

Step 1 – Exit value. 60 × 4 = 240 million.

Step 2 – Post-money value today. 240 / 1.40^5 = 240 / 5.3782 = **44.6 million**.

Step 3 – Ownership required at exit. 8 / 44.6 = 17.9%.

Step 4 – Adjust for dilution. Ownership needed today = 17.9% / (1 − 0.25) = **23.9%**.

Step 5 – Implied valuations today. Post-money = 8 / 0.239 = 33.5 million; pre-money = 33.5 − 8 = **25.5 million**.

Probability-weighted cross-check. Suppose the outcomes at year 5 are: failure (50%, value 0), modest success (35%, 120), and breakout (15%, 480). Expected exit value = 0 + 0.35 × 120 + 0.15 × 480 = 42 + 72 = 114. Discounted at a 15% rate reflecting a moderately high systematic risk: 114 / 1.15^5 = 114 / 2.0114 = 56.7 million.

Interpretation. The venture capital method's 44.6 million is below the scenario value of 56.7 million, because the 40% target return partly compensates for optimism in the management case, which already assumes success. The two approaches answer different questions, and the negotiated price lies between them. The founders should note that the 25.5 million pre-money implied by the VC method rests on a single success scenario; if they argue for a higher valuation, they must argue for the scenario weights.

21.8 Common Errors in Valuation

The most expensive valuation mistakes are rarely arithmetical. They are conceptual, and they tend to push the answer in the direction that the author or the client prefers. A checklist for review:

1. **Inconsistent numerator and denominator.** FCFF discounted at the cost of equity; FCFE discounted at WACC; EV/EBITDA applied to give equity value without subtracting net debt; P/E applied to EBITDA.
2. **Terminal growth above the economy's growth, or reinvestment inconsistent with growth.** The value-driver formula (Example 21.4) is the remedy.
3. **Terminal value overwhelms the answer.** If terminal value is 85 percent or more of EV, the explicit period is too short or growth in the final year is not yet at steady state.

4. **Book-value weights in WACC**, historical coupon as the cost of debt, or a company-wide WACC applied to a project in a different risk class.
5. **Tax-shield double counting.** Using after-tax cost of debt in the WACC and also including interest tax shields in cash flows (or using taxes after interest in FCFF).
6. **Mismatched currency, inflation, and time.** Nominal cash flows in a high-inflation currency discounted at a real or dollar rate, or the reverse.
7. **Circular reasoning from multiples.** Choosing the exit multiple from current peers, and then using it to "confirm" the DCF against the same peers.
8. **Cash and debt-like items.** Adding all cash, even trapped cash or operating cash; forgetting pensions, leases, restructuring provisions, litigation, or minority interests.
9. **Mid-year and timing errors.** Valuation dates that do not match balance-sheet dates for net debt; forgetting to roll the discount factors forward.
10. **False precision and anchoring.** Reporting a value to four significant figures from a model whose inputs are good to one; adjusting assumptions until the DCF equals the price somebody wants.
11. **Ignoring the control premium and minority discounts consistently.** DCF with management's plan values a firm under control; trading multiples value minority stakes; do not compare them as if they were the same concept.
12. **Ignoring option value and dilution.** Using basic rather than diluted shares, and neglecting convertible debt.

The remedy is largely procedural: reconcile the DCF to multiples, document every input, test the extreme cases, and ask what the market must believe to justify today's price. Reverse engineering the price is often the most informative exercise: solve for the growth or margin that justifies the current share price, and judge whether that expectation is plausible.

Case Study: Valuing Corvane Logistics for a Management Buy-in

Illustrative scenario; Corvane Logistics and its executives are fictional.

Corvane Logistics is a regional freight and warehousing company owned by a founding family that wishes to sell. It has revenue of 620 million, EBITDA of 93 million (15.0 percent margin), net debt of 210 million, and 40 million shares. An external management team, backed by a financial sponsor, is considering a buy-in and has asked its adviser, Ilse Brandt, to value the company and to recommend a bid.

Ilse builds a five-year plan. Revenue grows at 7 percent annually, driven by e-commerce volumes at three customers who together account for 41 percent of sales. Management believes EBITDA margin can expand from 15.0 to 17.5 percent through route optimisation and warehouse automation. Capex rises temporarily to 7 percent of revenue to fund automation (depreciation is 4.5 percent of revenue) and falls to 5.5 percent by year five. Tax is 26 percent. The adviser's base-case WACC is 8.6 percent, built from an unlevered beta of 0.80, a 5.5 percent equity risk premium, a 4 percent risk-free rate, and a target debt-to-capital ratio of 40 percent, reflecting what the sponsor expects to be able to raise.

The base-case DCF, with 2.5 percent terminal growth and an assumed ROIC of 11 percent on new capital, gives an enterprise value of 1,070 million, or 11.5 times LTM EBITDA. The terminal value is 82 percent of that total. Listed peers trade at 8.0 to 9.5 times EBITDA, with a median of 8.6 times; recent transactions in the sector were done at 9.5 to 11.0 times. The family's advisers have signalled an expectation of at least 11 times.

Ilse notices three things. First, the automation capex is a one-time bump, but the terminal reinvestment assumption (capex of 5.5 percent against depreciation of 4.5 percent) implies growth in the invested capital base of just 1 percent of revenue per year, which might be too low to support 2.5 percent growth. Second, the margin expansion produces more than 40 percent of the value gap between the DCF and the comps; if the margin stays at 15 percent, the DCF falls toward 9.5 times. Third, the concentration of three customers makes the value highly sensitive to the loss of any one, a risk not captured by the discount rate.

The sponsor's leverage plan envisages paying down debt from 40 percent to 25 percent of capital over five years. Ilse realises that a single WACC of 8.6 percent therefore overstates the value of the interest tax shield in later years and understates the cost of equity in the early years; she reruns the model as an APV, which reduces the value by about 3 percent. She also builds a downside scenario in which the largest customer (14 percent of revenue) leaves in year two, producing an EV of about 880 million (9.5 times LTM EBITDA).

Her recommendation is to bid at around 10 times LTM EBITDA (930 million) with an earn-out of up to 60 million payable if the margin target is achieved, rather than to pay the 11 times the sellers expect. The sponsor's partners are divided: one argues that competition from strategic bidders will force the price to 11 times anyway; another wants the earn-out structured on customer retention rather than margin.

Discussion Questions

1. Which of the three observations by Ilse (terminal reinvestment, margin dependence, customer concentration) would you test first, and how would you adjust the model to reflect each?
2. Why does an APV approach reduce the value when leverage is expected to fall, and when would a single WACC still be adequate?
3. Evaluate the proposed bid of 10 times LTM EBITDA with an earn-out against the football field (comps 8.0 to 9.5×, precedents 9.5 to 11.0×, DCF 11.5×, downside 9.5×). What does the structure of the offer say about who bears the risk of the margin plan?
4. The family's advisers anchor on 11 times. Construct a counter-argument using the sensitivity of EV to margin and WACC, and propose what information the sponsor should ask for in due diligence.

Key Takeaways

- Enterprise value belongs to all capital providers; equity value is what remains after debt-like claims. Match cash flow to discount rate: FCFF with WACC, FCFE with the cost of equity.
- Forecast value drivers (growth, margin, taxes, reinvestment), then check the outputs for ROIC, sales-to-capital, and margin plausibility, rather than extrapolating line items.
- Estimate WACC with a bottom-up relevered beta, target market-value weights, and a marginal cost of debt; the typical error of using an unlevered cost of equity with debt weights understates the cost of capital.
- Terminal value usually exceeds 60 percent of enterprise value, so it merits more scrutiny than the explicit period. Perpetual growth must not exceed long-run economic growth.
- Growth creates value only when ROIC exceeds WACC; the value-driver formula ties terminal reinvestment to growth and returns.
- Every exit multiple implies a perpetual growth rate, and every Gordon terminal value implies a multiple; compare the two to detect inconsistent assumptions.

- APV is preferable to WACC when leverage will change, when debt follows a schedule, or when financing side effects are material.
- Comparables work only when peers resemble the target in growth, risk, and returns; use median or harmonic averages, and always explain differences instead of averaging them away.
- A football field communicates that valuation is a range; disagreement among methods is diagnostic, not a failure.
- Start-ups require scenario thinking: treat failure as a cash flow, use the VC method as a negotiating reference, and account for dilution and preferences.

Review Questions

1. Explain why a multiple is a "compressed DCF" and what assumptions you import when you apply a peer's EV/EBITDA to your company.
2. Why must FCFF be taxed on EBIT rather than on earnings after interest? What would be double counted otherwise?
3. Describe how to build a bottom-up beta. Why is it preferred to a regression beta for a single company?
4. Why should the WACC use target market-value weights rather than book weights?
5. Under what conditions is the Gordon growth terminal value invalid or misleading? Give two examples.
6. Derive the value-driver formula and explain why growth is value-neutral when ROIC equals WACC.
7. Explain how you would convert an exit multiple to an implied perpetual growth rate and why this check is important.
8. When is FCFE preferred to FCFF? Name an industry in which FCFF is conceptually problematic.
9. Compare the Modigliani-Miller and Harris-Pringle treatments of the discount rate for interest tax shields. How does the financing policy determine which applies?
10. Why is the venture capital method's target IRR much higher than the CAPM cost of equity for a start-up, and what are the weaknesses of that approach?

Problems

1. **FCFF.** A firm has next-year revenue of 500, EBIT margin of 20%, a tax rate of 30%, D&A of 25, capex of 40, and an increase in net working capital of 10. Compute FCFF.
2. **WACC.** The risk-free rate is 3.5%, the ERP 5.0%, the unlevered beta 0.80, the target D/E 0.5, the pre-tax cost of debt 5.5%, and the tax rate 25%. Compute the relevered beta, cost of equity, and WACC.
3. **Gordon DCF.** FCFF is 60 in year 1 and grows at 4% forever. The WACC is 10%. Compute enterprise value. If net debt is 150 and there are 30 million shares, compute the value per share.
4. **Two-stage DCF.** FCFF for years 1 to 3 is 40, 46, and 52. After year 3, FCFF grows at 3% forever. WACC = 9%. Compute enterprise value.
5. **Value driver.** NOPAT next year is 200, WACC = 8%, g = 2%. Compute the terminal value if ROIC on new investment is (a) 8% and (b) 16%. What is the percentage difference?
6. **Implied growth.** Year-5 EBITDA is 150, and the exit multiple is 8×. The year-5 FCFF is 70, WACC = 9%. Find the implied perpetual growth rate using TV = FCFF × (1 + g)/(WACC − g).
7. **Multiples.** A target has EBITDA of 80, net income of 30, debt of 200, and cash of 20. Peers' median EV/EBITDA is 9.0× and median P/E is 14×. Compute implied equity value from both methods and comment.

8. **VC method.** An investor puts in 5 million, expects an exit value of 150 million in 4 years, requires a 35% IRR, and expects 20% future dilution. Compute the required ownership today and the pre-money valuation.

Answers to Selected Problems

1. EBIT = 100; NOPAT = 70; FCFF = 70 + 25 − 40 − 10 = **45**.
2. βL = 0.80 × [1 + 0.75 × 0.5] = 1.10. Ke = 3.5% + 1.10 × 5.0% = 9.0%. D/V = 0.5/1.5 = 33.3%; E/V = 66.7%. WACC = 0.6667 × 9.0% + 0.3333 × 5.5% × 0.75 = 6.0% + 1.375% = **7.375%**.
3. EV = 60 / (0.10 − 0.04) = **1,000**. Equity = 850; per share = 850 / 30 = **28.33**.
4. PV of years 1–3: 40/1.09 + 46/1.1881 + 52/1.295 = 36.70 + 38.72 + 40.15 = 115.57. TV = 52 × 1.03 / 0.06 = 892.67; PV = 892.67 / 1.295029 = 689.30. EV = **804.9**.
5. a. TV = 200 × (1 − 0.02/0.08) / 0.06 = 150/0.06 = **2,500**. (b) TV = 200 × (1 − 0.125) / 0.06 = 175/0.06 = **2,916.7**. Difference = +16.7%.
6. TV = 8 × 150 = 1,200. Solve 70(1 + g) = 1,200(0.09 − g): 70 + 70g = 108 − 1,200g; 1,270g = 38; g = **2.99%**.
7. EV method: 9.0 × 80 = 720; equity = 720 − 180 = **540**. P/E method: 14 × 30 = **420**. The EV/EBITDA value is higher by 120 (29%), which suggests the target is more heavily burdened by depreciation, interest, or tax than its peers; examine why before choosing a value.
8. Exit post-money today = 150 / 1.35^4 = 150 / 3.3215 = 45.16. Ownership at exit = 5 / 45.16 = 11.07%; today = 11.07% / 0.80 = **13.84%**. Post-money = 5 / 0.1384 = 36.13; pre-money = **31.1 million**.

Chapter 22: Mergers and Acquisitions

Learning Objectives

After studying this chapter, you should be able to:

- Distinguish value-creating motives for mergers from motives that benefit managers or advisers at the expense of shareholders.
- Value operating and financial synergies, net of integration costs, and compute the maximum premium a buyer can pay without destroying value.
- Decompose the division of merger gains between acquirer and target shareholders under cash and stock consideration, and explain how stock consideration shares valuation risk.
- Calculate EPS accretion and dilution, break-even synergies, exchange ratios and collar mechanics, and explain why EPS accretion is not a test of value creation.
- Describe tender offers, proxy contests, and takeover defences, and analyse the effect of a poison pill on a bidder.
- Explain the principal reasons integration fails, and summarise how antitrust review measures concentration and what remedies are available.

Opening Vignette

Illustrative scenario. Aventra Foods and Lumio Beverages are fictional.

Aventra Foods, a packaged-foods group with a market capitalisation of 16 billion, has just announced the acquisition of Lumio Beverages, a faster-growing premium drinks maker, for 3.12 billion. The price is 39 per share against an undisturbed price of 30, a premium of 30 percent. Aventra will pay entirely in its own shares, and its chief executive tells analysts that the deal will deliver 60 million of annual cost savings and open Lumio’s brands to Aventra’s distribution network.

On announcement day, Aventra’s stock falls 3 percent, wiping out about 480 million of market value, while Lumio’s stock jumps 27 percent. A fund manager asks the question that every acquirer should be able to answer: “You are paying 720 million more than Lumio was worth yesterday. Where is the 720 million of value coming from, and who is going to deliver it?”

The CFO of Aventra, Tobias Reyes, has done the work. He has valued the synergies at roughly 800 million after integration costs, so on paper the deal creates value even after the premium. But he also knows three things. First, about 275 million of that figure depends on revenue synergies, which he trusts the least. Second, because Aventra is paying in shares, Lumio’s shareholders will own 16 percent of the combined company and so will share both in the gains and in any shortfall. Third, EPS accretion, which the board has been fixated on, is slightly negative in the stock deal and slightly positive in a hypothetical cash deal, which says nothing about which is better for shareholders.

This chapter builds the toolkit to answer the fund manager: why firms buy other firms, how to put a value on synergies and a limit on the price, how consideration and deal structure allocate risk, how to calculate accretion and dilution without being misled by it, how hostile bids and defences work, how integration destroys or preserves value, and how regulators decide whether a combination may proceed.

22.1 Why Companies Merge

22.1.1 Terminology

A **merger** combines two companies into one legal entity; in an **acquisition** one company buys the shares or assets of another and may keep it as a subsidiary. A **tender offer** is a public offer to buy shares directly from shareholders, often bypassing the target's board. A **hostile** bid is one the target's board does not endorse; a **friendly** deal is negotiated. In practice, "M&A" covers all of these, plus divestitures, spin-offs, and leveraged buy-outs.

22.1.2 Motives that can create value

Economies of scale and scope. Combining plants, distribution, procurement, or overheads can cut unit costs. These are the most reliable synergies because they are within management's control.

Market access and revenue enhancement. Selling one firm's products through the other's channels, cross-selling, or entering new geographies. These are real but less certain, and take longer to realise.

Vertical integration. Acquiring a supplier or customer can reduce transaction costs and hold-up risk, especially where assets are specific and contracts are incomplete (Coase, Williamson).

Acquiring capabilities. Buying technology, talent, or brands is often faster than building them (the "buy versus build" decision), though retention of key people is a major risk.

Tax benefits. Use of tax losses or interest tax shields. These are limited by anti-abuse rules and rarely a primary rationale, and a deal that makes sense only because of tax should be examined sceptically.

Replacing poor management. The market for corporate control (Manne) lets outsiders buy underperforming firms and change strategy. The gain is the difference between value under better management and current value.

Market power. Combining competitors may let the firm raise prices. This is a private gain but a social cost, and it is exactly what antitrust law examines.

22.1.3 Motives that rarely create value

Diversification. Shareholders can diversify more cheaply than the firm can. Conglomerate acquisitions that diversify cash flows without improving them often lead to a "conglomerate discount."

EPS bootstrapping. A high-P/E company buying a low-P/E target with shares increases EPS mechanically, because it issues fewer shares than the target's earnings are "worth" at the acquirer's multiple. No economic value is created, and the market, if rational, adjusts the P/E downward on the combined earnings.

Empire building and hubris. Managers benefit from size (pay, prestige), and Jensen's free cash flow argument suggests that firms with excess cash tend to spend it on acquisitions. Roll's hubris hypothesis holds that bidders overestimate their ability to find synergies and so overpay. Both predict that acquirers' shareholders earn little or lose on average.

Misvaluation. Overvalued firms may use their shares as currency to acquire real assets. This is a rational choice for managers if it is correct, and a value-neutral transfer between old and new shareholders; it is not an operational synergy.

Merger waves. Bids cluster in time, with high valuations and abundant credit, and are often driven by industry shocks such as deregulation or technology change, or by the fear of being left behind.

22.1.4 What the evidence says

Empirical studies, across periods and countries, consistently report that target shareholders earn large positive abnormal returns around announcement, while acquirer shareholders earn returns close to zero or slightly negative on average, with wide dispersion across deals. Combined returns are typically positive, so deals create value in aggregate, but the premium transfers most of it to the target. Evidence also suggests that deals financed with stock tend to do less well for acquirers than cash deals, consistent with the idea that stock issuance signals overvaluation and shares risk. These are tendencies, not laws; carefully prepared acquisitions of focused scope, such as bolt-on deals in a core business, tend to do better than large transformational ones.

The managerial conclusion is simple: the buyer must be able to identify specific, controllable sources of value, quantify them, and compare them with the premium. If it cannot, the default is that the acquisition will not create value for the buyer's shareholders.

22.2 Valuing Synergies

22.2.1 The basic equation

The standard logic starts from the value of the combined firm:

```
V(A+B) = V(A) + V(B) + Synergies − Integration costs
Gain to acquirer's shareholders = Synergies − Integration costs − Premium
Premium = Price paid − Target's stand-alone market value
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Equivalently, the acquisition has a positive net present value (NPV) for the buyer only if the **net synergies** exceed the **premium**. Because the target's stand-alone value is already reflected in its market price, only the incremental value counts.

22.2.2 Types of synergies

Type	Source	Typical risk	Discount-rate view
Cost synergies	Overhead, procurement, footprint consolidation	Lowest	Rate for the combined business
Revenue synergies	Cross-selling, distribution, pricing	High; competitors respond	Higher rate, or probability-weight
Financial synergies	Lower cost of debt, debt capacity, tax shields	Moderate	Discount per financing risk
Dis-synergies	Customer losses, key-person exits, culture clashes	Often ignored	Include explicitly

Each synergy should be specified with a base, a driver, a timing profile, an owner, and a realisation cost. A statement such as "cost savings of 5 percent of combined SG&A" is a target; "elimination of two distribution centres and 180 positions, saving 60 per year after the second year, at a one-off cost of 150" is a plan.

22.2.3 Valuation steps

1. **Forecast synergies** by category, pre-tax, with the timing of ramp-up.
2. **Tax-effect** them at the marginal rate.
3. **Subtract integration costs** (severance, systems, rebranding, retention bonuses, advisory fees), which are real cash and typically come first.

4. **Discount** at a rate matching the risk: the combined firm's WACC for cost synergies, higher rates for revenue synergies, or equivalently probability-weight the cash flows.
5. **Capitalise** the sustainable run-rate at a growth rate no higher than inflation. Competition tends to pass on persistent synergies to customers, so perpetual growth assumptions on synergies should be conservative.

Example 22.1 – Valuing the Aventra-Lumio synergies. Assumptions: tax rate 25%; discount rate 9% for cost synergies and 12% for revenue synergies; terminal growth 2% for both.

Cost synergies. Run-rate pre-tax savings of 60, phased 50% in year 1 and 100% in year 2 (growing at 2% thereafter). After tax: full run-rate = 60 × 0.75 = 45.

- Year 1: 0.5 × 45 = 22.5. Year 2: 45 × 1.02 = 45.9.
- Terminal value at end of year 2: 45.9 × 1.02 / (0.09 − 0.02) = 46.82 / 0.07 = 668.8.
- PV = 22.5 / 1.09 + 45.9 / 1.09^2 + 668.8 / 1.09^2 = 20.6 + 38.6 + 562.9 = **622.2**.

Revenue synergies. Incremental revenue of 40, 80, 120, 140, 150 in years 1 to 5, at a contribution margin of 30%, taxed at 25%. After-tax cash flows: 9.0, 18.0, 27.0, 31.5, 33.75. Discount at 12%:

- PV of years 1–5 = 9.0/1.12 + 18.0/1.12^2 + 27.0/1.12^3 + 31.5/1.12^4 + 33.75/1.12^5 = 8.0 + 14.3 + 19.2 + 20.0 + 19.2 = 80.8.
- Terminal value at end of year 5 = 33.75 × 1.02 / (0.12 − 0.02) = 344.3; PV = 344.3 / 1.7623 = 195.3.
- Total = 80.8 + 195.3 = **276.1**.

Integration costs. Pre-tax 90 in year 1 and 60 in year 2; after tax 67.5 and 45.0. PV at 9% = 67.5 / 1.09 + 45.0 / 1.09^2 = 61.9 + 37.9 = **99.8**.

Net synergies. 622.2 + 276.1 − 99.8 = **798.5**.

Component	PV (millions)	Share of gross
Cost synergies	622.2	69.3%
Revenue synergies	276.1	30.7%
Gross synergies	898.3	100%
Less integration costs	(99.8)	
Net synergies	**798.5**	

Interpretation. Cost synergies form more than two-thirds of the value, and each 1 percentage point change in the terminal growth rate would change the cost-synergy value by more than 100 (raising g to 3% lifts it from 622 to about 729). If the revenue synergies were zero, net synergies would be 522.4, so a revenue-synergy shortfall alone does not eliminate value but strongly affects the maximum price.

22.3 Premiums and Who Captures the Gain

22.3.1 The maximum price

The most the buyer should pay is the target's stand-alone value plus the net synergies it can capture with certainty. At any higher price the buyer's shareholders transfer wealth to the seller. The target's board, conversely, should hold out for most of the synergies, which it will argue the buyer can only realise because of its asset. The negotiation is essentially about how

to divide the **synergy surplus**.

```
Maximum premium (%) = Net synergies / Target stand-alone market value
```

Example 22.2 – How much can Aventra pay? Lumio has 80 million shares trading at 30 (market capitalisation 2,400). Net synergies are 798.5 (Example 22.1).

Step 1 – Maximum premium. 798.5 / 2,400 = **33.3%**, corresponding to a maximum price of 30 × 1.3327 = 39.98, about 40 per share.

Step 2 – NPV to Aventra at various prices.

Offer price	Premium %	Premium (millions)	NPV to Aventra	Target's share of synergies
36.00	20.0%	480.0	+318.5	60.1%
39.00	30.0%	720.0	+78.5	90.2%
39.98	33.3%	798.5	0.0	100.0%
42.00	40.0%	960.0	−161.5	120.2%

Interpretation. At the offer of 39, Aventra pays away 90.2% of the synergies and keeps 78.5 (9.8%). A buyer in this position has almost no margin for error: a shortfall of 10% in the net synergies (about 80) would eliminate the entire gain. This is why acquirers in competitive auctions tend to experience the "winner's curse": the winning bidder is typically the one with the most optimistic estimate of synergies, and therefore the one most likely to overpay.

22.3.2 Cash versus stock: sharing risk and value

When the buyer pays in cash, the target's shareholders receive a fixed amount; the buyer's shareholders hold all the uncertainty about synergies. When the buyer pays in stock, the target's shareholders become owners of the combined firm and share in the outcome. Stock also conserves cash and debt capacity, and may be tax-deferred for target shareholders in some jurisdictions. It also signals management's view: a buyer offering shares may be implying its stock is fully valued.

The split of value follows directly. Let the acquirer's market value be V_A, the target's V_T, synergies S, and the price in cash be P.

```
Cash deal:   Acquirer shareholders' value = V_A + V_T + S − P
Stock deal:  Target shareholders own a fraction x = New shares / (Old shares + New shares)
             Acquirer shareholders' value = (1 − x) × (V_A + V_T + S)
```

Example 22.3 – Cash versus stock when synergies disappoint. Aventra has 400 million shares at 40 (V_A = 16,000). Lumio: V_T = 2,400. Offer: 39 per Lumio share, total 3,120. In a stock deal Aventra issues 3,120 / 40 = 78 million new shares, giving Lumio holders 78 / 478 = 16.3% of the combined firm. Evaluate for synergies of 0, 798.5, and 1,597 (assume the stock price reflects the true combined value).

Combined value = 16,000 + 2,400 + S = 18,400 + S.

Synergies S	Cash: acquirer holders' value	Gain vs. 16,000	Stock: acquirer holders' value	Gain vs. 16,000	Stock: target holders' value	Target gain vs. 2,400
0	15,280.0	−720.0	15,397.5	−602.5	3,002.5	+602.5
798.5	16,078.5	+78.5	16,065.7	+65.7	3,132.8	+732.8
1,597	16,877.0	+877.0	16,733.9	+733.9	3,263.1	+863.1

Calculation checks. Cash, S = 0: 18,400 − 3,120 = 15,280. Stock, S = 0: 18,400 × (1 − 0.16318) = 18,400 × 0.83682 = 15,397.5. Stock, S = 798.5: 19,198.5 × 0.83682 = 16,065.7. Target holders in the stock deal with S = 798.5: 19,198.5 × 0.16318 = 3,132.8.

Interpretation. If synergies are nil, the stock deal costs Aventra's shareholders 602.5 instead of 720, because Lumio's holders bear 16.3% of the shortfall. If synergies are double, Aventra's holders in the cash deal gain 877 against 733.9 in the stock deal: stock gives away part of the upside. Stock is thus a hedge against disappointment, for which the buyer pays by sharing the upside. Where the buyer is highly uncertain about synergies (large revenue-synergy content, cross-border, different business models), stock is the better risk-sharing instrument; where the buyer is confident, cash keeps all the upside.

An additional consideration is **signalling and information asymmetry** (Myers-Majluf; Hansen): the target's holders know that a buyer who is confident about large synergies would prefer cash, so a stock offer may be read as bad news about the acquirer or about synergies. The announcement-day decline of 3 percent in Aventra's shares in the vignette is consistent with that.

22.4 Deal Structure and Consideration

22.4.1 Asset deal versus share deal

In a **share purchase**, the buyer acquires the target company with all its assets and liabilities, known and unknown, and its contracts, licences and tax attributes (subject to change-of-control clauses). In an **asset purchase**, the buyer chooses the assets and liabilities it takes. Asset deals allow ring-fencing of liabilities and, in some tax systems, a step-up in the tax basis of assets, but they require re-papering contracts and employee transfers, and may be taxed twice for the seller. The choice reflects tax, liability exposure, and the need for third-party consents. Two further structures are the **merger** by operation of law (forward, reverse, or triangular, where the target merges into a subsidiary of the buyer to isolate liabilities) and the **tender offer** (section 22.6).

22.4.2 Accounting for the acquisition

Under IFRS 3 (and its US counterpart, ASC 805), acquisitions are accounted for under the **acquisition method**. The buyer measures the identifiable assets acquired and the liabilities assumed at fair value on the acquisition date. **Goodwill** is the excess of consideration transferred (plus the fair value of any non-controlling interest and any previously held interest) over the net identifiable assets at fair value. Goodwill is not amortised under IFRS but is tested for impairment at least annually; an impairment charge a few years later is the accounting trace of an overpayment. Transaction costs are expensed as incurred, so advisory fees reduce reported earnings, and contingent consideration is recognised at fair value, with changes in value flowing through profit or loss. Because goodwill sits on the balance sheet, a high-premium deal also affects leverage ratios (equity is unchanged but total assets grow) and later impairment risk.

22.4.3 Consideration choices and protections

Fixed exchange ratio. The target receives a fixed number of acquirer shares per target share. The headline value floats with the buyer's price; the target bears the price risk between signing and closing.

Fixed value (floating ratio). The target receives a fixed value in shares, with the number of shares adjusting to the acquirer's price. The acquirer bears the price risk through the ownership dilution it concedes.

Collar. A band combining the two: within the band the ratio is fixed, and outside it the ratio floats to keep the value at a floor or a cap. Collars limit both parties' exposure to share price swings during the (often months-long) period between signing and closing.

Earn-outs and contingent value rights. Part of the price is deferred and paid if the target meets targets (revenue, EBITDA, regulatory approval). They bridge differences in valuation and align interests of sellers who stay on, but invite disputes about how the metrics are measured and manipulated.

Deal protections. Break-up fees (commonly a few percent of deal value), no-shop clauses with "fiduciary out" exceptions, matching rights, and lock-ups deter competing bidders; courts examine them for reasonableness. *Material adverse change* (MAC) clauses allow the buyer to walk away in extreme circumstances, but the bar is high.

Example 22.4 – Fixed ratio versus fixed value and a collar. Aventra's price at signing is 40, the exchange ratio is 39 / 40 = 0.975 Aventra shares per Lumio share, and Lumio has 80 million shares. Compare outcomes at closing for different Aventra prices.

Aventra price at close	Fixed ratio: value per Lumio share	Premium to 30	Fixed value (39): ratio	Shares issued (m)	Lumio holders' ownership
44	42.90	43.0%	0.886	70.9	15.1%
40	39.00	30.0%	0.975	78.0	16.3%
36	35.10	17.0%	1.083	86.7	17.8%
32	31.20	4.0%	1.219	97.5	19.6%

Calculations. Fixed ratio at 36: 0.975 × 36 = 35.10; premium 35.10 / 30 − 1 = 17.0%. Fixed value at 36: ratio = 39 / 36 = 1.0833; shares = 80 × 1.0833 = 86.7; ownership = 86.7 / (400 + 86.7) = 17.8%.

A collar. Suppose the parties agree to a fixed ratio of 0.975 for Aventra prices between 36 and 44, a floor of value 35.10 below 36, and a cap of 42.90 above 44. At 32: the ratio floats to 35.10 / 32 = 1.0969, so Aventra issues 80 × 1.0969 = 87.75 million shares and Lumio holders own 87.75 / 487.75 = 18.0%. At 48: ratio = 42.90 / 48 = 0.894, Aventra issues 71.5 million shares.

Interpretation. Under a fixed ratio, Lumio's holders bear a swing from a 43% to a 4% premium. Under a fixed value, Aventra's holders concede ownership from 15.1% to 19.6%, a larger share of the synergy surplus when its stock has fallen. The collar splits the risk and provides a defined worst case for each party. Note that the share price may fall precisely because the market doubts the deal, so a fixed-value structure can create a "death spiral" in which a falling price leads to more issuance.

22.5 Accretion and Dilution Analysis

22.5.1 What it measures

Accretion/dilution analysis compares the acquirer's pro forma EPS with its stand-alone EPS. It is simple, quick, and the metric boards and analysts ask for first. Its use is natural for short-run reporting and for signalling to equity markets, but it is a poor guide to value creation, for two reasons: it ignores risk (a deal can be accretive and still destroy value if the target's earnings are riskier than the acquirer's), and it ignores the time value of money and the cost of equity (cheap debt makes almost any deal accretive).

22.5.2 The mechanics

```
Pro forma NI  = NI(acquirer) + NI(target) + after-tax synergies
                − after-tax interest on new debt − after-tax foregone interest on cash used
                − incremental D&A on asset write-ups (and amortisation of intangibles)
Pro forma shares = Acquirer shares + New shares issued
Pro forma EPS = Pro forma NI / Pro forma shares
Accretion (%) = Pro forma EPS / Stand-alone EPS − 1
```

22.5.3 A shortcut: earnings yield comparison

For a deal with no synergies and no write-ups, accretion depends on comparing the target's earnings yield at the offer price with the after-tax cost of the funding:

- **Cash funded by debt:** accretive if the target's earnings yield at the offer (NI_target / Purchase price) exceeds the after-tax cost of debt.
- **Stock-funded:** accretive if the target's offer-price P/E is lower than the acquirer's P/E (equivalently, the target's earnings yield at the offer exceeds the acquirer's earnings yield).
- **Cash from balance sheet:** accretive if the target's earnings yield exceeds the after-tax interest forgone on the cash.

Example 22.5 – Accretion and dilution for the Aventra-Lumio deal. Aventra: net income 800, 400 million shares, EPS 2.00, price 40 (P/E 20.0×; earnings yield 5.0%). Lumio: net income 150, 80 million shares, EPS 1.875, price 30 (P/E 16.0×). Offer: 39 per share, total 3,120; offer P/E = 39 / 1.875 = 20.8×. New debt costs 6.0% pre-tax; tax rate 25%.

Alternative 1 – All debt. After-tax interest = 3,120 × 0.06 × 0.75 = 140.4. Pro forma NI = 800 + 150 − 140.4 = 809.6. Shares = 400. EPS = 2.024. Accretion = 2.024 / 2.00 − 1 = **+1.2%**.

Alternative 2 – All stock. New shares = 3,120 / 40 = 78.0. Pro forma NI = 950. Shares = 478. EPS = 950 / 478 = 1.9874. Accretion = **−0.6%**.

Alternative 3 – 50% cash (debt), 50% stock. Debt = 1,560; after-tax interest = 1,560 × 0.045 = 70.2; NI = 879.8. New shares = 1,560 / 40 = 39.0; shares = 439. EPS = 2.0041. Accretion = **+0.2%**.

Earnings-yield check. Target's earnings yield at the offer = 150 / 3,120 = 4.81%. After-tax cost of debt = 4.50%, so debt financing is accretive (4.81% > 4.50%). The acquirer's earnings yield is 5.00%, higher than the target's 4.81%, so stock financing is dilutive (the target is bought at a higher P/E, 20.8×, than the acquirer's 20.0×).

Alternative 4 – With synergies. Run-rate cost synergies of 60 pre-tax = 45 after tax. All debt: EPS = (809.6 + 45) / 400 = 2.1365 (+6.8%). All stock: EPS = (950 + 45) / 478 = 2.0816 (+4.1%).

Break-even synergies for the stock deal. Required NI = 2.00 × 478 = 956; shortfall = 956 − 950 = 6.0 after tax, or 6.0 / 0.75 = **8.0 pre-tax**.

Financing	Pro forma NI	Shares (m)	EPS	Accretion/(dilution)	With 60 pre-tax synergies
All debt	809.6	400.0	2.024	+1.2%	+6.8%
50% debt / 50% stock	879.8	439.0	2.004	+0.2%	n/a
All stock	950.0	478.0	1.987	−0.6%	+4.1%

Interpretation. The ranking by EPS (debt, mixed, stock) is the opposite of the ranking by risk, and it says nothing about value. In Example 22.3 the stock deal was better for Aventra's holders if synergies disappointed, because the target holders share the downside. The EPS test tells us only that debt is cheaper than equity on an after-tax accounting basis. Cheap debt is not free: a 3.1 billion borrowing raises leverage, interest burden, and distress risk, which the EPS metric ignores. A prudent analysis also shows the **cash flow** and **credit statistics** (net debt/EBITDA, interest cover) before and after the deal.

22.5.4 Pitfalls

- **Write-ups and amortisation.** Fair-value step-ups of fixed assets and amortisable intangibles reduce pro forma earnings; many firms report "cash" or "adjusted" EPS excluding acquisition amortisation. Read the definitions.
- **Timing.** Synergies and integration costs are uneven; first-year EPS may be dilutive even in a value-creating deal.
- **Ratios of P/E are not constants.** A merger of a high-P/E growth company with a low-P/E mature one may lower the combined P/E because the growth mix changes.
- **Contribution analysis.** Comparing each party's contribution to revenue, EBITDA, and net income with its ownership share in the combined entity is a useful sanity check on the exchange ratio, but it should be adjusted for leverage, since net income and equity-value contributions differ with debt.

22.6 Tender Offers, Hostile Bids, and Takeover Defences

22.6.1 Routes to control

A bidder who does not obtain board support has three routes. In a **tender offer**, the bidder offers to buy shares directly from shareholders at a stated price for a limited period. In the United States, the Williams Act regulates these offers: a minimum offer period (20 business days), equal treatment of tendering holders, and disclosure when a holder crosses 5 percent of a class. In a **proxy contest**, the bidder or an activist solicits shareholder votes to replace directors who oppose the deal. In a **creeping acquisition**, the bidder purchases shares on the market, subject to disclosure rules and, in some jurisdictions, mandatory-bid rules (for example, under the UK Takeover Code an acquirer crossing 30 percent of the voting rights must bid for the remainder at not less than the highest price recently paid).

A **toehold**, a pre-bid stake below the disclosure threshold, lets the bidder buy shares at undisturbed prices and receive a share of the profit from a rival bid, thereby partly offsetting the premium.

22.6.2 The free-rider problem

Grossman and Hart showed that a tender offer for a widely held target faces a free-rider problem. If a bid at price b would raise the post-takeover value of the target to $V^* > b$, each atomistic shareholder reasons that her single share cannot determine the outcome, and so she holds out to enjoy V^* rather than sell at b. For the bid to succeed, it must offer essentially all of the improvement; the bidder then earns nothing on the target shares and must recoup its costs from other sources, such as a toehold, **dilution** (the ability to squeeze minority holders at a lower price after gaining control), or private benefits. This theory explains toeholds, two-tier and back-end offers, squeeze-out provisions, and why takeover law regulates the minority's treatment.

22.6.3 Defences

Poison pills (shareholder rights plans). The target issues rights that, when an outsider crosses an ownership threshold (typically 10 to 20 percent), allow all other holders to buy shares at a deep discount, massively diluting the bidder (a "flip-in"), or to buy the bidder's shares at a discount after a merger (a "flip-over"). The board can redeem the rights, so a bidder

must negotiate with the board or win a proxy contest to replace it. In the US, courts upheld pills in principle (Moran v. Household International) but subject their use to enhanced scrutiny (Unocal).

Staggered (classified) boards. Only a fraction of directors is elected each year, so a bidder cannot take control of the board in a single proxy contest. Combined with a pill, this makes a hostile bid very hard.

Supermajority provisions, dual-class shares, and golden shares. They place control in the hands of insiders or the state.

White knight and white squire. The target seeks a friendlier acquirer or a friendly investor who will take a block.

Pac-Man defence. The target bids for the bidder.

Restructuring defences. Selling the "crown jewels" (asset lock-ups), leveraged recapitalisations, large buy-backs, or special dividends, each reducing the target's attractiveness or making it more expensive.

Litigation and regulatory lobbying. Challenging the bid on antitrust or disclosure grounds to delay it.

Golden parachutes. Severance payments for executives on change of control. These can make bids more expensive but, more importantly, may reduce the incentive for managers to resist a good offer.

22.6.4 Law and governance

The legal environment differs materially. In **Delaware** (the main US corporate forum), directors may resist a bid if they reasonably perceive a threat and respond proportionately (Unocal), but once the company is "for sale" (a breakup or change of control), their duty is to obtain the best price reasonably available (Revlon). Under the **UK Takeover Code**, once an offer is imminent, the board may not take actions that could frustrate the offer without shareholder approval, which makes pills and other structural defences essentially unavailable, and a bidder must announce a firm intention or withdraw within a short deadline after a leak ("put up or shut up"). In **continental Europe**, regimes lie between those, with significant variation in the role of employees, supervisory boards, and the passivity rule.

Whether takeover defences help or harm shareholders is debated. Defenders argue that they give the board bargaining power to extract a higher price and to resist opportunistic low-ball bids, and that long-term investment requires insulation from short-term pressure. Critics argue that entrenchment protects underperforming managers, and that the market for corporate control is the ultimate discipline on them. The evidence is mixed, and good governance practice generally prefers defences that increase the board's bargaining power but remain subject to shareholder approval and removal.

Example 22.6 – What a poison pill does to a bidder. Target Ordinal Inc. has 100 million shares at 50 (market capitalisation 5,000). A hostile bidder has built a 15% stake (15 million shares). The pill is triggered at a 15% holding: each right entitles holders other than the bidder to buy, for an exercise price of 150, shares worth 300 at the pre-trigger market price (that is, 300 / 50 = 6 shares per right). Assume all other holders (85 million) exercise.

Step 1 – Shares issued and cash raised. 85 million × 6 = 510 million new shares. Cash received = 85 million × 150 = 12,750 million.

Step 2 – Post-exercise price. Value of the firm = 5,000 + 12,750 = 17,750 (assuming the cash does not change operating value). Shares = 100 + 510 = 610 million. Price = 17,750 / 610 = **29.10**.

Step 3 – Effect on the bidder. Ownership falls from 15 / 100 = 15.0% to 15 / 610 = **2.46%**. The value of the bidder's block falls from 15 × 50 = 750 to 15 × 29.10 = 436.5, a loss of **313.5** (41.8%).

Step 4 – Effect on exercising holders. Each pays 150 and receives 6 shares worth 6 × 29.10 = 174.6, a gain of 24.6. The gain is financed by the dilution of the bidder, not by the company.

Interpretation. The pill imposes a penalty on the bidder sufficient to deter an unfriendly accumulation. Since the board can redeem the rights, the bidder's only options are to negotiate or to remove the board. In practice, the pill is rarely triggered: it is a bargaining device. In real plans, the numbers are structured so that the dilution is serious but the cash raised is small relative to the firm, and exercise is not expected; the example's complete exercise is purely for illustration.

22.7 Post-Merger Integration

22.7.1 Why integration decides the outcome

The valuation assumes that synergies will be realised. That realisation depends on execution, which is why many analyses of acquisitions find the quality of integration a strong predictor of success. Value-destroying deals often fail because the buyer treats the signing as the finish line. The **integration agenda** typically includes:

- **Governance and leadership:** appointing the integration leader, naming the combined management team quickly (uncertainty drives talent loss), and setting a clear decision-making framework.
- **Synergy tracking:** assigning every synergy line a named owner, a budget, milestones, and a reporting cadence, in a dedicated integration management office. Synergies not tracked are not delivered.
- **Customers and revenue:** protecting the base. Revenue dis-synergies, from customers uncomfortable with the new owner or concerned about service continuity, are often larger than planned.
- **People and culture:** retaining key talent through retention bonuses and clarity about roles; resolving differences in compensation and decision styles. Cultural mismatch rarely shows in the model and often dominates the outcome.
- **Systems and operations:** ERP, supply chain, and IT integration, which are expensive and risky, with timelines that routinely exceed the plan.
- **Communication:** consistent messages to employees, customers, suppliers, regulators, and investors.

22.7.2 Speed versus care

Fast integration captures cost synergies sooner and limits the period of uncertainty, but it risks damaging customer relationships and losing key people. Two approaches are often seen: **absorption** (full integration of the target into the acquirer, suited to cost-driven deals in the same business) and **preservation** (keeping the target's brand, culture, and operations largely intact, suited to capability acquisitions, where autonomy is what preserves the value). The wrong approach (absorbing a creative business, or preserving a duplicate cost base) is a classic cause of failure.

Example 22.7 – What happens to value if integration under-delivers? In Example 22.1, gross synergies were worth 898.3 (cost 622.2 plus revenue 276.1) and integration costs 99.8. Suppose that, because of integration friction, only 70% of the planned synergies are realised, they are delayed by one year, and integration costs overrun by 25%.

Step 1 – Realised synergies. 0.70 × 898.3 = 628.8.

Step 2 – Delay. Discount one extra year at 9%: 628.8 / 1.09 = **576.9**.

Step 3 – Integration costs. 99.8 × 1.25 = **124.8**.

Step 4 – Net synergies. 576.9 − 124.8 = **452.1**, compared with 798.5 planned (a reduction of 43%).

Step 5 – Value to Aventra. NPV = 452.1 − 720.0 (premium at 39 per share) = **−267.9**.

Interpretation. A deal with an apparently comfortable 78.5 of NPV turns to a loss of 267.9 under a moderately disappointing integration. This shows that the buyer's margin of safety is the difference between the net synergies and the premium; where the premium consumes 90% of synergies, any execution slippage is value-destroying.

22.8 Antitrust and Regulatory Review

22.8.1 Why regulators care

Competition authorities ask whether the merger will substantially lessen competition or create or strengthen a dominant position, leading to higher prices, lower quality, reduced innovation, or less choice. Many jurisdictions require **pre-merger notification** above specified size thresholds: in the United States, under the Hart-Scott-Rodino Act, the parties file with the Federal Trade Commission and the Department of Justice and observe a waiting period during which the agencies may request more information (a "second request"); the legal substantive standard is in Section 7 of the Clayton Act. In the European Union, the European Commission reviews mergers with an EU dimension under the EU Merger Regulation, with a Phase I and, for serious concerns, a Phase II investigation. Other major jurisdictions (the UK, China, Japan, and many more) have their own regimes, and a large international deal may require a dozen or more clearances, with the slowest determining the timetable.

22.8.2 Theories of harm

Horizontal mergers (competitors) reduce the number of rivals and may allow **unilateral effects** (the merged firm raises prices because customers have fewer alternatives and diversion to the merging parties' products is high) or **coordinated effects** (the market becomes easier to collude in).

Vertical mergers may raise concerns about **foreclosure** (denying rivals inputs or customers) or access to competitors' sensitive information, but also deliver efficiencies by eliminating double marginalisation.

Conglomerate mergers may raise concerns about bundling or tying and portfolio power, though they are less often challenged.

Potential competition and "killer acquisitions". Buying a nascent competitor before it matures is scrutinised increasingly, particularly in technology and pharmaceuticals.

22.8.3 Measuring concentration

The **Herfindahl-Hirschman Index (HHI)** is the sum of squared market shares (in percent) of all firms in a market. A monopoly has an HHI of 10,000. If firms with shares s_1 and s_2 merge, the HHI increases by $2 \times s_1 \times s_2$. Under the 2023 US Merger Guidelines, markets with an HHI above 1,800 are considered highly concentrated, and a merger that produces such a market and raises the HHI by more than 100 points is presumed to substantially lessen competition, as is a merger creating a firm with a share above 30% when the HHI change exceeds 100. (Thresholds have changed over time; earlier guidelines used 2,500 and 200, and you should always check the current text.) The HHI is a screen, not a verdict; market definition, entry conditions, buyer power, and efficiencies are crucial.

Example 22.8 – Computing HHI before and after. Five firms in a regional market have shares of 30%, 25%, 20%, 15%, and 10%. The firms with 20% and 15% propose to merge.

Step 1 – Pre-merger HHI. $30^2 + 25^2 + 20^2 + 15^2 + 10^2 = 900 + 625 + 400 + 225 + 100 =$ **2,250**.

Step 2 – Change. $2 \times 20 \times 15 =$ **600**.

Step 3 – Post-merger HHI. 2,250 + 600 = **2,850**. Check: shares are 30, 25, 35, 10, giving 900 + 625 + 1,225 + 100 = 2,850.

Interpretation. The post-merger HHI exceeds 1,800 and the increase of 600 is far above 100; the market is already highly concentrated, and the merged firm would have the largest share (35%). The merger would be presumptively problematic. The parties would need to provide evidence of efficiencies, entry, or buyer power to rebut the presumption, or propose remedies.

22.8.4 Remedies and process

Authorities may clear the deal unconditionally, clear it with remedies, block it, or the parties may abandon it. **Structural remedies** require divestiture of overlapping businesses to a suitable buyer (preferred by most agencies as they restore competition directly). **Behavioural remedies** (price caps, access commitments, firewalls) are harder to monitor and so used more sparingly. Merger agreements typically contain *efforts covenants* (how far the buyer must go in remedies), a long-stop date, and *reverse break fees* if the deal fails on regulatory grounds. From a valuation perspective, regulatory risk lowers the expected value of the deal, since there is a probability of failure and the cost of delay, and requires scenario analysis: the cost of divesting the overlap and the probability that the deal is cleared.

Beyond competition review, deals may face foreign-investment screening on national security grounds, sector regulators (banking, telecoms, energy), employee consultation requirements, and shareholder approvals.

Case Study: Meridian Components and the Sunvale Bid

Illustrative scenario; Meridian Components, Sunvale Systems, and their executives are fictional.

Meridian Components is a listed manufacturer of industrial sensors with 150 million shares trading at 24, giving a market capitalisation of 3.6 billion, EPS of 1.20 (P/E 20.0×), and debt of 400 million. Its chief executive, Dalia Karimi, sees a strategic opportunity in Sunvale Systems, a smaller developer of embedded software for the same customers, with 40 million shares at 18 (market capitalisation 720 million), EPS of 0.90 (P/E 20.0×), no debt, and a high share of recurring licences. Sunvale’s founders hold 30 percent of the shares and are lukewarm: they like Meridian’s reach but worry about losing control of their product team.

Meridian’s team estimates that bundling hardware and software could add 40 million of annual after-tax operating profit from year four onward, with a 25 percent probability of failing to materialise because several of Sunvale’s customers compete with Meridian’s own clients. Cost savings of about 15 million after tax per year are more certain. Integration would cost 50 million after tax over two years. Using Meridian’s 8.5 percent cost of capital and a conservative terminal growth of 2 percent, the team values the cost savings at 15 / 0.065 = 231 million, the probability-weighted bundling profit (30 million expected, capitalised at 0.065 and discounted three years) at about 361 million, and integration costs at 44 million, so net synergies of roughly 548 million, or 76 percent of Sunvale’s market value.

The investment bankers propose an offer at 24.50 per share, a 36 percent premium (total 980 million), all in Meridian stock at an exchange ratio of 24.50 / 24 = 1.0208 Meridian shares per Sunvale share. That would give Sunvale holders 40.83 / 190.83 = 21.4 percent of the combined company, Although both companies trade at 20 times earnings today, the offer values Sunvale at 24.50 / 0.90 = 27.2 times earnings, so the stock deal is dilutive before synergies: pro forma EPS is (180 + 36) / 190.83 = 1.132 against 1.20, a dilution of 5.7 percent. A director points out that a cash offer financed by debt at 6.5 percent pre-tax (4.9 percent after tax at a 25 percent rate) would be no better: Sunvale’s earnings yield at the offer price is 0.90 / 24.50 = 3.7 percent, below the after-tax cost of debt, giving EPS of (216 - 47.8) / 150 = 1.12, a dilution of 6.5 percent, and Meridian’s leverage would also rise.

A second complication is that Meridian's largest customer has warned that it would view a combination with Sunvale as a competitive threat to its own software efforts, and a rival, Orbis Controls, with 5 percent of Sunvale's shares already, may make a competing bid. The Sunvale founders have signalled they would support a deal if the exchange ratio is fixed, retention packages are guaranteed for the product team, and the software unit is run as a separate division for at least three years. Meridian's lawyers consider the antitrust risk low (combined share below 15 percent in any relevant market) but caution that the customer's complaint might prompt a longer review.

Karimi must decide on the price, the form of consideration, the integration model, and how to respond if Orbis bids.

Discussion Questions

1. Compute the maximum premium Meridian can pay from the net synergy estimate. How much of the synergy surplus does the proposed 36 percent premium (260 million) hand to Sunvale's holders, and how sensitive is the conclusion to the 25 percent failure probability on the bundling profit?
2. Compare cash and stock consideration for this deal on three dimensions: EPS accretion, allocation of synergy risk, and signalling. Which would you recommend, and would you change your answer if Meridian were more confident in the bundling synergies?
3. The founders want the software unit to remain autonomous for three years. What does that imply for the realisable synergies, and how would you structure the integration (absorption versus preservation) and the incentives (retention, earn-out) to balance the two?
4. How should Meridian respond if Orbis bids higher? Discuss walk-away discipline, the winner's curse, and the deal-protection devices (break fees, matching rights) you would seek in the merger agreement.

Key Takeaways

- Acquisitions create value only through identifiable, controllable sources of gain (cost savings, capability, better management) that exceed the premium; diversification and EPS bootstrapping are not sources of value.
- The empirical regularity is that target shareholders capture most of the gain and acquirer shareholders earn roughly zero on average, with large variation; discipline in price is the main determinant of buyer returns.
- Value synergies by type, tax-effect them, net out integration costs, discount at risk-appropriate rates, and treat revenue synergies with scepticism and probability weights.
- The maximum premium equals net synergies divided by the target's stand-alone value; the closer the price to that bound, the smaller the margin of safety.
- Stock consideration shares both the upside and the downside with target holders, and also signals management's view; cash retains all upside and risk for the acquirer.
- Fixed-ratio, fixed-value, and collar structures allocate share-price risk between signing and closing; each party should know its worst case.
- EPS accretion is mechanical: cheap after-tax debt or a lower target P/E produces it. It is not a test of value creation, and it neglects risk and the cost of capital.
- Hostile bids face the free-rider problem; defences such as poison pills and staggered boards raise the bidder's cost and strengthen the board's bargaining position, but governance regimes differ greatly across jurisdictions.
- Integration is where synergies are delivered or lost; modest execution shortfalls can turn an apparent gain into a loss when the premium absorbs most of the synergies.

- Antitrust review screens concentration, for example with the HHI, and examines theories of harm; remedies and delay are part of the deal's expected value.

Review Questions

1. Why do empirical studies find that acquirers' shareholders earn approximately zero on average even though combined firm value tends to increase?
2. Explain why diversification is not generally a sound motive for a merger from the shareholders' perspective.
3. How should revenue synergies and cost synergies be valued differently, and why?
4. Derive the maximum premium a buyer can pay and explain what happens to the buyer's shareholders if it pays more.
5. Explain how stock consideration shares risk between acquirer and target holders. When might a buyer prefer cash?
6. Why is a high-P/E acquirer buying a low-P/E target with stock accretive, and why is that not evidence of value creation?
7. Compare a fixed exchange ratio, a fixed-value structure, and a collar. Which party bears the risk in each?
8. Describe the free-rider problem in tender offers and two mechanisms by which bidders mitigate it.
9. How does a flip-in poison pill work, and why can it be effective even if it is never exercised?
10. Why do many acquisitions fail at the integration stage? Illustrate with two examples of integration failure and what can be done to prevent them.

Problems

1. **Synergy value.** A merger will produce after-tax cost savings of 30 per year starting next year, growing at 2% forever. The discount rate is 8%. One-time after-tax integration costs of 80 are paid today. The target's stand-alone market value is 1,500. Compute the net synergies and the maximum premium in percent.
2. **Premium and NPV.** A target has 50 million shares at 20. An acquirer offers 25 per share in cash. The net synergies are estimated at 300. Compute the premium in percent and money, the NPV to the acquirer, and the share of synergies paid to the target.
3. **Accretion/dilution.** Acquirer: 100 million shares, EPS 3.00, price 45. Target: net income 54. The acquirer pays 900 for the target. If it uses debt at 5% pre-tax with a 20% tax rate, what is the pro forma EPS and accretion? If it uses stock at 45, what is the pro forma EPS and accretion?
4. **Exchange ratio.** An acquirer's shares trade at 45; it offers 54 per share of a target which has 25 million shares and trades at 40. Compute the exchange ratio, the premium, the new shares issued, and the percentage of the combined company owned by target holders if the acquirer has 100 million shares.
5. **Break-even synergies.** Continuing with Problem 3, compute the pre-tax synergies necessary for the all-stock deal to be EPS-neutral.
6. **HHI.** Five firms in a market have shares of 35%, 25%, 15%, 15%, and 10%. The firms with 25% and 15% merge. Compute the pre-merger HHI, the change, and the post-merger HHI.
7. **Poison pill.** A target has 200 million shares at 40. A bidder owns 20 million shares (10%). A flip-in right allows each other holder to buy, for 100, shares worth 200 at the market price of 40. Assuming all other holders exercise, compute the post-exercise price, the bidder's new ownership, and the value loss to the bidder.

8. **Collar.** A target is offered 0.8 acquirer shares per target share if the acquirer's price at closing is between 40 and 50. Below 40, the ratio floats to guarantee a value of 32 per target share; above 50, it floats to cap the value at 40 per target share. Compute the ratio and the value per target share if the acquirer's price at closing is 36, 45, and 56.

Answers to Selected Problems

1. PV of savings = 30 / (0.08 − 0.02) = 500. Net synergies = 500 − 80 = **420**. Maximum premium = 420 / 1,500 = **28.0%**.
2. Premium per share = 5, i.e. 25% (25 / 20 − 1); in money = 5 × 50 = 250. NPV to acquirer = 300 − 250 = **50**. Share of synergies paid to target = 250 / 300 = **83.3%**.
3. Debt: after-tax interest = 900 × 0.05 × 0.80 = 36; acquirer NI = 3.00 × 100 = 300; pro forma NI = 300 + 54 − 36 = 318; EPS = 3.18; accretion = **+6.0%**. Stock: new shares = 900 / 45 = 20 million; NI = 354; shares = 120; EPS = 2.95; accretion = **−1.7%**.
4. Exchange ratio = 54 / 45 = **1.20**. Premium = 54 / 40 − 1 = **35%**. New shares = 25 × 1.20 = **30 million**. Target holders' ownership = 30 / 130 = **23.1%**.
5. Required NI for EPS of 3.00 on 120 million shares = 360; shortfall = 360 − 354 = 6.0 after tax; pre-tax = 6.0 / 0.80 = **7.5** (using the 20% tax rate from Problem 3).
6. Pre-merger HHI = $35^2 + 25^2 + 15^2 + 15^2 + 10^2$ = 1,225 + 625 + 225 + 225 + 100 = **2,400**. Change = 2 × 25 × 15 = **750**. Post-merger HHI = **3,150**. Check: shares 35, 40, 15, 10: 1,225 + 1,600 + 225 + 100 = 3,150.
7. Shares per right = 200 / 40 = 5. Exercising holders = 180 million; new shares = 900 million; cash raised = 180 × 100 = 18,000. Post value = 200 × 40 + 18,000 = 26,000; shares = 1,100 million; price = **23.64**. Bidder ownership = 20 / 1,100 = **1.82%**. Value of the bidder's block falls from 800 to 472.7, a loss of **327.3**.
8. At 36: floor applies; ratio = 32 / 36 = **0.889**; value = **32**. At 45: fixed ratio 0.8; value = **36**. At 56: cap applies; ratio = 40 / 56 = **0.714**; value = **40**.

Chapter 23: Leveraged Buyouts, Private Equity and Restructuring

Learning Objectives

- Explain how a leveraged buyout is structured, financed and exited, and why the capital structure of a buyout differs from that of a comparable public company.
- Build a sources-and-uses table, a simple debt schedule and a sponsor returns analysis, and compute MOIC and IRR under alternative exit assumptions.
- Decompose private equity returns into earnings growth, multiple change and deleveraging, and evaluate the operational and financial levers that create value.
- Diagnose financial distress using coverage ratios and the Altman Z-score, and distinguish distress from economic failure.
- Describe liquidation and reorganization under bankruptcy, apply the absolute priority rule to a recovery waterfall, and identify the fulcrum security in a restructuring.
- Evaluate spin-offs, carve-outs and divestitures using sum-of-the-parts valuation, and explain when separation creates value for shareholders.

Opening Vignette

(Illustrative scenario; the company and all figures are fictional.)

Marlow Packaging is a family-founded maker of corrugated and specialty packaging with revenue of $500 million and EBITDA of $100 million. Its shares trade thinly on a regional exchange at roughly 7 times EBITDA, a discount to the 9 times that larger peers command. The founder’s children want liquidity; the board is tired of quarterly scrutiny over a business that earns steady, unexciting cash flows.

Northgate Capital, a mid-sized private equity firm, sees an opportunity. Its analysts note that Marlow’s plants run at 71 percent utilization, that purchasing is fragmented across six regional buyers, and that the company has never used more than 1.5 times leverage. Northgate proposes to acquire Marlow for 9 times EBITDA, financing more than half the price with debt that will be serviced from Marlow’s own cash flow. The sponsor will contribute roughly 41 percent of the capital in equity, install a new chief operating officer, consolidate procurement, close two inefficient plants and sell a non-core labeling division.

The founders’ advisers ask the question every board faces in such a transaction: if the buyer can pay a premium of roughly 29 percent and still earn a high return, where does the value come from? Is it tax shields on the new debt? Better management? A cheap entry price relative to the exit price? Or simply risk transferred to lenders and to employees? Northgate’s investment committee asks the mirror-image question: what happens if earnings fall 20 percent in a recession while $550 million of debt is outstanding?

This chapter answers both. We build the mechanics of a leveraged buyout from first principles, test where returns come from, examine what happens when leverage turns from friend to foe, and study the legal and financial machinery of restructuring and separation that sits at the other end of the corporate life cycle.

23.1 The Anatomy of a Leveraged Buyout

23.1.1 What an LBO is

A leveraged buyout (LBO) is the acquisition of a company, or a division, financed predominantly with borrowed money, where the target's own assets and cash flows support the debt. A private equity (PE) sponsor, usually a limited partnership fund, supplies a minority of the purchase price as equity, lenders supply the rest, and the sponsor takes control. The sponsor's goal is to improve the business, repay part of the debt from operating cash flow, and sell or list the company within roughly four to seven years at a higher equity value.

Three variants matter in practice. In a classic sponsor LBO, an outside fund buys the company. In a management buyout (MBO), the existing management team, usually backed by a PE fund, acquires the company. In a take-private, a listed company is acquired and delisted, typically at a premium of 20 to 40 percent over the undisturbed share price. Each shares the same engine: high leverage, concentrated ownership, strong incentives and a defined exit.

23.1.2 Why debt works here, and where it stops working

The theoretical case for high leverage draws on three strands of corporate finance covered earlier in this book.

First, the tax shield. Interest is deductible, so debt lowers the after-tax cost of capital as in Modigliani-Miller with taxes. For a firm paying a 25 percent tax rate, each dollar of interest saves 25 cents of tax, subject to the interest-deductibility limits that many jurisdictions impose (often a cap tied to a percentage of EBITDA or EBIT).

Second, agency costs of free cash flow, per Jensen. Mature companies with stable cash flows and few high-return projects tend to hoard cash or reinvest it in low-return expansion. A heavy debt load forces managers to disgorge cash through mandatory interest and principal payments. Debt acts as a disciplinary device.

Third, incentive alignment. The sponsor and management hold concentrated equity stakes, often with management investing meaningfully from personal wealth plus a pool of options, so they gain a large share of every dollar of value created.

Against these benefits stand the costs of financial distress: lost customers, supplier tightening, forced asset sales, management distraction and legal fees. The trade-off theory says the optimum lies where the marginal tax shield equals the marginal expected distress cost. LBO targets are chosen because their distress costs are lower than average, not because they are immune to distress. The ideal target has stable, predictable cash flow, modest capital expenditure needs, a strong market position, tangible assets that can serve as collateral, identifiable costs to cut and clear exit routes. Cyclical, capital-hungry or fast-changing businesses make poor LBO candidates.

23.1.3 The financing stack

LBO debt is layered by priority and cost, so each tranche appeals to a different investor.

Tranche	Typical seniority	Typical features	Indicative cost	Holders
Revolving credit facility	First lien	Undrawn at close; liquidity backstop	Floating rate + low spread	Banks
Term loan A / B	First lien	Amortizing (A) or bullet-like (B); secured on all assets; covenants (A) or covenant-lite (B)	Floating rate + moderate spread	Banks, institutional loan funds
Second lien / unitranche	Second lien or blended	Secured but junior; higher spread	Floating rate + high spread	Credit funds
Senior unsecured notes	Unsecured	Fixed coupon, 7 to 10 years, call protection	Fixed, high-yield	Bond investors
Subordinated / mezzanine / PIK	Subordinated	Cash or payment-in-kind interest, warrants	Highest debt cost	Mezzanine funds
Sponsor equity and management rollover	Residual	Common or preferred equity	Target return 20 percent plus	Fund, managers

Leverage is measured against EBITDA. Total debt of 5 to 6 times EBITDA was common in buoyant markets, and senior secured leverage of 3 to 4 times. Lenders also test coverage: EBITDA divided by cash interest, and fixed charge coverage including capex and mandatory amortization. Debt markets move in cycles, and the available leverage and the multiples sponsors can pay move with them. An important managerial implication: purchase multiples and returns are not independent of credit conditions, so a deal that works at 6 times leverage and a 4 percent base rate may fail at 4 times leverage and 8 percent.

23.1.4 Sources and uses

Every LBO analysis starts with a sources-and-uses table. Uses are what must be paid at closing: the equity purchase price (or enterprise value if existing debt is refinanced), fees (advisory, financing, legal) and any cash placed on the balance sheet. Sources are the debt tranches and the sponsor equity, which is the plug.

Example 23.1 – Sources and uses for the Marlow buyout. Northgate agrees to acquire Marlow at 9.0 times trailing EBITDA of $100 million. Existing debt is refinanced as part of the purchase, so the purchase price is the enterprise value. Transaction and financing fees total $27 million, and Northgate wants $10 million of minimum cash left on the balance sheet. Total debt is set at 5.5 times EBITDA, split into a $400 million term loan B and $150 million of senior notes.

Step 1: Enterprise value = 9.0 × $100m = $900m.

Step 2: Total uses = EV + fees + minimum cash = 900 + 27 + 10 = $937m.

Step 3: Total debt = 5.5 × 100 = $550m (TLB $400m + notes $150m).

Step 4: Sponsor equity (plug) = 937 − 550 = $387m, which is 387 / 937 = 41.3 percent of capitalization.

Uses	$m	Sources	$m
Purchase of enterprise	900	Term loan B	400
Fees and expenses	27	Senior notes	150
Cash to balance sheet	10	Sponsor equity	387
Total uses	**937**	**Total sources**	**937**

The equity cushion of 41 percent is typical of mid-cycle deals; in earlier eras equity was often 10 to 20 percent. Higher purchase multiples and tighter lending standards have raised the equity share. Notice that the sponsor's equity cheque is larger than the "price" of the business net of debt (900 − 550 = 350) because it also funds fees and cash. Fees are real value leakage, and they matter more at lower returns.

23.2 Projecting Cash Flow and Paying Down Debt

23.2.1 The cash sweep

The central mechanism of an LBO is the debt paydown. After servicing interest, paying taxes and funding capex and working capital, free cash flow is swept to repay the prepayable term loan. Each dollar repaid transfers a dollar of enterprise value from creditors to equity holders without any change in the enterprise value itself. Deleveraging is therefore a source of return even if the business does not grow, provided the exit multiple does not fall.

The free cash flow available for debt repayment each year is:

```
FCF to debt paydown = EBITDA − Cash interest − Cash taxes − Capex − ΔNWC
Cash taxes = tax rate × (EBITDA − D&A − Interest)
```

Interest in a simple model is computed on the beginning-of-year balance to avoid circularity; a more refined model uses average balances and iterates. The key behavioral feature is that interest falls as debt falls, so free cash flow accelerates even if EBITDA grows modestly.

Example 23.2 – Five-year debt schedule for Marlow. Revenue is $500m and EBITDA margin is 20 percent. EBITDA grows at 8, 8, 7, 6 and 5 percent in years 1 to 5, and the margin is held at 20 percent so revenue is EBITDA / 0.20. D&A is 3 percent of revenue, capex is 4 percent of revenue, and net working capital investment is 10 percent of the change in revenue. The tax rate is 25 percent. The term loan B ($400m) bears 9 percent and the notes ($150m) bear 11 percent fixed; the notes are non-callable and stay outstanding, while all free cash flow sweeps the term loan.

Year 1 detail:

- EBITDA = 100 × 1.08 = $108.0m; revenue = 108 / 0.20 = $540.0m.
- D&A = 3% × 540 = $16.2m; capex = 4% × 540 = $21.6m.
- ΔNWC = 10% × (540 − 500) = $4.0m.
- Interest = 400 × 9% + 150 × 11% = 36.0 + 16.5 = $52.5m.
- Taxable income = 108.0 − 16.2 − 52.5 = $39.3m; tax = 25% × 39.3 = $9.8m.
- FCF = 108.0 − 21.6 − 4.0 − 52.5 − 9.8 = $20.1m.
- Closing term loan = 400 − 20.1 = $379.9m.

Repeating for each year gives:

Year	EBITDA	Interest	Taxes	Capex	ΔNWC	FCF swept	Term loan (end)
0	100.0						400.0
1	108.0	52.5	9.8	21.6	4.0	20.1	379.9
2	116.6	50.7	12.1	23.3	4.3	26.2	353.7
3	124.8	48.3	14.4	25.0	4.1	33.0	320.8
4	132.3	45.4	16.8	26.5	3.7	40.0	280.8
5	138.9	41.8	19.1	27.8	3.3	47.0	233.8

Total debt falls from $550m to 233.8 + 150 = $383.8m. Leverage falls from 5.5 times to 383.8 / 138.9 = 2.76 times, and interest coverage (EBITDA / interest) rises from 100 / 52.5 = 1.9 times in the first year's base to 138.9 / 41.8 = 3.3 times in year 5. Cumulative debt repaid is $166.2m, only about 30 percent of the original debt. The sponsor is not relying on repaying the whole debt; it relies on the debt being refinanceable or repaid from sale proceeds.

23.2.2 Reading the schedule critically

Three features of this schedule deserve emphasis. First, year 1 free cash flow of $20.1m on $52.5m of interest is thin: a 15 percent EBITDA shortfall in year 1 (to $91.8m) would remove about $16.2m of EBITDA, and after tax (about $12.1m) leave roughly $8m of free cash flow, well under half of the plan. The model's cushion is small in the early years when leverage is highest. Second, the schedule assumes the margin holds; a margin squeeze from input cost inflation, which is the classic risk in packaging, would hit cash flow more than proportionally. Third, floating-rate term debt exposes the sponsor to base rate movements. A 200 basis point increase on $400m of floating debt adds $8m of annual interest, which is 40 percent of year 1 free cash flow. Sponsors therefore hedge, with swaps or caps, at least half of floating exposure in most deals.

Lenders protect themselves with covenants (maximum leverage and minimum coverage tests, restrictions on dividends and additional debt), excess cash flow sweeps (typically 50 percent of excess cash flow, stepping down as leverage falls) and security over assets. Covenant-lite loans drop the maintenance tests but keep incurrence tests that apply only when the borrower does something, such as taking on more debt.

23.3 Returns Mathematics: MOIC, IRR and Attribution

23.3.1 The two headline metrics

Sponsors report two numbers. The multiple of invested capital (MOIC, also called money multiple) is total proceeds divided by total equity invested. The internal rate of return (IRR) is the discount rate that sets the net present value of the equity cash flows to zero. For a single investment and a single exit,

```
MOIC = Exit equity value / Initial equity invested
IRR  = MOIC^(1/n) − 1        (single outflow, single inflow, n years)
```

MOIC ignores time; IRR rewards speed. A 2.0x in three years is a 26 percent IRR; a 2.0x in seven years is a 10.4 percent IRR. Sponsors target roughly 20 to 25 percent gross IRRs and 2.0x to 3.0x MOIC. Both metrics are gross of the fund's fees and carried interest, which we address in Section 23.4.

Exit equity value is exit enterprise value less net debt at exit, where net debt includes any residual term loan, notes, and accrued PIK interest, minus cash:

```
Exit EV     = Exit multiple × Exit-year EBITDA
Exit equity = Exit EV − (Term loan + Notes − Cash)
```

23.3.2 Where returns come from

Every dollar of equity gain can be attributed to three sources, which provides a disciplined check on whether a deal's thesis is realistic.

```
Equity gain = EBITDA growth + Multiple change + Net debt reduction − Fees and leakage
EBITDA growth  = (Exit EBITDA − Entry EBITDA) × Entry multiple
Multiple change = (Exit multiple − Entry multiple) × Exit EBITDA
Net debt reduction = Entry net debt − Exit net debt
```

The ordering of the first two terms (growth at the entry multiple, expansion on the exit EBITDA) is a convention that assigns the cross term to multiple change; other conventions exist, and an analyst should state the one used.

Example 23.3 – Marlow returns, sensitivity and attribution. Using Example 23.2, assume exit at the end of year 5 at 9.0 times EBITDA (no multiple change).

Step 1: Exit EBITDA = $138.9m; exit EV = 9.0 × 138.9 = $1,250.1m.

Step 2: Net debt at exit = term loan 233.8 + notes 150.0 − cash 10.0 = $373.8m.

Step 3: Exit equity = 1,250.1 − 373.8 = $876.3m.

Step 4: MOIC = 876.3 / 387 = 2.26x. IRR = 2.264^(1/5) − 1 = 17.8 percent.

Attribution of the $489.3m equity gain (876.3 − 387.0):

- EBITDA growth: (138.9 − 100.0) × 9.0 = $350.1m (72 percent of the gain).
- Multiple change: (9.0 − 9.0) × 138.9 = $0.
- Net debt reduction: entry net debt = 550 − 10 = $540m; exit net debt $373.8m; reduction = $166.2m (34 percent).
- Fees and leakage: −$27.0m (−6 percent).
- Check: 350.1 + 0 + 166.2 − 27.0 = $489.3m. ✓

The 17.8 percent IRR is below the sponsor's 20 to 25 percent target. Sensitivity to the exit multiple and to the holding period is shown below.

Exit multiple (year 5)	Exit EV	Exit equity	MOIC	IRR
8.0x	1,111.2	737.4	1.91x	13.8%
9.0x	1,250.1	876.3	2.26x	17.8%
10.0x	1,389.0	1,015.2	2.62x	21.3%

Exit year (at 9.0x)	Exit equity	MOIC	IRR
3	662.4	1.71x	19.6%
4	769.9	1.99x	18.8%
5	876.3	2.26x	17.8%

The first table shows how a one-turn multiple change swings the IRR by roughly 4 percentage points, comparable to the operational plan. The second shows the “IRR clock”: because equity compounds slower than the early years’ leveraged gains, extending the hold lowers the IRR even as MOIC rises. This explains why sponsors lean toward early exits, dividend recapitalizations and quick flips when markets are strong. For the sponsor to reach a 20 percent IRR on a five-year hold at a flat multiple, exit equity would have to be 387 × 1.2^5 = $963m, requiring either an exit multiple of about 9.6 times or year-5 EBITDA about $10m higher than planned.

23.3.3 Dividend recapitalizations

A dividend recapitalization (recap) raises new debt against a portfolio company and pays the proceeds to the sponsor as a dividend. It raises IRR by returning cash earlier, while leaving MOIC roughly unchanged or slightly lower because of the additional interest cost. It also raises risk, since leverage rises just as the sponsor has already recovered part of its capital.

Example 23.4 – A year-3 recapitalization. At the end of year 3, Marlow’s EBITDA is $124.8m and total debt is 320.8 + 150 = $470.8m (3.8x). The company issues $150m of additional senior notes at 10 percent and pays the sponsor a $150m dividend. For simplicity, the added after-tax interest cost reduces each later year’s debt paydown. Extra interest after tax is 150 × 10% × (1 − 25%) = $11.25m in year 4. In year 5 that foregone paydown also carries interest at the term loan rate: 11.25 × 9% × 75% = $0.76m extra, so year-5 foregone paydown = 11.25 + 0.76 = $12.0m.

Step 1: Additional net debt at exit = 150 + 11.25 + 12.01 = $173.3m (the new notes plus two years of foregone paydown).

Step 2: Exit equity = 1,250.1 − (373.8 + 173.3) = $703.0m.

Step 3: Sponsor cash flows: −387.0 (year 0), +150.0 (year 3), +703.0 (year 5).

Step 4: MOIC = (150 + 703.0) / 387 = 2.20x (versus 2.26x without the recap).

Step 5: IRR solves −387 + 150/(1+r)^3 + 703.0/(1+r)^5 = 0, giving r = 18.8 percent, compared with 17.8 percent without the recap.

The recap lifts IRR by about 1 point and reduces MOIC by 0.06x. Recaps transfer value from creditors and from the future only if the lenders mis-price the risk. After the recap, pro forma leverage is (470.8 + 150) / 124.8 = 5.0x, back near the original level, with less equity cushion. If EBITDA falls 20 percent, leverage becomes 6.2x. Sponsors who pay themselves a dividend funded by creditors leave management and employees to carry the downside, a reason recaps draw criticism when followed by distress.

23.4 Value Creation Levers and the Economics of a Private Equity Fund

23.4.1 Operational levers

The best sponsors insist that financial engineering is the smallest source of durable value. The main operational levers are:

- **Revenue growth:** pricing discipline, sales force reorganization, new products and geographic expansion, and bolt-on acquisitions at lower multiples than the platform (multiple arbitrage). In Marlow, a bolt-on bought at 6 times EBITDA and valued inside the platform at 9 times creates three turns of value per dollar of acquired EBITDA before synergies.
- **Margin improvement:** procurement consolidation, plant rationalization (Marlow’s utilization at 71 percent signals excess capacity), overhead reduction and automation. A one-point margin gain on $695m of year 5 revenue is $6.95m of EBITDA, worth $62.6m of EV at 9 times.

- **Capital efficiency:** shrinking working capital through payables terms and inventory control, reducing capex through asset-light models and selling non-core assets. Each dollar of released working capital reduces net debt by one dollar immediately.
- **Governance and incentives:** a small, expert board, monthly reporting, management equity, and the removal of public-company costs. Management ownership of 5 to 15 percent of equity changes behavior.
- **Financial levers:** capital structure optimization, the tax shield, refinancing at tighter spreads when credit improves, and recaps.

23.4.2 What the evidence and theory suggest

Academic and industry studies broadly find that buyouts of mature companies are associated with improvements in operating margins and productivity, though results vary with the sample and period, and that PE-backed firms are more exposed to distress in downturns. Gross returns have historically exceeded public equity benchmarks on average, but the net-of-fees premium is smaller and varies widely between funds; top-quartile funds persist more than in mutual funds. Treat any single-number claim with caution. The conceptual point is that a sponsor earns an excess return only by one of four mechanisms: buying below intrinsic value, improving the business, using leverage whose cost is below the return on assets (which transfers risk to equity rather than creating value in a world without frictions), or exiting at a higher multiple than entry. The first and last depend on market timing; the second is skill; the third is compensated risk-bearing, not a free lunch. A managerial rule of thumb: ask what an unlevered return would be. If a deal needs leverage to clear the hurdle and has no operational plan, it is a bet on credit markets.

23.4.3 Fund structure and fees

A PE fund is a limited partnership. Limited partners (LPs: pension funds, endowments, sovereign funds, insurers) commit capital that the general partner (GP) draws down over a three- to five-year investment period. Funds typically live ten years, with extensions. The GP earns a management fee, commonly around 2 percent of committed capital during the investment period and lower thereafter, and carried interest, commonly 20 percent of profits above a preferred return (hurdle) of about 8 percent. A catch-up provision lets the GP receive a larger share of distributions after the hurdle is met until it has received 20 percent of total profits.

Example 23.5 – A simple whole-fund waterfall. Ignore management fees for clarity. LPs contribute $1,000m, and over five years the fund distributes a total of $1,900m. The preferred return is 8 percent compounded annually, there is a 100 percent GP catch-up, and carry is 20 percent.

Step 1: Return of capital: $1,000m to LPs.

Step 2: Preferred return = 1,000 × (1.08^5 − 1) = 1,000 × 0.469328 = $469.3m to LPs.

Step 3: Catch-up: GP receives 100 percent until it holds 20 percent of cumulative profit. Let x be the catch-up: x = 0.20 × (469.3 + x), so x = 0.20 × 469.3 / 0.80 = $117.3m to GP.

Step 4: Remaining distributions = 1,900 − 1,000 − 469.3 − 117.3 = $313.4m, split 80/20: LPs $250.7m, GP $62.7m.

Step 5: Totals: LPs = 1,000 + 469.3 + 250.7 = $1,720.0m; GP carry = 117.3 + 62.7 = $180.0m = 20 percent of the $900m profit.

The gross multiple is 1.90x; the LP net multiple (before management fees) is 1.72x. Including a 2 percent annual fee on $1,000m would reduce LP net proceeds by several hundred basis points of IRR over the life of the fund. The structure creates incentive effects: GPs hold a call option on fund profits, which can encourage risk-taking, since the downside to the

GP is limited to lost fees, while the upside includes 20 percent of gains. LPs mitigate with hurdles, clawbacks (return of excess carry if later investments lose money), co-investment rights and key-person clauses.

23.4.4 Exit routes

Sponsors exit by (1) a trade sale to a strategic buyer, which can pay for synergies, (2) a secondary buyout to another sponsor, (3) an initial public offering, often with a staged sell-down because sponsors usually cannot sell their entire stake at the IPO, and (4) a recapitalization or continuation vehicle, in which a GP transfers an asset into a new vehicle for existing LPs to either cash out or roll over. Each route has a different valuation anchor and different execution risk; exit multiples are the most volatile input in the model.

23.5 Financial Distress and Bankruptcy

23.5.1 Distress, default and failure are not the same

Financial distress occurs when a firm cannot meet its obligations, or can meet them only by sacrificing the business. A firm can be economically viable, with its going-concern value exceeding liquidation value, yet financially distressed because its capital structure is too heavy. Conversely, an economically failed firm, with going-concern value below liquidation value, should be wound up even if its debt is small. Good restructuring practice separates the two questions: is the business worth more alive than dead (the operational question) and who bears the loss (the financial question)?

Distress produces direct costs (legal, advisory and administrative fees, which often total several percent of asset value for large cases, and more for complex ones) and indirect costs that are usually larger: customers who defect because they doubt warranty support, suppliers who demand cash in advance, key employees who leave, and managers who skip positive-NPV investments because the gains would go to creditors (debt overhang, per Myers). Equity holders in distress also have incentives to take excessive risk (asset substitution), since their payoff is an option on firm value, and to avoid raising new equity.

23.5.2 Early-warning indicators

Practitioners track:

```
Interest coverage    = EBITDA / Cash interest
Fixed charge cover   = (EBITDA − Capex) / (Interest + Scheduled principal)
Leverage             = Total debt / EBITDA
Liquidity runway     = (Cash + Undrawn committed facilities) / Monthly cash burn
```

Typical warning thresholds are coverage below 1.5 times, leverage above 6 times for cyclical businesses, and less than twelve months of runway. In Marlow’s case, if capex is 4 percent of revenue (about $21.6m in year 1) the year-1 ratio of (EBITDA − capex) / interest is (108.0 − 21.6) / 52.5 = 1.65x, not far above the warning zone.

The Altman Z-score, a multivariate discriminant model for public manufacturers, combines five ratios:

```
Z = 1.2·(Working capital / Total assets) + 1.4·(Retained earnings / Total assets)
  + 3.3·(EBIT / Total assets) + 0.6·(Market value of equity / Total liabilities)
  + 1.0·(Sales / Total assets)
```

Scores below about 1.81 indicate high distress risk, scores above about 2.99 indicate safety, and the range between is the grey zone. Variants exist for private firms and non-manufacturers. The score is a screening aid, not a verdict, and its coefficients reflect the data and period on which it was estimated.

Example 23.6 – Altman Z for a stressed manufacturer. Harlan Metals (fictional) reports total assets $800m, working capital $40m, retained earnings $60m, EBIT $48m, market value of equity $150m, total liabilities $620m and sales $900m.

- X1 = 40 / 800 = 0.050; 1.2 × 0.050 = 0.060.
- X2 = 60 / 800 = 0.075; 1.4 × 0.075 = 0.105.
- X3 = 48 / 800 = 0.060; 3.3 × 0.060 = 0.198.
- X4 = 150 / 620 = 0.2419; 0.6 × 0.2419 = 0.145.
- X5 = 900 / 800 = 1.125; 1.0 × 1.125 = 1.125.
- Z = 0.060 + 0.105 + 0.198 + 0.145 + 1.125 = 1.633.

The score is below 1.81, so Harlan lies in the distress zone, although its relatively high asset turnover contributes most of the score. The warning is real but not decisive; the next step is a liquidity forecast and an assessment of whether the business or the balance sheet is broken.

23.5.3 Out-of-court workouts

Before filing, firms and creditors negotiate: maturity extensions, covenant waivers, interest deferrals, debt-for-equity swaps, exchange offers (new securities in exchange for old, often at a discount), and sale of assets. Workouts are faster and cheaper than court processes, but they face a holdout problem: each creditor prefers that others make concessions, and under bond indentures a unanimous consent is often required to change payment terms. Distressed exchanges therefore often include exit consents that strip covenants from holdouts' bonds, or are combined with a prepackaged court filing in which creditors vote before filing to bind dissenters.

23.6 Bankruptcy: Liquidation and Reorganization

23.6.1 The legal frameworks

Insolvency regimes differ by country, but the common structure involves two paths. Liquidation (Chapter 7 in the United States) sells assets, and a trustee distributes proceeds to claimants. Reorganization (Chapter 11 in the United States) keeps the business operating under court supervision while a plan is negotiated; an automatic stay prevents creditors from seizing assets, the debtor typically continues in possession of its business, and it may obtain debtor-in-possession (DIP) financing that is given priority over existing debt. Other jurisdictions use administration procedures, schemes of arrangement or safeguard proceedings, with different balances of creditor and debtor power. Cross-border cases often hinge on where the company's centre of main interests lies. This chapter uses the US vocabulary for illustration and does not provide legal advice.

23.6.2 The absolute priority rule

Claims are paid in order of legal priority: secured creditors from their collateral, then administrative expenses (including professional fees and DIP loans), priority claims (certain wages and taxes), general unsecured creditors, subordinated creditors, preferred equity and finally common equity. Under the absolute priority rule (APR), a junior class receives nothing until all senior classes are paid in full. In practice, negotiated plans often deviate: junior holders may receive small recoveries ("gifts" or warrants) in exchange for consent, and a plan can be confirmed over a dissenting class if it is "fair and equitable" (the cramdown standard) and at least one impaired class accepts.

Example 23.7 – A liquidation recovery waterfall. Harlan Metals files for liquidation. Net proceeds from asset sales are $540m, shared as follows: the secured term loan has a $400m claim on collateral worth enough to cover it; administrative and priority claims total $40m; general unsecured claims are $300m of notes and $100m of trade debt (pari passu); subordinated debentures are $150m; equity is last.

Step 1: Secured creditors: take $400m from the proceeds. Recovery = 100 percent. Remaining = 540 − 400 = $140m.

Step 2: Administrative and priority claims: $40m, paid in full. Remaining = $100m.

Step 3: General unsecured: total claims = 300 + 100 = $400m. Available $100m. Recovery rate = 100 / 400 = 25 percent. Notes receive 25% × 300 = $75m; trade creditors receive 25% × 100 = $25m.

Step 4: Subordinated debentures and equity: nothing remaining. Recovery 0 percent.

Class	Claim (m)|*Recovery*(m)	Recovery rate	
Secured term loan	400	400	100%
Administrative and priority	40	40	100%
Senior unsecured notes	300	75	25%
Trade creditors	100	25	25%
Subordinated debentures	150	0	0%
Equity	n/a	0	0%
Total distributed		**540**	

In practice, administrative costs of the process reduce the distributable value further, and secured creditors often hold less than full collateral value, so recovery rates are typically lower and more uncertain than simple waterfalls suggest. Note that structural subordination also matters: creditors of an operating subsidiary are paid from that subsidiary's assets before a holding company's creditors share in the residual equity value.

23.6.3 The plan of reorganization and the fulcrum security

A reorganization plan must determine the going-concern value of the reorganized business, the new capital structure it can bear, and how claims are exchanged for new debt, equity or cash. The **fulcrum security** is the most senior class that is not paid in full; it is the class that, under the absolute priority rule, receives the equity of the reorganized company. Junior classes are out of the money, and senior classes are unimpaired or receive new debt equal to their claims. Valuation is therefore the battleground: a higher enterprise value moves the fulcrum down the capital structure and benefits junior creditors; a lower value moves it up and benefits the senior creditors who take the equity.

Example 23.8 – Finding the fulcrum and allocating a plan. Corvane Retail (fictional) emerges from Chapter 11 with a negotiated enterprise value of $540m. Its claims are first-lien term loan $450m, second-lien notes $200m, and general unsecured $300m.

Step 1: First lien: claim $450m versus EV $540m. It is covered; the residual after first lien = 540 − 450 = $90m.

Step 2: Second lien: claim $200m versus residual $90m. Recovery under APR = 90 / 200 = 45 percent. The second lien is the fulcrum security: it is the first class not paid in full.

Step 3: General unsecured and equity: residual after second lien is 0, so recovery is zero under strict APR.

Step 4: Plan design: first lien is reinstated (amended, with maturity extension) at $450m. New equity is valued at $90m. Second-lien holders receive 95 percent of new equity (value $85.5m, recovery 42.8 percent of their $200m claim), and unsecured creditors receive 5 percent of new equity as a settlement payment ($4.5m, recovery 1.5 percent of $300m) to secure their acceptance and avoid a costly valuation trial. That is a deviation from strict APR, which the second lien accepts as cheaper than litigation.

Class	Claim	Treatment	Value received	Recovery
First-lien term loan	450	Reinstated	450.0	100%
Second-lien notes	200	95% of new equity	85.5	42.8%
General unsecured	300	5% of new equity	4.5	1.5%
Old equity	n/a	Cancelled	0	0%

If a rival valuation puts enterprise value at $620m, the first lien still recovers 100 percent, the second lien equity pool becomes 170 and its recovery 85 percent. A $80m (15 percent) valuation difference moves the second lien’s recovery from 45 to 85 percent, which shows why valuation experts and discount rates become the center of the dispute. For junior claimants, valuation is an option: the lower the value the less the junior class gets, but the higher the volatility the more their out-of-the-money claim is worth (warrants and contingent value rights are used to bridge differences in view).

23.6.4 Restructuring beyond the balance sheet

Financial restructuring alone rarely saves a business. Successful turnarounds combine four elements: stabilizing liquidity (13-week cash flow forecasting, DIP financing, working capital discipline), fixing the operating model (closing loss-making sites, renegotiating leases and contracts, repricing), resolving the capital structure (debt-for-equity swaps, maturity extension, haircut) and rebuilding stakeholder confidence (a credible business plan, new management or a chief restructuring officer). A “amend and extend” can buy time, but if the underlying business is impaired it merely delays a filing. Sponsors in distress face a decision: inject additional equity to protect their existing investment (an option on recovery), hand over the keys to lenders, or negotiate partial support. Rational sponsors compare the option value of additional equity to the probability-weighted gain; sunk costs are irrelevant.

23.7 Spin-Offs, Carve-Outs and Divestitures

23.7.1 Forms of separation

Companies can reduce their scope through several mechanisms, each with different cash, control and tax consequences.

Form	What happens	Cash to parent	Control after	Typical purpose
Divestiture (sale)	Business sold to a strategic or financial buyer	Yes (sale price)	None	Raise cash, exit non-core
Spin-off	Subsidiary shares distributed pro rata to parent shareholders	No	Independent public company	Unlock focus, remove conglomerate discount
Split-off	Parent shareholders exchange parent shares for subsidiary shares	No (buyback effect)	Independent	Tax-efficient reduction of parent share count
Equity carve-out (IPO)	Minority stake of the subsidiary sold to public	Yes (IPO proceeds)	Parent keeps majority and consolidates	Raise capital, create currency and market price
Tracking stock	Parent issues shares linked to a division's performance	Possibly	Parent retains legal ownership	Separate valuation without legal separation

Tax treatment is crucial. In many jurisdictions, a spin-off that satisfies tests (active business, continuity of ownership, valid business purpose, no device to distribute earnings) can be tax-free for the company and shareholders, whereas a taxable sale generates a capital gain. These rules are technical and vary by jurisdiction; the managerial point is that the tax cost of separation can offset the strategic benefit and must be modeled.

23.7.2 Why separation might create value

The case for separation rests on several arguments.

- **Focus and managerial incentives:** a stand-alone company can tie executive pay to its own stock price rather than to a diversified average.
- **Information and valuation:** analysts specialized in one industry can value a pure-play more accurately; opaque conglomerates trade at a "conglomerate discount" if investors cannot value divisions or fear cross-subsidization.
- **Capital allocation:** the corporate center may starve high-growth units to fund legacy ones. Separation lets each unit raise capital at a cost reflecting its own risk.
- **Investor clientele:** a high-growth division and a high-dividend utility attract different shareholders.
- **Takeover readiness:** a stand-alone unit is easier for a bidder to acquire.

Against these, separation destroys any real synergies (shared purchasing, brand, distribution, tax capacity and the internal capital market), creates dis-synergies (duplicated corporate costs, loss of diversification that reduces debt capacity), and incurs one-time costs (legal, systems separation, rebranding). The conglomerate discount is also a measurement problem: part of the observed discount reflects the fact that firms diversify when their original industries are weak, not that diversification itself destroys value. The honest test is a sum-of-the-parts (SOTP) analysis net of costs.

23.7.3 Sum-of-the-parts valuation

SOTP values each segment at the multiple of comparable pure-play companies, capitalizes unallocated corporate costs, and subtracts consolidated net debt:

```
Equity value = Σ (Segment EBITDA × Peer multiple) − (Corporate costs × Multiple) − Net debt
```

Example 23.9 – SOTP and the net value of a spin-off. Stratus Holdings (fictional) has an Industrial division with EBITDA $200m (peer multiple 8.0x), a Consumer division with EBITDA $120m (peer multiple 11.0x), unallocated corporate costs of $30m capitalized at 8.0x, net debt of $900m and 100 million shares trading at $14.50.

Step 1: Industrial = 200 × 8 = $1,600m. Consumer = 120 × 11 = $1,320m. Corporate = −30 × 8 = −$240m.

Step 2: SOTP enterprise value = 1,600 + 1,320 − 240 = $2,680m. Equity value = 2,680 − 900 = $1,780m, or $17.80 per share.

Step 3: Current market EV = 14.50 × 100 + 900 = $2,350m, an implied multiple of 2,350 / 290 = 8.1x on combined EBITDA net of corporate costs (200 + 120 − 30 = 290). The conglomerate discount = (2,680 − 2,350) / 2,680 = 12.3 percent.

Step 4: Adjust for separation. A spin-off duplicates corporate functions, adding $15m of annual recurring costs (capitalized at 8.0x = $120m) and costs $60m one time. Net EV = 2,680 − 120 − 60 = $2,500m. Equity = 2,500 − 900 = $1,600m, or $16.00 per share.

Step 5: Value creation relative to market = 16.00 / 14.50 − 1 = 10.3 percent, or $150m in total, if investors re-rate each part to peer multiples. This depends on the key assumption that peer multiples apply. If the Consumer division would trade at 10x instead of 11x, its value falls by $120m and the net gain shrinks from $150m to $30m, which is within the uncertainty of the estimate. The decision should therefore hinge on credible evidence the discount is structural and removable.

23.7.4 Equity carve-outs

An equity carve-out sells a minority stake in a subsidiary through an IPO. It generates cash (primary proceeds to the subsidiary or secondary proceeds to the parent), establishes a market price that can be used to incentivize managers, and often serves as a stepping stone to a full spin-off or sale. Carve-outs often involve underpricing of 10 to 15 percent, which transfers value to the new investors, and create agency issues because the parent controls a company with minority public holders (related-party transactions, transfer pricing). Control and consolidation remain with the parent when it keeps more than 50 percent.

Example 23.10 – Pricing a carve-out. Stratus carves out 20 percent of Consumer. The division's EV is $1,320m (11x) and $300m of Stratus's net debt is pushed down to it. The IPO is priced at a 10 percent discount to fair value.

Step 1: Consumer fair equity value = 1,320 − 300 = $1,020m.

Step 2: 20 percent stake fair value = 0.20 × 1,020 = $204.0m.

Step 3: IPO proceeds at a 10 percent discount = 204.0 × 0.90 = $183.6m. The underpricing cost to Stratus shareholders is 204.0 − 183.6 = $20.4m. If the proceeds are used to repay debt, Stratus net debt falls to 900 − 183.6 = $716.4m (consolidated). The parent still consolidates 100 percent of Consumer's EBITDA but 20 percent of its equity value is now held by outside investors.

A carve-out thus raises funding at a cost of 10 percent of the amount sold, which compares with the cost of an equivalent equity issue at the parent. It makes sense when the subsidiary's valuation is expected to exceed that implied within the parent, or when it is a precursor to complete separation.

Case Study: Halden Foods and the Carved-Up Conglomerate

(Fictional company and figures, for teaching purposes only.)

Halden Foods is a diversified food group with three divisions: Ambient Grocery (shelf-stable products), Frozen & Chilled, and Foodservice Distribution. Combined revenue is $3.2 billion and EBITDA is $400 million (12.5 percent margin). Net debt is $1.6 billion, or 4.0 times EBITDA, a legacy of a leveraged recapitalization five years ago in which Halden's controlling shareholder, Brightwater Partners, a private equity firm, extracted a $500 million dividend. Brightwater holds 62 percent of the shares and the rest is publicly traded at 7.5 times EBITDA, below the 9.5 times average multiple of branded grocery peers and the 8.0 times of distributors.

Divisional EBITDA is: Ambient $210 million, Frozen & Chilled $120 million, Foodservice $110 million, less unallocated corporate costs of $40 million. Foodservice has been losing contracts to a larger rival, and a 12 percent margin decline in the last year pushed Halden's total interest coverage to 2.4 times. Its senior secured term loan carries a maximum leverage covenant of 4.5 times, so EBITDA can fall only to about $356 million, a cushion of roughly $44 million, before the covenant is breached. The $700 million of senior unsecured notes mature in 26 months, and refinancing them at current market yields would add roughly $25 million to annual interest.

The board is considering three options. Option A is to sell Foodservice to a strategic buyer, which has indicated a price of 7.0 times divisional EBITDA, and use the proceeds to repay the term loan. Option B is to spin off Ambient Grocery, the highest-quality business, as a separately listed company with a lower debt level of 2.5 times, leaving Frozen & Chilled and Foodservice together with the remainder of the debt. Option C is a standstill: operate as is, cut costs by $30 million, and refinance the notes at current rates.

Brightwater's fund is in year seven of a ten-year life and wants liquidity within three years. The CFO worries that option B leaves the remaining company too levered, and that the notes' covenants may require note holders' consent to the spin-off because it removes the largest asset pool. The treasury team estimates one-time separation costs of $90 million for the spin-off and $35 million in transaction costs for the sale, and annual dis-synergies of $20 million from duplicated corporate functions in option B.

Employees, suppliers and the controlling shareholder have competing interests. Ambient's managers want a listing so equity-based pay tracks their own performance. Foodservice's customers fear a buyer will rationalize product lines. The lenders prefer option A, which delevers directly. Minority shareholders suspect that Brightwater will favor the structure that gets its fund the earliest liquidity.

Discussion questions

1. Compute a sum-of-the-parts valuation for Halden using the multiples given (Ambient at 9.5x, Frozen & Chilled at 9.0x as an assumed peer multiple, Foodservice at 7.0x, corporate costs capitalized at 8.0x) and compare it with the market-implied enterprise value. How large is the apparent conglomerate discount, and how much of it could be explained by the leverage and the Foodservice problem rather than by opacity?
2. For option A, calculate pro forma leverage after using the sale proceeds to repay debt, and for option B, estimate the leverage of the remaining company after the Ambient spin-off with 2.5 times leverage on its EBITDA. Which structure leaves the more stable credit profile? Consider the covenant, the note maturity and the ability to refinance.
3. Brightwater's incentives differ from those of minority holders and lenders. Identify the conflicts of interest and recommend governance safeguards (independent committee, fairness opinion, minority approval) for whichever option the board chooses.
4. What early-warning indicators should Halden's treasury monitor, and what would trigger a pre-emptive restructuring discussion with lenders? Should the board consider a distressed exchange of the notes?

Key Takeaways

- An LBO finances a purchase mainly with debt serviced by the target's own cash flow; it works best for stable, cash-generative, asset-rich businesses with identifiable improvements and clear exits.
- The sources-and-uses table anchors every deal: sponsor equity is the plug after debt and fees, and fees are real value leakage.
- Debt paydown transfers enterprise value from creditors to equity holders, but early-year free cash flow is thin, so downside stress tests and interest hedging matter more than the base case.
- Sponsor returns decompose into EBITDA growth, multiple change and net debt reduction, less fees; a return that depends on multiple expansion is a market bet, not a business plan.
- IRR falls as the holding period lengthens even as MOIC rises, which explains the appeal of early exits and dividend recapitalizations; a recap raises IRR but increases leverage and risk.
- Fund economics (management fee, hurdle, catch-up, carry) create a call-option payoff for the GP; net LP returns are meaningfully below gross deal returns.
- Distress is a financial problem, economic failure is an operating problem; restructuring succeeds only when the going-concern value exceeds liquidation value and both the balance sheet and the business are fixed.
- Under the absolute priority rule, the fulcrum security is the first class not paid in full; valuation disputes decide how far down the capital structure value reaches.
- Spin-offs and carve-outs create value only when the discount to a sum-of-the-parts is structural and exceeds the dis-synergies, tax and separation costs.
- Delay is costly in distress and in separation: early, credible negotiation with creditors and a clear strategic rationale preserve value.

Review Questions

1. Why are stable cash flows, tangible assets and low capital intensity attractive characteristics for an LBO target? Which industries would you avoid and why?
2. Explain the three theoretical rationales for high leverage in a buyout (tax shield, free cash flow discipline, incentive alignment). Which is hardest to verify empirically?
3. What is the difference between a term loan B, a senior unsecured note and a mezzanine tranche in terms of seniority, cost and covenants?
4. Why is sponsor equity the plug in a sources-and-uses table, and how do fees affect returns?
5. Why does extending the holding period reduce IRR even when MOIC rises? What incentives does this create for the sponsor?
6. Describe the attribution of LBO returns into growth, multiple and deleveraging. Why is multiple expansion considered the least reliable source?
7. What is a dividend recapitalization, and how does it change IRR, MOIC and risk? Who bears the additional risk?
8. Explain the hurdle rate, catch-up and clawback in a PE fund. How do these terms affect the GP's incentives?
9. Distinguish financial distress from economic failure. How does debt overhang cause underinvestment?
10. What are the advantages and risks of a spin-off versus an equity carve-out, and what must be demonstrated before a break-up is announced?

Problems

1. **Sources and uses.** A sponsor buys a company with EBITDA of $60m at 8.0x. Fees are 3 percent of enterprise value, minimum cash funded is $5m, and debt is 5.0x EBITDA. Compute total uses, sponsor equity and the equity percentage of total capitalization.
2. **Simple returns.** The sponsor in Problem 1 sells the company four years later for exit equity of $600m after investing $200m of equity. Compute MOIC and IRR.
3. **Attribution.** In Problem 1, exit EBITDA is $80m, the exit multiple is 9.0x and exit net debt is $220m. Entry net debt is debt less minimum cash. Decompose the equity gain (using the fees from Problem 1 and entry equity of $199.4m) into EBITDA growth, multiple change, net debt reduction and fees, and verify that it sums to the total gain.
4. **Recap IRR.** A sponsor invests $200m. In year 3 the company pays a $100m recap dividend, and in year 5 the sponsor receives $450m at exit. Without the recap the exit proceeds in year 5 would have been $520m. Compute the IRR and MOIC with and without the recap.
5. **Coverage.** A company has EBITDA of $60m, debt of $300m at 9.5 percent interest and capex of $15m. Compute EBITDA / interest and (EBITDA − capex) / interest. Is either below a 1.5x warning threshold?
6. **Recovery waterfall.** A firm in liquidation has distributable value of $300m. Secured debt is $180m, unsecured debt is $220m and subordinated debt is $100m. Compute each class's recovery in dollars and as a percentage.
7. **Altman Z.** Total assets $500m, working capital $25m, retained earnings $30m, EBIT $20m, market value of equity $90m, total liabilities $400m, sales $450m. Compute Z and classify.
8. **SOTP.** Division A EBITDA $50m at 7.0x, Division B EBITDA $30m at 12.0x, corporate costs $10m capitalized at 8.0x, net debt $200m, 40 million shares. Compute the SOTP value per share, then subtract a separation with $4m of recurring dis-synergies (capitalized at 8.0x) and $25m one-time costs.

Answers to Selected Problems

1. EV = 8 × 60 = $480m; fees = $14.4m; uses = 480 + 14.4 + 5 = $499.4m; debt = $300m; sponsor equity = $199.4m, or 39.9 percent of capitalization.
2. MOIC = 600 / 200 = 3.0x; IRR = 3^(1/4) − 1 = 31.6 percent.
3. Exit EV = 9 × 80 = $720m; exit equity = 720 − 220 = $500m; gain = 500 − 199.4 = $300.6m. EBITDA growth = (80 − 60) × 8 = $160m; multiple change = (9 − 8) × 80 = $80m; net debt reduction = (300 − 5) − 220 = $75m; fees = − $14.4m. Sum = 160 + 80 + 75 − 14.4 = $300.6m.
4. With recap: IRR ≈ 24.8 percent, MOIC = (100 + 450) / 200 = 2.75x. Without: IRR = 2.6^(1/5) − 1 ≈ 21.1 percent, MOIC = 2.6x.
5. Interest = 300 × 9.5% = $28.5m. EBITDA / interest = 2.11x; (EBITDA − capex) / interest = 45 / 28.5 = 1.58x. Neither is below 1.5x, but the second is close.
6. Secured: $180m (100 percent). Remaining $120m to unsecured: 120 / 220 = 54.5 percent. Subordinated: $0 (0 percent).
7. X1 = 0.05, X2 = 0.06, X3 = 0.04, X4 = 90 / 400 = 0.225, X5 = 0.90. Z = 1.2 × 0.05 + 1.4 × 0.06 + 3.3 × 0.04 + 0.6 × 0.225 + 0.90 = 0.06 + 0.084 + 0.132 + 0.135 + 0.90 = 1.311. Below 1.81: distress zone.
8. EV = 50 × 7 + 30 × 12 − 10 × 8 = 350 + 360 − 80 = $630m; equity = 430; per share = $10.75. After separation: 630 − 4 × 8 − 25 = $573m EV; equity = 373; per share = $9.325, so the separation costs $1.43 per share unless the market re-rates the segments above the peer multiples used here.

Chapter 24: Sustainable Finance, Governance, Fintech and the Future of Corporate Finance

Learning Objectives

- Explain how environmental, social and governance (ESG) factors enter valuation through cash flows, discount rates and risk, and incorporate an internal carbon price into capital budgeting.
- Compare green, social and sustainability-linked bonds, quantify the greenium and step-up economics, and recognize the risks of greenwashing.
- Analyze corporate governance as a response to agency problems, including board structure, ownership, activism and takeover defenses.
- Evaluate executive compensation design, value stock options with the Black-Scholes model, and assess performance-share plans and clawbacks.
- Describe at a conceptual level how payments, fintech lending and digital assets change financing, treasury and business-model economics, and compute simple unit economics.
- Apply parity relationships to international capital budgeting, adjust for political risk, and integrate the book’s themes into a strategic value framework.

Opening Vignette

(Illustrative scenario; the company and all figures are fictional.)

Aldergate Industries is a $6 billion revenue manufacturer of heating and ventilation equipment, listed on a European exchange and active in thirty countries. On a single Tuesday its finance leadership faces four unrelated-looking requests.

The sustainability team wants $600 million of capex to electrify two foundries, cutting emissions by roughly 80 percent. The business case on a conventional discounted cash flow basis shows a slightly negative NPV. The treasurer proposes funding it with a green bond and says investors will pay a lower coupon. The head of investor relations warns that two large asset managers have asked for evidence that the bond’s use of proceeds is genuinely incremental.

The remuneration committee chair, meanwhile, is reviewing next year’s long-term incentive plan. A proxy adviser criticized last year’s grants because payouts depended on earnings per share, which management could lift through buybacks, and because the plan paid out at threshold performance. She wants a plan tied to relative shareholder return and return on invested capital, with a clawback.

A fintech start-up has approached the corporate development team offering a payments platform that would embed financing in Aldergate’s distributor network. The chief financial officer must decide whether to build, partner or buy, and what to do about customers who suggest paying in stablecoins.

Finally, the strategy team has proposed a plant in a country whose currency has lost half its value over a decade and whose government has restricted dividend remittances twice in the last fifteen years.

Each request is a corporate finance problem in disguise, and each touches the central question of this book: how do decisions about investment, financing and payout create or destroy value for those who own the firm and, increasingly, for the society in which it operates? This chapter reviews the tools used to answer these questions and ends with a synthesis for the finance leader of the next decade.

24.1 ESG, Materiality and the Valuation Channels

24.1.1 What ESG means for a financial decision-maker

ESG is a grouping of non-traditional factors: environmental (emissions, energy, water, biodiversity, climate transition and physical risk), social (workforce safety, labor practices, supply chain, product safety, data privacy, community) and governance (board independence, ownership rights, audit quality, pay, ethics). The label covers very different things, and for a CFO the useful question is not whether a firm is "ESG" but through which channel a particular factor affects value. There are four channels.

1. **Cash flows.** Carbon prices, energy-efficiency standards, product bans and customer preferences change revenues and costs. A firm that sells low-carbon products may gain share and pricing; one that depends on high-emission inputs faces cost increases.
2. **Risk and discount rates.** Physical climate risk (floods, heat, storms), transition risk (stranded assets) and litigation increase the variability of cash flows and may raise the systematic or idiosyncratic risk of the firm. Some investors screen out certain sectors, reducing the demand for those shares, which can raise their required return.
3. **Cost and availability of financing.** Lenders and insurers price climate exposure; sustainability-linked loans tie margins to targets; some investors have mandates that restrict holdings.
4. **Franchise value and talent.** Reputation, regulatory goodwill and employee engagement affect retention, productivity and the license to operate.

The concept of **materiality** distinguishes ESG factors that affect enterprise value (financial materiality) from those that affect society or the environment independently of financial effects. The term "double materiality" (used in some regulatory regimes) requires disclosure of both. A financial manager should focus first on financial materiality, which differs by industry: water is material to a beverage maker, data security to a payments company and workplace safety to a mining company.

24.1.2 What the evidence permits us to say

Empirical research on whether ESG performance leads to higher returns or lower cost of capital is mixed and sensitive to definitions, rating providers (whose scores diverge widely for the same company), sample periods and control variables. Reasonable and defensible claims are limited. Firms with serious governance and compliance failures tend to experience severe negative outcomes. Some material environmental and social issues do affect operating results over the long term. High-profile ESG ratings are only loosely correlated with each other. And in equilibrium theory, if many investors avoid certain stocks, those stocks should earn higher expected returns (a "sin stock" premium) while green stocks earn lower expected returns, even though green firms can still be good businesses. The last point is subtle and important: a lower cost of capital for green firms is the flip side of lower expected returns for green investors. It does not imply that every green project is a good investment.

24.1.3 Pricing carbon in capital budgeting

Many companies use an internal carbon price, a per-ton charge applied to project emissions in the NPV analysis, as a way to prepare for current and future regulation and taxes. It is a shadow price: no cash may change hands today, but the price approximates the expected cost of emissions through taxes, permits, customer pass-through or stranded-asset risk. The method is straightforward: forecast emissions each year, multiply by the forecast carbon price, treat the amount as an incremental cost and recompute NPV. Because the price path is uncertain, firms also compute break-even carbon prices and run scenarios.

Example 24.1 – Carbon-adjusted NPV of two foundry upgrades. Aldergate evaluates two ways of replacing a foundry's heat source. Project A (gas-fired upgrade) costs $40m, generates $9.5m of annual after-tax operating cash flow for 10 years and emits 60,000 tons of CO2 per year. Project B (electrification) costs $52m, generates $9.8m per year and emits 10,000 tons per year. The discount rate is 9 percent. The internal carbon price is $60 per ton in year 1, rising 6 percent per year. Cash flows are annual, end of year.

Step 1: Annuity factor, 10 years at 9 percent = (1 − 1.09^−10) / 0.09 = 6.4177.

Step 2: Conventional NPV. A: −40 + 9.5 × 6.4177 = −40 + 60.968 = **$20.97m**. B: −52 + 9.8 × 6.4177 = −52 + 62.893 = **$10.89m**. Without carbon, A wins by $10.1m.

Step 3: Carbon cost in year 1. A: 60,000 × $60 = $3.60m. B: 10,000 × $60 = $0.60m. Each later year grows by 6 percent.

Step 4: PV of carbon costs. For a growing annuity, PV = C1 / (r − g) × [1 − ((1 + g)/(1 + r))^n] = C1 / 0.03 × [1 − (1.06/1.09)^10]. The ratio 1.06/1.09 = 0.97248, and raised to the 10th power it is 0.7565, so the bracket is 0.2435 and the multiplier is 0.2435 / 0.03 = 8.118. A: 3.60 × 8.118 = **$29.22m**. B: 0.60 × 8.118 = **$4.87m**.

Step 5: Carbon-adjusted NPV. A = 20.97 − 29.22 = **−$8.26m**; B = 10.89 − 4.87 = **+$6.02m**.

The ranking reverses. The conventional analysis favors the gas upgrade by $10.1m; a carbon price of $60 growing 6 percent makes the electrified foundry better by $14.3m. The break-even initial carbon price at which the two NPVs equalize is approximately $60 × (10.08 / 24.35) = $25 per ton in year 1 (the difference in carbon PV, 29.22 − 4.87 = $24.35m at a $60 starting price, is linear in that price). Managers should also test the cost of electricity and the reliability of supply, since B's advantage depends on the power price assumption. The method is only as good as the carbon-price forecast; the right use is to show what carbon price makes the decision change.

Common mistakes. Using a carbon price only as a tie-breaker; applying it to project emissions but not to the firm's competitors, so the strategic effect is lost; and double counting if tax and permit costs are already in cash flows.

24.2 Sustainable Debt: Green, Social and Sustainability-Linked Instruments

24.2.1 Use-of-proceeds versus performance-linked structures

Sustainable debt comes in two broad structures.

Use-of-proceeds bonds (green, social, sustainability bonds) are ordinary debt obligations of the issuer, with proceeds earmarked for eligible projects such as renewable energy, energy efficiency, clean transport or affordable housing. Credit risk is that of the issuer, not of the project (they are not project finance). Market practice, reflected in voluntary principles administered by industry bodies, includes a framework describing eligible categories, a process for project selection,

management of proceeds in a separate account or sub-portfolio, and annual reporting on allocation and, where feasible, impact. Many issuers obtain a second-party opinion or external verification, and some jurisdictions have introduced taxonomies and standards for labeled bonds.

Sustainability-linked bonds (SLBs) and loans have no use-of-proceeds restriction. The issuer commits to improve measurable key performance indicators (KPIs), such as reducing greenhouse gas emissions by a stated percentage, against sustainability performance targets (SPTs) by a given date. If the target is missed, the financial characteristics change, usually through a coupon step-up of 12.5 to 25 basis points. Their credibility depends on the ambition of the target relative to the issuer's baseline and peers, materiality of the KPI to the business, and the size and certainty of the penalty.

Feature	Green bond	Sustainability-linked bond	Conventional bond
Use of proceeds	Earmarked to eligible green projects	General corporate purposes	General corporate purposes
Financial feature	Standard coupon (possible greenium)	Coupon step-up if KPI target missed	Standard coupon
Reporting	Allocation and impact reports	KPI performance verification	Financial reporting
Key risk to investors	Greenwashing, project quality	Weak targets; small penalty	Credit risk
Issuer risk	Reputational if misuse; reporting cost	Step-up cost; reputational if missed	None specific
Investor base	Wider (green mandates)	Wide, but some mandates exclude	Standard

24.2.2 The greenium and the economics of labeling

The **greenium** is the yield advantage of a labeled bond relative to an otherwise identical conventional bond from the same issuer. Where it exists it tends to be small, often a few basis points and not stable across markets, maturities and credit qualities; demand from mandate-driven investors can compress yields, but supply is large and many studies find a small or insignificant effect. The key point for treasurers is that the benefit should be compared with the costs: external review, ongoing reporting and tracking systems, and the legal risk of misstatement. The more relevant benefits may be a broader investor base, oversubscription, lower execution risk and strategic signaling, but these are hard to quantify.

Example 24.2 – Is a 6 basis point greenium worth the cost? Aldergate issues a $500m 10-year green bond with an estimated greenium of 6 bp. Incremental annual reporting and verification cost is $0.12m, and a one-time external review and framework design cost is $0.15m. The after-tax discount rate for these small flows is taken as the pre-tax bond yield of 5.0 percent (a simplification; ignore tax effects).

Step 1: Annual interest saving = 500m × 0.0006 = $0.30m.

Step 2: Annuity factor, 10 years at 5 percent = 7.7217.

Step 3: PV of savings = 0.30 × 7.7217 = $2.317m.

Step 4: PV of incremental reporting = 0.12 × 7.7217 = $0.927m.

Step 5: Net value = 2.317 − 0.927 − 0.150 = **$1.24m**.

The label is worth a modest $1.2m, about 0.25 percent of the amount raised. If the greenium were zero, the net value would be −$1.08m (−0.927 − 0.150), a small cost that must be justified by wider investor demand, a lower new-issue concession or the reputational value of the program. The calculation also shows the main sensitivity: halving the greenium to 3 bp

reduces savings to $1.16m and net value to +$0.08m. Because a green bond is an ordinary corporate obligation, the issuer's credit rating, not the label, still dominates pricing.

Example 24.3 – Expected cost of a sustainability-linked step-up. Aldergate considers a $500m 10-year SLB that pays a 4.50 percent coupon but steps up by 25 bp for years 6 to 10 if its emissions-intensity target, tested at the end of year 5, is missed. Management estimates a 40 percent chance of missing the target. The discount rate is 5 percent.

Step 1: Annual step-up cost if triggered = 500m × 0.0025 = $1.25m, paid at the end of years 6, 7, 8, 9 and 10.

Step 2: Value at year 5 of five-year annuity at 5 percent = 1.25 × 4.3295 = $5.412m.

Step 3: Discount to today: 5.412 / 1.05^5 = 5.412 / 1.27628 = $4.240m.

Step 4: Expected cost = 0.40 × 4.240 = **$1.70m**, equal to 0.34 percent of principal.

If the issuer is confident of meeting the target (probability 10 percent), the expected cost falls to $0.42m. Investors in turn consider the penalty trivial relative to the principal and may price the bond as a standard bond; critics say step-ups of 25 bp are too small to change behavior, since the cost of missing equals less than one year's savings from a more ambitious program. A credible SLB sets targets that exceed business-as-usual trends and a step-up that matters. Note the circularity: the issuer benefits from selecting an easy target, which is why external verification and benchmarking matter.

24.2.3 Greenwashing and the discipline of credibility

Greenwashing is the gap between claims and outcomes. For a financial manager, the risk shows up as regulatory enforcement and litigation, investor mistrust and exclusion from indices, and reputational damage. Controls include a documented framework approved by the board, auditable tracking of proceeds, independent verification, a clear and measurable KPI baseline, and alignment of the labeled financing with the strategic decarbonization plan, which should itself be credible enough to survive a skeptical read. Transition finance, funding to help high-emission businesses decarbonize, is an emerging category with high demand for credibility because it carries the greatest risk of being a label rather than a plan.

24.3 Corporate Governance

24.3.1 Governance as the management of agency conflicts

Governance is the set of mechanisms by which suppliers of capital ensure that managers act in their interests. It is needed because ownership is separated from control (Berle and Means; Jensen and Meckling), managers have private information and private benefits (empire building, perquisites, entrenchment), and contracts are incomplete. Governance exists at several levels.

- **Board of directors.** The board hires, fires and pays the CEO, approves strategy and capital allocation, and oversees risk and audit. Effectiveness depends less on formal independence than on information, expertise, time, and the willingness to dissent. Common practices include a majority of independent directors, a separate chair and CEO or a strong lead independent director, fully independent audit, compensation and nomination committees, regular executive sessions, and director refreshment. Evidence on the link between board features and performance is mixed because boards are endogenous: troubled firms change their boards.
- **Ownership structure.** Dispersed ownership creates free-riding by small holders; large blockholders can monitor but may extract private benefits at the expense of minorities. In many countries, controlling families or the state are the

main agency issue, and the conflict is between controlling and minority shareholders (tunneling, related-party transactions).

- **Control enhancements.** Dual-class shares, pyramids, cross-holdings and golden shares separate voting rights from cash-flow rights. Founders may argue that they protect long-term vision; critics say they entrench management and allow control with little capital at risk. Sunset provisions, which end the dual-class structure after a period, are a compromise.
- **Market discipline.** Product market competition, the market for corporate control (takeovers), analysts, creditors and the legal system impose discipline.
- **Disclosure and audit.** Accurate financial reporting, independent audits and effective internal controls are prerequisites for any monitoring.

24.3.2 Shareholder rights, activism and takeover defenses

Shareholders influence firms through voting (director elections, say-on-pay, mergers), engagement with management, proposals, and, if all else fails, selling. Activist investors take a stake, publicly push for change (divestitures, capital return, board seats, leadership change) and benefit if the share price rises. Research suggests that targets of activism often show operating improvements and positive average announcement returns, although the long-run effects are debated and activists may favor short-term payouts over long-term investment.

Takeover defenses include staggered boards, poison pills (rights plans that dilute a hostile acquirer), supermajority requirements, and restrictions on written consent. These protect the board's ability to negotiate a better price but also insulate poorly performing managers. A reasonable rule: the stronger the defenses, the more important independent oversight becomes. Proxy advisers and institutional investors generally favor annual director elections and the removal of pills unless approved by shareholders; no universal best practice exists because the right level of protection depends on ownership, the strategic position of the firm and its need for patient capital.

24.3.3 Stakeholders and purpose

A long-running debate concerns whether firms should maximize shareholder value or balance the interests of stakeholders. The disciplined position, consistent with the rest of this book, is that long-run shareholder value cannot be created by ignoring customers, employees, suppliers and regulators, and that stakeholder choices that are costly in the short run can be justified when they preserve long-run cash flows. The danger of an unlimited multi-stakeholder mandate is that managers become unaccountable: with no single metric, almost any decision can be defended. Practical governance responses include clear articulation of purpose, board oversight of material ESG risks, linking compensation to a small number of measurable outcomes and transparent reporting of trade-offs.

24.4 Executive Compensation

24.4.1 The design problem

Compensation is a contract designed to induce effort and appropriate risk-taking when the principal cannot observe effort. Standard components are base salary, annual bonus, long-term incentives (LTIs: stock options, restricted stock units, performance shares) and benefits and retirement plans. A well-designed package has the following characteristics.

- **Pay for performance.** A material share of pay is variable and linked to measures the executive can influence.

- **Long-term orientation.** Vesting periods of three to five years, holding requirements after vesting and minimum shareholding guidelines align the executive with long-term holders.
- **Risk calibration.** Pure option plans reward volatility because option value rises with volatility; plans with leverage in the payoff encourage risk-taking beyond what shareholders want. Debt-like instruments (pensions, deferred compensation) align executives with creditors, which is relevant for banks.
- **Relative measurement.** Relative total shareholder return (TSR) against peers filters out industry and market movements that executives do not control, reducing windfall gains.
- **Clawbacks and malus.** Provisions to recover or withhold pay after misstatements or misconduct strengthen accountability.
- **Transparency and simplicity.** Complex plans with many metrics blur incentives.

Typical pitfalls include metrics that can be manipulated (EPS through buybacks, adjusted EBITDA that excludes recurring costs), payouts for mediocre results because thresholds are low, repricing of underwater options, and benchmarking against peers' median or upper quartile, which ratchets pay up over time.

24.4.2 Choosing metrics

Metric	Strength	Weakness	Best used for
Earnings per share growth	Simple, observable	Manipulable via buybacks, accounting, leverage	Annual bonus with caution
ROIC or ROCE	Ties to value creation relative to capital	Can discourage good investments when measured on book value	Mature, capital-intensive firms
Relative TSR	Market-based, filters market effects	Noisy, short-term sentiment, peers selection	LTI plans, 3-year horizon
Free cash flow	Hard to manipulate	Can be boosted by underinvestment	Cash-generative businesses
Strategic or ESG KPIs	Links to long-run drivers	Measurement ambiguity, risk of box-ticking	Small weight, quantified targets

Many firms combine two or three metrics to balance their weaknesses, for example 50 percent relative TSR, 30 percent ROIC and 20 percent a quantified emissions or safety target.

24.4.3 Valuing options with Black-Scholes

The grant-date fair value of stock options is estimated with an option pricing model, typically Black-Scholes, adjusted for dividends. For a European call on a stock with spot S, strike K, time T, risk-free rate r, continuous dividend yield q and volatility σ:

```
C = S·e^(−qT)·N(d1) − K·e^(−rT)·N(d2)
d1 = [ln(S/K) + (r − q + σ²/2)·T] / (σ·√T)
d2 = d1 − σ·√T
```

Executive options are long-dated, subject to vesting and forfeiture, and often exercised early, so the model overstates value if applied naively; companies reduce the expected term to reflect early exercise behavior. We nevertheless use the formula to show the economics.

Example 24.4 – Valuing an at-the-money option grant. A CEO is granted 200,000 options with a 5-year term, struck at the current price of $50. Volatility is 30 percent, the risk-free rate is 4 percent and the dividend yield is 1 percent.

Step 1: d1 = [ln(50/50) + (0.04 − 0.01 + 0.30²/2) × 5] / (0.30 × √5) = [0 + (0.03 + 0.045) × 5] / 0.6708 = 0.375 / 0.6708 = **0.5590**.

Step 2: d2 = 0.5590 − 0.6708 = **−0.1118**.

Step 3: N(d1) = 0.7119; N(d2) = 0.4555.

Step 4: Discount factors: e^(−0.01×5) = 0.95123; e^(−0.04×5) = 0.81873.

Step 5: C = 50 × 0.95123 × 0.7119 − 50 × 0.81873 × 0.4555 = 33.86 − 18.65 = **$15.21 per option**.

Step 6: Grant value = 200,000 × 15.21 = **$3.04m**.

An equivalent economic grant in restricted stock units would be 3.04m / 50 = 60,800 units, ignoring the dividend (RSUs receive or accrue dividends; the option does not). The key managerial insights: an at-the-money option is worth about 30 percent of the share price at these parameters, so it is a levered bet on the share price; the value rises with volatility (at σ = 40 percent it is worth more), which means options reward risk-taking and are not neutral about strategic choices such as leverage or an acquisition that raises volatility; and they pay nothing if the share price ends below the strike, yet they also pay for market-wide gains that the executive did not cause, which is the argument for indexed or relative-performance designs. Executives compare an option's *value to them*, which is lower than the Black-Scholes fair value because they are undiversified and cannot hedge, with the *cost to the company*.

Example 24.5 – A relative TSR performance share plan. The CEO receives a target grant of 40,000 performance shares vesting after three years. The payout is 0 percent below the 25th percentile of peer TSR, 50 percent at the 25th percentile, 100 percent at the median, 150 percent at the 75th percentile and 200 percent at or above the 90th; payout is interpolated linearly between points. The company finishes at the 62nd percentile and the share price at vesting is $58.

Step 1: The 62nd percentile lies between the median (100 percent) and the 75th (150 percent). Interpolation fraction = (62 − 50) / (75 − 50) = 12 / 25 = 0.48.

Step 2: Payout % = 100 + 0.48 × 50 = **124 percent**.

Step 3: Shares earned = 40,000 × 1.24 = **49,600**.

Step 4: Value at vesting = 49,600 × $58 = **$2,876,800**.

The grant-date target value (40,000 × $50 = $2.0m) rose to $2.88m because of the above-median ranking and the 16 percent share price increase. If TSR had ranked at the 40th percentile, payout would be 50 + (15/25) × 50 = 80 percent. The shape of the payout curve matters: a payout at the 25th percentile rewards below-median performance, and a steep slope between the 50th and 75th percentiles raises the incentive for improvement but also the risk of paying for luck, since a three-year TSR rank is noisy. Some committees add a cap when absolute TSR is negative.

24.4.4 Say-on-pay, ratios and the political context

Many jurisdictions give shareholders an advisory or binding vote on pay. CEO-to-median-worker pay ratios are disclosed in some regimes. These measures create transparency but not necessarily moderation; they have been accused of encouraging benchmarking that increases pay. A defensible committee process includes using independent consultants, tying pay to a clear strategy, testing payout under multiple scenarios (including the worst case) before approval, and explaining outcomes to shareholders in plain language.

24.5 Fintech, Payments and Digital Assets

This section is deliberately conceptual. Technologies and regulation in this area change quickly; the aim is to give finance leaders a durable framework for evaluating proposals rather than a snapshot of products.

24.5.1 What fintech changes in the economics of finance

Financial services perform a few core functions: moving value (payments), pooling and allocating capital (lending, investing), managing risk (insurance, hedging) and producing information (credit assessment, price discovery). Fintech firms apply software, data and new distribution channels to these functions. They typically compete on one of four dimensions: lower unit cost through automation and API-based integration; better information, for example credit decisions using transaction or cash-flow data; superior user experience and distribution embedded in a non-financial product ("embedded finance"); and regulatory or structural arbitrage, which is the least durable. For a corporate finance leader, the implications fall into four areas: treasury and working capital, financing sources, customer-facing business models and risk management.

Treasury and working capital. Real-time payments, virtual accounts, automated reconciliation and API connectivity reduce float, reduce days sales outstanding by speeding collection, and enable dynamic discounting and supply chain finance. The value of faster cash can be computed directly: a firm with $2.4 billion revenue and 45 days sales outstanding that reduces DSO by 3 days releases 2,400 × 3 / 365 = $19.7m of cash, which at a 6 percent cost of funds is worth $1.18m annually.

Financing sources. Marketplace lenders, invoice platforms and revenue-based financing offer alternatives to bank credit for small and mid-sized firms. They can provide faster decisions but sometimes at higher all-in cost, and their funding can be pro-cyclical: platforms funded by wholesale investors may cut volume in a downturn. A treasurer should compare all-in annual percentage cost, including fees, and examine the reliability of the funding source.

24.5.2 Payments economics

Card payments involve several parties: the cardholder, the issuing bank, the card network, the acquiring bank or processor and the merchant. The merchant pays a **merchant discount rate** (MDR), usually a percentage of the transaction plus a fixed fee. Of this, the largest piece is **interchange**, paid to the issuer; the network charges assessment and processing fees; the acquirer or payment facilitator retains the remainder as its **net take**. Regulation of interchange differs by country, so the margin structure of the same business varies across markets. Payment facilitators aggregate small merchants and charge blended pricing (e.g., 2.9 percent plus a fixed fee), earning a spread over the underlying costs; they also bear fraud, chargeback and compliance risk.

Example 24.6 – Unit economics of a payment facilitator. A facilitator charges 2.9 percent + $0.30 per transaction on an average ticket of $50. Its costs per transaction are: interchange 1.65 percent of ticket; network and assessment fees 0.13 percent; processing and gateway cost $0.10; fraud and chargeback losses 0.08 percent of ticket; support and servicing $0.05. Fixed annual operating costs (technology, compliance, staff) are $40m.

Step 1: Revenue per transaction = 0.029 × 50 + 0.30 = 1.45 + 0.30 = **$1.75**.

Step 2: Variable cost per transaction = 0.0165 × 50 + 0.0013 × 50 + 0.10 + 0.0008 × 50 + 0.05 = 0.825 + 0.065 + 0.10 + 0.04 + 0.05 = **$1.08**.

Step 3: Contribution per transaction = 1.75 − 1.08 = **$0.67**, which is 1.34 percent of the ticket (67 cents on $50).

Step 4: Break-even volume = 40,000,000 / 0.67 = **59.7 million transactions per year**, equivalent to $2.99 billion of annual payment volume.

The model displays the main strategic characteristics of payments: tiny margins per transaction, high operating leverage (profit explodes after break-even), and heavy scale economies and network effects. Competition erodes the take rate, so sustainable advantage comes from lower cost, better risk management, bundled software or lending, or distribution. A 20 bp reduction in the take rate on a $50 ticket reduces contribution by $0.10 (15 percent of contribution), which would raise break-even volume by 17.5 percent. Strategically, a company deciding whether to build, partner with or buy a payments capability should model volume ramp-up, take-rate compression, fraud losses and regulatory capital, and compare the value created against the cost of simply using a third-party processor.

24.5.3 Embedded finance and fintech lending

Embedded finance integrates financial services into non-financial platforms: a marketplace offering working capital loans to sellers on the basis of sales data, or a manufacturer offering financing at the point of sale. The economic logic is that the platform has better information (cash flow visibility, ability to collect through the sales stream) and distribution at near-zero marginal acquisition cost. The risks are credit losses that arrive in downturns, balance sheet funding needs, regulatory perimeter (lending, deposit and consumer protection rules) and reputational spillovers. A corporate offering customer financing should decide whether to hold the receivables (earning the spread but bearing credit and funding risk) or sell them to a financial partner (earning fees but giving up the economics), and should evaluate the effect on its own leverage and rating if the receivables stay on its balance sheet.

24.5.4 Digital assets, stablecoins and tokenization

Digital assets are claims recorded on distributed ledgers. Several categories deserve different treatment.

- **Cryptocurrencies without an issuer** (for example, those with fixed algorithmic supply) have no cash flows, so their value rests on expectations of future demand. A corporate treasury holding such assets takes price risk akin to a speculative commodity, with accounting, custody and regulatory implications. Holding volatile assets as treasury reserves conflicts with the usual principle that corporate liquidity should be safe and liquid.
- **Stablecoins** are tokens designed to hold a stable value relative to a currency, usually backed by reserve assets (cash, short-term government securities). Their reliability depends on reserve quality, redemption rights, legal clarity, custody and the integrity of the issuer. They can enable faster, cheaper cross-border settlement, which may help corporate payments and treasury across time zones. Risks include runs if redemption is questioned, regulatory change, counterparty failure and limited legal recourse.
- **Tokenized assets**, such as tokenized money-market funds, bonds or deposits, represent traditional claims on a ledger. The potential benefits are faster settlement (reducing counterparty and capital costs), programmability and fractional ownership. These are improvements to market plumbing, not new sources of value in themselves; the underlying credit and legal rights remain what matter.
- **Central bank digital currencies (CBDCs)** are digital forms of central bank money under study or pilot in many jurisdictions; their design (retail versus wholesale, interest, limits) is still evolving.
- **Decentralized finance (DeFi)** replaces intermediaries with smart contracts. Its risks include coding errors, governance capture, oracle failures and weak consumer protection, and its relevance to the corporate sector is currently limited.

Whatever the technology, a CFO should test a digital-asset proposal with five questions. What problem does it solve better than existing rails? Who bears credit, legal and operational risk? How are assets custodied and audited? What is the regulatory and accounting treatment? And what is the exit route if the technology or counterparty fails?

Example 24.7 – Reserve economics of a stablecoin issuer. A fiat-backed stablecoin issuer has $10 billion of tokens outstanding, fully backed by short-term government securities yielding 4.0 percent. It pays 25 percent of reserve income to distribution partners (exchanges and wallets), and incurs operating and compliance costs of $60m a year.

Step 1: Gross reserve income = 10,000m × 4.0% = **$400m**.

Step 2: Distribution payments = 25% × 400 = **$100m**.

Step 3: Operating costs = $60m.

Step 4: Pre-tax profit = 400 − 100 − 60 = **$240m**, a 2.4 percent margin on reserves.

Step 5: Rate sensitivity. If yields fall to 2.0 percent, gross income = $200m, distribution (25 percent) = $50m, and profit = 200 − 50 − 60 = **$90m**, a decline of 62.5 percent for a 50 percent decline in yields.

The issuer's profit is therefore essentially a leveraged bet on short-term interest rates, with fixed operating costs and distribution-sharing, and it carries run risk on a balance sheet whose liabilities are redeemable at par on demand. Anyone assessing such a business, or considering using its tokens in payments, should evaluate the reserve composition and liquidity, redemption mechanics, regulatory status and the cyclical sensitivity of earnings.

24.5.5 Operational and cyber risk

Fintech increases dependence on software, data and third parties. Vendor concentration, cybersecurity, data privacy and algorithmic bias in credit models are core risks in the finance function's remit. Controls include concentration limits on critical vendors, tested contingency arrangements, segregation of duties for payment authorization and independent validation of models.

24.6 International Capital Budgeting

24.6.1 Two consistent methods

When a firm invests abroad, the cash flows are earned in a foreign currency while shareholders think in the home currency. There are two internally consistent ways to compute the NPV.

- **Method 1 (foreign currency approach):** forecast the foreign currency cash flows, discount at the foreign-currency cost of capital, and convert the resulting NPV to the home currency at the spot exchange rate.
- **Method 2 (home currency approach):** convert each foreign currency cash flow to home currency at the forecast (forward-equivalent) exchange rate for its date, and discount at the home-currency cost of capital.

The two agree when exchange rates follow the parity relationships. Under interest rate parity, the forward rate is F_t = S × [(1 + r_foreign)/(1 + r_home)]^t when exchange rates are quoted as foreign per home unit; and under relative purchasing power parity, expected exchange rates move with inflation differentials. The discount rates are linked by the international Fisher relationship:

```
(1 + r_foreign) = (1 + r_home) × (1 + i_foreign) / (1 + i_home)
```

where r denotes the nominal cost of capital and i expected inflation. In practice, the parity relationships hold only approximately, especially over short horizons, and a project's currency risk is a business risk that depends on whether its costs and revenues are in the same currency, how prices adjust, and the competitive structure.

Example 24.8 – A project in Country Z. A US company evaluates a plant in Country Z (currency ZED, quoted ZED per USD). Spot is 5.00. Expected inflation is 4.5 percent in Country Z and 2.5 percent in the US. The US-dollar cost of capital for this project is 9 percent. Expected ZED cash flows (millions) are: year 0 −1,000; year 1 +350; year 2 +400; year 3 +450; year 4 +450.

Step 1: Fisher: ZED cost of capital = 1.09 × (1.045 / 1.025) − 1 = 1.09 × 1.019512 − 1 = **11.13 percent**.

Step 2 (Method 1): NPV in ZED = −1,000 + 350/1.1113 + 400/1.1113² + 450/1.1113³ + 450/1.1113⁴ = −1,000 + 314.9 + 323.9 + 327.9 + 295.1 ≈ **ZED 261.9m**. In USD at spot 5.00: 261.9 / 5.00 = **$52.4m**.

Step 3 (Method 2): Expected ZED per USD each year = 5.00 × (1.045/1.025)^t = 5.000, 5.098, 5.197, 5.298, 5.402. USD cash flows: −1,000/5.000 = −200.0; 350/5.098 = 68.66; 400/5.197 = 76.97; 450/5.298 = 84.93; 450/5.402 = 83.31. Discount at 9 percent: −200.0 + 62.99 + 64.78 + 65.58 + 59.02 = **$52.4m**.

The two methods agree, as they must when parity holds. An analyst who discounted the ZED cash flows at the USD rate of 9 percent would compute a ZED NPV of about 324.0m, or $64.8m at spot, overstating value by $12.4m (24 percent) because it ignores the currency depreciation that offsets high local inflation. The error of discounting weak-currency flows at a strong-currency rate is among the most common in international budgeting.

24.6.2 Political risk, repatriation and taxes

Cash flows to the parent differ from cash flows of the project because of withholding taxes, restrictions on remittances (blocked funds), transfer pricing, and possible expropriation or contract renegotiation. Best practice is to adjust the cash flows rather than inflating the discount rate arbitrarily, because discount-rate premiums compound over time and penalize distant cash flows more than the underlying risk justifies. Adjustment methods include scenario analysis with probabilities, insurance (political risk insurance, multilateral guarantees), structuring (local debt, joint ventures with local partners, staged investment), and negotiating contractual protections such as stabilization clauses and international arbitration. Diversification does not reduce the risk if the risk is systematic to the parent's own operations; the correlation with the home market matters when using CAPM, and some firms use a country risk premium, estimated from sovereign spreads scaled for equity volatility, but these estimates are crude.

Example 24.9 – Adjusting cash flows for repatriation risk. Suppose management believes that, in expectation, only 92 percent of the ZED operating cash flows in Example 24.8 will reach the parent after withholding taxes, delays and negotiated concessions, while the initial investment is unaffected.

Step 1: Adjusted flows: 350 × 0.92 = 322; 400 × 0.92 = 368; 450 × 0.92 = 414; 450 × 0.92 = 414.

Step 2: Present values at 11.127 percent: 322/1.1113 = 289.7; 368/1.2350 = 298.0; 414/1.3724 = 301.7; 414/1.5251 = 271.5; sum = 1,160.9.

Step 3: NPV = 1,160.9 − 1,000 = ZED 160.9m; in USD at spot, **$32.2m**.

An 8 percent haircut on operating cash flows reduces NPV by 38.6 percent ($52.4m to $32.2m) because the project has high operating leverage in the sense that the initial investment is fixed. The project remains positive, but the margin of safety is thinner. The same analysis can find a break-even haircut: NPV hits zero when the PV of operating inflows equals 1,000, i.e., when the haircut reaches 1 − 1,000/1,261.8 = 20.7 percent.

24.6.3 Financing abroad and hedging

Foreign projects can be financed with local-currency debt, which creates a natural hedge for local cash flows and reduces transfer risk, albeit at the local interest rate, which reflects local inflation. Hedging the transaction exposure for committed flows (forwards, options, cross-currency swaps) is common; hedging long-dated economic exposure is difficult and costly, and often it is more effective to shape the operating footprint (matching currencies of costs and revenues, local sourcing, pricing flexibility). The value of hedging to shareholders is limited unless it reduces expected taxes, distress costs or underinvestment, which is the same logic as the corporate risk management discussion earlier in this book.

24.7 Strategic Synthesis: The Future of Corporate Finance

24.7.1 One value framework for many decisions

Throughout this book, every decision has been judged by its effect on the present value of cash flows available to capital providers. The topics in this chapter are extensions of the same logic, not departures from it. A compact way to see this is the value-driver formula for a firm growing at a constant rate g, with a return on new invested capital (RONIC) and a cost of capital WACC:

```
Value = NOPAT₁ × (1 − g / RONIC) / (WACC − g)
```

$NOPAT_1$ is next year's net operating profit after tax, and the fraction g / RONIC is the share of profit reinvested to sustain growth. Growth creates value only when RONIC exceeds WACC; when RONIC equals WACC, growth is neutral and value equals $NOPAT_1$ / WACC.

Example 24.10 – Which lever matters? A value-driver comparison. A firm has $NOPAT_1$ = \$120m, g = 3 percent, RONIC = 12 percent and WACC = 9 percent. Invested capital is \$1,000m.

Step 1 (base): Reinvestment rate = 3 / 12 = 25 percent; free cash flow = 120 × 0.75 = \$90m; value = 90 / (0.09 − 0.03) = 90 / 0.06 = **\$1,500m**, or 1.5 times invested capital.

Step 2 (lower cost of capital): Suppose better governance, disclosure and a broader investor base reduce WACC by 25 bp to 8.75 percent. Value = 90 / (0.0875 − 0.03) = 90 / 0.0575 = **\$1,565m**, an increase of \$65m (4.3 percent).

Step 3 (higher returns on new capital): Suppose the firm's electrification and digital investments lift RONIC to 15 percent at an unchanged WACC of 9 percent. Reinvestment = 3 / 15 = 20 percent; FCF = 120 × 0.80 = \$96m; value = 96 / 0.06 = **\$1,600m**, an increase of \$100m (6.7 percent).

Step 4 (both): FCF \$96m at WACC 8.75 percent: 96 / 0.0575 = **\$1,670m**, an increase of \$170m (11.3 percent).

Neither lever is free: a lower WACC from "ESG" labels is hard to prove and may be offset by lower expected returns to investors, whereas a higher RONIC requires actual projects that earn more than their capital cost. The example also shows why capital allocation discipline dominates: a three-point rise in return on new capital is worth more than a quarter-point fall in the discount rate. The tools of this book (NPV, WACC, capital structure, payout, M&A discipline, risk management) are the means by which a firm identifies and executes the investments behind the RONIC.

24.7.2 Decision checklist for the finance leader

Question	Tool from this book	Typical failure
Does the investment earn more than its risk-adjusted cost of capital?	NPV, IRR, real options, scenario analysis	Ignoring strategic or carbon costs; using one discount rate for all projects
Is the financing structure compatible with the cash-flow risk?	Capital structure trade-off, coverage tests, stress tests	Sizing debt on average, not downside, cash flow
Are managers' incentives aligned with long-run value?	Governance, pay design, board oversight	Pay metrics that can be manipulated
Is the project exposed to currency, political or technology risk?	Parity-consistent budgeting, cash-flow adjustment, hedging	Mixing currencies in discount rates
Is the claimed sustainability benefit verifiable?	Frameworks, KPIs, external assurance	Labels without commitments
Will separation or combination create value after costs?	SOTP, synergy valuation, tax analysis	Counting gross benefits without dis-synergies
What is the right response if things go wrong?	Contingent plans, liquidity runway, restructuring options	Late action

24.7.3 The enduring skill

Disclosure standards, automation, tokenized settlement, private capital and geopolitical fragmentation will keep changing the tools and the data available to finance leaders. The core skill remains judgment under uncertainty. Models quantify, but the quality of assumptions, the incentives of those who produce the forecasts, and the willingness to act on unwelcome results determine whether a firm creates value.

Case Study: Verdane Materials and the Sustainable Expansion Decision

(Fictional company and figures, for teaching purposes only.)

Verdane Materials is a listed specialty-chemicals and building-materials group with $4.5 billion revenue, EBITDA of $720 million, net debt of $1.8 billion (2.5 times EBITDA) and an investment-grade rating one notch above the cut-off. The company has a market capitalization of $5.4 billion and a dual-class structure in which the founding family holds 9 percent of the capital but 31 percent of the votes. The board has eleven directors, of whom six are independent; the CEO is also the chair. The long-term incentive plan pays out 70 percent on three-year EPS growth, which has been supported by $600 million of buybacks in two years.

Verdane's largest plant, in a coastal region, emits 1.2 million tons of CO2 a year, accounting for 40 percent of group emissions. The management team proposes an $800 million conversion to low-carbon heat and an electric kiln, financed with a $600 million green bond and $200 million of internal cash. The conversion would raise efficiency and lower other operating costs by an estimated $14 million a year after tax, but electricity costs more than gas, so that is the entire saving if carbon is free. It would also eliminate about 0.96 million tons of CO2 a year; treat the avoided carbon cost, at $70 per ton in year 1 rising 5 percent a year, as an after-tax saving. The plant is expected to operate for 20 years. Verdane's WACC is 8.5 percent. The sustainability team argues that two of its largest customers have indicated a preference for low-carbon inputs and may switch within three years if no progress is made; the sales team estimates the value at risk at $25 million a year of EBITDA contribution.

Meanwhile the company is considering a plant of similar scale in an emerging market where it would serve rapidly growing construction demand. The plant would cost the equivalent of $500 million and would generate local-currency cash flows. The local currency has depreciated about 4 percent annually against the dollar over the past decade, and local inflation runs 3 percentage points above US inflation. The government has twice delayed dividend payments abroad. A local partner proposes a joint venture with 40 percent ownership.

Finally, the CFO has been approached by a fintech firm offering a dynamic-discounting and embedded-financing platform for Verdane's 4,000 distributors, which would free perhaps $90 million of working capital and generate fee income of $6 million. Verdane's credit committee is concerned about being responsible for customer credit losses and about data security. At the same time, several customers have asked about paying in a dollar stablecoin, citing lower cross-border fees.

The board is divided. Some directors favor accelerating the low-carbon conversion in order to qualify as a leader in sustainable finance. Others note that the leverage target would leave little headroom for the foreign plant and that the share price could suffer if capital expenditure exceeds $1.2 billion over two years. A large pension investor has indicated that it will vote against the remuneration report unless EPS is replaced by return on invested capital and relative TSR.

Discussion questions

1. Evaluate the $800 million conversion. Compute the 20-year NPV at 8.5 percent with and without the carbon price (ignoring customer effects), then add the value of the customer risk avoided, treated as a level $25 million a year. What break-even carbon price or customer retention probability does the decision require? How should the green bond be assessed independently of the project?
2. Design a revised long-term incentive plan for the CEO. Which metrics and weights would you propose, what vesting period and clawback provisions, and how would you deal with the risk that a ROIC target discourages the conversion project?
3. Assess the emerging-market plant. Explain how to value it in USD using parity relationships, how to adjust for dividend repatriation risk, and what role the joint venture should play. How would you check that the local discount rate you use is consistent with inflation forecasts?
4. Evaluate the governance structure, including the dual-class shares, the combined CEO/chair role and the buyback-heavy EPS plan. What changes would you recommend, and how would you sequence them given the pension investor's threat? Should Verdane adopt the fintech platform or the stablecoin proposal, and what questions would you require answered first?

Key Takeaways

- ESG matters to valuation through identifiable channels: cash flows, risk and discount rates, financing access and franchise value. Begin with financial materiality by industry.
- An internal carbon price turns a regulatory risk into a cash-flow assumption; it can reverse project rankings, and break-even carbon prices are more useful than point forecasts.
- Green bonds are ordinary credit obligations with earmarked proceeds and, at best, a small greenium that must be weighed against reporting costs; sustainability-linked bonds need ambitious targets and meaningful step-ups, or they become labels.
- Governance mechanisms respond to agency problems between managers and shareholders, and between controlling and minority shareholders; stronger takeover defenses and control enhancements require stronger independent

oversight.

- Executive pay should link to long-run, hard-to-manipulate measures such as ROIC and relative TSR, with long vesting, shareholding requirements and clawbacks; options reward volatility.
- Black-Scholes values an at-the-money five-year option at roughly 30 percent of the share price at typical parameters, but executives value options at less than fair value because they cannot diversify.
- Fintech changes unit costs, information and distribution; payments are a scale game with thin per-transaction margins and high operating leverage.
- Digital assets should be evaluated with questions about the problem solved, risk allocation, custody, accounting and exit; stablecoin issuer profits are a leveraged function of interest rates.
- International capital budgeting must use consistent currencies: discount foreign cash flows at foreign rates or convert at parity-consistent forwards and discount at home rates; adjust cash flows for political risk rather than inflating the discount rate.
- Value ultimately depends on return on new capital relative to the cost of capital; governance, sustainability and technology matter to the extent they raise returns, lower risk or both.

Review Questions

1. Explain the four channels through which ESG factors can affect firm value. Why is the “cost of capital” argument for green firms ambiguous?
2. What is financial materiality, and why does it differ by industry? Give examples.
3. How does an internal carbon price work in capital budgeting, and what are its main limitations?
4. Compare green bonds and sustainability-linked bonds. Which has greater risk of greenwashing and why?
5. What determines whether a greenium is large enough to justify labeling costs?
6. Describe three mechanisms of corporate governance and one weakness of each.
7. Why do dual-class share structures create governance concerns, and what are the arguments in their favor?
8. Why does an option-heavy compensation plan encourage risk-taking? How can a committee mitigate this?
9. What are the economic advantages and risks of stablecoins for corporate treasury and cross-border payments?
10. Why is discounting foreign currency cash flows at a home-currency rate wrong, and how do the two consistent methods avoid the error?

Problems

1. **Green bond value.** A firm issues a $300m 7-year green bond with a greenium of 8 bp. Reporting and verification cost $0.10m a year and set-up cost is $0.12m at issue. Discount at 4.5 percent. Compute the net present value of the label.
2. **SLB step-up.** A $1,000m SLB pays an extra 20 bp coupon in years 4 to 8 if a KPI target tested at the end of year 3 is missed. The probability of missing is 50 percent; the discount rate is 5 percent. Compute the expected present value of the step-up cost today.
3. **Carbon-adjusted NPV.** A project costs $20m and yields $4.5m a year after tax for 8 years, with emissions of 25,000 tons a year charged at a flat $80 per ton. The discount rate is 10 percent. Compute the NPV before and after the carbon charge.

4. **Option valuation.** Using Black-Scholes, value a 4-year at-the-money call with S = K = $40, σ = 35 percent, r = 3 percent and no dividends.
5. **Performance shares.** A plan with a target of 30,000 shares pays 50 percent at the 25th percentile and 100 percent at the 50th percentile (linear in between, zero below the 25th). The firm ranks at the 40th percentile and the price at vesting is $45. Compute payout percentage, shares earned and value.
6. **International NPV.** Spot is 1.10 USD per EUR. USD and EUR risk-free rates are 5 percent and 3 percent. A EUR project has cash flows of −100, +60 and +70 (years 0 to 2) at a EUR discount rate of 8 percent. Compute the 1- and 2-year forward rates, the equivalent USD discount rate under the international Fisher relationship, the NPV in EUR, and verify in USD.
7. **Payments.** Average ticket is $80. A facilitator charges 2.6 percent + $0.25. Costs per transaction are interchange 1.55 percent of ticket, network 0.12 percent, processing $0.09, fraud 0.07 percent of ticket and support $0.04. Fixed costs are $30m. Compute contribution per transaction and break-even transaction count.
8. **Stablecoin issuer.** Reserves are $5,000m at a 3.5 percent yield, 20 percent of reserve income is shared with distributors, and operating costs are $40m. Compute pre-tax profit, and the reserve yield at which profit equals zero.

Answers to Selected Problems

1. Annual saving = 300 × 0.0008 = $0.24m; annuity factor (7 years, 4.5%) = (1 − 1.045^−7) / 0.045 = 5.8927. PV savings = 0.24 × 5.8927 = $1.414m; PV reporting = 0.10 × 5.8927 = $0.589m; NPV = 1.414 − 0.589 − 0.12 = **$0.705m**.
2. Annual step-up = 1,000 × 0.002 = $2.0m, paid in years 4 to 8. Value at end of year 3 = 2.0 × 4.3295 = $8.659m; discount to today: 8.659 / 1.05^3 = 8.659 / 1.157625 = $7.480m; expected cost = 0.5 × 7.480 = **$3.74m**.
3. Annuity factor (8 years, 10%) = 5.3349. NPV before carbon = −20 + 4.5 × 5.3349 = **$4.007m**. Carbon charge = 25,000 × 80 = $2.0m a year; PV = 2.0 × 5.3349 = $10.670m. NPV after = 4.007 − 10.670 = **−$6.66m**.
4. d1 = [0 + (0.03 + 0.35^2/2) × 4] / (0.35 × 2) = (0.03 + 0.06125) × 4 / 0.70 = 0.5214; d2 = 0.5214 − 0.70 = −0.1786; N(d1) = 0.6990, N(d2) = 0.4291; C = 40 × 0.6990 − 40 × e^(−0.12) × 0.4291 = 27.96 − 40 × 0.88692 × 0.4291 = 27.96 − 15.22 = **about $12.73 per option**.
5. Interpolation: 50 + (40 − 25)/(50 − 25) × 50 = 50 + 30 = **80 percent**; shares = 24,000; value = 24,000 × 45 = **$1,080,000**.
6. F1 = 1.10 × 1.05 / 1.03 = 1.1214; F2 = 1.10 × $(1.05/1.03)^2$ = 1.1431. USD rate = 1.08 × 1.05 / 1.03 − 1 = 10.10 percent. NPV in EUR = −100 + 60/1.08 + 70/1.08^2 = −100 + 55.56 + 60.01 = 15.57, or **$17.13m** at spot. In USD: cash flows −110.00, 60 × 1.1214 = 67.28, 70 × 1.1431 = 80.02; NPV = −110 + 67.28/1.1010 + 80.02/1.1010^2 = −110 + 61.11 + 66.02 = **$17.13m**, the same.
7. Revenue = 0.026 × 80 + 0.25 = $2.33; cost = 1.24 + 0.096 + 0.09 + 0.056 + 0.04 = $1.522; contribution = **$0.808** (1.01 percent of ticket); break-even = 30,000,000 / 0.808 = **37.1 million transactions**.
8. Gross income = 5,000 × 3.5% = $175m; distribution = $35m; profit = 175 − 35 − 40 = **$100m**. Break-even yield y: 0.8 × 5,000 × y = 40, so y = **1.0 percent**.

Appendices

Appendix A: Formula Sheet

This appendix collects the main formulas of the book, organised by chapter. Each formula is written in plain Unicode and followed by a one-line definition of its symbols. Use it as a quick reference when working problems, and remember that every formula rests on assumptions described in the relevant chapter. Notation conventions: r is a discount rate or required return per period, t or n a number of periods, CF or C a cash flow, g a growth rate, E[·] an expected value, σ a standard deviation, and ρ a correlation.

Chapter 1: The Corporation and the Finance Function

```
Shareholder wealth = Σ PV(future free cash flows to the firm's investors)
```

The market value of the firm’s claims equals the present value of the cash flows it is expected to generate; value maximisation is the guiding objective.

```
Agency cost = monitoring cost + bonding cost + residual loss
```

The total cost of conflicts between owners and managers: what owners spend to monitor, what managers spend to commit, and the value lost despite both.

```
Economic profit = NOPAT − (WACC × Invested capital)
```

Profit remaining after charging for all capital employed; positive economic profit signals value creation.

Chapter 2: Financial Statements and Ratios

```
Assets = Liabilities + Shareholders' equity
```

The accounting identity linking what the firm owns to how it is financed.

```
Net working capital = Current assets − Current liabilities
```

Short-term resources less short-term obligations.

```
Free cash flow to the firm (FCFF) = EBIT × (1 − T) + Depreciation − CapEx − ΔNWC
```

Cash generated by operations after taxes and reinvestment, available to all capital providers; T is the tax rate.

```
Current ratio = Current assets / Current liabilities
Quick ratio = (Cash + Marketable securities + Receivables) / Current liabilities
Cash ratio = (Cash + Marketable securities) / Current liabilities
```

Liquidity measures of increasing strictness.

```
Inventory turnover = Cost of goods sold / Average inventory
Receivables turnover = Credit sales / Average receivables
Payables turnover = Cost of goods sold / Average payables
Total asset turnover = Sales / Average total assets
Fixed asset turnover = Sales / Average net fixed assets
```

Efficiency measures showing how many times a balance is “turned over” in a period.

```
Days inventory = 365 / Inventory turnover
Days sales outstanding (DSO) = 365 / Receivables turnover
Days payables outstanding (DPO) = 365 / Payables turnover
Cash conversion cycle = Days inventory + DSO − DPO
```

Time measures; the cash conversion cycle is the days between paying suppliers and collecting from customers.

```
Gross margin = (Sales − COGS) / Sales
Operating margin = EBIT / Sales
Net profit margin = Net income / Sales
EBITDA margin = EBITDA / Sales
```

Profitability measures as a share of sales.

```
ROA = Net income / Average total assets
ROE = Net income / Average shareholders' equity
ROIC = NOPAT / (Debt + Equity − Excess cash)
```

Returns on assets, equity and invested capital; NOPAT is net operating profit after tax.

```
DuPont: ROE = (Net income / Sales) × (Sales / Assets) × (Assets / Equity)
```

Decomposes ROE into profit margin, asset turnover and the equity multiplier (financial leverage).

```
Extended DuPont: ROE = (NI/EBT) × (EBT/EBIT) × (EBIT/Sales) × (Sales/Assets) × (Assets/Equity)
```

Five-factor version separating the tax burden, interest burden, operating margin, turnover and leverage.

```
Debt-to-equity = Total debt / Total equity
Debt ratio = Total debt / Total assets
Equity multiplier = Total assets / Total equity
Interest coverage = EBIT / Interest expense
Fixed-charge coverage = (EBIT + Lease payments) / (Interest + Lease payments)
Net debt / EBITDA = (Debt − Cash) / EBITDA
```

Leverage and solvency measures.

```
EPS = (Net income − Preferred dividends) / Weighted average shares
P/E = Price per share / EPS
Market-to-book = Market value of equity / Book value of equity
Dividend yield = Dividend per share / Price per share
Payout ratio = Dividends / Net income
Retention ratio (b) = 1 − Payout ratio
```

Per-share and market-based measures.

```
Sustainable growth rate = ROE × b / (1 − ROE × b)
Internal growth rate = ROA × b / (1 − ROA × b)
```

Growth achievable without new external equity (sustainable) or without any external financing (internal).

```
Enterprise value = Market capitalisation + Debt + Preferred + Minority interest − Cash
EV/EBITDA = Enterprise value / EBITDA
```

Value of the operating business to all capital providers, and a common multiple.

Chapter 3: Time Value of Money

```
FV = PV × (1 + r)^n
PV = FV / (1 + r)^n
```

Future value and present value of a single sum compounded over n periods at rate r.

```
FV = PV × (1 + r/m)^(m×n)
EAR = (1 + APR/m)^m − 1
FV (continuous) = PV × e^(r×n)
EAR (continuous) = e^r − 1
```

Compounding m times per year; EAR is the effective annual rate and APR the quoted nominal rate.

```
PV of ordinary annuity = C × [1 − (1 + r)^−n] / r
FV of ordinary annuity = C × [(1 + r)^n − 1] / r
PV of annuity due = PV of ordinary annuity × (1 + r)
```

Level payments C at the end (or, for annuity due, the start) of each of n periods.

```
PV of perpetuity = C / r
PV of growing perpetuity = C₁ / (r − g)    (r > g)
PV of growing annuity = C₁ / (r − g) × [1 − ((1 + g)/(1 + r))^n]
```

Payments forever, or growing at g, where C_1 is the first payment.

```
Loan payment = P × r / [1 − (1 + r)^−n]
Interest in period k = r × Balance at start of period k
Principal repaid in period k = Payment − Interest in period k
```

Level-payment amortising loan on principal P.

```
Real rate: (1 + r_real) = (1 + r_nominal) / (1 + π)
Approximate (Fisher): r_nominal ≈ r_real + π
```

Converts between nominal and inflation-adjusted rates; π is inflation.

```
Rule of 72: years to double ≈ 72 / (rate in %)
Exact: n = ln(2) / ln(1 + r)
```

Approximate and exact doubling time.

```
NPV = Σ CF_t / (1 + r)^t, t = 0 … n
IRR: the r satisfying Σ CF_t / (1 + r)^t = 0
```

Net present value and internal rate of return of a cash flow series.

Chapter 4: Bonds

```
Bond price = Σ C / (1 + y)^t + F / (1 + y)^n
           = C × [1 − (1 + y)^−n] / y + F / (1 + y)^n
```

Price equals the PV of coupons C and face value F discounted at the yield y per period.

```
Current yield = Annual coupon / Price
Yield to call: solve Price = Σ C/(1+y)^t + Call price/(1+y)^T for y
```

Simple income measure and the yield assuming the bond is called at time T.

```
Zero-coupon price = F / (1 + y)^n
Spot rate: (1 + s_n)^n = F / P
Forward rate: (1 + s_n)^n = (1 + s_(n−1))^(n−1) × (1 + f_n)
```

Pure discount bond pricing and the relationship of spot and one-period forward rates.

```
Macaulay duration D = Σ [t × PV(CF_t)] / Price
Modified duration D* = D / (1 + y)
ΔP/P ≈ −D* × Δy + ½ × Convexity × (Δy)²
Convexity = [1 / (P × (1 + y)²)] × Σ [CF_t × t × (t + 1) / (1 + y)^t]
```

Duration is the PV-weighted average time to cash flows; modified duration measures price sensitivity to yield; convexity refines the approximation.

```
DV01 ≈ D* × Price × 0.0001
```

Dollar change in price for a one-basis-point change in yield.

```
Yield = Risk-free rate + Default premium + Liquidity premium + Maturity premium
Expected loss = Probability of default × Loss given default × Exposure
Credit spread ≈ Probability of default × Loss given default (annual)
```

Components of bond yield and a simple decomposition of credit risk.

```
Floating-rate note price at a reset date ≈ Par
Duration of a portfolio = Σ w_i × D_i
```

A floating-rate note reprices to par at resets; portfolio duration is the value-weighted average.

Chapter 5: Equity Valuation

```
P₀ = Σ D_t / (1 + r_e)^t + P_n / (1 + r_e)^n
```

Share price equals the PV of dividends and the terminal price.

```
Gordon growth: P₀ = D₁ / (r_e − g) = D₀ × (1 + g) / (r_e − g)
r_e = D₁ / P₀ + g
```

Constant-growth dividend model and the implied cost of equity.

```
Two-stage: P₀ = Σ[t=1..N] D₀(1+g₁)^t/(1+r)^t + [D_(N+1)/(r − g₂)] / (1+r)^N
```

High growth g_1 for N years followed by stable growth g_2.

```
g = b × ROE
P₀ = EPS₁ × (1 − b) / (r − g)
PVGO = P₀ − EPS₁ / r
```

Growth from retained earnings, price from earnings, and the present value of growth opportunities.

```
Justified P/E = (1 − b) / (r − g)
Justified P/B = (ROE − g) / (r − g)
PEG = (P/E) / (growth rate in %)
```

Fundamental multiples linked to payout, growth, and return on equity.

```
Residual income model: V₀ = B₀ + Σ [(ROE_t − r_e) × B_(t−1)] / (1 + r_e)^t
```

Value equals book equity plus the PV of earnings above the required return on book equity.

```
Preferred stock value = Dividend / r_p
```

Perpetuity valuation of non-growing preferred dividends.

Chapter 6: Net Present Value and Investment Criteria

```
NPV = −I₀ + Σ CF_t / (1 + r)^t
Profitability index (PI) = PV of future cash flows / Initial investment = 1 + NPV / I₀
Payback period = Years before recovery + Unrecovered cost / Cash flow in recovery year
Discounted payback: payback on discounted cash flows
```

Core capital-budgeting criteria; accept when NPV > 0, PI > 1, IRR > r.

```
Average accounting return (AAR) = Average net income / Average book value
```

An accounting-based criterion that ignores timing and is not recommended as a primary rule.

```
MIRR = (FV of positive flows at r_reinvest / PV of negative flows at r_finance)^(1/n) − 1
```

Modified IRR assuming explicit reinvestment and financing rates.

```
Crossover rate: r at which NPV_A(r) = NPV_B(r), i.e. IRR of (CF_A − CF_B)
Equivalent annual annuity (EAA) = NPV × r / [1 − (1 + r)^−n]
```

Tools for ranking mutually exclusive projects with conflicting NPV and IRR, or with unequal lives.

```
Capital rationing: choose the set of projects maximising ΣNPV subject to ΣI ≤ Budget
```

When funds are limited, rank by PI where projects are divisible.

Chapter 7: Capital Budgeting

```
Operating cash flow (OCF) = EBIT × (1 − T) + Depreciation
                          = (Sales − Costs) × (1 − T) + Depreciation × T
FCF = OCF − CapEx − ΔNWC
```

Project cash flow under the bottom-up and tax-shield approaches; T is the tax rate.

```
Straight-line depreciation = (Cost − Salvage) / Life
Depreciation tax shield = Depreciation × T
After-tax salvage = Salvage − T × (Salvage − Book value)
```

Depreciation and its tax effect, and the after-tax proceeds from asset sale.

```
Terminal value (growing) = FCF_(n+1) / (r − g)
ΔNWC_t = NWC_t − NWC_(t−1)
```

Value beyond the forecast horizon, and the investment in working capital.

```
Break-even (accounting): Q = (Fixed costs + Depreciation) / (Price − Variable cost)
Break-even (financial): Q = (Fixed costs + EAC − Depreciation × T) / [(Price − Variable cost) × (1 − T)]
Degree of operating leverage (DOL) = 1 + Fixed costs / OCF = %ΔEBIT / %ΔSales
EAC = PV of costs / Annuity factor
```

Break-even analysis, operating leverage, and equivalent annual cost for cost-minimising decisions.

```
Replacement chain: compare EAC (or EAA) across lives, or repeat each project to a common horizon
Lease-versus-buy: NAL = Cost of asset − PV of after-tax lease payments + PV of lost tax shields
```

Approaches to unequal lives and the net advantage to leasing, discounted at the after-tax cost of debt.

Chapter 8: Uncertainty and Real Options

```
Expected NPV = Σ p_i × NPV_i
Standard deviation of NPV = √[Σ p_i × (NPV_i − E[NPV])²]
```

Scenario analysis summarising outcomes by probability.

```
Sensitivity = ΔNPV / ΔInput
Expected value of perfect information (EVPI) = E[value with information] − E[value without]
```

How much NPV changes per unit change in an input, and the value of resolving uncertainty.

```
Value of project with option = Static NPV + Option value
Expansion option value ≈ max(0, PV of expansion cash flows − Cost of expansion), probability-weighted
Abandonment value = max(Continuing value, Salvage value)
```

Real-option logic: flexibility has value beyond static NPV.

```
Binomial: u = e^(σ√Δt), d = 1/u, q = (e^(rΔt) − d) / (u − d)
Option value = e^(−rΔt) × [q × V_up + (1 − q) × V_down]
```

Risk-neutral valuation on a binomial tree; q is the risk-neutral probability.

```
Decision tree: roll back from terminal nodes, taking the expectation at chance nodes and the maximum at decision nodes
```

The procedure for sequential decisions under uncertainty.

Chapter 9: Risk and Return

```
Holding period return = (P₁ − P₀ + D₁) / P₀
Arithmetic mean = (1/N) Σ R_t
Geometric mean = [Π (1 + R_t)]^(1/N) − 1
```

Single-period return and the two averages; the geometric mean measures compound growth.

```
E[R] = Σ p_i × R_i
σ² = Σ p_i × (R_i − E[R])²
Sample variance = Σ (R_t − R̄)² / (N − 1)
Coefficient of variation = σ / E[R]
```

Expected return and dispersion measures.

```
Cov(A,B) = Σ p_i (R_Ai − E[R_A])(R_Bi − E[R_B])
ρ_AB = Cov(A,B) / (σ_A × σ_B)
```

Covariance and correlation ($-1 \le \rho \le +1$).

```
Risk premium = E[R] − R_f
Sharpe ratio = (E[R_p] − R_f) / σ_p
Value at Risk (normal) = −(μ − z × σ) × Portfolio value
```

Reward for bearing risk, reward per unit of total risk, and a parametric loss threshold with z = 1.645 (95%) or 2.326 (99%).

```
Variance of an N-asset equal-weighted portfolio = (1/N) × avg variance + (1 − 1/N) × avg covariance
```

Diversification removes firm-specific risk but not the average-covariance floor.

Chapter 10: Portfolio Theory and the CAPM

```
E[R_p] = Σ w_i E[R_i]
σ_p² = Σ_i Σ_j w_i w_j σ_i σ_j ρ_ij
Two assets: σ_p² = w_A²σ_A² + w_B²σ_B² + 2 w_A w_B ρ_AB σ_A σ_B
```

Portfolio return is the weighted average; portfolio risk depends on correlations.

```
Minimum-variance weight in A = (σ_B² − ρσ_Aσ_B) / (σ_A² + σ_B² − 2ρσ_Aσ_B)
```

Weight of asset A in the two-asset global minimum-variance portfolio.

```
Capital allocation line: E[R_c] = R_f + w × (E[R_p] − R_f); σ_c = w × σ_p
Tangency portfolio maximises the Sharpe ratio
```

Combining the risk-free asset with a risky portfolio; w is the weight in the risky portfolio.

```
β_i = Cov(R_i, R_m) / σ_m² = ρ_im × σ_i / σ_m
CAPM: E[R_i] = R_f + β_i × (E[R_m] − R_f)
Security market line: E[R] plotted against β
```

Systematic risk and the equilibrium required return.

```
Jensen's alpha: α_i = R̄_i − [R_f + β_i × (R̄_m − R_f)]
Treynor ratio = (R̄_p − R_f) / β_p
Information ratio = α_p / Tracking error
Portfolio beta = Σ w_i β_i
```

Performance measures relative to risk.

```
Total variance = β²σ_m² + σ²(ε)
R² = β²σ_m² / σ_i²
Single-index model: R_i − R_f = α_i + β_i (R_m − R_f) + ε_i
Fama-French: R_i − R_f = α + β_m(R_m − R_f) + s × SMB + h × HML + ε
```

Decomposition into systematic and idiosyncratic components, and a multifactor extension.

```
APT: E[R_i] = R_f + Σ β_ik × λ_k
```

Expected return as exposure to each factor k times that factor's risk premium λ_k.

Chapter 11: Cost of Capital

```
WACC = (E/V) × r_e + (D/V) × r_d × (1 − T_c) + (P/V) × r_p
```

Weighted average cost of capital with market-value weights; V = D + E + P and T_c is the corporate tax rate.

```
r_e (CAPM) = R_f + β_e × MRP
r_e (dividend growth) = D₁ / P₀ + g
r_e (bond yield plus premium) = r_d + risk premium
```

Three estimation approaches for cost of equity; MRP is the market risk premium.

```
Unlevering (no debt beta, no taxes): β_U = β_E × E / (E + D)
Hamada (with taxes): β_L = β_U × [1 + (1 − T) × D/E]
Debt beta adjustment: β_A = β_D × D/V + β_E × E/V
```

Relationships between equity beta, asset beta and leverage.

```
Pre-tax cost of debt: solve Price = Σ C/(1+y)^t + F/(1+y)^n
After-tax cost of debt = r_d × (1 − T)
Flotation-adjusted cost = Net proceeds method: Price × (1 − f) = Σ CF_t/(1+y)^t
```

Cost of debt estimated from yield to maturity and adjusted for flotation f.

```
Project (divisional) discount rate = R_f + β_project × MRP, using a pure-play unlevered beta re-levered to
the target structure
Adjusted present value (APV) = Base-case NPV (all-equity) + PV(financing side effects)
Capital cash flow approach: discount at pre-tax WACC
```

Matching discount rates to risk, and the alternative to WACC.[468]

Chapter 12: Market Efficiency and Behavioural Finance

```
Abnormal return AR_t = R_t − E[R_t]
CAR = Σ AR_t over the event window
Market model: E[R_t] = α + β × R_m,t
```

Event-study building blocks; the cumulative abnormal return (CAR) measures the information effect.

```
Test statistic t = CAR / (σ_AR × √N_days)
Autocorrelation ρ_k = Cov(R_t, R_(t−k)) / Var(R_t)
Variance ratio VR(q) = Var(q-period return) / (q × Var(1-period return))
```

Tests for event significance and predictability; VR = 1 under a random walk.

```
Prospect-theory value: v(x) = x^a for gains; −λ(−x)^b for losses, λ > 1
Net present value of mispricing = Intrinsic value − Market price
```

Loss aversion makes losses weigh more than equal gains; mispricing is the gap between value and price.

Chapter 13: Raising Equity

```
Pre-money valuation + New investment = Post-money valuation
Investor ownership = Investment / Post-money valuation
Price per share = Pre-money valuation / Pre-money fully diluted shares
```

Venture financing arithmetic.

```
IPO underpricing = (First-day close − Offer price) / Offer price
Gross spread = Offer price × Shares × Spread %
Net proceeds = Gross proceeds − Underwriting fees − Other expenses
```

Costs of going public.

```
Rights offering: Ex-rights price = (N × P_cum + Subscription price) / (N + 1)
Value of one right = P_cum − Ex-rights price = (P_cum − Subscription price) / (N + 1)
```

N rights are needed to buy one new share at the subscription price.

```
Shares issued = Proceeds required / Issue price
Dilution = New shares / (Old shares + New shares)
EPS accretion/dilution = EPS_after − EPS_before
Value per share after issue = (Old shares × P + New proceeds) / (Old shares + New shares)
```

Mechanics of equity issuance.

```
Dual-class voting control = Votes held / Total votes; Cash-flow rights = Shares held / Total shares
```

Wedge between control and economic ownership.

Chapter 14: Debt Financing

```
Effective cost of a fee-bearing loan: solve Net proceeds = Σ Payments_t / (1 + y)^t
Yield of a term loan: solve Net proceeds = Σ Payments_t / (1 + y)^t
Commitment fee cost = Fee % × Undrawn amount
```

All-in cost of borrowing.

```
Floating-rate coupon = Reference rate + Margin
Swap fixed rate: Σ [df_t × S] = Σ [df_t × Forward_t]
```

Floating coupons and the par swap rate set so that fixed and floating legs have equal present value, with discount factors df.

```
Debt service coverage ratio (DSCR) = Cash available for debt service / (Interest + Principal)
Loan-to-value = Loan / Asset value
Lease liability = PV of lease payments at the incremental borrowing rate
Convertible: Conversion ratio = Par / Conversion price
Conversion value = Conversion ratio × Share price
Conversion premium = Bond price / Conversion value − 1
Convertible value ≥ max(Straight-bond value, Conversion value)
```

Credit capacity, lease capitalisation and convertible bond relationships.

```
Call-adjusted yield: solve Price = PV(coupons to call date) + PV(Call price)
Refunding NPV = PV(interest savings after tax) − (Call premium + Issue costs) after tax
```

Evaluating whether to refinance outstanding debt; discount savings at the after-tax cost of new debt.

Chapter 15: Capital Structure

```
MM Proposition I (no taxes): V_L = V_U
MM Proposition II (no taxes): r_e = r_0 + (D/E) × (r_0 − r_d)
```

In perfect markets capital structure does not affect firm value; the cost of equity rises with leverage; r_0 is the unlevered cost of capital.

```
MM with corporate taxes: V_L = V_U + T_c × D
MM Proposition II (taxes): r_e = r_0 + (D/E) × (r_0 − r_d) × (1 − T_c)
```

The interest tax shield adds T_c × D (permanent debt).

```
PV of tax shield (fixed debt, perpetual) = T_c × D
PV of tax shield (debt rebalanced to target ratio) = Σ T_c × r_d × D_t / (1 + r_d)^t, using suitable
discount rate
Miller: Gain from leverage = [1 − (1 − T_c)(1 − T_e) / (1 − T_d)] × D
```

Value of tax shield, and Miller’s model including personal taxes on equity (T_e) and debt (T_d).

```
V_L = V_U + PV(Tax shield) − PV(Financial distress costs) − PV(Agency costs of debt) + PV(Agency benefits
of debt)
```

Trade-off theory of capital structure.

```
Expected distress cost = Probability of distress × Cost of distress
Financial leverage effect on EPS: EPS = (EBIT − Interest)(1 − T) / Shares
Break-even EBIT between plans: (EBIT − I₁)(1 − T)/N₁ = (EBIT − I₂)(1 − T)/N₂
Degree of financial leverage (DFL) = EBIT / (EBIT − Interest)
Degree of total leverage = DOL × DFL
```

Practical leverage analysis.

```
Altman Z (public manufacturing) = 1.2 X₁ + 1.4 X₂ + 3.3 X₃ + 0.6 X₄ + 1.0 X₅
```

A well-known distress score where X_1 is working capital/assets, X_2 retained earnings/assets, X_3 EBIT/assets, X_4 market equity/liabilities, X_5 sales/assets.

Chapter 16: Payout Policy

```
Ex-dividend price drop ≈ Dividend × (1 − T_d) / (1 − T_g)
Total shareholder yield = (Dividends + Net repurchases) / Market capitalisation
Dividend payout ratio = DPS / EPS
```

Price adjustment on the ex-date and total cash returned to owners.

```
Repurchase: Shares retired = Cash spent / Repurchase price
Post-buyback EPS = Net income / (Shares − Shares retired)
Price after repurchase (perfect markets) = Equity value before − Cash spent, divided by remaining shares
Homemade dividends: sell shares worth (Desired cash − Dividend received)
```

Mechanics showing payout irrelevance in perfect markets (Modigliani-Miller).

```
Residual dividend policy: Dividend = Net income − Equity needed for positive-NPV projects
Equity needed = Target equity ratio × Capital budget
Lintner partial adjustment: ΔDiv_t = a + c × (target payout × EPS_t − Div_(t−1))
Target dividend = Target payout × EPS_t; c is the speed of adjustment
Stock split ratio: New shares = Old shares × Split factor; New price = Old price / Split factor
Stock dividend: New shares = Old shares × (1 + Dividend %)
```

Dividend policy rules and share-count adjustments.

Chapter 17: Working Capital

```
Operating cycle = Days inventory + DSO
Cash conversion cycle = Operating cycle − DPO
Working capital investment = Daily sales × Cash conversion cycle (approx.)
```

Time and capital tied up in operations.

```
Cost of forgoing a trade discount = [d / (1 − d)] × [365 / (Full period − Discount period)]
EAR of forgoing discount = [1 + d/(1 − d)]^(365/(Full − Disc)) − 1
```

Implied cost of not taking a d% early-payment discount.

```
Baumol EOQ-for-cash: C* = √(2 × F × T / k)
Miller-Orr: Spread = 3 × [(3/4) × F × σ² / k]^(1/3); Upper limit = Lower limit + Spread; Return point = Lower limit + Spread / 3
Economic order quantity: EOQ = √(2 × D × S / H)
```

Optimal cash conversions (F trading cost, T total cash need, k opportunity rate, σ^2 daily variance), and inventory ordering (D annual demand, S order cost, H holding cost per unit).

```
Credit policy NPV = PV(incremental contribution) − Δ Receivables investment − Δ Bad debts − Δ Collection costs
Expected cash from credit sale = (1 − Default probability) × Price / (1 + r)^t − Cost
Average receivables = Credit sales per day × DSO
Float = Average daily receipts × Days delayed
```

Credit decisions and cash-management benefits.

```
Effective rate on a line of credit = Interest + Fees / (Usable funds), annualised
Effective cost with compensating balance = Rate / (1 − Compensating balance %)
```

Short-term financing costs.

Chapter 18: Options

```
Call payoff at expiry = max(S_T − K, 0)
Put payoff at expiry = max(K − S_T, 0)
Put-call parity: C + K × e^(−rT) = P + S    (no dividends)
Put-call parity (dividends): C + PV(K) + PV(Dividends) = P + S
```

Payoffs and the no-arbitrage relationship between European calls (C) and puts (P).

```
Intrinsic value (call) = max(S − K, 0); Time value = Option price − Intrinsic value
Bounds: C ≥ max(S − K e^(−rT), 0); P ≥ max(K e^(−rT) − S, 0)
```

Lower bounds on option prices.

```
Binomial (one step): Δ = (C_u − C_d) / (S_u − S_d)
Option value = e^(−rΔt) [q C_u + (1 − q) C_d]
q = (e^(rΔt) − d) / (u − d)
```

Replicating portfolio and risk-neutral pricing, where Δ is the hedge ratio.

```
Black-Scholes call: C = S × N(d₁) − K × e^(−rT) × N(d₂)
Black-Scholes put:  P = K × e^(−rT) × N(−d₂) − S × N(−d₁)
d₁ = [ln(S/K) + (r + σ²/2) × T] / (σ × √T)
d₂ = d₁ − σ × √T
```

The Black-Scholes model for European options on non-dividend-paying stock; N(·) is the standard normal cumulative distribution.

```
With continuous dividend yield q: replace S by S × e^(−qT)
Greeks: Delta = N(d₁) (call), N(d₁) − 1 (put); Gamma = n(d₁) / (S × σ × √T); Vega = S × n(d₁) × √T
Theta = time decay; Rho = K × T × e^(−rT) × N(d₂) (call)
```

Sensitivities of option value to price, delta, volatility, time and rates; n(·) is the standard normal density.

```
Equity as a call on firm assets: E = Call(V, K = Face value of debt, T)
Debt = V − E = Risk-free debt − Put(V, K, T)
Warrant value dilution adjustment: Warrant = [N / (N + M)] × Call
```

Merton's model of corporate securities, and dilution-adjusted warrant value (N shares outstanding, M warrants).

```
Strategy payoffs: Covered call = S_T − max(S_T − K, 0)
Protective put = S_T + max(K − S_T, 0)
Straddle = |S_T − K|; Collar = S_T + Put(K₁) − Call(K₂)
```

Common option combinations at expiry (before premiums).

Chapter 19: Derivatives and Hedging

```
Forward price (no income): F₀ = S₀ × e^(rT)    or  S₀ × (1 + r)^T
Forward price (income yield q): F₀ = S₀ × e^((r − q)T)
Forward price (storage costs and convenience yield): F₀ = S₀ × e^((r + u − y)T)
```

Cost-of-carry pricing, where u is storage cost rate and y is convenience yield.

```
Value of long forward at time t = S_t − K × e^(−r(T − t))
Futures P&L = (F_t − F_(t−1)) × Contract size per day (marked to market)
Basis = Spot price − Futures price
```

Forward valuation, daily futures settlement and basis.

```
Minimum-variance hedge ratio h* = ρ × σ_S / σ_F
Number of contracts N* = h* × (Position size / Contract size)
Beta hedge: N = (β_target − β_portfolio) × Portfolio value / (Futures price × Multiplier)
Duration hedge: N = (D_target − D_portfolio) × P_portfolio / (DV01 of futures × 10,000 basis)
```

Hedge-ratio formulas: ρ is the correlation of spot and futures changes, σ_S and σ_F their standard deviations.

```
Swap value (fixed receiver) = PV(fixed leg) − PV(floating leg)
Fixed-rate swap payment = Notional × Fixed rate × Day-count fraction
Floating leg value at reset ≈ Notional
Cap payoff per period = Notional × max(Reference − Strike, 0) × τ
Floor payoff per period = Notional × max(Strike − Reference, 0) × τ
Forward rate agreement (FRA) payoff = Notional × (Ref − K) × τ / (1 + Ref × τ)
```

Interest-rate swap, cap, floor and FRA relationships; τ is the accrual fraction.

```
Credit default swap spread ≈ Annual default probability × (1 − Recovery rate)
Option-based hedge cost = Premium × Notional
```

Rough CDS pricing and the upfront cost of options hedges.

Chapter 20: Risk Management and Foreign Exchange

```
Interest rate parity (covered): F / S = (1 + r_d) / (1 + r_f)
Forward premium/discount ≈ r_d − r_f
```

Forward exchange rate F and spot S, quoted as domestic currency per unit of foreign currency; r_d and r_f are domestic and foreign interest rates.

```
Purchasing power parity (absolute): S = P_d / P_f
Relative PPP: E[S₁]/S₀ = (1 + π_d)/(1 + π_f)
International Fisher: (1 + r_d)/(1 + r_f) = E[S₁]/S₀
```

Parity conditions linking exchange rates, inflation and interest rates.

```
Cross rate: S(A/C) = S(A/B) × S(B/C)
Percentage change in foreign currency = (S₁ − S₀) / S₀
Triangular arbitrage profit = ending amount − starting amount after three conversions
```

Cross rates and arbitrage tests for quote consistency.

```
Transaction exposure = Foreign cash flows × Spot rate sensitivity
Money-market hedge: Borrow PV of foreign payable at foreign rate … convert at spot … invest domestically
Forward hedge cash flow = Foreign amount × Forward rate
```

Hedging a foreign-currency receivable or payable.

```
Foreign project NPV = Σ [E[S_t] × CF_t^foreign] / (1 + r_d)^t − I₀ × S₀
Foreign discount rate: (1 + r_d) = (1 + r_f) × E[S₁]/S₀
Value at Risk (portfolio) = z × σ_p × √(horizon days) × Portfolio value
Expected shortfall = Average loss beyond VaR
```

International capital budgeting (consistent domestic or foreign approach) and tail-risk measures.

Chapter 21: Enterprise Valuation

```
Enterprise value = Σ FCFF_t / (1 + WACC)^t + TV / (1 + WACC)^n
Equity value = Enterprise value − Net debt − Preferred − Minority interest
Value per share = Equity value / Diluted shares
```

Discounted cash flow valuation of the operating business.

```
Gordon terminal value: TV_n = FCFF_n × (1 + g) / (WACC − g)
Exit-multiple terminal value: TV_n = Multiple × EBITDA_n
Implied multiple = TV_n / EBITDA_n; Implied g = (TV × WACC − FCFF_n) / (TV + FCFF_n)
Value-driver formula: TV = NOPAT_(n+1) × (1 − g/ROIC) / (WACC − g)
```

Terminal value methods and cross-checks; ROIC is the return on new invested capital.

```
FCFE = Net income + Depreciation − CapEx − ΔNWC + Net borrowing
Equity value = Σ FCFE_t / (1 + r_e)^t
Net operating profit after tax (NOPAT) = EBIT × (1 − T)
Invested capital = Operating NWC + Net fixed assets + Other operating assets
```

Equity-based cash flow valuation and operating profit measures.

```
Relative valuation: Implied value = Peer median multiple × Target metric
EV multiples: EV/EBITDA, EV/EBIT, EV/Sales
Equity multiples: P/E, P/B
Multiples for cash flows: EV/FCFF
Mid-year discounting: PV = CF / (1 + WACC)^(t − 0.5)
Sum-of-the-parts = Σ (Segment metric × Segment multiple) − Net debt
Control premium = (Offer price − Unaffected price) / Unaffected price
```

Market-based valuation and adjustments.

Chapter 22: Mergers and Acquisitions

```
Synergy = V_AB − (V_A + V_B)
NPV to acquirer = (V_B + Synergy) − Price paid = Synergy − Premium
Premium = Price paid − V_B (stand-alone)
NPV to target = Premium
```

Value created by a combination and its split between acquirer and target.

```
Cash offer: Cost = Cash paid − V_B
Stock offer: Ownership of target = Shares issued / (Shares_A + Shares issued)
Exchange ratio = Offer price per target share / Acquirer share price
Cost (stock offer) = α × V_AB − V_B,  where α = target's post-merger ownership share
Shares issued = Exchange ratio × Target shares
```

Comparing payment methods; in a stock deal the target shares in synergy and risk.

```
Pro forma EPS = (Net income_A + Net income_B + After-tax synergies) / (Shares_A + New shares)
Accretion/(dilution) = Pro forma EPS / Acquirer EPS − 1
Break-even synergies: solve Pro forma EPS = Acquirer standalone EPS
Combined P/E (weighted) = Combined market value / Combined earnings
Bootstrap effect: EPS rises when acquirer P/E > target P/E in a stock deal without synergy
```

Earnings effects, which do not by themselves indicate value creation.

```
Merger arbitrage spread = (Offer value − Target price) / Target price
Implied probability of completion = (Price − Downside) / (Offer − Downside)
Maximum acquirer price = V_B + Synergy
Takeover premium paid in synergy share = Premium / Synergy
Contingent value rights, earn-outs: payoff = max(0, performance − threshold) × multiple
```

Deal-risk pricing and structuring tools.

Chapter 23: LBOs, Private Equity and Restructuring

```
Sources = Uses
Uses = Purchase equity + Refinanced debt + Fees
Sources = Senior debt + Subordinated debt + Sponsor equity (+ Management rollover)
```

The balance of funding in a leveraged buyout.

```
Exit equity = Exit multiple × EBITDA_exit − Net debt_exit
MOIC (multiple of invested capital) = Exit equity / Sponsor equity
IRR = MOIC^(1/years) − 1    (single inflow and outflow)
```

Sponsor return measures; for interim cash flows solve the IRR equation directly.

```
Value creation attribution:
  EBITDA growth = (EBITDA_exit − EBITDA_entry) × Entry multiple
  Multiple expansion = (Exit multiple − Entry multiple) × EBITDA_exit
  Debt paydown = Net debt_entry − Net debt_exit
```

Sources of equity gain in a buyout.

```
Cash sweep: Debt repayment_t = max(0, Cash available for debt service_t − Mandatory amortisation) × Sweep
%
Leverage = Total debt / EBITDA;  Interest cover = EBITDA / Cash interest
Free cash flow after debt service = EBITDA − Taxes − CapEx − ΔNWC − Interest − Mandatory amortisation
Maximum debt capacity ≈ Multiple × EBITDA, constrained by interest cover and DSCR
```

Debt capacity and cash-flow waterfall analysis.

```
Fund economics: Management fee = Fee % × Committed (or invested) capital
Carried interest = Carry % × (Profits above hurdle, after catch-up)
Net IRR < Gross IRR because of fees and carry
DPI = Distributions / Paid-in capital; TVPI = (Distributions + Residual value) / Paid-in capital
Preferred return (hurdle) compounding: Hurdle amount = Capital × (1 + h)^t
```

Private equity fund measures and the distribution waterfall; DPI is distributions to paid-in and TVPI total value to paid-in.

```
Liquidation value vs going-concern value: restructure if Going-concern value > Liquidation value
Recovery rate = Recovery / Claim; Absolute priority: senior claims paid in full before junior
Debt-for-equity swap: New equity share = Converted claim / Post-reorganisation equity value
Equity value in distress ≈ Call option on firm assets with strike equal to debt face value
```

Restructuring and bankruptcy logic.

Chapter 24: Sustainable Finance, Governance and Fintech

```
Carbon-adjusted NPV = NPV − PV(Carbon price × Emissions_t)
Shadow carbon price: r-adjusted escalation P_t = P₀ × (1 + g_c)^t
Green premium ("greenium") = Yield (conventional bond) − Yield (green bond)
Weighted average carbon intensity (WACI) = Σ w_i × (Emissions_i / Revenue_i)
```

Climate-aware valuation and portfolio carbon metrics; w_i are portfolio weights.

```
Sustainability-linked loan margin = Base margin ± Step-up/Step-down if KPI target is missed or met
Expected coupon step-up cost = Probability of missed KPI × Step-up × Notional
Social return on investment = PV(Social benefits) / PV(Investment)
```

Pricing features tied to ESG targets, and a simple social-return measure.

```
Board independence ratio = Independent directors / Total directors
Pay-for-performance sensitivity = ΔCEO wealth / ΔShareholder wealth
Governance cost-benefit: Value of monitoring = Reduction in agency loss − Cost of monitoring
Ownership wedge = Voting rights % − Cash-flow rights %
```

Basic governance diagnostics.

```
Payment platform revenue = Transaction volume × Take rate
Customer lifetime value (LTV) = Σ [Margin_t × Retention^t] / (1 + r)^t = Margin × (1 + r) / (1 + r − Retention) (annual)
LTV / CAC (customer acquisition cost) ratio ≥ 3 is a common rule of thumb
Payback on CAC = CAC / Monthly margin per customer
Net interest margin (NIM) = (Interest income − Interest expense) / Earning assets
Loan expected loss = PD × LGD × EAD
Token or platform network value (Metcalfe-style heuristic) ∝ n²
```

Unit economics and credit metrics for fintech platforms; PD, LGD and EAD are probability of default, loss given default and exposure at default.

Quick Reference: Key Decision Rules

```
Accept project if NPV > 0
Accept project if IRR > required return (conventional cash flows)
Accept project if PI > 1
Price a security as the PV of its expected cash flows at a risk-adjusted rate
Required return for risk: R_f + β × MRP
Optimal capital structure minimises WACC and maximises firm value
Create value only when ROIC > WACC
Hedge to reduce unwanted variance, not to speculate
In M&A, pay no more than stand-alone value plus the share of synergies you can deliver
```

A one-glance summary of the book’s central principles.

Common Constants and Conversions

```
Basis point = 0.01% = 0.0001
Rule-of-thumb market risk premium (illustrative) = 4% to 6%
Days per year convention = 365 (or 360 for money-market instruments)
z-scores: 90% one-tailed 1.282; 95% one-tailed 1.645; 97.5% 1.960; 99% 2.326
Continuous to discrete rate: r_discrete = e^(r_cont) − 1; r_cont = ln(1 + r_discrete)
Standard normal: N(0) = 0.5; N(1) ≈ 0.8413; N(−1) ≈ 0.1587; N(2) ≈ 0.9772
```

Handy numbers for calculations; market risk premiums vary with era and method and should be treated as assumptions, not facts.

Notes on Use

Three cautions apply to every formula in this appendix. First, check the timing convention: most annuity and bond formulas assume cash flows at the end of the period, and a mismatch between the rate period and the cash-flow period is the most frequent source of error. Second, keep consistency between cash flows and discount rates: nominal cash flows take nominal

rates, real cash flows take real rates, after-tax flows use after-tax rates, and flows to equity use the cost of equity. Third, treat formulas as models: each depends on assumptions such as constant growth, perfect markets, normal returns, or no default, and the manager’s task is to judge when those assumptions are acceptable and when to adjust.

Appendix B: Glossary

Terms are listed alphabetically. Chapter numbers in parentheses point to the main discussion in the book.

Abandonment option – The right to shut down a project and recover salvage value if conditions deteriorate, which sets a floor on project value (Ch. 8).

Abnormal return – The actual return on a security minus the return expected from a benchmark model; the basic measure in event studies (Ch. 12).

Accounts payable – Amounts owed to suppliers for goods and services received on credit, a spontaneous source of short-term financing (Ch. 17).

Accounts receivable – Amounts owed to the firm by customers who bought on credit; an investment in working capital (Ch. 17).

Accretion/dilution – The change in the acquirer’s earnings per share caused by an acquisition; it describes earnings effects, not value creation (Ch. 22).

Accrual accounting – Recording revenue when earned and expenses when incurred rather than when cash changes hands (Ch. 2).

Activist investor – A shareholder who takes a significant stake to press management for changes in strategy, capital allocation, or governance (Ch. 24).

Adjusted present value (APV) – A valuation method that adds the value of financing side effects, such as interest tax shields, to the all-equity value of a project (Ch. 11).

Agency conflict – A divergence of interests between a principal, such as shareholders, and an agent, such as management, that creates costs (Ch. 1).

Agency cost – The sum of monitoring, bonding, and residual losses arising from agency conflicts (Ch. 1).

Alpha – The return on an asset in excess of what its systematic risk would predict under a pricing model such as the CAPM (Ch. 10).

American option – An option that may be exercised at any time up to and including expiry (Ch. 18).

Amortisation – The gradual repayment of a loan through scheduled payments, or the allocation of an intangible asset’s cost over its useful life (Ch. 3).

Annuity – A finite series of equal cash flows occurring at regular intervals (Ch. 3).

Annuity due – An annuity whose payments occur at the beginning of each period rather than the end (Ch. 3).

Anti-takeover provision – A charter or contractual device, such as a poison pill or staggered board, that makes hostile acquisition more difficult (Ch. 22).

Arbitrage – Earning a riskless profit with no net investment by exploiting price differences between equivalent assets (Ch. 18).

Arbitrage pricing theory (APT) – A model in which expected return is a linear function of exposures to several systematic risk factors (Ch. 10).

Asset beta – The beta of a firm's operating assets, free of the effect of financial leverage; also called unlevered beta (Ch. 11).

Asset turnover – Sales divided by assets, measuring how intensively assets generate revenue (Ch. 2).

At-the-money – Describes an option whose strike price equals the current price of the underlying (Ch. 18).

Backwardation – A market condition in which futures prices are below the spot price or decline with maturity (Ch. 19).

Balance sheet – A statement of a firm's assets, liabilities, and equity at a point in time (Ch. 2).

Bankruptcy – A legal process for resolving a firm's inability to pay its debts, through liquidation or reorganisation (Ch. 23).

Basis – The difference between the spot price of an asset and the price of a related futures contract (Ch. 19).

Basis point – One hundredth of one percent, or 0.0001, used to quote small changes in rates and spreads (Ch. 4).

Basis risk – The risk that a hedging instrument and the exposure being hedged do not move perfectly together (Ch. 19).

Behavioural finance – The study of how psychological biases and limits to arbitrage affect investors and prices (Ch. 12).

Benchmark – A reference portfolio or index against which performance is measured (Ch. 10).

Beta – A measure of an asset's sensitivity to market movements, equal to its covariance with the market divided by market variance (Ch. 10).

Bid-ask spread – The difference between the price at which a dealer will buy and the price at which it will sell, a transaction cost (Ch. 12).

Binomial model – An option pricing approach that models the underlying price as moving up or down in each discrete step (Ch. 8, 18).

Black-Scholes model – A closed-form formula for pricing European options, derived by Black and Scholes and extended by Merton (Ch. 18).

Board of directors – The body elected by shareholders to oversee management and approve major strategic and financial decisions (Ch. 24).

Bond – A debt security promising fixed or floating payments of interest and principal (Ch. 4).

Bond covenant – A contractual restriction or requirement in a debt agreement that protects lenders (Ch. 14).

Bond indenture – The legal contract between issuer and bondholders that specifies terms, covenants, and the trustee's role (Ch. 14).

Book building – A process in which underwriters collect investor orders to determine the offer price of a new issue (Ch. 13).

Book value – The accounting value of an asset or of equity, based on historical cost less accumulated depreciation (Ch. 2).

Bootstrap effect – An apparent rise in EPS after an acquisition caused by a high-P/E acquirer buying a low-P/E target, without any real synergy (Ch. 22).

Break-even analysis – Determining the sales volume at which a project's profit or NPV equals zero (Ch. 7).

Bridge loan – Short-term financing that provides funds until permanent financing is arranged (Ch. 14).

Business risk – The variability of operating income arising from the firm's industry, cost structure, and demand (Ch. 15).

Call option – A contract giving the holder the right, but not the obligation, to buy an asset at a fixed strike price (Ch. 18).

Call provision – A feature allowing the issuer to redeem a bond before maturity at a stated price (Ch. 14).

Cap – A series of interest rate call options that pays when a floating rate exceeds a strike rate (Ch. 19).

Capital asset pricing model (CAPM) – The model stating that expected return equals the risk-free rate plus beta times the market risk premium (Ch. 10).

Capital budgeting – The process of evaluating and selecting long-term investments (Ch. 7).

Capital gains yield – The expected percentage price appreciation of a security, equal to g in the constant-growth model (Ch. 5).

Capital market line – The line showing the best attainable risk-return combinations from the risk-free asset and the market portfolio (Ch. 10).

Capital rationing – A situation in which a firm has more positive-NPV projects than it can fund (Ch. 6).

Capital structure – The mix of debt, equity, and other securities used to finance a firm (Ch. 15).

Carried interest – The share of fund profits, commonly 20 percent, paid to private equity general partners above a hurdle (Ch. 23).

Cash conversion cycle – The days between paying suppliers and collecting from customers; inventory days plus receivable days minus payable days (Ch. 17).

Cash flow statement – A statement reporting cash generated and used in operating, investing, and financing activities (Ch. 2).

Cash sweep – A loan provision requiring excess cash to be applied to debt repayment (Ch. 23).

CDS (credit default swap) – A contract in which a buyer pays a periodic premium in exchange for compensation if a reference entity defaults (Ch. 19).

Certainty equivalent – The guaranteed amount that makes an investor indifferent to a risky payoff (Ch. 9).

Clientele effect – The tendency of investors to prefer firms whose dividend policy suits their tax position and income needs (Ch. 16).

Collar – A strategy combining a long position with a purchased put and a written call to bound outcomes (Ch. 18).

Collateral – Assets pledged to secure a loan, which the lender may seize on default (Ch. 14).

Commercial paper – Short-term unsecured promissory notes issued by creditworthy corporations (Ch. 17).

Commodity risk – Exposure to changes in the prices of raw materials or energy (Ch. 20).

Common stock – Equity ownership that carries residual claims on cash flows and assets and usually voting rights (Ch. 5).

Comparable company analysis – Valuing a firm by applying multiples observed in similar publicly traded firms (Ch. 21).

Compensating balance – A minimum deposit a borrower must keep at a bank, raising the effective cost of a loan (Ch. 17).

Compounding frequency – The number of times per year interest is added to principal (Ch. 3).

Compound interest – Interest earned on both the original principal and previously accumulated interest (Ch. 3).

Conglomerate discount – The tendency of diversified firms to trade below the sum of the values of their parts (Ch. 21).

Contango – A market condition in which futures prices exceed the spot price or rise with maturity (Ch. 19).

Contingent claim – A security whose payoff depends on the value of another asset, such as an option (Ch. 18).

Control premium – The excess over the unaffected market price paid to gain control of a company (Ch. 21).

Conversion ratio – The number of common shares received on conversion of one convertible bond or preferred share (Ch. 14).

Convertible bond – A bond that the holder may exchange for a fixed number of shares (Ch. 14).

Convexity – The curvature of the price-yield relationship of a bond, which refines duration estimates of price changes (Ch. 4).

Corporate governance – The system of rules, practices, and processes by which a company is directed and controlled (Ch. 24).

Correlation – A standardised measure, from −1 to +1, of how closely two variables move together (Ch. 9).

Cost of capital – The return required by investors on a firm’s capital, used as the discount rate for projects of similar risk (Ch. 11).

Cost of debt – The effective rate a firm pays on its borrowing, usually measured as the yield to maturity before tax (Ch. 11).

Cost of equity – The return shareholders require for bearing the risk of owning the firm’s stock (Ch. 11).

Coupon rate – The annual interest payment on a bond expressed as a percentage of face value (Ch. 4).

Covariance – A measure of how two random variables move together, unscaled by their standard deviations (Ch. 9).

Covenant-lite loan – A loan with few maintenance covenants, offering borrowers more flexibility and lenders less protection (Ch. 14).

Covered call – Holding an asset while writing a call on it to earn premium income (Ch. 18).

Credit rating – An independent opinion on an issuer’s creditworthiness, expressed on a letter scale (Ch. 4).

Credit risk – The risk that a borrower or counterparty fails to meet its payment obligations (Ch. 4).

Credit spread – The yield on a risky bond minus the yield on a comparable risk-free bond (Ch. 4).

Cross rate – An exchange rate between two currencies derived from each one’s rate against a third currency (Ch. 20).

Cumulative abnormal return (CAR) – The sum of abnormal returns over an event window (Ch. 12).

Current ratio – Current assets divided by current liabilities, a measure of liquidity (Ch. 2).

Current yield – A bond’s annual coupon divided by its market price (Ch. 4).

Days sales outstanding (DSO) – The average number of days it takes to collect receivables (Ch. 17).

Dead-weight cost – A loss of value, such as the expense of financial distress, that benefits no party (Ch. 15).

Debenture – An unsecured bond backed only by the issuer’s general creditworthiness (Ch. 14).

Debt capacity – The amount of debt a firm can carry while retaining acceptable credit quality and flexibility (Ch. 15).

Debt overhang – A situation in which existing debt discourages new investment because gains would accrue mostly to creditors (Ch. 15).

Debt service coverage ratio (DSCR) – Cash available for debt service divided by scheduled interest and principal (Ch. 14).

Debt-to-equity ratio – Total debt divided by total equity, a measure of financial leverage (Ch. 2).

Default – Failure to make a promised payment or to comply with a debt covenant (Ch. 4).

Default risk premium – The extra yield investors demand for bearing the chance that the issuer will not pay in full (Ch. 4).

Delta – The change in an option's price per unit change in the underlying asset's price (Ch. 18).

Depreciation – The allocation of the cost of a tangible asset over its useful life, a non-cash expense that creates a tax shield (Ch. 7).

Depreciation tax shield – The reduction in taxes from deducting depreciation, equal to depreciation times the tax rate (Ch. 7).

Derivative – A financial contract whose value derives from an underlying asset, rate, or index (Ch. 19).

Dilution – The reduction in existing shareholders' percentage ownership or per-share value when new shares are issued (Ch. 13).

Discounted payback – The time needed to recover an investment from discounted cash flows (Ch. 6).

Discount factor – The present value of one unit received in a future period, equal to 1/(1+r)^t (Ch. 3).

Discount rate – The rate used to convert future cash flows to present value, reflecting their risk and timing (Ch. 3).

Diversification – Spreading investments across assets to reduce risk that is specific to individual holdings (Ch. 9).

Dividend – A distribution of earnings by a company to its shareholders, usually in cash (Ch. 16).

Dividend discount model – A valuation model that sets a stock's price equal to the present value of expected dividends (Ch. 5).

Dividend irrelevance – The Modigliani-Miller result that, in perfect markets, dividend policy does not affect firm value (Ch. 16).

Dividend payout ratio – The fraction of earnings paid out as dividends (Ch. 16).

Dividend yield – Annual dividends per share divided by share price (Ch. 5).

Dual-class shares – A structure in which classes of stock carry different voting rights, separating control from cash-flow ownership (Ch. 24).

Due diligence – The investigation of a target or issuer's financial, legal, and operational condition before a transaction (Ch. 22).

DuPont analysis – A decomposition of return on equity into profit margin, asset turnover, and leverage (Ch. 2).

Duration – A measure of a bond's price sensitivity to yield changes, equal to the weighted average time to receipt of cash flows (Ch. 4).

Earnings before interest and taxes (EBIT) – Operating profit before financing costs and income taxes (Ch. 2).

Earnings per share (EPS) – Net income available to common shareholders divided by weighted average shares outstanding (Ch. 2).

Earn-out – A deal feature in which part of the purchase price depends on the target's future performance (Ch. 22).

EBITDA – Earnings before interest, taxes, depreciation, and amortisation, a common proxy for operating cash earnings (Ch. 2).

Economic profit – Operating profit after tax minus a charge for the capital employed; also called economic value added (Ch. 1).

Effective annual rate (EAR) – The annual rate that accounts for compounding within the year (Ch. 3).

Efficient frontier – The set of portfolios offering the highest expected return for each level of risk (Ch. 10).

Efficient market hypothesis – The proposition that security prices fully reflect available information to a specified degree (Ch. 12).

Enterprise value – The market value of a firm's operating business: equity plus debt and other claims minus cash (Ch. 21).

Environmental, social and governance (ESG) – A framework for assessing a firm's environmental impact, social relationships, and governance quality (Ch. 24).

Equity multiplier – Total assets divided by equity, a measure of leverage used in DuPont analysis (Ch. 2).

Equity risk premium – The expected excess return of equities over the risk-free rate (Ch. 11).

Equivalent annual annuity – The level annual cash flow with the same present value as a project's NPV over its life, used to compare projects of unequal lives (Ch. 6).

Eurobond – A bond issued in a currency other than that of the country where it is sold (Ch. 14).

European option – An option that may be exercised only at expiry (Ch. 18).

Event study – An empirical method measuring the effect of an announcement on security prices through abnormal returns (Ch. 12).

Exchange rate – The price of one currency in terms of another (Ch. 20).

Ex-dividend date – The first day on which a buyer of a stock is not entitled to the declared dividend (Ch. 16).

Exercise price – The fixed price at which an option holder may buy or sell the underlying; also called strike price (Ch. 18).

Expected return – The probability-weighted average of possible returns (Ch. 9).

Expected shortfall – The average loss in the tail beyond the value at risk threshold (Ch. 20).

Expiration date – The date on which an option or derivative contract ceases to exist (Ch. 18).

Face value – The principal amount a bond repays at maturity; also par value (Ch. 4).

Fair value – The price at which an asset would exchange between informed, willing parties in an orderly transaction (Ch. 2).

Fama-French model – An asset pricing model adding size and value factors to the market factor (Ch. 10).

Financial distress – A condition in which a firm has difficulty meeting obligations to creditors (Ch. 15).

Financial flexibility – The ability of a firm to raise capital or adjust operations to meet unforeseen needs (Ch. 15).

Financial lease – A long-term lease that transfers substantially all the risks and rewards of ownership to the lessee (Ch. 14).

Financial leverage – The use of debt to finance assets, which magnifies returns and risk to equity (Ch. 15).

Firm commitment underwriting – An arrangement in which the underwriter buys the whole issue and bears the risk of resale (Ch. 13).

Fisher effect – The relationship that the nominal interest rate approximately equals the real rate plus expected inflation (Ch. 3).

Fixed costs – Costs that do not vary with output in the short run (Ch. 7).

Floating-rate note – A bond whose coupon resets periodically relative to a reference rate (Ch. 4).

Floor – A series of interest rate put options that pays when a floating rate falls below a strike rate (Ch. 19).

Flotation cost – The expense of issuing new securities, including underwriting fees and legal costs (Ch. 13).

Forward contract – A customised agreement to buy or sell an asset at a fixed price on a future date (Ch. 19).

Forward rate – An interest rate for a future period implied by current spot rates (Ch. 4).

Free cash flow – Cash generated by operations after taxes and reinvestment in capital and working capital (Ch. 2, 7).

Free cash flow to equity (FCFE) – Cash flow available to shareholders after debt service and reinvestment (Ch. 21).

Free cash flow to the firm (FCFF) – Cash flow available to all capital providers before financing payments (Ch. 21).

Free-rider problem – The tendency of dispersed shareholders to leave costly monitoring to others (Ch. 24).

Futures contract – A standardised, exchange-traded agreement to buy or sell an asset at a set price on a future date, settled daily (Ch. 19).

Future value – The amount to which a present sum grows when invested at a given rate (Ch. 3).

Gamma – The rate of change of an option’s delta with respect to the underlying price (Ch. 18).

General partner – The manager of a private equity or venture fund who makes investments and bears unlimited liability (Ch. 23).

Geometric mean return – The compound average growth rate of an investment over multiple periods (Ch. 9).

Golden parachute – A contractual payment to executives upon loss of their jobs after a change in control (Ch. 22).

Goodwill – The excess of an acquisition price over the fair value of identifiable net assets acquired (Ch. 22).

Green bond – A bond whose proceeds are earmarked for environmentally beneficial projects (Ch. 24).

Greenium – The lower yield, relative to comparable conventional bonds, that investors sometimes accept on green bonds (Ch. 24).

Gross margin – Sales minus cost of goods sold, as a percentage of sales (Ch. 2).

Growth opportunities – Future investments with positive NPV, whose value is included in a stock price beyond current earnings (Ch. 5).

Hamada equation – A formula relating levered beta to unlevered beta, the debt-equity ratio, and the tax rate (Ch. 11).

Hedge ratio – The quantity of a hedging instrument per unit of exposure needed to offset risk (Ch. 19).

Hedging – Taking an offsetting position to reduce exposure to price, rate, or currency movements (Ch. 19).

Herding – The tendency of investors to imitate the actions of others rather than act on their own analysis (Ch. 12).

High-yield bond – A bond rated below investment grade, offering higher yield for higher default risk (Ch. 14).

Holding period return – The total return from owning an asset for a given period, including income and price change (Ch. 9).

Hostile takeover – An acquisition attempt made without the agreement of the target's board (Ch. 22).

Hubris hypothesis – The idea that acquirers overpay because managers overestimate their ability to create value (Ch. 22).

Hurdle rate – The minimum return required for an investment to be accepted, or the preferred return in a private equity fund (Ch. 6, 23).

Idiosyncratic risk – Risk specific to a single firm or asset that can be diversified away; also unsystematic risk (Ch. 9).

Implied volatility – The volatility of the underlying asset that makes a model price equal to the observed option price (Ch. 18).

Income statement – A statement of revenues, expenses, and profit over a period (Ch. 2).

Incremental cash flow – The change in a firm's cash flows resulting from accepting a project (Ch. 7).

Information asymmetry – A situation in which one party, typically management, knows more than another, such as investors (Ch. 13).

Initial public offering (IPO) – The first sale of a company's shares to the public (Ch. 13).

Insider trading – Trading securities on material non-public information in breach of a duty, which is illegal in most jurisdictions (Ch. 12).

Intangible assets – Non-physical assets such as patents, brands, and software (Ch. 2).

Interest coverage ratio – EBIT divided by interest expense, a measure of ability to service debt (Ch. 2).

Interest rate parity – The condition that the forward exchange rate premium reflects the interest rate differential between two currencies (Ch. 20).

Interest rate risk – The risk that changes in interest rates affect the value of assets or the cost of liabilities (Ch. 4).

Interest rate swap – An agreement to exchange fixed-rate and floating-rate interest payments on a notional principal (Ch. 19).

Interest tax shield – The tax saving from the deductibility of interest expense (Ch. 15).

Internal rate of return (IRR) – The discount rate that makes a project's NPV equal to zero (Ch. 6).

International Fisher effect – The proposition that expected currency changes reflect nominal interest rate differentials (Ch. 20).

In-the-money – Describes an option that would have positive payoff if exercised immediately (Ch. 18).

Intrinsic value – The value of an asset based on fundamentals, or for an option the payoff if exercised immediately (Ch. 5, 18).

Inventory turnover – Cost of goods sold divided by average inventory (Ch. 2).

Investment banker – A financial intermediary that advises on and underwrites securities offerings and transactions (Ch. 13).

Investment grade – Credit ratings at or above BBB− (S&P scale) or Baa3 (Moody's scale), indicating relatively low default risk (Ch. 4).

Jensen's alpha – The excess return of a portfolio over its CAPM-predicted return (Ch. 10).

Junk bond – Informal term for a high-yield bond (Ch. 14).

Key performance indicator (KPI) – A measurable value used to track progress toward an objective, often linked to sustainability-linked financing (Ch. 24).

Lease – A contract in which the owner of an asset grants another party the right to use it for payments (Ch. 14).

Letter of credit – A bank's guarantee of payment on behalf of a customer, common in trade finance (Ch. 17).

Leveraged buyout (LBO) – An acquisition financed largely with debt secured on the target's assets and cash flows (Ch. 23).

Leverage ratio – A measure of the extent to which debt finances a firm, such as debt to EBITDA (Ch. 2).

Limited partner – An investor in a private equity fund who provides capital and has limited liability and no management role (Ch. 23).

Limits to arbitrage – Frictions such as risk and costs that prevent arbitrageurs from eliminating mispricing (Ch. 12).

Line of credit – A bank commitment to lend up to a stated amount, drawn as needed (Ch. 17).

Lintner model – A description of dividend policy in which firms gradually adjust dividends toward a target payout (Ch. 16).

Liquidation value – The net amount realised by selling a firm's assets piecemeal (Ch. 23).

Liquidity – The ease with which an asset can be converted to cash without significant loss of value (Ch. 2).

Liquidity premium – The extra return demanded for holding assets that are hard to sell quickly (Ch. 4).

Lock-up period – A period after an IPO during which insiders may not sell their shares (Ch. 13).

Long position – Ownership of an asset or a contract that gains when the underlying price rises (Ch. 19).

Macaulay duration – The present-value-weighted average time until a bond's cash flows are received (Ch. 4).

Management buyout (MBO) – An acquisition of a company by its own managers, usually with outside financing (Ch. 23).

Management fee – The annual fee a fund manager charges, typically a percentage of committed or invested capital (Ch. 23).

Marginal tax rate – The tax rate on the next dollar of income (Ch. 7).

Margin call – A demand to deposit additional funds when a margin account falls below the required level (Ch. 19).

Market capitalisation – Share price multiplied by shares outstanding (Ch. 2).

Market efficiency – The degree to which prices reflect information (Ch. 12).

Market model – A regression relating a security's return to the market return, used in event studies (Ch. 12).

Market portfolio – The portfolio of all risky assets weighted by market value (Ch. 10).

Market risk – Systematic risk that affects all assets and cannot be diversified away (Ch. 9).

Market risk premium – The expected return on the market portfolio minus the risk-free rate (Ch. 10).

Market-to-book ratio – The market value of equity divided by its book value (Ch. 2).

Maturity – The date on which a debt instrument's principal becomes due (Ch. 4).

Mean-variance analysis – Portfolio selection based on expected return and variance, introduced by Markowitz (Ch. 10).

Merger – A combination in which one firm absorbs another and only one legal entity survives (Ch. 22).

Mezzanine financing – Subordinated debt or preferred equity that ranks between senior debt and common equity (Ch. 23).

Minimum-variance portfolio – The portfolio on the efficient frontier with the lowest possible risk (Ch. 10).

Modified duration – Macaulay duration divided by one plus the yield per period, giving percentage price change per unit yield change (Ch. 4).

Modified internal rate of return (MIRR) – An IRR variant that assumes explicit reinvestment and financing rates (Ch. 6).

Modigliani-Miller theorems – Propositions showing that, in perfect markets, capital structure and dividend policy do not affect firm value (Ch. 15, 16).

Momentum – The tendency of recent winning securities to continue outperforming in the short to medium term (Ch. 12).

Money market – The market for short-term, highly liquid debt instruments (Ch. 17).

Moral hazard – The temptation for an insured or protected party to take more risk because someone else bears the cost (Ch. 1).

Mortgage bond – A bond secured by a lien on specific real property (Ch. 14).

Multiple of invested capital (MOIC) – Total value returned to investors divided by capital invested (Ch. 23).

Mutually exclusive projects – Alternatives for which accepting one precludes accepting the others (Ch. 6).

Net advantage to leasing (NAL) – The present value of the benefits of leasing versus buying, after taxes (Ch. 7).

Net debt – Total debt minus cash and equivalents (Ch. 21).

Net income – Revenue minus all expenses, including interest and taxes (Ch. 2).

Net present value (NPV) – The present value of a project’s cash inflows minus the present value of its outflows (Ch. 6).

Net working capital – Current assets minus current liabilities (Ch. 2, 17).

Nominal rate – A stated interest rate not adjusted for compounding or inflation (Ch. 3).

Non-recourse debt – Debt for which the lender’s claim is limited to specified assets or project cash flows (Ch. 14).

NOPAT – Net operating profit after tax: EBIT multiplied by one minus the tax rate (Ch. 21).

Normal distribution – A symmetric bell-shaped probability distribution fully described by its mean and standard deviation (Ch. 9).

Notional principal – The reference amount on which payments in a swap or other derivative are calculated (Ch. 19).

Operating cycle – The time from acquiring inventory to collecting cash from its sale (Ch. 17).

Operating lease – A lease that does not transfer most of the risks and rewards of ownership (Ch. 14).

Operating leverage – The degree to which fixed operating costs magnify the effect of sales changes on operating income (Ch. 7).

Opportunity cost – The value of the best alternative forgone when a resource is used for a project (Ch. 7).

Option premium – The price paid by an option buyer to the seller (Ch. 18).

Out-of-the-money – Describes an option that would have no payoff if exercised immediately (Ch. 18).

Overconfidence – A bias in which individuals overestimate their knowledge or the precision of their forecasts (Ch. 12).

Over-the-counter (OTC) – Describes trades negotiated directly between parties rather than on an exchange (Ch. 19).

Par value – The face value of a bond or the nominal value assigned to a share (Ch. 4).

Payback period – The time required for a project's cumulative cash flows to recover the initial investment (Ch. 6).

Pecking order theory – The theory that firms prefer internal funds, then debt, and issue equity only as a last resort due to information asymmetry (Ch. 15).

Perpetuity – A stream of equal cash flows continuing indefinitely (Ch. 3).

Poison pill – A defensive measure that dilutes a hostile bidder's stake if it crosses a stated ownership threshold (Ch. 22).

Political risk – The risk that government actions harm the value of an investment abroad (Ch. 20).

Portfolio – A collection of investments held by an investor or institution (Ch. 9).

Portfolio weight – The proportion of a portfolio's value invested in a given asset (Ch. 10).

Post-money valuation – The value of a company immediately after a financing round, including new money (Ch. 13).

Preferred stock – An equity security with a fixed dividend and priority over common stock in dividends and liquidation (Ch. 5).

Premium – The amount by which a price exceeds a reference value, such as an offer above market price (Ch. 22).

Pre-money valuation – The value of a company before a new financing round (Ch. 13).

Present value – The current worth of a future cash flow discounted at an appropriate rate (Ch. 3).

Price-earnings ratio (P/E) – Share price divided by earnings per share (Ch. 5).

Primary market – The market in which new securities are sold by issuers to investors (Ch. 13).

Prime rate – A benchmark rate that banks use as a base for loans to creditworthy customers (Ch. 17).

Principal-agent problem – The difficulty of motivating an agent to act in the principal's interest (Ch. 1).

Private equity – Investment in companies not traded on public markets, typically through funds (Ch. 23).

Private placement – A sale of securities directly to a small group of institutional investors rather than the public (Ch. 14).

Probability distribution – A listing of the possible outcomes of a random variable and their probabilities (Ch. 9).

Profitability index – The present value of future cash flows divided by the initial investment (Ch. 6).

Pro forma statements – Projected financial statements prepared under specified assumptions (Ch. 7).

Project finance – Financing of a standalone project primarily from its own cash flows, usually through a special-purpose vehicle (Ch. 14).

Prospect theory – A descriptive theory of choice, by Kahneman and Tversky, in which people value gains and losses relative to a reference point and fear losses more (Ch. 12).

Prospectus – A legal document describing a securities offering to prospective investors (Ch. 13).

Proxy fight – An effort by dissident shareholders to win board seats or votes by soliciting proxies (Ch. 22).

Put-call parity – The no-arbitrage relationship linking the prices of European calls and puts with the same strike and expiry (Ch. 18).

Put option – A contract giving the holder the right, but not the obligation, to sell an asset at a fixed strike price (Ch. 18).

Quick ratio – Liquid current assets, excluding inventory, divided by current liabilities (Ch. 2).

Random walk – A process in which price changes are unpredictable and independent of past changes (Ch. 12).

Real option – The flexibility to alter a real investment, such as to expand, delay, or abandon, valued like a financial option (Ch. 8).

Real rate – An interest rate adjusted to remove the effect of inflation (Ch. 3).

Recapitalisation – A significant change in a firm's capital structure, such as a leveraged payout (Ch. 23).

Recovery rate – The fraction of a claim recovered by creditors after default (Ch. 4).

Reinvestment risk – The risk that cash flows from an investment must be reinvested at lower rates than expected (Ch. 4).

Repurchase (buyback) – A transaction in which a company buys its own shares from shareholders (Ch. 16).

Residual dividend policy – A policy of paying dividends only from earnings left over after funding all positive-NPV investments (Ch. 16).

Residual income – Earnings in excess of the required return on book equity, used in valuation (Ch. 5).

Restructuring – A significant change to a firm's operations, assets, or financing to improve value or avoid failure (Ch. 23).

Return on assets (ROA) – Net income divided by total assets (Ch. 2).

Return on equity (ROE) – Net income divided by shareholders' equity (Ch. 2).

Return on invested capital (ROIC) – NOPAT divided by invested capital, measuring operating returns to all capital providers (Ch. 2).

Revolving credit facility – A committed credit line that the borrower may draw, repay, and redraw (Ch. 14).

Rights offering – An issue of new shares offered first to existing shareholders in proportion to their holdings (Ch. 13).

Risk aversion – A preference for a certain outcome over a risky one with the same expected value (Ch. 9).

Risk-free rate – The return on an asset with no default risk, usually proxied by government securities (Ch. 10).

Risk-neutral probability – An artificial probability under which expected payoffs discounted at the risk-free rate give correct prices (Ch. 18).

Risk premium – The return earned over the risk-free rate as compensation for bearing risk (Ch. 9).

Roadshow – A series of presentations by company management and underwriters to prospective investors before an offering (Ch. 13).

Salvage value – The estimated resale value of an asset at the end of its useful life (Ch. 7).

Scenario analysis – Evaluating a project's outcomes under a small number of coherent combinations of assumptions (Ch. 8).

Seasoned equity offering (SEO) – A sale of additional shares by a firm that is already publicly listed (Ch. 13).

Secondary market – The market in which investors trade previously issued securities among themselves (Ch. 13).

Secured debt – Debt backed by specific collateral (Ch. 14).

Security market line (SML) – The graph of the CAPM relating expected return to beta (Ch. 10).

Senior debt – Debt with priority over other claims in payment and liquidation (Ch. 14).

Sensitivity analysis – Examining how a project's NPV changes when one input varies while others are held constant (Ch. 8).

Shareholder value – The value of the firm's equity to its owners, the central objective in corporate finance (Ch. 1).

Sharpe ratio – A portfolio's excess return over the risk-free rate divided by its standard deviation (Ch. 9).

Short position – A position that gains when the underlying price falls, often created by borrowing and selling an asset (Ch. 19).

Signalling – Actions, such as dividend changes, that convey private information to investors (Ch. 16).

Simulation (Monte Carlo) – A technique that evaluates outcomes by repeatedly sampling inputs from probability distributions (Ch. 8).

Sinking fund – A provision requiring the issuer to retire a portion of a bond issue periodically (Ch. 14).

Social bond – A bond whose proceeds fund projects with positive social outcomes (Ch. 24).

Sovereign wealth fund – A state-owned investment fund financed from national surpluses such as commodity revenues (Ch. 23).

Special purpose vehicle (SPV) – A separate legal entity created to isolate financial risk or hold specific assets (Ch. 14).

Spot rate – The interest rate or exchange rate for immediate settlement (Ch. 4, 20).

Spread – The difference between two prices or rates, such as the bid-ask spread or credit spread (Ch. 4).

Stakeholder – Any group with an interest in the firm, including employees, customers, suppliers, and communities (Ch. 1).

Standard deviation – The square root of variance, a common measure of total risk (Ch. 9).

Statement of cash flows – See Cash flow statement (Ch. 2).

Stock dividend – A distribution of additional shares to existing shareholders in proportion to holdings (Ch. 16).

Stock split – A division of existing shares into a larger number, lowering the price per share (Ch. 16).

Straddle – An option strategy combining a call and a put with the same strike and expiry (Ch. 18).

Strategic buyer – An acquirer motivated by operational synergies in its own business, as opposed to a financial sponsor (Ch. 22).

Strike price – See Exercise price (Ch. 18).

Structured finance – Techniques that pool assets and issue tranched securities with different risk profiles (Ch. 14).

Subordinated debt – Debt that ranks below senior debt in claims on assets (Ch. 14).

Sunk cost – A cost already incurred that cannot be recovered and should be ignored in decisions (Ch. 7).

Supply chain finance – Arrangements in which buyers' credit strength is used to give suppliers cheaper early payment (Ch. 17).

Sustainability-linked loan – A loan whose interest margin varies with the borrower's performance against sustainability targets (Ch. 24).

Sustainable growth rate – The maximum growth rate a firm can sustain without raising new equity or changing leverage or payout (Ch. 2).

Swap – An agreement to exchange cash flows according to a formula over time (Ch. 19).

Swaption – An option to enter into a swap at a specified rate in the future (Ch. 19).

Synergy – The additional value created by combining two firms beyond the sum of their stand-alone values (Ch. 22).

Systematic risk – Risk common to all assets in the economy that cannot be eliminated by diversification (Ch. 9).

Takeover – The acquisition of control of a target firm, friendly or hostile (Ch. 22).

Tangency portfolio – The risky portfolio with the highest Sharpe ratio, the optimal risky holding when a risk-free asset exists (Ch. 10).

Tax shield – A reduction in taxable income from a deductible item such as interest or depreciation (Ch. 7, 15).

Tender offer – A public offer to buy shares directly from shareholders at a stated price (Ch. 22).

Terminal value – The value of cash flows beyond an explicit forecast horizon (Ch. 21).

Term structure of interest rates – The relationship between yields and maturities of bonds of similar quality (Ch. 4).

Theta – The rate at which an option loses value as time passes (Ch. 18).

Time value of money – The principle that a sum received today is worth more than the same sum received later (Ch. 3).

Tobin’s q – The ratio of the market value of a firm’s assets to their replacement cost (Ch. 5).

Tracking error – The standard deviation of the difference between a portfolio’s return and its benchmark’s (Ch. 10).

Trade credit – Credit extended by suppliers allowing delayed payment for goods (Ch. 17).

Trade-off theory – The theory that firms balance the tax benefits of debt against the costs of financial distress (Ch. 15).

Tranche – A slice of a securitised or structured instrument with a specific risk and priority (Ch. 14).

Transaction exposure – The risk that exchange-rate changes alter the value of contracted foreign-currency cash flows (Ch. 20).

Translation exposure – The effect of exchange-rate changes on reported consolidated statements (Ch. 20).

Treasury bill – A short-term government security issued at a discount (Ch. 17).

Treasury function – The part of the finance organisation managing cash, funding, and financial risk (Ch. 1).

Treasury stock – Shares repurchased by the issuing company and held in its treasury (Ch. 16).

Triangular arbitrage – Profiting from inconsistent exchange rates among three currencies (Ch. 20).

Underpricing – The tendency of IPO shares to be offered below their first-day trading price (Ch. 13).

Underwriter – A financial institution that markets a securities offering and often guarantees its sale (Ch. 13).

Unlevered beta – See Asset beta (Ch. 11).

Value at Risk (VaR) – The loss that is not expected to be exceeded over a given horizon at a given confidence level (Ch. 9, 20).

Value driver formula – A valuation expression linking terminal value to growth, return on invested capital, and the cost of capital (Ch. 21).

Variable costs – Costs that vary in proportion to output (Ch. 7).

Variance – The expected squared deviation of a random variable from its mean (Ch. 9).

Vega – The sensitivity of an option’s price to changes in the volatility of the underlying (Ch. 18).

Venture capital – Financing provided to early-stage, high-potential companies in exchange for equity (Ch. 13).

Volatility – The degree of variation in an asset’s returns, usually measured as annualised standard deviation (Ch. 18).

Warrant – A long-dated option issued by a company to buy its own shares at a fixed price (Ch. 18).

Weighted average cost of capital (WACC) – The blended after-tax cost of a firm’s debt, equity, and other capital, weighted by market values (Ch. 11).

White knight – A friendly acquirer sought by a target to avoid a hostile bid (Ch. 22).

Working capital management – Planning and control of current assets and liabilities to balance liquidity and profitability (Ch. 17).

Yield curve – A graph of yields against maturities for bonds of similar credit quality (Ch. 4).

Yield to call – The yield earned if a callable bond is held until its call date and redeemed (Ch. 4).

Yield to maturity (YTM) – The discount rate that equates a bond’s price with the present value of its promised cash flows (Ch. 4).

Zero-coupon bond – A bond that pays no periodic interest and is issued at a discount to face value (Ch. 4).

Z-score – Either a count of standard deviations from a mean, or the Altman score used to predict bankruptcy (Ch. 9, 15).

Appendix C: Study Guide and Course Designs

This appendix helps students and instructors turn the book into a working course. It offers semester plans for a 14-week and an 8-week format, exam-preparation checklists, a team project template with a grading rubric, and suggested reading categories for going further. The plans are starting points; adapt them to class size, prior preparation, and the mix of full-time, part-time, and executive students.

C.1 Principles Behind the Course Designs

Four principles shaped the plans below. First, **valuation is the spine**. Chapters 3 to 6 build the machinery of discounting, and every later topic, from cost of capital to M&A, is an application of it. Students who are shaky here will struggle later, so the early weeks deserve more time than their page count suggests. Second, **risk is priced, not avoided**. Chapters 9 to 11 explain why discount rates differ across projects, and this idea must be re-used continuously. Third, **decisions come in three families**: investment (what to buy), financing (how to pay for it), and payout or risk management (what to return or protect). Each class session should be placed in one of these families so students see the whole. Fourth, **judgement matters as much as calculation**. Models are only as good as their assumptions, so every numerical session should end with the question: what would change my mind?

Suggested workload is roughly two to three hours of preparation per hour of class. Students from non-financial backgrounds benefit from a pre-course refresher on algebra, logarithms, and basic statistics (mean, variance, correlation, regression), plus a spreadsheet tutorial covering NPV, IRR, and goal seek functions.

C.2 Fourteen-Week Semester Plan

The 14-week plan assumes one three-hour session per week or two 90-minute sessions. It covers all 24 chapters, with some chapters paired. Weeks 7 and 14 hold assessments or review.

Week 1: The Firm and Its Financial Statements (Chapters 1 and 2)

Introduce the corporation, the goal of value maximisation, agency problems, and the roles of the CFO and treasurer. Then read financial statements as a manager: link the balance sheet, income statement, and cash flow statement, and use DuPont analysis to diagnose performance. In-class activity: students receive statements for a fictional firm and write a half-page diagnosis of strengths and weaknesses. Deliverable: ratio workbook template completed.

Week 2: Time Value of Money and Bonds (Chapters 3 and 4)

Teach present value, future value, annuities, perpetuities, effective rates, and loan amortisation, then apply them to bond pricing, yields, duration, and credit spreads. In-class activity: price a bond from a coupon schedule and shock its yield to compare duration-based and exact price changes. Deliverable: problem set 1 on discounting and bonds.

Week 3: Equity Valuation and NPV (Chapters 5 and 6)

Value stocks with dividend models, earnings multiples, and residual income; then establish NPV as the master rule and compare it with IRR, payback, profitability index, and capital rationing. In-class activity: a debate on a project where IRR and NPV conflict. Deliverable: problem set 2.

Week 4: Capital Budgeting and Uncertainty (Chapters 7 and 8)

Build project cash flows with taxes, depreciation, working capital, and terminal values; run sensitivity, scenario, and break-even analysis; introduce decision trees and real options. In-class activity: a full spreadsheet capital budget for a fictional plant, with an abandonment option added. Deliverable: capital budgeting mini-case.

Week 5: Risk, Return, Portfolios and CAPM (Chapters 9 and 10)

Define risk and return, correlation, diversification, the efficient frontier, the capital allocation line, beta, the security market line, and multifactor models. In-class activity: a two-asset portfolio exercise showing how correlation changes risk. Deliverable: problem set 3.

Week 6: Cost of Capital and Market Efficiency (Chapters 11 and 12)

Estimate WACC, unlever and relever betas, and select project-specific rates; then discuss what efficiency means, the evidence on anomalies, and the lessons of behavioural finance for managers and investors. In-class activity: estimate a divisional hurdle rate from comparable firms. Deliverable: cost of capital worksheet; project proposal due (Section C.6).

Week 7: Midterm Examination and Review

A 120-minute closed-book examination covering Chapters 1 to 12, with a formula sheet permitted (Appendix A without worked content). The balance should be about 40 percent calculations, 30 percent short interpretation, and 30 percent a mini-case.

Week 8: Raising Capital (Chapters 13 and 14)

Cover venture financing and valuation rounds, IPOs, seasoned offerings and rights issues, then bank loans, bonds, covenants, leases, convertibles, and project finance. In-class activity: structure the financing of a growth company and justify the mix. Deliverable: problem set 4.

Week 9: Capital Structure and Payout Policy (Chapters 15 and 16)

Teach Modigliani-Miller, taxes, distress costs, trade-off and pecking-order views, and the practical determinants of leverage; then dividends, repurchases, signalling, and clientele. In-class activity: choose a target capital structure for a fictional firm and recommend a payout policy. Deliverable: capital structure memo (one page).

Week 10: Working Capital and Options (Chapters 17 and 18)

Study the cash conversion cycle, credit policy, inventory, and short-term financing, then option payoffs, put-call parity, the binomial model, Black-Scholes, and corporate securities as options. In-class activity: compute the cost of forgoing a trade discount and price an option on a tree. Deliverable: problem set 5.

Week 11: Derivatives, Hedging and Foreign Exchange (Chapters 19 and 20)

Cover forwards, futures, swaps, caps and floors, hedge ratios, exchange-rate parity, exposure types, and hedging policy. In-class activity: design a hedging programme for a fictional exporter, comparing a forward with an option. Deliverable: hedging recommendation.

Week 12: Enterprise Valuation and M&A (Chapters 21 and 22)

Perform a full discounted cash flow valuation with terminal value and cross-checks against multiples; then evaluate synergies, payment methods, accretion and dilution, and takeover defences. In-class activity: a bid simulation in which teams decide how much of the synergy to concede. Deliverable: valuation exercise.

Week 13: LBOs, Restructuring, and Sustainable Finance (Chapters 23 and 24)

Study leveraged buyout models, private equity economics, distress and restructuring, then sustainable finance, governance, and fintech business models. In-class activity: a simple LBO model with returns attribution, and a debate on whether green financing lowers the cost of capital. Deliverable: team project drafts exchanged for peer review.

Week 14: Integration and Final Presentations

Teams present their projects (Section C.6); the instructor leads a closing discussion tying together investment, financing, and payout decisions. The final examination is cumulative, with weight on Chapters 13 to 24 but including integrative valuation questions.

Suggested Assessment Weights (14 Weeks)

Component	Weight
Problem sets (five, best four count)	15%
Midterm examination	25%
Team project (written plus presentation)	25%
Final examination	30%
Class contribution	5%

C.3 Eight-Week Intensive Plan

The 8-week format suits an accelerated or executive course, with one four-hour session per week or two two-hour sessions. It cannot cover all 24 chapters in equal depth, so it designates **core chapters** (taught fully) and **reading chapters** (assigned for self-study with a brief class discussion). Core chapters are 3, 4, 6, 7, 9, 10, 11, 15, 16, 18, 19, 21, and 22. Reading chapters are 1, 2, 5, 8, 12, 13, 14, 17, 20, 23, and 24.

Week 1: Foundations (Chapters 1, 2, 3)

Frame value maximisation and agency, read the financial statements quickly (a pre-class self-study with a short quiz), and teach discounting thoroughly. Students leave able to compute present values, annuities, effective rates, and loan schedules.

Week 2: Bonds and Investment Decisions (Chapters 4, 6, 7)

Price bonds and measure interest rate risk; adopt NPV as the decision rule and compare it with IRR; build a project cash flow forecast with taxes, depreciation, and working capital. Chapter 5 is assigned as reading; students apply the dividend discount model in a short exercise.

Week 3: Risk and Return (Chapters 9, 10, with 8 as reading)

Teach diversification, the efficient frontier, and CAPM. Use Chapter 8 reading to introduce sensitivity analysis and real options in a brief class discussion rather than full technical development. Quiz covering Weeks 1 to 3 at the start of Week 4.

Week 4: Cost of Capital and Capital Structure (Chapters 11, 15)

Estimate WACC and project rates, then examine how leverage affects value and risk. Include a short discussion of market efficiency (Chapter 12 reading) as it bears on issuing decisions and the cost of equity. Midterm exam (take-home or in-class, 90 minutes) covering Weeks 1 to 4.

Week 5: Financing and Payout (Chapters 16, with 13 and 14 as reading)

Cover dividends and repurchases in class. Students read the financing chapters, then complete a short instrument-choice exercise: given a firm's profile, justify equity, bank debt, bonds, or convertibles.

Week 6: Options and Derivatives (Chapters 18, 19, with 20 as reading)

Teach payoffs, put-call parity, binomial and Black-Scholes pricing, and the logic of forwards, futures, swaps, and hedge ratios. Exchange-rate exposure and the parity conditions in Chapter 20 are assigned as reading and examined through a short case.

Week 7: Valuation and M&A (Chapters 21, 22)

Conduct a full DCF valuation with terminal value and multiples cross-check, followed by an M&A session on synergy, premium, payment method, and accretion/dilution. Teams begin integrating their project analysis.

Week 8: LBOs, Sustainability, and Presentations (Chapters 23, 24 as reading)

Teams present projects. A closing session uses a short LBO illustration and a discussion of sustainable finance and governance to show how valuation concepts extend to new settings. The final examination is a 120-minute cumulative test.

Suggested Assessment Weights (8 Weeks)

Component	Weight
Weekly problem sets and quizzes	20%
Midterm examination	25%
Team project (shortened version)	25%
Final examination	30%

Notes for Executive Audiences

Executive students usually come with strong judgement and limited tolerance for algebra. Replace some derivations with decision-focused cases drawn from their own firms. Emphasise interpreting outputs: what a WACC implies, why a hedge may create accounting volatility, what the market is saying through a multiple. Keep calculation practice in spreadsheets, and give a one-page "what would change my mind" prompt with each case.

C.4 Exam-Preparation Checklists

Use these checklists in the final week before an exam. A topic is "ready" when you can solve a problem on it without notes, explain the intuition in two sentences, and name the most common mistake.

Checklist 1: Foundations and Valuation (Chapters 1 to 8)

- Distinguish value maximisation from profit maximisation and explain at least three agency problems with a remedy for each.
- Compute and interpret liquidity, turnover, margin, leverage, and return ratios; carry out a three-factor DuPont decomposition.
- Build a free cash flow from financial statement data.
- Compute present values and future values for single sums, annuities, growing annuities, and perpetuities, including cases with non-annual compounding.

- Convert between nominal, effective, and real rates.
- Construct a loan amortisation schedule and identify the interest and principal in a given payment.
- Price a bond, find its yield to maturity, and estimate the price effect of a yield change using duration and convexity.
- Explain the relationship between bond prices, coupons, and yields, including why prices of long-maturity, low-coupon bonds are most sensitive.
- Value a stock with constant-growth and two-stage dividend models; decompose price into assets-in-place and growth opportunities.
- Compute NPV, IRR, payback, discounted payback, and profitability index; explain when IRR misleads (multiple IRRs, mutually exclusive projects, scale and timing differences).
- Choose between projects of unequal lives using equivalent annual annuity.
- Develop incremental after-tax cash flows, including depreciation tax shields, working capital, salvage, and opportunity costs; ignore sunk costs and financing flows.
- Run sensitivity, scenario, and break-even analysis and interpret operating leverage.
- Value a simple option to delay, expand, or abandon using a decision tree or a binomial approach.

Checklist 2: Risk, Return and Cost of Capital (Chapters 9 to 12)

- Calculate expected return, variance, standard deviation, covariance, and correlation from a probability table or a return history.
- Compute two-asset portfolio risk and return, and show how correlation alters the benefit of diversification.
- Distinguish systematic from idiosyncratic risk and explain why only the former earns a premium.
- Locate the minimum-variance portfolio and the tangency portfolio; use the capital allocation line.
- Estimate beta by regression or from covariance, and compute required return with CAPM.
- Compute Sharpe ratio, Treynor ratio, and Jensen’s alpha and say what each concludes.
- Explain what multifactor models add and what the Fama-French factors represent.
- Compute WACC with market-value weights and after-tax debt costs; justify using a project-specific rate.
- Unlever and relever betas using the Hamada relation; state the assumptions behind it.
- Estimate the cost of equity by three methods and explain why they differ.
- Define weak, semi-strong, and strong market efficiency and describe the type of evidence for each.
- Perform and interpret a simple event study.
- Explain at least four behavioural biases and the limits to arbitrage that allow mispricing to persist.

Checklist 3: Financing and Policy (Chapters 13 to 17)

- Compute pre- and post-money valuations, ownership shares, and dilution through a venture round.
- Calculate IPO proceeds, underwriting costs, and underpricing; explain why underpricing occurs.
- Compute the ex-rights price and the value of a right.
- Compare bank loans, public bonds, private placements, leases, and convertibles on cost, flexibility, covenants, and risk.
- Value a convertible bond using conversion value and straight-bond value.
- Evaluate a refunding decision or a lease-versus-buy decision with after-tax cash flows.

- State Modigliani-Miller propositions with and without taxes and compute the value of the interest tax shield.
- Describe trade-off and pecking-order theories and the evidence for each.
- Analyse the EPS effects of alternative financing plans and find the break-even EBIT.
- Explain dividend irrelevance, signalling, clientele, and taxes; compare dividends and buybacks in a numerical example.
- Compute the cash conversion cycle and the effect of a policy change on investment in working capital.
- Calculate the cost of forgoing a trade discount and compare it with bank borrowing.
- Evaluate a credit policy change using incremental contribution, receivables investment, and bad debts.

Checklist 4: Options, Hedging and Valuation (Chapters 18 to 24)

- Draw payoffs and profits for calls, puts, and common combinations.
- Apply put-call parity to find an arbitrage or a missing price.
- Price an option with a one- or two-step binomial tree and by the Black-Scholes formula.
- Interpret delta, gamma, vega, and theta.
- Explain equity as a call option on the firm's assets and warrants as dilutive options.
- Price a forward using cost of carry; compute the variance-minimising hedge ratio and number of futures contracts.
- Value a plain-vanilla interest rate swap and explain how swaps convert exposures.
- Use covered interest parity and purchasing power parity; identify triangular arbitrage.
- Compare forward, money-market, and option hedges of a foreign-currency receivable.
- Value an enterprise with free cash flow to the firm, a terminal value, and a bridge to equity value per share.
- Cross-check a DCF with comparable multiples and explain the sources of divergence.
- Estimate synergies, the premium paid, and who captures value in cash and stock deals; compute accretion or dilution and explain why it misleads.
- Build a simple LBO sources-and-uses table and compute MOIC and IRR; attribute returns to EBITDA growth, multiple change, and debt paydown.
- Explain recovery in bankruptcy under absolute priority and when restructuring beats liquidation.
- Discuss how carbon pricing, green financing, governance, and fintech unit economics affect valuation.

General Examination Strategy

1. **Read the whole exam first** and mark the questions you can answer quickly. Begin with them to secure marks and confidence.
2. **State assumptions explicitly.** When a problem is ambiguous (timing of cash flows, whether to use book or market weights), write your assumption and proceed; partial credit often depends on it.
3. **Show formulas before numbers.** A correct setup with an arithmetic slip earns more than a bare wrong answer.
4. **Check units and periods.** Match the rate period to the cash-flow period and keep nominal with nominal, real with real.
5. **Sanity-check results.** A bond priced above par with a yield above its coupon, or a beta of 15, signals an error.
6. **Interpret.** Close each calculation with a sentence on what it means for the decision. Many exams reward the recommendation as much as the arithmetic.

7. **Manage time** in proportion to marks, and leave a few minutes for review.

Common Errors to Avoid

Error	Correction
Discounting flows at a rate that does not match their frequency	Convert the rate to the period of the cash flows
Including sunk costs or interest payments in project cash flows	Use incremental, after-tax, unlevered cash flows; capture financing in the discount rate
Using book-value weights in WACC	Use target market-value weights
Applying the firm's WACC to a project of different risk	Use a rate that reflects the project's own risk
Using IRR to choose between mutually exclusive projects	Use NPV, or incremental IRR with care
Treating EPS accretion as proof of value creation	Compare price paid to stand-alone value plus achievable synergy
Forgetting that debt raises equity risk even when it lowers WACC mechanically	Remember the equity cost rises with leverage
Mixing continuously compounded and discrete rates	Convert explicitly
Ignoring working capital recovery at project end	Add the release of working capital to the terminal year
Describing hedging as profit-seeking	Describe it as variance reduction, with a clear policy

C.5 Using the Book in Class

Active-Learning Formats

Cold call and warm call. Assign a worked example in advance and ask a student to walk the class through it, announcing the order beforehand for anxious students.

Two-minute drills. At the beginning of a session, pose a single quantitative question that students answer individually, then compare with a neighbour. This reveals misconceptions quickly.

Decision memos. Replace some problem sets with a one-page memo addressed to a named fictional executive, requiring a recommendation, a computation, and a risk. This builds the communication skill that distinguishes leaders from analysts.

Debates. Useful topics include whether a firm should pay dividends or repurchase shares, whether to hedge, whether market prices can be trusted, and whether green financing creates value. Assign sides at random to force students to argue positions they may not hold.

Spreadsheet labs. Dedicate part of a session to building a model live: an amortisation schedule, a capital budget, a WACC, a DCF, or an LBO. Provide a clean template and require students to audit one another's formulas.

Case Method

Each chapter's case study can anchor a class discussion. A workable sequence is: five minutes to frame the decision, fifteen to establish the facts and calculations, twenty to debate alternatives, and ten to extract lessons. Encourage students to separate **what the numbers say** from **what the numbers leave out**, such as strategic fit, execution risk, and stakeholder reactions.

Differentiating for Mixed Backgrounds

Pair students with financial and non-financial backgrounds for problem sets, rotating partners each week. Provide optional "bridge" exercises on accounting and statistics for those who need them and "stretch" tasks, such as adding a real option to a capital budget, for those who want more. Hold weekly office hours devoted to spreadsheet troubleshooting.

C.6 Team Project Template

The team project integrates the course into a single piece of analysis. Teams of four or five students act as an advisory group to the chief financial officer of a company, real (using public information only) or fictional (using a data pack supplied by the instructor). The deliverables are a written report of 3,000 to 4,000 words with a spreadsheet model appended, and a 15-minute presentation followed by questions.

Project Options

1. **Investment and financing recommendation.** Evaluate a major expansion, then recommend how to finance it.
2. **Valuation and strategic options.** Value the firm, test the sensitivity of value to key assumptions, and recommend whether to acquire, divest, or remain independent.
3. **Capital structure and payout review.** Diagnose the existing structure and payout policy, benchmark against peers, and recommend changes.
4. **Risk management programme.** Identify the firm's exposures, quantify them, and design a hedging policy.
5. **Transaction analysis.** Assess a merger or leveraged buyout, including synergy, price, financing, and returns.

Required Structure of the Report

1. **Executive summary (one page).** The decision, the recommendation, the key numbers, and the main risks. A reader should be able to act on this page alone.
2. **Company and context.** Business model, strategy, industry position, and the decision at hand. Include a short ratio analysis using Chapter 2 tools.
3. **Analysis.** The core quantitative work, with every assumption stated. Depending on the option, this includes cash flow forecasts, cost of capital, valuation, capital structure analysis, hedging calculations, or transaction modelling.
4. **Uncertainty and alternatives.** Sensitivity and scenario analysis, a discussion of real options, and at least one realistic alternative to the recommendation, evaluated on the same basis.
5. **Implementation and risks.** How the recommendation would be executed, timing, governance requirements, and what monitoring signals would trigger a change of course.
6. **Appendices.** Model documentation, data sources, and detailed tables.

Model Requirements

The spreadsheet should separate inputs, calculations, and outputs; include a sheet listing every assumption with its source or rationale; use no hard-coded numbers inside formulas; and include an error check, such as a balance check or sources-equal-uses test. Teams should be able to change one assumption and see all results update.

Milestones

Milestone	Timing (14-week course)	Content
Proposal	Week 6	Project option, company, data plan, team roles (one page)
Draft analysis	Week 10	Key models and preliminary results
Draft report for peer review	Week 13	Complete draft exchanged with another team
Final report and presentation	Week 14	Polished submission and presentation

In the 8-week course, shorten the report to 2,000 words, reduce the model scope, and set the proposal in Week 3 and the presentation in Week 8.

Team Project Rubric

Each criterion is scored from 1 (unsatisfactory) to 5 (excellent) and multiplied by its weight.

Criterion	Weight	Excellent (5)	Satisfactory (3)	Unsatisfactory (1)
Problem framing	10%	Decision, objective, and constraints crisply defined; value criterion explicit	Decision identified but objective or constraints vague	Decision unclear or confused with description
Technical accuracy	25%	Correct methods, consistent assumptions, accurate calculations	Mostly correct with minor errors or inconsistencies	Fundamental errors such as wrong discount rate logic
Use of data and assumptions	15%	Assumptions sourced, reasonable, and tested	Assumptions stated but weakly justified	Assumptions missing or unrealistic
Treatment of risk and uncertainty	15%	Sensitivity, scenarios, and options integrated into the recommendation	Sensitivity done but not linked to the decision	Risk ignored or only qualitative
Quality of recommendation	15%	Clear, actionable, and consistent with the analysis; alternatives compared	Reasonable but under-argued	Missing or contradicts analysis
Communication, written	10%	Concise, well-organised, professional tables	Understandable but uneven	Disorganised or hard to follow
Presentation and questions	10%	Confident, within time, handles questions with evidence	Adequate delivery, partial answers	Poor structure, unable to defend work

A total score is computed as the weighted sum divided by five, giving a percentage. Instructors may adjust individual marks using a peer-evaluation form in which each member rates others’ contribution; a common approach is to limit adjustments to plus or minus ten percentage points.

Guidance on Using Real Companies

Use only public information, such as annual reports and market data, and state the date of the data. Avoid inventing facts about a real firm; where information is unavailable, say so and state an assumption. Teams must treat their work as an academic exercise, not investment advice, and must not claim to represent the views of the company studied.

C.7 Suggested Reading Categories

The following categories point to the types of sources that deepen each part of the course. No specific titles are listed here; instructors and students should consult library catalogues and current syllabi for up-to-date editions and recommendations.

Foundational corporate finance textbooks. Standard MBA and graduate texts offer alternative explanations, additional problems, and broader coverage. Reading a second treatment of a difficult topic often unlocks it.

Valuation handbooks. Practitioner-oriented books on enterprise valuation, multiples, and cash flow forecasting complement Chapter 21 and provide industry-specific guidance.

Asset pricing and portfolio theory. Graduate-level treatments of mean-variance theory, CAPM, factor models, and market efficiency help students who wish to understand the foundations and the empirical debates.

Original academic papers. Foundational contributions on capital structure and dividend irrelevance, portfolio selection, the capital asset pricing model, option pricing, agency theory, signalling, and prospect theory are accessible to motivated MBA students, and reading excerpts of the originals shows how the ideas were first argued.

Behavioural finance and decision-making. Popular and academic accounts of judgement under uncertainty, heuristics and biases, and the limits of arbitrage provide context for Chapter 12 and for corporate decision-making generally.

Fixed income and derivatives references. Specialist texts on bond markets, interest rate modelling, options, futures, and swaps support Chapters 4, 18, and 19, and show how markets work in practice.

Financial statement analysis and accounting. Guides to interpreting statements, quality of earnings, and accounting standards help students apply Chapter 2 and avoid misreading reported numbers.

Mergers, acquisitions, and private equity. Works that combine deal mechanics, valuation, negotiation, and empirical evidence on acquisition outcomes and buyout returns enrich Chapters 22 and 23.

Corporate governance and sustainable finance. Regulatory codes, stewardship principles, standards for sustainability disclosure and reporting, and academic evidence on ESG and cost of capital are relevant to Chapter 24. Because standards and regulations change quickly, consult the current versions published by the relevant bodies.

Case collections. Case books from business schools provide realistic, ambiguous decisions for discussion and can be combined with the chapter cases here.

Data sources and tools. Company annual reports and regulatory filings, central bank and statistical agency publications, and market data providers supply the inputs for projects. Spreadsheet and statistical software manuals help with modelling.

Financial press and professional journals. Regular reading of business newspapers, magazines, and practitioner journals links theory to events. Encourage students to bring one current news item per week and classify it as an investment, financing, or payout or risk decision.

How to Read Academic and Professional Sources Critically

When reading any source, ask five questions. What is the claim? What evidence or reasoning supports it? What are the assumptions, and do they apply to the problem at hand? Who benefits from the conclusion? What would a sceptic say? Treat statistics found in secondary sources with caution, check the date and the definition, and verify against primary data when the figure matters to a decision.

C.8 Self-Study Roadmap

Independent learners can use the following sequence. First, read Chapters 1 to 3 and make sure you can discount cash flows fluently; use a spreadsheet to reproduce every worked example. Second, read Chapters 4 to 8, then try capital budgeting on a personal or small-business decision. Third, complete Chapters 9 to 12, and estimate the cost of capital of a listed company, noting every judgement you make. Fourth, study Chapters 13 to 17, and read the financing section of a company's annual report to identify its capital structure and payout policy. Fifth, study Chapters 18 to 20 and write one paragraph explaining why a firm in your industry would or would not hedge. Sixth, finish Chapters 21 to 24 by valuing a company and writing a one-page recommendation on a hypothetical acquisition.

After each chapter, spend ten minutes on three tasks: write the key formula from memory, restate the main idea in two sentences, and list one situation where the idea would not apply. Return to Appendix A before exams, and to Appendix B whenever a term is unfamiliar.

C.9 Instructor Notes on Pacing and Common Difficulties

Several topics reliably cause difficulty and merit extra time. **Compounding conventions** confuse students who have not met continuous rates; spend a short session converting among nominal, effective, and continuous rates. **Cash flow forecasting** is a skill, not a formula; model one forecast live and let teams repeat it. **Beta and WACC** involve many moving parts; give a complete numerical walk-through from regression output to final rate. **Capital structure theory** is abstract; anchor it in a numerical example with and without taxes before introducing distress costs. **Option pricing** benefits from the binomial tree before Black-Scholes, since the tree reveals the logic of replication and risk-neutral valuation. **M&A** invites enthusiasm for accretion; consistently return to the question of who pays for, and who receives, the synergies.

Throughout the course, keep returning to the book's central message: finance is a toolkit for deciding where to invest, how to fund investment, and how to manage the risks and returns that follow. The tools are quantitative, but the decisions are about people, incentives, and judgement under uncertainty.